Apprehensional constructions in a cross-linguistic perspective

Edited by

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Chapter 1

Introduction: Apprehensional constructions in a cross-linguistic perspective

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In this introduction, we discuss recurrent issues of interest raised in the contributions to this volume, and in the wider literature, with respect to apprehensional constructions, broadly defined as constructions which convey a possibility that is also undesirable. The domain of apprehensionality is taken here to encompass constructions at various levels, including main clause modal constructions (apprehensives), subordinate clause constructions akin to English *lest*-clauses (precautioning clauses), and 'fear' predicates, which are often found to overlap in their formal manifestation with precautioning clauses. Apprehensional case markers (timitive) are included here only insofar as their marking may overlap with that of precautioning subordinate clauses. We discuss the frequently ambiguous syntactic status of apprehensional clauses and link it to their potential syntactic or pragmatic relationship with a second preemptive clause, encoding an action to be taken to avoid the potential undesirable event. We further outline some interesting language-specific constraints attested for constructions in the apprehensional domain, including constraints on the combination with specific grammatical persons, tense/aspect values, and negation. We also review various semantic and pragmatic issues like the status of the undesirability component (conventional or context-dependent, i.e. a pragmatic implicature), and the performative nature of apprehensionals, which links to the well-documented overlap of apprehensionals with prohibitives or optatives in a number of languages. Finally, we examine the remarkable diversity of diachronic origins and patterns of polyfunctionality attested for apprehensionals. This and, more generally, their cross-linguistic semantic and grammatical diversity, can be taken as a correlate of their particularly complex semantics and the interactional nature of their use.

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1 Introduction

1.1 Apprehensionality

The functional domain of APPREHENSIONALITY, broadly speaking, encompasses grammatical markers and constructions which conventionally encode, or pragmatically implicate, that the situation described by the clause is an undesirable possibility. As such, apprehensionality is a subtype of modality. We use the cover term *apprehensional* for a variety of such constructions, illustrated with examples from Northern Australian Kriol, Kalaallisut and English below.[1]

(1) Kriol (English-lexified creole; Northern Australia; Hodgson 2017)
Flowa im wetwan. ***Bambai*** *yu slip en boldan.*
floor is wet APPR 2SG slip and fall
'(Oh no baby girl, don't run around inside!) The floor is wet. You might slip and fall.'

(2) Kalaallisut (Eskimo-Aleut; Greenland; Fortescue 1984: 27)[2]
*Nakka-**qina**-vutit.*
fall-APPR-2SG.IND
'[Be careful], do not fall!'

(3) *He tied his daughter's laces,* ***lest*** *she fall.*

Apprehensionals have been labelled "apprehensive", "(ad)monitive", "preventative", "evitative", "timitive", "precautioning" or "*lest*-markers", among other terms. Apprehensional clauses are often used to alert or warn the addressee of a potentially undesirable situation as in (1) and (2). They may be accompanied by a clause describing a way to avoid the undesirable situation, like 'don't run inside' in the context preceding (1) or 'he tied his daughter's laces' in (3). Following Evans (1995: 265), we label such clauses "preemptive clauses".

Although wide-spread cross-linguistically, apprehensional markers have received little attention in the linguistics literature. Until recently, their discussion has been mostly restricted to brief sections in reference grammars and other language-specific discussions of some lesser studied languages (e.g. in languages from the Australian, Papuan, South Pacific, and Amazonian areas). There exist

[1] Croft (2022: 1333) uses the term "apprehensional" to refer to a semantic relation between two clauses, which actually represents only a subdomain of the functional domain of apprehensionality as defined above. The subdomains are detailed in §1.2.

[2] Cited in Dobrushina (2006: 31).

very few works that address the phenomenon from a cross-linguistic perspective. With the exception of the seminal paper by Dobrushina (2006), these works tend to focus on an individual language, a small set of languages or a language family (Lichtenberk 1995, Pakendorf & Schalley 2007, Vuillermet 2018b), or a specific subtype of apprehensional (as in the discussion of subordinate 'lest' clauses in direct comparison with purpose clauses in Schmidtke-Bode 2009: 129–144). In contrast to grammatical markers of desirability and volitionality, which are discussed alongside general possibility and necessity modals, markers of undesirability only receive passing mention – if at all – in works on modality from a typological-functional perspective (Bybee et al. 1994: 211; Palmer 2001: 22) and have mostly been ignored entirely in the formal semantic literature on modality until very recently (AnderBois & Dąbkowski 2021, Phillips 2021, Tahar 2021, Dąbkowski & AnderBois 2023). Even more strikingly, the very existence of apprehensionals has only very recently been acknowledged for colloquial varieties of well-studied languages such as Dutch (Boogaart 2009, 2020) and German (Angelo & Schultze-Berndt 2018, Faller & Schultze-Berndt 2018, Fenk-Oczlon 2018). The description, translation and teaching of all languages as well as theories of modality of all traditions therefore stand to benefit from cross-linguistically informed explorations of this category.

The present volume aims to showcase the variety of linguistic manifestations of apprehensionality, and explore the parameters by which this broader domain can be subcategorised. It includes 12 selected papers from two workshops on the topic of apprehensionality, one at the 2018 European Linguistics Society meeting,[3] co-organised by Schultze-Berndt and Faller, and the other at the 2018 Syntax of the World Languages conference,[4] co-organised by Vuillermet and Schultze-Berndt. Each of the papers responds to a set of issues that were raised during the workshops, and presents in-depth discussions of apprehensionals in an individual language, language group or area, by an expert on these languages and based on fieldwork or corpus data. The languages discussed in this volume represent all major linguistic areas and the following language families/subgroups: Dene, A'ingae, Mande, Semitic, Nakh-Daghestanian, Slavic, Sino-Tibetan (with Kiranti and Sinitic subgroups), Korean, Oceanic, and Pama-Nyungan. In addition, we have included a questionnaire designed by Vuillermet for the in-depth exploration of variation within apprehensional constructions (based on Vuillermet 2018c). The volume is aimed at typologists, semanticists interested in modality, and general linguists, especially those engaged in fieldwork-based research on lesser studied languages.

[3] https://societaslinguistica.eu/wp-content/uploads/2020/03/SLE-2018-Book-of-abstracts.pdf
[4] https://swl8.sciencesconf.org/resource/page/id/6.html

Some of the main issues addressed by the papers in this volume will be introduced below: §1.2 is a brief and general presentation of the apprehensional subtypes, while §2 more specifically targets morphosyntax, §3 semantics, and §4 diachrony. §2.1 expressly explores the complexity of determining the syntactic status of an apprehensional clause, and, consequently, the exact function(s) encoded by the apprehensional morpheme or construction. §2.2 investigates the interactions of apprehensionals in general with other grammatical features, especially grammatical person. §3.1 draws attention to the varying degree of semanticisation for all apprehensional (sub)functions, while §3.2 and §3.3 respectively explore the type(s) of modality instantiated by apprehensives, and their different pragmatic functions. §4 examines the diversity of the diachronic sources for apprehensionals. Finally, §5 is a brief summary of the highlights of each contribution.

1.2 Apprehensionality subtypes

We use APPREHENSIONAL as the cover term for grammatical markers or constructions of the entire functional domain, with the definition given in §1.1: grammatical markers or constructions which conventionally encode or pragmatically implicate that the situation described by the clause is an undesirable possibility. We refer to apprehensionals that conventionally encode undesirability as DEDICATED, and those that only pragmatically imply it as NON-DEDICATED. Most authors in this volume adopt this definition and the terminological choices summarised in Table 1 and discussed in detail below.

A subdivision that has been influential for existing descriptions of apprehensional constructions is Lichtenberk's (1995), who distinguishes between three broad construction types. The first type, which he terms APPREHENSIONAL-EPISTEMIC, is typically realised as a construction at the level of the main clause. The other two are realised at the level of subordinate constructions. These in turn are subdivided into PRECAUTIONING ADVERBIAL CLAUSES and COMPLEMENT CLAUSES OF 'FEAR' PREDICATES. Lichtenberk's discussion centres on the Oceanic language To'abaita, where according to his description, the same marker occurs in all three functions, as illustrated in (4).[5]

[5]We have slightly changed the glosses in some of the examples for the sake of clarity. Lichtenberk uses the gloss LEST while we use APPR here as a more general label for the apprehensional construction available in To'abaita.

Table 1: Terminology for the apprehensional constructions and their subtypes, alternative format

	DEDICATED APPREHENSIONAL (undesirability semantically encoded)	NON-DEDICATED APPREHENSIONAL (undesirability pragmatically implied)
MAIN CLAUSE	**Apprehensive**	**Possibility modal**
	Specialised marker Transparent neg. optative	
ADVERB. CLAUSE	**Precautioning**	**Possible consequence**
	Negative purpose Specialised marker Transparent negative purpose *Possible undesirable consequence* ('in-case')	
COMPL. CLAUSE	**'Fear' complementiser**	**General complementiser**

(4) To'abaita (Austronesian: Oceanic; Solomon Islands; Lichtenberk 1995: 296–8)

a. Apprehensional-epistemic

Ada *wane 'eri ka riki nau.*

APPR man that he:SEQ see me

'[I fear] the man might see me.'

b. Precautioning (negative purpose subfunction)

Nau ku agwa 'i buira fau ***ada*** *wane 'eri ka riki nau.*

I I:FACT hide at behind rock APPR man that he:SEQ see me

'I hid behind the rock so that the man might not see me / lest the man see me.'

c. Complement of ‘fear’ predicate
Nau ku ***ma’u*** *‘asia na’a* ***ada*** *laalae to’a baa ki keka*
I I:FACT **be.afraid** very **APPR** later people that PL they:SEQ
lae mai keka thaungi kulu.
go hither they:SEQ kill us(INCL)
‘I am scared the people might come and kill us.’ / ‘I am scared lest the people should come and kill us.’

Languages like To’abaita, which display a single apprehensional marker for the three semantically related functions outlined above, are not infrequent. However, languages exhibiting a clear morphological distinction between main clause and subordinate apprehensional constructions also exist. For example, Ese Ejja, discussed in detail in Vuillermet (2018b), has different markers for the (main clause) apprehensive and the (subordinate) precautioning function as illustrated in (5a)–(5b) below. In addition, Ese Ejja also has a distinct apprehensional marker at the level of the noun phrase, i.e. a timitive (‘for fear of’) case marker/adposition, illustrated in (5c).

(5) Ese Ejja (Pano-Tacanan; Bolivia/Peru; Vuillermet 2018b: 270, 257)

a. Apprehensive
*A’a María wowi-jji, poki-**chana**!*
PROH María tell-PROH go-**APPR**
‘Don’t tell Maria, she might go!”

b. Precautioning
*Owaya ekowijji shijja-ka-ani [**e**-jja-saja-ki **kwajejje**].*
3ERG rifle clean-3A-PRS **PREC**-MID-block-MID **PREC**
‘He is cleaning his rifle [lest it get blocked].’

c. Timitive
*Akwi iña-kwe iñawewa=**yajjajo**!*
stick grab-IMP dog=**TIM**
‘Grab a stick for fear of the dog!’

Timitives are known to show overlap with other apprehensional markers. For instance, Dixon (2009: 74, 88) mentions the frequent use of what he calls aversive cases as complementisers of ‘fear’ predicates, and as precautioning subordinators in Australian languages. Timitives are excluded from the scope of this volume unless they show such formal overlap, as is the case e.g. in a number of Ngumpin-Yapa languages (Browne et al. 2026 [this volume]), in A’ingae (aka. Cofán; isolate; Dąbkowski & AnderBois 2026 [this volume]), and in Upper Tanana Dene

(Athabaskan-Eyak-Tlingit; Lovick 2026 [this volume]). The formal overlap is motivated by the fact that a timitive-marked entity can often be interpreted as metonymically standing for an undesirable event (e.g. in (5c): 'for fear of the dog attacking').

In line with wide-spread usage (e.g. Dobrushina 2006, Pakendorf & Schalley 2007, Angelo & Schultze-Berndt 2016, Vuillermet 2018b) we adopt the related term APPREHENSIVE for constructions that can occur at the main-clause level, i.e. Lichtenberk's (1995) apprehensional-epistemic use, corresponding to the Ese Ejja construction illustrated in (5a).[6] Apprehensives mark the proposition in their scope as both possible and undesirable, though whether the possibilities considered are epistemic, as claimed by Lichtenberk (1995), remains to be established rather than assumed, and may be language-specific. In some languages, apprehensives are part of a grammatical modal or mood paradigm (e.g. Ese Ejja, see (5a), or Kambaata (Afro-Asiatic: Cushitic), Treis 2026 [this volume]), while in others, they pertain to a different syntactic class. A negative marker may be part of a main-clause level construction to transparently indicate the speaker's desire for the situation described in the clause not to take place, in what can be described as a negative-optative apprehensional construction (see Wiemer 2026 [this volume]) on Slavic languages).

Where a language-specific construction pragmatically implicates rather than semantically encodes undesirability, a more general label, e.g. POSSIBILITY MODAL, may be more appropriate.

For subordinate apprehensional clauses within a complex sentence construction, corresponding to the Ese Ejja construction illustrated in (5b), we will adopt Lichtenberk's (1995) term PRECAUTIONING clause. Like apprehensives, precautioning clauses convey both possibility and undesirability of the situation described, but require in addition the presence of a PREEMPTIVE clause, which encodes an action to be taken or omitted in order to avoid the undesirable event expressed in the precautioning clause. (We deliberately chose Evans 1995: 264 term *preemptive* over Lichtenberk's *precautionary clause* because the latter is easily confused with *precautioning*.) In these constructions, it is often a subordinator, e.g. English *lest*, that contributes the semantics of undesirable possibility. Preemptive clauses are frequently (according to Dixon 2009: 24, "most frequently") directives, i.e. commands or prohibitions. Preemptive clauses are also frequently found to co-occur with apprehensives as defined above, which can contribute to

[6] As will be discussed below (§2.1), the main clause status is not necessarily easy to establish, and there may be non-syntactic, though orthogonal, relevant criteria (like the necessary speaker's perspective) to distinguish apprehensive from precautioning uses.

blurring the boundary between main clause apprehensive and subordinate precautioning clauses (see §2.2).

Precautioning clause is in turn a cover term for NEGATIVE PURPOSE adverbial clauses (Lichtenberk's 1995: 298 *avertive clauses*[7]) and 'IN-CASE' adverbial clauses, following Lichtenberk's (1995: 298) term. Negative purpose clauses are characterised by having a direct causal link with the preemptive clause, and, consequently, the realisation of the preemptive action has an impact on the non-realisation of the undesirable event. For example, in (6), not climbing up likely avoids the consequence of falling (from the wall or the tree) altogether. By contrast, a causal link is not required between an 'in-case' precautioning clause and the preemptive clause: the undesirable event may happen anyway, but the preemptive action rather targets the (implicit) consequences of the event encoded by the 'in-case' clause. In (7), the undesirable event avoided by the realisation of the preemptive event is not the lightning strike itself but the potential harm to the climber. As shown by these examples, *lest* is acceptable in both types of clauses, whereas the construction *so that … not* transparently marks negative purpose only.

(6) Negative purpose precautioning clauses: causal link
   a. *Don't climb up there **lest** you fall!*
   b. *Don't climb up there **so that** you do**n't** fall!*

(7) 'In-case' precautioning clauses: no causal link
   a. *She didn't climb up the tree **lest/in case** lightning struck.*
   b. # *She didn't climb up the tree **so that** lightning would **not** strike.*

Note that the presence or absence of a causal relationship between the preemptive clause and the precautioning clause is related to the overall (lack of) preventability of the event: assuming that preventing lightning itself is impossible, taking precautions may at most avoid the undesirable potential consequences of this event. A number of contributions to this volume discuss the semantic and pragmatic relationship between negative purpose and 'in-case' clauses for languages where both can be encoded by the same marker, e.g. *pen* in Classical Hebrew (Afro-Asiatic: Semitic; Francez 2026 [this volume]), *sa'ne* in A'ingae

[7]We avoid the term *avertive* for the negative purpose subtype (used e.g. by Lichtenberk 1995, Schmidtke-Bode 2009, Vuillermet 2018b), since it has also been adopted for the unrelated function of expressing an action that is narrowly averted (e.g. *almost*, *be about to*) (e.g Kuteva 2000, Kuteva et al. 2019). We retain Lichtenberk's term '*in-case*' for the other clause type (an alternative is "possible undesirable consequence") despite the fact that the English subordinator *in case* is not restricted to undesirable events (see below).

(aka. Cofán; isolate; Dąbkowski & AnderBois 2026 [this volume]), *-lkut* in Archi (Nakh-Daghestanian; Daniel & Dobrushina 2026 [this volume]), *tiple* in Mwotlap (Austronesian: Oceanic; François 2026 [this volume]); see also§2.2 in Vuillermet 2026 [this volume]'s questionnaire.

Negative purpose adverbial clauses, in turn, can be transparently marked as negative or employ a specialised marker like English *lest* (Schmidtke-Bode 2009: 130). If the subordinate clause type under investigation does not entail undesirability, a more general term than precautioning may be appropriate, e.g. POSSIBLE CONSEQUENCE (cf. Dixon 2009: 2) for clauses comparable to *in case* clauses in English (7a).[8]

Clauses serving as the complement of 'fear' verbs can be described transparently as 'FEAR' COMPLEMENTS (the 'fear' function in Lichtenberk's 1995: 297 terms). Here the undesirability reading of the subordinate clause is also conveyed by the semantics of the main clause predicate. The reason for including these in the apprehensional domain is that in many languages the same subordinator as in negative purpose or precautioning 'in-case' clauses may be used, as illustrated for English *lest* in (8) and for To'abaita *ada* in (4c). Very often a general complementation construction is used alongside or instead of an apprehensional construction in complements of 'fear' verbs, as illustrated in (8) for English and discussed for Ngumpin-Yapa languages by Browne et al. (2026 [this volume]).

(8) *Tom was worried* ***lest*** *he fail /* ***that*** *he might fail the exam.*

2 Morphosyntactic aspects

2.1 Syntactic status of the apprehensional clause

In some languages the syntactic status of the apprehensional clause is straightforwardly identified. Example (5) showed that three of the main apprehensional functions discussed in §1.2 align with clearly differentiated syntactic constructions and different morphological marking in Ese Ejja: The Apprehensive[9] *-chana* is a main clause modal marker, the Precautioning *e-…kwajejje* is a subordinate construction, and the timitive clitic *=yajajjo* is a case marker (Vuillermet

[8] Note that 'in-case' clauses of possible consequence are semantically distinct from conditional clauses (pace Schmidtke-Bode 2009: 136) – *in case* in English covers both functions. The semantic distinction is formally reflected in other languages, cf. the two German equivalents *im Fall, dass* 'if, in the case that' for conditionals and *für den Fall, dass* 'for the case that' for possible consequence clauses.

[9] We capitalise the labels of individual language-specific morphemes.

2018b). However, for many languages the identification of the syntactic status of an apprehensional construction is a thorny issue, as already discussed in the literature (e.g. Austin 2021: 230). Language-specific descriptions often remain obscure about the reasons for identifying an apprehensional marker e.g. as a subordinate marker or mood marker. Moreover, a segmental marker of subordination may be absent (in which case the semantic relationship between the two clauses also remains underspecified). Verstraete (2006b: 219) emphasises that the absence of a relational marker does not exclude "two clauses [to be] actually integrated via intonation, with intonational information taking over from segmental information as a relational marker." As a matter of fact, the role of intonation in creating clausal dependencies is typically underreported in grammatical descriptions.

One of the difficulties for distinguishing between main and subordinate apprehensional clauses more specifically is that their semantics frequently, possibly even typically, triggers the presence of another (preemptive) clause that encodes an action (to be) taken to avoid the undesirable consequence encoded or implied by the apprehensional clause. Thus, in language-specific discussions of apprehensionals, their subordinate status is often linked to the (near-)obligatory presence of a preemptive clause (e.g. Smith-Dennis 2021). This is however not sufficient, as illustrated by the Kriol example in (1): While a preemptive clause is present, and the apprehensional marker *bambai* has its origin in a temporal connective, Angelo & Schultze-Berndt (2016) show that Kriol apprehensional clauses have the status of main clauses, and that *bambai* is a sentence connective in both its temporal and its apprehensive function. Thus *bambai* serves to establish a pragmatic (rather than syntactic) dependency between the preemptive clause (if present) and the apprehensional clause, without being a subordinator (see François 2026 [this volume] for a similar argument for Mwotlap). Similarly, the Ese Ejja example (5a) shows that the mere presence of a preemptive clause cannot be a sufficient criterion for the subordinate status of the apprehensional clause. The Ese Ejja Apprehensive -*chana* is syntactically a main clause marker for two reasons: it belongs to the obligatory main clause tense/mood paradigm, and only the main clause pronoun sets are available with this paradigm (Vuillermet 2018b: 267–268).

The appearance of subordination may also arise from the use of a main clause apprehensional construction (i.e. an apprehensive) in a reported speech/thought construction, as in 'She did not go to the party, saying "I might get bored"'. This construction is illustrated in (9) for Tariana. Similar examples of reported speech constructions are discussed by Daniel & Dobrushina (2026 [this volume]) for Mehweb (aka. North Central Dargwa, Nakh-Daghestanian), Rhee & Kuteva

(2026 [this volume]) for a number of the Korean apprehensional constructions discussed in their paper, and Jacques (2021: 1170) for Japhug (Sino-Tibetan: Gyalrongic).

(9) Tariana (Arawakan; Brazil/Colombia; Aikhenvald 2003: 384)
[*Nesu-**da*** *na-swe* *di-**a-ka***]
3PL+shit-**VIS.APPR** 3PL-stay+CAUS 3SG.NF-**say-SUB**
di-piti-naka.
3SG.NF-drive.away-PRS.VIS
'He is driving (the hens) away for fear they might shit.' (lit. '… saying they might shit.')

Even apparent complements of 'fear' verbs may in reality be syntactically independent clauses. According to Tsunoda (2011: 619), a Warrongo speaker translated the English prompt 'The woman is afraid that her child might be drowned' with the utterance in (10) (glosses adjusted).

(10) Warrongo (Pama-Nyungan; Australia; Tsunoda 2011: 619)
Warrngo wanbali-n galbin jojorri-ngga.
woman fear-NFUT child be.drowned-APPR
'The woman is afraid – the child might be drowned.'

The translation that we provided with (10) reflects the observations that the verb *wanbali-* 'fear, be afraid' can occur in an intransitive clause without a complement (Tsunoda 2011: 401), that the Warrongo Apprehensive marker is found in independent main clauses (Tsunoda 2011: 286–287), and that there is no marker of subordination in such complex constructions.

In other cases, there may be genuine arguments for claiming that an apprehensional construction has subordinate status while not showing many signs of deranking (i.e. formal correlates of subordination),[10] or that the same construction can alternatively have main clause and subordinate clause status, as discussed in some detail by Lichtenberk (1995) and Dobrushina (2006). This syntactic ambiguity can come about through syntactic tightening of two juxtaposed clauses (see Wiemer 2026 [this volume]). Conversely, a main clause use of an apprehensive marker could arise through insubordination of a subordinate precautioning clause (an incipient case is arguably the English expression *lest we forget*), with potentially formal remnants of such an origin (the subjunctive in the case of *lest*).

[10] Schmidtke-Bode (2012: 439) observes that cross-linguistically, subordinate precautioning/negative purpose clauses show fewer hallmarks of de-ranking than positive purpose clauses.

For example, the Apprehensive in To'abaita illustrated in (4a) does not belong to the same paradigm as other moods and still displays a subject/tense marker from the sequential paradigm, typical of subordinate clauses (Lichtenberk 1995: 296).

Apart from general language-specific correlates of subordination, one potential semantic criterion for disentangling the two syntactic levels is the judge of undesirability: The judge in main clause apprehensionals is always the speaker, the speaker-addressee dyad, or another contextually determined judge (Green 1989: 164–170, Fedotov 2026 [this volume]; François 2026 [this volume], Vuillermet 2018b: 272–273, Vuillermet 2026 [this volume]: §2.3).[11] By contrast, in a precautioning clause, the judge has to be the subject of the main clause. In (3), the judge is the father who tied his daughter's laces, and in (5b), it is the hunter cleaning his rifle. One exception to the generalisation that the judge of undesirability in a precautioning clause has to be the subject of the main clause arises when this main clause is a directive. In ***Don't swim*** *in this lake, lest the crocodiles eat you!* or *Put your jacket on, lest you catch a cold!*, the judge is typically the speaker who utters the directive.

As already mentioned above, if a language does not have a precautioning construction, the functional need for assigning the judge role to someone other than the speaker can be met in languages like Tariana by using an apprehensive clause in a reported speech construction as in (9): this binds the judgement of undesirability to the main clause agent whose thoughts or speech are reported.

In some of this volume's contributions, the syntactic status of the apprehensional clause can straightforwardly be allocated to a single construction type. Many of the contributions, however, are dealing with the use of the same construction in what appear to be both apprehensive and precautioning clauses. Some of these languages transparently use the reported speech strategy mentioned above, while others seem to have developed the independent from the dependent construction via an insubordination pathway (Dąbkowski & AnderBois for A'ingae; Daniel & Dobrushina for Archi, Francez for Hebrew). On the other hand, Wiemer cautions against the assumption of insubordination for Slavic languages, and François (cf. also François 2003) considers apprehensional clauses in Mwotlap to have a pragmatically rather than syntactically dependent status.

[11]See also the speaker's vs agent's perspective in Verstraete's (2006a: 68ff.) discussion of mood use in past intentional constructions. Verstraete (2006a: 68–69) also discusses grammaticalised "agent-binding" morphemes, which bind the modal feature (e.g. apprehensive) to the agent of the main clause (e.g. yielding a precautioning clause), in the Australian languages Nyikina (Nyulnyulan) and Ngiyambaa (Pama-Nyungan).

2.2 Apprehensionals and their interaction with other grammatical domains

Another fertile area of investigation is the interaction of apprehensional constructions with other grammatical domains. We will briefly discuss the interaction with grammatical person, tense/aspect, and negation. Interaction with grammatical person concerns the restriction of apprehensional markers to specific grammatical persons, or to specific combinations of grammatical persons (Vuillermet 2018a). (Grammatical persons may also more or less directly impact the semantic interpretation of apprehensives, which is dealt with in §3.2.) For instance, the Andoke (isolate; Colombia) Apprehensive is only available with 2nd person (Landaburu 1979: 89), while the Harakmbut (aka Amarakaeri, isolate; Peru) Apprehensive seems to only be available with 1st and 3rd person (Helberg Chávez 1989: 240, Tripp 1995: 222, An Van linden, p.c. August 2019). Matsés displays another type of restriction, in that it has several Apprehensive markers specialised for distinct persons. Among them are *-mane* (11a), which exclusively occurs with 1st person subjects, and *-nunda* (11b), with third person subjects.

(11) Matsés (Pano-Tacanan; Brazil/Peru; Fleck 2003: 439–440)

a. 1st person subject Apprehensive marker
Mibi cues-mane.
2:ABS hit-**APPR:1**
'I might accidentally hit you.'
[e.g. if you stand behind me as I split firewood]

b. 3rd person subject Apprehensive marker
chononda
*cho-**nunda***
come-**APPR:3(>3)**
'(he) might come'
[e.g. "don't go to your field, your lover might come", or "don't go there with your 1st lover, the other lover might come."] (D. Fleck p.c., September 2017)]

Another two Matsés Apprehensive markers contrast in having different grammatical persons as objects: *-panondac*, illustrated in (12), only occurs with 1st and 2nd person objects, and *-nunda* with 3rd person objects (see also (11b)).

(12) Matsés (Pano-Tacanan; Brazil/Peru; Fleck 2003: 440)

a. 3>1 or 2 Apprehensive marker
Nisi-n *pe-**panondac**.*
snake-ERG bite-**APPR:3>1-2**
'Now maybe a snake will bite me.'
[Said, e.g. as the speaker is going off to hunt on a rainy day.]

b. 3>3 Apprehensive marker
Nisi-n *pe-**nunda**.*
snake-ERG bite-APPR:3(>3)
'(Be careful), a snake might bite (e.g. our son).'
Not: '… bite you/me.'

Ye'kwana (Cariban; Venezuela; Cáceres 2011: 232–233) is another language whose apprehensive construction has two forms depending on the grammatical person combination: the negative marker *an-* is necessarily present if a third person patient is involved, while it is absent in all other configurations.

Apprehensionals may also differ with respect to how they interact with tense or aspect. The impact of aspect on the availability of the prohibitive vs preventive reading is well documented in Slavic languages (see §4 and (Wiemer 2026 [this volume]). In Ese Ejja, no tense marker can occur together with the Apprehensive since they belong to the same paradigmatic slot (Vuillermet 2012: 367, 449; Vuillermet 2018b: 268). In Hup (Naduhup; Epps 2008: 607–608), the Future marker *-teg* ~ *-te-*, which indicates a certain future event, can also not co-occur with the Apprehensive (nor with the Imperative, the Negative, the Habitual and some conditional expressions). However, the Future Contrast marker *tán*, which indicates a "potential or hypothetical future", can (see (23) further below). Here it seems that possibility of co-occurrence is determined by the semantics of the two Future markers and that of the Apprehensive.

As already stated in §1, some apprehensional constructions (negative optatives, negative purpose clauses) include a transparent marker of negation (Wiemer 2026 [this volume]). Other apprehensional constructions, on the other hand, may include expletive negation, that is, a negation marker which does not transparently negate the clause it occurs in, but rather emphasises the undesirability of the potential situation. Expletive negation is quite common in the complement of 'fear' predicates (Jin & Koenig 2020, Dobrushina 2021). Olguín Martinéz (2023: 601–602; 2024: 16) also finds that the presence of negation in precedence ('before') clauses – in languages where it is optionally possible – tends to mark these clauses as non-factive and apprehensional, as opposed to factive clauses with a purely temporal reading. Expletive negation in French in a 'fear' complement and a precedence construction is illustrated in (13) and (14), respectively. In both constructions, the (optional) negative marker *ne* does not negate the undesirable event; the full French negative construction ***(ne)*** … ***pas*** would be necessary to achieve this.

(13) French (Indo-European: Italic; Jin & Koenig 2020: 39)

J'ai	*peur*	*qu'il*	(***ne***)	*pleuve*	*demain.*
I.have	fear	that.it	EXPL.NEG	rain:3SG.SBJV	tomorrow

'I fear that it will rain tomorrow.'
Not: 'I fear that it will **not** rain tomorrow.'

(14) French (Indo-European: Italic; Tahar 2021)

Jim	*doit*	*attraper*	*le*	*vase*	*grec*	*avant*	*qu'il*	(***ne***)	*tombe.*
Jim	must	catch	the	vase	greek	before	that.it	EXPL.NEG	fall:-3SG.SBJV

'Jim must catch the Greek vase before it falls.'
Not: '… before it does **not** fall.'

Whether a formally negative apprehensional construction transparently negates the clause it occurs in or is to be interpreted as expletive negation may depend on the linguistic context. For example, Fedotov (2026 [this volume]) argues that the negation marker in Gban constitutes transparent negation in negative purpose constructions, but is expletive in all other apprehensional uses (see also Francez 2026 [this volume]).

Finally, there are apprehensionals which cannot occur with negation at all (e.g. in Archi and Akhvakh, Daniel & Dobrushina 2026 [this volume] – or more precisely, they cannot change polarity if they already include an expletive or transparent negation marker. In Upper Tanana Dene, precautionary clauses are only compatible with affirmative main clauses (Lovick 2026 [this volume]).

3 Semantics of the apprehensional marker

3.1 Degree of semanticisation of the undesirability component

A major issue to consider in the domain of apprehensionality is whether the construction in question semantically encodes undesirability (*dedicated apprehensional*, cf. §1.2) – or whether this is merely pragmatically implied in certain contexts (*non-dedicated apprehensional*), for example as a result of its frequent use in warnings and other typical apprehensional contexts. Diachronically, semanticisation of a pragmatic implicature may of course lead from non-dedicated to dedicated apprehensionals.

The distinction between dedicated apprehensionals and more general constructions which are also frequently used in apprehensional contexts can be illustrated with the contrast between *in case* and *lest* in English. *Lest* is a subordinator specialised for undesirability (López-Couso 2007), while *in case* has a general meaning of possible consequence (Dixon 2009: 23–25; see also §1.2). Although often used in contexts of undesirability, *in case*, but not *lest*, can introduce subordinate clauses that describe desirable events, as illustrated in (15).

(15) *You should keep your lottery ticket in a safe place,* ***in case*****/#*****lest*** *you win.*

An example from the modal domain for an apprehensional strategy that relies on pragmatic inference comes from the Australian language Jaminjung-Ngaliwurru (Schultze-Berndt & Caudal 2019). In affirmative clauses, the irrealis modal prefix (which contrasts with a second modal prefix used in desiderative and deontic contexts) is overwhelmingly used in warnings such as (16a), and in most contexts implies an undesirable possibility even in the absence of a preemptive clause. However, examples such as (16b) show that it is still a modal of general possibility which does not semantically encode undesirability[12] (cf. also Browne et al. 2026 [this volume]) for a similar pattern in some neighbouring Australian languages).

(16) Jaminjung-Ngaliwurru (Mirndi; Australia; fieldwork Schultze-Berndt)

a. *Gurrany mard ya-nth-angu, guyug-burru,* ***ya****-nth-irna!*
NEG touch IRR-2SG>3SG-get/handle fire-PROPR **IRR**-2SG-burn
'Don't touch it, it is hot like fire, you might get burned!'

b. *Ning* ***ya****-nth-ina lidburrg-ni.*
break.off **IRR**-2SG>3SG-chop axe-INST
(Echidnas are found under rocks.) 'You might/can kill one with an axe.'

If a marker still conveys undesirability in pragmatic contexts that are typically considered desirable (or else is judged unacceptable in such contexts), it can be concluded that it conventionally codes this meaning component and is thus a dedicated apprehensional marker. An example is the Ese Ejja Apprehensive -*chana*. Even if it occurs with a typically positive event like the verb *swa-* 'laugh', as in (17), the event of laughing can only be interpreted in the sense of a mockery.

[12] Additional evidence comes from the fact that the IRR modal is the only modal appearing in negative contexts, i.e. it expresses general impossibility without an undesirability component.

(17) Ese Ejja (Pano-Tacanan; Bolivia/Peru; fieldwork Vuillermet)
*Marina swa-**chana**, mi='ba-majje!*
Marina laugh-**APPR** 2SG.ABS=see-TMP.SS
'Marina might laugh at seeing you!'

Another example is the German connective *nachher*, which in its original temporal sense means 'afterwards, later', but has in colloquial German acquired a second, apprehensional sense. In this sense, it is a dedicated apprehensional marker according to this test. Thus, the apprehensional clause in (18) can only be interpreted as describing an undesirable situation; combining it with an expression of positive evaluation as in (18b) leads to infelicity.

(18) German (Indo-European: Germanic; Faller & Schultze-Berndt 2018)

a. *Ich glaube, ich nehme lieber nicht am Gewinn-spiel*
I think:1SG.PRS I take:1SG.PRS better not at:DEF win-game
*teil. **Nachher** gewinne ich noch ....*
part **APPR/later** win:1SG.PRS I ADD
'I think I better not take part in this prize draw. I might actually win [a trip] ....'
Context: ... and would not be able to watch the Champions League final at home.[13]

b. # ***Oh super! Nachher*** *gewinne ich noch ...*
oh great **APPR/later** win:1SG.PRS I ADD
Intended: 'Great! I might actually win ....'

If undesirability has been established as a semantic, that is, conventional meaning component of the construction, a further question arises regarding the type of semantic meaning. Is it part of the at-issue meaning, that is, the main point of the utterance, or is it backgrounded in some way? One test to distinguish at-issue from not-at-issue meaning relies on the observation that only at-issue meaning is accessible to direct expressions of assent/dissent (Faller 2002, Papafragou 2006, Tonhauser 2012, Murray 2021). As shown in (19), this test establishes that the proposition denoted by the apprehensive clause in (19) is at-issue, while its undesirability is not. In contrast, with *I fear* in (20), both the proposition denoted by the complement clause and its undesirability may be at-issue.

[13] http://www.hifi-forum.de/viewthread-144-7707-4.html (last accessed 2024-03-18)

(19) German (Indo-European: Germanic; Faller & Schultze-Berndt 2018)

A : *Spiel* *nicht* *mit* *der* *Vase,* ***nachher*** *geht* *sie*
play:IMP NEG with DEF.DAT.FEM vase **APPR/later** go:3SG.PRS 3SG.FEM
noch kaputt.
ADD broken
'Don't play with the vase, it might break (and that would be undesirable).'

B : 'No, it won't break.' (⇒ breaking at-issue)

B': # 'No, that wouldn't be a bad thing – it's ugly anyway.' (⇒ undesirability not at-issue)

(20) English

A : ***I fear*** *the vase might break.*

B : *No, it won't break. (*⇒ breaking at-issue*)*

B': *(Don't worry,) that wouldn't be a bad thing – it's ugly anyway. (*⇒ undesirability at-issue*)*

Francez (2026 [this volume]) also claims for the Hebrew apprehensionals that the undesirability component is not at-issue and specifically analyses it as a conventional implicature.

3.2 Apprehensives as a type of modal

While all apprehensionals have modal semantics as markers of possibility and undesirability, apprehensives, that is those markers that can occur in main clauses, in many languages are part of the modal system, and specifically of the temporal/modal/mood paradigm (see, in this volume, Browne et al. for a subset of Ngumpin-Yapa languages, Fedotov for Gban, Francez for one of the Hebrew apprehensionals, François for Mwotlap, Treis for Kambaata, and also Vuillermet 2012 2018b for Ese Ejja). That apprehensives are a subtype of modal is also supported by a diachronic relationship between possibility modals of various types and apprehensives (see further §4.)

A careful analysis of such markers regarding the paradigmatic oppositions and/or syntagmatic co-occurrence with other modals or mood will go towards elucidating their semantic contribution to the clause. So far, it is an unresolved question what subtype(s) of modality can be instantiated by apprehensive modals. They are described as a mixed modal category with both epistemic and attitudinal components (potentiality and undesirability) by Lichtenberk (1995:

293), but it remains to be established whether apprehensives are always truly epistemic. For example, apprehensives may be used to mark general undesirable dispositions such as *Dogs may bite* or statistical undesirable likelihoods such as *Your lottery ticket may/will likely not win*, and these are arguably not epistemic. It is also conceivable that an apprehensive could be used to mark obligations as undesirable. A diagnostic for potential epistemic status is interaction with tense. It has been observed that non-epistemic modals are necessarily future-oriented (Condoravdi 2002, Rullmann & Matthewson 2018), that is, the state of affairs described must hold after the relevant circumstances are established (the circumstances themselves can be established in the past, present or future). Epistemic modals, however, can also be past- or present-oriented, that is, the possible state of affairs may hold not only after but also before or at the time at which the epistemic agent's relevant beliefs hold. For example, the speaker of (21) considers it a possibility that their oral exam date is in the past based on their current knowledge (other students have already had their oral exam). This suggests that German *nachher* is indeed used epistemically (similarly for Hebrew, Francez 2026 [this volume]).

(21) German

Ich	*will*	*auch*	*meinen*	*mündlichen*	*termin!!!!*	*nachher*	
1SG	want.1SG.PRS	also	my	oral	date	**APPR/later**	
war	*er*	*schon*	*und*	*ich*	*bin*	*nicht*	*informiert*	*worden!*
be.3SG.PST	3SG	already	and	1SG	AUX.1SG	NEG	informed	PASS.PST

'I want the date for my oral [exam] too! It may have taken place already and I wasn't informed!'[14]

However, some Apprehensives (e.g. in Gban (Fedotov 2026 [this volume]), in Akhvakh (Daniel & Dobrushina 2026 [this volume]) and in Kambaata (Treis 2026 [this volume]) cannot occur with the past at all, which indicates that they cannot be past-oriented, and, consequently, that they may be non-epistemic.

3.3 Functions

Since apprehensives are strongly associated with contexts of warnings and associated directives, such utterances are often performative. Bybee et al. (1994: 179) therefore subsume apprehensives (their "admonitives") under "speaker-oriented modality" (together with imperative, optative, etc.). Cross-linguistic evidence for

[14] https://www.medi-learn.de/foren/archive/index.php/t-43472-p-66.html (last accessed 2024-03-15).

the performative nature of apprehensive modals comes from their attested diachronic relationship with prohibitives (see below for discussion and references) and optatives (Dobrushina 2006, Lahaussois 2026 [this volume], Wiemer 2026 [this volume]), and their occurrence in the same paradigmatic slot as the imperative, e.g. in Kotiria/Wanano (Tucanoan; Brazil and Colombia; Stenzel 2013: 268, 309). Treis (2026 [this volume]) discusses to what extent the Apprehensive paradigm in Kambaata may be considered a subtype of the directive mood, as it shares some semantic features with negative imperatives and jussives, but also considers the possibilities that it is a subtype of the indicative mood or constitutes its own category.

In this context, an interesting but underexplored semantic parameter for apprehensives is whether they can occur in both warning and threats, or only warnings; in other words, whether or not they are restricted to unintentional undesirable events (Vuillermet 2017, 2018c, 2018b: 273–4). For example, Matsés is a language where only the warning reading, illustrated in (22), is available, not the threat reading (D. Fleck p.c., September 2017).[15]

(22) Matsés (Pano-Tacanan; Brazil/Peru; Fleck 2003: 439)
*Mibi cues-**mane**.*
2:ABS hit-**APPR:1**
'I might accidentally hit you.' [e.g. if you stand behind me as I split firewood] – not: 'Watch out I'm going to hit you!'

In languages where it is available, the threat reading seems to be favoured by a number of factors. One is grammatical person: Epps (2008: 231) observes that a "threat is […] the default interpretation when the subject is in the first person" in her Hup corpus, as illustrated in (23).

(23) Hup (Naduhup; Brazil/Colombia; Epps 2008: 231)
ʔám-ăn ʔãh yɔmɔ̆y yók tán-ãh!
2SG-OBJ 1SG anus stab.APPR FUT.CNTR-DECL
'I'll stab you in the anus!'

Vuillermet (2018b: 273–4) suggests that the feature of control in the verb semantics is another factor: in *(Watch out) I might fall on you!*, the typical reading without context is a warning rather than a threat. These two factors (1st person subject and +control verb) are typical correlates of the necessary broader conditions that must hold for a threat reading (Searle 1975, Salgueiro 2010), but these

[15] Fleck (2003) calls the Matsés Apprehensive a *Future Potential*. The number following the adapted gloss APPR stands for the person of the subject, namely '1st person' in example (22).

can also be met in other ways. Thus, *He might burn your house!* is a threat if the speaker commits to bringing about the event, which requires them to have direct or indirect control over it, and the event is (or believed to be) undesirable primarily to the addressee (rather than to the speaker or to both). Often, threats are conditional on the addressee fulfilling a request, for example, handing over money, that is, the addressee is given a way of avoiding the undesirable event or its consequences (see also Fedotov 2026 [this volume]).

A prohibitive reading is also frequently available with second person apprehensives, typically with events that the addressee can control, as has been discussed in the literature (Dobrushina 2006: 57, Pakendorf & Schalley 2007, see also Wiemer 2026 [this volume]). In Wintu, the 2nd person Apprehensive, which originally comes from an optative 'may, might' (Pitkin 1984: 193), receives an apprehensive or a prohibitive interpretation, depending on the degree of control the addressee has over the event: getting drowned or getting hurt typically occur unwillingly (lack of control), resulting in a warning interpretation of (24a), while telling a lie is typically controllable by the addressee, and therefore more likely intended as a prohibitive as in (24b).

(24) Wintu (Wintuan; United States; Pitkin 1984: 115)

a. Apprehensive reading
*tuku**ken*** 'You might get drowned, watch out!'
*podo'lu**ken*** 'Look out, you might get hurt!'

b. Prohibitive reading
*bala**ken*** 'Don't tell a lie!'

More generally, the preventability of the undesirable event plays a role in determining the function of apprehensives (Fedotov 2026 [this volume], Vuillermet 2026 [this volume]). This semantic variation is discussed further in the next section on diachrony.

4 Diachrony

Lichtenberk (1995) makes the strong claim that apprehensive main clause markers have their origin in precautioning clauses via the function of complement of 'fear' predicates (assuming a pathway of insubordination):

> The implication of a situation being undesirable, or indeed feared, becomes explicit by virtue of clauses encoding such situations as subordinate to predicates of fearing. […] Over time, through metonymy based on cooccurrence

> of linguistic forms, the notion of apprehension comes to be associated with the LEST element, and the predicates of fearing become expendable. (Lichtenberk 1995: 319)

However, Lichtenberk (1995: 303) also notes that the Toba'a'ita apprehensional marker *ada* has a likely origin in a verb of perception used as an attention getter ('look (out)!'). Attention getters are indeed attested in other languages as source constructions for apprehensionals, e.g. (possibly) in Akhvakh (Daniel & Dobrushina 2026 [this volume]); see also the discussion of Spanish *ten cuidado* 'be careful' as the source for the apprehensional *kuyra* in Huallaga Quechua (see (27)). An attention getter as grammaticalisation source however suggests a main clause origin of the apprehensive construction rather than insubordination via a complement of 'fear'.

Subsequent studies have shown a wide range of possible source constructions for apprehensional constructions; a number of them are discussed by the contributors to this volume, to the extent that the data for a given language permit conclusions about the origin of the construction. For example, Rhee & Kuteva (2026 [this volume]) report, for Korean, the grammaticalisation of 'fear' predicates (rather than insubordination of a 'fear' complement clause) as one possible pathway for the semanticisation of undesirability.

The link between apprehensionals and prohibitives – already alluded to and illustrated with examples from Wintu in §3.3 – is well attested; it has been discussed in a typological perspective by Dobrushina (2006: 50–63) and Pakendorf & Schalley (2007), and is confirmed by a number of contributions to this volume (Lahaussois, Treis, Wiemer) and elsewhere (Smith-Dennis 2021). Pakendorf & Schalley (2007) posit a pathway from general possibility marker to prohibitive via an apprehensive modal: A sentence describing a potential undesirable event can be interpreted as a warning (as in (25a)) and therefore be reinterpreted as a directive to avoid the undesirable event (25b).

(25) a. (*Be careful,*) *you might slip!*
 b. (*Be careful,*) *don't slip!*

Dobrushina (2006: 62) considers both this and the reverse pathway plausible. As pointed out by her (2006: 61–63), and discussed in some detail by Wiemer (2026 [this volume]), an important link between the apprehensive and prohibitive function are a subtype of prohibitives termed *preventives*, prominent in the literature on Slavic languages since here they are distinguished from canonical prohibitives in terms of aspectual marking (perfective vs. imperfective) (see also e.g. Xrakovskij 1990, Kuehnast 2008). The hallmark of preventives is that they are formally prohibitives but encode an (undesirable) event that is not directly controllable by the addressee, as in (25b). Rather, their pragmatic function is to invite the addressee to take measures to prevent the event in question – in other words, they warn of a potential negative consequence rather than directly prohibiting the action leading to it. In either case, pragmatic enrichment is at work: a prohibitive can be pragmatically employed to encode a warning (if avoiding an event is in the interest of the addressee rather than another party), as in (25b), leading to a re-analysis of the prohibitive as an apprehensive. The reverse could ultimately lead to a re-analysis of the apprehensive as a prohibitive.

The pathway from a more general possibility modal (which itself will have ultimate lexical sources such as motion verbs or expressions of uncertainty) to an apprehensive modal, via pragmatic strengthening of the undesirability component (Bybee et al. 1994: 211; Dobrushina 2006: 34–36; Pakendorf & Schalley 2007), was already briefly discussed above and in §3.1 and §3.2. Several contributions to this volume examine the degree of semanticisation of the undesirability component for apprehensive strategies originating in lexical verbs of cogitation via markers of epistemic possibility/uncertainty (Rhee & Kuteva for Korean), in a modal adjective encoding propensity (Francez for Modern Hebrew), in a general possibility marker (Browne et al. for various Ngumpin-Yapa languages), and in a prospective/future marker with and without negation (Fedotov for Gban; Song for Baoding). Pragmatic strengthening of an undesirability component is also involved in the development of general purpose/manner markers into precautionary markers, likewise discussed in this volume by Rhee & Kuteva for Korean and by Treis for Kambaata.

An apprehensional strategy which perhaps most clearly conveys the undesirability to the speaker of an event is the use of an optative, i.e. a modal or construction which conveys the wish of the speaker, in combination with negation (Dobrushina 2006; see also §1.2, §3.3). In this volume, negative optatives as the source of apprehensionals are discussed for Slavic languages by Wiemer and for Thulung by Lahaussois. In Thulung the optative marker itself demonstrably originates in a lexical verb 'be auspicious, be good'; Lahaussois also shows how a subordination of the negative optative construction can give rise to a precautionary construction.

Conditional constructions, including pseudo-conditional conjunction or disjunction, make a preemptive or enabling event explicit in relation to the event to be prevented, as in (26), and thus have been discussed as an apprehensional strategy (e.g. van der Auwera 1986, Dominicy & Franken 2002, Narrog 2012: 35–40, Clancy et al. 1997, Knoob 2008). Lahaussois (2026 [this volume]) describes a conditional construction as one of the apprehensional strategies of Thulung.

(26) a. ***If** you do not start revising now, you might fail.*
b. *Start revising now **or else/otherwise** you're going to fail.*
c. *Keep on idling **and** you'll fail.*

A recurrent grammaticalisation source which gives rise to apprehensionals are ablative spatial cases or adpositions, represented in this volume in Upper Tanana Dene (Lovick 2026 [this volume]) and in some Oceanic languages of Vanuatu (François 2026 [this volume]). In this case, the semantic change arises through metaphorical extension (spatial 'away from' > 'avoidance').

Another apparently cross-linguistically recurrent source construction are temporal connectives like 'before' (see (14) in §2.2), or conversely 'soon', 'later', as illustrated in the Kriol example in (1) and the German examples in (18) and (19). Equivalents are attested in Dutch (Boogaart 2009, 2020), in a number of traditional Australian languages, in a variety of Hawaiian that was used for interethnic communication also known as Pidgin Hawaiian (Roberts 2013 cited in Angelo & Schultze-Berndt 2016: 283) and in the Gyalrongic (Sino-Tibetan) language Japhug (Jacques 2021: 1419–1420). In this volume, Francez (2026 [this volume]) briefly discusses a modern Hebrew marker with an origin in a temporal connective.

Other grammaticalisation sources for (components of) apprehensionals are more rarely discussed. Curiously, some languages make use of so-called alternative-sensitive particles in their apprehensional constructions: German *nachher* 'later/APPR' tends to co-occur with the temporal additive/modal particle *noch* 'still' (see (1)), and in Cuzco Quechua, warnings typically involve a contrastive marker as well as a possibility modal (fieldwork Faller). Francez (2026 [this volume]) shows how the temporal additive/modal particle *od* 'still' in Hebrew on its own can invite apprehensional inferences, and invokes the same association of a subsequent event with an undesirable event that presumably facilitates the development from a temporal connective to an apprehensional marker.

Two further grammaticalisation paths not attested in the languages covered in the contributions to this volume will be mentioned here since they shed some light on how grammatical person restrictions may come into existence (see §2.2). The first case comes from Huallaga Huánuco Quechua. Weber (1989: 304) shows how *kuyra:* (from the original Spanish attention verb *(ten) cuidado* 'be careful') plus a subordinate clause expressing an undesirable event gave way to two distinct constructions. One (27a) requires a same-subject adverbial subordinator (if the logical subject is 2nd person), in line with the fact that *kuyra:* was originally a main clause 2nd person imperative to which the clause expressing the undesirable event is subordinated. The other (27b) requires a nominalised clause plus the comitative.

(27) Huallaga Huánuco Quechua (Quechuan; Peru; Weber 1989: 304)

a. For 2nd person subject
***Kuyra**: tuny-r.*
APPR fall-SS.ADV
'Be careful you might fall!'

b. For persons other than 2nd person subject
***Kuyra**: pay pusha-shu-**na**-yki-**wan**.*
APPR 3 lead-2-NMZR-2P-COM
'He may lead you astray.'

If one imagined a kind of Jespersen cycle where *kuyra:* disappeared, one can see how two very distinct constructions may arise for different grammatical persons.

Another attested grammaticalisation path explaining a grammatical person restriction is that of the Matsés apprehensive construction. Fleck (2003: 443) discusses how the Apprehensive *-me-enda*, originally a Causative followed by a Prohibitive, is regularly shortened to *-me*. Example (28) below is currently best translated as a proper apprehensive, but the full form would be literally translated as 'don't let the jaguar kill you!'.

(28) Matsés (Pano-Tacanan; Brazil/Peru; Fleck 2003: 443, our glosses)
Bëdi ac-me.
jaguar kill-CAUS/APPR
'A jaguar might kill you.'

Interestingly, Fleck (2003: 443) describes this construction as being restricted to third person subjects, and most frequent with second person objects (first and third person objects are however grammatically possible). This restriction has a diachronic explanation, since a third person negatively affecting a second person is the most expected scenario for a warning, assuming that a warning use was the most frequent use of this construction at some point.

The bewildering variety of potential sources for apprehensionals just discussed may well be due to their particularly complex semantics (Kuteva et al. 2019, Dąbkowski & AnderBois 2025) and the interactional nature of their use. Possibly for the same reason, apprehensional constructions do not seem to be a diachronically stable phenomenon, an observation that is underlined by several contributions to this volume. The individual languages from the Ngumpin-Yapa

(Browne et al. 2026 [this volume]), Oceanic (François 2026 [this volume]; see also Smith-Dennis 2021) and Nakh-Daghestanian (Daniel & Dobrushina 2026 [this volume]) language groups have quite distinct strategies for the expression of apprehensionality. The apprehensional markers in Gban (Fedotov 2026 [this volume]) and Kambataa (Treis 2026 [this volume]) appear to be recent innovations in these languages which are not shared in the wider family or area. Not uncommonly, languages have multiple co-existing strategies for conveying apprehensional meaning (Francez 2026 [this volume] for Hebrew, Lahaussois 2026 [this volume] for Thulung, Rhee & Kuteva 2026 [this volume] for Korean, Wiemer 2026 [this volume] for Slavic languages). Apprehensional markers are also found in new languages i.e. creoles, e.g. Tok Pisin (briefly discussed by Smith-Dennis 2021) and Northern Australian Kriol (Angelo & Schultze-Berndt 2016, Phillips 2021).

We hope that this volume lays some groundwork for future research on the origins and the syntactic, semantic and pragmatic variety of apprehensional constructions in the world's languages. The last section concludes this introduction on the apprehensional domains covered in this volume with brief overviews of each of the contributions. They are geographically ordered from West to East, starting in North America and ending in Australia.

5 Overview of this volume

In her paper *Speaking and acting with care: The grammar of circumspection in Upper Tanana Dene*, Olga Lovick casts both a linguistic and anthropological perspective on two apprehensional constructions in Upper Tanana Dene (Athabaskan-Eyak-Tlingit), spoken in the Alaska/Yukon border region between the US and Canada. The avertive clause construction in precautioning function involves the marker *ch'à'* and the "urgent request" construction the marker *dè'* (+ Optative). The origin of the avertive marker, which has cognates in the same function in several other Dene languages, is likely the Proto-Dene postposition **k'ya·n* 'away, apart from'. It exclusively encodes a direct causal relation with the preemptive main clause. Interestingly, the preemptive clause can never be negated. The "urgent request" construction is used to form expressions with translations like 'make sure X happens', and likely originated in an insubordinated conditional request. The construction is linked to the apprehensional domain in that non-compliance with a request expressed in this way is likely to have severe consequences, although they are never spelled out. Having discussed how traditional Upper Tanana speakers (and other Northern Dene peoples) "are very circumspect both in words and actions", Lovick demonstrates how the two constructions allow the speaker to help the addressee prepare for or avoid undesirable eventualities while still respecting the strong cultural principle of non-interference.

Maksymilian Dąbkowski and Scott AnderBois present a detailed unified analysis of the multi-functional apprehensional morpheme *=sa'ne* in A'ingae (aka. Cofán), an isolate language spoken in the Northeastern Ecuador/Southern Colombia border region in Amazonia. Considering semantic and syntactic evidence as well as the paradigmatic status of this morpheme, they argue for a semantically conventionalised status of the undesirability component for *=sa'ne*. Its primary function is that of a subordinator, i.e. a precautioning morpheme, with both avertive and 'in-case' uses. Dąbkowski and AnderBois propose a unified semantic analysis for these two uses of *=sa'ne* (and comparable markers cross-linguistically): *=sa'ne* always invokes an undesirable situation which is to be avoided. In the avertive reading this situation is the one described by the subordinate clause itself, whereas in the in-case reading it is a larger situation that contains the one described. The marker moreover covers the three other apprehensional uses identified in the literature: it can function as a complementiser with 'fear' verbs (and some other verbs with negative semantics), and has timitive uses with noun phrases which are interpretable as metonymically invoking an undesirable situation. It also has more marginal main-clause uses as an apprehensive, which are analysed as incipient insubordination.

Maksim Fedotov discusses the apprehensional construction in Gban, a South Mande language spoken in Côte d'Ivoire. This construction covers the three apprehensional core functions attested cross-linguistically. In independent clauses with an apprehensive function, the apprehensional construction is compatible with all three grammatical persons, in a pure apprehension, warning, or threat reading. In dependent clauses with a precautioning or 'fear' complementation function, the construction requires additional subordinators. Fedotov argues that the origin of this formally complex construction is plausibly a negative purpose construction, in turn grammaticalised from the combination of a negator and a prospective morpheme, and that the precautioning construction still has a residual negative semantics, as indicated by the availability of negative polarity items. The paper pays detailed attention to constraints on all three uses, e.g. that in the apprehensive function, the construction is always future-oriented and is never applied to events that cannot be in principle prevented, e.g. weather events (though it can be used for avoidable events involving weather events, such as rain falling on one's car).

Yvonne Treis discusses apprehensionality in Kambaata, a Cushitic language of Ethiopia, which seems to be the only Afroasiatic language to display a verbal paradigm of dedicated apprehensive forms. These forms only occur in main clauses, and Treis provides detailed evidence showing that they do not exactly fit the verb forms of either the indicative aspectual paradigm or that of the directive. When the apprehensive clause co-occurs with a preemptive clause, the latter is either an independent clause or (unexpectedly from a cross-linguistic perspective) subordinated to the former. The apprehensive is available with all grammatical persons and shows cross-linguistically expected variation in the resulting pragmatic interpretation: a warning interpretation is available for all persons, while threat and prohibition uses are only attested for 1^{st} and 2^{nd} person, respectively. Treis methodically demonstrates that, like other verbal markers in the language, the apprehensive forms have a phrasal origin, which in this case involves a purposive converb and an existential copula.

Itamar Francez' overview of apprehension in Hebrew covers five markers with apprehensional interpretations. Two of these (the complementiser *pen* and the modal *alul*) are argued to conventionally encode (or to have encoded at a previous diachronic stage) that the core proposition is possible and undesirable. Francez argues that the undesirability component is a conventional implicature, while the possibility meaning is an entailment. Both these markers can be used in the apprehensive function in matrix contexts, and in the precautioning function when the clause they appear in occurs in hypotaxis or parataxis with another clause. Apprehensional interpretations also arise with two temporal adverbials, one having an additive meaning (*od* 'still'), and the other a sequential meaning (*axarkax* 'afterwards/later'). The future orientation of both arguably provides the modal component of their apprehensional interpretations. The undesirability component however is not always present, and Francez therefore analyses it as a pragmatic inference that arises only in contexts which establish the associated clause as undesirable. Finally, a complex complementiser *še lo* – consisting of a complementiser that can also stand on its own and a negation marker – has acquired an apprehensional interpretation, including when introducing the complement of 'fear' predicates. Francez argues that negation in this construction is expletive, i.e. is no longer semantically interpreted.

In *Two major apprehensional strategies in Slavic: A survey of their areal and grammatical distribution*, Björn Wiemer offers a comprehensive survey of two apprehensional strategies widely attested in Slavic languages, and their genealogical and areal distribution in all three branches (East, West and South Slavic). The discussion is based on data from both major corpora and linguistic publications often addressed to Slavic specialists. The first strategy foregrounds the speaker's apprehension of some undesirable event. It is dubbed the NEGATIVE-OPTATIVE strategy since it is mostly expressed by clause-initial connectives which are formally negated optatives, but it can also be realised by constructions associated with negative purpose clauses. The second strategy alerts the addressee to an action to be taken to avoid an undesirable situation. This apprehensional strategy includes what is known as *preventives* in the literature on Slavic languages but is more generally dubbed NEGATIVE-DIRECTIVE in this paper. It most commonly involves a negated imperative (or equivalent clause type) with perfective verbs, whereas the imperative with imperfective verbs tends to express prohibition, at least in North Slavic. Wiemer also offers an account of the diachronic development and grammaticalisation of the language-specific constructions and their synchronic interpretations.

MICHAEL DANIEL and NINA DOBRUSHINA's study considers dedicated apprehensionals in three distantly related NAKH-DAGHESTANIAN (East Caucasian) languages, ARCHI (Lezgic), MEHWEB (Dargwa) and NORTHERN AKHVAKH (Andic). The markers in the three languages have distinct forms with different syntactic properties and probably arose through different pathways of grammaticalisation. The Akvakh morpheme exclusively expresses the apprehensive function, i.e. it is a "discourse interactional category". The Mehweb morpheme primarily expresses the apprehensive function, i.e. it occurs in main clauses, but it extends to subordinate uses. In those uses, a (perfective) converb form of a 'say' verb as subordinator is optional in precautioning function but obligatory in 'fear' complement function. By contrast, the Archi morpheme, for which the most data were available, primarily expresses the precautioning and 'fear' complementation function, and its main clause functions seem to have developed from 'fear' complement clauses via insubordination. The syntactic status of Archi and Mehweb apprehensional clauses is tested and discussed in detail, as well as their semantics, showing that at least in the usage of the Archi apprehensional, the emotion of fear is not necessarily present, but only sheer undesirability. Whether the functional convergence of the apprehensional constructions in the three languages – despite their different historical pathways – is the result of areal pressure remains to be investigated.

While THULUNG (Kiranti, Sino-Tibetan) has not developed a dedicated apprehensional construction – seemingly in line with other South Asian languages, which have been remarkably absent in the apprehensional literature so far – AIMÉE LAHAUSSOIS, in her paper *Inauspicious events in Thulung,* explores how the negative purpose construction fulfills the precautioning function. The construction rather transparently originated in a negative optative verb form, which systematically associated with a subordinator typically introducing reported speech elsewhere in the language, originally a grammaticalised ‘say/ think’ converb. Such a developmental pathway is not only found in many Tibeto-Burman languages, but in the larger region. The optative itself goes back to a lexeme *nʉmu* ‘be good, auspicious’, which gave rise to two distinct grammatical constructions, the impersonal negative deontic necessity construction on the one hand, and the precautionary construction on the other. This article provides useful information on how a language with no dedicated apprehensional construction may develop a strategy to express the function. It also convincingly links the semantics of the original lexical concept with apprehension and deontic necessity, and highlights its conceptual relation with Thulung cultural practices.

Na Song examines the apprehensive function of a sentence-final particle in Baoding, a Mandarin (Sinitic) dialect spoken in Northern China. The Apprehensive enclitic *=lɛ.ia* is atypical for an apprehensive in several regards: syntactically, it can occur in interrogative clauses; semantically, it largely conveys unexpectedness (on top of undesirability); and pragmatically, it requires the addressee’s involvement in avoiding the potential undesirable situation. The particle systematically expresses apprehension with change of state verbs, and tends to express imminent future with activity verbs. The grammatical first person however seems to force an apprehensive reading in the latter case. Song provides phonological, morphosyntactic, semantic and typological arguments for a diachronic source of the marker in the univerbation of two final particles, *=lɛ* denoting a change of state, and *=ia* denoting the intention of the speaker about an imminent future action.

Songha Rhee and Tania Kuteva’s contribution addresses the diachronic development and synchronic distribution of no fewer than 13 markers in apprehensional function in Korean (also known as preventives in Korean linguistics). Using both historical and contemporary corpora, Rhee and Kuteva are able to trace the first attestations of these markers as well as differences in their frequency and in their use over time, including the degree to which each marker appears to be restricted to apprehensional contexts. They therefore posit multiple layers in the functional domain of apprehensionality for present-day Korean. They also provide an in-depth discussion of the grammaticalisation processes that gave rise to each marker, and argue that some of the markers preserve a degree of semantic transparency and analysability (lack of erosion) due to not being fully grammaticalised. The source constructions of one set of markers – in a predominantly precautionary function – involve combinations of a nominaliser and either a mode/manner or purposive marker. The source of the others are various combinations of lexical verbs expressing fear but also perception (‘see’) and cognition ('not know', 'think') with existing grammaticalised future and interrogative markers. The grammaticalisation of perception and cognition verbs in these contexts provides strong support for a cognitive link between uncertainty about the future and anxiety, which can lead to the development of apprehensional markers from expressions of mere possibility via pragmatic strengthening. These forms are also often ambiguous between a main clause use and a subordinate/connective use, which is attributed to processes of insubordination and incoordination, also attested elsewhere in the history of Korean. Interestingly, for those markers, the apprehensional function is more prominent in the corpus attestations in their subordinate (connective) use than in their main clause use.

In *Explicit apprehensions, implicit instructions: An indirect speech act in the grammar,* Alexandre François offers a survey of apprehensional strategies in the Oceanic languages of north Vanuatu, followed by an in-depth study of the syntactic, semantic and pragmatic properties of a dedicated apprehensive marker in one of these languages, Mwotlap. Despite their close genealogical relationship, the languages of the Torres-Banks area of north Vanuatu exhibit a fascinating diversity of apprehensionals (subordinators and/or modal markers), with interesting patterns of polyfunctionality in that they overlap with either an ablative case marker or a prohibitive marker (and in the case of Lemerig, with both). Out of 8 languages surveyed, only Mwotlap has a distinct apprehensive mood marker (*tiple* and allomorphs). Clauses featuring *tiple* can function as complements of 'fear' verbs, and are also frequently found in combination with a preemptive clause in both avertive and 'in-case' functions. However, François argues that *tiple* is not a subordinator and that the relationship of apprehensive clauses with preemptive clauses is one of pragmatic rather than syntactic dependency. Thus, apprehensive clauses on their own can be exploited to convey an implicit instruction (that of taking appropriate action to avoid the situation described by the apprehensive clause), i.e. an indirect speech act, which just like other cases of indirect speech act serves as a politeness strategy: an explicit directive can be avoided by merely pointing out to the addressee the consequences of not taking action.

Mitchell Browne, Thomas Ennever and David Osgarby explore both dedicated and non-dedicated apprehensional expressions at all levels of grammar in a dozen languages of the Ngumpin-Yapa subgroup of Pama-Nyungan (Australia), in a comparative Australianist perspective. Despite their relatively close genealogical relationship, the languages examined exhibit an interesting diversity in the apprehensional domain. Most Ngumpin-Yapa languages surveyed have an aversive case suffix with the specialised function of indicating that a nominal referent is to be feared or avoided. This can have a subordinating function in non-finite clauses, and thus form precautioning subordinate constructions. It can also mark (nominal or non-finite clause) complements of 'fear' predicates in some languages, and even displays insubordinate uses with the illocutionary force of a directive. The second major strategy for encoding apprehensional meanings in Ngumpin-Yapa languages involves grammatical markers variously analysed as auxiliaries or complementisers/subordinators in existing descriptions. The authors favour an analysis as auxiliaries, even when a preemptive clause is mostly present in the context, i.e. they analyse the combination of preemptive clause and apprehensional clause as cases of juxtaposition, with a pragmatic rather than syntactic relationship between the two, although they consider the potential of a diachronic development to subordination constructions. The semantic interpretation of the auxiliaries may also depend on the verbal inflection, e.g. apprehension with future inflection but past counterfactual with past inflection. The authors also demonstrate that such auxiliaries are dedicated apprehensives in some of the languages surveyed but have a broader irrealis, habitual or capability meaning in others. Such semantic specialisation likely arose through pragmatic strengthening, from a general possibility modal to an apprehensive marker.

Marine Vuillermet's contribution differs significantly from the others as it is a questionnaire intended as a guide to fieldworkers. It systematically explores the semantic, syntactic and pragmatic specificities of the four main construction types in the apprehensional domain – namely apprehensive, precautioning, 'fear' complementation and timitive. It outlines the potential variation in the properties of the respective constructions that is expected based on a cross-linguistic survey of apprehensionals, and provides contexts targeting the corresponding parameters in the form of over 80 prompts in English, to be translated into the language of study (and/or adapted to the cultural context). It is hoped that this questionnaire will enable the systematic investigation of the fine-grained semantics of apprehensional constructions, which will in turn contribute to a better understanding of the intricacies of the apprehensional functional domain.

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Abbreviations

1/2/3	first/second/third person	IRR	irrealis
A	agent	MID	middle
ABS	absolutive	NEG	negation
ADD	additive	NF	non-feminine
ADV	adverb	NFUT	non-future
APPR	apprehensive	NMZR	nominaliser
CAUS	causative	OBJ	object
CNTR	control	P	patient
COM	comitative	PL	plural
DAT	dative	PREC	precautioning
DECL	declarative	PROH	prohibitive
DEF	definite	PROPR	proprietive ('having')
ERG	ergative	PRS	present
EXPL.NEG	expletive negation	SBJV	subjunctive
FACT	factive	SEQ	sequential
FEM	feminine	SG	singular
FUT	future	SS	same subject
IMP	imperative	SUB	subordinator
INCL	inclusive	TIM	timitive
IND	indicative	TMP	temporal subordinate
INST	instrumental	VIS	visual evidential

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Chapter 2

Speaking and acting with care: The grammar of circumspection in Upper Tanana Dene

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Upper Tanana Dene has two constructions falling into the domain of circumspection: avertive clause linkings and urgent requests. In avertive linkings, the avertive clause expresses an undesirable consequence that can be averted by the action expressed in the main clause. Somewhat unusual typologically, avertive linkings with a request as main clause are rare in Upper Tanana. I argue in this paper that this is likely due to the fact that requests are generally problematic in Northern Dene, and that spelling out the undesirable consequences to be avoided could suggest to the addressee that their cultural knowledge is inadequate. Instead, there is a dedicated strategy for urgent requests resulting from an insubordinate conditional construction, which draws attention to the undesirable consequences of non-compliance without spelling them out explicitly. Together, these two constructions reflect the circumspect mindset of the Upper Tanana Dene, which requires them to act and speak with care.

1 Introduction

In this paper I describe two constructions falling into the domain of circumspection in Upper Tanana, a Dene (Athabascan) language spoken in eastern interior Alaska and the western Yukon. *Avertive linkings* are adverbial clause linkings where the subordinate clause expresses an undesirable event that can be averted or prevented by taking the action expressed in the main clause. I choose the label *avertive* over that of *negative purposive*, since in Upper Tanana, this clause

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type differs in important ways from negated purposive clauses, as discussed in §2.3. *Urgent requests* are a special type of request signaled by the presence of the marker *dè'* 'if, when.FUT, UR'. This particle suggests that this construction arose via insubordination of a conditional clause construction. Non-compliance with these requests usually has dire consequences, but these are never spelled out explicitly. I will also show how the existence and use of these constructions reflect the circumspect mindset of the Upper Tanana Dene.

In §1.1, I provide cultural background on the Upper Tanana, focusing in particular on the need for circumspect action and careful speaking. §1.2 provides linguistic background on the Upper Tanana language. The avertive clause linking is described in detail in §2, urgent requests in §3. §4 situates both constructions in the cultural context. The findings are summarized in §5.

In my discussion of cultural traits of the Upper Tanana throughout this paper, I draw on ethnographic resources on other Northern Dene groups, in particular on the work by Nelson (1973) on Gwich'in, Nelson et al. (1982) and Nelson (1983) on Koyukon, and Scollon & Scollon (1981) on Tanacross (all these groups are located in Alaska) as well as Rushforth (1985) and Rushforth & Chisholm (1991) on Bearlake (Northwest Territories). The ethnographic literature on Upper Tanana itself focuses on topics that do not relate directly to the focus of the present paper. McKennan (1959) and McKennan (1981) provide broad cultural overviews while McKennan (1969) is concerned with population development during the contact period. The early work by Guédon (1974, 1981) describes the potlatch, one of the most important cultural institutions; her most recent work (Guédon 2005) focuses on the spiritual life and particularly on shamanism.

Even though all of the above cultures differ from each other in many details, the fundamental patterns are very similar, and, as has been argued by McKennan (1981) for the groups known today as Lower, Middle, and Upper Tanana as well as Tanacross, and by Krauss (2000) for Alaskan Dene languages, the group distinctions made by ethnographers and linguists do not reflect the perspective of the Dene themselves, who view their communities as part of an intricate network covering a large portion of North America. In taking this approach, I follow the practice of Scollon & Scollon (1981) in their work on interethnic communication, who focus on two groups they label *Athabaskans* and *speakers of English*. By *Athabaskan*, they "mean anyone who has been socialized to a set of communicative patterns which have their roots in the Athabaskan languages" (p. 12). By the same token, I use the term *Dene* to refer to anyone socialized into any (Northern) Dene culture, while also paying attention to customs specific to the Upper Tanana area.

It is also important to note that many of the cultural traits described here are less salient to younger individuals, likely due to the influence of mainstream American culture.

1.1 The need for circumspection: Careful speaking, careful action

The interactions of Alaskan Dene are characterized by circumspect behaviour, linguistic and non-linguistic. Some aspects of linguistic circumspection have been described by Scollon & Scollon (1981). Of particular importance for our purposes are two points raised there: reluctance to talk about the future (coupled with a need for preparedness) and the principle of non-interference.

For an Alaskan Dene person, to talk with any degree of certainty about the future constitutes a violation of *įįjih*, the moral system governing an individual's behavior (cf. Guédon 2005: chap. 3; Lovick 2016, Lovick & Tuttle 2019 for more discussion of Upper Tanana *įįjih*; similar concepts are described for many Alaskan Dene groups). Making plans for the future, or expressing an expectation about what the future will be like (or that the future will be good), is to court bad luck, and is thus consistently avoided (Scollon & Scollon 1981: 20). Nelson (1973: 194–195) notes the unreliability of the weather in interior Alaska, and the difficulty even experienced hunters encounter when trying to predict the weather. The natural world is thought to be too large to be a safe discussion topic; Nelson (1983: 45) reports that Koyukon children are often reminded of the need for humility in the face of nature using the admonition "Don't talk; your mouth is too small."

As a consequence, the Upper Tanana Dene place a lot of cultural value on the concept of *ts'edòònih*, an uninflectable word of uncertain lexical category often translated as 'you don't know what's gonna happen' (author's fieldnotes) or 'you never know' (Kari 1997). Implicit in the concept of *ts'edòònih* is the recognition that there are many things beyond an individual's control, and the awareness that it is each person's responsibility to be prepared for all of them at all times. Part of this preparedness is the need to look out for one another, as articulated by Mrs. Avis Sam (N) in a recent text about *ts'edòònih* (1).[1]

[1] All data is presented in the practical orthography. Tone is not usually indicated in the practical orthography but marked in the bulk of the data in this paper, with the exception of the published data from the Scottie Creek dialect (Tyone 1996), which is presented as in the original source. High tone, associated with standard negation and several lexical items, is marked for all data from the Beaver Creek, Northway, and Tetlin dialects. Low tone originating from Proto-Dene constriction (Krauss 2005) is marked for the Northway and Beaver Creek data but not for the Tetlin data, as low tone is not phonemic in that variety (Minoura 1994, Lovick 2020a).

(1) *Nił-'èh na-ts'i-dek dè' ts'iikeey iin*
recp.po-with prmb-1pl.s:ø.ipfv:ø-pl.go:ipfv:cust when children pl
ts'edòònih xà' hǫǫsú'
never.know because well
nił-k'à-nal-tà'. (N)
recp.po-care-qual:ø.ipfv:2pl.s:l-take.care:ipfv
'When we walk around together, children, take good care of each other because of *ts'edòònih*.' [UTOLAF19Jan0501:019]

In addition to this general mindset of circumspection, Alaskan Dene traditionally perform numerous small, often ritualistic acts that serve to prevent undesirable events or to ensure favorable outcomes. This is illustrated in (2), which describes the proper placement of whitefish on a drying rack in the Upper Tanana area.[2]

(2) a. *A-dógn da-ts'e-łeeg tah chih*
ntrl-up:ar up-1pl.s:ø.ipfv:h-handle.pl.o:ipfv:cust:nmz when also
k'at'eey né' nts'ą' k'a
neg backward:all advzr neg
da-tsaa-dláál u-tthi' ts'ąy
up-1pl.s:inc:aa.ipfv:ø-keep.pl.o:fut:neg 3pl.psr-head side
a-nóó ts'ą' shyį́į́' da-tsaa-łaał
ntrl-ahead:all advzr only up-1pl.s:inc:aa.ipfv:ø-keep.pl.o:fut
u-tthi'. (T)
3pl.psr-head
'Also when we put [whitefish] up [on the drying rack], we may not keep it up there backward, we have to keep it there with the sides of the head facing forward, their heads.' [UTOLVDN11Jul2903:003]

b. *Ay tah chih k'at'eey keł na-tat-dlą́ą́*
3sg when also neg ? it-3pl.s:inc:aa.ipfv:d-be.many:fut:neg
he-ndiik na-né' nts'ą'
3pl.s:ø.ipfv:ø-say:ipfv med-backward:all advzr
da-hiiy-eh-łeek tah.
up-3pl>3pl-ø.ipfv:ø-handle.pl.o:ipfv:cust when
'There won't be as many, they say, when they put them up facing backward.' [UTOLVDN11Jul2903:004]

[2]Whitefish are caught during their annual migration to their spawning grounds, hence there is a 'forward' and a 'backward' relative to the direction of their migration.

There are countless acts such as these; Nelson et al. (1982: chs. 3, 4, 9, 14) list a large number of them for the Koyukon people, but are careful to note that there are many more. Lists of these acts, and of the consequences they are intended to avert, are a rich source for avertive linkings (see also §4).

Knowledge of these acts is part and parcel of being an Alaskan Dene; in the Upper Tanana area, this is one component of "knowing one's *į̨įjih*" (author's field-notes), i.e. having a solid grasp of cultural knowledge. A competent individual is expected to know (and ideally adhere to) their *į̨įjih*, and Upper Tanana speakers are very careful not to suggest that an individual's knowledge of *į̨įjih* is deficient (cf. Lovick 2016: 273).

This last point is linked to another fundamental tenet of Northern Dene culture: non-interference with another's autonomy (cf. Scollon & Scollon 1981 for Alaskan Dene, Rushforth & Chisholm 1991 for Bearlake Dene, or Basso 1979, 1990 for Western Apache). A direct consequence of this attitude is a general reluctance to utter questions and requests (Scollon & Scollon 1981). These authors note that both types of speech act are routinely avoided, while Rushforth (1985) and Rushforth & Chisholm (1991) observe a marked preference for indirect requesting strategies.[3] Lovick (2016) finds that, while direct affirmative requests are common in Upper Tanana, their form depends on the relationship between the interlocutors as well as on the imposition resulting from complying with the request. Negative requests are strongly avoided in Upper Tanana (cf. Lovick 2016, Lovick & Tuttle 2019), not merely because of the principle of non-interference, but also because they could be interpreted as suggesting that the addressee's knowledge of *į̨įjih* is deficient.

Thus, traditional Upper Tanana speakers are very circumspect both in words and actions. Verbally, they avoid interfering in another's autonomy, making predictions about the future, or talking about things for which their "mouth is too small". Practically, they value careful, often symbolic, interactions with animate and inanimate entities, as well as preparedness for all matters outside their control.

1.2 Upper Tanana language

Upper Tanana is a Northern Dene language spoken by 30–50 mostly elderly speakers in eastern interior Alaska and the Yukon Territory. Minoura (1994) identifies five dialects; this paper contains data from my own fieldwork on the North-

[3]This is also the reason for my terminological choice to talk about requests rather than commands.

way (N), Tetlin (T), and Beaver Creek (B) dialects as well as from a story collection (Tyone 1996) in the Scottie Creek (S) dialect.

Morphologically, Upper Tanana can be classified as polysynthetic, incorporating, and fusional. It is characterized by particularly rich verbal morphology which can be represented by the template indicating morpheme order in Table 1.

Table 1: The Upper Tanana verb, represented as a template

12	11	10	9	8	7	#	6	5	4	3	2	1	0	-1
Postpositional object	Postposition	Adverbial-derivational	Iterative	Incorporate	Distributive	– Disjunct boundary –	Pronominal	Qualifier	Conjugation	Mode	Subject	Voice/valence	**Stem**	Suffix

The basic lexical unit in the verbal domain is the *verb theme* which consists minimally of a verb stem and a voice/valence marker. Many verb themes contain additional lexical prefixes and form the basis for further derivations. There are no nonfinite verb forms; each verb is obligatorily inflected for subject (positions 2 or 6 in the template), conjugation (position 4) and mode (position 3). Transitive verb themes are inflected for direct object (position 6). Morphemes in positions 6–1 often fuse into a single syllable, which I gloss as a portmanteau here and in other work, with zero expression indicated in the gloss (see Lovick 2020a: 6–10 for background on Upper Tanana glossing specifically and Hargus et al. 2020 for more information on glossing conventions in Dene languages). Other lexical classes have much more limited morphology; nouns can be inflected for possessor and (free) postpositions for postpositional object.

Upper Tanana distinguishes four *modes*: imperfective, perfective, optative, and future.[4] The first two encode viewpoint aspect, while the latter two have modal functions. The optative expresses primarily deontic modality such as desire or obligation, while the future has mostly epistemic functions (Lovick 2017). The formation of all modes involves prefixes (conjugation and mode in Table 1) as well as stem changes originating in historic suffixation (Leer 1979); the future additionally requires the presence of the inceptive prefix (one of the qualifiers in

[4]The label *modes* is somewhat misleading, but for lack of a better term, I continue to use it here.

Table 1). Upper Tanana also has productive negative inflection as illustrated in (3). Negative inflection is used only in standard negation and involves changes in the conjugation/mode domain as well as a negative suffix which is absorbed into the stem syllable with sometimes idiosyncratic effects; in (3b), it voices the stem-final consonant. Standard negation also involves tonal changes; the unmarked pre-stem syllable in (3a) becomes low in the negative form, while all stems (unmarked and low-marked alike) become high. Standard negation additionally requires the presence of the negator *k'à(t'eey)* under most circumstances.

(3) a. *shyì' įį-'ààł* (N)
meat 3SG.S:AA.PFV:Ø-eat:PFV
's/he ate the meat' [UTCBAF13Nov0802:008]

b. *k'àt'eey shyì' ì-'áál* (N)
NEG meat 3SG.S:NEG.PFV:Ø-eat:PFV:NEG
's/he didn't eat the meat' [UTCBAF13Nov0802:009]

Upper Tanana is generally verb-final, although some syntactic material may follow the verb. Arguments of verbs and postpositions as well as nominal possessors may be expressed by free noun phrases or marked on the verb, postposition, or noun.

Important for the discussion in this paper is the structure of adverbial clauses, which are characterized by the presence of a clause-final subordinator. Many subordinators are formally identical to postpositions and require nominalization of the verb (marked formally by a historical vocalic suffix which is absorbed into the verb stem; common effects of this suffix include voicing of stem-final consonants, deletion of stem-final *h*, and lengthening of the stem vowel);[5] many also require the verb in either clause to be in a particular mode. The order of adverbial clause and main clause is generally free but often follows iconic principles. Two types of adverbial clauses are illustrated in (4).

(4) a. *Ay èh dhįį-t'eh dè' shit*
and DH.PFV:2SG.S:Ø-get.burned:PFV if scar
taa-'aał. (N)
3SG.S:INC:AA.IPFV:Ø-classify.CO:FUT
'And if you get burned, there will be a scar.'
[UTOLVDN14Apr2601:051]

[5] In his description of Dëne Sułiné, Cook (2004: 375–381) argues in favor of treating such clauses as nominal objects of postpositions.

b. *… ishyiit nii-ts'in-deeł*
there TERM:IT-1PL.S:N.PFV:D-PL.go:PFV
na-nts'-uu-'aal *xah* … (N)
CONT-1PL>INDF-OPT:Ø-eat:OPT:NMZ PURP
'… we went back there in order to eat …' [UTOLVDN14Apr2602:115]

Example (4a) illustrates a conditional linking formed using the subordinator *dè'* 'if, when.FUT', which is formally identical to the postposition PO+*dè'* 'at a time PO in the future'. This subordinator does not generally cause nominalization. The verb in the condition may be in any mode, while the verb in the consequent is almost always in the future. The condition usually, but not necessarily, precedes the consequent.

(4b) illustrates a purposive clause formed using the subordinator *xah* 'in order to, PURP', related to the postposition PO+*xah* 'for PO, because of PO'.[6] This subordinator usually triggers nominalization of the subordinate verb. Purpose clauses are always in the optative; there are no mode restrictions on the verb in the main clause. The main clause usually precedes the purpose clause.

Also important for this paper is the formation of (affirmative) requests, originally described in Lovick (2016). The two most common types described there are requests in the imperfective and optative modes (5).

(5) a. *"Jah shyi' ay iin tl'a-'ah-łeey,"*
this meat 3 PL to-Ø.IPFV:2PL.S:Ø-handle:PL.O:IPFV:IMP
nee-'eh-niik.
1PL.O-3SG.S:Ø.IPFV:H-say:IPFV:CUST
' "Give (PL) them the meat here," he would say to us.'
[UTCBAF13Nov1203:039]

b. *Mb-aa n-che' ski-dǫǫ-teey* …
3SG.P-for 2SG.PSR-tail across-QUAL:OPT:2SG.S:Ø-handle.LRO:OPT
'Put your (SG) tail across for him …' [UTOLVDN14Nov2301:359]

Generally, requests in the imperfective are used when the speaker is comfortable making a request, either because of the relationship between the interlocutors (e.g. a father addressing his children as in (5a), or an interaction between close friends or siblings) or because of the trivial nature of the action to be performed (Lovick 2016: 280–283). Requests in the optative are used, by contrast,

[6]This subordinator is also used in causal clauses with the meaning 'because', which have a slightly different structure from the purpose clauses described here.

in situations where the request is an imposition, either because of the nature of the action to be performed or because of a less easy relationship between the speaker and the addressee. In (5b), the speaker is addressing a Fox—a stranger to her. Optative requests are used in situations where the speaker does not feel entitled to making a request at all (Lovick 2016: 283–285).

1.3 Data

The bulk of the data for this study comes from a corpus comprising over 8,000 utterances of mostly narrative text recorded and processed by the author between 2006–2019. About 15 speakers are represented there, but the majority of text comes from two speakers of the Tetlin dialect and four speakers of the Northway dialect. Another 2,000 or so utterances come from the story collection by Mary Tyone (1996) in the Scottie Creek dialect. I mined this corpus for instances of avertive clauses using a string search for the avertive subordinator *ch'à'* and the urgent request marker *dè'*. I also searched for common translations of both construction, e.g. "so that … not", "to prevent that", or "make sure". Selected examples were discussed with speakers during the transcription and translation process; the notes I took during those discussions also fed into this paper.

The narrative corpus is supplemented by targeted elicitation. Elicitation of avertive clauses followed the methodology outlined in Vuillermet (2018). Using the contact language (English), I developed a scenario, e.g., walking about in the bush. I then asked for a translation of a target sentence into Upper Tanana, e.g., 'I am going to boil the water so I don't get sick from it' and discussed the result with the speaker(s): is this the only way of saying it? What other ways of expressing this are there? Why did you phrase it this way? I found that the best results were obtained when speakers could work in pairs, since they could discuss the translation with each other, but this was not always feasible.

Direct elicitation of apprehensional requests is more challenging (see Lovick 2016, Lovick & Tuttle 2019 for discussion of the challenges of request elicitation), so instead, I selected examples from the corpus and discussed those with speakers, paying attention to how changing the form of the request would impact its meaning and use.

All data collected by the author or her team is cited by its corpus identifier.

2 Avertive linkings

Avertive linkings are a type of adverbial linking. The structural properties of avertive linkings and some structural restrictions are described in §2.1. Their semantics is discussed in §2.2; alternative constructions with similar semantics are described in §2.3. In §2.4 I show that avertive linkings with similar structural properties are attested in several other Dene languages. The origin of the avertive subordinator used in all of these languages is discussed in §2.5. This section closes with a discussion of the relative frequency of avertive linkings in different speech acts.

2.1 Structural properties

Avertive linkings always involve the subordinator *ch'à'* 'AVERT, lest' in clause-final position. The verb in the avertive clause is usually in the future (6a)–(6c) or in the inceptive perfective mode, which often has the meaning of 'immediate future' ((6d), cf. Lovick 2017). The avertive clause (bolded throughout this paper) usually follows the main clause (6a)–(6c), but the opposite order is also attested (6d).

(6) a. *Huu-ła' tah chih tl'uuł net-chüh*
3PL.PSR-hand among also rope 3SG.S:QUAL:DH.PFV:D-be.tied:PFV
huu-łaa-güü taa-łeel ch'a'. (S)
3PL.PSR-finger-forked 3SG.S:INC:AA.IPFV:Ø-be:FUT:NMZ AVERT
'Rope was tied between their fingers **to prevent them from [having] spaces between their fingers**.' (Tyone 1996: 19)

b. *Di-nįį' t'eey hiiy-ee-k'ia dégn' nts'ą'*
REFL.PSR-face ADVZR 3PL>3PL-DH.PFV:Ø-exercise:PFV up:ALL ADVZR
na-h-nal-xon ch'a' … (T)
down-3PL.S-INC:QUAL:AA.IPFV:L-sag:FUT:NMZ AVERT
'They train their faces upward (i.e. they wash their faces with upward motions) **to prevent them from sagging down** …'
[UTOLVDN11Jul2901:033]

c. *Ay xah* ***doo iin nįį-deel iin***
3SG because who PL 3PL.S:N.PFV:Ø-PL.go:PFV:NMZ PL
i-tįl-nüü ch'à' *ay xah ch'à'*
PP-INC:AA.IPFV:2SG.S:L-forget:FUT:NMZ AVERT 3SG because FOC
hu'èh dhįįdah. (N)
3PL.PO-with DH.PFV:2SG.S:Ø-SG.sit:IPFV
'That's why, **so you don't forget those who came**, that is why you sit with them.' [UTOLVDN14May0508:038]

d. *Ay tl'ààn* ***tuu tèt-shyoo ch'à'***
and.then water 3sg.s:inc:dh.pfv:d-get.splashed:pfv:nmz avert
hà-nì-t'ah, dits'än. (N)
up/out-3sg.s:n.ipfv:ø-sg.fly:ipfv duck
'And **to avoid getting splashed**, the duck flies up.'
[UTOLVDN07Aug2205:048]

The main clause in an avertive linking may have any person subject: third person in (6a), (6b), (6d), second person in (6c), and first person in (7).

(7) *Nts'ą̀ą̀'* ***tat-xòl ch'à'*** *nts'ą̀ą̀' łaan*
how 3pl.s:inc:aa.ipfv:d-break:fut:nmz avert how truly
ts'uh-'į̀į̀? (N)
1pl.s:opt:h-act:opt
'What should we do **so [the eggs] won't break?**'
[UTOLVDN07Aug2205:054]

Similarly, there are no person restrictions on the avertive clause's subject. While third person seems the most common ((6a)–(6c), (7)), second person (6d) and first person (8) are attested. There also is no co-reference requirement between the two clauses' subjects ((6)–(8)).

(8) *"Sh-taay, tl'uuł sh-tl'ädn hi-'į̀h-shüh*
1sg.psr-paternal.uncle rope 1sg.psr-waist pp-ø.ipfv:2sg.s:h-tie:ipfv
sta-tih-haał ch'a'," … (S)
away-inc:aa.ipfv:1sg.s:ø-sg.go:fut:nmz avert
'"My uncle, tie a rope around my waist **lest I run away**," …'
(Tyone 1996: 26)

While the avertive clause is always in the future or the inceptive perfective, there appear to be no mode restrictions on the main clause.

2.1.1 Structural restrictions

The only structural restriction on avertive linkings is that the main clause has to be affirmative. Avertive linkings with negated main clauses are not attested in the corpus, nor could they be elicited.[7] Speakers consistently employ different constructions, e.g. (9a), where the undesirable consequence of tooth decay is expressed by a causal clause, or (9b), where Raven's feared theft of the sun is expressed by simple juxtaposition.

[7] This is true for standard negation, negative requests, and negation of existence.

(9) a. *… k'à hii-nel-déél,* ***hu-ts'änh***
neg 3pl.s>3pl.o-qual:ø.ipfv:l-eat:pl.o:ipfv:neg 3pl.po-from
hu-xù' ***tah-jiit*** ***xah.*** (B)
3pl.psr-teeth 3pl:inc:aa.ipfv:h-rot:fut because
'… they do not eat [cranberries], because their teeth will rot from them.' [UTOLVDN10Jul2602:063]

b. *Sǫ̀' ay chih èh dul-xoo,* ***ą̈́n***
proh 3sg too with 3sg.s:opt:l-play:opt outside:all
y-èh ***tthi-taa-haał!*** (N)
3sg.s>3sg.po-with out-inc:aa.ipfv:ø-sg.go:fut
'Don't let him play with [the sun], he will go outside with it!' [UTOLAF18Jun1001:009]

The reason for this is unclear. The small overall number of avertive linkings in the corpus (see §2.6) raises the possibility that this is an accidental gap, but the fact that elicitation of avertive linkings with negative or prohibitive main clauses consistently yielded constructions similar to those in (9) suggests that this is a true structural restriction.

2.2 Semantic properties

Semantically, the avertive clause expresses an undesirable event that can be prevented by taking the action expressed in the main clause. Pubescent girls traditionally were not supposed to spread their fingers; to avoid this, their fingers were tied together (6a).[8] (6b) describes a measure to prevent skin aging. (6c) is about courtesy to potlatch visitors; the person putting on a potlatch has the responsibility of spending a little time with each visitor so they will remember who came (and give gifts appropriately). (6d) describes duck behavior that is exploited for egg gathering.

Undesirability is a semantic component of this construction, not a pragmatic implicature. Avertive clauses cannot be used in situations where the possible consequence is not undesirable. In most cases, the event in the avertive clause is intrinsically undesirable, but this is not required. (10) comes from elicitation. The scenario given to the speakers was that several children are playing hide-and-seek. Two of them are sharing a hiding spot. When the seeker comes close,

[8] One anonymous reviewer suggests that this may be to avoid the appearance of witchcraft, which in some Dene groups involves hand gestures. To my knowledge, nothing of the sort is reported for the Upper Tanana (or related Alaskan Dene) area, but this may be due to a general reluctance to discuss such topics.

the girl is about to make a noise: either a scream (intrinsically undesirable; (10a)) or a giggle (situationally undesirable; (10b)). The speakers originally suggested the negated purposive clause in (10c) to express the scenario involving laughter, but decided after some discussion that the avertive clause in (10b) was preferable. (The problematic status of negated purposive clauses in Upper Tanana is discussed in §2.3.)[9]

(10) a. *D-i̧i̧łà'* *u-thaat*
REFL.PSR-hand 3SG.PSR-mouth
da-dì̧h-'ąą
closed-3SG.S:QUAL:AA.PFV:H-handle.CO:PFV
taa-säl ***ch'à'*** . (N)
3SG.S:INC:AA.IPFV:Ø-scream:FUT:NMZ AVERT
'He put his hand over her mouth **so she wouldn't scream**.'
[UTOLAF18Jun0805:002]

b. [*D-i̧i̧łà'* *u-thaat*
REFL.PSR-hand 3SG.PSR-mouth
da-dì̧h-'ąą]
closed-3SG.S:QUAL:AA.PFV:H-handle.CO:PFV
taa-dlòò ***ch'à'***. (N)
3SG.S:INC:AA.IPFV:Ø-laugh:FUT:NMZ AVERT
'[He put his hand over her mouth] **so she wouldn't laugh**.'
[UTOLAF18Jun0805:004]

c. *? D-i̧i̧łà'* *u-thaat*
REFL.PSR-hand 3SG.PSR-mouth
da-dì̧h-'ąą ***k'à***
closed-3SG.S:QUAL:AA.PFV:H-handle.CO:PFV NEG
uu-dlóó ***xah***. (N)
3SG.S:OPT:Ø-laugh:OPT:NEG PURP
'He put his hand over her mouth **so she wouldn't laugh**.'
[UTOLAF18Jun0805:001]

The undesirability is evaluated from the perspective of the subject of the main clause. The girl who utters (11a) wants to escape, but the addressee, her evil "uncle", wants her to stay. In (11b), people are trying to prevent a man whose only son has just drowned in a boat accident from taking his own life by walking out into the lake.

[9]In (10b), the speaker did not repeat the main clause, hence it is included here in brackets.

(11) a. *"Sh-taay, tl'uuł sh-tl'ädn hi-'įh-shüh*
1SG.PSR-paternal.uncle rope 1SG.PSR-waist PP-Ø.IPFV:2SG.S:H-tie:IPFV
sta-tih-haał ch'a', *"* ... (S)
away-INC:AA.IPFV:1SG.S:Ø-SG.go:FUT:NMZ AVERT
'"My uncle, tie a rope around my waist **lest I run away**," ...'
(Tyone 1996: 26)

b. *Ay du' nahugn diniign thüh tl'uul' heh-tsįį*
and CT there moose skin rope:POSS 3PL.S:DH.PFV:H-make.SG.O:PFV
ay hii-'i-nih-tl'ǫǫ eł nóó
and 3PL>3SG-PP-N.PFV:H-tie:PFV and ahead:ALL
ch'aa-'i-mbeek
out:PRMB-3SG.S:Ø.IPFV:Ø-swim:IPFV:CUST
ti-taa-haal ch'a'. (T)
underwater-3SG.S:INC:AA.IPFV:Ø-SG.go:FUT:NMZ AVERT
'And they made a moose skin rope and tied it around him [when] he walked out into the water, **so he would not go under [and drown himself]**.'
[UTOLAF10Jul0801:023]

2.2.1 Semantic restrictions on avertive clauses

The most important semantic restriction on avertive clauses in Upper Tanana is that the event expressed in the avertive clause actually can be prevented by the action in the main clause; i.e. the subject of the main clause has control over the event in the avertive clause.[10] One can prevent a person from running away or drowning themselves by physically restraining them (11), and screaming or giggling can be prevented, or at least subdued, by placing a hand over a person's mouth (10), but, as the elicited example in (12) shows, one cannot prevent a child from grabbing a knife by merely watching him or her, and thus an avertive linking cannot be used.

[10] This contrasts sharply with the situation described by Wiemer (2026 [this volume]) for several Slavic languages, where uncontrollability, i.e. the fact that "neither a discourse participant (in particular the addressee) nor a third party can influence the course of events", is an integral component of what Wiemer terms "apprehensional strategies". This is illustrated abundantly by the examples brought by Wiemer, where possible events to be avoided or prevented, include weather events such as rain or droughts, catching cold or other diseases, or war breaking out.

(12) *Ts'iiniin k'à-nįh-tà',* ***seeyh***
child care-QUAL:Ø.PFV:2SG.S:H-take.care:IPFV knife
uu-taa-nik! (N)
3SG:CON-INC:AA.IPFV:Ø-grab:FUT
'Take care of/watch the baby, he might grab the knife!'
[UTOLAF18Jun0803:003]

The scenario in the elicited example (13) involves the speaker getting into an argument with an obnoxious stranger at the grocery store. The cue used was 'Shut up so I don't beat you up!'. A causal linking is used instead of an avertive one.

(13) *Da-dhį̀į̀-nąy,*
LEX-DH.IPFV:2SG.S:Ø-say:IPFV:NEG
n-nak-gòt ***xah!*** (N)
2SG.O-INC:QUAL:AA.IPFV:1SG.S:H-punch:FUT because
'Shut up, because I might punch you!' [UTOLAF18Jun0803:020]

The speakers rejected my suggestion of *nnakgot ch'a'* 'lest I punch you' with the rationale that your shutting up may not be sufficient to keep me from punching you: a rude gesture or facial expression might, after all, provoke me just as much as continued talking would. Another reason for the avoidance of an avertive linking could be that the verb in the main clause is formally negative.[11] I am however uncertain whether speakers recognize this as a negative form due to the degree of fossilization and the lack of negative particles.

In 'in-case' scenarios like the elicited example in (14), avertive clauses can never be used; see §4 for more discussion.

(14) *Eh Bill: Tay' t'eey tah-chaan ay xah*
and Bill again ADVZR 3SG.S:INC:AA.IPFV:H-rain:FUT:NMZ 3SG because
ch'ale u-shyii chąą taathüh di-dhih-dlay
FOC 3SG.PO-in umbrella QUAL-DH.PFV:1SG.S:Ø-keep.PL.O:IPFV:NMZ
tthiishyoh eh tl'aan dii na-ti-dhag-niign
hat and and what PRMB-INC-DH.PFV:1SG.S:L-wear:IPFV:CUST:NMZ

[11] *Dadhįįnąy* 'shut up' is one of a handful of fossilized morphologically formed prohibitives containing the negative prefix *dh-* otherwise lost in Upper Tanana (Kari 1993, van Gelderen 2008); Lovick (2020a: 472). The structure of *dadhįįnay* 'shut up' is discussed in some more detail in Lovick (2020b).

tat'eey u-shyii dhih-dlah. (T)
also 3sg.po-in dh.pfv:1sg.s:ø-keep.pl.o:ipfv
'And Bill: It might rain again, that's why I keep the umbrella in [the backpack] and the hat and my clothes I also keep in it.'
[UTCBAF13Nov1510:035]

2.3 Alternative constructions

A number of strategies can be used when an avertive linking is not possible due to semantic or structural restrictions. As shown in the preceding sections, both juxtaposition (9a) and causal clauses (9b), (13) and (14) occur with some frequency, but there are additional options. In 'in-case' scenarios, temporal linkings with *ttheh* 'before' are a common strategy (15).

(15) ***"Nah-ógn tah-shyǜǜdn ttheh***
med-out:ar inc:aa.ipfv:h-snow:fut:nmz before
na-tsut-dèèł," he-nih. (N)
it-1pl.s:inc:opt:d-pl.go:opt 3pl.s:ø.ipfv:ø-say:ipfv
'"Let's go back before it snows," they said.' [UTCBAF15May0403:039]

In the speech of at least one speaker, *ttheh* has been extended to have avertive meaning as well (16).

(16) ***Ni-h-teedak ttheh*** *"Nah-'óg*
up-3pl.s-inc:dh.pfv:ø-pl.go:pfv before med-out:all
na-h-tet-dag iin," nih
prmb-3pl.s-inc:ø.ipfv:d-pl.go:ipfv:cust:nmz pl say
hu-'i-chi-jel-nih. (T)
3pl.po-pp-lie-indf:3sg.s:qual:ø.ipfv:l-lie:ipfv
'**To prevent them getting up**, "There's people walking about outside," she said, she lied to them.' [UTOLVDN13May2909:067]

(16) is taken from a narrative where a young woman has been abducted by warriors. Her brothers have come to rescue her and are killing her captors outside the tent where she is being held. The men inside the tent hear the noises of the killing, but to prevent them from discovering the slaughter outside, she lies to them, pretending that the noises outside are normal and harmless. Temporal linkings with an avertive interpretation also occur in elicitation (17):

(17) *U-ch'a'* *ch'i-nag-'įį*
3SG.PO-from INDF-N.PFV:1SG.S:L-hide.self:PFV
sh-oo-dah-'aal ***ttheh***. (T)
3SG.S>1SG.O-CON-INC:QUAL:AA.IPFV:H-find:CO:FUT:NMZ before
'I'm hiding from him **to prevent him from finding me**.'
[UTOLAF18Jun0201:016]

It seems however that the undesirability of the event in the subordinate clause is implicated rather than part of the semantic meaning of the construction. The snow in (15), for example, is not necessarily undesirable, although it is preferable to travel in dry conditions; nor is being found in a game of hide-and-seek in (17) generally problematic.

In the discussion of the semantics of avertive linkings, the potential existence of a negative purposive linking such as (10c) was noted. Such linkings are attested primarily in elicited examples such as that in (18); to facilitate structural comparison, an affirmative purposive clause is provided in (19).

(18) *? Olga da-dégn'* *hǫǫ-heeyh* ***k'àt'eey***
Olga PROX-up:ALL 3SG.S:QUAL:QUAL:Ø.IPFV:Ø-talk:IPFV:NMZ NEG
uh-ts'íígn ***xah***. (N)
3SG.S:OPT:H-be.sick:OPT:NEG PURP
'Olga prays **so she won't get sick**.' [UTOLAF14Nov2203:025]

(19) *Olga da-dégn'* *hǫǫ-heeyh*
Olga PROX-up:ALL 3SG.S:QUAL:QUAL:Ø.IPFV:Ø-talk:IPFV:NMZ
na-'ut-shye' ***xah***. (N)
IT-3SG.S:OPT:D-heal:PASS:OPT PURP
'Olga prays **so she'll get healed**.' [UTOLAF14Nov2203:024]

Comparison of (18) and (19) suggests that standard negation of the verb in a purposive clause results transparently in a negated purposive clause. I suspect however that negated purposive clauses such as (10c) or (18) are due to influence from English. (18) was elicited immediately after (19), using the cue 'so she won't get sick'. Both the temporal proximity of the two examples (only one minute elapses between them on the recording) and the English cue may have primed the speaker towards producing an English calque. Looking through other elicited instances of negated purposive linkings, all of them occur following cues containing 'so (that) … not'. Even more intriguing is the only clear textual example of this construction, given in (20).

(20) ***Ch'its'ą̀ą̀' tsaa-nį̂l*** ***xah*** *hǫ̀ǫ̀'*
wrong 1PL.S:INC:AA.IPFV:Ø-happen:FUT:NEG PURP thus
nił-ts'et-niik. (N)
RECP.PO-1PL.S:Ø.IPFV:D-say:IPFV:CUST
'**So we won't do it wrong**, we discuss it with each other.'
[UTOLVDN14May0508:114]

This utterance occurred about halfway through a 30 minute recording of four elders. For the first ten minutes, one elder talks about how to put on a traditional potlatch, and which pitfalls to avoid. When she is done, a second elder picks up the thread of proper potlatching. (20) is an utterance produced by the second elder. The subordinate clause is clearly negative despite the absence of the negative particle *k'à(t'eey)*, as evidenced by the tonal pattern. During a transcription and translation session in July 2019 with this same (second) elder, she confirmed that the verb form was formally negative (exaggerating the high tone) and suggested adding the negative particle.[12] I also asked her to evaluate the avertive clause *ch'its'ą̀ą̀' ts'aanį̀l ch'à'* 'lest we do it wrong' (without the tonal pattern associated with standard negation) instead of the negated purposive clause in (20). The speaker responded that the avertive clause would be vastly preferable to both the original version and the version including *k'àt'eey*, but that it sounded like a rare construction to her.

It is intriguing to compare (20) with (21), an utterance from earlier in the recording, produced by the first elder.[13] This structure clearly contains a causal clause with the subordinator *xah* 'because', which always triggers nominalization of the subordinate verb. Crucially, the form *hihtaanį̀l* 'they might do/happen' is *not* formally negative as evidenced by the low stem tone: *xah* is produced at slightly higher pitch than the stem *nìl*, which bears a lexical low tone.

(21) *Hǫ̀ǫ̀' hudahnih,* ***ayaa nts'ą̃' t'eey***
thus 3PL.PO-QUAL:2PL.S:Ø-say:IPFV bad ADVZR
hihtaanį̀l ***xah***. (N)
3PL.S:INC:AA.IPFV:Ø-happen:FUT:NMZ because
'Tell (PL) them, **because they might do it badly.**'
[UTOLVDN14May0508:053]

After perusing the corpus carefully multiple times, I tentatively conclude that utterances such as (10c), (18), or (20) are English calques.

[12]The negative particle is almost always present, but there are occasional instances without it, as discussed in Lovick (2023).

[13]Both utterances are uttered in similar contexts: the need to consult with one's close relatives to ensure that gifts at the potlatch are given appropriately. Proper gift-giving at potlatch is a very involved process; see Simeone (1995).

2.4 Beyond Upper Tanana

Avertive linkings seem to be attested all across Northern Dene (22)–(25),[14] with subordinators that are clearly cognate to Upper Tanana *ch'à'*. The constructions in the two Alaskan languages Dena'ina (22) and Ahtna (23) are identical to that in Upper Tanana.

(22) *Q'uch'a yadi ninya qyeghusłtish dghu qyeł*
how whatever animal 3PL>3SG:skin:IPFV:CUST when 3PL>3SG:with
dnish ***tutnitutq'ashi*** ***k'u***. (Dena'ina)
3PL.S:know:IPFV:CUST 3PL.S:cut.skin:FUT:NMZ AVERT
'They knew how to skin any animal **so as not to damage the skin.**'
(Kalifornsky 1991: 260–261)

(23) *Naltsiin 'iin du' xuts'enekaey stanxuhneł'iin* ***xuhnał'iił***
Naltsiin PL CT 3PL.PSR:children 3PL>3PL:hide.away:PFV 3PL>3PL:see:FUT
c'a', … (Ahtna)
AVERT
'The Naltsiin had hidden their children **so [the Tailed People] wouldn't see them**, …'
(Kari 1986: 55)

In the two Canadian languages, the construction looks a little different. This is partially due to differences between the language groups; while future and negative polarity are inflectional categories in the Alaskan languages, they are expressed by postverbal particles or clitics in the Canadian ones. This likely explains the imperfective form in the avertive clause in the Dëne Sųłiné example (24). The Slave example in (25) uses an optative form in the avertive clause, but most surprising is the fact that this form contains the negative enclitic *=híle* (this enclitic is partly absorbed into the stem). The same is true for all other avertive linkings cited by Rice (1989), suggesting that the construction in Slave has undergone some changes, possibly under the influence of English contact.

(24) *Hustón=ë* ***yëkéth*** ***ch'á***. (Dëne Sųłiné)
1SG>3SG:hold:IPFV=FUT 3SG:fall.over:IPFV AVERT
'I'll hold [the swingset] **so it doesn't fall over**.'
[deslas-ANR-2016-07-10-C]

[14] Due to my insufficient familiarity with all of these languages, I only provide word-level glosses containing participant and mode/aspect information in this section.

(25) ***Daniel yegúhʔále*** ***ch'á*** *goghádehk'a.* (Slave)
Daniel 3SG>3SG:find.CO:OPT:NEG AVERT 1SG>3SG:throw.CO:PFV
'I threw it **so Daniel wouldn't find it.**' (Rice 1989: 1262)

2.5 Origin and other uses of *ch'à'*

The origin of the avertive subordinator is likely the Proto-Dene postposition PO+ **k'ʸa·n̠'* 'away, apart from PO' (Leer 1996).[15] In many daughter languages, cognates occur as a free or bound postposition with meanings such as 'separating PO, emanating from PO', or adverbial-derivational verb prefixes with the meaning 'out and away'. In the Alaskan languages (apparently not beyond), the meanings 'warding off PO, protecting against/from PO, keeping PO away' are also common.

In Upper Tanana, cognates of the Proto-Dene morpheme include the bound adverbial-derivational morpheme *ch'a-* 'out and away' which is restricted to motion verbs (26). This morpheme has purely spatial meaning without connotations of avoidance or fear.

(26) a. *Ch'a-y-dih-tį̨į̨ …* (T)
out-3SG>3SG-QUAL-AA.PFV:H-handle.AO:PFV
'She took him out [from between two spruce trees] …'
[UTOLAF09Jun2202:015]

b. *Nóó* *ch'a-ts'i-nį̨į̨-kį̨į̨* *eh* … (T)
ahead:ALL out-1PL.S-N.PFV:Ø-go.by.boat:PFV and
'We start out by boat and …' [UTOLAF09Jun2902:011]

The free postposition PO+*ch'à'* 'away from PO', by contrast, does have strong connotations of avoidance (27a, 27b), hiding (27c), or escape (27d). This last example was a prompt by me that was flagged as problematic by the speakers, who suggested (27e) instead. A sentence like (27d) would be appropriate if the postpositional object were some sort of evil creature, but not if it is a respected elder like Sherry—in that situation, the (purely spatial) ablative postposition like PO+*ts'änh* 'from PO' in (27e) has to be used.

(27) a. *Nts'ą'* *na-tthi-ts'el-dak* *shyį̨į̨'* *t'eey,* *nih,*
and CONT-out-1PL.S:Ø.IPFV:L-PL.go:IPFV:CUST only ADVZR say:IPFV
huk'an ***ch'a'***. (T)
burn-out from
'And we were just running, she said, **from that burn-out**.'
[UTOLAF09Jun2402:028]

[15] A similar origin is observed by François (2026 [this volume]) for a comparable subordinator in several Torres-Banks languages.

b. ***Hu-ch'à'*** *i-tal-ch'ee …* (N)
3PL.PO-from ?-3SG.S:INC:AA.IPFV:L-be.aloof:FUT
[Money, success] will stay **away from them** …'
[UTOLAF12Jul1203:067]

c. *S-seeyy'* *sta-tnįh-'įįl*
1SG.PSR-knife:POSS away-FUT:QUAL:AA.IPFV:2SG.S:H-hide:FUT:NMZ
sh-ch'a' *…* (T)
1SG.PO-from
'You (SG) must have hidden my knife **away from me** …'
[UTOLVDN13May2804:077]

d. *? **Sherry shyah ch'à'** ih-haał.* (N)
Sherry house from AA.IPFV:1SG.S:Ø-SG.go:IPFV:PROG
(Intended: 'I'm coming **from Sherry's house**')
Comment: 'Sounds like you're getting away from her.'
[Prompt by OL for UTOLAF18Jun0101:087]

e. ***Sherry shyah ts'änh*** *ih-haał.* (N)
Sherry house from AA.IPFV:1SG.S:Ø-SG.go:IPFV:PROG
'I'm coming **from Sherry's house**.' [UTOLAF18Jun0101:087]

PO+*ch'à'* also has non-spatial meanings as demonstrated in (28), but it retains its negative connotations.

(28) a. *Chih nah-'ógn ni-'eeł-ts'e-łeek*
also MED-out:AR TERM-trap-1PL.S:Ø.IPFV:L-handle.PL.O:IPFV:CUST
nts'ą' t'eey t'axoh ti-ts'at-ndak ***u-ch'a'***. (T)
ADVZR finally LEX-1PL.S:AA.PFV:D-be.tired:IPFV 3SG.PO-from
'We would be setting traps too out there; finally we'd get tired **of it**.'
[UTCBVDN13Nov1501:096]

b. ***Ch'indziidn ch'à'*** *dhih-säł.* (N)
bee from DH.PFV:1SG.S:Ø-scream:PFV
'I screamed **for fear of the bees**.' [UTOLAF18Jun0805:037]

(28b) comes from elicitation; the scenario given to speakers was that I was stung twelve times by bees the last time I stole honey: now the bees are after me again and I scream with fear.

The textual example (29) shows that the postposition PO+*ch'à'* can also be used with clearly avertive function. The speaker explains why silence is expected during condolence visits. (29a) paints a scenario where the visitor is not silent;

the free pronoun *ay* in (29b) refers to that scenario. (Basso (1990: 90–92) reports a similar injunction against unnecessary conversation in encounters with the recently bereaved in the Western Apache context.)

(29) a. *N-èh* *stoo-tnèl-nay* *èh*
2sg.po-with away:ar.s-inc:qual:dh.pfv:l-co.moves:pfv:nmz with
dįį-nįįł *nts'ą̀', shyą̀ą̀'* *t'eey*
inc:qual:aa.ipfv:2sg.s:ø-say:fut and for.nothing advzr
ha-tha-dįh-nay *dè'*
out-mouth-qual:aa.pfv:2sg.s:h-move.co.carelessly:pfv:nmz if
ch'its'ą̀ą̀' tah *dįį-nįįł.* (N)
wrong among inc:qual:ø.ipfv:2sg.s:ø-say:fut
'You (sg) will talk when [your mind] is not all the way there, and if you (sg) just blab for no reason, you (sg) will say something wrong.'[16]
[UTOLVDN14May0508:013]

b. ***Ay ch'à'*** *shyą̀ą̀'* *dhįį-dah.* (N)
3sg avert for.nothing dh.pfv:2sg.s:ø-sg.sit:ipfv
'**To avoid that [i.e., saying the wrong thing]**, sit (sg) for nothing [without talking]. [UTOLVDN14May0508:014]

Although the postposition po+*ch'à'* may have connotations of fear, the subordinator does not. Unlike what Lichtenberk (1995: 296–297) describes for the Austronesian language To'aba'ita, *ch'à'* is not attested in the corpus as a subordinator when the verb of the main clause is a predicate of fearing; instead, the complementizer *t'eey* is used (30):

(30) ***Tee-te'*** ***t'eey*** *neljit!* (T)
3sg.s:inc:dh.pfv:ø-sg.sleep:pfv comp 3sg.s:qual:ø.pfv:l-be.scared:ipfv
'She was scared **to fall asleep**!' [UTCBAF13Nov0501:086]

Ch'à' could also not be elicited with this function as shown in (31); the cues were 'I'm afraid the dog might bite me', 'I'm afraid they might kill us', and 'I'm afraid he will find me'.

[16]This sentence contains two fascinating morphological idioms. *Neh stootnelnay eh* 'when your mind is not all the way there' can be glossed more literally as 'when a compact object has moved independently away, affecting you', while *hathadįhnay* 'you just blab' could be glossed as 'you move a compact object carelessly out using your mouth'.

(31) a. *Łįį sh-taa-'aał*
dog 1SG.O-3SG.S:INC:AA.IPFV:Ø-bite:FUT
m-i-nag-jit. (T)
3SG.PO-PP-QUAL:Ø.PFV:1SG.S:L-be.afraid:IPFV
'The dog might bite me, I'm afraid of it.' [UTOLAF18Jun0201:001]

b. *Han iin hu-'i-nagn-jit,*
DEM PL 3PL.PO-PP-QUAL:Ø.PFV:1SG.S:L-be.scared:IPFV
nee-h-taa-xąą k'eh hel-tsįį. (N)
1PL.O-3PL.S-INC:AA.IPFV:Ø-kill.PL.O:FUT like 3PL.S:DH.PFV:L-seem:IPFV
'I'm afraid of them, they look like they might kill us.'
[UTOLAF18Jun0101:008]

c. *Idzii sh-oo-dah-'aał.* (T)
scary 1SG.O-3SG.S:QUAL-INC:QUAL:AA.IPFV:H-find.CO:FUT
'Scary, he will find me.' [UTOLAF18Jun0201:015]

2.6 Avertive clauses and speech acts

Dixon (2009: 24) notes that typologically, the main clause in an avertive construction ("possible consequence linking" in his terminology) is "most often a positive imperative … or a negative one".[17] In Upper Tanana, this is not the case. As mentioned above, avertive clauses cannot be used at all with negative requests. With affirmative requests, they are rare. Table 2 shows that almost all naturalistic instances of avertive linking occur in declarative speech acts.

Table 2: Speech act types in avertive linkings

declarative	question	request	hortative	total
14	2	1	1	18

The total number of naturalistic examples of avertive clauses is so small (18 instances in about 10,000 utterances) that the relative frequencies have to be taken with a grain of salt. Yet it does appear as though Dixon's claim about the predominance of avertive linkings in requests cannot be maintained for Upper Tanana. Since avertive linkings in requests can however be elicited without difficulty, the rarity of this construction is likely to have pragmatic rather than grammatical reasons. I suggest here that this is due to the availability of an urgent request

[17]The basis for his claim is unknown to me.

construction that points to, but does not spell out, potential undesirable consequences of non-compliance. These urgent requests are described in §3.

3 Urgent requests

The structure of urgent requests is described in §3.1. This is followed by a discussion of the origin of this construction in §3.2, and of its semantics in §3.3.

3.1 Structure

Urgent requests are in the optative and always contain the particle *dè’* (32).

(32) *Jin dǫh-’à’ dè’.* (B)
this qual:opt:2sg.s:vv-sing:opt please?
‘Sing (sg) this [song], you (sg) should sing this [song], please sing (sg) this [song].’ [UTOLAF10Jul0202:196]

In discussing (32) and others like it, Lovick (2016: 269) suggests that the translation ‘please’ might be appropriate for this particle and that it serves to mitigate the request. More recent discussions with speakers, however, suggest a different function for this particle: that of adding urgency to a request, resulting in the translation ‘make sure you sing this [song]’ for (32), as discussed in §3.3.

3.2 Origin

The origin of urgent requests with *dè’* is likely a conditional request, which are a special type of conditional linking. In declarative conditional linkings, the conditional clause is marked by *dè’* ‘if, when.fut’. (This polysemy is common in Dene languages, see also Anisman 2019.) The consequent clause is usually in the future.

(33) *Ay and ay tl’ààn udzih dhèh-xį̀į dè’*
and and and then caribou 3sg.s:qual:dh.pfv:h-kill.sg.o:pfv if
nah-’ǫ́ǫ́ nìì-taa-łeeł. (N)
med-out:abl term:it-3sg.s:inc:aa.ipfv:ø-handle.pl.o:pfv
‘And if he’d kill a caribou, he’d bring it from out there.’
[UTOLVDN07Nov2901:020]

Conditional requests differ from conditional declaratives in two respects: the consequent is usually in the optative, and both clauses contain the particle *dè’* (34).

(34) a. *"K'àt'eey k'àt'eey hǫǫsú' m-i-hǫl-náy* ***dè'*** *t'eey,*
NEG NEG good 3SG.P-PP-AR:Ø.PFV:L-taste:IPFV:NMZ if even
at-naah ***dè',"*** *nih.* (N)
OPT:2PL.S:D-drink:OPT UR say:IPFV
'"Even if it doesn't taste good, make sure you (PL) drink it," he said.'
[UTOLAF18Jun0401:013]

b. *Ch'i-dhih-'eeł* ***de', k'a' ǫǫ-keet*** ***de',***
INDF.O-DH.PFV:2SG.S:H-trap:PFV if gun OPT:2SG.S:Ø-buy:OPT UR
huugn ta-'ǫl-shyeek ***de',*** *…* (T)
etc. distribute-OPT:2SG.S:L-handle.PL.O:OPT:DIST UR
'And if you (SG) trap something, make sure you (SG) buy guns with it, make sure you (SG) distribute it [at potlatch] …'
[UTCBAF13Nov0401:070]

The presence of *dè'* 'if, when.FUT' in both clauses in conditional requests suggests that the consequent clause sets up a further condition with an unspoken consequence: even if the broth doesn't taste good, make sure you drink it; if you do not drink it, you will die of starvation.[18] If you have good luck trapping, make sure to distribute your wealth through a potlatch; if you fail to do so, you will no longer be successful when trapping. Full conditional requests like those in (34) thus are instances of insubordination (Evans 2007), where the main clause is unspoken, and where the (second) conditional clause is modified by the first conditional clause. Urgent requests such as (32) lack the first conditional clause.

3.3 Semantics

Urgent requests with *dè'* imply the presence of undesirable consequences of non-compliance, such as snow out of season in the case of (32), starvation in (34a), or poor luck in trapping in (34b). More examples are given in (35).

(35) a. *Tl'aan hu-na-chin-'ah-deeł* *tl'aan*
and 3PL.P-surround-group-OPT:2PL.S:Ø-PL.move:OPT and
hu-b-e-kon'-nah-shyiił *de'*
3PL.P-?-PP-fire-QUAL:OPT:2PL.S:H-strike.LRO:OPT UR
hu-'ah-xąą *de', nee-he-nih.* (T)
3PL.O-OPT:2PL.S:H-kill.PL.O:OPT UR 1PL.P-3PL.S:Ø.IPFV:Ø-say:IPFV
'Then gang (PL) up on them and make sure you (PL) set fire to them and kill them, they said to us.' [UTCBVDN14Jul1801:127]

[18]The broth in question is prepared from moose droppings and consumed only under the most dire circumstances.

b. *Nts'ą̀ą̀' n-dih-nih niign t'eey*
whatever 2sg.p-qual:1sg.s:ø-say:ipfv comp advzr
dǫǫ-dį̀' dè', u-dih-nih. (N)
qual:opt:2sg.s:ø-do:ipfv appr 3sg.p-qual:1sg.s:ø-say:ipfv
'Make sure you (sg) do [exactly] whatever I tell you (sg), I said to him.' [UTOLVDN13May2004:067]

(35a) is taken from a text about World War II, where US army representatives instructed the people of Tetlin village on what to do if they were attacked by the Japanese. Non-compliance here would have resulted not only in the death of many villagers, but also in the Japanese establishing a foothold on the Alaskan mainland, something the US army desperately wanted to avoid. (35b) (discussed at length in Lovick 2016) is from a text where the speaker and her brothers are attacked by a bear, but only have a single shot between them. The speaker is trying to protect her brothers from the bear; when her older brother starts arguing, she shuts him up with (35b). In this example, the consequence of non-compliance would be being killed by the bear. Generally, *dè'* is used in urgent requests where non-compliance would result in very undesirable consequences.

It should be noted that this modified analysis of *dè'* as an urgency marker does not necessarily invalidate the claim that it mitigates a request, as put forward in Lovick (2016: 269). As Lovick (2016) notes, reiterating arguments also made by Scollon & Scollon (1981) or Rushforth & Chisholm (1991), uttering requests is considered to be interfering with another's freedom of action and thus undesirable in and of itself. Using *dè'* in a request draws attention to the undesirable consequences implicit in non-compliance, which in turn provides motivation for uttering the request in the first place – a form of mitigation.

Thus requests containing *dè'* serve the speaker's need to point to the necessity of compliance but crucially never spell out the possible consequences of non-compliance, trusting the addressee to be able to work this out for themselves.[19] Spelling out the desirable consequences of compliance is not compatible with the Upper Tanana practice of speaking carefully about the future described in §1.1.

3.4 Beyond Upper Tanana

Similar constructions do not seem to be reported for related languages.

[19]This looks like the reversal of the indirect requests described by François (2026 [this volume]) for Mwotlap, where the undesirable consequence is spelled out, but no request is uttered.

4 Speaking carefully

In this section, I discuss several aspects of avertive linkings and urgent requests in greater detail, showing how they reflect general cultural patterns in Alaskan Dene: the keen awareness of which things are beyond an individual's control, and the need to be prepared for those things that are not (§4.1) as well as the principle of non-interference and respecting the competence of others (§4.2).

4.1 Control and preparedness

In §2.2, I showed that avertive linkings cannot be used in 'in-case' scenarios but only when the event in the avertive clause is actually under the control of the subject of the main clause. Speakers proved to be quite sensitive about this, consistently rejecting examples where control was absent or inadequate. As discussed in §2.3, there are several other constructions that can be used under these circumstances.

Yet, precisely because the Upper Tanana are aware of their limited control over their environment, they employ a circumspect mindset requiring them to be prepared for any eventuality. This had some surprising consequences for fieldwork. One of the examples of 'in-case' scenarios comes from a storyboard[20] (Vander Klok 2013); it is given in (36), repeated from (14) above.

(36) *Eh Bill: Tay' t'eey tah-chaan ay xah*
and Bill again ADVZR 3SG.S:INC:AA.IPFV:H-rain:FUT:NMZ 3SG because
ch'ale u-shyii chąą taathüh di-dhih-dlay
FOC 3SG.PO-in umbrella QUAL-DH.PFV:1SG.S:Ø-keep.PL.O:IPFV:NMZ
tthiishyoh eh tl'aan dii na-ti-dhag-niign
hat and and what PRMB-INC-DH.PFV:1SG.S:L-wear:IPFV:CUST:NMZ
tat'eey u-shyii dhih-dlah. (T)
also 3SG.PO-in DH.PFV:1SG.S:Ø-keep.PL.O:IPFV
'And Bill: It might rain again, that's why I keep the umbrella in [the backpack] and the hat and my clothes I also keep in it.'
[UTCBAF13Nov1510:035]

Bill's behavior – he arrives at work wearing a coat, a hat, and carrying an umbrella on a fine day – may strike someone socialized in the mainstream American

[20] Storyboards are sequences of images serving as stimulus for a speaker to tell the story in their own words. They allow the researcher to control the context of an utterance without having to use the contact language. For discussion of this methodology, see Burton & Matthewson (2015).

or Canadian worldview as ridiculous. Upper Tanana Dene, however, consider this perfectly sensible behavior due to the emphasis on *ts'edòònih* discussed in §1.1. Preparedness as that shown by Bill is valued and in fact encouraged as shown in (37), which is taken from a text about *ts'edòònih*:

(37) a. *Nithaad dą̀' sh-nąą iin èh sh-tà' eh*
long.time when:PST 1SG.PSR-mother PL and 1SG.PSR-father and
"Hah-'ógn elih tah hǫǫsú', hǫǫsú'
NTRL-outside:AR cold when well well
din-tl'ǫǫk hah'ógn ts'edòònih
QUAL:Ø.IPFV:2SG.S:D-dress:IPFV:CUST NTRL-outside:AR never.know
xah," nee-he-niik. (N)
because 1PL.PO-3PL.S:Ø.IPFV:Ø-say:IPFV:CUST
'A long time ago, our parents used to say to us, "When it is cold outside, dress (SG) well, because you never know."'
[UTOLAF19Jan0501:010]

b. *"Elók hǫǫ-łįį nįį-thänh dè'*
hot AR.S:Ø.PFV:Ø-be:IPFV QUAL:Ø.PFV:2SG.S:Ø-think:IPFV if
n-ch'ìl' tįį-łeek." (N)
2SG.PSR-clothes:POSS INC:Ø.IPFV:2SG.S:Ø-handle.PL.O:IPFV:CUST
'"Even if you (SG) think it is hot, take your (SG) clothes."'
[UTOLAF19Jan0501:011]

c. *N-èh n-èh tįį-łeek*
2SG.PO-with 2SG.PO-with INC:Ø.IPFV:2SG.S:Ø-handle.PL.O:IPFV:CUST
nah-'ógn nts'ą̀ą̀' u-taa-nį̀ł nts'ą̀'
MED-outside:AR ADVZR AR.S-INC:AA.IPFV:Ø-happen:FUT ADVZR
hi-ts'-in-nííg̱n xah. (N)
PP-PEJ-Ø.PFV:2SG.S:D-not.know:IPFV because
'"Take them with you because you don't know what will happen out there."' [UTOLAF19Jan0501:012]

In a way, the cultural emphasis on preparedness could be considered one reason for the rarity of avertive linkings in the narrative corpus.

4.2 Respecting the competence of others

Looking at the distribution of avertive linkings and urgent requests by genre (Table 3), it becomes apparent that they are used with very different audiences.

Table 3: Avertive linkings and urgent requests by genre

	avertive	urgent requests
past events	12	44
teachings	6	3
total	18	47

"Past events" includes the three narrative genres of old-time stories, histories, and personal narratives identified in Lovick (2012). As pointed out by Jones & Thompson (1990: ix) in the context of old-time stories, these stories are told "to entertain, educate and inspire – to cause the reader or listener to think" (see also Bergelson (2019) on lessons implicit in personal narratives). The typical audience for this type of narrative is composed of community members. Crucially, the lessons contained in these stories are expressed implicitly and require the audience to unpack them, using the cultural knowledge at their disposal. "Teachings" refers to a non-traditional genre where speakers explain cultural knowledge to an audience consisting of community outsiders such as linguists and anthropologists, or to young people who did not grow up in the traditional way. The explanations in such texts are very explicit and often contain rationales for the ritualistic acts discussed in §1.1. It is thus not surprising that a third of all examples of avertive linkings come from teachings: these are explanations of cultural practices aimed at those who are unfamiliar with them ((6a)–(6b) fall into that category). It is also not surprising that there are so few urgent requests in these texts: leaving aside the relative rarity of non-declaratives in this text type, speakers are also cognizant of the fact that the audience of a "teaching" text may not be able to interpret the request correctly. Instead, urgent requests occur predominantly in stories where characters well-versed in Upper Tanana culture interact with one another (see Lovick (2016) for a discussion of using stories for the investigation of requests). The traditional audience for these texts does need to be told the consequences of (non-)compliance. Explicitly stating the (desirable) outcome of compliance with the urgent request would be a violation of the practice of careful speaking. Stating the (undesirable) consequence of non-compliance, on the other hand, could suggest that the addressee is not able to think for themselves or "does not know their *į̨įjih*", which is not compatible with the principle of non-interference and respect. In either case, verbalizing the potential consequences in an urgent request would collide with fundamental principles of Upper Tanana culture.

Together, avertive linkings and requests with *dè'* fulfill an Upper Tanana speaker's need to remind the addressee of circumspect action, while at the same time treating them as a competent person who "knows their *įįjih*".

5 Conclusion

In this paper, I described avertive linkings and urgent requests in Upper Tanana Dene, and discussed how they reflect the circumspect mindset of Upper Tanana people.

Avertive linkings are a type of adverbial linking involving the subordinator *ch'à'* 'AVERT, lest'. The avertive clause is usually in the future, more rarely in the inceptive perfective mode. It is always formally affirmative. The main clause has no mode restrictions and also has to be formally affirmative. There are no other formal restrictions on avertive linkings. Semantically, avertive linkings are limited to situations where an undesirable consequence (expressed in the avertive clause) can be avoided by taking the preemptive action expressed in the main clause. This linking is very sensitive to the notion of what can and cannot be controlled by human action and thus cannot be used in 'in-case' scenarios. It also cannot be used in linkings where the main predicate is one of fear.

Somewhat unusually typologically, avertive linkings with a request as main clause are rare in Upper Tanana. I argue in this paper that this is likely due to the fact that requests are problematic in Northern Dene in general, and that spelling out the undesirable consequences to be avoided could suggest to the hearer that they are thought to be lacking in cultural knowledge. Instead, there is a dedicated strategy for urgent requests formed using the particle *dè'*. This type of request likely has arisen as a insubordinate conditional structure. The undesirable consequence of non-compliance is implicated, but never spelled out.

The existence and usage patterns of these constructions reflect the circumspect mindset of the Upper Tanana people, i.e., the need to act and speak mindfully. Avertive linkings spell out the undesirable consequences that can be avoided by taking the action in the main clause. They occur primarily in texts directed at community outsiders who often are unaware of the many, often ritualistic, actions performed to ensure a favorable outcome. Requests formed with *dè'* point to the existence of undesirable consequences, but crucially never spell these out: the speaker assumes that the hearer has the required cultural knowledge to do so. Thus, such requests usually are directed towards a member of the community, who can be relied on to have this knowledge. In this fashion, avertive linkings and requests with *dè'* complement each other in the domain of circumspection, while respecting the addressee's cultural competence.

Abbreviations

AA	*aa*-conjugation	N	*n*-conjugation
ABL	ablative	NMZ	nominalizer
ADVZR	adverbializer	NEG	negative
ALL	allative	NTRL	neutral distance
AR	areal	O	direct object
AVERT	avertive	OPT	optative
B	Beaver Creek dialect	PO	postpositional object
COMP	complementizer	PEJ	pejorative
CO	compact object	PFV	perfective
CONT	continuative	PL	plural
CT	contrastive topic	POSS	possessed
CUST	customary	PP	postposition
D	D-voice/valence maker	PRMB	perambulative
DH	*dh*-conjugation	PROG	progressive
FOC	focus	PROH	prohibitive
FUT	future	PSR	possessor
H	H-voice/valence marker	PURP	purposive
IMP	imperative	QUAL	qualifier
IPFV	imperfective	RECP	reciprocal
INC	inceptive	REFL	reflexive
INDF	indefinite	S	Scottie Creek dialect
IT	iterative	S	subject
L	L-voice/valence marker	SG	singular
LEX	lexical morpheme	T	Tetlin dialect
LRO	long rigid object	TERM	terminative
MED	medial distance	UR	urgent request
N	Northway dialect		

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Chapter 3

The apprehensional domain in A'ingae (Cofán)

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This paper provides the first detailed description of the apprehensional domain in A'ingae (Cofán, ISO 639-3: con), with the cross-linguistic category of *apprehension* defined as a mixed modality encoding both undesirability and epistemic possibility. We contribute to the study of apprehensional typology by reporting on a language of a so far unattested profile: one apprehensional morpheme *=sa'ne* 'APPR' spanning robust precautioning uses (both avertive and 'in-case'), negative-verbal complementizer uses, restricted timitive uses, and marginal apprehensive uses. Lastly, we contribute to the study of apprehensional semantics by arguing that the particular functional range of the A'ingae apprehensional clitic *=sa'ne* 'APPR' is not entirely a question of diachronic development, but rather that much of its polyfunctionality emerges from a single meaning. We propose that the situation denoted by the *=sa'ne* 'APPR' clause is contained within a possible undesirable situation. If that containment is proper (i.e. the situation denoted by the *=sa'ne* 'APPR' clause is not identical to the negative situation), the 'in-case' function obtains. If the containment is improper (i.e. the situation denoted by the *=sa'ne* 'APPR' clause is the undesirable situation), the avertive function obtains.

1 Introduction

In this paper, we contribute to the cross-linguistic understanding of the apprehensional domain through a detailed exploration of apprehensional forms in A'ingae (or Cofán, ISO 639-3: con), an isolate language of Amazonia, spoken by approximately 1,500 speakers in Northeastern Ecuador and Southern Colombia (Dąbkowski 2021a).

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While the domain of *apprehension* is characterized by cross-linguistic variability in both form and function, its distinguishing characteristics are high probability combined with undesirability (Vuillermet 2018a). Although the category has varied and robust manifestations across languages, it has been largely overlooked by descriptive grammarians, typologists, and formal analysts alike until quite recently (for extant descriptions, see Lichtenberk 1995, Green 1989, François 2003, Dobrushina 2006, Pakendorf & Schalley 2007, Angelo & Schultze-Berndt 2016, Vuillermet 2018a).

In line with the nascent understanding of the apprehensional domain emergent from cross-linguistic research (Vuillermet 2018b), we distinguish four different main uses of apprehensional morphology.

First, the *apprehensive proper*[1] encodes a highly probable undesirable situation and is typically associated with matrix-clausal uses. It is often employed to convey warnings. No English construction directly corresponds to it; it can be best rendered by *beware* (possibly in combination with *lest*), *watch out* (both encoding undesirability), *might* (encoding high likelihood), or negative imperative (encoding a warning).

Second, the *precautioning* – prototypically biclausal – function hypotactically relates an apprehension-causing event to a preemptive event aimed to counteract it. Previous literature has distinguished two subtypes of the precautioning function. Following Lichtenberk (1995), we will label them *avertive* and *'in-case'*. The avertive subfunction refers to uses where the preemptive situation is aimed at forestalling the apprehension-causing one, while the 'in-case' subfunction applies to uses where the preemptive situation is aimed only at mitigating its negative consequences. In English, the negative purpose constructions *in order not to* or *so as not to* express only the avertive semantics, although the largely archaic *lest* can be used to express both the avertive and 'in-case' subfunctions.

Third, the *timitive* introduces an NP in a manner similar to case or adpositions. The timitive relates a feared entity to the matrix-clausal situation; the latter is presented as having been triggered by the former. These uses are best translated by English constructions involving *for fear of.*

Fourth, apprehensionals in *complementizer* function head the complements of certain negative verbs, most often associated with the emotion of fear. Here again, English translations are not straightforward, as the complementizer of 'fear' predicates is most often *that* or null, although *lest* can also be archaically used.

[1]Observe the distinction between *apprehensional*, which refers to the domain of functional morphemes encoding fear and related emotions, and *apprehensive (proper)*, which refers to just one among several *apprehensional* functions.

While some languages have distinct morphological manifestations for various of these different categories, others have morphemes that can be used across several of them. So is the case in A'ingae, whose only apprehensional morpheme *=sa'ne* 'appr' is multifunctional. It is used most robustly as a head of subordinate clauses introducing apprehension-causing situations (precautioning function, 1), and to mark complements of certain verbs (complementizer function, 2). It also occurs in a somewhat restricted fashion as a timitive (3), and even more marginally as an apprehensive (4). For clarity, constituents headed by *=sa'ne* 'appr' and some subordinate clauses in the examples given across the paper will be bracketed. We will refer to the clauses and NPs headed by *=sa'ne* 'appr' as the *arguments* of *=sa'ne* 'appr'.

(1) *Phuraen kan-ñakha* [*amphi ja=sane.*]
touch try-iter fall go=appr
'He felt it with his hand so as not to fall down.'
(20170803_dyandyaccu_LC: 40)

(2) *Tsama ña* [*dañu=sane=khe*] *dyuju-je=ya.*
but 1sg be hurt=appr=mann.dem be.afraid-ipfv=ver
'I was afraid I would get hurt.' (20170731_yaje2_MM: 53)

(3) *Anae'ma=ni=ngi phi* [*thesi=sa'ne.*]
hammock=loc=1 sit jaguar=appr
'I'm in a hammock for fear of a jaguar.'

(4) [*Tsai-ye=sa'ne.*]
bite-pass=appr
'You might get bitten.'

An immediate question that arises in such cases is that of the relationship between various apprehensional functions. Is it solely diachronic? Does it arise from a uniform semantics which is more general, or from a covert ambiguity or polyfunctionality? Do some functions – or aspects of their semantics – pattern together? For example, does the availability of 'in-case' uses of one apprehensional morpheme entail anything about its reading in an apprehensive role? Furthermore, what is the relation of apprehension to other domains, such as purpose constructions (encoded by the infinitive in A'ingae), which also tend to have prospective or irrealis modalities and a variety of formally distinct adjunct and argument uses?

Beyond providing the first detailed description of the A'ingae *=sa'ne* 'appr', we add to the study of the apprehensional domain a language with a previously unreported typological profile: robust precautioning and 'fear' complement uses, restricted timitive uses, and marginal apprehensive uses. We argue that the apprehensive uses of *=sa'ne* 'APPR' are instances of partially conventionalized uses of subordinate clauses. They occupy, therefore, an intermediate stage in the diachronic trajectory of insubordination proposed by Evans (2007). Lastly, although a proper formal semantic analysis is beyond the scope of this paper, we gesture at common threads across the different functions of *=sa'ne* 'APPR' to point out how the semantic commonalities underlying all of them are responsible for the range of functions attested.

The road map for the remainder of the paper is as follows: §2 briefly presents background on A'ingae and the data used here; §3 examines the precautioning use of *=sa'ne* 'APPR' as a subordinator; §4 examines the use of *=sa'ne* 'APPR' as a complementizer of *dyuju* 'be afraid' and other negatively valenced predicates; §5 examines the timitive use; §6 examines the apprehensive use; §7 concludes.

2 Background

A'ingae (or Cofán, ISO 639-3: `con`) is an indigenous language spoken by around 1,500 people in the province of Sucumbíos in northeast Ecuador as well as southern Colombia (Dąbkowski 2021a). Despite being an isolate, a number of aspects of its grammar, both phonologically and morphosyntactically, point to membership in the Amazonian sprachbund (Fischer & Hengeveld 2023, Repetti-Ludlow et al. 2019, AnderBois et al. 2019, Sanker & AnderBois 2024).

Outside of a few brief word lists, the first contributions to the systematic study of A'ingae were made by Marlytte Bub Borman and Roberta Bobbie Borman, missionary linguists first active in the Cofán communities in 1950s. Borman (1976) provides the first (and only) substantial dictionary; Borman & Criollo (1990) – a collection of cosmological narratives. Other notable works include a grammatical sketch by Fischer & Hengeveld (2023), a traditional story collection (Blaser & Umenda 2008), and various descriptive and theoretical papers, including, but not limited to, AnderBois & Silva (2018), Repetti-Ludlow et al. (2019), Pride et al. (2020), Dąbkowski (2019, 2021a,b, 2023, 2024), Sanker & AnderBois (2024), AnderBois et al. (2023), Dąbkowski & AnderBois (2023), AnderBois & Dąbkowski (2021, 2025), Zheng & AnderBois (2023), Morvillo & AnderBois (2022).

An orthography for the language was first developed by the Bormans, and recently revised by members of the Cofán communities themselves. The present chapter makes use of the revised orthography. For details, see Fischer & Hengeveld (2023) and Repetti-Ludlow et al. (2019).

While phonological and orthographic details are generally not relevant here, one slight exception is the presence of glottal stops. Glottal stops are frequently contrastive (at least in some positions) and represented by apostrophes orthographically. Nevertheless, they are not transcribed consistently by native speakers. Furthermore, glottal stops influence the position of lexical stress, and lexical stress, conversely, influences the surfacing of glottal stops: in unstressed positions, glottal stops tend to be realized suprasegmentally or not realized at all. This interplay feeds back into the orthography, as apostrophes end up being used to cue lexical stress and, therefore, morphological boundaries (Dąbkowski 2023).

Across all of its uses, the apprehensional *=sa'ne* 'APPR' – like many other morphemes – shows variation between variants with the glottal stop: *=sa'ne* 'APPR', and without it: *=sane* 'APPR'. Since this difference does not appear to be semantically important and the phonetic/phonological reality is somewhat unclear, we retain the forms of previously published works and native speaker transcriptions in naturalistic data. We transcribe the glottal stops in elicited data.[2]

The preponderance of our data comes from the fieldwork conducted by the authors. The naturalistic interviews and elicitation sessions that form the basis of our analyses come from our work with speakers representing three Ecuadorian communities: Zábalo, Sinangoé, and Dureno. If naturalistic, the citation accompanying the example contains the identifier (file name) and line number in the collection deposited as AnderBois & Silva (2018) with the Endangered Languages Archive at SOAS University of London. If elicited, no citation is given. A minority of the data sourced from written texts is cited as such, but updated to the revised orthography.

2.1 Typological profile

A'ingae is a head-final language, with predominantly SOV basic word order. In matrix clauses, word order is largely free, though with a preference for SOV and subject to a variety of pragmatic demands (Fischer & Hengeveld 2023). Subordinate clauses are strictly verb-final, a fact which we make use of below (see §2.3 for more detail).

The language's functional morphology is dominated by enclitics, with a lesser role of suffixation. The verbal paradigm is quite complex with many verbal and

[2]Older sources (Borman & Criollo 1990, Borman 1990, Borman 1976; and collaborators) often show glottal stops in places where modern-day transcribers do not and for which phonetic support is not immediately clear. This is especially true in unstressed positions, and is suggestive of a phonological reduction process. These complexities, however, are not at all unique to *=sa'ne* 'APPR' and we refer the interested reader to Dąbkowski (2023) for more detailed discussion and analysis.

clausal morphemes typically present, significant ordering restrictions between them, as well as morphophonological interactions with stress and glottalization. Verbal morphology is discussed more fully in §2.2.

The language is consistently dependent marking; verbal dependents are marked for case with nominative-accusative alignment for core arguments. The four cases most commonly used to introduce verbal arguments include the nominative (unmarked), the dative *=nga* 'DAT', as well as two accusatives: *=ma* 'ACC', marking the prototypical affected object, and *=ve* 'ACC2' (nasal allomorph *=me* 'ACC2'), marking the unaffected or absent object.

Case is expressed via clitics. The clitichood of A'ingae case markings is corroborated on prosodic grounds (they stand outside of the phonological noun, Dąbkowski 2019) and by their NP-final position, regardless of its internal order (Fischer & Hengeveld 2023), as seen in (5)–(6).

(5) *Rande tsa'u=ma athe.*
large house=ACC see
'I saw a large house.'

(6) *Tsa'u rande=ma athe.*
house large=ACC see
'I saw a large house.'

2.2 Verbal template

A'ingae has several dozen inflectional morphemes that can attach to verbs across a dozen or so slots.[3] A fragment of the verbal template is given in Table 1.[4] For the full template and its justification, see Dąbkowski (2019, 2021b, 2023).[5] The template captures the ordering of inflectional morphemes as well as the co-occurrence restrictions that obtain among them. As such, it is a visual representation of a generative algorithm for A'ingae conjugation. To generate a legal verbal form, go from left to right picking at most one morpheme per column along the way. Do not cross the horizontal lines.

[3] All the morphemes discussed in this section have been classified by Fischer & Hengeveld (2023) as clitics. Since the co-occurrence of these morphemes is arbitrarily restricted (Zwicky & Pullum 1983), they do not change the syntactic category of their hosts, and some display morphophonological idiosyncrasies (Dąbkowski 2019), many of them should likely be reclassified as inflectional suffixes. Nevertheless, for consistency with previous work, we gloss them with equal signs and refer to them as clitics throughout. For an extensive discussion of A'ingae suffixhood and clitichood and a different glossing convention, see Dąbkowski (2019).

[4] Only those morphemes are listed which will be relevant to the discussion of the paradigmatic status of the apprehensional clitic *=sa'ne* 'APPR'. Valence, aspect, and associated motion suffixes, which all come before the plural subject *='fa* 'PLS' number clitic, are omitted. So are the second-position clitics (the polar interrogative *=ti* 'INT' clitic, the reportative evidential *=te* 'RPRT' clitic, as well as the person subject clitics), which all come after the information structure clitics.

[5] See Fischer & Hengeveld (2023) for an alternative template.

Table 1: Verbal inflectional template, a fragment.

…	NUM	MOD	POL	TAX	INFO-STRUCTURE			…
				=’ya VER				
				=pa SS				
		=ya IRR	*=mbi* NEG	*=si* DS	*=’yi* EXCL	*=’ta* NEW	*=’khe* ADD	
			=’ma FRST					
				=’ni LOC				
	=’fa PLS			*=sa’ne* APPR		*=’ja* CNTR		
				=ye INF				
				=ja IMP				
				=kha IMP2				
				=’se IMP3				
				=jama PROH				

The number NUM slot lists the only number clitic, the plural subject *=’fa* ‘PLS’. The modality MOD slot lists one modal clitic, the irrealis *=ya* ‘IRR’, although other clitics (e. g. the imperatives given in the taxis TAX slot) also arguably express modal semantics. The polarity POL slot lists one negative indicative *=mbi* ‘NEG’, although negativity can also be expressed in the semantics of the frustrative *=’ma* ‘FRST’ and the prohibitive *=jama* ‘PROH’.

The TAX slot lists all the clitics which appear in a clause-final position (except when followed by the information structure clitics or a few clitics not listed in

Table 1), do not co-occur with other TAX clitics, and which establish its status as independent or dependent. Parts of the table which contain subordinating clitics are colored in grey.

Among subordinating clitics figure the same subject *=pa* 'SS', which signals identity between subjects of two clauses, the different subject *=si* 'DS', which signals non-identity between subjects of two clauses,[6] the frustrative *='ma* 'FRST', which signals a frustration of otherwise anticipated consequences of the encoded clause, the locative *='ni* 'LOC', the apprehensional *=sa'ne* 'APPR' (our focus here), as well as the infinitive *=ye* 'INF'.

Among matrix clausal clitics figure the three imperatives *=ja* 'IMP', *=kha* 'IMP2', and *='se* 'IMP3', the semantic differences among which are not well understood, the prohibitive *=jama* 'PROH', expressing negative commands – or prohibitions, and the elusive veridical *='ya* 'VER', as epithetized by Fischer & Hengeveld (2023: 35), whose semantics is likewise unclear.

The INFO-STRUCTURE slot lists the exclusive focus *='yi* 'EXCL', new topic *='ta* 'NEW', contrastive topic *='ja* 'CNTR', and additive focus *='khe* 'ADD' clitics. One of the uses of topic clitics is to mark conditional antecedents.

2.3 Subordination

A *subordinate* clause forms a part of another clause. The subordinate status of an A'ingae clause can be ascertained via a number of diagnostics. Below, we present three such diagnostics, one of which is semantic, while the other two are syntactic. Although the syntactic diagnostics are proposed in Fischer & van Lier (2011), apprehensional clauses are not explicitly discussed.

First, a subordinate clause can be identified via scopal means. For example, the negation scoping over the infinitival clause in (7) testifies to its subordinate status. A paratactic analysis here would predict a clearly incorrect meaning.[7]

(7) *In'jan=mbi=gi* [*panza=ye.*]
want=NEG=1 hunt=INF
subordinate analysis: 'I don't want to hunt.'
paratactic analysis: '#I don't want to. I'm off to hunt.'

[6] The same and different subject clitics *=pa* 'SS' and *=si* 'DS' can be employed in subordinate as well as co-subordinate constructions. The distinction is immaterial for our purposes. For the definition and discussion of A'ingae subordination, see Fischer (2007).

[7] The hypothetically available adjunct reading ('I don't want it in order to hunt') is impossible or very difficult to get here.

Second, the subordinate status can be corroborated by restrictions on word order. While word order in matrix clauses is quite flexible, subordinate clauses are strictly verb-final (Fischer & Hengeveld 2023, Fischer & van Lier 2011), as in (8–9).

(8) [*ûnjin tûi='ni=nda*] *avûja=ya.*
rain splash=LOC=NEW rejoice=IRR
'I will be happy if it rains.'

(9) *[*Tûi='ni=nda ûnjin*] *avûja=ya.*
splash=LOC=NEW rain rejoice=IRR
intended: 'I will be happy if it rains.'

Third, the subordinate status can be established by a restriction on the occurrence of sentence-level clitics. These include the reportative evidential *=te* 'RPRT' (nasal allomorph: *=nde*) and the polar interrogative *=ti* 'INT' (nasal allomorph: *=ndi*), as well as the optional first person *=ngi* '1', second person *=ki* '2', and third person *=tsû* '3' clitics, which encode, sometimes redundantly, the sentential subject. All of the sentence-level clitics occur close to the left edge of the clause, often right after the first clausal constituent. Thus, they are second-position clitics (although information structure-dependent permutations of word order may obscure their second-position nature). In (8), the subordinate clause as a whole is treated as the first constituent in a matrix clause, whereas attaching a clitic to the first constituent within the subordinate clause is ungrammatical (10).

(10) *[*ûnjin {=ngi, =tsû} tûi='ni=nda*] *avûja=ya.*
rain {=1, =3} splash=LOC=NEW rejoice=IRR
intended: 'I will be happy if it rains.'

Subordinate clauses can have the function of verbal *arguments* or *adjuncts*. Just as infinitives in a language like English have both argument and adjunct uses, we demonstrate below that A'ingae *=sa'ne* 'APPR' clauses do too. We therefore summarize these two classes of subordinate clauses in A'ingae, noting key similarities and differences.

2.3.1 Argument clauses

In A'ingae, various types of subordinate clauses can serve as verbal arguments. All subordinate clauses carry enclitics on their main verbs which, due to the rigidly verb-final word order of subordinate clauses, are at the same time clause-final. The argument clause can appear to the left or right of the matrix clause.

One general strategy for sentential subordination involves the nominalizing subordinator *='chu* 'SBRD'. Since *='chu* 'SBRD' creates formal nominalizations, *='chu* 'SBRD' clauses can appear in all the same environments as NPs. Other sentential complementizers include the infinitival *=ye* 'INF', the manner deictic *=khen* 'MANN.DEM', the adverbial *=e* 'ADV', attributive *='sû* 'ATTR', and apprehensional *=sa'ne* 'APPR'.

The category of verbs selecting for infinitival *=ye* 'INF' clauses includes, among others, attitude verbs such as *in'jan* 'want' (11) or *chi'ga* 'not want' (12) and aspectual verbs such as *tsun* 'do' (13), prospective semantics) or *atesû* 'know' (14), habitual or acquired ability semantics, Fischer & Hengeveld 2023).

(11) *In'jan=gi [panza=ye.]*
want=1 hunt=INF
'I want to hunt.'

(12) *Chi'ga=fa [thûthû=ye.]*
not want=PLS fell=INF
'They did not want to chop down the trees.'
(20170731_building_house_sapohe_mmemq_jc_: 15)

(13) *[Ya jañu=ngi asha-en=ñe] tsun-jen=fa.*
already now=1 beginning-CAUS=INF do-IPFV=PLS
'Now we're going to start.' (20170801_autobiography_CLC: 1)

(14) *[Tsa=ma tshe'tshe=pa] yaya'khashe'ye=ye [ujun=ñe] atesû.*
ANA=ACC mash=SS grandfather=HONR bathe=INF know
'My grandfather, having mashed it, would use it to bathe.'
(20170807_autobiography_JWC: 72)

Verba dicendi, *in'jan* 'think' (polysemous with 'want'), *tsun* 'do' and *iyikhu* 'fight' select for manner deictic *=khen* 'MANN.DEM' clauses. With verba dicendi and *in'jan* construed cognitively (i. e. to mean 'think'), the reading is that of speech or thought report. With *tsun* 'do' and *iyikhu* 'fight', the reading is that of desire or intention. The verb *da* 'become' selects for accusative 2 nominalized clauses (marked by *='chuve* 'SBRD.ACC2') or adverbial clauses (marked by *=e* 'ADV'). Motion verbs can select for attributive *='sû* 'ATTR' clauses to express the purpose of the motion.

Finally, verbs such as *dyu* 'be scared', *dyuju* 'be afraid',[8] *ansa'nge* 'be ashamed', *se'pi* 'prohibit', and *chi'ga* 'not want', select for *=sa'ne* 'APPR'. The complemen-

[8] Given the semantic relatedness and formal similarity, the verb *dyuju* 'be afraid' is presumably morphologically related to *dyu* 'be scared'. However, the semantic contribution of the unproductive formant *-ju* is not clear, as no other instances of it have been identified.

tizer function of *=sa'ne* 'APPR' is discussed more fully in §4, especially since it is not immediately clear that these are complements as opposed to adjuncts with precautioning *=sa'ne* 'APPR' (15).

(15) *Dyuju=ngi [thesi ña=ma mandian=sa'ne.]*
be.afraid=1 jaguar 1=ACC chase=APPR
argument paraphrase: 'I am afraid that the jaguar would chase me.'
adjunct paraphrase: '#I would be afraid in case a jaguar would chase me.'

2.3.2 Adjunct clauses

There are many available strategies in the language for adjunction. As adjunct clauses are not selected for by the matrix verb, but rather modify the predicate or the clause, there are no particular restrictions on which types of adjunct clauses can go together with which verbs. Like argument clauses, adjunct clauses carry enclitics and can appear on either side of the matrix clause.

Common strategies for adjunction include the nominalizing *='chu* 'SBRD' with oblique case marking, the adverbializing clitic *=e* 'ADV' for circumstance clauses, the locative clitic *='ni* 'LOC'[9] to signal a temporal relation between two clauses, the infinitive clitic *=ye* 'INF' to express positive purpose semantics, the new *='ta* 'NEW' and contrastive *='ja* 'CNTR' topic clitics to signal a conditional relation between two clauses, as well as the apprehensional clitic *=sa'ne* 'APPR' for undesirable outcome clauses in precautioning sentences (16). The precautioning function of *=sa'ne* 'APPR' is discussed more fully in §3.

(16) *Pa'khu a'ta ja-je=ya tsa undikhûje=ma khani=nde*
every day go-IPFV=VER ANA robe=ACC elsewhere=RPRT
tsa'u-ña=mba ambian=ña [tise khashe athe=pa ja=sane.]
house-CAUS=SS have=VER 3SG old woman see=SS go=APPR
'Every day he would go to see the clothes because he had them in another house far away so that his wife does not see them.'
(`20170730_kunsiana_cuento_VC2`: 77-79)

3 Precautioning function

Having reviewed the landscape of other subordinate clauses in A'ingae, we turn now to our main focus, the apprehensional *=sa'ne* 'APPR'. One typologically common apprehensional function is what Lichtenberk (1995) has dubbed *precautioning*. The precautioning function involves two clauses: one *apprehension-causing*

[9]The locative 'LOC' can express location in time or space, hence the gloss.

clause, which expresses a negative potential situation, and one *preemptive* clause, which expresses the precaution taken to either avert the apprehension-causing situation expressed in the other clause or else to be prepared for it, in case it should occur. Lichtenberk (1995) has labeled these two cross-linguistically attested subfunctions of precautioning morphemes the *avertive* and the *'in-case'* function, respectively. In English, the former may be expressed with the somewhat archaic conjunction *lest* or a negative purpose clause (17).[10] The latter can be expressed with *lest*, but not a negative purpose clause (18).

(17) I took a rifle {lest a jaguar kill me, so that a jaguar does not kill me}.

(18) I took a rifle {lest I see a jaguar, #so that I do not see a jaguar.}

The precautioning use of the apprehensional clitic *=sa'ne* 'APPR' is its most common one. In (19), *=sa'ne* 'APPR' has the avertive subfunction; we will illustrate the 'in-case' subfunction of *=sa'ne* 'APPR' below.

(19) *Tse=fan* *khi⟨'⟩tsha=jama* [*khitsha thûña=sane.*]
ANA=PEJ.ACC pull⟨PLV⟩=PROH pull break=APPR
'Don't pull it so that you don't break it!'
(`20170801_river_contamination_ARLQ`: 8)

In principle, the syntactic relation between the apprehension-causing and the preemptive clause could be that of parataxis, coordination, or subordination. In A'ingae, the apprehension-causing *=sa'ne* 'APPR' clauses are subordinate to the preemptive clauses, as we demonstrate below. They are adjuncts, their presence is optional – they are not selected by particular verbs and have a similar distribution to other adjuncts, as shown in §2.3.2.

3.1 Syntactic and semantic status

The apprehensional clitic *=sa'ne* 'APPR' scopes over a full clause, whose subject as well as object can be overt. Whereas some subordinators in A'ingae encode switch reference, the subject of the apprehension-causing *=sa'ne* 'APPR' clause can be the same (20) or different (21) from that of the preemptive clause.

(20) *Sema-'je=ngi* [*khiphue'sû=sa'ne.*]
work-IPFV=1 starve=APPR
'I am working lest I starve.'

[10] The semantics literature on English dating back to Faraci (1974) typically reserves the term *purpose clause* for a very specific subtype of such clauses, ones where the clause specifically encodes the purpose of the direct object. Here, we broaden the usage in accord with its less technical sense.

(21) *Sema-'je=ngi* [*dû'shû=ndekhû khiphue'sû=sa'ne.*]
work-IPFV=1 child=PLH starve=APPR
'I am working lest my children starve.'

The apprehension-causing *=sa'ne* 'APPR' clauses are subordinate. This can be demonstrated via the diagnostics introduced in §2.3.

First, we consider the interaction between *=sa'ne* 'APPR' and scope-taking operators such as negation. A paratactic analysis of the apprehension-causing *=sa'ne* 'APPR' clauses more or less approximates the apparent meaning when no such operator is present (22). However, once we add in negation to the preemptive clause, we see that the paraphrase the paratactic analysis would provide is no longer even approximately right (23) – the constituent in scope of negation is the preemptive clause as modified by the *=sa'ne* 'APPR' clause.

(22) *Tise=ta=tsû tsakhû=ma guathian-'jen* [*iyufa jin=sa'ne.*]
3SG=NEW=3 water=ACC boil-IPFV worm be=APPR
subordinate analysis: 'He is boiling water lest there be germs.'
paratactic analysis: 'He is boiling water. There might be germs.'

(23) *Tise=ta=tsû tsakhû=ma guathian-'jen=mbi* [*iyufa jin=sa'ne.*]
3SG=NEW=3 water=ACC boil-IPFV=NEG worm be=APPR
subordinate analysis: 'He is not boiling water lest there be germs. (He is boiling it for chicha.)'
paratactic analysis: '#He is not boiling water. There might be germs.'

Second, the subordinate status of the undesirable outcome *=sa'ne* 'APPR' clauses is corroborated by strict-verb finality (24)–(25).

(24) [*ña dû'shû=ndekhû khiphue'sû=sa'ne*] *sema-'jen.*
1SG child=PLH starve=APPR work-IPFV
'I am working lest my children starve.'

(25) * [*Khiphue'sû=sa'ne ña dû'shû=ndekhû*] *sema-'jen.*
starve=APPR 1SG child=PLH work-IPFV
intended: 'I am working lest my children starve.'

Third, we find that second-position subject clitics in the apprehension-causing clause are ungrammatical (26). In sum, precautioning *=sa'ne* 'APPR' clauses display all of the major syntactic and semantic properties associated with A'ingae subordinate clauses more generally.

(26) * [*ña dû'shû=ndekhû {=ngi, =tsû} khiphue'sû=sa'ne*] *sema-'jen.*
1SG child=PLH {=1, =3} starve=APPR work-IPFV
intended: 'I am working lest my children starve.'

In addition, the apprehensional clitic *=sa'ne* 'APPR' is paradigmatically related to and in complementary distribution with other subordinating enclitics in the language (same subject *=pa* 'SS', different subject *=si* 'DS', frustrative *='ma* 'FRST', and locative *='ni* 'LOC') in that it combines with subject number NUM, modal MOD, [11] and polarity POL clitics to its left; and topic TOP and focus FOC clitics to its right, as shown in Table 1.

The apprehensional clitic *=sa'ne* 'APPR' differs from the infinitive clitic *=ye* 'INF', which does not combine with modal MOD and polarity POL clitics. The infinitive clitic *=ye* 'INF' creates purpose clauses (27), subject argument clauses, or object argument clauses when selected for by matrix verbs.

(27) *Yaje=ma kû'i=pa [tse='an fi'thi=ye.]*
ayahuasca=ACC drink=SS ANA=PEJ.ACC kill=INF
'They drank ayahuasca to kill it.' (20170807_tshararukuku_RJCL: 39)

Finally, the apprehensional clitic *=sa'ne* 'APPR' also stands apart from matrix clausal clitics, i. e. the veridical *='ya* 'VER' and the four directive clitics, which include the three poorly understood imperative clitics *ja* '=IMP', *=kha* 'IMP2', and *='se* 'IMP3', and the prohibitive *=jama* 'PROH'. The matrix-clausal clitics do not combine with the topic TOP and focus FOC clitics.

In terms of linear order, the apprehension-causing *=sa'ne* 'APPR' clauses can appear before or after preemptive clauses (28)–(29).

(28) [*ña chan=ma iyikha'ye=sa'ne*]*=ngi shu'khaen.*
1SG mother=ACC annoy=APPR=1 cook
'I cooked so that my mother does not get mad.'

(29) *Ka'shi=ngi apishu'thu=ma* [*chan ña=ma iyû'û=sa'ne.*]
wash=1 dish=ACC mother 1SG=ACC scold=APPR
'I washed the dishes so that my mother does not scold me.'

The content of the apprehension-causing *=sa'ne* 'APPR' clauses contributes to the sentence's main point, i. e. is *at-issue* content in the sense of Simons et al. (2010) and related work. This is demonstrated by showing that it can be directly dissented to (30). The embeddability of *=sa'ne* 'APPR' clauses further supports their at-issueness. The at-issueness was illustrated with negation above (23). In

[11] The semantic effect of coupling the apprehensional *=sa'ne* 'APPR' with the irrealis *=ya* 'IRR' is not clear (i). If any, the difference is likely very subtle.

(i) *In'jan=ngi panza(=ya)=sa'ne.*
think=1 hunt(=IRR)=APPR
'I'm thinking (what to do) so that he doesn't kill it.'

(31), it is illustrated with the antecedent of a conditional clause (cf. Tonhauser 2012).

(30) A: *Tise=ta=tsû tsa'khû=ma guathian-'jen* [*iyufa jin=sa'ne.*]
3sg=new=3 water=acc boil-ipfv worm be=appr
'He is boiling water in case there are germs.'

B: *Me'in guathian-'jen=tsû* [*kûnapecha=ma mandyi=ye.*]
no boil-ipfv=3 chicha=acc squeeze=inf
'No, he's boiling it for chicha.'

(31) [[*Iyufa jin=sa'ne*] *tayu tsa'khû=ma gua'thian='chu=ni*] *khase*
worm be=appr already water=acc boil=sbrd=loc again
gua'thian=ñe injienge=mbi.
boil=inf be important=neg
'If the water has already been boiled in case there are germs, there is no reason to boil it again.'

Lastly, the undesirability of the apprehension-causing *=sa'ne* 'appr' clauses is uniformly subject-oriented. In (32), the judges of undesirability are the invadees (the subject), not the invaders (the speaker). In (33), rain is undesirable to the elder (the subject), not the speaker, who overtly expresses his contrary preference.

(32) *Tise'pa putaen'gu=ma am'bian='fa* [*ingi ja='fa=sa'ne.*]
they rifle=acc have=pls we go=pls=appr
'They got their shotguns ready in case we come.'

(33) *ûn'jin tûi=ye=ngi in'jan tsa'ma kuenza yaje=ma kûi* [*ûn'jin*
rain splash=inf=1 want but elder ayahuasca=acc drink rain
tûi=sa'ne.]
splash=appr
'I want it to rain, but the elder drank yaje so that it does not rain.'

3.2 Avertive and 'in-case' subfunctions

The apprehensional clitic *=sa'ne* 'appr' used in a precautioning fashion displays the two typologically attested subfunctions: *avertive*, with the preemptive clause expressing an action undertaken to avoid the event expressed in the apprehension-causing clause, and *'in-case'*, with the preemptive clause expressing an action undertaken to avoid the undesirable consequences of the event expressed in the apprehension-causing clause. Both readings are available with agentive verbs (34)–(35), non-agentive verbs (36)–(37), as well as weather verbs (38)–(39).

(34) *Ka'shi=ngi apishu'thu=ma* [*chan ña=ma iyû'û=sa'ne.*]
wash=1 dishes=ACC mother 1SG=ACC scold=APPR
'I washed the dishes so that my mother does not scold me.'

(35) *Putaen'gu=ma am'bian* [*tetete=ndekhû ji='fa=sa'ne.*]
rifle=ACC have Tetete=PLH come=PLS=APPR
'I got my rifle ready in case the Tetetes come.'

(36) *Upûi=ngi* [*cha'ndi'sû=sa'ne.*]
cover up=1 be.cold=APPR
'I covered myself so that I don't get cold.'

(37) *Vasûi=ngi tsûi* [*iyu khûi=sa'ne.*]
slowly=1 walk snake lie=APPR
'I walked slowly in case snakes be there.'

(38) *Kuenza=ja yaje=ma kû'i* [*ûnjin tûi=sa'ne.*]
old=CNTR ayahuasca=ACC drink rain splash=APPR
'The elder drank ayahuasca for rain not to come.'[12]

(39) *Chaketa=ma=ngi undikhû* [*ûnjin tûi=sa'ne.*]
jacket=ACC=1 don rain splash=APPR
'I put on a jacket in case it rains.'

Undesirability associated with the precautioning *=sa'ne* 'APPR' clauses is conventional (semantic). The avertive *=sa'ne* 'APPR' clauses might contain predicates of negative (40), neutral (41), or positive (42) emotional connotation, though the subject of the matrix clause is always presented by the speaker as being the judge of the undesirability of the prospective situation expressed by the avertive *=sa'ne* 'APPR' clause (or the larger situation containing it in the 'in-case' use). This distinguishes the 'in-case' precautioning uses from the ostensibly similar English *in case* construction, which need not have any undesirability associated with it.[13]

[12] Felicitous weather-averting scenarios often implicate shamanic training.

[13] The apparent similarity between the 'in-case' function of the A'ingae *=sa'ne* 'APPR' and the English *in case* is equivocal. On one hand, English *in case* constructions need not involve undesirability (i), which would mean that their semantics extends beyond apprehension. On the other hand, as observed by Eva Schultze-Berndt, they admit avertive semantics in some (though not all) dialects (ii), which might testify to a permeability between the avertive and 'in-case' functions.

(i) I will be happy in case I win the lottery.

(ii) I put the mug out of reach in case I knock it over.

We remain agnostic about the status of English *in case* pending further evidence.

(40) [*ña chan=ma iyikha'ye=sa'ne*]*=ngi shu'khaen.*
1SG mother=ACC annoy=APPR=1 cook
'I cooked so that my mother does not get mad.'

(41) *Jûnde ja* [*tise faengae ji=sa'ne.*]
soon go 3SG together come=APPR
'I hurried up to leave so that he doesn't come with us.'

(42) *Pûshesû tsû tsandie aya'fa=ma phikhu* [*feña=sa'ne.*]
woman 3 man mouth=ACC cover laugh=APPR
'She covered his mouth so that he does not laugh.'

Likewise, the 'in-case' *=sa'ne* 'APPR' clauses may appear with verbs of negative (43), neutral (44), or positive (45)–(46) emotional connotation. When the situation referred to by the *=sa'ne* 'APPR' clause is unambiguously positive, only 'in-case' readings are pragmatically available, i. e. ones in which a larger potential situation including that described is deemed undesirable. For example, my father bringing home the tapir and there being no hot water in (45) or my friends coming and the house being dirty in (46).[14]

(43) *Seje'pa=ma=ngi tsun-'jen* [*ña dû'shû iyu=nga tsei-ye=sa'ne.*]
medicine=ACC=1 do-IPFV 1SG child snake=DAT bite-PASS=APPR
'I'm preparing medicine in case my son gets bitten by a snake.'

(44) *Jayi=mbi=ngi fiesta=nga* [*tsetse'pa jin=sa'ne.*]
go.PRSP=NEG=1 party=DAT alcohol be=APPR
'I'm not going to the party in case there is alcohol.'

(45) *Tsa'khû=ma=ngi guathian-'jen* [*ña yaya khuvi=ma i=sa'ne.*]
water=ACC=1 boil-IPFV 1SG father tapir=ACC bring=APPR
'I am boiling water in case my father brings a tapir.'

(46) *Tsa'u=ma=ngi giyaen-'jen* [*faengasû=ndekhû ji='fa=sa'ne.*]
house=ACC=1 clean-IPFV friend=PLH come=PLS=APPR
'I am cleaning my house in case my friends come.'

[14] In many cases, this larger situation can be trivially thought of as a union of the *=sa'ne* 'APPR' clause and the negation of the matrix clause. Nevertheless, hardwiring that into the semantics of *=sa'ne* 'APPR' clauses would miss the intuition that there are many things the agent can do to avoid the undesirable outcome. In other words, the undesirable situation in (45) is not that of one's father bringing home a tapir and water not having been boiled, but rather that of having to deal with a decaying tapir. The latter, in turn, can be addressed via a multiplicity of means, e.g. by boiling water, making fire, sharpening knives, etc.

3.3 Precautioning and negative purpose clauses

We define *purpose clauses* as adjuncts that express the purpose of the action given by the matrix clause. In doing so, we deviate from the definition used in some previous literature (see Footnote 10). English has several constructions capable of expressing purpose semantics (47).

(47) I took a rifle {to, in order to, so as to} hunt a jaguar.

In A'ingae, positive purpose clauses are most typically introduced with the infinitive clitic *=ye* 'INF' (48).

(48) *Ciendo dolar=khû=ki ja=ya Lago=ni [chava=ye.]*
hundred dollar=INST=2 go=IRR Lago Agrio=LOC buy=INF
'You're going to Lago Agrio with $100 to buy something.'
(20170801_escuela_CLC: 194)

The subjects of the matrix and subordinate purpose clauses may or may not be the same (49)–(50); if no subject is overtly given in the purpose clause, it is interpreted as co-referential with the subject of the matrix clause (49).

(49) *Sema-'je=ngi [(ña) ankhe'sû=ma a'mbian=ñe.]*
work-IPFV=1 1SG food=ACC have=INF
'I am working to have food.'

(50) *Sema-'je=ngi [dû'shû ankhe'sû=ma a'mbian=ñe.]*
work-IPFV=1 children food=ACC have=INF
'I am working so that my child can have food.'

There is, to a large extent, semantic parallelism between *=ye* 'INF' and *=sa'ne* 'APPR': the infinitival *=ye* 'INF' purpose clauses are the positive counterpart to the avertive *=sa'ne* 'APPR' clauses; the purpose *=ye* 'INF' clauses express a desirable outcome which is intended to be brought about by the matrix clause, whereas the avertive *=sa'ne* 'APPR' clauses express the apprehension-causing situation that is supposed to be forestalled by the matrix clause.

This parallelism may be the reason why Fischer & Hengeveld (2023) gloss *=sa'ne* as a negative purpose clause clitic 'NEG.PURP'. Yet, although one function of the clitic *=sa'ne* 'APPR' is to head negative purpose clauses, this does not capture its versatility, as it can also head precautioning 'in-case' clauses, complements of certain verbs (§4), and timitive adjuncts (§5). A better candidate for a properly negative purpose operator is the complex *=mbe kañe* 'NEG.ADV AUX.INF' with exclusively avertive semantics.

The complex operator *=mbe kañe* 'NEG.ADV AUX.INF' provides a periphrastic means by which to express a combination that violates the syntactic restrictions on clitic co-occurrence discussed above. As shown in Table 1, the infinitive clitic *=ye* 'INF' does not combine with the negative polarity clitic *=mbi* 'NEG' (51).

(51) * *Sema-'je=ngi* [*vana=mbi=ye.*]
work-IPFV=1 suffer=NEG=INF
intended: 'I'm working to not be in trouble.'

The dummy auxiliary verb *kan* 'AUX' originates as a lexical verb *kan* 'watch' (52) and simultaneously functions as a productive modal auxiliary of tentative (i. e. 'try') semantics (53). Nevertheless, its use in the complex construction *=mbe kañe* 'NEG.ADV AUX.INF' is distinct from the other two, as evidenced by the fact that encoding an infinitival negative tentative (i. e. 'not to try') requires employing both the tentative *kan* 'try' and the auxiliary *kan* 'AUX' (54).

(52) *Ke=ma kan=ña=mbi.*
2SG=ACC look=IRR=NEG
'He is not going to look for you.' (`20170731_attembi_a'i`: 18)

(53) *Me'in ña an kan=mbi=ngi ña.*
no 1SG eat try=NEG=1 1SG
'No, I have not tried it.' (`20170801_fishing_CLC`: 22)

(54) *In'jan=ngi panza kan=mb=e kan=ñe.*
want=1 hunt try=NEG=ADV AUX=INF
'I want not to try to hunt.'

In forming *=mbe kañe* 'NEG.ADV AUX.INF', *=mbi* 'NEG' is first combined with *=e* 'ADV' to yield the negative adverbial clitic *=mbe* 'NEG.ADV'. Aside from the periphrastic negative purpose clauses to be discussed, *=mbe* 'NEG.ADV' is used to form negative circumstance clauses, as shown in (55).

(55) *Tsa'kan=nda* [*u⟨'⟩ya=mb=e*] *dyai=ye.*
thus=NEW move⟨PLV⟩=NEG=ADV sit=INF
'Then I will sit still.'
(Fischer & Hengeveld 2023; `20170801_cuiccu_chicha_ARLQ`: 206)

Second, the auxiliary *kan* 'AUX' combines with the negative adverbial clause. The dummy verb is there to carry the infinitive clitic *=ye* 'INF', which conveys the purpose semantics.

The complex operator *=mbe kañe* 'NEG.ADV AUX.INF' is used to create negative purpose clauses proper. Negative purpose *=mbe kañe* '=NEG.ADV AUX.INF' clauses have the same interpretation as the avertive *=sa'ne* 'APPR' clauses. This is to say, in (56), the meaning is the same regardless of whether *ansa'ne* or *ambe kañe* is used. The uses of the operator *=mbe kañe* 'NEG.ADV AUX.INF' are limited to the avertive.

(56) *Putaen'gu=ma=ngi am'bian [thesi ña=ma {an=sa'ne, an=mb=e*
rifle=ACC=1 have jaguar 1=ACC {eat=APPR, eat=NEG=ADV
kan=ñe.}]
AUX=INF}
'I have a rifle so that a jaguar does not eat me.'

Negative purpose *=mbe kañe* '=NEG.ADV AUX.INF' clauses and precautioning *=sane* 'APPR' clauses diverge in contexts where *=sa'ne* 'APPR' clauses receive 'in-case' readings. Compare (45)–(46) with the pragmatically aberrant (57)–(58), whose infelicity is readily signalled by native speakers.

(57) # *Tsa'khû=ma=ngi guathian-'jen [ña yaya khuvi=ma i=mb=e*
water=ACC=1 boil-IPFV 1SG father tapir=ACC bring=NEG=ADV
kan=ñe.]
AUX=INF
'#I am boiling water so that my father does not bring a tapir.'

(58) # *Tsa'u=ma=ngi giyaen-'jen [faengasû=ndekhû ji='fa=mb=e*
house=ACC=1 clean-IPFV friend=PLH come=PLS=NEG=ADV
kan=ñe.]
AUX=INF
'#I am cleaning my house so that my friends do not come.'

3.4 Relating the subfunctions

With two precautioning strategies, one capable of expressing both precautioning subfunctions (*=sa'ne* 'APPR'), the other restricted to the avertive subfunction (*=mbe kañe* 'NEG.ADV AUX.INF'), A'ingae parallels To'aba'ita exactly (Lichtenberk 1995). To'aba'ita's first strategy involves the apprehensional conjunction *ada* 'APPR', analogous to *=sa'ne* 'APPR' (59). Its second strategy involves the purpose clause conjunction *fasi* 'PURP', which in combination with a grammatically negative clause yields negative purpose semantics (60).

(59) To'aba'ita (Austronesian; Lichtenberk 1995: ex. (12), glossing simplified)
Nau ku agwa 'i buira fau ada [*wane 'eri ka riki nau.*]
I I:fact hid at behind rock appr man that he see me
'I hid behind a rock so that the man might not see me.'

(60) To'aba'ita (Austronesian; Lichtenberk 1995: ex. (17), glossing simplified)
Ngali-a kaleko 'aa'ako [*fasi 'osi gwagwari 'afa rodo.*]
take-them clothes warm purp you.neg be cold at night
'Take warm clothes so that you are not cold at night.'

Discussing the two precautioning strategies in To'aba'ita, Lichtenberk 1995 raises the question of how the avertive and 'in-case' subfunctions are related. Having considered ambiguity and polysemy, he concludes that they are polysemous ("semantically rather than pragmatically ambiguous", p. 302), thus granting the two uses equal status. His three main pieces of evidence for the avertive and the 'in-case' functions being "conceptually distinct from each other" (p. 302) are:

> (a) that an element used to encode negative purpose need not have an 'in-case' function; (b) that there is a formal difference between negative-purpose and 'in-case' clauses in at least one language; and (c) that there are differences in paraphrase possibilities between negative-purpose and 'in-case' clauses (Lichtenberk 1995: 302).

The idea that avertive and 'in-case' clauses have equal status, with morphemes like A'ingae *=sa'ne* 'appr' and To'aba'ita *ada* 'appr' simply being ambiguous between the two, cannot be rejected without a more thorough typology. There are, however, reasons for skepticism. First, we can note that the one language with a formal difference between negative-purpose and 'in-case' clauses Lichtenberk (1995) references is Martuthunira, where both avertive and 'in-case' uses deploy the same apprehensional morpheme, *-wirri* 'appr', and differ only in that the avertive use combines with an accusative *-i* 'acc' case marker, whereas the 'in-case' use combines with a locative *-la* 'loc' case marker or no case marker at all (Dench 1988). Without a more detailed understanding of case marking in Martuthunira, then, it is not clear how precisely to interpret this data.

More generally, there appears to be an asymmetry in the attested precautioning morphemes. While we find a number of elements like A'ingae *=sa'ne* 'appr' and To'aba'ita *ada* 'appr', which have both uses, as well as elements like A'ingae

=mbe kañe ‘NEG.ADV AUX.INF’ and To’aba’ita *fasi* ‘PURP’, which only have avertive uses, we are not aware of precautioning morphemes which only have the ‘in-case’ use, but cannot be used in avertive cases as well.[15]

In contrast to Lichtenberk (1995) and what seems to have been at least implicitly assumed in subsequent literature, we have at times above discussed the two uses in a somewhat different, asymmetrical way which we make explicit here. In our analysis, the situation denoted by the *=sa’ne* ‘APPR’ clause is contained within a possible undesirable situation which is contextually salient or otherwise recoverable. If the containment is proper (i. e. the undesirable situation contains, but is not identical to the argument of *=sa’ne* ‘APPR’), the ‘in-case’ function obtains. For example, this is the case in (35), where the arrival of the Teteté is contained within a larger undesirable situation (the Teteté coming and killing the subject). If the containment is improper (i. e. the undesirable situation is identical to the argument of *=sa’ne* ‘APPR’), the avertive function obtains. This is, for example, the case in (34), where the undesirable situation of being scolded by one’s mother is encoded in the *=sa’ne* ‘APPR’ clause. In short, elements like A’ingae *=sa’ne* ‘APPR’ require a salient undesirable situation containing the one they introduce, whereas elements like A’ingae *=mbe kañe* ‘NEG.ADV AUX.INF’ require the situation they introduce itself to be undesirable.

Since the situation explicitly stated in the *=sa’ne* ‘APPR’ clause is necessarily salient, this approach therefore captures the apparent typological asymmetry between the avertive and ‘in-case’ uses. Moreover, it illuminates why the two subfunctions are expressed in the same way in so many languages in a way that the ‘ambiguity’ account does not. Finally, as we will argue in §5, the same mechanism that allows for the ‘in-case’ uses also explains the semantics of the timitive.

[15] Another component distinguishing between the avertive and ‘in-case’ functions proposed in the previous literature, as Eva Schultze-Berndt (p.c.) observes, is that of control of the main clause subject over avoiding the event encoded by the precautioning clause. In the avertive uses, the subject has control over the precautioning clause, while in the ‘in-case’ uses, the subject need not have control over the precautioning clause. Observe that this is not the case in A’ingae, as ‘in-case’ precautioning readings are available even when the subject has full control over whether the events in the *=sa’ne* ‘APPR’ clause take place (i).

(i) *Tise tsû am’bian putaen’gu=ma* [*tsampi=ni ja=sa’ne.*]
(s)he 3 have rifle=ACC forest=LOC go=APPR
‘He$_x$ got his shotgun in case he$_x$ goes hunting.’

4 Complementizer function

It has been observed in previous descriptive work that the morphemes which serve apprehensional functions might also act as complementizers with verbs of fearing (Lichtenberk 1995, Dobrushina 2017, Wiemer 2018). We refer to this as the *complementizer* function. In English, for example, *lest* can be a somewhat archaic complementizer of the predicate *fear* (61).

(61) I fear lest a jaguar eat me.

In To'aba'ita, complements of 'fear' predicates are introduced by the apprehensional morpheme *ada* 'APPR' discussed above (62).

(62) To'aba'ita (Austronesian; Lichtenberk 1995: ex. (8), glossing simplified)
Nau ku ma'u 'asia na'a [*ada laalae to'a baa ki keka lae mai*
I I:FACT be.afraid very APPR later people that they go hither
keka thaungi kulu.]
they kill us
'I am scared the people might come and kill us.'

In A'ingae, the apprehensional clitic *=sa'ne* 'APPR' can introduce complements of 'fear' predicates as well (63).

(63) *Tsama ña* [*dañu=sane=khe*] *dyuju-je=ya.*
but 1SG be.hurt=APPR=MANN.DEM be.afraid-IPFV=VER
'But I didn't want to get hurt.' (20170731_yaje2_MM: 53)
literally: 'But I was afraid to get hurt.'

Nevertheless, the formal status of the so-called 'fear' complementizer uses is often far from obvious and its analysis is fraught with difficulties. The complement status of the apprehensional clauses when used with fear predicates is difficult to discern since they can also be interpreted as 'in-case' precautioning uses (64). The extant literature rarely provides explicit arguments for the genuine complement status of such uses.

(64) I fear it might rain. ≈ I (would) fear (it) in case it rained.

On the other hand, other researchers report apprehensional morphemes acting as complementizers of a wider range of predicates (Yallop 1997, François 2003). A'ingae fits in the latter category: the apprehensional *=sa'ne* 'APPR' clauses can function as complements and their distribution is not limited to 'fear' predicates. Since there are strong parallelisms between the apprehensional and the infinitival constructions, this finding is not implausible. Like the apprehensional *=sa'ne*

'APPR' clauses, the infinitival *=ye* 'INF' clauses have both complement uses and purpose-like adjunct uses cross-linguistically, including in A'ingae. Furthermore, both clause types can be arguments of the switch-reference subordinating conjunction *kûintsû* 'SRCN', possibly to the exclusion of all other clausal types.

We analyze 'fear' complementizer uses as involving genuine complementation and argue against the other *a priori* available alternatives: the adjunct and paratactic analyses. While we argue that *=sa'ne* 'APPR' has uses as a complementizer, there are also many cases which have the superficial appearance of complements, but whose interactions with operators such as negation do not support this conclusion. Moreover, the two categories can be distinguished on semantic grounds: *=sa'ne* 'APPR' functions as a complementizer to verbs which have the component of undesirability and as an adjunct to verbs which do not.

To see how the alternative analyses yield incorrect meanings, first consider the case of negated 'fear' predicates. Although all three paraphrases (complement, adjunct, and paratactic) are sensible semantic approximations when the polarity of the matrix clause is positive (65), the adjunct and paratactic paraphrases fail to properly reflect the meaning of A'ingae sentences when negated (66).

(65) *Ansa'nge=ngi* [*ña=ma feña=sa'ne.*]
be.ashamed=1 1SG=ACC laugh=APPR
complement paraphrase: 'I am afraid that he might laugh at me.'[16]
adjunct paraphrase: 'I am afraid in case he laughs at me.'
paratactic paraphrase: 'I am afraid. He might laugh at me.'

(66) *Ansa'nge=mbi=ngi* [*ña=ma feña=sa'ne.*]
be.ashamed=NEG=1 1SG=ACC laugh=APPR
complement paraphrase: 'I am not afraid that he might laugh at me.'
adjunct paraphrase: '#I am not afraid in case he laughs at me.'
paratactic paraphrase: '#I am not afraid. He might laugh at me.'

The paratactic paraphrase of (66) asserts that the speaker is not afraid[17] and

[16]The semantics of the A'ingae *ansa'nge* 'be ashamed', 'be afraid' differs from the English *be ashamed* in that the English predicate takes a past or present situation but not a potential future situation as a complement. Thus, the meaning of *ansa'nge* can perhaps be more adequately paraphrased as 'be fearful of a situation that would cause one to feel shame.'

[17]A'ingae is a pro-drop language, where most verbal complements need not be overtly expressed if they are recoverable from context. Thus, (i) is by itself a well-formed sentence.

(i) *Ansa'nge=ngi.*
be.ashamed=1
'I am afraid/ashamed.'

that someone might laugh at them. In this paraphrase, the negation of the first clause fails to scope over the second clause, yielding incorrect semantics.

The adjunct paraphrase of (66) asserts that the speaker is not afraid with the prospect of someone else laughing at them. Importantly, it does not assert that it is being laughed at specifically that the speaker fears, failing to adequately capture the meaning of the sentence. Further evidence against the adjunct paraphrase will be provided below with *se'pi* 'prohibit' and *chi'ga* 'not want'. These verbs do not have sensible intransitive paraphrases (one can simply be afraid but not prohibiting or not wanting), which supports the assumption about the truth conditions here.

As complements, the apprehensional *=sa'ne* 'APPR' clauses are subordinate and pass the three subordination diagnostics introduced in §2.3. They pass the first semantic test (65). They pass the second test of strict verb-finality (67)–(68). Lastly, they pass the third test of second-position clitic-shunning (69).

(67) [*Thesi ña=ma an=sa'ne*] *dyuju.*
jaguar 1SG=ACC eat=APPR be.afraid
'I fear the jaguar might eat me.'

(68) *[*ña=ma an=sa'ne thesi*] *dyuju.*
1SG=ACC eat=APPR jaguar be.afraid
intended: 'I fear the jaguar might eat me.'

(69) *[*ña=nda{=ngi, =tsû} tise=ma tshai=sa'ne=tsû*] *dyuju.*
1SG=NEW{=1, =3} 3SG=ACC hit=APPR=3 be.afraid
intended: 'He's afraid I'll hit him.'

As seen above, the verb *ansa'nge* 'be ashamed' is one verb with 'fear'-like semantics which can take *=sa'ne* 'APPR' complements. Two more such 'fear'-like verbs are *dyuju* 'be afraid' (70) and *dyu* 'be scared' (71).

(70) *Dyuju=ngi* [*thesi=nga mandia-ñe=sa'ne.*]
be.afraid=1 jaguar=DAT chase-PASS=APPR
'I'm afraid of being chased by a jaguar.'

(71) *Kani=ngi dyu* [*thesi=nga mandia-ñe=sa'ne.*]
yesterday=1 be.scared jaguar=DAT chase-PASS=APPR
'Yesterday I got scared that a jaguar would chase me.'

Other verbs with negative semantics which can take *=sa'ne* 'APPR' complements include *se'pi* 'prohibit', where *=sa'ne* 'APPR' expresses the object of prohibition (72), and *chi'ga* 'not want' where it expresses the object of distaste (73).

(72) *Yaya=tsû se'pi* [*dûshû=ndekhû phi='fa=sa'ne.*]
father=3 prohibit child=PLH sit=PLS=APPR
'The father prohibited the children from sitting (in the hammock).'

(73) *Yaya=tsû chi'ga* [*dûshû=ndekhû phi='fa=sa'ne.*]
father=3 not.want child=PLH sit=PLS=APPR
'The father does not want the children to sit (in the hammock).'

The complement status of the *=sa'ne* 'APPR' clauses in these cases (72)–(73) is corroborated by their semantics under negation (74)–(75).

(74) *Yaya=tsû se'pi=mbi* [*dûshû=ndekhû phi='fa=sa'ne.*]
father=3 prohibit=NEG child=PLH sit=PLS=APPR
'The father did not prohibit the children from sitting (in the hammock).'

(75) *Yaya=tsû chi'ga=mbi* [*dûshû=ndekhû phi='fa=sa'ne.*]
father=3 not,want=NEG child=PLH sit=PLS=APPR
'The father does not mind the children sitting (in the hammock).'

As such, the negatively-valenced verb *chi'ga* 'not want' contrasts starkly with the positively-valenced *in'jan* 'want, think'. While we can find sentences with *in'jan* and a *=sa'ne* 'APPR' clause, careful consideration of them shows that these are in fact adjuncts headed by 'in-case' precautioning uses of *=sa'ne* 'APPR' rather than complements. First, looking at the simple sentences, we see that – unlike in the case of intuitively negative predicates – the object of *in'jan* is coreferential with something from prior discourse rather than the content of the *=sa'ne* 'APPR' clause, which is reflected in back-translations (76). Second, we find a quite different interaction with negation than what we have seen above for *chi'ga* 'not want' (77). Taken together, these observations confirm that whereas *=sa'ne* 'APPR' can serve as a complementizer for negative-valenced predicates, sentences that are superficially similar except for having a positive predicate have a quite different structure, one not involving complementation.

(76) *Yaya=tsû in'jan* [*dûshû=ndekhû phi='fa=sa'ne.*]
father=3 want child=PLH sit=PLS=APPR
'The father wants it in case the children sit (in the hammock).'

(77) *Yaya=tsû in'jan=mbi* [*dûshû=ndekhû phi='fa=sa'ne.*]
father=3 want=NEG child=PLH sit=PLS=APPR
'The father does not want it in case/because the children might sit (in the hammock).'

To account for the complementizer uses, we propose a pathway of diachronic development from precautioning uses. In the absence of relevant historical data, we hypothesize that this development can be attributed to the pragmatic near-equality between the two uses in simple positive unembedded statements. This is to say, we observe that being afraid in case of an event is nearly equal to being afraid of that event. This, we posit, facilitated a syntactosemantic tightening of the 'in-case' precautioning adjuncts into proper complements of verbs of fearing, such as *dyuju* 'be afraid', *dyu* 'be scared', and *ansa'nge* 'be ashamed'. Analogous reasoning applies to *chi'ga* 'not want' and *se'pi* 'prohibit'. As for the former, we observe that a distaste in case of an event is near equal to a distaste of that event; as for the latter – that a prohibition given lest an event occur is near equal to a prohibition of that event.

Our account is thus close to that of Lichtenberk (1995)'s account of To'aba'ita's analogous data. In relating the complementizer function to other apprehensional functions, Lichtenberk (1995: p. 305) observes that "[a]n undesirable future situation is likely to be feared" and proposes that "[t]hrough this metonymy, *ada* clauses began to be embedded under predicates of fearing." Our proposal, however, goes beyond Lichtenberk (1995)'s in that it extends to other verbs of negative emotional valence (i. e. *chi'ga* 'not want' and *se'pi* 'prohibit') and provides syntactosemantic tests for the genuine complement status of this function.[18]

5 Timitive function

A third apprehensional function proposed in previous literature is the *timitive*. The timitive function picks out a noun phrase that refers to a feared entity and relates it to the matrix-clausal situation triggered by the feared entity. Since timitive morphemes prototypically attach to NPs, this function is sometimes referred to as the timitive case marker or adposition (Vuillermet 2018b,a). Timitive phrases can function as either adjuncts or arguments. When they convey non-essential information, they are understood to be adjuncts. When they are selected for by verbs of 'fearing' (and other negative verbs), they are understood to be arguments. Although there is no dedicated timitive morphology in English, the timitive can often be approximated with the periphrastic *for fear of* (78).

[18]Given the range of predicates which we find take *=sa'ne* 'APPR' complements in A'ingae, we might wonder whether Lichtenberk (1995)'s characterization of the To'aba'ita data in terms of 'fear' specifically is correct, or whether there too, we might find other negative desire predicates in this category. If not, some further explanation would seem to be needed since undesirable future situations are not only feared, but also likely not wanted, or prohibited.

(78) I ran for fear of the jaguar.

In A'ingae, *=sa'ne* 'APPR' can attach to nominal phrases in the function of a timitive, although it appears quite rarely in naturalistic speech (79).

(79) *Tuyakaen ña ambian setsani=da=tsû jin=ña ña=mbe ushachu,*
and 1SG have downriver=NEW=3 be=VER 1=BEN everything
[*ayafakhupi=sane*] *kûpakhu.*
mouth.sore=APPR prayer.plant
'I have prayer plant downriver [at my old house] in case of mouth sores.'
(20170803_garden_medicinal_plants_LC: 34)

As discussed by Vuillermet (2018a), there appears to be considerable cross-linguistic variation with respect to the semantic properties of the timitive. For example, the timitive in Ese Ejja does not require that the feared entity be avoided (Vuillermet 2018a), while the analogous morpheme in Marrithiyel does (Green 1989). In Ese Ejja, stand-alone uses of the timitive are not attested (Vuillermet 2018a), while in Manambu, they are (Aikhenvald 2008). It is therefore desirable to outline the parametrization of the A'ingae timitive, which we will later relate to other uses of *=sa'ne* 'APPR'.

To begin with, the timitive *=sa'ne* 'APPR' can combine with non-human entities such as weather conditions (80) and with inanimate entities such as mycosis (81).

(80) *Tsa'u=ni=ngi jayi* [*ûnjin=sa'ne.*]
house=LOC=1 go.PRSP rain=APPR
'I'm going home for fear of rain.'

(81) *Tsumba tsu'the, thenangu='ki, shamandakhû=ma='khe santshe*
then foot leg=2 armpit=ACC=ADD drily
san'jan=ña='chu [*asapa'chu=sa'ne.*]
dry=IRR=SBRD mycosis=APPR
'You must then thoroughly dry your feet, legs, and armpits to avoid mycosis.' (Pederson & Cooper 1982: p. 11)

The timitive *=sa'ne* 'APPR' can introduce the object of 'fear' predicates (82)–(83).[19] It coexists alongside strategies with the accusative *=ma* 'ACC' marking the object of *dyuju* 'be afraid' (84) and the dative *=nga* 'DAT' marking the stimulus of *dyu* 'be scared' (85). Intriguingly, the two strategies tend to be back-translated

[19] Analogous to the 'fear' complement uses discussed in §4, some or all of the *=sa'ne* 'APPR'-marked objects could potentially be best analyzed as precautioning-like adjuncts rather than arguments per se. We leave demonstrating their genuine object status to future work.

differently. Accusative and dative objects are translated with nominal phrases; timitive objects – with full clauses.[20]

(82) *Dyuju=ngi* [*thesi=sa'ne.*]
be.afraid=1 jaguar=APPR
'I am afraid there could be a jaguar.'
'I am afraid I'll encounter a jaguar.'

(83) *Dyu=ngi* [*thesi=sa'ne.*]
be.scared=1 jaguar=APPR
'I got scared there would be a jaguar.'

(84) *Dyuju=ngi thesi=ma.*
be.afraid=1 jaguar=ACC
'I am afraid of the jaguar.'

(85) *Dyu=ngi thesi=nga.*
be.scared=1 jaguar=DAT
'I got scared of a jaguar.'

Timitive phrases are available in narratives (86), conveying the fear of the sentence subject, not speaker, and co-occur with any main sentence type, e. g. the interrogative (87).

(86) *Kuenza=ndekhû uke='fa uvepa'chu tsau'pa=ma* [*anchan=sa'ne.*]
elder=PLH burn=PLS termite nest=ACC mosquito=APPR
'The elders burn termites' nests to avoid mosquitoes.'

(87) *Kuenza=ndekhû=ti uke='fa uvepa'chu tsau'pa=ma* [*anchan=sa'ne.*]
elder=PLH=INT burn=PLS termite nest=ACC mosquito=APPR
'Do the elders burn termites' nests to avoid mosquitoes?'

A timitive phrase can stand on its own, but only when there is a strong cultural association between its object and the threat it poses and it can be interpreted elliptically in a pragmatically rich context (88)–(89).

[20] The distribution and back-translation of timitive objects (discussed later in this section) suggests that the A'ingae timitive might be a result of semantic coercion of a noun phrase into a clausal reading or clausal ellipsis. While the account we propose for the timitive is consistent with semantic coercion, there are two considerations that argue against an analysis with syntactic ellipsis. First, there is no tendency for the elided material to appear in previous linguistic context. Second, *=sa'ne* 'APPR' cannot attach to a case-marked DP (i).

(i) * *Kuenza=ndekhû uke='fa uvepa'chu tsau'pa=ma* [*anchan=nga=sa'ne.*]
elder=PLH burn=PLS termite nest=ACC mosquito=DAT=APPR
'The elders burn termites' nests to avoid (getting stung by) mosquitoes.'

(88) [*Thesi=sa'ne.*]
jaguar=APPR
'In case of jaguars.' [uttered upon handing in a rifle]

(89) [*ûnjin=sa'ne.*]
rain=APPR
'In case of rain.' [uttered upon handing in an umbrella]

Nevertheless – aside from its 'fear'-complement function (82)–(83) – the timitive clitic *=sa'ne* 'APPR' is restricted to entities which make salient a situation avoided by the main clause event, paralleling the 'in-case' semantics of the precautioning function (90). In this respect, A'ingae *=sa'ne* 'APPR' timitive phrases differ from the English *for fear of* phrases, as the latter can also introduce the cause or stimulus of the main clause event (91).

(90) *Tsampi=ni ja=mbi=ngi* [*thesi=sa'ne.*]
forest=LOC go=NEG=1 jaguar=APPR
'I did not go to the forest for fear of a jaguar.'

(91) # *Juva=tsû* {*i'na, fûndu*} [*unkumari=sa'ne.*]
DIST=3 {cry, scream} bear=APPR
intended: 'He {cried, screamed} for fear of the bear.'

The timitive cannot generally combine with nouns of positive emotional connotation, such as *chan* 'mother' (92). Instead, its semantics is expressed with a precautioning *=sa'ne* 'APPR' that makes explicit the nature of the avertive situation (93) or a periphrastic *dyu* 'be scared' construction (94).

(92) # *Shu'khaen=ngi ña* [*chan=sa'ne.*]
cook=1 1SG mother=APPR
intended: 'I cooked for fear of my mother.'

(93) [*ña chan=ma iyikha'ye-en=sa'ne*]*=ngi shu'khaen.*
1SG mother=ACC annoy-PASS=APPR=1 cook
'I cooked so that my mother does not get mad.'

(94) [*ña chan=ma dyu='chu=i'khû*]*=ngi shu'khaen.*
1SG mother=ACC be.scared=SBRD=INST=1 cook
'I cooked for fear of my mother.'

Finally, it can be combined with neutral nouns, such as *tsetse'pa* 'chicha', though its distribution is restricted. Such uses are judged as felicitous only when the preceding linguistic context explicitly sets up the unwelcome situation (95). Even then, though, including a verb makes it better, with *=sa'ne* 'APPR' preferably heading a sentence, rather than a noun phrase (96).

(95) # *(Pûi fiesta=nga tsû tsetse'pa jin=ñe atesû.)* $^?$*jayi=mbi=ngi fiesta=nga*
each party=DAT 3 chicha be=INF know go.PRSP=NEG=1 party=DAT
[*tsetse'pa=sa'ne.*]
chicha=APPR
'There is alcohol at every party. I'm not going to the party for fear of alcohol.'

(96) *Pûi fiesta=nga tsû tsetse'pa jin=ñe atesû. jayi=mbi=ngi fiesta=nga*
each party=DAT 3 chicha be=INF know go.PRSP=NEG=1 party=DAT
[*tsetse'pa* $^?$*(jin)=sa'ne.*]
chicha be=APPR
'There is alcohol at every party. I'm not going to the party to avoid (for fear of) alcohol.'

We understand the avertive, 'in-case', and timitive functions of the apprehensional *=sa'ne* 'APPR' to be underpinned by uniform semantics: *=sa'ne* 'APPR' encodes the avoidance of a potential undesirable situation which includes the situation (or entity) expressed by its argument. That is to say, the semantics of the timitive is that laid out in §3, where the clitic *=sa'ne* 'APPR' is posited to encode the apprehension of a situation which contains its argument. This accounts for all the discussed properties of its timitive function.

First, it accounts for the timitive *=sa'ne* 'APPR''s rarity, as noun phrases do not prototypically denote situations (*=sa'ne* 'APPR' preferably combines with clauses).

Second, it accounts for its occurrences with eventive nouns, as the timitive use of *=sa'ne* 'APPR' is common with eventive nouns such as *ûnjin* 'rain' (80) or *tsanda* 'thunder' (97), which make salient such situations.

(97) *Chaketa=ma undikhû=ja* [*tsanda=sa'ne.*]
jacket=ACC don=IMP thunder=APPR
'Put on a jacket in case of thunder.'

Third, it accounts for its occurrences with negatively connoted nouns and events stereotypically associated with them. The timitive use of *=sa'ne* 'APPR' is likewise common with nouns such as *asapa'chu* 'mycosis' (81), *thesi* 'jaguar' (88) or *anchan* 'mosquito' (86), where an association between the noun phrase and the undesired situation is immediate (i. e. being eaten by a jaguar, ravaged by mycosis, or stung by mosquitoes). On the other hand, when such an association is lacking, as is the case with mothers not inherently perilous (92), the timitive construction is deemed infelicitous. (Thus, the timitive is different from the precautioning function where the undesirability is conventionally coded by *=sa'ne* 'APPR'.)

Fourth, it accounts for its availability for complementation with negative verbs (82)–(83) and the sensitivity of back-translations to the complementation strategy used. The timitive objects tend to be rendered with full clauses, which brings in close correspondence to their situational semantics.

Fifth, it accounts for its availability in narratives and with any main sentence type (86)–(87), by analogy with precautioning uses which encode subject, not speaker, fear.

Sixth, it accounts for its occasional ability to stand on its own (88), possibly when interpretable as ellipsis in a pragmatically loaded context, by analogy with monoclausal uses elaborated in §6.

Seventh, it accounts for its restriction to entities which make salient a situation avoided by the main clause event, paralleling the 'in-case' semantics of the precautioning function (90)–(91). The argument of *=sa'ne* 'APPR' maps to a situation avoided by the main clause event, paralleling the precautioning 'in-case' uses.

And finally, eighth, it accounts for the amelioration of certain infelicitous examples in rich contexts (95), since making the dispreferred situation explicit makes it more easily recoverable.

6 Apprehensive function

The precautioning, complementizer, and timitive uses are the main functions of the apprehensional clitic *=sa'ne* 'APPR'. These main functions all involve subordination (with precautioning and sentential complementizer uses) or NP-complementation (with timitive uses). The last use of the A'ingae *=sa'ne* 'APPR' clitic is the apprehensive proper, although this function is attested only marginally.

The *apprehensive* function is used to mark potential undesirable future events. It is most closely rendered by the English *watch out* (98), *might* (99), or the negative imperative (100). Unlike the other apprehensional uses, the negativity with apprehensives is typically speaker-, not subject-, oriented (the situation is undesirable in the judgment of the speaker).

(98) *Watch out for the curb.*

(99) *You might trip.*

(100) *Don't trip!*

The apprehensive is prototypically used with warning speech acts where the speaker is worried about some potential negative situation, often but not always one the addressee can take actions to avoid (101).

(101) [*Tsa'khû=ma sefa-en*]*=sa'ne.*
water=ACC end-CAUS=APPR
'Don't use up all the water.' (Borman 1990: p. 37)

Although functionally independent, apprehensive uses have the formal properties of subordinate clauses and disallow second-position clitics (102).

(102) [*Ke(*=ki) ana=sa'ne.*]
2SG=2 sleep=APPR
'You might fall asleep.'

The data above suggest insubordination, defined by Evans (2007: p. 367) as "conventionalized main clause use of what, on *prima facie* grounds, appear to be formally subordinate clauses." The development of the apprehensive use out of the precautioning use is a typologically attested pathway, first proposed by Lichtenberk (1995).

While the matrix clausal apprehensive uses here discussed are attested, they are strongly dispreferred. Native speakers characterize them as most appropriate as elliptical answers, conceptualizing them within larger discourse (103). When a discourse-initial position is demanded of them, often paraphrases are offered where the lacking matrix clausal verb *in'jan'jen* 'be careful' is supplied (104).

(103) A: *Jungueje=ngi yuku=ma kû'i=ya.*
why=1 yoco=ACC drink=IRR
'Why should I drink yoco?'

B: [*Ke anae'sû=sa'ne.*]
2SG be.sleepy=APPR
'Because you might fall asleep.'
'So that you don't fall asleep.'

(104) [*Tsai-ye=sa'ne*] *in'jan-'jen=jan.*
bite-PASS=APPR watch.out-IPFV=IMP
'(Watch out) you might get bitten.'

This suggests that the insubordination is in its early stages where the elided material is recoverable, and the ellipsis is preferably altogether avoided given insufficient contextual priming.

The transitional nature of A'ingae insubordination is further evident from the fact that native speakers differ in whether they allow it. Some translate the below in a manner congruent with the insubordination hypothesis. For others, the insubordination reading is unavailable even when the context is very loaded. The translation offered by those suggests they interpret it as an ellipsis of linguistically recoverable material in a rich enough context. That is to say, for those speakers, the ellipsis has not reached the stage where it is conventionalized.

(105) [*Tshipa=sa'ne.*]
be.wet=APPR
back-translation A: 'Don't get wet.'/'Avoid getting wet.'
back-translation B: 'So that you don't get wet.' [uttered upon handing in an umbrella]

Lastly, the ongoing insubordination hypothesis is supported by the availability of third-person-oriented apprehension (106) given a sufficiently rich context.

(106) A: *Khuvi mingae=tsû da-'je?*
tapir how=3 become-IPFV
'What's up with the tapir?' [uttered upon seeing a running tapir]

B: [*Thesi tise=ma fi'thi=sa'ne.*]
jaguar 3SG=ACC kill=APPR
'It's afraid the jaguar will kill it.'

All this testifies to the fact that insubordination in A'ingae is in its first stage, where both first-person (101) and third-person (106) monoclausal 'fear' uses can be analyzed as "underlying subordinate clauses whose main clauses have been ellipsed but can plausibly be restored for analytic purposes" (Evans 2007: 430). Plausible restorations for (101) and (106) are given in (107) and (108), respectively.

(107) [*Tsa'khû=ma sefa-en=sa'ne*] *in'jan-'jen=jan.*
water=ACC end-CAUS=APPR watch.out-IPF=IMP
'Pay attention lest you use up all the water.'

(108) [*Thesi tise=ma fi'tti=sa'ne*] *dyuju.*
jaguar 3SG=ACC kill=APPR be.afraid
'It's afraid the jaguar will kill it.'

The insubordination of the apprehensive uses has not reached its second stage, whereby "the structure itself may still be adequately described by treating it as an underlying subordinate clause", but only at the cost of "turning a blind eye to the greater semantic specificity associated with the insubordinated clause, and ignoring the fact that certain logically possible 'restored' meanings or functions are never found with the insubordinated construction" (Evans 2007: 430–431).[21] In relationship to the apprehensive function, this refers to the narrowing of the pool of potential grammatical persons towards which the apprehension is oriented. In prototypical apprehensives, it is only the speaker's fear that can be thus encoded. Since in A'ingae monoclausal uses of the *=sa'ne* 'APPR' clitics can express both first and third persons' fear, we know this stage has not been achieved.

7 Conclusions

The A'ingae apprehensional clitic *=sa'ne* 'APPR' has robust precautioning uses, both avertive and 'in-case', restricted timitive uses, and marginal apprehensive uses. Furthermore, it can serve as a complementizer to a number of negatively-valenced verbs. The apprehensional clitic *=sa'ne* 'APPR' thus presents us with novel typological properties, as this particular range of function has not been reported in previous literature. For a more fully fleshed-out typological framework that facilitates describing and comparing apprehensional synchrony and diachrony, see AnderBois & Dąbkowski (2025).

For several languages, attempts have been made to explicate the ranges of usage for their respective apprehensional morphologies on diachronic grounds. (among others, Lichtenberk 1995, Dobrushina 2017, Wiemer 2018). We argue that the particular functional range of the A'ingae apprehensional clitic *=sa'ne* 'APPR', and in particular the range of available precautioning and timitive uses, should be accounted for on synchronic – both semantic and syntactic – grounds. For a formalization of our argument, see Dąbkowski & AnderBois (2023).

Semantically, *=sa'ne* 'APPR' expresses the avoidance of a possible undesirable situation which contains that of its argument. If the containment is proper (i. e. the undesirable situation contains and is not identical to the argument of *=sa'ne* 'APPR', the 'in-case' function obtains. If, on the other hand, the undesirable situation is identical to the argument of *=sa'ne* 'APPR', the avertive function obtains.

[21]Trivially, therefore, apprehensive insubordination has not reached its third stage either, in which insubordinate "clauses have been so nativized as main clauses that the generalizations gained by drawing parallels with subordinate structures are outweighed by the artificiality of not including them in the muster of main clause types" (Evans 2007: 431).

Semantically, then, the possibility for a timitive use (at least given its semantics in A'ingae) automatically[22] follows from the presence of the 'in-case' use. The timitive use is limited due to the recoverability of the apprehensional situation from a noun phrase. For further elaboration of the argument and cross-linguistic evidence, see AnderBois & Dąbkowski (2021).

Syntactically, *=sa'ne* 'APPR' is a subordinator, a status which we have supported by both semantic evidence as well as through examining language-particular syntactic properties of subordinate clauses (see §2.3 and §3). The marginal apprehensive uses are therefore understood as contextual ellipsis or incipient insubordination of the precautioning or 'fear'-complementation uses, given their dependence on context and their retention of A'ingae subordinate clause hallmarks.

Abbreviations

1	first person subject clitic
1SG	first person singular pronoun
1PL	first person plural pronoun
2	second person subject clitic
2SG	second person singular pronoun
2PL	second person plural pronoun
3	third person subject clitic
3PL	third person plural pronoun
ABL	ablative case
ACC	accusative case
ACC2	accusative case 2
ADD	additive focus
ADJ	adjectivizer
ADV	adverbializer
ANA	anaphoric demonstrative
AND	andative direction
APPR	apprehensional marker
ATTR	attributive marker
AUX	dummy auxiliary verb
BEN	benefactive case
3SG	third person singular pronoun
CAUS	causative voice
CMP	comparative marker
CNTR	contrastive topic
DAT	dative case
DMN	diminutive aspect
DS	different subject
ELAT	elative case
EXCL	exclusive focus
FACT	factive
FOC	focal clitics
FRST	frustrative marker
HES	hesitative particle
HONR	honorific marker
IMP	imperative mood
IMP2	imperative mood 2
IMP3	imperative mood 3
INF	infinitive marker

[22] Of course, other languages may have precautioning morphemes with 'in-case' uses which lack timitive uses altogether for syntactic reasons. We therefore do not predict that any precautioning morpheme with 'in-case' uses must have timitive uses, but rather that if it lacks such uses, it is for syntactic reasons rather than semantic ones.

INST	instrumental case	POL	polarity clitics
INT	polar interrogative	PRCM	precumulative aspect
IPFV	imperfective aspect	PROH	prohibitive mood
IRR	irrealis mood	PROX	proximal demonstrative
ITER	iterative aspect	PRSP	prospective aspect
LOC	locative case	PURP	purpose clause marker
MANN.DEM	manner demonstrative	QUAL	qualitative marker
		RECP	reciprocal voice
MOD	modal clitics	RPRT	reportative evidential
NEG	negative polarity	SBRD	nominal subordinator
NEW	new topic	SRCN	switch-reference conjunction
NUM	subject number clitics	SS	same subject
PASS	passive voice	TAX	dependency (taxis clitics)
PEJ	pejorative marker		
PLH	human plurality	TOP	topical clitics
PLS	subject plurality	VEN	venitive direction
PLV	pluractionality (verbal plurality)	VER	veridical mood

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Chapter 4

The Apprehensional construction in Gban

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This paper focuses on the dedicated Apprehensional construction in Gban (South Mande, Côte d'Ivoire), which expresses several meanings related to fear or undesirability. The Apprehensional construction is used in main clauses and in dependent clauses. In main clauses, three readings or functions can be identified: expressing pure apprehension, warning, and threat. Two different dependent uses have been attested: in fear predicate complements and in negative purpose clauses. The paper describes the formal make-up of the construction, its semantic and syntactic properties, its relation to negative polarity and to the grammatical system of Gban in general. Diachronic sources for the elements comprising the construction and possible scenarios of its grammaticalization are proposed.

1 Introduction

Gban (*gbą́*; ISO 639-3 code: ggu, Glottocode: gagu1242) (Fedotov 2016, 2017, 2021) is a South Mande language spoken by about 60,000 people in the central part of Côte d'Ivoire. This study is based on the Bovo dialect.

Gban has a dedicated grammatical construction that expresses meanings belonging to the apprehensional semantic domain (the domain of fear and undesirability) (Lichtenberk 1995, Dobrushina 2006, Vuillermet 2018, Vuillermet et al. 2026 [this volume]) – henceforth the *Apprehensional construction.*

The Apprehensional construction in Gban covers almost all meanings/uses that are cross-linguistically associated with the apprehensional domain. It has

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main-clause uses (expressing pure apprehension, warning, or threat), uses in fear predicate complements, and uses in negative purpose (= precautioning) clauses.

Both the construction and the semantic domain may be provisionally illustrated by examples (1)–(3).

(1) Main-clause pure-apprehension use
*E̋! W-**ǎ** **nṵ̀** ké ḭ̀ mɔ̰́ wȍtló kṵ̋a̰̋!*
oh 3PL-**APPR** **APPR** EMPH I for car steal[INF]
[The speaker has left his/her car outside for the night unattended. He/she is now in bed, talking to him/herself, just expressing his/her anxiety:] '— Oh, my car **might** get stolen!'

(2) Fear predicate complement
*Ɛ̏ gbɛ̰̏ [nḭ́ Flȁ̰sóá (y)-**ǎ** **nṵ̀** gà].*
3SG IPFV\be_afraid that François 3SG-**APPR** **APPR** die[INF]
'She's **afraid** that François **might** die.'

(3) Negative purpose
*Ɔ̏ nṵ̏ kwà zɔ̰̏ ɛ̀ dı̋ [ḭ́ gwȁ y-**ǎ***
3PL IPFV\come arm descend[INF] you_SG under that stone 3SG-**APPR**
***nṵ̀** ɛ̀ kpɛ̋ kplɔ̰̏ lɛ̌].*
APPR you_SG foot scratch[INF] because
'[T]hey will lift you up in their hands, **so that** you will **not** strike your foot against a stone.' (Luke 4:11)

This paper will explore in detail the different types of uses, as well as other properties of the construction. Some typologically relevant parameters will be discussed for each use/reading, including its compatibility with different grammatical persons, with situations located in the past, the present, and the future, and with preventable and unpreventable situations.

In §2, I will describe the formal properties of the Apprehensional construction. In §3, I will discuss its paradigmatic status with respect to other core grammatical constructions of Gban. In §4, the usage of the construction in main clauses will be discussed, with the three attested readings/functions: expression of pure apprehension (§4.1), warnings (§4.2), and threats (§4.3). In §5, I will describe the two types of syntactically dependent uses: fear predicate complement uses (§5.1) and negative-purpose uses (§5.2). In §6, I will present the most probable diachronic sources for the elements comprising the Apprehensional construction and propose possible scenarios of its grammaticalization as a whole. Finally, in §7, I will separately discuss the synchronic relation of the Apprehensional construction to polarity.

The study is based on field data (elicitation and spontaneous texts) from the Bovo dialect collected in Côte d'Ivoire in 2011–2020, as well as on the text of the New Testament translation (Howard et al. 1998). Elicitation was partially guided by Marine Vuillermet's "Questionnaire on the apprehensional domain" (Vuillermet 2026 [this volume]). Most of the description is built on elicited data.[1] Examples from spontaneous texts and from the New Testament will be marked accordingly ("Narrative text", "2 John 1:8", etc.), while elicited examples will be left unmarked.

In the examples throughout the paper, I use the following notation. A superscript "OK" before an example indicates that the whole example was constructed by me in Gban and then was approved by the speaker as felicitous (while an asterisk "*" or one or several "?" indicate a greater or lesser degree of infelicitousness). Examples without "OK" were translated from French into Gban, suggested by the speaker in the course of the discussion, or taken from texts. The symbol "#" before an example indicates that the sentence is either a) grammatically correct, but was said to be nonsensical, or b) cannot be used with the given interpretation (but may be absolutely felicitous with some other interpretation, which is irrelevant to the discussion). Square brackets "[ ]" are used in some of the translations to indicate those semantic components (or a more general context) which are not explicitly expressed in the example. Curly brackets "{ }" are used to enclose the translation of the omitted parts of the example (which had been present in the Gban text, but which I removed for reasons of space). A tilde "~" is placed before some translations to mark them as approximate (including those cases where the translation was missing from the data and I produced a provisional one myself). Additionally, for some examples I provide not only the English translation, but also, in parentheses, the original (vernacular) French translation given by the speaker.

2 Form

The Apprehensional construction in Gban contains a dedicated *predicative marker*[2] *-ǎ̋/-ǎ* 'APPR' at the beginning of the clause and a dedicated invariable

[1] This is partially due to the fact that the construction is very infrequent in the texts (except for the negative-purpose uses, see §5.2), and partially because only elicitation made possible a fine-grained study of the semantics of the construction and of different combinations of the examined parameters. Still, it would have been desirable, of course, to have more examples from natural texts in the data.

[2] Predicative markers are special grammatical markers that are placed right after the subject NP and that participate in the expression of TAM and other sentential grammatical categories (i.e.

auxiliary verb *nṵ̀* (glossed here as ‘APPR’, too), which introduces the zero-marked infinitive form of the main verb:

(4) *N-a̰̋* ***nṵ̀*** *ɛ̀* *blɛ̋* *ɛ̀* *kwà* ***ylɛ̏*.**
2SG-**APPR** **APPR** you_SG self you_SG arm break[**INF**]
‘[**It would be bad if**] you (sg.) broke your arms [this way]’,
‘You **might** break your arms [this way]’ (to a child playing on a staircase);
‘[**Be careful not to**] break your arms!’

A sample paradigm of the Apprehensional construction with the lexical verb *nṵ̀* ‘come’ and different person-number markers is presented in Table 1. (While formally identical to the Apprehensional auxiliary, this second *nṵ̀* is not part of the apprehensional construction, but the infinitive form of ‘come’, see further §6.)

Table 1: Apprehensional construction in combination with different person-number markers: ‘[I’m afraid that] I/we/you (sg./pl.)/(s)he/they might come’.

	SG			PL		
1st	*m-a̰̋* 1SG-APPR	***nṵ̀*** APPR	*nṵ̀* come[INF]	*w-a̋* 1PL-APPR	***nṵ̀*** APPR	*nṵ̀* come[INF]
2nd	*n-a̰̋* 2SG-APPR	***nṵ̀*** APPR	*nṵ̀* come[INF]	*a̋-a̋* 2PL-APPR	***nṵ̀*** APPR	*nṵ̀* come[INF]
3rd	*y-ǎ* 3SG-APPR	***nṵ̀*** APPR	*nṵ̀* come[INF]	*w-ǎ* 3PL-APPR	***nṵ̀*** APPR	*nṵ̀* come[INF]

It is important to distinguish the predicative marker *-a̋/-ǎ* ‘APPR’ from a quasi-homonymous (although most certainly diachronically related, see §6) predicative marker *-a̋/-ǎ* ‘SBJV.NEG’ in the Negative Subjunctive construction. The latter, too, has two allomorphs *-a̋* and *-ǎ*, but these have a slightly different distribution. A sample paradigm is given in Table 2. Note that this construction does not involve an auxiliary and features a different form of the verb with a different tone pattern.

The subtle difference in the form of the marker itself can be seen in the 2nd person. For the Apprehensional predicative marker, an allomorph *-a̋* with an

categories pertaining to the whole predication, hence the term *predicative*). See §3 for more details on predicative markers and the grammatical system of Gban in general.

Table 2: Negative Subjunctive construction in combination with different person-number markers: ‘I/we/we and you (pl.)/you (sg./pl.)/(s)he/they shouldn’t come.’

	sg		pl/pl_incl	
1st	*m-**a̰̋***	*nṵ̋*	*w-**a̋***	*nṵ̋*
	1sg-sbjv.neg	ipfv\come	1pl-sbjv.neg	ipfv\come
			*à(ù)w-**a̋***	*nṵ̋*
			1pl_incl.sbjv-sbjv.neg	ipfv\come
2nd	*n-**ǎ̰***	*nṵ̋*	***ȁa̋***	*nṵ̋*
	2sg-sbjv.neg	ipfv\come	2pl:sbjv.neg	ipfv\come
3rd	*y-**ǎ***	*nṵ̋*	*w-**ǎ***	*nṵ̋*
	3sg-sbjv.neg	ipfv\come	3pl-sbjv.neg	ipfv\come

extra-high tone is chosen, while the Negative Subjunctive marker has an allomorph *-ǎ* with a rising tone.[3]

At the same time, a predicative marker that is formally fully identical to *-a̋/-ǎ* ‘appr’ is used in another construction, the Enimitive construction,[4] which has the totally unrelated meaning/function ‘… isn’t it?, … right?; indeed, certainly’. A sample paradigm for the construction is presented in Table 3 and an illustration of its use in example (5) below.

(5) *N-**a̰̋*** *yȅ̋ȅ̋* *yɔ̰̈* ***nḭ̀**!*
2sg-**enim** ipfv\be there **enim**
[To a person who is standing close to two children fighting, but isn’t trying to pull them apart:] ‘— [Hey!] You’re [standing] there, **aren’t you!**’ [Why aren’t you doing something?]

As can be seen, while sharing the predicative marker *-a̋/-ǎ* with the Apprehensional construction, the Enimitive construction formally differs from it by the

[3]In the case of the 2nd-person plural marker, this is realized as spanning over two vowels: *ȁa̋*.

[4]In this construction, this marker will be glossed as ‘enim’ for convenience, although it could be glossed in both constructions as ‘appr/enim’ (or with some simplex gloss that would somehow cover the functions of both constructions). Any such decision is arbitrary, since it is not clear whether *-a̋/-ǎ* ‘appr’ and *-a̋/-ǎ* ‘enim’ should be considered synchronically as one and the same marker or as two different homonymous ones.

In using this label, I follow Panov (2020), who suggested the term *enimitive* for a very close range of meanings (the term itself is derived from the Latin particle *enim* expressing similar semantics).

Table 3: Enimitive construction (in Non-past) in combination with different person-number markers: ‘I/we/you (sg./pl.)/(s)he/they come(s) indeed.’

	SG			PL		
1st	*m-**a̰̋***	*nṵ̋*	***nḭ̀***	*w-**ā̋***	*nṵ̋*	***nḭ̀***
	1SG-ENIM	IPFV\come	ENIM	1PL-ENIM	IPFV\come	ENIM
2nd	*n-**a̰̋***	*nṵ̋*	***nḭ̀***	***a̋a̋***	*nṵ̋*	***nḭ̀***
	2SG-ENIM	IPFV\come	ENIM	2PL:ENIM	IPFV\come	ENIM
3rd	*y-**ǎ***	*nṵ̋*	***nḭ̀***	*w-**ǎ***	*nṵ̋*	***nḭ̀***
	3SG-ENIM	IPFV\come	ENIM	3PL-ENIM	IPFV\come	ENIM

lack of the invariable auxiliary *nṵ̀* and the presence of a special clause-final particle *nḭ̀* and also by the fact that the Enimitive construction combines with finite (inflecting) forms of the main verb.

3 Paradigmatic status

The Apprehensional construction is an integral part of the grammatical tense-aspect-mood system of Gban, and can be shown to be one of its core sentential constructions.[5]

As mentioned above, the Apprehensional construction contains a predicative marker *-a̋/-ǎ* ‘APPR’. Let us first discuss the predicative markers in general.

Gban has an SOV word order. The predicative markers make up five rigid consecutive slots at the beginning of the sentence, right after the subject NP (if present) and before the direct object NP and the verb. In each of these slots, only one marker can appear at a time; all five of them can be occupied simultaneously by non-zero markers. An overview of the predicative marker system is given in Table 4, with the Apprehensional predicative marker in bold.

[5] They are regarded as sentential and not verbal because they are not expressed (solely) by finite verb forms, but by sentence-level constructions that combine predicative markers and finite verb forms; see below.

By a *core* grammatical construction (opposed to a *peripheral* one) I understand a construction that belongs to a certain closed set of grammatical constructions from which it is obligatory to choose and employ one (and only one) construction whenever one utters a finite clause in this language.

Table 4: The structure of the predicative marker slots and their linear position in the sentence. *Notes:* 1. The predicative markers (PMs) are located between the subject (S) and the direct object (DO) or the (intransitive) verb (V). 2. The person-number markers in Slot 1 have complex allomorphy; only the first few allomorphs are listed. 3. The Past predicative marker -$^{\uparrow}$ is a tonal morpheme that shifts the tone of the preceding syllable two levels up. 4. Two of the glosses for zero markers, 'NCOND' and 'NIPFVPST', are simplified: for 'NCOND', see Footnote 8; 'NIPFVPST' stands for 'everything else but Imperfective Hesternal/Prehesternal Past'. 5. There are a few restrictions on combinations, but in general, most semantically coherent combinations of markers in different slots are possible. 6. Several non-obligatory predicative markers can occur after Slot 5; they are not presented here.

S	PMs					(DO)	V …
	Slot 1	Slot 2	Slot 3	Slot 4	Slot 5		
	ḭ̀/m(ḭ̀) '1SG'	-∅ 'NEUTR'	-∅ 'NCOND.AFF'	-∅ 'NPST'	-∅ 'NIPFVPST'		
	ù/w(ì)/wù '1PL'	*-lí/-lè* 'FOC'	*-kè/-à* 'IND.NEG'	-$^{\uparrow}$ 'PST'	*-è* 'IPFV.HEST'		
	àù/à(ù)w '1PL_INCL.SBJV'	*-lè(è)* 'PRSV'	*-kȅ˵* 'COND.AFF'		*-é* 'IPFV.PREH'		
	ɛ̀/n(ḭ̀) '2SG'		*-kě* 'COND.NEG'				
	bȅ '2SG.SBJV'		*-a̋/-ǎ* 'SBJV.NEG'				
	∅ '2SG.IMP'		***-a̋/-ǎ*** '**APPR**/ENIM'				
	bë '2SG.IMP_POL'						
	à '2PL'						
	ɛ̏˵/˵/∅/y(ȅ) '3SG'						
	ɔ̏˵/˵/∅/w(ȍ) '3PL'						

Table 5: The four finite verb forms

Form	Example	and gloss
Unmarked (zero) forms	*gò*	‘buy[...]’[a]
Imperfective	*sa̋* ***gő***	‘house **IPFV**\buy’
Perfective Hesternal Past	*g**ȍ***	‘buy**PFV.HEST**’
Perfective Pre-hesternal Past	*g**ö***	‘buy**PFV.PREH**’

[a]The three dots stand for a number of different glosses for different zero forms, see Footnote 6.

As for the verb forms, the (finite) verb can have four different tonal forms, see Table 5. First, the verb can retain its lexical tone (i.e. no tonal changes occur). Such unmarked forms express the Affirmative Subjunctive, the Imperative, the Polite Imperative, and the Perfective Hodiernal Past (and also the non-finite form of the Infinitive).[6] Second, the verb can become a tonal enclitic, copying onto itself the tone of the last syllable of the preceding word (such as the direct object *sa̋* ‘house’ in the example below). This form expresses the Imperfective; it is also used in all Non-past and Negative Subjunctive constructions. Third and fourth, there are two forms where the verb acquires a fixed grammatical tone. The extra-low tone expresses the Perfective Hesternal Past, and the rising tone the Perfective Pre-hesternal Past.

When combined, the paradigms of bound predicative markers described above, together with different forms of the verb (and sometimes with some other additional elements) form Gban’s seven sentential grammatical categories: person-number, informational status, mood, polarity, tense, temporal distance, and aspect. Examples (6) and (7) illustrate combinations of members of these categories.

(6) ***Ɛ̏-lè-kɛ̋-↑-è*** *ä ̏* *tä* *sɔ̰̏.*
3SG-FOC-IND.NEG-PST-IPFV.HEST he/she/it IPFV\sow today
‘It’s him who wasn’t going to sow it today.’
(3rd-p. sg. + Subject focus + Indicative + Negative + Past + Hesternal + Imperfective)

[6]In this paper I follow the convention of glossing these zero verb forms differently, according to the construction they occur in, i.e. ‘[SBJV]’, ‘[IMP]’, ‘[IMP_POL]’, ‘[PFV.HOD]’, and ‘[INF]’.

(7) *Sɔ̀kŭ ɛ́-Ø-Ø-ꜛ-Ø*[7] *yì zɛ̀ tűɛ̋*
Soku 3SG-NEUTR-NCOND.AFF-PST-NIPFVPST sleep kill[PFV.HOD] at_night
sàkàà.
well
'Soku slept (lit. "killed sleep") well this night.'
(3rd-p. sg. + Neutral informational status + Indicative + Affirmative + Past + Hodiernal + Perfective)

Returning now to the Apprehensional construction, we can see that the predicative marker *-ā́/-ă* 'APPR' puts it in contrast with several other sentential constructions that cumulatively express different mood and polarity meanings. These constructions and their formal makeup are presented in Table 6. Examples of Negative Indicative (6) and Affirmative Indicative (7) were already given above.

In this paradigmatic set of constructions, the choice of one or another is obligatory. Even in Slot 3, the zero marker can be attributed a clear-cut, albeit idiosyncratic, semantics: it simultaneously expresses Affirmative and either Indicative, Subjunctive, Imperative, or Polite Imperative.[8]

Given that the Apprehensional construction is included in the same set, it is coherent to regard the Apprehensional as a special mood in Gban. As for its relation to polarity, at this point I will confine myself to stating that the Apprehensional construction cannot be negated/does not have a different-polarity version in any of its uses; for more details, see §7.

It is also justifiable to assign the Apprehensional construction the status of one of Gban's core grammatical constructions (see Footnote 5), since it belongs to a closed set of obligatorily-chosen constructions constituting the grammatical categories of mood and polarity.

[7] In this example, the null predicative markers are shown and glossed for illustrative purposes, but henceforth in the paper they will not be.

[8] I.e. Slot 3 has a zero marker only in Affirmative Indicative, Affirmative Subjunctive, (Affirmative) Imperative, and (Affirmative) Polite Imperative constructions, while having different (non-zero) markers in all Conditional, Enimitive, and Apprehensional constructions and in all Negative constructions. The choice of the gloss 'NCOND.AFF' for this zero marker, denoting "Affirmative Non-Conditional", is a simplification.

Table 6: The Mood and Polarity constructions. *Notes:* 1. In the constructions represented with "V (different forms)", the verb can have many forms according to other grammatical categories. 2. In contrast to them, those constructions for which a specific verbal form is indicated, are characterized by this form as their constant feature. 3. The dots ("…") before the verb stand for the direct object NP, if present, and/or for other minor units that can occur in this part of the clause.

Meaning	Form
Affirmative Indicative	-∅ 'NCOND.AFF' $^{\text{(Slot 3)}}$ … + V (different forms)
Negative Indicative	*-kè/-à* 'IND.NEG' $^{\text{(Slot 3)}}$ … + V (different forms)
Affirmative Conditional	*-kë̏̏* 'COND.AFF' $^{\text{(Slot 3)}}$ … + V (different forms)
Negative Conditional	*-kě* 'COND.NEG' $^{\text{(Slot 3)}}$ … + V (different forms)
Affirmative Subjunctive	-∅ 'NCOND.AFF' $^{\text{(Slot 3)}}$ … + V $^{\text{(lexical tone)}}$ 'SBJV'
Negative Subjunctive	*-ấ/-ǎ* 'SBJV.NEG' $^{\text{(Slot 3)}}$ … + V $^{\text{(tonal enclitic)}}$ 'IPFV'
Imperative	∅ '2SG.IMP' $^{\text{(Slot 1)}}$ + -∅ 'NCOND.AFF' $^{\text{(Slot 3)}}$ … + V $^{\text{(lexical tone)}}$ 'IMP'
Polite imperative	*bë* '2SG.IMP_POL' $^{\text{(Slot 1)}}$ + -∅ 'NCOND.AFF' $^{\text{(Slot 3)}}$ … + V $^{\text{(lexical tone)}}$ 'IMP_POL'
Enimitive	*-ấ/-ǎ* 'APPR/ENIM' $^{\text{(Slot 3)}}$ … + V (different forms) + final particle *nḭ̀* 'ENIM'
Apprehensional	*-ấ/-ǎ* 'APPR/ENIM' $^{\text{(Slot 3)}}$ + auxiliary verb *nṵ̀* 'APPR' … + V $^{\text{(lexical tone)}}$ 'INF'

4 Semantics and syntax: Usage in main clauses

The Apprehensional construction can be used in main clauses as well as in dependent clauses.

In this section, I will describe the readings/functions available in main clauses: expression of pure apprehension (§4.1), warnings (§4.2), or threats (§4.3). These are illustrated in (1), (8) and (9) respectively.

(8) Warning
N-a̰̋ ***nṵ̀*** *ké* *sı̏ȅ* *ɛ̀* *í* *tɔ̏=ɛ̋.*
2SG-**APPR** **APPR** EMPH spoil[INF] you_SG GEN cloth=with
[Mother to a child who is playing with a knife, about to spoil his/her clothes:] '— **Don't** spoil your clothes! / You **might** spoil your clothes!'

(9) Threat
M-a̰̋ ***nṵ̀*** *ké* *ɛ̀* *zɛ̀!*
1SG-**APPR** **APPR** EMPH you_SG kill[INF]
[In the street, a robber who wants to rob a person:] '— I **might** kill you / I'**ll** kill you!'

One of the characteristics dividing these readings is that the first one (pure apprehension) presumes undesirability for the speaker and the other two (warning and threat) undesirability for the addressee.[9] More detailed definitions will be given in the respective subsections. It is also important to note that these readings, although they are treated separately here, are not always easily distinguishable and probably even not mutually exclusive.

A passing remark to be made here is that in Gban, main clauses with the Apprehensional construction cannot be used as questions:

[9] I do not assert here that the inverse combinations (a pure-apprehension use with a situation undesirable for the addressee and a warning/threat use with a situation undesirable for the speaker) are logically impossible. However, it seems reasonable to assume at least that in pure-apprehension uses the situation is always undesirable **at least** for the speaker and that in warning/threat uses the situation is always undesirable **at least** for the addressee. In the first case, it seems that even if we could refer (using an apprehensional marker) to an unpreventable situation that is bad mostly for the addressee (cf. [In court, before the verdict is announced:] *Oh dear, you might get a big fine*), we would still express our own "undesiring" of it, which arises from empathy. Similarly, in the second case, even if we uttered a warning about a situation that is bad mostly for us (cf. *Watch out, you might run me over!*), it would be the addressee's empathic "undesiring" of it that we would appeal to.

(10) * *Nɔ̰̌ bȅbèɛ̀ dò y-ǎ nṵ̀ bɔ̀ ȁ nɛ̰̏* ***lìì****?*
thing bad one 3sg-appr appr arrive[inf] he/she/it on q
Intended meaning: 'Might something bad happen to him?'

4.1 Expression of pure apprehension

First, in main clauses, the Apprehensional construction can express *pure apprehension* – i.e. convey just the speaker's emotion of fear, apprehension, or anxiety (or, probably, more neutral undesirability) with respect to some possible situation.[10] It is assumed that such uses do not convey any additional illocutionary component other than expressivity.

In Gban, this pure apprehension reading is available in all three persons (which is not the case in some other languages, where person can affect the interpretation of the apprehensional construction, Vuillermet 2026 [this volume]).

Consider a series of examples with a nervous mother expressing her anxieties: with a human 3rd-person subject (11, 13b), an inanimate 3rd-person subject (12), a 1st-person subject (13a), and a 2nd-person subject (13b).

(11) OK *Ù nḭ̋ bɛ̏ y-****ǎ*** ***nṵ̀*** *gṵ̀à̰.*
we child there 3sg-**appr** **appr** get_lost[inf]
[A boy has gone for a walk in a forest, alone. His parents discuss this, knowing that at the moment they can do nothing to prevent bad things from happening to him. His mother:] '— [I fear that] our son **might** get lost.'

(12) OK *Nɔ̰̌ bȅbèɛ̀ dò y-****ǎ*** ***nṵ̀*** *bɔ̀ ȁ nɛ̰̏!*
thing bad one 3sg-**appr** **appr** arrive[inf] he/she/it on
[A constantly nervous mother, after the departure of her son, is worrying and expresses her fears aloud, talking to herself:] '— Something bad **might** happen to him.'

(13) a. *M-****a̰̋*** ***nṵ̀*** *ȁ gɔ̰̀ wò vɔ̰̏ṵ̏ṵ̏!*
1sg-**appr** **appr** he/she/it stay[inf] put[inf] cold
[In the same context as in (12):] '— [My son might get lost and] I **might** become sorrowful/anxious (lit. "stay cold/sad")!'

[10]This function is basically what Dobrushina calls *aprexensiv* 'apprehensive', contrasting it with the term *preventiv* 'preventive' for the warning function (Dobrushina 2006).

b. *Ɛ̀ nḭ̋ bɛ̏ y-**ǎ** **nṵ̀** gṵ̀à̰. N-**a̰̋** **nṵ̀***
you_SG child there 3SG-**APPR** **APPR** get_lost[INF] 2SG-**APPR** **APPR**
ȁ gɔ̰̀ wò vɔ̰̏ṵ̏ṵ̏!
he/she/it stay[INF] put[INF] cold
[In the same context, but now she speaks to the father:] '— Your son **might** get lost [and] you **might** become sorrowful/anxious!'

Consider also example (14) with an overcautious or squeamish person speaking.

(14) *N-**a̰̋** **nṵ̀** gà dɔ̀ ḭ̀ nɛ̰̀.*
2SG-**APPR** **APPR** illness put[INF] I at
[The addressee is sick. The speaker does not want to touch him/her. (S)he is explaining why (s)he does not offer his/her hand (this is the first utterance in the dialogue):] '— You **might** infect me.' / '— **For** you **not to** infect me.'[11]

All clear examples of pure-apprehension uses in my data are elicited. However, there is one example of such use from the New Testament translation:

(15) [...] *Ɛ̏-kě tȁ nɔ̰́, ȁ mɔ̰́ nṵ̏=ȅ*
3SG-COND.NEG IPFV\be.NEG like_that he/she/it GEN come=LOC2
*nṵ̀a̰=le y-**ǎ** **nṵ̀** ḭ̀ gbí bò.*
constantly?=NMLZ 3SG-**APPR** **APPR** I breast remove[INF]
{... yet because this widow keeps bothering me, I will grant her justice.}
'Otherwise, she will keep coming and wear me out.' (Luke 18:5)
(More precisely: '... Otherwise (lit. "If it's not like that"), her continual coming **might** upset me.')

All the examples presented above feature situations that are inherently undesirable, so it may not yet be obvious that the Apprehensional construction itself expresses undesirability here. However, there are several examples in my data where the situations expressed by the predicates are not inherently undesirable, but are interpreted as undesirable in the context of the Apprehensional construction. This is illustrated in example (16) with 'come', in example (18) with 'be/become cold', and especially in example (75) with 'see my son again'.[12]

[11]This second version of the translation (Fr. 'Pour ne pas me contaminer') hints at a different interpretation, with a detached negative purpose clause (see §5.2), answering to a silent question 'Why aren't you offering me a hand?'.

[12]Examples of neutral situations are also attested in the dependent uses of the Apprehensional construction; compare (42) and (54)–(56) in the following sections.

(16) ^OK^ *M-ą̰̋ nṵ̀ nṵ̀.*
1SG-**APPR** **APPR** come[INF]
'[I **fear** that] I might come [there].'

As for the *temporal* localization of the situation, the Apprehensional construction in pure-apprehension uses is compatible only with future situations, and not with past (17) or present (18) situations. This is similar, for example, to the apprehensive marker in Ese Ejja (Vuillermet 2018: 265), but different from the one in To'abaita, which is temporally-neutral (Lichtenberk 1995: 296).

(17) ^OK^ *M-**ą̰̋*** ***nṵ̀*** *gṵ̀ą̰̀.*
1SG-**APPR** **APPR** get_lost[INF]
1. [Worrying:] '— [I'm afraid that] I might get lost [in the future].'
2. *[Worrying:] '— [I'm afraid that] I might have got lost [already].'

(18) ^OK^ *Y-**ǎ*** ***nṵ̀*** *yè fɔ̰̀ṵ̈fɔ̰̀ṵ̈.*
3SG-**APPR** **APPR** be[INF] cold
1. '[It would be bad for / We don't want] it [e.g. the water] to be/ become cold [in the future].'
2. *'[I'm afraid that] it [e.g. the water] is cold [now, already].'

An interesting characteristic of the Apprehensional construction in pure-apprehension uses is that it seems to be compatible only with situations that are preventable at least in principle.

Preventability[13] of a situation P can be defined as the possibility of a controllable (by the interlocutors) situation Q that would cause non-P. For example, getting lost in a forest (P) is an uncontrollable situation itself, but it is preventable: the person can be or could have been persuaded (Q) not to go there. Preventability can be generic (situations that are preventable in principle, theoretically) or actual (situations that are currently preventable).

In all previous examples, the situations were at least in principle preventable (although in actual circumstances the situation may be already unpreventable). Getting infected (14) can still be prevented by not touching the sick person. The situations of one's son getting lost (11), getting into trouble (12), or making his parents mournful (13) could all have been prevented by not letting the son go, although now it is already too late.

[13] I introduce here the notion of preventability, which is related to control(lability), but should be differentiated from it. A similar opposition "avertible–non-avertible" is mentioned in (Lichtenberk 1995: 311).

But consider now an example about raining, which is in principle unpreventable. In Gban, the Apprehensional construction is infelicitous in such a context:

(19) # *Gwlɛ̰́ y-**ǎ** **nṵ̀** kấ.*
rain 3sg-**appr** **appr** fall[inf]
Intended meaning: [A person comes out of the house in the morning and sees dark clouds in the sky. As today is the day of a feast (s)he is preparing, (s)he does not want rain. (S)he begins to worry:] '— It might rain.'
(But a commentary was made that this could be possible for "mystic people who speak with the rain" and who thus may indeed **prevent** the rain from falling.)

At the same time, this unpreventable situation of raining (in general) can be converted into a very similar but preventable one – by introducing an additional participant (which is under the control of the interlocutors). The Apprehensional construction becomes felicitous in this case:

(20) OK *Gwlɛ̰́ y-**ǎ** **nṵ̀** kấ=lɛ̋ ḭ̀ mɔ̰́ wȍtló nɔ̰́ là.*
rain 3sg-**appr** **appr** fall[inf]=loc2 I for car that on

There is also a similar example in my data about the weather getting cold (which is in principle unpreventable as well). A constructed version of it with the Apprehensional construction was, too, considered infelicitous by the speaker with the commentary "It's as if you prevent the cold from coming".

This behaviour of the Apprehensional construction in Gban is rather peculiar, if we compare it to what is known about apprehensional markers in (some) other languages of the world. For example, the Russian apprehensional construction with *kak by … ne* is perfectly compatible even with situations that are in principle unpreventable (21). The same is reported for Ese Ejja (Vuillermet 2018: 274). Analogous examples are found in To'abaita (22) and in a variety of Tatar (23).

(21) Russian (personal knowledge)
Kak by ***ne*** *pošël dožd'.*
how sbjv neg go.pfv.pst.m.sg rain
'[I'm worried that] It might rain.'

(22) To'abaita (Austronesian; Lichtenberk 1995: 295)
Ada *dani ka 'aru.*
appr rain it.seq fall
'It may rain.'

(23) Mordovian varieties of Mishar Tatar (Turkic; Dobrushina 2006: 33)
*Jaŋɣyr jau-**ma-ɣaj**.*
rain fall-NEG-OPT
'It might rain.'

Consider also the claim by Dobrushina that the apprehensional ("apprehensive") "does not have a (…) restriction on controllability" and "expresses controllable as well as fully uncontrollable situations" (Dobrushina 2006: 57–59).

4.2 Warnings

Another possible use of the Apprehensional construction in main clauses is in *warnings*[14] – i.e. in utterances that inform the addressee of an undesirable situation which the addressee can prevent if (s)he pays attention to it.

One illustration of such a use is example (4) above. Consider also examples (24b), (25b), and (26) below.

It can be seen that in 2nd-person warning contexts, the Negative Subjunctive (which is, among its different uses, the standard negative counterpart to the Imperative) can also be used. In this section, the Negative Subjunctive versions are always given before the Apprehensional versions, in the (a) subexamples.

(24) a. *(Súɛ́ɛ́,) n-ǎ̰ gbɛ̋ɛ̋ ké!*
watch_out 2SG-SBJV.NEG IPFV\fall EMPH

b. OK *N-**ã̰̋** **nṵ̀** ké gbɛ̏ɛ̋!*
2SG-**APPR** **APPR** EMPH fall[INF]
[A warning to a child:] '— (Watch out,) **don't** fall!'
(Fr. '[Attention,] faut pas tomber') / '— (Watch out,) you **might** fall!'

(25) a. *Súɛ́ɛ́, n-ǎ̰ tő=lɛ̏ ké mı̰̏ɛ̰̏ là!*
watch_out 2SG-SBJV.NEG IPFV\leave=LOC2 EMPH snake on

b. *Súɛ́ɛ́, n-**ã̰̋** **nṵ̀** tő=lɛ̋ ké mı̰̏ɛ̰̏ là!*
watch_out 2SG-**APPR** **APPR** leave[INF]=LOC2 EMPH snake on
'(Watch out,) **don't** step on a snake! / (Watch out,) you **might** step on a snake!' (*Súɛ́ɛ́* can also be omitted.)

[14] Dobrushina (2006) uses the term *preventiv* 'preventive' for this warning function. The term "admonitive" (Russ. *admonitiv*) is also sometimes used in the literature (see Vuillermet 2018: 39–40; Dobrushina 2006: 29).

(26) *N-a̰̋* ***nṵ̀*** *ké* *sȉȅ* *ɛ̀* *í* *tɔ̏=ɛ̋.*
2sg-**appr** **appr** emph spoil[inf] you_sg gen cloth=with
[Mother to a child who is playing with a knife, about to spoil his/her clothes:] '— **Don't** spoil your clothes! / You **might** spoil your clothes!'

The distribution of the Apprehensional and the Negative Subjunctive in these uses is not exactly clear, but there are hints that the Negative Subjunctive is more felicitous than the Apprehensional in imminent warnings, while the Apprehensional is rather used in more neutral, less urgent warnings with potential danger. In example (24), for instance, (24a) was considered by the speaker to be more felicitous than (24b) if we are trying to prevent an imminent falling, while for (24b) a more suitable context would be when the child is running or walking in the mud or any other behaviour that only can lead to a falling event. At the same time, the Apprehensional seems infelicitous in more general warnings, without a current direct motivation – see (27)–(28).

(27) a. *N-ǎ̰* *gbɛ̋a̋* *ké* *dı̋=à.*
2sg-sbjv.neg ipfv\fall emph road=on
[A mother is warning her son before sending him to school in the morning:] '— **Don't** fall along the way!'

b. # *N-a̰̋* ***nṵ̀*** *gbȅa̋* *ké* *dı̋=à.*
2sg-**appr** **appr** fall[inf] emph road=on
Intended meaning: 'Don't fall along the way!'

(28) a. *N-ǎ̰* *sı̋ɛ̋* *ké* *ɛ̀* *í* *tɔ̏=ɛ̋.*
2sg-sbjv.neg ipfv\spoil emph you_sg gen cloth=with
[A mother is warning her son before sending him to school in the morning:] '—**Don't** spoil your clothes!'

b. # *N-a̰̋* ***nṵ̀*** *ké* *sȉȅ* *ɛ̀* *í* *tɔ̏=ɛ̋.*
2sg-**appr** **appr** emph spoil[inf] you_sg gen cloth=with
Intended meaning: 'Don't spoil your clothes!'
(But example (28b) was said to be felicitous as a more direct warning, see (26) above.)

Apart from 2nd-person uses (that can be viewed as canonical for warnings), Gban also allows the Apprehensional construction with 1st- and 3rd-person subjects in warnings. An example with a 1st-person subject:

(29) OK *M-ạ̋* ***nụ̀*** *vı̏* *dı̋* *ɛ̀* *là!*
1SG-**APPR** **APPR** throw[INF] down you_SG on
[A person climbing down from a tree:] '— [Watch out there,] I **might** fall down on you!'

And a version of example (4) with a 3rd-person subject:

(30) OK *Ɛ̀* *kwà [y]-**ǎ*** ***nụ̀*** *ylɛ̏!*
you_SG arm 3SG-**APPR** **APPR** break[INF]
'[Be careful,] don't break your arms!' (lit. "Your arms **might** break!")

Consider also a 3rd-person example (36) from the next section, in which the threat reading was found implausible, but which is felicitous with the reading of a friendly warning or advice.

It follows from the definition of a warning given above that, from a theoretical point of view, warnings (as well as threats, see §4.3) must introduce situations that are currently preventable, unlike pure-apprehension contexts. For example, if I am trying to warn a person that (s)he might fall (*Watch out, you might fall!*), this would only make sense if the fall is still preventable. However, it should be noted that the event *explicitly* referred to in a warning utterance needs not itself be preventable, but may just imply the possibility of some undesirable consequence which, in turn, can be prevented[15] (compare 'in-case' contexts, discussed in the end of §5.2). For example, we could utter *Hey, it might rain today* to a person leaving the house, meaning it as a warning, for the person to take an umbrella. In this case, the warning still semantically introduces a currently preventable situation (of getting wet), but does so implicitly (as a possible consequence of raining).

Returning to Gban, here such "indirect" warnings that explicitly introduce an unpreventable situation are infelicitous. Consider example (31).

(31) # *Yı̋* *y-**ǎ*** ***nụ̀*** *yèè* *fɔ̰̏ṵ̏fɔ̰̏ṵ̏.*
water 3SG-**APPR** **APPR** be[INF] cold
Intended meaning: [A person wants to swim in a lake tomorrow, but we know that the forecast is bad and we warn him/her:] '— The water might be [too] cold [tomorrow].'
(But the example was said to be felicitous in the case of some small, controllable amount of water: '[Don't throw ice into water,] it might be/become too cold.')

[15] I am grateful to the editors of the volume for bringing this to my attention.

This situation of water in a lake being cold at the time of swimming is unpreventable (regardless of the fact that it implies a possible preventable situation: a possibility for a swimmer to get cold). However, if we rather speak, for instance, about some water in a glass, this situation is preventable, and the warning becomes felicitous.

4.3 Threats

The Apprehensional construction can also be used in *threats*, i.e. in utterances that express a currently preventable situation undesirable for the addressee that might be brought about (directly or indirectly) by the speaker, if the addressee does not carry out what (s)he is asked to.

There are clear examples of threat usage for the 1st and 2nd persons:

(32) OK (*N-ă̰ gbı̋ɛ̋=ɛ̈ ḭ̀ nɛ̰̀!*)
2SG-SBJV.NEG IPFV\drag=LOC1 I at
M-ă̰̋ ***nṵ̀*** *ɛ̀ zɛ̀ gbɛ̰̏ yɛ̋!*
1SG-APPR APPR you_SG kill[INF] knife with
[On encountering a robber in the street, a person pulls out a knife and says:] '— (Don't come any closer!) **I'll** stab you (sg.)!'

(33) a. ***M-ă̰̋ nṵ̀*** *ké ɛ̀ zɛ̀!*
1SG-**APPR APPR** EMPH you_SG kill[INF]

b. ***W-ă̋ nṵ̀*** *ké ɛ̀ zɛ̀!*
1PL-**APPR APPR** EMPH you_SG kill[INF]
[In the street, a robber or several robbers who want to rob a person:]
'**I might** kill you / **I'll** kill you!'[16] / '**We might** kill you / **We'll** kill you!'

(34) ***N-ă̰̋ nṵ̀*** *gà! (Bɛ̈ ḭ̀ gbà̰ lắlà yɛ̋!)*
2SG-**APPR APPR** die[INF] 2SG.SBJV I give[SBJV] money with
[A robber:] '— **You (sg.) might** die! (Give me the money!)'

(35) ***Ắ nṵ̀*** *ké gà gȍgȍ!*
2PL:**APPR APPR** EMPH die[INF] empty
[A robber:] '— **You (pl.)'re gonna/might** die in vain!' [Give me that money!]

[16] My consultant commented that this threat sounded "advice-like".

In the 3rd person, the threat interpretation of the Apprehensional construction seems to become more problematic, although it is still available.

For example, in (36), the use of the Apprehensional construction was considered felicitous not with a threat reading, but rather with an advice or friendly warning reading.

(36) # *N-ǎ̰ tɛ̋=kpɛ̏ ɔ̏ nɛ̰̏, w-**ǎ** **nṵ̀** ɛ̀*
2sg-sbjv.neg ipfv\approach=loc1 they on **3pl-appr appr** you_sg
kɪ̋ɔ̏ʊ̋.
bite[inf]
Intended meaning: [On encountering a robber in the street, a person with dogs threatens:] '— Don't come closer! They'll bite you!'
(The following comment was given for the example: "It can't be a threat; it's like advice, as if we help the robber".)

However, in example (37), the threat interpretation was considered possible.

(37) *Y-**ǎ** **nṵ̀** gà! (Bɛ̏ ḭ̀ gbà̰ la̋là̏ yɛ̋!)*
3sg-**appr appr** die[inf] 2sg.sbjv I give[sbjv] money with
[A robber has caught the wife of the man he wants to rob and threatens him:] '— **She might** die! (Give me the money!)'

Another observation on threat uses is that, in general (for all persons), the threat interpretation seems to be most felicitous when the Apprehensional construction is accompanied by a preceding imperative/prohibitive clause, as in (32). However, the problem for the analysis of such examples is that in this case they become indistinguishable from syntactically dependent negative-purpose uses without an overt subordinator (see §5.2).

5 Semantics and syntax: Dependent uses

In this section, I will describe the two attested dependent uses of the Apprehensional construction, i.e. uses in complements of fear predicates (and probably in negative irrealis complement clauses, too) (§5.1) and in negative purpose clauses (but not in 'in-case' contexts) (§5.2).

When the Apprehensional construction is used in dependent clauses, it is combined with certain markers of subordination. These additional subordinators will also be discussed, in the respective subsections.

5.1 Fear predicate complement

The Apprehensional construction can be used in the complement clauses of fear predicates such as *gbɛ̰̏* ‘be afraid’, where it simply introduces the mental content associated with the fear predicate; see examples (38)–(44) and (49)–(50) below.

Here the Apprehensional competes with the Negative Subjunctive (38b), (40b) and with the Indicative (40c), (43) with no obvious difference in semantics, although with some restrictions (see below).

The complement clauses are introduced by the complementizer *nḭ́* ‘that’ (in some cases considered optional, see (49)). This marker *nḭ́* (of uncertain etymology) is the default complementizer in Gban, used with verbs of speech and thought and in irrealis clauses with matrix predicates such as ‘make (somebody do something)’, ‘prevent’, ‘have to’, ‘need’, and ‘want’. It is also used as a subordinator introducing purpose clauses (see §5.2). The same *nḭ́* can be used as a quotative marker to introduce direct speech.

(38) a. *Ḭ̀ **gbɛ̰̀*** [***nḭ́*** *ù nḭ̋ bɛ̏ y-**ǎ** **nṵ̀** gṵ̀à̰*].
1SG IPFV\be_afraid **that** we child there 3SG-**APPR** **APPR** get_lost[INF]

b. *Ḭ̀ **gbɛ̰̀*** [***nḭ́*** *ù nḭ̋ bɛ̏ y-ǎ gṵ̋a̰̋*].
1SG IPFV\be_afraid **that** we child there 3SG-SBJV.NEG IPFV\get_lost
[A boy has gone for a walk in a forest, alone. His parents discuss this, although at the moment they can do nothing to prevent anything that can happen to him. His mother:] ‘— I **fear that** our son **might** get lost.’

(39) [...] *wö̀-é **gbɛ̰́*** [***nḭ*** *y-**ǎ** **nṵ̀** bato*
3PL-IPFV.PREH IPFV\be_afraid **that** 3SG-**APPR** **APPR** ship
dɛ̰̀-ka=le Siite yɛ̰-tu kpaa=nṵ][...]
stop-CAUS[INF]=LOC2 Syrtis sand-heap big=PL
‘Because they **were afraid** they **would** run aground on the sandbars of Syrtis,’ {they lowered the sea anchor and let the ship be driven along}. (Acts 27:17)
(More precisely: ‘They **feared that** it [= the wind] **might** make the ship run aground at the large sandbank of Syrtis [...]’)

(40) a. *Ɛ̏ **gbɛ̰̏*** [***nḭ́*** *Flà̰sóá (y)-**ǎ** **nṵ̀** gà*].
3SG IPFV\be_afraid **that** François 3SG-**APPR** **APPR** die[INF]

b. OK *Ɛ̏ **gbɛ̰̏*** [***nḭ́*** *Flà̰sóá y-ǎ gắ*].
3SG IPFV\be_afraid **that** François 3SG-SBJV.NEG IPFV\die

c. $^{\text{OK}}$ *Ɛ̏* ***gbɛ̰̏*** **[*nḭ́*** *Flə̰̏sóá* *̏* *sɛ̏* *ɛ̏* *gȁ*].
3SG IPFV\be_afraid **that** François 3SG IPFV\can 3SG IPFV\die
'She's **afraid that** François **might** die.'
(A commentary was made that the Apprehensional version (40a) conveys here a preventive meaning, for example if François wants to go to a place where there is a mortal danger.)

The dependent status of these clauses is at least partially confirmed by a) the possible occurrence of a clause-final subordinator *lɛ̌* (see below) and b) the observed absence of deictic shift, as in (41). The absence of deictic shift manifests itself in that pronominal elements in the clause are chosen according to the primary speaker's perspective, as in traditional indirect speech/thought: *You're afraid lest **you** might fall ill* (which, in case of clauses expressing mental content, is associated with subordination and not with main clause status). They are not shifted according to the internal perspective of the participant, as they would be in direct speech/thought, i.e. in a quote: *You're afraid [saying/thinking to yourself]: "**I**/#You might fall ill"*.

(41) $^{\text{OK}}$ ***À*** ***gbɛ̰̀*** **[*nḭ́*** ***à*** *bò* *y-***ǎ*** ***nṵ̀*** *kpǒ*
2PL IPFV\be_afraid **that you_PL** head 3SG-**APPR APPR** break[INF]
(***lɛ̌***)].
because
'**You (pl.)** are afraid that **you (pl.)** might fall ill (lit. '… that your (pl.) head might break').'

Interestingly, in several of the translated examples, the fear predicate complement clause was marked not only with the subordinator *nḭ́* 'that', but also, optionally, with the dedicated clause-final *reason/purpose marker lɛ̌* 'because, in order to'; see examples (41)–(42). In these cases, the marking was thus identical to the one used in negative purpose clauses (see §5.2). Moreover, such marking seems to be possible even in fear complement clauses that do not contain the Apprehensional construction, see example (43) with the Indicative.

(42) *Ḭ̀* ***gbɛ̰̀*** **[*nḭ́*** *w-***ǎ*** ***nṵ̀*** *à* *vȍ* *bò* *bɛ̋*
1SG IPFV\be_afraid **that** 3PL-**APPR APPR** he/she/it already head take[INF]
^{OK}(***lɛ̌***)].
because
[A person who is late:] '— I'm afraid they might already start [without me].'

(43) *Ḭ̀* ***gbɛ̰̀*** *nɔ̰̀* [***nḭ́*** *yɛ̏ɛ̏* *fɔ̰̏ṵ̏fɔ̰̏ṵ̏* *OK*(***lɛ̌***)].
1SG IPFV\be_afraid here **that** 3SG:IPFV\be cold **because**
'I'm afraid it [water from the fridge] might be too cold.'

Apart from *gbɛ̰̏* 'be afraid', similar complement-clause uses have been attested with some other, synonymous, predicates, such as 'worry':

(44) *Ḭ̀* ***gbí*** *̏* ***vṵ̏*** [*nḭ́* *y-**ǎ*** ***nṵ̀*** *gṵ̀à̰*].
I breast 3SG IPFV\stir that 3SG-**APPR** **APPR** get_lost[INF]
'I **worry/fear** (lit. "my breast is stirring") that he might get lost.'

As for the *temporal* localization of the situation, the Apprehensional construction in a fear predicate complement is compatible only with future/posterior readings (illustrated by any of the examples above). For past/anterior and present/simultaneous situations, other constructions must be used, like the Indicative:

(45) a. *Ḭ̀* *gbɛ̰̀* [*nḭ́* *ɛ́* *yȁ* *bàà*
1SG IPFV\be_afraid that 3SG\PST go_away\PFV.HEST village
wíɛ̏].
yesterday
'I'm afraid that he might have gone to town **yesterday**.'

b. * *Ḭ̀* *gbɛ̰̀* [*nḭ́* *y-**ǎ*** ***nṵ̀*** *yà* /
1SG IPFV\be_afraid that 3SG-**APPR** **APPR** go_away[INF]
yȁ *bàà* ***wíɛ̏***].
go_away\PFV.HEST village yesterday
Intended meaning: 'I'm afraid that he might have gone to town yesterday.'

(46) a. *Ḭ̀* *gbɛ̰̀* *nɔ̰̀* [*nḭ́* *ɪ̰́* *gṵ̀à̰*].
1SG IPFV\be_afraid here that 1SG\PST get_lost[PFV.HOD]
'I'm afraid that I might have got lost [already].'

b. OK *Ḭ̀* *gbɛ̰̀* *nɔ̰̀* [*nḭ́* *m-**ǎ̰*** ***nṵ̀*** *gṵ̀à̰*].
1SG IPFV\be_afraid here that 1SG-**APPR** **APPR** get_lost[INF]
1. 'I'm afraid that I might get lost [at a later time].'
2. *'I'm afraid that I might have got lost [already].'

(47) a. OK *Ḭ̀* *gbɛ̰̀* *nɔ̰̀* [*nḭ́* *yɛ̏ɛ̏* *fɔ̰̏ṵ̏fɔ̰̏ṵ̏*].
1SG IPFV\be_afraid here that 3SG:IPFV\be cold
'I'm afraid that it [the water from the fridge] might be cold [now, already].'

b. OK *Ḭ̀ gbɛ̰̀ nɔ̰̀ [nḭ́ y-**ǎ** **nṵ̀** yè fɔ̰̏ṵ̏fɔ̰̏ṵ̏].*
1sg ipfv\be_afraid here that 3sg-**appr** **appr** be[inf] cold
1. 'I'm afraid that it [the water from the fridge] might be cold [at a later time].'
2. *'I'm afraid that it [the water from the fridge] might be cold [now].'

Just like the pure-apprehension uses (§4.1), fear predicate complement uses seem (unexpectedly) not compatible with *unpreventable* situations. Example (48) is infelicitous because the raining event is not preventable in principle.

(48) # *Ḭ̀ gbɛ̰̀ [(nḭ́) gwlɛ̰́ y-**ǎ** **nṵ̀** kắ].*
1sg ipfv\be_afraid that rain 3sg-**appr** **appr** fall[inf]
Intended meaning: [A person comes out of the house in the morning and sees dark clouds in the sky. As today is the day of a festival (s)he is preparing, (s)he does not want rain. (S)he says:] '— I'm afraid it might rain.'

In this case, a different construction is used, the (Affirmative) Indicative:

(49) *Ḭ̀ gbɛ̰̀ [(nḭ́) gwlɛ̰́ ̏ kȁ nḭ̋nɛ̰́].*
1sg ipfv\be_afraid that rain 3sg ifpv\fall just_now
'I'm afraid it might rain.'

If the situation of raining is converted into a preventable one by introducing a controlled participant (50), the Apprehensional construction becomes felicitous again.

(50) OK *Ḭ̀ gbɛ̰̀ [(nḭ́) gwlɛ̰́ y-**ǎ** **nṵ̀** kắ=lɛ̋ ḭ̀ mɔ̰́*
1sg ipfv\be_afraid that rain 3sg-**appr** **appr** fall[inf=loc2] I for
wȍtló nɔ̰́ là].
car that on
[Oh, I left my car outside in the open air.] 'I'm afraid it might rain on my car.' [I need to put it away.]

Consider also example (40a) with the situation of dying, where the Apprehensional construction was said to be felicitous in a case where dying is preventable (by not going to a dangerous place).

This restriction, again, makes the Apprehensional construction in Gban different from, for instance, the Russian construction with *kak by … ne* or the English *lest*-construction:

(51) Russian (personal knowledge)
On pereživaet, **kak by ne** *pošël dožd'.*
he worry.IPFV.NPST.3SG **how SBJV NEG** go.PFV.PST.M.SG rain
'He's worried that/lest it might rain.'

(52) American English
I would scan the heavens anxiously on the morning of the great event, fearful **lest** *it might rain, in which case the picnic would be called off.*
(P. G. Bernard (comp.). *Years Gone By* (1967), p. 73)

The Apprehensional construction seems to be also used in complementation of some predicates other than 'fear'. Let us consider example (53) with the predicate *lè (w)ɔ̰́* 'be good'.

(53) *Ȁ nɛ̰̀ ɛ̏* **lȅ ɔ̰** *à mɔ̰, ɛ̏-li yɔ̰̏* **nḭ́** *mṵ*
he/she/it which 3SG IPFV\be_good you_PL for 3SG-FOC there that person
doo gà mṵ fofo gɔ vɛ̰ɛ̰, ku lɛ̏ gbḭ vi do mɔ̰
only die[SBJV] person all stomach? after so then tribe whole one for
*mṵ fofo bo y-***ǎ** **nṵ̀** *vò yɛ̰̀.*
person all head 3SG-**APPR APPR** finish[INF] do_all[INF]
'[I]t is better for you that one man die for the people than that the whole nation perish.' (John 11:50)
(More precisely: 'What **is good** for you is **that** one man would die for all people and [that] so all people of a whole nation **would not** perish.')[17]

Since such main predicates (there are also two similar, albeit less reliable, examples in my data with the predicate *kɛ̀* 'say/want') do not express undesirability nor fear, such examples seem to exhibit a different phenomenon. It appears that the Apprehensional construction has expanded its use to some extent onto the more general function of marking negative[18] irrealis complement clauses, in this case with a predicate of desire/positive evaluation (the normal option for this function in Gban being the Negative Subjunctive). Unfortunately, the data are too scarce here yet to make any solid claims.

[17] In this example, there are two conjoined subordinate clauses, sharing a single complementizer *nḭ́*, both relating to *lè (w)ɔ̰́* 'be good' (in a cleft construction). Cf. the French translation (from *La Bible en français courant* (Société biblique française 1997)) of this passage, which is closer to Gban: "Ne saisissez-vous pas qu'**il est de votre intérêt qu**'un seul homme meure pour le peuple **et qu**'ainsi la nation entière **ne soit pas** détruite?".

[18] For the ability of the Apprehensional construction to express negation of the dependent proposition, see §7.

5.2 Negative purpose

Finally, the Apprehensional construction is used in Gban as the *principal* means of expressing *negative purpose* in subordinate clauses.[19] By default, it is accompanied in such clauses by an explicit subordination marker: the clause-final reason/purpose marker *lɛ̌* 'because, in order to'[20] as in (54) and, in addition, also sometimes the clause-initial subordinator *nḭ́* 'that', see (55a) and (56a), or the clause-initial subordinator *nɛ̰̀* 'which; when; because', see (57). However, negative purpose clauses without an explicit subordinator are also possible if they follow the main clause, see (55b),[21] but not if they precede it, see (56b).

(54) *Ma̰̋-à̰ nṵ̀ yà wòtló yɛ̋, [n-a̰̋ **nṵ̀** nṵ̀ **lɛ̌**].*
1SG-IND.NEG IPFV\come go_away[INF] car with 2SG-**APPR** **APPR** come[INF] **because**
'I won't go in a car **in order** for you **not to** come.'

(55) a. *Ɛ́ la̋là nɔ̰̀ ḭ̀ nɛ̰̀ [**nḭ́** m-a̰̋ **nṵ̀** bȅa̋ (kȅkpa̋) wò (^OK^**lɛ̌**)].*
3SG\PST money give[PFV.HOD] I at **that** 1SG-**APPR** **APPR** work another put[INF] **because**

b. ^OK^ *Ɛ́ la̋là nɔ̰̀ ḭ̀ nɛ̰̀ [m-a̰̋ **nṵ̀** bȅa̋ kȅkpa̋ wò].*
3SG\PST money give[PFV.HOD] I at 1SG-**APPR** **APPR** work another put[INF]
'He gave me money, **so that** I would**n't** work (anymore).'

[19] It is systematically used to translate stimuli with negated purpose clauses. The Negative Subjunctive has sometimes been tolerated in such contexts, too, but was considered less felicitous.

[20] This marker's central use is to mark dependent reason clauses (usually together with the clause-initial subordinator *nɛ̰̀* 'which; when; because'):

(i) *Wö-é ɔ̋ mɔ̰́ sɔwi wḭ̋ yḭ̋=o [**nɛ̰̀** zɔ̰kɛɛ=nṵ Ø-lí-é nḭ́ ɔ yɛ **lɛ̌**].*
3PL-IPFV.PREH they for net IPFV\throw water=in **when** fisher=PL 3PL-FOC-IPFV.PREH IPFV\be_that they with **because**
'They were casting a net into the sea, **because** they were fishermen.' (Matt. 4:18)

[21] Such unmarked dependent clauses are often hard to distinguish from main clauses expressing pure apprehension, warning, or threat. Because of this, in this section I will mostly consider examples with subordinators, which are unambiguous.

(56) a. [OK] [*(**Nḭ́**) m-a̰̋* ***nṵ̀*** *bȅa̋ kȅkpa̋ wò* ***lɛ̌***], *ɛ́*
that 1SG-**APPR** **APPR** work another put[INF] **because** 3SG\PST
la̋lȁ nɔ̰̀ ḭ̀ nɛ̰̀.
money give[PFV.HOD] I at
'**In order that** I would**n't** work anymore, that's why he gave me money.'

b. * [*M-a̰̋* ***nṵ̀*** *bȅa̋ kȅkpa̋ wò*], *ɛ́ la̋lȁ*
1SG-**APPR** **APPR** work another put[INF] 3SG\PST money
nɔ̰̀ ḭ̀ nɛ̰̀.
give[PFV.HOD] I at
Intended meaning: 'In order that I would**n't** work anymore, that's why he gave me money.'

(57) [*Kɛ̋* ***nɛ̰̀*** *w-**ǎ*** ***nṵ̀*** *ù lɛ̏ zè* ***lɛ̌***], [*w-**ǎ*** ***nṵ̀***
but **when** 3PL-**APPR** **APPR** we fate say[INF] **because** 3PL-**APPR** **APPR**
yě ka̋ ú nɛ̰̀], *ȁ lì lɛ̌ wű ù blɛ̋ ù*
laughter put[INF] we at he/she/it FOC_OBJ because 1PL\PST we self we
gbí kṵ̏, w a̋ bȅ.
breast catch\PFV.HEST 1PL\PST he/she/it eat\PFV.HEST
'But **so that** they would**n't** criticize us and [**so that**] they would**n't** mock us, that's why we forced ourselves and ate it.' (Narrative text)

It is important that in these uses the Apprehensional construction can be shown to indeed express negative purpose and not just 'or else …/(because) otherwise …'. In other words, they can be shown to convey a direct *negation* of the proposition expressed by the dependent clause. This issue will be discussed and argued for separately in §7.

In addition, as can be seen from the examples, the (non-negated) situation expressed in such uses is not necessarily feared – it is just undesirable.

The *syntactically dependent* status of such clauses can be demonstrated a) by the possible usage of an explicit subordinator (55a), b) by the observed absence of deictic shift (the pronominal elements are chosen according to the primary speaker's perspective, see example (55)), and c) by the possibility of their internal (central) position in the main clause. In example (58), the negative purpose clause is inside an indirect speech clause, itself embedded in an interrogative utterance with the clause-final question particle *lìì*; the negative purpose clause thus occupies a central linear position in the main clause.

(58) *Y ä zè [nḭ́ ɛ̏-kè nṵ̀ ȁ*
3sg\pst he/she/it say[pfv.hod] that 3sg-ind.neg ipfv\come he/she/it
*zè ḭ̀ nɛ̰̀, [m-a̰̋ **nṵ̀** nṵ̀ **lɛ̌**]] **lìì?***
say[inf] I at 1sg-**appr** **appr** come[inf] **because** **q**
'Did he say that he wouldn't inform me **so that** I would**n't** come?

In (59), the negative purpose clause is placed after a topic constituent related to the main clause.

(59) a. [OK] *(Sɔ̰̏,) Sɔ̀kȕ bɛ̏, [**nḭ́** m-a̰̋ **nṵ̀** wò **lɛ̌**], ɛ́*
today Soku there **that** 1sg-**appr** **appr** cry[inf] **because** 3sg\pst
yě kű nṵ̀-kȁ ḭ̀ nɛ̰̀.
laughter put[inf] come-caus[pfv.hod] I at

b. [OK] *Sɔ̀kȕ bɛ̏, [**nɛ̰̀** m-a̰̋ **nṵ̀** wò **lɛ̌**], ɛ́*
Soku there **when** 1sg-**appr** **appr** cry[inf] **because** 3sg\pst
yě kű nṵ̀-kȁ ḭ̀ nɛ̰̀.
laughter put[inf] come-caus[pfv.hod] I at
'(Today) Soku, **so that** I did**n't** cry, made me laugh.'

The dependent status of these clauses is also reflected in their ability to be used in preposition to the main clause while containing a cataphoric pronoun:

(60) *[**Nɛ̰̀** / [OK] **Nḭ́** bísȉ y-**ǎ** **nṵ̀** **ȁ** tő **lɛ̌**], **Sɔ̀kȕ***
when **that** bus 3sg-**appr** **appr** he/she/it leave[inf] **because** Soku
ɛ́ vȍ sȍ.
3sg\pst early pass[pfv.hod]
'So that **he**$_i$ didn't miss the bus, **Soku**$_i$ left [home] early.'

As it is expected for negative-purpose (precautioning) uses in contrast to main-clause (apprehensive) uses, in Gban the negative purpose clauses allow situations that are clearly undesirable only for the subject referent of the main clause, but not for the speaker, as in (55)–(56), (58), and also (61) with a generic subject.

(61) *Ɛ̏ yȅȅ / W ȁ kȍ bòò [**nḭ́** mṵ̋ mḭ̀ y-**ǎ***
3sg ipfv\be 3pl he/she/it ipfv\cover top **that** person body 3sg-**appr**
***nṵ̀** yɛ̀kɛ̋ vɔ̰̏ṵ̏vɔ̰̏ṵ̏ **lɛ̌**].*
appr do[inf] wet **because**
{—Umbrella, what is it for?} '— It is (/ One covers oneself with it) **for not** having one's body wet.'

The semantically affirmative counterpart to this use of the Apprehensional construction is a dedicated *Purpose construction* with the predicative marker *wò* 'purp' and an unmarked form of the main verb, which combines with the same set of subordinators and may have different linear positions. See examples (62), (63), and (64). This construction is used only in affirmative purpose clauses and cannot be combined with a negative marker.

(62) *Ɛ̏ kṵ̋ wő tȅnṵ̋ɛ̰̋ sésé là* [***nḭ́*** *ɔ̏* ***wò*** *lɛ̌*
3sg catch[inf] ipfv\put child:pl small on **that** 3pl **purp** matter
dɔ̀].
know[sbjv?][22]
'He keeps an eye on the kids **so that** they behave.'

(63) [***Nɛ̰̀*** *ḭ̀* ***w*** *à vòtè* ***lɛ̌***], *ɛ́ kɛ̏ nḭ́*
when 1sg **purp** he/she/it vote[sbjv?] **because** 3sg\pst say\pfv.hest that
ɛ̏ wȍ lắlȁ nɔ̰̀ ḭ̀ nɛ̰̀.
3sg pstr money ipfv\give I at
'He offered me money **so that** I would vote for him.'

The negative-purpose use is arguably the most frequent use of the Apprehensional construction in Gban. At least, in the text of the New Testament, the absolute majority of examples with the Apprehensional construction – around 80% of all (149) tokens – are clear examples of negative-purpose uses.[23]

Consider the following example from the New Testament, where the Apprehensional construction is paired with the standard affirmative Purpose construction:

(64) *Ke ɛ́ yȅkɛ́ yɔ̰̈ nɔ̰́* [***nḭ́*** *w-**ắ*** ***nṵ̀*** *kɛ̀ ù*
but 3sg\pst do\pfv.preh there that **that** 1pl-**appr** **appr** say[inf] 1pl
gbi dɔ́ u blɛ u li nɛ̰], ***ke*** [*ù* ***wò*** *a*
breast ipfv\put we self we foc_obj at but 1pl **purp** he/she/it
dɔ̀ Kwlɛ̰ li nɛ̰].
put[sbjv?] God foc_obj at
{Indeed, we felt we had received the sentence of death.} 'But this happened that we might not rely on ourselves but on God,' {who raises the dead.} (2 Cor. 1:9)

[22]The unmarked form in the Purpose construction is probably the Subjunctive form; however, its status is still unclear.

[23]This might, however, still be an artifact of the general underrepresentation of interactional contexts (to which belong all other uses of the Apprehensional in Gban, apart from negative-purpose uses) in most linguistic corpora, see Vuillermet (2018: 259–260).

(More precisely: '… But it happened this way **so that** we **would not** want[**appr**] to rely on ourselves, **but** that we **would** rely[**purp**] on God …')

Moreover, in my elicitation data, French negative-purpose clauses with *pour ne pas (que)* have often been the default translation candidates even for isolated clauses with the Apprehensional construction, in the absence of context. For example, the phrase *Ù nı̰́ bɛ̏ yǎ nṵ̀ gwà̰* in (11) was accepted first with the translation 'For our son not to get lost' ('**Pour ne pas que** notre fils se perde'), and only then with the translation '[I fear that] our son might get lost'. Consider also the translation 'For you not to infect me' ('**Pour ne pas** me contaminer') proposed for (14).

For negative-purpose uses, the issue of combinability with preventable or unpreventable situations is irrelevant: negative purpose semantics by definition requires the situation to be currently preventable (at the reference time).

However, there is a related class of contexts where the issue becomes relevant: the (undesirable) 'in-case' contexts (Lichtenberk 1995: 297–298; Vuillermet 2018: 279–280). They consist in adverbial dependent clauses with possible undesirable consequence semantics. Such clauses express situations that may be themselves currently unpreventable, but might cause some currently preventable undesirable effects. In some languages, apprehensional markers can have such 'in-case' uses as well: for example, in To'abaita (Lichtenberk 1995: 301) and in English: *Take an umbrella with you* ***lest*** *it (should) rain.* Gban, however, does not allow using the Apprehensional construction in these contexts:

(65) # *Ɛ́ tɔ̈ dı̀gblı̏ı̏ wò, nı̰́nı̰́ y-ǎ* ***nṵ̀*** *bɔ̀.*
3sg\pst cloth heavy put[pfv.hod] chill 3sg-**appr** **appr** arrive[inf]
Intended meaning: 'He put on warm clothes in case the weather gets cold (lit. "… the chill arrives").'

To make it felicitous, my consultant altered the sentence, changing the unpreventable situation into a preventable one, thus making it a negative purpose clause and no longer an 'in-case' clause:

(66) *Ɛ́ tɔ̈ dı̀gblı̏ı̏ wò, nı̰́nı̰́ y-ǎ* ***nṵ̀*** *ȁ*
3sg\pst cloth heavy put[pfv.hod] chill 3sg-**appr** **appr** he/she/it
zɛ̀.
kill[inf]
~'He put on warm clothes **so that** he would**n't** freeze (lit. "… the chill wouldn't beat him").'

Instead, Gban resorts to the use of other, more general constructions to express ‘in case’, none of which is dedicated to undesirable situations.

6 Diachrony

There are no data on earlier varieties of Gban, and I am also not aware of any similar-looking constructions existing in related languages. The only dedicated apprehensional construction in a closely-related language that I know of is found in Loma (Toma) (Southwestern Mande, ISO 639-3 code: tod, Glottocode: toma1245) and is based on the (unrelated) marker *míná*, which is also used in Loma in the prohibitive construction (Mishchenko 2017: 378, 379).

However, the present-day formal makeup of the Gban Apprehensional construction gives us hints about what it grammaticalized from.

The invariable auxiliary verb *nṵ̀* used in the Apprehensional construction is most likely connected with the lexical verb of motion *nṵ̀* ‘come’, but most probably indirectly, through the Prospective construction, that must have previously grammaticalized from this lexical verb. The Prospective construction features a variable (inflecting) auxiliary verb *nṵ̀* plus an infinitive form of the main verb and expresses prospective/intentional/avertive/future semantics. See (67) and (75).

(67) *Gwȁ* ˵ ***nṵ̏*** ***vȉ*** *dı̋.*
stone 3SG[NPST] IPFV**come** fall[**INF**] below
[Watch out!] ‘The stone **is going to** fall!’

The predicative marker *-ă̋/-ă* ‘APPR’ is apparently related to an older *negative* marker that can be reconstructed as **-a*. The evidence in favour of this hypothesis is the following. Firstly, the Apprehensional construction itself can be argued to express negation at least in some of its uses even synchronically (see §7). Secondly, the predicative marker used in it is similar to several other Gban markers connected to negation. One is the Enimitive construction, containing an identical predicative marker (see §2), which, too, is likely to be diachronically connected to a negative morpheme: although synchronically the construction does not convey negation in any of its uses, its grammaticalization path is most probably linked to rhetorical constructions with (formal) negation (compare English *Is**n’t** she lovely ≈ She’s lovely*). Next, as was described in §3, the Negative Subjunctive predicative marker is a very similar looking morpheme *-ă̋/-ă* (with a slightly different morphophonology). Finally, the Negative Indicative predicative marker *-kè* ‘IND.NEG’ has an additional variant (seemingly used in rapid speech) that looks like *-à*, see (68) and (75).

(68)

*Mȁ̰-**ạ̀***	*nṵ̀*	*yà*	*wȍtló*	*yɛ̋,*	[...]
1SG-**IND.NEG**	IPFV\come	go_away[INF]	car	with	

'I won't go in a car ...' (the "full" version would be ***Ḭ̀-kè** nṵ̀ ...*)

Combining these source elements, the prospective-like one and the negative one, does not immediately and easily give us the apprehensional semantics that the construction now has. But it could acquire this semantics via its usage in a dependent clause (fear predicate complement or negative purpose, see below).

There are two (most plausible) paths that can be proposed for the emergence of the construction and of its different attested uses/readings. Both are based on the hypothesis that the original contexts where the construction began its grammaticalization were negative purpose clauses.

The negative morpheme **-a* would be initially used in negative purpose clauses as a "normal" negation (to express negation of the dependent proposition), and a prospective/future marker *nṵ̀* would accompany it simply to express posteriority: 'I'll leave early, so that I **won't** miss the bus'. Here the idea of (un)desirability would at first be introduced externally, by the context and by the reason/purpose subordinators *nḭ́*/*lɛ̌* 'so that' (combined with the negation in the dependent clause), but later it could become associated with the **-a* and *nṵ̀* elements, thus giving rise to the new non-compositional construction expressing undesirability.

According to the first scenario, after that, it could undergo insubordination (notice that even the dependent negative-purpose uses may lack overt markers of subordination, see (55b)) and spread to main-clause warning/threat uses ('Hurry up not to miss your bus' → 'Hurry up, you might miss your bus') and pure-apprehension uses. Finally, the latter uses could naturally start occurring in dependent clauses under fear predicates. So the path would be as in (69):

(69) (Negative prospective →) negative purpose → warning/threat → pure apprehension → fear predicate complement

Next, a similar, but slightly different second scenario can be proposed. It would start off similarly to the first one, but involve, at a later stage, a direct transition from negative-purpose uses to fear predicate complement uses, similar to what Lichtenberk (1995: 319) (see also Schmidtke-Bode 2009: 186–187) proposes as a general path for apprehensional markers (although it is not self-evident how precisely such a transition should proceed). In this case, the main part of the path would look like (70).

(70) (Negative prospective →) negative purpose →
fear predicate complement (...)

(As to the remaining warning/threat and pure-apprehension uses, there are several equal possibilities in the second scenario, all involving insubordination. They could: a) be both derived second after first from the negative-purpose use (as in the first scenario), b) both be derived second after first from the fear predicate complement use, or c) be separately derived from the negative-purpose use and from the fear predicate complement use, respectively.)

There are several arguments in favour of the two proposed scenarios, i.e. in favour of paths that start from negative-purpose uses rather than from fear predicate complements (which would be the main alternative). To begin with, the Apprehensional construction seems to be compatible only with situations that are at least in principle preventable, in all its uses. This preventability requirement is perfectly natural for warning, threat, and negative-purpose uses, but it is not expected for pure-apprehension and fear predicate complement uses (see the discussion in §4.1 and §5.1). So, this requirement is best explained as a remnant of the precedent, diachronically original use. And this points to negative-purpose and/or warning/threat uses as being the earliest ones. There is also a similar persistence in future-/posterior-orientedness of the construction, even in pure-apprehension and fear predicate complement uses (see §4.1 and §5.1), which, too, points to negative-purpose and/or warning/threat uses as the earliest ones, because they always refer to future/posterior situations (compare Lichtenberk 1995: 320). And synchronically, the negative-purpose uses seem to be the most frequent, which might also be explained by their status as diachronically original.

There is also a separate possible argument in favour of a direct connection between negative-purpose and fear predicate complement uses (the second scenario, similar to Lichtenberk's one). It is that sometimes fear predicate complement examples were translated in my data with an additional reason/purpose marker *lɛ̌*, the same one that is typical for negative-purpose uses (see (41) and (42) in §5.1).

7 The Apprehensional construction and polarity

Let us now discuss the separate issue of the relation between the Apprehensional construction and polarity/negation. As has been argued for in §6, etymologically, the predicative marker *-ɐ̋/-ɐ̌* used in it can be rather confidently traced back to a negative morpheme. Typologically, there is also a well-established link between apprehensional semantics/undesirability and negation, manifested in many languages.[24] So the question arises, what is the *synchronic* relation of the Apprehensional construction to negation in Gban?

If we start by looking at English translations (however unreliable this first approach may be), we will see that they do not contain the negative marker *not*[25] for most uses/readings of the Apprehensional construction. Compare for pure apprehension: ‘I might become sorrowful’, for threat: ‘I might/will kill you!’, for fear predicate complement: ‘She’s afraid that François might die’. As for warnings, their translations frequently contain *not*, as in ‘Watch out, do**n’t** fall!’, but can always be paraphrased without negation: ‘Watch out, you might fall!’.

But if we turn to dependent **negative-purpose uses** (described in §5.2), *not* always appears in their translations, which allows us to hypothesize that (in such uses) the Apprehensional construction in Gban still truly conveys semantic negation. Let us repeat example (54) here in (71) as an illustration.

(71) *Mȁ̰-à̰* *nṵ̀* *yà* *wȍtló* *yɛ̋,* [*n-a̰̋* ***nṵ̀***
1sg-ind.neg ipfv\come go_away[inf] car with 2sg-**appr** **appr**
nṵ̀ *lɛ̌*].
come[inf] because
‘I won’t go in a car **in order** for you **not to** come.’

These translations alone are not enough to make such a claim, of course. For many negative-purpose examples, paraphrases of the translations are possible that do not feature *not* in the dependent clause while maintaining the same general sense. For example, for (71): ‘I won’t go in a car, **because** (**otherwise**) you **might** come’ or ‘I won’t go in a car, **or else** you **might** come’.

However, one can form in Gban such negative-purpose sentences with the Apprehensional construction that would be impossible to translate without negation, which would show that in such uses it must indeed convey semantic nega-

[24]Negative forms of moods expressing semantics of desire (optative, subjunctive, and imperative) are one of the frequent historical sources for apprehensional constructions in the languages of the world (Dobrushina 2006: 49). Often apprehensional semantics is even synchronically expressed by constructions that are grammatically negative. For example, by a specialized negator *me:* in Classical Greek (Lichtenberk 1995: 311–313); by the negative optative in Mishar Tatar (see example (23)) and Balkar; by the negative imperative in Shor, Altai, and Khakas; by the negative subjunctive with an additional marker in Russian (see examples (21) and (51)) and Urdu (Dobrushina 2006: 32–33, 36–39). In addition, fear predicate complements are one of the typologically-relevant contexts for the so-called expletive negation (Jin & Koenig 2020) (see below).

[25]The following holds equally for the original French translations and the occurrence of *ne … pas* in them.

tion. We are talking about sentences with indefinite pronominal expressions such as (72).

(72) ^OK^ *M* *ă̰̋* *yɛ̀kɛ̋* [*nḭ́* ***mṵ̋*** ***dò*** *y-ǎ* *nṵ̀*
1SG\PST he/she/it do[PFV.HOD] that **person one** 3SG-APPR APPR
yà].
go_away[INF]
[I decided to go to town with the children. But there are five of them, and in the taxi there were only places for me and four children. So I suggested that the children compete in a game; the one who loses stays at home.] 'I did it, so that **someone** [out of the five] would**n't** go.'

As can be seen from the context, the existential quantifier introduced by the indefinite expression is interpreted here as having scope over a negative operator, at the same time being under the scope of the hypothetical desire/intention predicate – which we can assume to be part of the purpose semantics: '(intended (someone (NOT (go))))'.[26]

With this scopal configuration, all paraphrases without *not* in the dependent clause (that were possible before) result in wrong interpretations. *I did it, because (otherwise) someone might have gone*, would ultimately denote a different state of affairs, corresponding roughly to 'I did it so that **no one** would go'.

Another argument in the same vein is the availability of sentences with the exclusive focus-sensitive particle *dídòő* 'only' that demonstrate a similar scopal configuration (73).

(73) ^OK^ *M* *ă̰̋* *yɛ̀kɛ̋* [*nḭ́* *Zá̰* ***dídòő*** *y-ǎ* *nṵ̀*
1SG\PST he/she/it do[PFV.HOD] that Jean **only** 3SG-APPR APPR

[26]The semantics of purpose can be analyzed as including a component of desire/intention (together with a reason component and a modalized causal component): '*X P*-ed in order to *Q*' ≈ '*X P*-ed because *X* **wanted** *Q* to happen and because *X* believed that *P* may cause *Q*'.

Cf. the formulation by Goddard (1991: 44) (after Wierzbicka 1989): "the meaning of these constructions [= *purposive constructions* – M.F.] involves the notion of *because*, in combination with the complex notion of *someone wanting something to happen* …". And a following analysis by Wierzbicka (1992: 409): "*I went out to get some meat = I went out because I thought: I want: I get some meat.*"

Consider also the definitions of purpose and especially of negative purpose by Kortmann (1996: 86): "Purpose (PURPOS): 'in order to *p*, *q*' – *p* is an intended result or consequence of *q* that is yet to be achieved"; "Negative Purpose (N_PURP): 'lest *p*, *q*', '*q*, in order that not *p*' – not-*p* is an intended result or consequence of *q* that is yet to be achieved". Note the low position of negation with respect to desire/intention in the latter definition.

nṵ̀].
come[INF]
'I did [all I could], so that **only** Jean would**n't** come [… but that others would].'

Such examples demonstrate that in negative-purpose uses the Apprehensional construction conveys a direct *negation* of the proposition expressed by the dependent clause.

All other types of uses of the Apprehensional construction (i.e. non-negative-purpose uses), on the other hand, do not seem to convey such negation. Compare an intended pure-apprehension context in (74).

(74) # *M-**ä̰̋*** ***nṵ̀*** *ḭ̀ nḭ́ bɛ̏ kɛ̀kpä̋ yò wä̰̋.*
1SG-**APPR** **APPR** I son there another see[INF]
'(I'm afraid) I might see my son again.'

This example cannot denote a *negated* dependent proposition '[I'm afraid] I **might not see** my son again'. A commentary was given by the speaker: "[It sounds like] 'I don't want to see my son anymore'; this is strange, it's as if we wished his death". The sentence thus provides at most just the bizarre reading '[I'm afraid] I **might see** my son again [which is undesirable]', which leads to it being considered anomalous.

Periphrastic means[27] are used to express pure-apprehension semantics if the dependent proposition needs to be negated, compare (74) and (75).

(75) *Ḭ̀* ***gbɛ̰̀*** *nɔ̰̀* [***nḭ́*** *mä̰̏-**à̰*** ***nṵ̀*** *ḭ̀ nḭ́ bɛ̏*
1SG **IPFV\be_afraid** here **that** 1SG-**IND.NEG** **IPFV\come** I child there
kɛ̀kpä̋ yò wä̰̋].
another see[INF]
[A constantly nervous mother, after the departure of her son, is worrying and expresses her fears aloud, talking to herself:] '— I'm afraid I **might not see** my son again.'

Compare also a fear predicate complement example such as (76), where an interpretation of the Apprehensional construction with a *negated* dependent proposition (i.e., 'I'm afraid I might not work anymore') is again unavailable (and the same periphrastic means as in (75) would be used to get this interpretation).

[27] In (75), we see a combination of an explicit fear predicate and a complement clause with the (Negative) Indicative plus an instance of the peripheral Prospective construction with the variable (inflecting) auxiliary *nṵ̀* 'come' (see §6).

(76) [OK] *Ị̀ gbɛ̰̀ [nɪ̰́ m-ā̰̋ **nṵ̀** bɛ̏ā̋ kɛ̏kpā̋ wɔ̀].*
1SG IPFV\be_afraid that 1SG-**APPR** **APPR** work another put[INF]
'I'm afraid to work again [i.e. I don't want to work anymore].'

We can see that semantically these uses of the Apprehensional construction do not convey negation of the dependent proposition. However, the question remains whether the construction still has some (weaker) connection to negation even in these (non-negative-purpose) uses.

First, if we accept the above analysis for negative-purpose uses, where it does express negation, then the whole construction can be seen as *formally* (grammatically) negative.[28] Then in non-negative-purpose uses it is formally negative, too. In other words, the construction in these uses may still represent an instance of the grammatical category of (Negative) Polarity, even though semantically "deactivated". Such a state of affairs corresponds to the phenomenon of the so-called expletive negation.[29] At its extreme, such an analysis may imply that there is a formal negation, but it is fully devoid of negative force and semantically vacuous – this would correspond to a traditional understanding of expletive negation, cf. Espinal (1992).

Second (and regardless of its connection to the grammatical category of Polarity), the Apprehensional construction in these uses may express some semantic component(s) indirectly related to negation. Such a state of affairs would be similar to the way Abels (2005) analyzes the expletive negation in Russian apprehensional sentences with *kak by … ne*: as mirroring a negative operator somewhere higher in the semantic structure than the proposition in which it occurs. In our case, this negative operator would be related to the notion of undesirability itself (given that undesirability can be seen as a contrary negation of desire). Another (typologial) analysis of expletive negation by Jin & Koenig (2020) states, in a similar vein, that expletive negation reflects an entailed (or implied) negated version

[28]This is a possibility because, as was shown in §3, mood and polarity categories are always expressed together in Gban by a single paradigm of cumulative predicative markers. Therefore the predicative marker *-ā̋/-ǎ* 'APPR' could be analyzed as always expressing not only the Apprehensional Mood, but also the Negative Polarity, i.e. 'APPR.NEG' (just like, e.g. *-kě* 'COND.NEG' cumulatively expresses in Gban the Conditional Mood and the Negative Polarity).

[29]Expletive negation can be defined as "a construction where a negator is grammatically licensed but does not contribute to the polarity of the proposition which contains it" (cf. French *J'ai peur qu'il **ne** pleuve demain.* 'I fear that it will rain tomorrow') (Jin & Koenig 2020: 1).

In fact, we find an unambiguous case of expletive negation in Gban with a different construction, the Negative Subjunctive, which competes with the Apprehensional construction in fear predicate complements. It is (this time undoubtedly) grammatically negative and in these uses it is devoid of negative force. Compare (38b) and (40b).

of the dependent proposition that is activated alongside it. Fear predicates are analyzed as activating two contrasting propositions at the same time: *p*, corresponding to the Speaker's attitude (what is feared), and ¬*p*, corresponding to the Speaker's desires (what is wanted) (Jin & Koenig 2020: 22). For example, in *I fear that I might miss the train*, the dependent proposition *p* is 'I miss the train', while its negated version ¬*p*, 'I don't miss the train', is introduced via an entailed desire predicate.

Support for there being at least some synchronic connection to negation even in non-negative-purpose uses of the Apprehensional construction comes from the fact that Gban *negative polarity items* such as *nɔ̰̌ dònḭ̋* 'anything (at all)' or *mṵ̋ dònḭ̋* 'anyone (at all)'[30] can occur in its scope both in its negative-purpose uses (77) (with negation of the dependent proposition) and its other uses (without such negation). My data provide such examples for pure-apprehension (78) and fear predicate complement uses (79), (80).

(77) OK *Ɛ̏ kṵ̋ wő tȅnṵ̋ɛ̰̋ sésé là* [*nḭ́ w-ǎ nṵ̀*
3sg catch[inf] ipfv\put child:pl small on that 3pl-appr appr
sı̏ȅ ***nɔ̰̌ dònḭ̋*** *yɛ̋*].
spoil[inf] **thing any** with
'He keeps an eye on the kids **so that** they do**n't** spoil **anything**.'
(a negative-purpose use)

(78) OK *Tȅnṵ̋ɛ̰̋ sésé, w-ǎ nṵ̀ sı̏ȅ* ***nɔ̰̌ dònḭ̋*** *yɛ̋!*
child:pl small 3pl-appr appr spoil[inf] **thing any** with
~ '[I'm afraid that] the kids **might** spoil something (lit. "**anything**").'
(a pure-apprehension use)

(79) OK *Ḭ̀* ***gbɛ̰̀*** [*nḭ́ w-ǎ nṵ̀ sı̏ȅ* ***nɔ̰̌ dònḭ̋*** *yɛ̋*]*!*
1sg ipfv\be_afraid that 3pl-appr appr spoil[inf] **thing any** with
'I'm **afraid** that they might spoil something (lit. "**anything**").'
(a fear predicate complement use)

In fact, in these latter (non-negative-purpose) uses, these NPIs compete with indefinite expressions such as *nɔ̰̌ dò* 'something' or *mṵ̋ dò* 'someone', compare (80a) and (80b).

[30] These items are also used as emphatic irrealis non-specific indefinites in contexts such as 'Find **at least someone**!' (but not as free-choice indefinites).

(80) a. OK *Ḭ̀ **gbɛ̰̀** [nḭ́ mṵ̋ dò y-ǎ nṵ̀ ḭ̀ mɛ̰̏].*
1sg ipfv\be_afraid that person one 3sg-appr appr I hit[inf]
'I'm **afraid** that someone might hit me.'
(a fear predicate complement use)

b. OK *Ḭ̀ **gbɛ̰̀** [nḭ́ **mṵ̋** **dònḭ́̋** y-ǎ nṵ̀ ḭ̀ mɛ̰̏].*
1sg ipfv\be_afraid that **person any** 3sg-appr appr I hit[inf]
'I'm **afraid** that someone (lit. "**anyone**") might hit me.'
(a fear predicate complement use)

8 Conclusions

This paper described the Apprehensional construction in Gban, including its formal properties, paradigmatic status, different main-clause and dependent uses, its relation to negation, and possible diachronic sources.

The Apprehensional can be regarded as a special mood in Gban and the Apprehensional construction as one of Gban's core grammatical constructions. It is compatible with all three persons, both in dependent and in main-clause uses. It is also compatible not only with agentive ([+control], e.g. 'kill'), but also with some non-agentive ([−control], e.g. 'get lost') situations.

In main clauses, the following readings/functions of the Apprehensional construction have been identified: pure apprehension, warning, and threat. In the former function, it expresses undesirability for the speaker, in the latter two undesirability for the addressee.

In dependent clauses, two different uses have been attested. First, the construction can be used (together with certain subordination markers) in complement clauses of fear predicates. Second, it is used in negative purpose clauses (and probably also in some types of negative irrealis complement clauses).

In the negative-purpose uses, the dependent clause is frequently marked by explicit subordinators, although it may remain unmarked if it follows the main clause. The latter makes some examples of negative-purpose uses hard to distinguish from main-clause uses, compare 'Hurry up not to miss your bus' (negative purpose) and 'Hurry up, you might miss your bus' (warning). The negative-purpose use seems to be the most frequent use or at least one of the most frequent uses (around 80% of all occurrences in the New Testament) of the Apprehensional construction in Gban.

In all of its uses, the Apprehensional construction seems to have a peculiar general restriction: even in pure-apprehension uses and fear predicate complement uses, it appears to be compatible only with situations that are at least in principle preventable (i.e. that in principle allow another, controllable, situation that would cause the situation in question not to take place). The lack of 'in-case' uses, which would allow situations that are in principle unpreventable, also can be attributed to this restriction.

Regarding the diachronic sources of the Apprehensional construction, it can be conjectured to derive from a combination of an older negative morpheme and a marker expressing prospective/future semantics. The most plausible paths for its grammaticalization are ones starting with negative-purpose uses, such as: "(negative prospective →) negative purpose → warning/threat → pure apprehension → fear predicate complement". Or, alternatively, a similar path that would feature a direct transition "negative purpose → fear predicate complement".

Finally, concerning the (synchronic) relation of the Apprehensional construction to negation, it can be shown to convey negation of the dependent proposition in negative-purpose uses and it also most probably retains some weaker formal and/or semantic connection to negation in other uses (as expletive negation). In all of its uses, the construction is compatible with negative polarity items such as *nǒ̰ dònḭ̋* 'anything (at all)'.

Abbreviations

1/2/3	1st/2nd/3rd person
1PL_INCL	1st person plural inclusive
AFF	affirmative
APPR	apprehensional
CAUS	causative
COND	conditional
EMPH	emphasis
ENIM	enimitive
FOC	focus
FOC_OBJ	object focus
GEN	genitive
HEST	hesternal (past)
HOD	hodiernal (past)
IMP	imperative
IMP_POL	polite imperative
INCL	inclusive
IND	indicative
INF	infinitive
IPFV	imperfective
LOC1/LOC2	locative applicative markers
N-	non-
NEG	negation
NEUTR	neutral informational status
NIPFVPST	non-imperfective-past
NMLZ	nominalization
NCOND	non-conditional
OPT	optative (Mishar Tatar)
PFV	perfective
PL	plural
PREH	pre-hesternal (past)
PRSV	presentative
PST	past
PURP	purpose
Q	interrogative particle
SBJV	subjunctive
SEQ	sequential (To'abaita)
SG	singular

Acknowledgements

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Dedication

I am endlessly grateful and would like to dedicate this paper to my late friend Taki Oya Robert (Tȁkĭ Wȅyȁ Wlȍbɛ́), a speaker of the Bovo dialect, who worked with me as a consultant throughout these ten years and was the main source of data for all my studies on Gban, including this one. May his soul rest in peace.

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Chapter 5

The apprehensive in Kambaata (Cushitic): Form, meaning and origin

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Kambaata, a Cushitic language, has a dedicated, fully grammaticalised apprehensive paradigm without known parallels in related or neighbouring languages. This chapter presents an analysis of the morphology, syntax, meaning and origin of the apprehensive. Morphological and syntactic criteria demonstrate its main clause status. Data from a variety of sources show that the apprehensive is employed in direct dialogue. It encodes that a situation is unrealised at reference time, considered possible in the future and judged by the speaker to be undesirable, if not dangerous for any discourse participant. The primary function of the Kambaata apprehensive, in any person, is to express warnings to the addressee, who is alerted to avert the danger. Apprehensive forms of the first person may also serve as a threat. In the second person, the apprehensive has come to express prohibition. Based on detailed language-internal evidence, this chapter demonstrates that the apprehensive paradigm is likely to have resulted from the fusion of a periphrastic verb form in the recent history of the language. The source construction consisted of an affirmative same subject purposive converb plus the existential copula and may have first served to express intentional/imminent future.

1 Introduction

This chapter presents a detailed analysis of the fully grammaticalised apprehensive main verb paradigm in Kambaata,[1] a Cushitic language of Ethiopia. The apprehensive, as illustrated by *agókkookke* 'it might drink you, it might make you drown' in (1) and *eebbókkoont* 'you might bring' in (2), expresses apprehension

[1]Iso-code 639-3: ktb, Glottolog code: kamb1316.

Yvonne Treis. 2026. The apprehensive in Kambaata (Cushitic): Form, meaning and origin. In Martina Faller, Marine Vuillermet & Eva Schultze-Berndt (eds.), *Apprehensional constructions in a cross-linguistic perspective*, 163–195. Berlin: Language Science Press. DOI: ??

on the part of the speaker that a potential, undesirable situation may arise and serves as a warning to the addressee. The addressee is alerted so that they take care or counteract any adverse effects. The apprehensive is often translated as negative (‘beware that SUBJECT does not VERB’) but does not contain any overt negative morphology.

(1) *Wó’-u* *ag-**ókkoo**-kke!*
water-M.NOM drink-3M.APPR-2SG.OBJ
(Husband alerts his wife when approaching a dangerous river) ‘(Watch out!) The water might “drink” you. / Beware that the water does not “drink” you (i.e. that you don’t drown).’ [Dialogue in a narrative, Field notes 2004]

(2) *Aador-á* *úl-t* *tíg-unta.*
rock-M.ACC touch-3M.PFV.CVB tumble.down-3M.PURP.DS
áabb-a, *eeb-**bókkoont*** *reh-úta!*
son-M.VOC bring-2SG.APPR death-F.ACC
(In a playful competition, a girl warns a flirty boy of undesirable consequences of his advances) ‘(Watch out!) When you touch the rocks and cause a landslide, (my) son, you might bring death!’ / ‘Don’t touch the rocks to cause a landslide, (my) son, and (thus) bring death!’ (Double entendre, Alemu Banta, personal communication, 2019)[2]]

Kambaata is one of only two known African languages with a dedicated main verb form for warnings and threats (see Fedotov 2026 [this volume] on the apprehensive construction in the Mande language Gban).[3] This chapter envisages a detailed synchronic and diachronic description of the Kambaata apprehensive based on field notes and data from local publications. The first section of this chapter is a brief introduction into the language; it provides information on its classification, sociolinguistic aspects and official orthography (§1.1) and on typological features (§1.2). In §1.3, the type of linguistic data used for this study and the number of occurrences of apprehensive examples in the corpus are tabulated. §2 is dedicated to formal aspects of the apprehensive paradigm: §2.1 discusses where the apprehensive fits into the typology of Kambaata verb forms,

[2] Genre: *Qaanqúta* ‘double entendre’. Note the alliteration and the rhyme.

[3] As pointed out to me by Ronny Meyer (personal communication, 2020), Muher, an Ethiosemitic language of the Gunnän Gurage branch, may also have a yet undescribed dedicated apprehensive paradigm. Muher is unrelated to Kambaata but spoken in the proximity of Highland East Cushitic languages in southwestern Ethiopia.

while §2.2 analyses the individual apprehensive forms and explains their morphological makeup. Semantic aspects of the apprehensive are the focus of §3, which divides into sections on third, first and second person forms (§3.1-§3.3). The syntax of apprehensive sentences and the expression of pre-emptive measures are elaborated on in §4. Finally, §5 traces the historical development from a periphrastic imminent future verb to an apprehensive on the basis of detailed language-internal evidence. The paper is concluded in §6.

Terminological note: In earlier publications and conference papers on Kambaata, I have glossed and labelled the verb forms for warning and threats in various ways. As I was unsure about the functional range and typological parallels of the *-ókkoo*-paradigm, I was torn between the labels "intimidative", "admonitive", "preventive" and "advertive". The comparison with functionally equivalent verb forms in other languages shows that "apprehensive" is the most appropriate name for the *-ókkoo*-paradigm.

1.1 Classification, sociolinguistics and orthography

Kambaata (endonym: *Kambaatissáta*) is a Cushitic language of the Afroasiatic phylum. Together with six other languages it constitutes the Highland East Cushitic language group; Alaaba and K'abeena are its closest relatives. Kambaata is spoken in southwestern Ethiopia in the Kambaata-Xambaaro Zone of the Southern Nation, Nationalities, and Peoples' Region, about 300km's distance from the Ethiopian capital Addis Ababa. According to the last census data (Ethiopia Central Statistical Agency 2007: 91), the language has (at least) 615,000 speakers. Amharic, the Ethiopian lingua franca and federal working language, is the most important second language of Kambaata speakers.

The official Kambaata orthography is based on the Roman script (Treis 2008: 73–80; Alemu 2016) and is largely phonemic: It represents the 5 vowel and 27 consonant phonemes as well as phonemic length. In this contribution, the official orthography is adopted with only minor adaptations: phonemic stress is consistently marked throughout the paper by an acute accent, and the phonemic glottal stop is consistently written wherever it occurs in word-medial and word-final position. The following graphemes of the official orthography are not in accordance with the IPA conventions: <ph> /p'/, <x> /t'/, <q> /k'/, <j> /dʒ/, <c> /tʃ'/, <ch> /tʃ/, <sh> /ʃ/, <y> /j/ and <'> /ʔ/. Geminate consonants and long vowels are marked by doubling, e.g. <shsh> /ʃ:/ and <ee> /e:/. Consonant clusters consisting of a glottal stop and a sonorant are written as trigraphs, e.g. <'rr> /ʔr/, to distinguish them from laryngealised sonorants, e.g. <'r> /r'/.

1.2 Typological profile

Kambaata is an agglutinating-fusional language and strictly suffixing. Its constituent order is head-final; hence all modifiers precede the noun in the noun phrase, and all dependent clauses precede independent main clauses. The last constituent in a sentence is usually a fully finite verb or a copula.[4] The following open word classes can be defined on morphosyntactic grounds: nouns, adjectives, verbs, ideophones and interjections. The word class of adverbs is marginal, there are a small number of independent discourse markers, no adpositions and only two true conjunctions. Kambaata is head- and dependent-marking with an elaborate case system and subject indexing on verbs. The case system is nominative-accusative: The nominative marks the grammatical subject, whereas the accusative marks direct objects and certain adverbial constituents and serves as the citation form. Nouns and certain pronouns distinguish nine case forms,[5] all of which are marked by a segmental suffix and a specific stress pattern; the case system of adjectives is reduced to three forms (accusative, nominative, oblique). Nouns, pronouns and adjectives are not only obligatorily marked for case but also for gender (masculine vs. feminine). The assignment of gender is mostly arbitrary and only sex-based in the case of nouns referring to human beings and higher animals. Adjectives are a macro-word class of case-/gender-agreeing lexemes that also encompasses cardinal numerals and adnominal demonstratives. Pronouns form a heterogeneous closed word class that subdivides into personal, interrogative and demonstrative pronouns. Ideophones and interjections are morphologically invariant. The former are syntactically integrated and inflected with the help of light verbs (*y-* 'say', *ih-* 'become', *ass-* ~ *a'-* 'do'), the latter constitute utterances of their own. Major features of the verbal system are highlighted in §2 to compare the morphology of the apprehensive with that of other verb forms.

1.3 Data

My corpus contains altogether 136 apprehensive examples, which come from a variety of sources that were collected since 2002: recorded conversations and stories, overheard examples, solicited mock dialogues, elicited data and local publications (see Table 1).[6]

[4]In poetry – see e.g. (2) – the word order is more flexible.

[5]None of these case forms is used with apprehensional meanings.

[6]The written sources consulted are: Saint-Exupéry (2018); schoolbooks: Kambaata Education Bureau (1989: grade 1–8), (2007: grade 9–10), (2008a: grade 1–4), (2008b: grade 11) and (2010: grade 12); biblical and religious texts: Kambaata and Hadiyya Translation Project Hosaina (2005) and (2023), and Brook & Yonathan (2013); dictionary: Alemu (2016); proverb collection: Alamu & Alamaayyo (2017).

Table 1: Sources of apprehensive examples

	1st person	2nd person	3rd person	All
Recorded: Conversations	-	2	1	3
Recorded: Stories	-	-	1	1
Field notes: Overheard	-	4	-	4
Field notes: Mock dialogues	3	5	13	21
Field notes: Elicitation	12	17	19	48
Written: Little Prince	-	7	1	8
Written: Schoolbooks	2	27	11	40
Written: Gospel of John	-	1	-	1
Written: Deuteronomy (ms.)	-	1	5	6
Written: Religious text	-	1	-	1
Written: Dictionary	-	1	1	2
Written: Proverb collection	-	-	1	1
Total	16	66	53	**136**

Mock dialogues are dialogues of a small number of turns (often 2–4) that are invented by a native speaker under minimal meta-language influence. The speaker is either given a certain communicative setting in English, as in (3), or a Kambaata word form as a point of departure, as in (4).

(3) Example instruction
Imagine a natural dialogue between a mother and a daughter about a dangerous situation for their chicken. (For the apprehensive form thus obtained see (28).)

(4) Example instruction
Imagine a natural dialogue between two people in which the word form *osa'llókkoombe* (= 2SG.APPR of *osa'll-* 'laugh') is naturally used by one of the speakers. (For the example sentence thus obtained see (12).)

Elicited data was prompted by apprehensive examples extracted from oral and written texts, or examples sentences were generated by native speakers after I had proposed potential verb forms to them. Elicited examples are also found as by-products in questionnaires on the tense-aspect system or on subordination. The elicited examples were discussed with native speakers to solicit descriptions of possible natural contexts in which they could occur. Most of the data on which this contribution is based was again verified during a field trip in February 2019.

2 Morphology

2.1 The apprehensive in the typology of Kambaata verbs forms

Kambaata makes a primary morphological distinction between verbs that are used in main vs. subordinate clauses (Figure 1). Fully finite main verbs sub-divide into indicative and directive verb forms. Indicative main verb forms are marked for four different aspects.[7] The directive is unmarked for aspect and splits up into imperative, jussive and benedictive moods. As the apprehensive is exclusively used as a final verb in main clauses and can as such govern all types of subordinate clauses (§4), it definitely belongs to the class of main verbs.

<table>
<tr><td></td><td></td><td>Affirmative</td><td>Negative</td><td></td></tr>
<tr><td rowspan="10">Main verbs</td><td rowspan="5">Indicative</td><td>Imperfective</td><td>Negative Imperfective</td><td></td></tr>
<tr><td>Perfective</td><td rowspan="3">Negative Non-Imperfective</td><td></td></tr>
<tr><td>Perfect</td><td></td></tr>
<tr><td>Progressive</td><td></td></tr>
<tr><td colspan="2">←</td><td>Apprehensive?</td></tr>
<tr><td rowspan="4">Directive</td><td>Imperative</td><td>Negative Imperative</td><td></td></tr>
<tr><td>Jussive</td><td rowspan="2">Negative Jussive</td><td></td></tr>
<tr><td>Benedictive</td><td></td></tr>
<tr><td colspan="2">←</td><td>Apprehensive?</td></tr>
<tr><td colspan="3">←</td><td>Apprehensive?</td></tr>
</table>

<table>
<tr><td></td><td></td><td>Affirmative</td><td>Negative</td></tr>
<tr><td rowspan="9">Subordinate verbs</td><td rowspan="4">Relative</td><td>Imperfective</td><td rowspan="4">Negative Relative</td></tr>
<tr><td>Perfective</td></tr>
<tr><td>Perfect</td></tr>
<tr><td>Progressive</td></tr>
<tr><td rowspan="4">Converb</td><td>Perfective(-DS)</td><td rowspan="2">Negative Converb</td></tr>
<tr><td>Imperfective(-DS)</td></tr>
<tr><td>SS Purposive</td><td rowspan="2">(Periphrasis: Negative Relative + =g ‘like’)</td></tr>
<tr><td>DS Purposive</td></tr>
<tr><td>Non-finite</td><td>Verbal noun</td><td>(Periphrasis with hoog- ‘not do’)</td></tr>
</table>

Figure 1: Classification of Kambaata verb forms

Determining the exact place of the apprehensive in this classification is not trivial. If the criterion of morphological structure is applied, then the apprehensive clearly patterns with affirmative indicative main verbs. Like indicative verbs, it has two slots of subject morphology (Figure 2), whereas directive verbs have

[7]Tense is not an inflectional category of the Kambaata verb system. The post-predicate particle *íkke* is used to mark past tense (if not clear from the linguistic context) or counterfactuality.

only one subject slot (Figure 3) preceding the aspect/mood (A/M) slot. The bipartite, discontinuous subject indexes of indicative verbs are likely the result of the fusion of a periphrastic verb form consisting of a subordinate verb and a superordinate auxiliary, each of which contributed its subject index slot to the fused verb.

<table>
<tr><td colspan="2"></td><td colspan="3">Inflection</td><td></td></tr>
<tr><td colspan="2">Stem</td><td rowspan="2">Subject Index 1</td><td rowspan="2">A/M</td><td rowspan="2">Subject Index 2</td><td rowspan="2">(Object Suffix)</td></tr>
<tr><td>Root</td><td>(Derivation)</td></tr>
</table>

Figure 2: Structure of inflected verbs with two subject index slots (all affirmative indicative verbs, **apprehensive**, negative imperfective, affirmative relative)

<table>
<tr><td colspan="2"></td><td colspan="2">Inflection</td><td></td></tr>
<tr><td colspan="2">Stem</td><td rowspan="2">Subject Index 1</td><td rowspan="2">A/M</td><td rowspan="2">(Object Suffix)</td></tr>
<tr><td>Root</td><td>(Derivation)</td></tr>
</table>

Figure 3: Structure of inflected verbs with one subject index slot (all affirmative and negative directive verbs, negative non-imperfective, negative relative, all affirmative and negative converbs)

Other criteria suggest that the apprehensive is better considered to be a type of directive verb together with the imperative and the jussive. The discussion of the meaning of the apprehensive in §3 shows that it shares semantic features with the negative imperative and jussive. Furthermore, two formal criteria set the apprehensive apart from indicative verbs and align it with directive verbs: (i) The apprehensive cannot be relativized, unlike all indicative verbs (Treis 2012a: 222–226) and (ii) The apprehensive may not combine with the past and counterfactual particle *íkke* – again unlike all indicative verbs (Treis 2015). Finally, the combinability of the apprehensive with pragmatically determined verbal suffixes (e.g. mitigators, markers of (non-)shared knowledge, speaker attitude) seems to be similar to that of directives.[8] Finally, one may also consider establishing a third sub-category of main verb forms for the apprehensive alone – see the third option in Figure 1. Unlike all other main verbs, the apprehensive does not come in

[8]The last statement is to be taken with due care as the pragmatically determined verbal suffixes, which are a characteristic trait of natural conversations, are still little investigated and would make an interesting subject for future research.

an affirmative-negative polarity pair and cannot be morphologically negated.[9] Instead, (near) antonymic lexical pairs may express warnings with opposite polarity[10] or speakers resort to periphrastic means to express apprehension that something may *not* happen. In (5), the danger of not noticing is rendered as a danger of passing by without noticing, in which case the negation (see the morpheme NEG4) is marked on the subordinate converb rather than on the apprehensive verb.

(5)

Háy	*láshsh=y-ít*	*otá'-i!*	*Xuu<n>du'nnáan*
please.INTJ	slow.IDEO=say-2SG.PFV.CVB	drive-2SG.IMP	see<1PL>.NEG4

hi<n>gókkoomm.
pass<1PL>.APPR

'Please, drive slowly! We might pass (the sign) without noticing (it)!' [Elicited, Field notes 2021]

2.2 The morphological structure of the apprehensive

The apprehensive paradigm distinguishes between seven bipartite, discontinuous subject indexes (Table 2), of which the first element precedes and the second element follows the apprehensive morpheme *-ókkoo*: 1SG, 2SG, 3M, 3F/3PL, 3HON, 1PL, and 2PL/2HON. The subject indexes in the first slot (SBJ1) are identical across all finite and partially finite verb forms of Kambaata, those in the second slot (SBJ2) are only found in indicative affirmative main verbs and in the indicative negative imperfective paradigm.[11] The apprehensive morpheme that is wedged between the two parts of the discontinuous subject marker occupies the same position as aspect morphemes (imperfective, perfective, perfect, progressive); aspect-marking and apprehensive morphology is thus mutually exclusive.[12]

At the boundary between the verb stem and the first subject index slot, predictable morphophonological processes are observable, as the exemplary paradigms of *ub-* 'fall' and *torr-* 'throw' in Table 2 illustrate. Assimilation, epenthesis and metathesis help prevent illicit consonant clusters when 2SG or 3F/3PL *-t* or

[9] The use of any of the five negative morphemes the language has (Treis 2012b) is judged ungrammatical by native speakers.

[10] The verb *kot-* 'not suffice, be (too) little', for instance, may be employed as the opposite of *ih-* 'suffice' and *bata'-* 'be (too) much'. The semantically fairly general verb *fa'is-* 'save, leave behind, leave out, skip, not impact' can stand in as the antonym of various verbs expressing events that causally affect a participant (e.g. *ba'is-* 'destroy, damage', *sh-* 'kill', *woqqar-* 'beat').

[11] There are slight differences in the 3M, 3HON and 2PL/2HON morphemes of the SBJ2 slot across the main verb paradigms; see Table 5.

[12] The aspect/mood slot in Figure 2 and Figure 3 may be filled by one morpheme only.

Table 2: The apprehensive paradigm

				Examples	
	-sbj1	*-ókkoo*	sbj2	*ub-* ‘fall’	*torr-* ‘throw’
1sg	-∅	*-ókkoo*	*-mm*	*ub-ókkoomm*	*torr-ókkoomm*
2sg	*-t*	*-ókkoo*	*-nt*	*ub-bókkoont*	*torr-i-tókkoont*
3m	-∅	*-ókkoo*	*-’u*	*ub-ókkoo’u*	*torr-ókkoo’u*
3f/3pl	*-t*	*-ókkoo*	*-’u*	*ub-bókkoo’u*	*torr-i-tókkoo’u*
3hon	*-een*	*-ókkoo*	*-mma*	*ub-eenókkoomma*	*torr-eenókkoomma*
1pl	*-n*	*-ókkoo*	*-mm*	*u<m>b-ókkoomm*	*torr-i-nókkoomm*
2pl/2hon	*-t-een*	*-ókkoo*	*-nta*	*ub-beenókkoonta*	*torr-i-teenókkoonta*

1pl *-n* meet C- and CC-final verb stems. In the examples given in this contribution, the discontinuous subject index and the apprehensive morpheme are not segmented from each other, but the inflectional complex (see Figure 2) is glossed as if it was a portmanteau-morpheme, namely as 1sg.appr, 2sg.appr, etc. Predictable morphophonological changes are not indicated in the glosses either. Dependent object pronouns are suffixed after the second subject index slot – see, for instance, (1), (6), (7) and (20). Pragmatically determined discourse suffixes may occur in the slot after the object pronouns – see, for instance, (12). The morphological structure of the apprehensive verb is sketched in Figure 4.

Root	(-Derivation)	-sbj1	*-ókkoo*	-sbj2	(-obj)	(-prag)

Figure 4: Morphological structure of the apprehensive verb

3 Semantics

The discussion of the semantics of the apprehensive divides into three sections that treat third (§3.1), first (§3.2) and second person forms (§3.3). The division of §3 into subsections for different persons is motivated by Vuillermet’s (2018b) observation that languages may impose person restrictions on apprehensives, that some person forms may be less frequent than others, and/or that certain persons favour default readings.

3.1 Third person apprehensive forms

In the third person, the apprehensive forms express warnings of potential events that the speaker is worried about, that they consider undesirable if not dangerous and that they expect the addressee to avert. Pre-emptive measures do not need to be expressed overtly – but they are often mentioned in adjacent sentences or in clauses that are subordinate to the apprehensive clause (see details in §4). As shown in Table 2, Kambaata distinguishes three third person forms: masculine, feminine/plural and honorific/impersonal.[13] In the introductory example (1), a husband warns his wife of a dangerous river, which might make her drown. In (6), from a recorded conversation about birth traditions, the quoted impersonal speakers (= the Kambaata people) fear that a young mother falls victim to a disease. As a pre-emptive measure, the mother leaves the house with a knife and a bunch of sooty straw in her hand. The addressees of the reported speech event are unexpressed but can be assumed to be her family members.

(6) (…) *shum-áa* *ful-taní-i* (…) *billaww-ahá-a*
urine-F.DAT go.out-3F.IPFV.CVB-ADD knife-M.ACC-ADD
ka *xit-ahá-a* *áf-f* *ful-táa'u,*
A_DEM1.M.ACC soot-M.ACC-ADD seize-3F.PFV.CVB go.out-3F.IPFV
*michch-**ókkoo**-se* *y-éeni-yan* (…)
cause.disease.sp-3M.APPR-3F.OBJ say-3HON.PFV.CVB-DS
(Speaking about a woman who has recently given birth) '(…) and when she goes out to pee, she grabs a knife and this (bunch of straw smeared with) soot, (because one) says (among the Kambaata): "She (as a young mother) might get attacked by the *michcha*-disease".' (lit. 'it might *michcha* her') [Recorded conversation, EK2016-02-23_002]

Also the following apprehensive examples from the written corpus are found in direct speech reports. Example (7) is a self-quotation, in which the speaker reports a warning from an internal monologue.[14] Example (8) is here presented to demonstrate that the quoted speaker (and not the actual speaker/writer) is the source of the evaluation of the situation as undesirable and as apprehension-causing.[15]

[13]By accident, all apprehensive forms in (near-)natural examples have 3M subjects; 3F and 3HON forms are only attested in elicited data.

[14]I refer here to the form *kar-ókkoo-he*; second person forms such as *waal-tókkoont* and their prohibitive pragmatic force are discussed in §3.3.

[15]All examples taken from publications in the Kambaata language are stress-marked, segmented, glossed and translated to English by the present author.

(7) (…) *án* *waal-tókkoont* *y-áayyoommi-i*
1SG.NOM come-2SG.APPR say-1SG.PROG.REL-NMLZ1.M.NOM
worr-íichch-u *kar-**ókkoo**-he* *y-í-ne-eb-be.*
snakes-SGV-M.NOM sting-3M.APPR-2SG.OBJ say-1SG.PFV.CVB-L-COP3-PRAG5
(Little Prince speaking to the pilot) '(…) I said "(Better) you don't come!" (because) I thought (lit. said) "The snake might bite you".' (Saint-Exupéry 2018: 87)

(8) *Gag-á-s* *sh-eenáni-yan* *"Arg-é* *oddishsh-áta*
self-M.ACC-3M.POSS kill-3HON.IPFV.CVB-DS loan-F.GEN clothes-F.ACC
qég-u *ba'-is-**ókkoo'u**"* *y-ée'u.*
blood-M.NOM become.spoilt-CAUS1-3M.APPR say-3M.PFV
(Proverb) 'When he is being killed, he says, "The blood might spoil the borrowed clothes."' (i.e. 'He is more worried about his borrowed clothes than his own life.') (Alamu & Alamaayyo 2017: 52)

Kambaata does not seem to impose any semantic restrictions on the verbs that can serve as input for the apprehensive form. One finds, for instance, also inchoative-stative property verbs in warnings (9).

(9) *Juus-áan* *hoolam-á* *wo'-á* *wór-tooti,*
juice-M.LOC much-M.ACC water-M.ACC put.in-2SG.IMP.NEG2
*qac-**ókkoo'u**.*
become.thin-3M.APPR
'Don't pour too much water into the juice, it might become (too) thin.' [Elicited, Field notes 2019]

Furthermore, Kambaata does not exclude verbs that typically express positive states of affairs, such as *bajig-* 'be(come) happy', from apprehensive clauses. However, when such verbs are apprehensive-marked, the event is evaluated as negative for the speaker and/or the addressee. In (10), the speaker is worried that the addressee's enemy is happy about the addressee's failure.

(10) *Qáar-t* *huját-i!*
become.strong-2SG.PFV.CVB work-2SG.IMP
Bux-xoontí=da *díin-u* *bajig-**ókkoo'u**.*
become.poor-2SG.PFV.REL=COND enemy-M.NOM become.happy-3M.APPR
'Work hard! If you sink into poverty, the enemies might be happy.' [Elicited, Field notes 2019]

3.2 First person apprehensive forms

First person singular and plural apprehensive forms are the least frequent forms in my corpus (see Table 1), hence I supplemented the two examples from the schoolbooks, which are quoted in (14) and (17), by elicited sentences and invented dialogues. In the first person, as in the third person (§3.1), the apprehensive is used to utter warnings of looming dangers and potential accidental events, such as a dangerous fall (11), an unintended breach of social conventions (12), or a car crash (14). The speaker assumes the addressee to be able to take pre-emptive steps in order to avert the undesirable event.

(11) *Ub-**ókkoomm**, ka masalaal-á áf-i!*
fall-1SG.APPR A_DEM1.M.ACC ladder-M.ACC seize-2SG.IMP
(Possible context provided: Speaker is standing on a ladder that is held by the addressee) '(Take care!) I might fall. Hold the ladder!' [Elicited, Field notes 2019]

(12) *Osa'll-**ókkoom**-be, sá!*
fall-1SG.APPR-PRAG5 shush.INTJ
(Context: On the way to visit a mourning family, Speaker$_i$ asks Addressee$_j$ to keep his$_j$ mouth shut so that he$_i$ wouldn't burst out laughing. However, Addressee keeps chatting away. Speaker interrupts him:) '(Watch out!) I might laugh (as I said earlier), shush!' [Mock dialogue, DW2019-02-17]

The worried speaker and the alerted addressee can be the same person, as in (13), which is a warning to self.

(13) *Ub-**ókkoomm**, qoraphph-eemmí=da xúm-a-a.*
fall-1SG.APPR take.care.MID-1SG.PFV.REL=COND good-M.PRED-M.COP2
(Possible context provided: Speaker spots a slippery spot ahead of them and tells themself) 'I might fall, it's good to be careful.' [Elicited, Field notes 2003, 2019]

The potential event is undesirable for the speaker (11)–(13), or the speaker and their group (14).

(14) *Háy láshsh=y-ít otá'-i! Ka'llixx-áan*
please.intj slow.ideo=say-2sg.pfv.cvb drive-2sg.imp accident-m.loc
***aa<n>gókkoomm*.**
enter<1pl>.appr
(Speaker warns a speeding driver) 'Please, drive slowly! We might have (lit. enter) an accident.' (Kambaata Education Bureau 2008a: 4.55)

Apart from expressing the speaker's fear of an undesirable event with a negative impact on themself, the first person apprehensive form is also used for threats. Here the potential danger is not accidental but inflicted on the addressee by the speaker. The event expressed in the apprehensive clause can be straightforwardly undesirable for the addressee, such as the kicks in (15) and the speaker's betrayal of a secret in (16), or have undesirable consequences, such as the inevitable punishment that would follow if the speaker saw the addressee breaking the rules in (17).

(15) *Ool-**ókkoon**-ke!*
kick-1sg.appr-2sg.obj
(Possible context provided: Addressee has been teasing Speaker for a while; Speaker threatens) '(Stop it, otherwise) I kick you!' [Elicited, Field notes 2019]

(16) *Mát-o bar-e-'ée=bii kul-**ókkoomm**.*
one-m.obl day-?-assoc.f.gen=nmlz2.m.acc tell-1sg.appr
(Possible context provided: Speaker knows a secret of Addressee that they promised to keep. Now that Addressee annoys them, they threaten) '(Stop it, otherwise) I might tell/reveal that (secret) of the other day!' [Elicited, Field notes 2004]

(17) *Íi béet-o, lankíi kánn haqq-í al-í*
1sg.gen son-m.voc again a_dem1.m.obl tree-m.gen top-m.acc
*ful-táni-yan xuud-**ókkoon**-ke.*
go.up-2sg.ipfv.cvb-ds see-1sg.appr-2sg.obj
(Context: Although the mother has strictly forbidden it, a boy (= the addressee) keeps on climbing onto the tree in the front yard. The mother gets angry and threatens him) '(Stop it,) my son, I might see you climbing this tree again (and this will have negative consequences)!' (Kambaata Education Bureau 1989: 4.45)

Vuillermet (2018b) shows in her cross-linguistic study of 46 South American and Australian languages that only 63.6% of the languages with apprehensive morphology have first person forms attested. Of the 17 languages that have first person transitive subject forms of [+control] verbs, one language, Matsés, permits only a warning reading but excludes a threat reading in the first person. For the other 16 languages, the apprehensive forms are found in threats or both in threats and warnings. Kambaata, like e.g. Ese Ejja (Vuillermet 2018b: 273–274), does not impose a warning or threat reading with [+control] in the first person, as the two possible translations of the 'hit'-verb in (18) show.

(18) *Woqqar-**ókkoon**-ke!*
hit-1SG.APPR-2SG.OBJ
(i) (Threat) '(Stop it, or) I hit you!'
(ii) (Warning) '(Watch out! Step aside, otherwise) I might accidentally hit you!' (Deginet Wotango Doyiso, personal communication, 2020)

3.3 Second person apprehensive forms

The second person apprehensive forms are not only the most frequently attested forms in the database (Table 1) but also commonly overheard in conversations. In (19), the speaker expresses concern that the addressee might finish the coffee to his (= the addressee's) disadvantage. In (20), the speaker is worried about a potential infection. In an example from oral literature in (2), a girl warns a boy about the potential undesirable consequences of his advances. The looming, dangerous or disadvantageous situation is undesirable for the speaker and/or the addressee – or to phrase it more generally, apprehensives express events that can be evaluated negatively by any discourse participant.

(19) Overheard in a Kambaata household (2019)
*Kesáa xoof-**fókkoont**.*
2SG.DAT finish-2SG.APPR
(Observed context: Addressee has boiled coffee for three people, now he is generously filling two cups for Speaker and a third person. Speaker is worried that not enough will be left in the pot to also fill Addressee's own cup.) '(Take care!) You might finish it to your disadvantage!'/'Don't finish it off on yourself!'

(20) *Gansh-ú-kk* *hig-is-**sókkonte**-'e,* *shiin-á-kk*
cold-m.acc-2sg.poss pass-caus1-2sg.appr-1sg.obj side-m.acc-2sg.poss
waal-áam-ba'a.
come-1sg.ipfv-neg1
'You might pass on the cold to me, I won't be coming near you.' [Elicited, Field notes 2019]

The two alternative translations in (2) and (19) reflect the recurring difficulty to decide whether second person apprehensive forms are better translated as warnings of a worried speaker or as simple negative commands. The use of the conative interjection *háy* 'please, I beg you' with the apprehensive form in (21) shows that the utterance is possibly less a warning of an imprudent, unintentional realisation of an event ('you might hurry on') than a request ('do not hurry on').

(21) *Ta* *ma'nn-ichch-ú<n>ta-ma* *iill-iteentáachch*
a_dem1.f.acc place-sgv-f.acc<emp>-cf reach-2pl.pfv.rel.abl
háy *sarb-an-**teenókkoonta**.*
please.intj hurry-pass-2pl.appr
(Pilot to the readers of the Little Prince:) 'If you should come upon this very spot, please, do not hurry on.' (Saint-Exupéry 2018: 95)

In contexts such as (21), the semantic differences between second person apprehensive and negative imperative forms (singular: *´-tooti*, plural: *-téenoochche*) are subtle and hard to pinpoint, and native speakers often accept a swap of the apprehensive for regular negative imperative forms and vice versa. In discussions about the meaning differences, speakers voice conflicting intuitions. The verb forms are sometimes qualified as synonymous, or the apprehensive is interpreted as a reinforced (22), occasionally even as a mitigated command (21). It seems safe to assume that mitigation and reinforcement are not inherent to the second person apprehensive forms but contributed by other elements in the linguistic context, such as the polite interjection in (21), the adverb *hináten* 'by any means, at all' in (22) and possibly various other factors.

(22) *Buchch-á* *ít-i!* [Definition:] *mat-ú* *íkk-ee*
soil-m.acc eat-2sg.imp one-m.acc become-3m.prf.rel
xaw-á *huj-íta* *hinát-e-n* *agur-**tókkoont***
issue-m.acc work-f.acc totality-f.obl-emp abandon-2sg.appr
y-éen *qaar-s-íi* *y-eennó*
say-3hon.pfv.cvb become.strong-caus1-m.dat say-3hon.ipfv.rel

yann-á.
saying-M.ACC
(Dictionary entry) 'Eat soil! (i.e. Persevere!)' = (Definition) 'Saying that is uttered to encourage (lit. strengthen) (somebody), meaning "Do not, by any means, give up on a problem (or) a job!"' (Alemu 2016: 158)

The near synonymy of the apprehensive and the negative imperative is also reflected in the written Kambaata literature where the two verb forms may occur in identical contexts; see the Bible excerpt in (23).

(23) *Bookk-íta it-**téenoochche**;* (...) *resh-á-ssa-n*
pigs-F.ACC eat-2PL.IMP.NEG2 carcass-M.ACC-3PL.POSS-EMP
*ul-**teenókkoonta**.*
touch-2PL.APPR
(Deuteronomy 14,8) 'Don't eat pigs; (...) don't touch their carcasses.' (The Bible Society of Ethiopia. 2023)

Despite examples such as (22) and (23),[16] the apprehensive tends to express warnings of unintended, undesirable actions and the negative imperative prohibitions of actions that the addressee carries out wilfully, as can also be seen in "minimal pairs" such as (24) and (25). All speakers I consulted agreed that (24) is a warning of an unintentional step into the mud, while (25) is a prohibition of an intentional step into the mud. Similarly, the imperative *úb-booti* (fall-2SG.IMP.NEG2) 'don't fall' is only considered felicitous when the addressee is intentionally falling down, e.g. to practice it, and the speaker wants them to stop, while *ub-bókkoont* (fall-2SG.APPR) '(watch out!) don't fall' is used as a warning of an unintended fall.

(24) *Orc-áan aag-**gókkoont**.*
mud-M.LOC enter-2SG.APPR
(Possible context provided: Addressee approaches a muddy spot. Speaker warns Addressee of a danger) '(Watch out!) You might step into the mud!'/'Don't step (accidentally) into the mud!' [Elicited, Field notes 2005]

(25) *Orc-áan áag-**gooti**!*
mud-M.LOC enter-2SG.IMP.NEG2
(Possible context provided: Addressee, e.g. a child, is happily jumping into every mud pit on the way) 'Don't step (intentionally) into the mud!' [Elicited, Field notes 2005]

[16] See also the second person apprehensive form used for the prohibition of a willful action in (7).

Dobrushina (2006) and Pakendorf & Schalley (2007) have documented a grammaticalization path from markers of possibility to prohibitives via intermediate stages of apprehension and warning. Kambaata gives further evidence for this diachronic development. The apprehensive verb form, which is used to express the speaker's concern about the realisation of an undesirable potential event, as is still clearly seen in the first (§3.2) and third persons (§3.1), becomes more and more interpreted as a prohibitive in the second persons. The pre-emptive measure originally only implicit in the apprehensive verb (or expressed in adjacent clauses) has become its central meaning component, whereas the apprehensional component is backgrounded or lost: 'You might finish it' (implicit: 'Take care not to'). > 'Do not finish it!' However, the existence of contexts such as (24) and (25) where either only second person negative imperative or second person apprehensive forms are felicitous shows that the verb forms are not yet fully synonymous.

4 Syntax

The verbal paradigms of Kambaata split up into those used in main clauses and those used in subordinate clauses (Figure 1); main clause verbs can constitute a complete sentence on their own. Apprehensive verbs are main clause verbs and, due to Kambaata's fairly strict head-finality, they usually occur sentence-finally (cf. §1.2 and §2.1). It is only in reported speech constructions that we find apprehensive verbs in a sentence-medial position, see (6)–(8) and (22).[17] Apprehensive verbs may head complex sentences consisting of several clauses. While the apprehensive main clause – the final clause – expresses the apprehension-causing situation, dependent subordinate clauses – non-final clauses – express the temporal context or the condition under which the apprehension-causing situation is realised. It is fairly common to find the pre-emptive action expressed in a negative subordinate clause, either a converb or a conditional clause ('if not [pre-emptive action]'). Example (26) consists of three clauses, of which clause 3, the main clause, contains the apprehension-causing situation, namely the subject's burning; clause 2 expresses an event that precedes the burning in time; clause 1 states the condition under which the apprehension-causing situation becomes true, namely under the condition that the addressee fails to carry out the pre-emptive action.[18] The subordinate clauses 1 and 2 in (26) are headed by

[17] Another exception to the head-finality rule are examples in which constituents follow the main verb as an afterthought.

[18] While it is clear that clause 1 and 2 are dependent on clause 3, the main clause, I am unable to say whether clause 1 is dependent on clause 2 or whether the two clauses are on the same level.

perfective converbs (PFV.CVB), which leave the semantic relation with the subsequent clause vague. Here the perfective converbs are interpreted as expressing a condition (*hóoggiyan*) and a relation of anteriority (*aphphítiyan*).

(26) [*Giir-áta danáam-o=gg-a tú'mm=a'-ú*
fire-F.ACC good-M.OBL=SIM-M.OBL narrow.down.IDEO=do-M.ACC
hóog-gi-yan]$_{\text{CLAUSE 1}}$ [*mát-oa=rr-áan*
not.do-2SG.PFV.CVB-DS one-M.OBL=NMLZ4-M.LOC
aphph-íti-yan]$_{\text{CLAUSE 2}}$ [***bu<m>bókkoomm***.]$_{\text{CLAUSE 3}}$
seize.MID-3F.PFV.CVB-DS burn<1PL>.APPR
'(Watch out!) If you don't narrow down the fire properly (to the small spot in the centre of the fireplace), it (lit. she = the fire) might light something, and we might burn.' [Elicited, Field notes 2004]

In (27), the pre-emptive action is expressed in a different type of subordinate clause, in a negative conditional/temporal clause (formally a headless, ablative-marked participial clause).

(27) *Rubbat-aan-ch-ú áf-f waal-tumb-óochch*
guarantee-AG-SGV-M.ACC take-2SG.PFV.CVB come-2SG.NEG5-ABL
*gizz-á kam-**ókkoon**-ke.*
money-M.ACC deny-1SG.APPR-2SG.OBJ
'If you don't come with a guarantor, then, (I am afraid,) I might deny you the money (i.e. the loan).' [Elicited, Field notes 2004]

Most frequently, however, we find the pre-emptive action expressed in a sentence adjacent to (i.e. syntactically independent of) the sentence expressing the apprehension-causing situation. A typical example is given in (28): here a following imperative sentence tells the addressee how to avert the danger.

(28) *Tees-ó antabee'-ú súl-u oróoshsh*
now-F.GEN chicken-M.ACC predator.sp-M.NOM go.out.CAUS1.3M.PFV.CVB
*kam-**ókkoo'u**, háy xúud-i!*
do.completely-3M.APPR please.INTJ look-2SG.IMP
'A *sula*-predator (*genetta abyssinica*) might snatch these chickens away, please, look (after them)!' [Mock dialogue, DW2016-04-01]

In much the same way as in (28), pre-emptive situations are expressed in syntactically independent imperative, prohibitive and indicative sentences in other examples of this chapter; see Table 3 for links to the relevant examples.

Table 3: The expression of pre-emptive situations in independent sentences

Preceding indicative sentence	(6)
Following indicative sentence	(13), (20)
Preceding imperative sentence	(9), (10), (14)
Following imperative sentence	(11), (28)
Following imperative interjection	(12)
Preceding prohibitive sentence	(7)

It is important to note that the Kambaata apprehensive is only found in main clauses but used neither in negative purpose clauses nor for direct embedding under ‘fear’-predicates, unlike apprehensives in other languages that the literature reports on (cf. Lichtenberk 1995). Instead, as the following excursus shows, ‘fear’-predicates in Kambaata govern reported speech constructions. Either a direct speech report, as in (29), or an indirect speech report, as in (30), followed by the quotative verb *y-* ‘say’, is dependent on the ‘fear’-predicate.

(29) (…) [*haqq-i-sí* *al-íichch* *úbb* *biix-am-áno*]$_{\text{DIRECT REPORT}}$
tree-m.gen-def top-m.abl fall.3m.pfv.cvb break-pass-3m.ipfv
y-éen *abb-is-éen* *waajj-éemma.*
say-3hon.pfv.cvb become.big-caus1-3hon.pfv.cvb fear-3hon.pfv
‘(…) she (hon) was very afraid that he might fall down from the tree and break (a limb).’ (lit. “‘He will fall …,” (she) having said, she was very afraid).’ (Kambaata Education Bureau 1989: 4.45)

The indirect speech construction specific to ‘fear’-predicates consists of a quote which is marked like a negative purpose clause and which is followed by a quotative verb (30). Negative purpose clauses are negative relative clauses headed by the similative morpheme *=g* ‘like; manner’, i.e. ‘so that the subject does not verb’ is expressed as ‘like the subject does not verb’ (see Treis 2010 and 2017 for details). Thus while the direct quote in (29) ends in a main verb form and could in principle correspond to the original statement made by the fearful subject, the indirect quote in (30) takes a subordinate (relative) form that cannot constitute an independent utterance.

(30) *J-eechch-ú-s* [*makíin-u*
time-SGV-F.ACC-DEF car-M.NOM
fushsh-aqq-úmbo-nne=g-a]$_{\text{INDIR. REPORT}}$ *y-ín*
go.out.CAUS1-MID-3M.NEG5-1PL.OBJ.REL=SIM-M.OBL say-1PL.PFV.CVB
báa<m>beemm íkke.
fear<1PL>.PRF PST
'At the time we were afraid that the bus might leave without us.' (lit. 'So that the bus does not leave on us, (we) saying, we were afraid).' [Elicited, Field notes 2021]

5 Historical origin

The Kambaata apprehensive must have been grammaticalised fairly recently, because – to the best of my knowledge – no similar paradigm is found in other Cushitic languages. Not even the grammars of the most closely related languages, Alaaba (Schneider-Blum 2007), K'abeena (Crass 2005), Hadiyya (Tadesse 2015) and Sidaama (Kawachi 2007, Anbessa 2014), mention a verbal paradigm of similar form or function.[19] The discussion of the origin of the Kambaata apprehensive paradigm can thus not be based on comparative but only on language-internal evidence.

I argue in this section that the apprehensive paradigm is of phrasal origin and the result of the merger of a complex verb form consisting of a same subject affirmative purposive converb[20] plus the existential copula *yoo* – 'exist, be (located)' – both verb forms still exist independently in the language. The diachronic development of the apprehensive is likely to have proceeded in the scenario sketched in Figure 5, with Stage 4 representing the synchronic stage.

At Stage 1, the purposive converb and the copula were independent words forming a complex predicate. At Stage 2, the tight syntactic link between the two words led to phonological reductions. First, the unstressed final vowel (*a*) of the converb was dropped, then the resulting illicit consonant cluster simplified. The initial consonant (*y*) of the copula assimilated to the abutting final consonant

[19] Even if it seems unlikely, I can, of course, not exclude that the apprehensive paradigms escaped the attention of Cushitic grammaticographers. I could not find examples expressing warnings or threats in the grammars of any closely related language to investigate possible alternative means of expressing apprehension.

[20] Kambaata makes a distinction between same subject and different subject affirmative purposives (Figure 1); see Treis (2010) for details.

(1)	Purposive converb + existential copula {Verbal stem}-SBJ1-*ó-ka* + *yóo*-SBJ2	'[SBJ] is about to VERB'
(2)	{Verbal stem}-SBJ1-*ó-k* + *yóo*-SBJ2	
(3)	{Verbal stem}-SBJ1-*ó-k*(-)*koo*-SBJ2	
(4)	{Verbal stem}-SBJ1-*ókoo*-SBJ2	'[SBJ] might VERB (= undesirable)'

Figure 5: The four stages of the apprehensive paradigm

(*k*) of the converb.[21] Thus, at Stage 3, the periphrastic verb became interpreted as a single word, and the stress on the copula was lost. In the synchronic Stage 4, there is no longer any indication of a morpheme boundary between the converb and the copula.

In the following, I present evidence for the plausibility of the diachronic scenario in Figure 5. First, I show that the existential copula is likely to have contributed the second syllable of today's apprehensive morpheme (*koo*) and the second subject index slot (SBJ2). Then, I argue that the first subject index slot (SBJ1) and the first syllable of the apprehensive morpheme (*ok*) can be traced back to a purposive converb. It is important to note that the assumed components of the apprehensive, i.e. the existential copula and the purposive converb, are also attested in closely related Cushitic languages and thus not themselves innovations of Kambaata.

Table 4 demonstrates that the second subject indexes of the apprehensive paradigm are identical to the subject morphemes of the existential copula,[22] but slightly different from the second subject indexes of other main verb forms in the third person masculine, third person honorific/impersonal and in the second person plural/honorific (see the lines formatted in bold in Table 5). Furthermore, the vowel in the second syllable of the apprehensive morpheme *-ókkoo* is a direct reflex of the *oo* of the existential verb stem *yoo-*. As Kambaata does not allow sequences[23] of *k* and *y* and repairs illicit consonant clusters through assimilation, the *y* of the copula can reasonably be assumed to be at the origin of the second *k* in *-ókkoo*.

The first subject indexes of the apprehensive paradigm are identical to those

[21]Progressive (perseveratory) place and manner assimilation processes are observed elsewhere in the language, both synchronically and historically (Treis 2008: 65, 71).

[22]The existential copula is a defective verb. It only inflects for perfective aspect in the indicative mood (see the paradigm in Table 4). Unlike all other indicative main verbs in the language, it only has a single subject index slot.

[23]Kambaata allows essentially only three types of consonant clusters, (i) nasal stop/approximant + obstruent other than glottal stop (e.g. *nt*), (ii) glottal stop + nasal stop/approximant (e.g. /ʔn/ <'nn>) and, exclusive to the causative derivation, (iii) C + *s* (e.g. *fs*).

Table 4: The apprehensive and the existential paradigm compared

		Apprehensive			Existential	
	V	-SBJ1	-APPR	-SBJ2	‘exist’	SBJ
1SG	V	-∅	*-ókkoo*	*-mm*	*yóo*	*-mm*
2SG	V	*-t*	*-ókkoo*	*-nt*	*yóo*	*-nt*
3M	V	-∅	*-ókkoo*	*-’u*	*yóo*	*-’u*
3F/3PL	V	*-t*	*-ókkoo*	*-’u*	*yóo*	*-’u*
3HON	V	*-een*	*-ókkoo*	*-mma*	*yóo*	*-mma*
1PL	V	*-n*	*-ókkoo*	*-mm*	*yóo*	*-mm*
2PL/2HON	V	*-t-een*	*-ókkoo*	*-nta*	*yóo*	*-nta*

Table 5: The second subject indexes (SBJ2) compared across the main verb paradigms

	‘exist’ APPR PROG	PFV	PRF	IPFV
1SG	*-mm*	*-mm*	*-mm*	*-mm*
2SG	*-nt*	*-nt*	*-nt*	*-nt*
3M	***-’u***	-∅/-***’u***	***-’u***	***-no***
3F/3PL	*-’u*	*-’u*	*-’u*	*-’u*
3HON	***-mma***	***-ma***	***-maa’u***	***-no***
1PL	*-mm*	*-mm*	*-mm*	*-mm*
2PL/2HON	***-nta***	***-ta***	***-taa’u***	***-nta***

found in all other verb forms, including those of the same subject purposive converb; compare SBJ1 in Table 4 and Table 6.

It now remains to be argued that the *ók*-sequence in *-ókkoo* goes back to a purposive converb ending – even though, synchronically, the purposive is not marked by **-ó-ka* (as assumed in Figure 5) but by *-ó-ta* (as seen in Table 6), with *-ó* being the purposive morpheme in the narrow sense and *-ta* a case/gender suffix.[24] The *-o* and the *-ta* element of the purposive can synchronically still be separated by an object suffix; see *-’é* ‘(for/to/from) me’ in (31).

[24]Most subordinate verb forms (apart from imperfective and perfective converbs) are case- and gender-marked because Kambaata’s subordinating morphology has its origin in (pro)nouns.

Table 6: The paradigm of the same subject purposive converb

	V	-SBJ1	-PURP.SS
1SG	V	-∅	*-ó-ta*
2SG	V	*-t*	*-ó-ta*
3M	V	-∅	*-ó-ta*
3F/3PL	V	*-t*	*-ó-ta*
3HON	V	*-een*	*-ó-ta*
1PL	V	*-n*	*-ó-ta*
2PL/2HON	V	*-t-een*	*-ó-ta*

(31) *Át esáa m-á fushsh-it-o<'é>ta*
2SG.NOM 1SG.DAT what-M.ACC go.out.CAUS1-2SG-PURP.SS<1SG.OBJ>
kul-áan-ke-la?
tell-1SG.IPFV-2SG.OBJ-PRAG1
'What is my aim of telling you this?' (lit. 'I tell you (this) so that you take out what from me?)' [Recorded conversation, EK2016-02-23_003]

The first argument in favour of a purposive origin concerns the prosody of the apprehensive forms. The morpheme *-ókkoo* imposes the same stress rules as the bare purposive ending *-ó-ta* (i.e. without an inserted object), namely consistent stress on the *ó*, irrespective of the shape of the preceding subject morpheme.

In a second step, the link between the synchronic purposive marking *-ó-ta* and the assumed diachronic input of the apprehensive, i.e. **ó-ka*, needs to be clarified. For this, we have to undertake an excursus on case/gender marking in Kambaata.

From a historical perspective, nouns in certain case forms or of a certain morphological makeup mark case and gender twice ("multiple exponence"), by a primary and a secondary case/gender suffix (Treis 2008: 100). In (32)–(33), the secondary case/gender suffix is formatted in bold.[25]

(32) Feminine noun

a. *áng-a-**t(i)***
hand-F.NOM-F.NOM
'hand'

[25] Note that I do not segment the (historically) primary and secondary case/gender suffixes in other examples of this chapter, i.e. I write *ang-áta* hand-F.ACC, rather than *ang-á-ta* hand-F.ACC-F.ACC/OBL 'hand'.

b. *ang-á-**ta***
hand-F.ACC-F.ACC/OBL
'hand'

c. *ang-áa* ~ *ang-áa-**ha***
hand-F.DAT hand-F.DAT-M.ACC/OBL
'for a/the hand'

d. *ang-áan-**ta**-se*
hand-F.ICP-F.ACC/OBL-3F.POSS
'with her hand'

(33) Masculine noun

a. *adab-óo-**hu***
boy-M.NOM-M.NOM
'boy'

b. *adab-áa-**ha***
boy-M.ACC-M.ACC/OBL
'boy'

c. *adab-ée* ~ *adab-ée-**ha***
boy-M.DAT boy-M.DAT-M.ACC/OBL
'for a/the boy'

d. *adab-éen-**ta**-s*
boy-M.ICP-F.ACC/OBL-3M.POSS
'with his boy'

The *-ta* of the purposive (Table 6) is the same element that is used as a secondary case/gender suffix for the accusative form of most feminine noun declensions (32b). The masculine counterpart of *-ta* is *-ha*, as seen in (33b). The secondary case/gender suffixes, which are likely to have arisen from proximal demonstratives (Treis 2008: 128–130), are no longer gender-sensitive in all contexts; in synchronic Kambaata, dative nouns usually carry a masculine *-ha*, even when the noun is feminine – compare (32c) and (33c). In contrast, possessive-modified instrumental-comitative-perlative and locative nouns take the feminine *-ta* as a secondary suffix, even on masculine nouns – compare (32d) and (33d). In Alaaba, Kambaata's closest relative, secondary suffixes have retained their gender-sensitivity everywhere; see, especially the nominal dative forms in (34).

(34) Alaaba (Cushitic; Ethiopia; Schneider-Blum 2007: 128)
*jaal-áa-**ta*** – *jaal-íi-**ha***
friend-F.DAT-F.ACC/OBL friend-M.DAT-M.ACC/OBL
'for a (female) friend' – 'for a (male) friend'

Having demonstrated that there is a clear functional link between the elements *-ta* and *-ha* (*< *ka*), with both going back to demonstratives and with both presently being used as secondary case/gender markers, among others on accusative and dative nouns, it seems legitimate to hypothesize that the Kambaata same subject purposive converb formerly ended, at least in certain environments, in *-*ó-ka* and not, as at present, (exclusively) in *-ó-ta*.

Additional supporting evidence for the diachronic scenario proposed in Figure 5 comes from the observation that Kambaata has – repeatedly throughout its history – fused periphrastic verb forms into morphologically complex main verb forms. The present-day imperfective, perfective and perfect main verb forms with their two subject index slots (recall Figure 2) can be assumed to go back to a combination of a converb plus an auxiliary – even though not all the forms in the respective paradigms can solely be explained as resulting from the merger of periphrastic verb forms, as pointed out by Tosco (1996). The present-day progressive (Table 7), another recent innovation of Kambaata that is not shared with its most closely related languages, has undergone a development that is parallel to the scenario proposed for the apprehensive (Table 3). A periphrastic verb form consisting of an imperfective converb, marked by *-án*, and the existential copula *yoo-* fused into the progressive morpheme *-áyyoo*. Unlike the source construction of the apprehensive, the periphrastic form that has given rise to the progressive is still occasionally in use today.

Table 7: The origin of the progressive

	V	-SBJ1	-PROG	SBJ2	←V	-SBJ1	-IPFV.CVB	+'exist'	-SBJ2
1SG	V	*-∅*	*-áyyoo*	*-mm*	V	*-∅*	*-án*	*yóo*	*-mm*
2SG	V	*-t*	*-áyyoo*	*-nt*	V	*-t*	*-án*	*yóo*	*-nt*
3M	V	*-∅*	*-áyyoo*	*-'u*	V	*-∅*	*-án*	*yóo*	*-'u*
3F/3PL	V	*-t*	*-áyyoo*	*-'u*	V	*-t*	*-án*	*yóo*	*-'u*
3HON	V	*-een*	*-áyyoo*	*-mma*	V	*-een*	*-án*	*yóo*	*-mma*
1PL	V	*-n*	*-áyyoo*	*-mm*	V	*-n*	*-án*	*yóo*	*-mm*
2PL/2HON	V	*-t-een*	*-áyyoo*	*-nta*	V	*-t-een*	*-án*	*yóo*	*-nta*

After the discussion of the formal development, it is necessary to broach the issue of the semantic changes through time. The proposed scenario for these changes is tentative, but it seems plausible to depart from a periphrastic verb form that expresses intentional/imminent future ('SUBJECT intends to/is about to VERB'). Although the combination of a same subject purposive converb and an existential copula does not express futurity in the synchronic stage of the language,[26] Kambaata uses a parallel combination of the same subject purposive converb and the identificational (non-verbal) copula -V*t* to express 'SUBJECT intends to/is about to VERB' (Treis 2011: 139–141), as illustrated in (35).

(35) *Án íi beet-í qenef-á*
1SG.NOM 1SG.GEN son-M.GEN ceremony.type-M.ACC
aass-aqq-óta-at.
give-MID-1SG.PURP.SS-COP3
'For my own benefit, I intend to/am about to provide (food for) my son's *qenefa*-ceremony.' [Recorded conversation, EK2016-02-23_002]

The semantic change from intentional/imminent future to apprehensive must have proceeded via an intermediate step in which the future event became interpreted as a potential and then as a potential and apprehension-causing undesirable event (Figure 6). In a last step, the implied pre-emptive action against potential, undesirable VERB-ing, i.e. making sure not to VERB, becomes the central meaning component of the second person forms and the apprehensional component is backgrounded or lost.

(1)	'[SBJ] intends to/is about to VERB'	**POSSIBILITY**
(2)	'[SBJ] might VERB'	**FUTURE**
(3)	'[SBJ] might VERB, this is undesirable'	**POSSIBILITY+APPREHENSION**
(4)	'[SBJ] might VERB, this is undesirable, addressee take care or take precautionary measures!'	**POSSIBILITY+APPREHENSION +WARNING**
(5)	'[2nd person SBJ] don't VERB'	**PROHIBITION**

Figure 6: The five stages of the diachronic semantic development of the apprehensive

At the present state, I am missing language-internal or comparative evidence for the change of intentional/imminent future markers into markers expressing possibility (potentiality). The development from a marker along the chain POSSIBILITY > APPREHENSION > WARNING > (2nd person) PROHIBITION is, however, well established through the typological works by Dobrushina (2006) and Pakendorf & Schalley (2007).

[26]No such complex verb forms are attested in my corpus.

6 Conclusion

Kambaata has a dedicated, fully grammaticalised apprehensive paradigm without known parallels in related languages or in languages of the Ethiopian Linguistic Area. Like all Kambaata indicative main verb forms, the apprehensive is marked for seven different subjects by a bipartite, discontinuous subject index. The apprehensive morpheme is found in the same slot as aspectual markers and is thus incompatible with them. The apprehensive verb can carry pronominal object suffixes and pragmatically determined morphology, whose interaction with the apprehensive meaning remains to be investigated. The apprehensive is the only paradigm in Kambaata that does not occur in an affirmative-negative polarity pair. The morphological makeup of the apprehensive is one of several pieces of evidence for its main clause status; other proofs pertain to its sentence-final position and to its ability to govern complex subordinate clauses. The apprehensive is not used in subordinate clauses except in direct speech complements of the verb *y-* 'say', recall (6)–(8) and (22).

The apprehensive is mainly employed in direct dialogue. It conventionally encodes that a situation is unrealised at reference time, considered possible in the future and judged by the speaker (or the reported speaker) to be undesirable, if not dangerous for a discourse participant. The Kambaata apprehensive is always future-oriented and cannot be used to express apprehension about an undesirable situation that was potentially realised in the past (the future orientation of the apprehensive may have been inherited from its source construction; cf. §5). The Kambaata apprehensive is indexical, i.e. anchored in the speech act: The (reported) speaker and not the grammatical subject is the source of the evaluation of a situation as undesirable and potential. Undesirability is part of the conventional meaning of the Kambaata apprehensive, it cannot refer to a potential, pleasant situation. If an apprehensive form of an inherently positive verb such as 'be(come) happy' is formed, it is necessarily understood as bringing an unwelcome type of happiness to the addressee's attention, e.g. the happiness of an enemy (10). I am unaware of any semantic restrictions that would exclude certain verbs from apprehensive clauses.

Although the potential, undesirable situation of which the apprehensive warns is often imminent to the speech situation, imminence is not part of the conventional meaning of the apprehensive, as (36) shows. The undesirable situation, i.e. being dependent on other people, is set in the far future, at the addressees' old age (the addressees are fifth-graders).

(36) *Malees-í roshsh-áta qoorim-áan awwann-é.*
wisdom-M.GEN education-F.ACC intelligence-F.ICP follow-2PL.IMP
Mann-í ang-áta xuud-ú ih-ókkoo'u
people-M.GEN hand-F.ACC look-M.ACC become-3M.APPR
zákk-u-'nne.
old.age-M.NOM-2PL.POSS
'Educate yourself wisely, (otherwise) you (PL) run the risk of being dependent on others.' (lit. '…your old age might be looking at people's hands).' (Excerpt from a poem, Kambaata Education Bureau 1989: 5.90)

The primary function of the Kambaata apprehensive is the warning function, which is attested for first, second and third person forms. In addition, the apprehensive form of the first person may serve as a threat; in this case the undesirable situation is not accidental but potentially inflicted on the addressee by the speaker. The warning/threat polysemy of apprehensives has been reported for a number of languages in the apprehensive literature; see, e.g., Lichtenberk (1995), Faller & Schultze-Berndt (2018), Vuillermet (2018a, 2026 [this volume]), Vuillermet et al. (2026 [this volume]).

The apprehensive solicits a response (in action) of the addressee, who is alerted so as to avert the danger or to prepare themself to alleviate its negative effects. The pre-emptive actions can remain unexpressed. If they are expressed overtly, we most commonly find them in sentences that are adjacent but syntactically independent from the apprehensive sentence (Table 3). If the avertive function of the apprehensive is foregrounded, the interpretation as a directive is facilitated. As in many languages of the world (Dobrushina 2006, Pakendorf & Schalley 2007), the second person form of the Kambaata apprehensive has come to be interpreted as a prohibition, 'you might VERB (watch out! counteract!)' > 'do not VERB!'. However, to date, the prohibition function of second person form has not yet ousted the warning function, and second person apprehensive forms continue to be used to alert the addressee of their unintended, undesirable actions.

Based on detailed language-internal evidence, the final section of this chapter has demonstrated that the apprehensive paradigm is likely to have resulted from the fusion of a periphrastic verb form in the recent history of the language. The source construction consisted of an affirmative same subject purposive converb plus the existential copula, which may have first served to express intentional/imminent future. Via the intermediate stage of epistemic possibility, the verb form came to express apprehension of potential, undesirable events and acquired a warning function. In a last step, the second person apprehensive form developed a prohibition function.

Abbreviations

1	first person
2	second person
3	third person
A_DEM	adnominal demonstrative
ACC	accusative
ADD	additive
AG	agentive
APPR	apprehensive
ASSOC	associative plural
CAUS1	simple causative
CF	contrastive focus
COND	conditional
COP2	identificational, ascriptive copula *-(h)a(a)* (M)/*-ta(a)* (F)
COP3	identificational, ascriptive copula -V*t*
CVB	converb
DAT	dative
DEF	definite
DS	different subject
EMP	emphasis
F	feminine
GEN	genitive
HON	honorific, impersonal
ICP	instrumental-comitative-perlative
IDEO	ideophone
IMP	imperative
INTJ	interjection
IPFV	imperfective
L	linker
LOC	locative
M	masculine
MID	middle
NEG1	standard negator
NEG2	imperative negator
NEG4	converb negator
NEG5	relative negator (negative participle)
NMLZ1	nominalizer -V
NMLZ2	nominalizer *-bii*
NMLZ4	nominalizer =*r*
NOM	nominative
OBJ	object
OBL	oblique
PL	plural
P_DEM	pronominal demonstrative
PASS	passive
PFV	perfective
POSS	possessive
PRAG1	mitigator −*la*
PRAG5	*-be*-suffix (function yet to be determined)
PRED	predicative
PRF	perfect
PROG	progressive
PST	past
PURP	purposive converb
REL	relative
SBJ1	first subject index
SBJ2	second subject index
SG	singular
SGV	singulative
SIM	similative, manner nominalizer
SS	same subject
VOC	vocative

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Chapter 6

Apprehension in Hebrew

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This paper describes five constructions that give rise to apprehensional meaning in Hebrew. These include a modal complementizer and a modal adjective *alul* that encode apprehension conventionally, a temporal additive that gives rise to apprehensive readings pragmatically in particular contexts and a complex complementizer formed with negation. Apprehension in Hebrew is situated within the broader context of apprehensive marking across languages, preliminary remarks are made about diachronic changes, and the outline of an analysis is proposed.

1 Introduction

This paper provides a description, and sketches an analysis, of five constructions that give rise to apprehensional meaning in Hebrew at various diachronic stages, with the aim of contributing both to the understanding of apprehension as an interpretational category and to the description and analysis of Hebrew grammar.

The first construction investigated is the Biblical Hebrew apprehensive marker *pen* 'lest'. This marker occurs at the left periphery of a clause and, intuitively speaking, conveys that the eventuality described by the clause it marks is possible and undesirable from the perspective of the speaker, as well as, potentially, others (the listener, or individuals mentioned in the clause). This marker was lost in the later stage of the language known as Mishnaic Hebrew (roughly, 70BC to 400CE, see e.g. Bar Asher 1999), brought back to usage during the revival period, but fell out of use again in the modern language, where it survives today mostly as a bookish expression used in purposefully high-register writing.

Contemporary spoken and written Hebrew has several other ways of expressing apprehension that I describe here. The first is the apprehensive possibility

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modal *alul* ‘liable’ inherited from Mishnaic Hebrew. The other three are constructions in which an apprehensive interpretation arises inferentially in particular contexts, by the use of expressions that do not conventionally encode either possibility or undesirability. These are the additive/temporal particle *od* ‘still/yet/more/another’, the temporal adverbial *axarkax* ‘afterwards’, and the matrix and non-matrix use of clauses introduced by *še-lo* ‘that not’.

The apprehensive marker *pen* resembles *lest*-type markers found across languages and discussed in the literature (Lichtenberk 1995, Angelo & Schultze-Berndt 2016, Puskás 2017 *inter alia*), including several studies in this volume. It appears in two of the three contexts distinguished by Lichtenberk (1995): apprehensive (marking a matrix clause, as in *lest we forget …*) and precautioning (marking the second clause in a paratactic structure, as in *I will remind you lest you forget*). It does not, however, seem to appear in Lichtenberk’s ‘fear’ context, complementing verbs like *fear*. While *pen*-marked clauses occur with such verbs, there is no clear evidence that they complement them. The apprehensional use of the additive/temporal *od* can arise in matrix and embedded clauses as well as in the complement clauses of verbs of fear. Both *pen* and the apprehensional use of the complementizer + negation sequence *še-lo* raise the problem, discussed in the literature and in several papers in this volume (Vuillermet et al. 2026 [this volume], Wiemer 2026 [this volume]), of lexemes occurring both in paratactic (and sometimes also embedded) structures and in matrix contexts.[1] In the case of *še-lo*, as in the Slavic cases discussed by Wiemer (see also Baydina 2016), the problem is complicated further by the presence of negation which, as shown below, is arguably not clausal negation occurring in a subordinate clause.

The rest of this paper is structured as follows. After describing the distribution of *pen* in §2, §3 argues that the apprehensive marker *pen* is semantically a possibility modal associated with a conventional implicature of bouletic dispreference for the prejacent proposition. Precautioning contexts, I argue, do not involve distinct interpretations of the apprehensive marker. Rather, *precautioning context* should be viewed as a name for a distributional environment, namely for the occurrence of an apprehensively marked clause in construction with another clause. The ‘negative purpose’ or ‘in-case’ inferences that arise in precautioning contexts, i.e. the inference that the eventuality described in the main clause can prevent the one described in the prejacent or some consequence of it, are argued to arise from the textual interpretative effects of parataxis. The analysis I propose for ‘in-case’ and avertive inferences closely resembles that

[1] In the case of *še-lo* this is particularly clear, because, like *lest*, *še* is clearly a subordinator and its occurrence in matrix clauses is both relatively new and highly restricted (cf. *lest we forget*, the only insubordinate occurrence of *lest* commonly attested). See Schwarzwald & Shlomo (2015) and Francez (2015) for discussion.

proposed in AnderBois & Dąbkowski (2021). The section ends with a brief discussion of the diachronic trajectory of *pen* in post-Biblical Hebrew. §4 briefly describes the apprehensive modal adjective *alul*, whose interpretation is essentially identical to that of *pen*. §5 describes and analyzes apprehension with the additive particle *od* 'too/another/still/yet' and with the temporal adverb *axarkax* 'afterwards'. Apprehension in these cases is shown to be a pragmatic inference that arises only in specific contexts. For *od*, I argue that it arises from a special future-oriented, temporal-additive interpretation. In other contexts, this same interpretation leads to an inference that the clause introduced by *od* describes a bouletically *preferred* possibility, an inference I call *promissive*, and which is the opposite of apprehension.

Finally, for the use of complementizer *še* + negation, I discuss some suggestive evidence for the possibility that this complex marker is, or is gradually developing into, a grammaticalized new apprehensive marker. Throughout the paper, data from contemporary Hebrew come from naturally occurring examples from spoken and written genres whenever possible, complemented by my own judgments as well as those of consulted first language users.

2 Biblical Hebrew *pen*

The marker *pen*, exemplified in (1), is one of a class of clause-initial lexemes found in Biblical Hebrew.

(1) *… kax et ištexa ve-et šte bnotexa*
… take.IMP ACC wife.CS.2M.SG and-ACC two.F.CS daughter.PL.CS.2M.SG
ha-mica'ot ***pen*** *tisape ba-avon ha-ir.*
the-present APPR perish.MOD.2M.SG in.the-iniquity.CS the-city
'… take your wife and your two daughters who remain with you, lest you be wiped out in the punishment of the city.' (Genesis 19:15)

Because clauses marked with *pen* are usually in construction with other clauses, *pen* seems at first to be a subordinator, syntactically similar to English *that*. However, as shown below, *pen* clauses also serve as matrix, independent clauses which might be viewed as instantiating "insubordination" in the sense of Evans (2007). Doron (2019), however, classifies *pen* together with a host of other similar clause-initial items as complementizers.[2] The evidence that *pen* is a complemen-

[2]In contemporary generative usage, the term *complementizer* does not entail subordination. Rather, *complementizer* is a label for a distributional category. For example, *wh*-words in English, such as *which* or *who*, are standardly taken to be complementizers. Belonging to this category does entail various properties, such as occupying (or being generated in) a certain position in the clause relative to other elements.

tizer comes from its complementary distribution with a class of elements which function to introduce subordinate clauses. As far as I am aware, however, there are no clear syntactic tests that can determine whether *pen*-clauses involve subordination or coordination, and as Doron notes, both hypotheses can be found in the literature. I assume in this paper that *pen* is not a subordinator, but a left-periphery element occurring in matrix clauses, which in many cases occur as second clauses in a coordinate structure.

Clauses introduced by *pen* are predominantly irrealis, featuring verbs in the *yiqtol* template (i.e. the *yi*-CCoC template). The exact nature of this template is a matter of much controversy. In Modern Hebrew, this is a future tense form used for future time reference, but also has various other irrealis uses that intuitively involve modality. I therefore follow Doron here in glossing this verbal form as MOD for 'modal', without committing to what exactly its semantics should be. In a few cases, *pen* marks a clause with present or past temporal reference.

2.1 Coordination

Clauses marked with *pen* most often occur in construction with another clause, making *pen* a kind of coordinating or perhaps subordinating conjunction. I know of no evidence that *pen* clauses are subordinate clauses, and will assume throughout that they are coordinate matrix clauses in all cases. The *pen* marked clause is, in the vast majority of cases, preceded by a directive clause. A typical example is given in (2).[3]

(2) *U-mi-pri ha-ec ašer be-tox ha-gan amar elohim lo*
and-from-fruit.CS the-tree that in-inside the-garden said God NEG
toxlu mimenu ve-lo tig'u bo ***pen***
eat.MOD.2.M.PL from.3M.SG and-NEG touch.MOD.2M.PL in.3M.SG APPR

[3] All translations of Biblical examples in this paper are taken from Robert Alter's recent translation of the Hebrew Bible (Alter 2018). For simplicity and ease of reading, Hebrew examples in this paper are transliterated as if they were Modern Hebrew. This entails a great simplification and misrepresentation of phonological and morphological information about the Biblical and Mishnaic language as well as the early modern revivalist language. Since representing this information involves complex and often theoretically laden choices which are entirely immaterial to the goals of this paper, I omit it here and indicate only as much information as is needed to understand the examples and relevant aspects of their structure. The reader should bear in mind, however, that as far as pre-Modern Hebrew is concerned, the transliterated examples do not accurately render phonology, morphology, and some of the morphosyntactic structure of verbs. In transliteration and glossing, I use a hyphen (-) to separate non-inflectional morphology, and in glossing I use a full stop (.) to indicate inflectional information.

temutun.
die.MOD.3MP
'But from the fruit of the tree in the midst of the garden God has said, "You shall not eat from it and you shall not touch it, lest you die".' (Genesis 3:3)

As discussed by Azar (1981), in this as in many other occurrences in the Hebrew Bible, the *pen*-clause is preceded by a clause with directive force. While in (2) the preceding clause is not imperative, but rather is also in the *yiqqtol* template, its force is nevertheless directive. An example in which the preceding clause is in the imperative is (1) above. The sentence preceding a *pen*-marked clause need not have directive force, and can be a simple declarative, as in (3).[4]

(3) *Ba-anoxi lo 'uxal lehimalet ha-hara* ***pen***
and-I NEG can.MOD.1SG escape.INF the-mountain.ALL APPR
tidbakani ha-ra'a va-matti.
catch.MOD.3F.SG.1SG the-evil.F and-die.1SG
'But I cannot flee to the high country, lest evil overtake me and I die.' (Genesis 19:19)

In some cases, the *pen*-marked clause could, potentially, be analyzed as the antecedent of a conditional (and is indeed thus analyzed by Azar 1981), though this is not the only plausible analysis. An example is (4).

(4) *Arba'im yakenu lo yosif* ***pen*** *yosif*
forty hit.MOD.3M.SG:3M.SG NEG add.MOD.3M.SG APPR add.MOD.3M.SG
lehakoto al ele maka raba ve-nikla axixa
hit.INF.3M.SG on these blow great and-dishonour.3M.SG brother.CS.2M.SG
le-eynexa.
to-eyes.CS.2M.SG
'Forty blows he may strike him, he shall not go further, lest he go on to strike him beyond these a great many blows, and your brother seem of no account in your eyes.' (Deuteronomy 25:3)

While Robert Alter's translation does not render these verses as conditionals, other translations do. For example, the JPS Bible translates this passage as follows: *If he should exceed, and beat him above these with many stripes, then thy*

[4] The form *va-matti* in (3) is an instance of the *ve-qatal* verbal form, which is, very roughly, an irrealis form occurring in narrative sequence. See Hatav (1997) for a discussion of the complicated system of indicating time and modality in Biblical Hebrew within a generative framework.

brother should be dishonoured before thine eyes. Another possibility is that the entire complex clause following *pen* is in its syntactic scope. I cannot determine here with any certainty whether, in this and similar cases (such as (2) above), a conditional analysis should be adopted. If such an analysis is correct, however, the result, while broadly compatible with the informal idea that *pen* marks its prejacent as expressing a dispreferred possibility, has potential consequences for the analysis of conditionals.[5]

While *pen* mostly introduces clauses with a future or modal interpretation, in a few cases it marks a clause with a present or past temporal interpretation. This is shown in (5), where the form *yeš* has only present temporal reference, and in (6) where the verbal forms have only a past temporal reference.

(5) ***Pen*** *yeš baxem iš o iša o mišpaxa o švet ašer levavo*
APPR exist in.3M.PL man or woman or family or clan that heart.CS.3M.SG
pone ha-yom me-im YHWH elohenu … lo yove YHWH
turns the-day from-with YHWH God.CS.1PL … NEG agree YHWH
salo'ax lo.
forgive.INF to.3M.SG
'Should there be among you a man or a woman or a clan or a tribe whose heart turns away today from the LORD our God to go worship the Gods of those nations … The LORD shall not want to forgive him.' (Deuteronomy 29:17)

(6) *Yelxu na v-yvakšu et adoneyxa* ***pen***
go.MOD.3PL DIR and-search.3PL ACC master.CS.2SG APPR
nesa'o ru'ax YHWH va-yašlixehu be-axad
carried.3M.SG.3M.SG spirit.CS YHWH and-throw.3M.SG.3M.SG in-one.CS
he-harim o be-axat ha-geva'ot.
the-mountain.PL or in-one.F.CS the-valley.PL
'Let them go, pray, and seek your master, lest the spirit of the LORD has borne him off and flung him down on some hill or into some valley.' (2 Kings 2:16)

[5] For example, conditional antecedents are not generally assumed to have the assertive content of a modal, though they are plausibly assumed to carry a modal presupposition (see e.g. Leahy 2011). Furthermore, sentences like (4) carry an inference of conditional perfection, and the question arises whether this inference is cancellable or not. For example, (4) implies that if the blows do not exceed a certain threshold, the resulting dishonouring of the victim is avoided. Given the nature of the corpus, it is difficult to determine whether this implication is cancellable or not. If it is not, then a conditional analysis of the sentence would have to account for why it is not, as conditional perfection is not generally uncancellable.

The interpretation of (5), as Alter's translation indicates, is another case in which a *pen* clause is interpreted similarly to the antecedent of a conditional. However, it could equally be seen as a matrix occurrence of *pen*, simply asserting the possibility of something undesirable. Sentence (6) is an instance of what Lichtenberk called the 'in-case' interpretation, where the *pen*-marked clause describes an issue which is already metaphysically settled at the time of utterance, but remains epistemically open. In this case, searching for the master (the prophet Elijah in the context) cannot possibly affect the already determined facts about whether the Lord has flung him on a hill or into a valley, but can avert some undesirable consequence of this possibility. These examples are discussed in more detail in §3.

2.2 Verbs of fear and precaution

Examples (7) and (8) demonstrate occurrences of *pen* in construction with verbs of precaution or fear, respectively.

(7) *Hišamer lexa* ***pen*** *tašiv et bni šama.*
beware to.2M.SG APPR return.MOD.2M.SG ACC son.CS.1SG there.ALL
'Watch yourself, lest you bring my son back there.' (Genesis 24:6)

(8) *Ki yare anoxi oto* ***pen*** *yavo ve-hikani*
for fear.1M.SG I ACC.3M.SG APPR come.MOD.3M.SG and-strike.3M.SG.1SG
em al banim.
mother on son.PL
'For I fear him, lest he come and strike me, mother with sons.' (Genesis 32:12)

Cases like (7) can be viewed as involving a *pen*-marked complement clause. In the context, the *pen*-marked clause in (7) arguably describes the content of what is to be avoided, rather than merely the justification for the directive 'watch yourself'. The directive issued by the speaker (Abraham in the context) is an answer to a question: 'should I bring your son back'? The issued directive is to be careful not to bring the son back. In other words, the interpretation of this verse could equally well be paraphrased as 'be careful not to bring my son back from there' (though the *pen*-marked clause contains no negation). While the verb *hišamer* 'beware' does not require a complement, and can occur intransitively, optional complements are by no means an oddity. Examples like (7) can, however, also be viewed as involving a directive and a *pen*-marked clause in parataxis, instantiating what Lichtenberk (1995) calls the precautionary function of apprehensive markers (see §3.2 below). Semantically speaking, an analysis in which the apprehensive clause is a semantic argument seems to me more straightforward, but I leave the question undecided here.

In (8) as well, one might argue that the *pen*-marked clause is a complement clause indicating exactly what the speaker, Jacob, fears about his brother Esau. This line of analysis is reflected in the English Standard Version Bible translation of the verse: *Please deliver me from the hand of my brother, from the hand of Esau, for I fear him, that he may come and attack me, the mothers with the children.*

However, as Azar (1981) argues, there is not a single clear instance of a *pen* marked clause occurring as the direct complement of a verb of fear in the Bible. This makes an analysis in which the *pen* clause in (8) is an adjunct clause providing the motivation for the matrix assertion ('I fear him') seem more plausible.

2.3 Stand alone matrix occurrences

In a few cases, *pen* can also occur on a matrix clause that is not in construction with a preceding clause, in which case the sentence asserts that something undesirable is possible. Examples are given in (9)[6] and (10).

(9) *Ve-ata* ***pen*** *yišlax* *yado* *ve-lakax* *gam*
and-now APPR send.MOD.3M.SG hand.CS.3M.SG and-take.3M.SG also
me-'ec *ha-xayim* *ve-axal* *va-xay* *le-olam.*
from-tree.CS the-life and-eat.3M.SG and-live.3M.SG to-eternity
'He may reach out and take as well from the tree of life and live forever.' (Genesis 3:22)

(10) ***Pen*** *yasit* *etxem* *hizkiyahu* *leemor* *YHWH*
APPR incite.MOD.3M.SG ACC.3M.PL Hezekiah say.INF YHWH
yacilenu *…*
save.MOD.3M.SG.1PL
'Lest Hezekiah mislead you, saying the LORD will save us …' (Isaiah 36:18)

[6] In the Biblical context, this sentence describes God's reaction to Adam and Eve having eaten from the tree of knowledge, acquiring the divine trait of moral judgement. The undesirable possibility here is that they would also acquire the divine trait of immortality, thus achieving a God-like status.

3 *pen* as a modal complementizer and the semantics/pragmatics of apprehension

The data presented above show that all occurrences of *pen* in the Biblical corpus always involve both of the characteristic inferences of apprehension: possibility and bouletic dispreference. This, coupled with the fact that *pen* can occur in a matrix clause and hence is not straightforwardly a subordinating (or coordinating) conjunction, makes it natural to analyze it as a modal complementizer.[7] Semantically, I assign *pen* the meaning of a possibility modal that furthermore carries a conventional implicature that the prejacent proposition is bouletically dispreferred. Since Karttunen & Peters (1979), conventional implicatures are standardly understood to be automatic updates of the common ground between interlocutors. *Pen*, therefore, takes a proposition, asserts its possibility, and, automatically adds to the common ground that it is undesirable, i.e. false in all the worlds in which the speaker's desires are met.[8] While the limited corpus of the Hebrew Bible does not afford the possibility of testing that undesirability is backgrounded, rather than at issue, content, this seems a reasonable assumption given the nature of similar markers in other languages. For example, the dispreference inference associated with English *lest* is clearly not at issue.[9] The semantics proposed for *pen* is given in (11). That the undesirability content is not presupposed is clear from the contexts in which *pen*-marked clauses appear, in which it is not generally already established in the common ground that the eventuality described by the clause is undesirable. For example, in the context for (7) above, the information that bringing the son back is undesirable is not only not common ground, but entirely new to the addressee.

(11) **The semantics of *pen*:**
A clause with the complementizer *pen* expressing a proposition *p*:
a. asserts the possibility of *p*

[7] See Footnote 2 above.

[8] The assumption that it is the speaker's desires, rather than the addressee's, for example, is difficult to establish within the relatively small corpus of the Biblical text. I have not seen any clear cases where the context clearly distinguishes the speaker's bouletic preferences from the addressee's and where the *pen*-clause clearly alludes to the latter rather than the former.

[9] This is evidenced, for example, by the fact that the dispreference inference cannot be targeted by explicit denial moves. In the dialogue in (i), B's denial can either target the proposition that he was quiet, or the causal relation asserted to hold between that proposition and the dispreference for waking up the children, but it cannot be used to only deny that dispreference.
(i) A: He was quiet lest he wake up the children. B: No, that's not true.

b. conventionally implicates that there is a contextually salient q that is causally dependent on p and bouletically dispreferred.

Clause (11b) does not directly state that the event described in the prejacent of *pen* is what is dispreferred, but rather that some causal consequence of this event is dispreferred. This feature of the analysis, which seems strange at first, is what allows it to capture the difference between the three "functions" identified by Lichtenberk (1995), as explained in the rest of this section.

3.1 The apprehensional-epistemic function

What Lichtenberk calls the apprehensional-epistemic function is the case in which an apprehensive marker occurs in a matrix context, as exemplified in the To'aba'ita example (12).

(12) To'aba'ita (Austronesian; Lichtenberk 1995: 294)
Ada *'oko* *mata'i.*
lest you.SG.SEQ be.sick
'You may be sick.'

That Biblical Hebrew *pen* can occur in matrix contexts with the same kind of meaning was shown in examples (9) and (10) in §2.3 above. As the translations of those examples indicate, these sentences assert the possibility of an eventuality taken to be undesirable by the speaker. In terms of the semantics proposed in (11), these are cases in which the proposition q conventionally implicated to be undesirable is identical to the proposition p expressed by the prejacent of *pen*. Since every proposition is causally dependent on, and a consequence of, itself, a special case of (11b) is the case in which $p = q$, and it is this case which comprises the so-called apprehensional-epistemic function.[10]

3.2 The precautionary function

What Lichtenberk calls the precautionary function of apprehensive markers are those cases in which an apprehensive clause occurs in hypotaxis and/or parataxis with another clause. These cases, exemplified for *pen* in §2.1 above, give rise to inferences beyond undesirability, which Lichtenberk calls the avertive and 'in-case' inferences. The avertive inference is the inference, seen in examples like (1),

[10]This is a point where the proposed analysis diverges from the one developed in AnderBois & Dąbkowski 2021, where the apprehensive item analyzed, the A'ingae morpheme *=sa'ne*, does not have stand-alone matrix occurrences, and is therefore analyzed as always relating two propositions that are compositionally supplied.

repeated in (13), that the eventuality described by the non-apprehensive clause would avert the undesirable possibility described by the *pen*-marked clause.

(13) *… kax et ištexa ve-et šte bnotexa*
… take.IMP ACC wife.CS.2M.SG and-ACC two.F.CS daughter.PL.CS.2M.SG
ha-mica'ot ***pen*** *tisape ba-avon ha-ir.*
the-present APPR perish.MOD.2M.SG in.the-iniquity.CS the-city
'… take your wife and your two daughters who remain with you, lest you be wiped out in the punishment of the city.' (Genesis 19:15)

The 'in-case' inference is the inference, seen in examples like (6), that the eventuality described by the non-apprehensive clause cannot avert the possibility described by the *pen*-marked clause (in this case, because the question of the realization of this possibility is already settled), but can prevent some undesirable consequence of it. In example (6), repeated here as (14), it is already settled, at utterance time, whether the possibility that God has carried the addressee's master somewhere has been actualized. It nevertheless remains epistemically undetermined for the interlocutors. The relevant inference is that searching for him might avert some undesirable consequence of this possibility (e.g. that the master will perish if left unattended.)

(14) *Yelxu na v-yvakšu et adoneyxa* ***pen***
go.MOD.3PL DIR and-search.3PL ACC master.CS.2SG APPR
nesa'o ru'ax YHWH va-yašlixehu be-axad
carried.3M.SG.3M.SG spirit.CS YHWH and-throw.3M.SG.3M.SG in-one.CS
he-harim o be-axat ha-geva'ot.
the-mountain.PL or in-one.F.CS the-valley.PL
'Let them go, pray, and seek your master, lest the spirit of the LORD has borne him off and flung him down on some hill or into some valley.' (2 Kings 2:16)

In terms of the semantics of *pen* proposed in (11) above, the 'in-case' inference is just a special case of the avertive inference. In both cases, what is to be averted is the contextually salient proposition *q* that is conventionally implicated to be bouletically dispreferred and causally dependent on the proposition. In the case of avertive inferences, this *q* is just the prejacent itself (as was the case in matrix cases discussed in the previous subsection), and in the case of 'in-case' inferences, the two are distinct, with *q* being a potential consequence of the prejacent proposition.

Given that avertive and 'in-case' inferences are not present when *pen* occurs in a matrix context, they cannot be part of the conventional meaning of *pen*, raising the question why they arise when *pen* clauses occur in construction with another clause. My suggestion is that these are inferences due to discourse coherence principles, which may be ultimately rooted in Gricean reasoning.

If avertive/'in-case' inferences are not linked to the conventional meaning of *pen*, then they are expected to arise quite independently of the presence of this marker, which they indeed do, as shown in (15).[11]

(15) *Take your family and leave. You might perish with the destruction of Sodom.*

As stated, the idea that avertive inferences are linked to textual/discourse coherence principles is not much more than an intuition. Turning it into an analysis requires a worked out theory of such coherence principles, which I cannot offer, but on which there is an extensive literature (see for example Hobbs 1985, Lascarides & Asher 1993, 2003, Asher & Lascarides 2003). However, I suggest here a rough outline of how standard Gricean pragmatic reasoning might account for how these inferences arise.

Generally speaking, I propose that the avertive inferences associated with parataxis are parasitic on an *explanation* discourse relation conventionally associated with parataxis.[12] The apprehensive clause, which is the second element in the parataxis, stands in an explanatory relation to the first one, and the explanation has to do with an avertive relation between events.

How this happens seems to be fairly straightforward when the first sentence in the paratactic relation is a directive prescribing an action. In this case, the undesirable possibility named by the *pen*-marked clause explains the motivation for issuing the directive. Specifically, what motivates the directive is precisely that taking the prescribed action can avert the undesired possibility. The reasoning involved is illustrated in (16) ($\rightsquigarrow$ indicates a conclusion that follows from the listed premises, a conclusion that the hearer is likely to draw and that the speaker expects her to draw).

(16) **Deriving avertive inferences**
- A directive clause S_1 expresses a preference for the addressee to do A, rather than $\neg A$.

[11]Of course, the reverse does not hold – the fact that an inference arises independently of the presence of a marker does not preclude that it is also encoded conventionally by that marker. For example, causal inferences encoded by expressions like *because* often arise in the absence of this marker, as in a discourse like *I left. It was too noisy.*

[12]*Explanation* is one of the discourse cohesion relations explored more elaborately and formally by Asher & Lascarides (2003).

– An apprehensively marked clause S_2 asserts that an undesirable eventuality B, which is a consequence of the eventuality described by the clause and possibly that eventuality itself, is a possibility.
– The paratactic relation S_1, S_2 indicates that the undesirable possibility B is an *explanation* for the preference for A over $\neg A$ expressed by S.
$\rightsquigarrow$ Speaker recommends doing A because she believes not doing A leads to B, whereas doing A averts B.

In other cases, however, such as (17), the first clause, a declarative, does not describe an eventuality that can be inferred to avert anything.

(17) *Va-anoxi lo 'uxal lehimalet ha-hara* ***pen***
and-I NEG can.MOD.1SG escape.INF the-mountain.ALL APPR
tidbakani ha-ra'a va-matti.
catch.MOD.3F.SG.1SG the-evil and-die.1SG
'But I cannot flee to the high country, lest evil overtake me and I die.' (Genesis 19:19)

In (17), this first clause is modal, and asserts that Lot cannot flee to the high country. Not being able to flee obviously does not avert the undesired possibility, named in the *pen*-marked second clause, that Lot will die prematurely. The paratactic relation, however, relates this undesired possibility to the assertion of Lot's inability to flee to the high country as an explanation. Presumably, the reason that fleeing to the high country is not an option is precisely the possibility of being overtaken by evil and dying there. If this, however, is the explanation for the assertion, then the hearer can safely conclude that not fleeing to the high country will avert this undesired possibility.

The avertive inference in (17) arises differently than the one in (15) above, which involves neither an apprehensive marker nor syntactic parataxis. In (15), the fact that the second sentence describes an undesirable eventuality is entirely a fact of world knowledge and a specific context, rather than conventional marking. Replacing that sentence with one describing a stereotypically desirable eventuality, as in (18), destroys the avertive inference (though the causal inference based on explanation remains).

(18) *Take your family and leave. You might find a better place to live.*

So-called 'in-case' inferences are just avertive inferences in which world knowledge rules out a causal relation between the eventuality described by the main clause and the one described by the prejacent of the apprehensive marker, leading the hearer to infer that it is some other consequence of the prejacent that is to be averted.[13] The pragmatic reasoning in this case works in the same way as in avertive inferences. The hearer takes the conventionally implied undesirable consequence of the eventuality described by the prejacent to be the explanation for the speaker's assertion (or her recommendation to do A rather than $\neg A$), but in this case this consequence is not the prejacent-eventuality itself.

In summary, this section proposed that the Hebrew apprehensive marker *pen* is a complementizer or a left-periphery particle with the semantics of a possibility modal, and a conventional implicature that some contextually inferable consequence of the prejacent is disprefered by the speaker. This makes the occurrence of *pen*-clauses in stand-alone matrix contexts entirely unremarkable. Since *pen*-marked clauses give rise to so-called avertive and 'in-case' inferences only when they appear in construction with another clause, and since such inferences arise also when no apprehensive marker is present, it is natural to view them as linked to textual structure. This remains, however, an indication of a potential direction rather than an analysis. Specifically, the idea is that these inferences arise pragmatically from reasoning about an explanatory relation between the two clauses, with the prejacent interpreted as providing an explanation for the assertion of the non-apprehensive clause, and world knowledge determining whether what is to be averted is the eventuality described by the *pen* clause or some other causal consequence of it.

Finally, the status of avertive and 'in-case' inferences emerges here as a point of cross-linguistic variation. In languages which have an apprehensive marker that is always subordinating (or coordinating, if there are languages with purely coordinating apprehensive markers), these inferences are conventional and are always present when the marker is present. An example of such a language seems to be A'ingae as analyzed by AnderBois & Dąbkowski (2021). In languages like Hebrew, the apprehensive marker occurs in stand-alone matrix contexts and is essentially a possibility modal. In such languages, the relevant inferences must be derived through the interplay of the conventional meaning of the marker and something else, which, if my proposal is in the right direction, is the coherence effects of parataxis.

[13]In the Biblical corpus, 'in-case' inferences arise only when the *pen*-marked clause is in the present or past tense. This makes it tempting to try and derive 'in-case' inferences from the fact that the undesirable possibilities involved are metaphysically settled, and present only epistemic possibilities. However, the general availability of 'in-case' inferences with future-oriented clauses, as in *take an umbrella lest it rain*, argues against this line of analysis. I thank Eva Schultze-Berndt for pointing this issue out to me.

3.3 The trajectory of *pen* in post-Biblical Hebrew

This section provides a brief and preliminary discussion of *pen* in post-Biblical varieties of Hebrew. Biblical Hebrew ceased to be a spoken language around the 2nd century CE. In the subsequent variety of Mishnaic Hebrew, *pen* was entirely replaced by the more general particle *šema* 'whether/if/maybe', used also in matrix and embedded questions, which marks possibility and can, but does not have to, convey undesirability (see Segal 2001).

(19) *Hevey zahir bi-dvarexa* ***šema*** *mi-toxam*
be careful in-words.CS.2M.SG maybe from-inside.CS.3M.PL
yilmedu lešaker.
learn.MOD.3M.PL lie.INF
'Be careful with your words lest from them they will learn to lie.' (Avot 1:9)

After the 2nd century, Hebrew no longer had native speakers, and subsequent varieties of Hebrew were written varieties, and the language did not have a spoken variety with a stable community of native speakers until the early 20th century. In Renaissance-era and early Modern Hebrew texts, *pen* is sometimes used as a modal/mood marker, co-occurring with a complementizer.

(20) *Hodi'enu ma šimxa lada'at mi nexabed* ***ki pen***
tell.2M.SG.3PL what name.3M.SG know.INF who honor.1PL.MOD for APPR
lo nakirxa az.
NEG know.1PL.3M.SG then
'Tell us your name so that we know who we are honoring for we might not recognize you then.' (Rabbi David Alteschueler, Mecudat David, 1750s)

During the *haskala*, the Jewish Enlightenment movement, Hebrew revivalists, like Abraham Mapu (1808–1867), began using Biblical Hebrew in writing modern, secular, novelistic prose. In Mapu's writing, *pen* is used largely as described above for Biblical Hebrew. For later revivalists, *pen* starts losing its conventional implicature of undesirability, and is used in positive as well as in the neutral contexts in which Mishnaic *šema* is used.[14] This is exemplified in (21)–(23). In (21), the maid, who is lusting over the speaker, desires the possibility that he come

[14] All examples in this section are taken from the cited texts as they appear in the Ben-Yehuda Project, accessed using the repository created and kindly shared by Aynat Rubinstein for the Jerusalem Corpus for Emergent Modern Hebrew (see Rubinstein 2019). I thank Noam Tedelis for his invaluable help with the corpus work. The translations are my own.

close to her. In (22), the lepers desire the possibility that they be healed. In (23), which is a line from a poem, the narrating voice describes parents who are going out to the street to see if anyone has seen or heard their son, who is late coming home. In this case, the possibilities are just that, possibilities, neither desired nor undesired (hence the translation with *maybe*).

(21) *Le-tum-i lo yadati az, ki nefeš ha-ama*
to-innocence.CS-1SG NEG know.PST.1.SG then, that mind.CS the-maid
ogevet alay bišvil cniut-i, ve-hi šoha be-xadr-i
lust.3F.SG on.1SG for modesty.CS-1SG and-she stay.3F.SG in-room.CS-1SG
yoter mi-day ***pen*** *ve-'ulay 'etkarev 'ele-ha.*
more than-enough APPR and-maybe approach.1SG to-3F.SG
'In my innocence, I did not then understand that the maid is lusting over me for my modesty, and is lingering in my room, to see whether I might come close to her.' (Berdichevsky, *ˁorva parax*, 1900)

(22) *Va-yišmeu ha-mecora-im va-yismexu al ha-davar va-yomru*
and-heard.3PL the-leper-PL and-rejoiced.3PL on the-thing and-said.3PL
iš el axiv: nelxa gam anaxnu el ha-arec ha-hi, ***pen***
man to brother.CS:3M.SG go.IMP.PL also we to the-land the-that, APPR
niga'el gam 'anaxnu.
be.saved.1PL also we
'And the lepers heard this and were glad, and said to each other: let us also go to that country, we might also be saved.' (Berdichevsky, *ve-anu ba cidkata*, 1908)

(23) ***Pen*** *šama šome'a /* ***pen*** *ra'u ha-roim / lo ra'ata ayin / ozen lo*
APPR heard hearer APPR saw.PL the-seeing NEG saw eye ear NEG
šam'a
heard.
'Maybe someone has heard something / Maybe someone has seen something / No eye has seen anything / No ear has heard anything.' (Bialik, *ha-na'ar ba-ya'ar*, 1933)

In contemporary spoken Hebrew, *pen* has, together with all the other Biblical and Mishnaic left-periphery items, been abandoned in favor of a single complementizer/subordinator *še*. In writing, *pen* is still used, but mostly in high register prose style. Modern Hebrew does, however, have a more productive dedicated apprehensive marker, namely the modal *alul* 'can', discussed in the next section.

4 The modal *alul*

Contemporary Hebrew inherited from Mishnaic Hebrew the two adjectives *asuy* and *alul*, the literal meaning of which is roughly 'made/done', but both of which have a modal sense of being prone or inclined by nature towards something. Examples of this modal sense, which is the one of interest here, are given in (24) and (25). I gloss both adjectives as 'can' throughout.

(24) *ktav še-hu* ***asuy*** *lehištanot*
writing that-it can change.INF
'a script that is prone to change' (Palestinian Talmud, 71, b-c)

(25) ***alul*** *lekabel tum'a*
can receive.INF impurity
'prone to become impure' (Tosefta, Oholot 15)

In the Mishnaic sources, *asuy* is used to describe propensity towards undesirable as well as desirable or neutral things, whereas *alul* appears only once, in the example cited, which involves a propensity towards something undesirable.

In the early revivalist period, both *asuy* and *alul* are used with a more general possibility sense 'can', not clearly specific to inherent propensity, and with no bouletic implications, as in (26), where *alul* takes an infinitive describing a presumably bouletically preferred positive possibility.

(26) *kol ma še-eyno* ***alul*** *lehavi lo revaxim*
all what that-not can bring.INF to.3M.SG profit.PL
'everything that can't bring him profits' (Yosef Brener, *šxol ve-kišalon*, 1922)

In the usage of many contemporary speakers, myself included, *alul* and *asuy* belong to a higher register and are not used as commonly in speech as the more colloquial modal adjective *yaxol* 'can'. When used, however, *alul* is associated with a negative bouletic implication.[15] For such speakers, the (constructed) sentence in (27) must be read as construing the addressee's success as undesirable, e.g. in a context in which the speaker and addressee are adversaries, or one in which the speaker is being ironic, implying that the addressee is scared of the consequences of success (cf. *of course you don't want to try to run for office, you might succeed and then you'll have to put your money where your mouth is*).

[15] The public perception is that this difference between *alul* and *asuy*, with the former carrying a negative bouletic implication and the latter neutral, is a historically "correct" one that has been lost, see e.g. the discussion in the online forum of the Academy of the Hebrew Language at https://hebrew-academy.org.il/2018/12/17/עשוי-ועלול/

(27) *Ata* ***alul*** *lehacliax.*
you can succeed.INF
'You might (God forbid) succeed.'

This use of *alul* as a possiblity modal adjective can be analyzed in exactly the same way as proposed above for *pen*, the difference between the lexemes being combinatoric. While *pen* introduces a finite clause, *alul* is a "raising" adjective which takes a nominal subject and an infinitival complement, similarly to English modal adjectives like *likely*, *liable*, etc. In terms of interpretation, *alul* takes as its prejacent the proposition formed by applying the meaning of the infinitive to that of the subject, and the semantic analysis is the same as proposed above for *pen*.

5 The apprehensive use of additive *od* and temporal *axarkax*

This section describes the apprehensive use of two temporal expressions that is common in contemporary spoken Hebrew. The first is the additive particle *od*, the second is the temporal adverb *axarkax* 'afterwards/later'.

Generally, *od* has the meanings expected of an additive particle (see Greenberg 2012). When modifying a clause, it is interpreted as an aspectual 'continuative' adverbial equivalent to English *still* and German *noch*.[16] When modifying a nominal, *od* is interpreted as 'more'/'another'. This is shown in (28).[17]

(28) a. *Šarti* ***od*** *širim.*
sang.1sg ADD song.PL
'I sang more/other songs.'

b. *Hu* ***od*** *po.*
he ADD here
'He's still here.'

When *od* modifies a clause in the future tense, however, it has an interpretation different from the aspectual adverbial one, and on this interpretation, which I henceforth call *boding*, it can give rise to apprehensive inferences, as in the naturally occurring examples in (29).

[16] On aspectual adverbials, see Löbner (1989) and subsequent literature.
[17] Hebrew *od* does not have the interpretation 'also/too', for which *gam* 'also' is used.

(29) a. *Al tedabri ito, hu* ***od*** *yaxšov*
neg.imp speak.fut.2f.sg with.3m.sg he add think.fut.3m.sg
še-at meunyenet.
that-2f.sg interested
'Don't talk to him, he'll end up thinking you're interested.'

b. *Amarti lo liyot be-šeket, hu* ***od*** *haya meir et*
told.1sg him be.inf in-quiet he add was.3m.sg wake.m.sg acc
ha-banot.
the-girl.pl
'I told him to be quiet, he would have woken up the girls.'

The boding reading of *od* is not new, though its use in contexts of apprehension seems to be. In the Hebrew Bible, the boding reading of *od* occurs a few times in prophecy, always in the genre called "prophecies of consolation", where the prophet prophecises a redemptive remote future contrasting with a wretched present and immediate future, as in (30) from Jeremiah.

(30) ***Od*** *evnex ve-nivnet betulat yisrael.*
add build.mod.1sg.2f.sg and-be.built.2f.sg maiden.cs Israel
'Yet will I rebuild you and you will be built, O Virgin Israel.' (Jeremiah 31:3)

In the early revival literature, boding *od* starts to appear also in contexts where it leads to apprehensive inferences, as in the following examples from author Avraham Mapu.

(31) *Da lexa ki lo arev ani lexa, ve-****od*** *tišlax*
know.imp to.you that neg assuring I to.you and-add send.mod.3m.sg
yad be-nafšexa.
hand in-soul.cs.2m.sg
'Know that (then) I cannot assure you and you might yet kill yourself.' (Mapu, *ayit cavua*, 1857)

(32) *Kaspi ve-zehavi yaase acabim, ve-****od*** *yefarek*
money.cs.1sg and-gold.cs.1sg make.mod.3m.sg idols, and-add take.off
nizmex ve-xelyatex ve-herimam le-vošet.
jewelry.cs.2f.sg and-ring.cs.2f.sg and-offer.mod.3m.sg.3pl for-idols
'He uses my money and my gold to make false idols and will yet remove your jewelry and rings and offer them for idols.' (Mapu, *ašmat šomron*, 1865)

The boding reading of *od* is easily distinguishable from the aspectual one. On the temporal 'still' reading, *od* is always paraphrasable with the temporal adverbial *adayin* 'still/yet', a paraphrase not available in (29). As mentioned in the introduction, however, when *od* modifies a future verb phrase, it can also give rise to promissive inferences, which are the opposite of apprehension in that the actuation of an anticipated possibility is desired. This is exemplified in (33), a line of poetry, and (34), which is naturally occurring. In both of these examples, *od* clearly is not interpreted as 'still'.

(33) ***Od*** *omar lax et kol ha-milim ha-tovot še-yešnan,*
ADD say.MOD.1SG to.you ACC all.of the-words the-good that-exist.3F.PL
še-yešnan adayin.
that-exist.3F.PL still
'I will say to you yet all the beautiful words that there are, that there are still remaining.' (Natan Alterman, *od ašuv el sipex*, 1938)

(34) *Al tafsik lehitamen! Ba-sof* ***od*** *tenaceax oti.*
NEG.IMP stop.2M.SG practice.INF in.the-end ADD beat.FUT.2M.SG me
'Don't stop practicing! In the end you will yet[18] beat me.'

The boding reading of *od* is also distinguishable from its aspectual 'still' reading in its interaction with negation. The temporal adverbial *od*, when occurring with negation, predictably means *not yet*, as in (35). When *od* occurs with negation on its boding reading, as in the naturally occurring example (36), it cannot be interpreted as *not yet*. Instead, it communicates that the negative eventuality described by the modified verb phrase is possible and (un)desirable. In example (36), which occurs in the context of a story about failures to deal with sexual harassment in organized sports, the writer is being sarcastic.

(35) *Be-eser hu* ***od*** *lo yagia.*
in-ten he ADD NEG arrive.FUT
'At ten he will not yet arrive.'

[18] There is an interesting parallel to be drawn between additives with boding readings like *od* and the positive polarity use of *yet* exemplified in this translation. Positive polarity *yet* also bears some interesting connection to additives, as seen in locutions like *yet again* and *yet another*. Positive polarity *yet* seems to me to be amenable to the same analysis proposed below for *od*, but an exploration of this item and of this hypothesis must be left for a future occasion.

(36) *Im ze yimašex kaxa, ba-sof* ***od lo*** *tiye*
if this continue.FUT.3M.SG thus in.the-end ADD NEG be.FUT.3F.SG
lahem brera ela lemanot iša la-tafkid.
to.3M.PL choice except appoint.INF woman to.the-position
'If it goes on like this, in the end they will have no choice but to appoint a woman to the position'.[19]

Finally, the two uses also differ in prosody. While (35) can have pitch accent on *od*, this is not possible in (36).

The fact that *od* on its boding interpretation need not lead to apprehensive inferences shows that apprehension is not part of the conventionally encoded meaning of *od*. Similarly, while (29) demonstrates that the boding reading of *od* can give rise to precautioning readings of the kind discussed above for *pen*, (34) clearly shows that it need not do so.

These observations raise the question of what the actual interpretation of *od* is in this context, and why it gives rise to apprehensive and promissive inferences when it does. In what follows I propose a preliminary, informal analysis of *od* as a temporal additive, which takes as its prejacent a future-oriented proposition, and the interpretation of which, like that of other additives, is sensitive to alternatives. Intuitively speaking, my analysis of such temporal additives is that sentences that feature them assert that something that is not currently a historical necessity will become, if things continue as they are, a historical necessity in the future.

Before stating the required semantics, it is useful to situate *od* in a broader context of temporal expressions that give rise to apprehension inferences. The harnessing of future-oriented expressions to express apprehension is a known phenomenon across unrelated languages. Angelo and Schultze-Berndt (2016, 2018) describe apprehensive uses of the German adverbial *nachher* 'afterwards', as well as apprehensive uses of the adverbial *bambai* in Kriol, which expresses a relation of subsequentiality (see also Phillips 2018, 2021 for an extensive analysis). The apprehensive use of *nachher* is exemplified in (37).

(37) German (Indo-European; Angelo & Schultze-Berndt 2018: ex. 12)
Context: Offer of a last-minute start place at a motor race (on a racing forum)
Ne, lass mal! ***Nachher*** *haue ich da noch jemanden raus!*
no let PART APPR/later hit.1SG.PRS 1SG there ADD someone out
'No, (let's) leave it. I might end up kicking someone out!'[20]

[19] https://www.haaretz.co.il/gallery/mejunderet/.premium-1.2089786
[20] https://www.well-rc.de/include.php?path=forumsthread&threadid=1300&entries=0

In fact, the Hebrew equivalent of the German *nachher*, the temporal adverbial *axarkax* 'afterwards/later', has the same apprehensive use, as exemplified in (38). In all three examples, the content of the clause headed by *axarkax* is not inherently undesirable, but is necessarily understood as such in the context of *axarkax*.

(38) a. *"Al tidxi et ze", odeda et acma,*
NEG.IMP postpone.FUT.2F.SG ACC it coax.3F.SG ACC self.CS.F
*"**axarkax** tiškexi ve-ha-sipur yaxzor al*
afterwards forget.FUT.2F.SG and-the-story return.FUT.3M.SG on
acmo".
self.CS.3M.SG
'"Don't leave it for later" she coaxed herself, "In the end you'll forget and the same story will happen all over again".'[21]

b. *Im kvar lixtov šir, az adif lifney ha-milxama.*
if already write.INF song, then preferable before the-war.
***axarkax** lo iye mi še-yišma.*
afterwards not be.FUT.3M.SG who that-hear.FUT.3M.SG
'If one is going to write a song, then better before the war, otherwise there might be nobody to hear it.' (Tom Shin'an, *šir lifney ha-milxama*, song lyrics).

c. *Ani bekoshi mar'a lahem bubot xadashot **Axarkax** hem*
I hardly show.3F.SG them doll.PL new.PL afterwards they
yircu et kulam.
want.FUT.3PL ACC all.CS.3M.PL
'I hardly ever show them new dolls, otherwise they might want all of them'.[22]

Angelo and Schultze-Berndt demonstrate that German *nachher*, on its apprehensive use, can also introduce undesirable possibilities that are metaphysically already settled at the time of utterance, but epistemically open for the speaker, and can also give rise to Lichtenberk's 'in-case' readings discussed above. My own intuition is that this is not the case with Hebrew *axarkax*, but determining this would require a more thorough empirical investigation of this expression than I can offer here. What is important in the current context is that, as put forth by Angelo and Schultze-Berndt, the future-orientation of temporal adverbs

[21] https://www.ynet.co.il/articles/0,7340,L-5573412,00.html
[22] Facebook comment

entails a modal component in their interpretation, which seems to be linked to their propensity to give rise to apprehensive inferences.

The case of *od* is interesting against this general picture, because, as discussed above, *od* is an additive that doubles as a temporal aspectual adverbial. A similar additive used both as an aspectual adverbial and with a boding interpretation is the German additive particle *noch*. German *noch* functions as an aspectual particle meaning 'still' as well as an additive meaning 'another', and it also has a boding reading in which it states the prospective occurence of a future event. Löbner (1989) provides the example in (39).[23]

(39) German (Indo-European; Löbner 1989: 199)
Sie kommt noch.
she comes ADD
'She will yet/eventually come.'

This boding use of *noch* (which often accompanies *nachher*, see example (37) above) also gives rise to apprehensive interpretations, as in the naturally occurring example in (40).

(40) *Jakob, geh Dir eine trockene Hose anziehen, Du wirst noch krank.*
Jakob go.IMP you.DAT a.F dry.F pants wear.INF you be.FUT.2SG ADD sick
'Jakob, go put on some dry pants, you might get sick.'[24]

As with Hebrew *od*, *noch* on its boding interpretation only gives rise to apprehensive inferences in particular contexts, and can also be used in context in which the future possibility expressed by the prejacent is a positive one.[25]

(41) *Ich werde noch gewinnen.*
I FUT.1SG ADD win.INF
'I will win yet/eventually.'

[23]Löbner briefly discussed the difficulties of accounting for it under an analysis in which the prejacent of *noch* contains a hidden prospective operator. As far as I can tell, the boding reading of *noch* is not discussed in Beck's (2020) overview.

[24]https://kinder-getrappel.de/geschichten/wuttroll/hans-der-wuttroll-jakobs-wut-mit-pipi-in-der-hose/

[25]According to Angelo & Schultze-Berndt (2016), the future-oriented use of *nachher* discussed above cannot easily be used in positive contexts. Its apprehensive component therefore seems to be more conventionalized.

The issue for an analysis of the boding reading of additives like *od* and *noch* is determining their lexical meaning in a way that captures their additive nature and allows an explanation for their propensity to give rise to apprehensive inferences in some contexts and to promissive inferences in others.

Combining Angelo and Schultze-Berndt's observation that apprehensive inferences can be linked to future-orientation with an alternative-sensitive additive semantics, I propose the analysis in (42), the basic insight of which is due to Ashwini Deo (in conversation, August 28, 2018).

(42) The semantics of boding additives:
- PRESUPPOSES: the prejacent is not instantiated anywhere within the interval *I* that runs from a contextual left boundary *lb* up until and including now.
- ASSERTS: there is an alternative interval *j* starting at *lb* at which the prejacent is instantiated.

That the alternative interval *j* asserted to instantiate the prejacent is a future one, not one that ends at utterance time, is ensured by the presupposition. If it were a subinterval of *I*, then, necessarily, *I* would also verify the prejacent, contrary to what is presupposed.

On this analysis, a sentence with the boding *od*/*noch* ends up asserting that something that is now not a metaphysical certainty (hence might never happen), will nevertheless be a metaphysical necessity in the future. For example, while the interval that runs up to now does not secure that *we win* will become true in all possible futures, a longer interval that extends into the future DOES ensure that. In a context in which our winning is taken to be desirable for the interlocutors, the assertion leads to a promissive inference (it communicates that something desirable is possible). In a context in which it is undesirable, the assertion is also a warning.[26]

6 *še-lo*: + negation

In this section, I very briefly describe another strategy of apprehension involving the interaction of the subordinator *še* with negation. Rubinstein et al. (2015) show that this strategy is attested already in Mishanaic and Rabbinical Hebrew and

[26]Angelo and Schultze-Berndt (2018), however, make the very interesting observation that German *nachher* cannot be used to perform the speech act of a threat. My intuition is that this is equally true for Hebrew *od*. The eventualities depicted by the prejacent of *od* tend to not be under the control of the speaker, and have more to do with inertial development.

survived through the various written stages of the language. It is very common, and informal, in Modern Hebrew.

Normally, sentential complements of verbs of apprehension describe the eventuality the speaker is worried might occur. For example, in (43), the speaker is worried about recognition when no negation is present, and about non-recognition when it is present.

(43) *I worried/feared that they would (not) recognize me.*

However, in Hebrew, as in many other languages, a negative sentential complement of a verb of fear can also behave as if it lacked negation altogether, as in the naturally occurring (44).

(44) *Paxadeti še-**lo** yagidu še-ani tipša.*
feared.1SG that-NEG say.FUT.3PL that-I stupid.F
'I was scared that they would say I'm stupid.'

Sentences like (44) are ambiguous. On one reading, which is pragmatically odd, (44) means that the speaker was afraid that people would not say that she is stupid. On the other reading, the one actually conveyed by (44) in the context in which it occurred and reflected in the translation, this sentence conveys that she speaker was afraid people would say she *was* stupid. On this reading, the complementizer and negation sequence *še-lo* can be said to be much like an apprehensive marker in a language in which such markers introduce complements of verbs of fear. What the speaker fears in (44) is that people will say she is stupid, not that they would not say so. Similar cases in Romance have been argued to involve "expletive negation", but there is no consensus as to what expletive negation is, whether or not the contexts in which negation seems to be expletive or superfluous form a natural class, and, of course, how such negation should be analyzed (see for example van der Wurff 1999, Abels 2005, Eilam 2009, Yoon 2011, Makri 2013, Puskás 2017, Dobrushina 2021 among many others).

I suggest here tentatively that in Hebrew, *še+lo* has been reanalyzed as a complex modal complementizer, interpreted much like *pen*. The fact that negation is not morphologically fused with the complementizer and can appear inside the prejacent seems to be an immediate and strong argument against this suggestion. Nevertheless, I put forth here a few considerations that support it. I suggest further that *še+lo* freely alternates with *še* in the complement of fear verbs, since such verbs already lexically encode possibility and undesirability, obviating the need for marking apprehension.

Some evidence that negation is not interpreted semantically in the embedded clause comes from the fact that *še-lo* can co-occur with another negation inside that clause, as in the naturally occurring (45) (in which *še-lo* could equally well be replaced with the complementizer *še*).

(45) *Ani gam mamaš hayiti be-laxac lifney ha-mikve* ***še-lo***
I also really was.1.SG in-stress before the-ritual.bath that-NEG
pitom lo argiš tov ve-lo uxal litbol.
suddenly NEG feel.FUT.1SG well and-NEG can.FUT.1SG dip.INF
'I was also really stressed out before the ritual bath that I might suddenly not feel well and won't be able to dip.'[27]

Another piece of evidence is that, unlike sentential negation, *še-lo* negation can, for many speakers, precede the subject. In the examples in (46), the verb *yadati* 'I knew' contrasts with the verb *paxadeti* 'I feared' in that the former, being factive, selects for clausal complements introduced by the complementizer *še* and does not allow *še-lo* clauses. The negation in the complement clause of 'know', like Hebrew sentential negation generally, must therefore follow the subject, as shown in (46b). The verb 'fear', in contrast, can occur with both *še-lo* clauses, as in (46c), and with *še* clauses, as in (46d). In (46c), the indefinite subject follows the complementizer + negation sequence.

(46) a. *Yadati še mišehu* ***lo*** *yavo.*
knew.1SG that someone NEG come.FUT.3M.SG
'I knew that someone wouldn't come.'

b. **Yadati še-**lo** mišehu yavo.*
knew.1SG that-NEG someone come.FUT.3M.SG
Intended: 'I knew that someone wouldn't come.'

c. *Paxadeti še-**lo** mišehu yavo.*
feared.1SG that-NEG someone come.FUT.3M.SG
'I was scared that someone would come.'

d. *Paxadeti še mišehu* ***lo*** *yavo.*
feared.1SG that someone NEG come.FUT.3M.SG
'I was scared that someone wouldn't come.'

These examples also demonstrate the semantic parallelism between *še-lo* and apprehensive modal complementizers like *pen* and *lest*. As discussed above, *pen* clauses occurring with verbs of precaution and fear describe the potential eventuality to be feared or avoided. The same is true of the *še-lo* clause in (46c), where what is feared is that someone will come, as well as of the naturally occurring

[27] https://www.inn.co.il/Forum/Forum.aspx/t1190779

(47) (in which negation does not precede the indefinite subject). In (47), what the addressee is asked to be careful about is the possibility that somebody will beat him up, not that someone will not beat him up.[28]

(47) *Tizaher še mišehu #(lo) yarbic lexa.*
careful.FUT.2M.SG that someone NEG hit.FUT.3M.SG to.you
'Watch out that someone #(doesn't) beat you up.'

Similarly, the naturally occurring example in (48), where negation again precedes the indefinite subject, shows that *še-lo* clauses parallel *pen* clauses in parataxis, giving rise to avertive inferences.

(48) *Sim zxuyot yocrim ba-šir,* ***še-lo*** *mišehu*
put.IMP right.CS.PL artist.PL in.the-song that-NEG someone
yavo ve-yagid še-hu katav et ha-šir
come.FUT.3M.SG and-say.FUT.3M.SG that-he wrote.3M.SG ACC the-song
ha-ze.
the-this
'Put copyright on the song, lest someone come and say that he wrote this song.'[29]

A third piece of evidence comes from the interpretation of the aspectual adverb *kvar* 'already'. This expression cannot, or at least not easily, occur in the scope of regular sentential negation and receive the intended *not yet* interpretation. To the degree that (49b) is acceptable, it can only be a metalinguistic response to an assertion that it is already too late. Instead, the negation of (49a) requires the adverbial *adayin* 'still/yet', similarly to negating English sentences with 'already'.

(49) a. *Meuxar miday kvar.*
late too already
'It's already too late.'

b. *?? Lo meuxar miday kvar.*
NEG late too already
'It's not (the case that it is) too late already.'

c. *Lo meuxar miday adayin.*
NEG late too yet
'It's not too late yet.'

[28]In modern Hebrew, as these examples show, an analysis of *še-lo* clauses as complement clauses seems straightforward.

[29]https://www.fxp.co.il/showthread.php?t=12627978

In this respect, the negation in *še-lo* sentences does not behave like sentential negation, in that *kvar* is licensed in *še-lo* sentences in the complement of fear verbs, and is interpreted as expected if that clause does not contain negation. While not all my consultants like (50), which is very colloquial, all have very clear intuitions about what it means.

(50) *Paxadti* ***še-lo*** *meuxar miday kvar.*
feared.1SG that-NEG already late too
'I was scared that it might already be too late.'

Finally, similar complementizers arguably exist in other languages. For example, the Hungarian *nehogy* (Szabolcsi 2002, Puskás 2017), which is composed of modal negation *ne* and complementizer *hogy*, has been argued to be a modal complementizer. The example in (51) is similar to the Hebrew (48).[30]

(51) Hungarian (Uralic; Puskás 2017: ex. 15b)
Taxi-val ment, ***ne-hogy*** *le-késse a vonatot.*
taxi-INSTR go.PST.3S MOD.NEG-that PART-miss.SUBJ the train.ACC
'She took a taxi, lest she miss the train.'

As Puskás shows, *nehogy* differs from occurrences of standard embedded negative clauses of the form *hogy … ne* in detectable ways, some of which are the same ones that distinguish *še+lo* from *še … lo. Nehogy*, unlike *hogy … ne*, does not allow prefixes that can follow the verb to do so (see Puskás' extensive discussion), and, just like *še-lo*, it licenses positive but not negative polarity items.

These arguments, taken together, make an at least *prima facie* plausible case for viewing *še-lo* as a complex complementizer with conventionally encoded apprehensive semantics, or as moving towards becoming one in Modern Hebrew. Future research will have to decide whether this is ultimately a plausible route of analysis, and if so, what it implies for the typology of apprehensive markers. For example, stand-alone matrix sentences with *še-lo*, unlike those with *alul* and *pen*, cannot have assertive force, asserting that the prejacent is possible and undesirable. Instead, they always have the force of a wish, as in (52).[31]

(52) *Še-lo yigamer li ha-kesef!*
that-NEG end.FUT.3M.SG to.me the-money
'May I not run out of money!'

[30]The glossing here is Puskás'

[31]This undoubtedly has to do with the fact that matrix *še* clauses have a kind of directive or optative force; see Schwarzwald & Shlomo (2015) and Francez (2015) for discussion.

Furthermore, it does not seem that *še-lo* can be used with sentences that give rise to 'in-case' inferences. (53) represents my own judgment. While it is perfectly grammatical and interpretable, it only has the pragmatically odd avertive interpretation, namely it gives rise to the inference that taking an umbrella might avert the rain.

(53) *#Kax* *mitriya* ***še-lo*** *yered* *gešem.*
take.IMP umbrella that-NEG go.down.FUT.3M.SG rain
#'Take an umbrella so that it doesn't rain.' (Intended: 'Take an umbrella, in case it rains.')

7 Conclusion

This paper surveyed five key ways in which Hebrew expressed and expresses apprehension at various stages of its highly non-linear history. I proposed that apprehensive complementizers like Biblical Hebrew *pen* are, semantically, possibility modals that carry a non-at issue bouletic dispreference content (which I labeled, for convenience, a conventional implicature). The kinds of inferences that Lichtenberk (1995) calls avertive and 'in-case' were argued to have a different status across languages, being conventional in languages in which the apprehensive marker takes two arguments (i.e. does not have pure matrix occurrences), and to be derived, pragmatically or as a matter of discourse structure, in a language like Hebrew, in which the the apprehensive marker is, semantically, a modal with one propositional argument.

The paper also proposes that there is a class of temporal additive markers that have what I called a *boding* interpretation, Hebrew additive *od* and German *noch* being instances. Boding additives are future-oriented additives that carry the presupposition that their future-referring prejacent is not a historical necessity (i.e. true in all possible futures) at utterance time, and are used to assert that their prejacent will become a historical necessity at some future point (and hence, by entailment, will someday be true).

Finally, the paper suggested that Modern Hebrew has, or perhaps is coming to have, a morphologically complex apprehensive complementizer, fusing a subordinating complementizer with negation. This hypothesized complementizer functions in some ways like the obsolete Biblical complementizer *pen*, and finds a close parallel in Hungarian and in the Slavic languages, but overall, the plausibiliy of such a complementizer, and its putative relation to the broader landscape of apprehensive marking, remains to be studied in more detail.

Abbreviations

1, 2, 3	first, second, third person	DIR	directive particle	MOD	modal
ACC	accusative	F	feminine	NEG	negation
ADD	additive	FUT	future	PART	particle
ALL	allative	IMP	imperative	PST	past
APPR	apprehensive	INF	infinitive	PL	plural
CS	construct state	INSTR	instrumental	Q	question particle
DAT	dative	NEG	negation	SEQ	sequential
		M	masculine	SG	singular

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Chapter 7

Two major apprehensional strategies in Slavic: A survey of their areal and grammatical distribution

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The paper gives a first comprehensive survey of two types of apprehensional strategies which are widely attested in Slavic languages. The first type foregrounds the speaker's apprehension that some undesirable event may occur (or might have occurred), the second one builds on the first one: the speaker, assuming that an undesirable event is imminent, admonishes the addressee to undertake (or to abandon) some action in order to prevent that undesirable event. The latter type is known as preventive and most commonly expressed by perfective verbs in the negated imperative (or equivalent clause types). Since this strategy is a variety of directive speech acts, it is here called NEGATIVE-DIRECTIVE. The former variety is here dubbed NEGATIVE-OPTATIVE strategy, since it is marked by clause-initial connectives most of which are used as negated optatives, but it subsumes also patterns associated with negative purpose clauses. This article shows how both strategy types complement each other conceptually, how variable the means of expression are and to which extent their components have lost compositionality, how the patterns are distributed over Slavic languages, which ones can occur in clausal complements and whether usage in independent clauses can be considered a result of insubordination.

1 Introduction

Slavic languages lack dedicated morphosyntactic markers of apprehensional meanings, but they have at their disposal constructions which can be regarded

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as apprehensional strategies, i.e. as patterns that are regularly associated with apprehensional semantics.[1] Apart from them, there are various lexical means, usually referred to as particles.[2] Many of them are employed as clause-initial connectives, with different degrees of syntactic integration on a coordination—subordination cline, but there is only one strategy, the negative optative strategy, which can clearly be considered as part of clausal complementation after suitable predicates denoting apprehension (see §5.1 below).

Here I will focus on two strategies which are widespread across the Slavic landscape, which can occur in isolated utterances and which complement each other conceptually. One of them is derived from negative optative or negative purpose clauses (via extension into epistemic modality) (1)–(2), the other from imperatives or equivalent constructions coding directive speech acts (3)–(4).

Negative-optative strategy

(1) Russian
__Kak by__/__Liš' by__ ego ne oštrafova-l-i.
how IRR/only IRR 3M.ACC NEG fine[PFV]-PST-PL
'[I'm afraid] they might fine him.'

(2) Bulgarian (Ivanova 2014: 147)
T-oj približi s nebrežn-o
3-NOM.SG.M approach[PFV].AOR.3.SG with indifferent-SG.N
izraženi-e na lic-e-to – __da ne bi__ tija, malki-te,
expression[N]-SG on face-SG.N-DEF.SG.N IRR NEG SBJV 3PL small-DEF.PL
da si pomislj-at nešto.
IRR REFL.DAT think[PFV]-PRS.3.PL something
'He came closer with an indifferent expression on his face – lest the little ones think something (uncomfortable for him).'

Negative-directive strategy

(3) Russian
(Smotr-i,) ne poskol'zn-i-s'!
look[IPFV]-IMP.SG NEG slip[PFV]-IMP.SG-REFL
'(Be careful,) don't slip!'

[1] I will employ 'apprehensive' only with reference to dedicated markers and use 'apprehensional' only for the semantics. For a brief terminological overview cf. Mitkovska et al. (2017: 59).

[2] Cf. Dobrushina (2006: 46f.), Arkadiev [Arkad'ev] (2007), Plungian [Plungjan] (2011: 448), Dobrovol'skij & Mikaėljan (2021) for Russian and Ivanova (2014) for Russian and Bulgarian.

(4) Bulgarian
(Pazi se) da ne padn-eš!
look[IPFV]:IMP.SG REFL IRR NEG fall[PFV]-PRS.2SG
'(Look) that you don't fall!'

The first pattern primarily marks just apprehension, i.e. the speaker evaluates a situation both as likely to occur (or have occurred) and as undesirable.[3] From a conceptual point of view, the second pattern builds on the first one, but encourages the addressee to do (or to abandon) something in order to prevent this undesirable situation. Correspondingly, this pattern has been labelled *preventive*, and it has been treated in connection with directive speech acts.[4] For the sake of brevity I will call the latter pattern NegDir and the former NegOpt. The label NegOpt comprises constructions related either to optative or to purpose clauses. This joint treatment of negative optative and negative purpose sources seems justified because in their contemporary stages they are more similar to each other in structure and distribution than either of them to NegDir. This also holds true for their provenance, even though in comparison to NegDir (§4) the relevant cases of NegOpt and their diachronic sources are more diversified (§3 and §5). If not stated otherwise, claims concerning NegOpt will pertain to structures irrespective of their provenance from properly negative-optative or from negative-purpose clauses. Both NegDir and NegOpt constructions occur as main clauses; NegDir does so exclusively, while NegOpt can occur embedded as a clausal complement. Either of them can also be uttered in isolation, i.e. out of the blue in dialogue, whereas most of the less salient apprehensional strategies cannot.

As for the latter, it is worth emphasizing that at least in some Slavic languages we find strategies which are based on metalinguistic justification. These strategies cannot be uttered in isolation, as they depend on preceding verbal discourse. A prominent case in point is the Polish causal conjunction *bo* 'because' (cf. Kawka 1988), illustrated in (5).

(5) Polish (Ożóg 1990: 135)
*Nie rusza-j t-ego **bo** się oparz-ysz.*
NEG touch[IPFV]-IMP.SG DEM-GEN.SG because REFL burn[PFV]-FUT.2SG
'Don't touch this, **or** [lit. 'since'] you'll burn yourself.' (≅ '… lest you burn yourself.')

[3]This is probably the most widespread understanding of the terms apprehensive/apprehensional in Slavic Linguistics. Many (e.g., Ivanova 2014, Bužarovska & Mitkovska 2015, Mitkovska & Bužarovska 2017, Mitkovska et al. 2017) restrict them to this type. Typically, these researchers focus on Balkan Slavic, while researchers dealing with the negative-directive strategy (see next footnote) have focused on Russian (and other North Slavic languages).

[4]Cf. Laskowski (1998), Xrakovskij (2001), Jary & Kissine (2014), Zorikhina Nilsson (2012, 2013), to name but a few.

The causal connective introduces the second conjunct in a clause chain; the preceding conjunct usually expresses a directive speech act. As a rule, the predicate of this conjunct with the causal connective is a perfective verb in the present tense stem (by default interpreted as future). In any case, the conjunct following the directive speech act is interpreted as a warning (or admonition) and verbalizes the consequences that are deemed to follow if the addressee does not comply with this speech act. Such consequences can include concerns for the interlocutor's benefit, in which case the expression acquires an apprehensional function.

Here causal and equivalent connectives[5] will be left out of consideration. The motivation to concentrate on the aforementioned apprehensional constructions, NegOpt and NegDir, is as follows. First, to my knowledge, there has not been so far any comprehensive account of apprehensional strategies in Slavic at all. To start with (especially in view of space restrictions), it seems justified to choose those constructions which are most salient in terms of areal spread, conventionalization and syntactic independence, and for which there already exists enough reliable research. Second, these constructions show an interesting areal distribution which can be described in a more fine-grained manner if we take into account their dependence on core distinctions in the grammar of Slavic languages (TAM distinctions). Moreover, subareas of Slavic differ considerably in the degree with which they distinguish these two strategies in structural and functional terms. Third, only for the two strategies chosen can we supply some limited diachronic background, whereas for the others hardly any such research has been carried out (at least I am not aware of any), whether corpus-based or not. One of the reasons certainly lies in the fact that apprehensional strategies are often difficult to discern, since they are tightly bound to discourse pragmatics, and their degree of conventionalization is low. Jointly with this, and fourth, I want to avoid discussions concerning the coordination-subordination gradient, which would arise for many of the clause-initial particles mentioned in the literature. In my conviction, discussions about where such structures are to be located on a coordination-subordination cline do not yield much for an understanding of apprehensional strategies. The NegOpt strategy is probably the only one in Slavic languages which really can occur as a syntactically embedded structure. For this reason I will pay attention to the issue of insubordination in connection with NegOpt (see §5.1).

[5] Functional equivalents are metadiscursive uses of the Bulgarian conjunction *če* (today the standard THAT-complementizer, but deriving from a causal function) and the Russian adversative connective *a to*. All these connectives have other functions outside the domain of apprehension and their apprehensional interpretation is highly dependent on the discourse context.

The article has the following structure. I will first discuss conceptual matters important to the understanding of how the two main strategies are related on the semantics-pragmatics interface. For this purpose I will introduce templates (§2). Subsequently, I will analyze the two patterns one by one, first NegOpt (§3), then NegDir (§4). Each of these sections will account for the areal spread and differentiation in the contemporary languages. This account will be somewhat uneven, because for many Slavic languages sufficiently reliable data are lacking or not easy to access. In particular, I will not have anything to say here on Ukrainian and Belarusian, nor on Sorbian. In §5 I will discuss some diachronic background, in particular the issue of insubordination, which is relevant only for NegOpt (§5.1). The last section gives a summary of findings and provides an outlook (§6).

I will be referring to subareas of Slavic following commonplace divisions into East, West and South Slavic, as well as Balkan Slavic. North Slavic unites West and East Slavic. Balkan Slavic comprises Bulgarian, Macedonian and Torlak and Timok dialects in southeastern Serbia. I will gloss clause initial connectives according to their etymological sources; however the reader should have in mind that as apprehensional markers (or in similar functions) they have lost transparency (see §3). con is used to gloss connectives if no further specification is warranted. Examples given in the running text will be glossed to a minimum, aspect will then be marked by upper case indexes ($^{\text{PFV}}$, $^{\text{IPFV}}$).

2 Conceptual premises

Apprehensional strategies can be classified in various ways. From a conceptual point of view, some researchers (e.g., Dobrushina 2006, Ivanova 2014) regard the NegOpt pattern as basic and the NegDir pattern as secondary (see Footnote 3). Other researchers take the opposite view, since they approach apprehensional semantics from imperatives and directive speech acts (e.g., Birjulin 1992, 1994; Aikhenvald 2010; Gusev 2013). After all, apprehensives and apprehensional strategies are conceptually very complex. When descriptions by various scholars are compared,[6] three components belonging to different domains become salient. See Table 1, designed to apply to the analysis of Slavic apprehensional strategies, which I focus on here. Thus, it is not meant as a general conceptual schema of apprehensive markers and/or apprehensional strategies; not all scholars distinguish, or acknowledge, all three components.

[6] Cf. Lichtenberk (1995: 293–296); Dobrushina (2006: 30, 49, 2021); Plungian [Plungjan] (2011: 448f.); Angelo & Schultze-Berndt (2016: 258f.).

Table 1: Components of apprehensional strategies (at least in Slavic languages)

1. Epistemic assessment	The speaker lends some knowledge- or belief-based support that a certain eventuality E has obtained, is obtaining or can be predicted: ‘I consider E likely to occur/ to have occurred/ to be the case.’
2. Undesirability	The speaker considers E, or its consequences, undesirable and doesn’t want them to occur; this presupposes an element of volition: ‘I do not want E, or its consequences, to occur.’
3. Uncontrollability	Neither a discourse participant (in particular the addressee) nor a third party can influence the course of events.

Components 1 and 2 are separated from each other in Birjulin (1992, 1994: 114f.) and Zorikhina Nilsson (2013: 98); they are also discussed by Holvoet (2010: 83–88) and Gusev (2013: 62). All these authors concentrate on preventives, but cf. also Mitkovska et al. (2017). Preventives conjoin two elements: (i) an epistemic (‘P may/is bound to occur’) and (ii) a directive one (‘do something so that P doesn’t occur’). This is why Gusev characterizes them by features typical for both indicative and imperative forms.

Component 2 appears to be uncontroversial, in fact it spells out the core of apprehensional marking. It can however be implied in a warning and, in this sense, be somewhat downgraded in communicative salience.

The epistemic component (Component 1) has been questioned by some researchers discussing co-occurrence and replacement conditions of apprehensive modals vis-à-vis other modals. In particular, one may question whether apprehensive markers, especially when they are exclusively future-oriented, meet a restrictive definition of epistemic modality (Vuillermet 2026 [this volume]). Regardless of how apprehensives relate to modals, the requirement of unrestricted occurrence (in all tenses) applies to the NegOpt strategy in a conceptual perspective, and as for the NegOpt strategy in South Slavic there are also no grammatical restrictions concerning tense (see §3). More importantly, from a conceptual point of view, one cannot decompose the notion of apprehension without recurrence to epistemic attitudes (similarly to the concept ‘hope’), because one can only feel fear of something which one considers to be possible. Typically, the epistemic assessment relates to a situation posterior to a reference interval, but this is not necessarily the case.[7]

Component 3 is likewise controversial, at least as a general feature of apprehensional patterns, but for a different reason. It has been downplayed in many discussions (e.g., Lichtenberk 1995: 297, 312–314, Zorikhina Nilsson 2013: 95, 98–100, Mitkovska et al. 2017: 59f., but see p. 65), while other linguists have paid quite some attention (e.g., Holvoet 1989: 95f., 103–105, Xrakovskij 1990: 213–215, Łuczków 1997: 35f.). Disputes can be solved if we ask what is considered to be the object of control: the observed situation or an action to be undertaken. Moreover, it is not the speaker, but the addressee who is considered to (be able to) control that action. This becomes evident with preventives, which are realized by the NegDir strategy. Thus, Birjulin (1992: 5, 7; 1994) claims that preventives imply uncontrollability as their most pervasive feature, but he then insists that this construction prompts the addressee to take measures to avert consequences (cf. also Aikhenvald 2010: 225). Control(lability) is thus presupposed, although it often relates not to the event denoted by the verb, but to another, unnamed action (see §4). As for NegOpt-strategies, controllability may or may not hold (see §3).[8] This settled, Gusev (2013) divides the meaning of preventives into two components as shown in Table 2.[9]

Table 2: Two components of the meaning of preventives (Gusev 2013: 61f. my translation, BW)

(a)	The speaker states that some undesirable event P may occur.
(b)	By stating this, the speaker tries to cause the addressee or some third person to undertake some intermediate action (P_{aux}) which would prevent P from occurring (i.e. which would cause non-P).

Dobrushina (2006: 30, 51, 58) bases her typology on the fore-/backgrounded status of these two components, and this yields what I have dubbed the NegOpt and the NegDir pattern. Gusev (2013: 61) makes this relation more explicit: the

[7] Cf. Dobrushina (2021: §2.1) on complements of ‘fear’ verbs.

[8] Closely related ‘in case’-readings, in turn, are rather indifferent to both controllability (Dobrushina 2006: 58) and undesirability. In an ‘in-case’ reading the speaker’s assumption about a possible event need not be undesirable (e.g. *Take a bag, in case they have food left*), and often the event is neither under the speaker’s nor the addressee’s control (*Take an umbrella, in case it starts raining*); cf. Lichtenberk (1995: 298–302, 311).

[9] More precisely, an alternative should be added to subcomponent (b): ‘… or to not undertake some intermediate action that would/might lead to P’ (cf. Birjulin 1992, 1994, Birjulin & Xrakovskij 1992: 38, 2001: 37f. Dobrushina 2006: 30).

two constructions differ depending on which of these two components is focused. If the focus lies on component (a), i.e. an apprehension (= epistemic+undesirable), component (b), representing a directive illocution, remains as an implicature; under favorable circumstances it can, however, be interpreted as the proper illocutionary point. This is shown in Table 3 (here and in the following, grey shading indicates asserted parts of meaning or preferred interpretations, respectively). The foregrounded part can also be circumscribed as 'I wish that something likely doesn't/won't happen'; for this reason, this strategy is termed the Negative Optative (NegOpt) strategy here.

Table 3: Apprehensional Template 1: Negative-optative strategy

Component	Russian	English
Background	*Segodnja skol'zko*	'It's slippery today.'
FOREGROUND (=a) apprehension	*Ty mož-eš' upas-t'/upad-eš'*	'You might/will fall [but I don't want this to happen.]' = 'If only you don't fall / I don't wish you to fall!'
Implicature (=b) pre-emptive action; can become implicit illocutionary point	*Xodi ostorožnee!*	'Walk more carefully!'

If the directive component (b) is foregrounded, component (a) is presupposed. However, if the directive predicate denotes an event that is out of the addressee's control, it does not call the intermediate, pre-emptive, action (P_{aux}) by its name; in this case P_{aux} remains vague and the illocutionary point implicit. If, on the other hand, the directive predicate denotes an event that can be controlled by the addressee, then the illocutionary point is named directly – compare the directive in Table 4 (component (b)) with *Oden'sja teplo!* 'Put on warm clothes!' In order to tease apart these components, we have to extend Gusev's schema (Table 2) as in Table 4.[10]

[10] Cf. Laskowski (1998: 19) for a similar template. Birjulin (1992: 15f.) did not fully consider the possibility that the directive predicate may denote a controllable action directly, but in other respects the reasoning presented here is congruent with his view.

Table 4: Apprehensional Template 2: Negative-directive strategy

Component	**Russian**	**English**
Assumption	*Na ulice xolodno*	'It's cold outside.'
Presupposition (=a) Apprehension	*Prostudiš'sja. / Ty možeš' prostudit'sja*	'You'll/You might catch a cold.'
FOREGROUND illocutionary point (=b) directive [uttered]	*Ne prostudis'!*	'Don't catch a cold!'
If the verb in the directive denotes an uncontrollable event, illocutionary point shifts to another implied controllable action (= Gusev's P_{aux}):		
[not uttered]	'do something about it (when leaving the house)', e.g. *Oden'sja teplo!*	'Put on warm clothes!'

The often implicit character of the illocutionary purpose and the more complex conceptual structure of the NegDir pattern causes a certain asymmetry in comparison with the NegOpt pattern, but nonetheless both strategies complement each other in terms of their illocutionary purpose. Crucially, components (a) and (b) can be captured in this fore-backgrounding relation.

Notably, negation plays a key role in this relation. Although the "positive" counterparts of both apprehensional strategies, optatives and unnegated directives, are volition-based, they differ with respect to control; jointly with this, they differ in their association with interlocutors' roles: directives imply the addressee's control, whereas optatives rather exclude control and are often speaker-oriented (although the speaker may express concern for the addressee or some other person; see §3.1, §4.1). Control is maintained for negated directives and remains weak (or absent) for apprehensional expressions, but negation brings both into connection because it marks what should not happen; the difference is that in NegOpt strategies negation relates to the volitional evaluation (emotion), while in NegDir strategies it relates to what should be prevented (action).

Moreover, one gets the general impression that constructions with a negative-purpose source (see 6) seem to acquire a preventive (NegDir) meaning more easily than constructions with a negative-optative source (as, e.g., with Russian *kak by … ne*; see (7)). However, this remains to be investigated. Preventives are often accompanied by adverbials emphasizing that the respective event might occur inadvertently, such as *czasem* 'accidentally, occasionally' in (6).

(6) Polish (Danielewiczowa 2014: 182)
Aby czasem nie *zdepta-l-i moich tulipan-ów!*
con.irr occasionally neg trample.down[pfv]-pst-pl my tulip-gen.pl
'If only they don't accidentally trample down my tulips!'

Furthermore, apprehensives and apprehensional strategies are inherently time-located. This is most obvious from their by far most predominant use, which relates to events that can be foreseen in the immediate future, or are even on the verge of happening (see Component 1 in Table 1). Apprehensional constructions sound weird with repeated (or habitual) events. They share these two features with avertive grams, which mark that an (usually unpleasant) event was first considered imminent but nonetheless did not occur, e.g. *I was about to fall (but didn't)* (Kuteva 1998; Kuteva et al. 2019; Kytö & Romaine 2005; however, Schmidtke-Bode 2009: 130 et passim identifies *avertive* with negative purpose). Since, contrary to apprehensional constructions, avertive grams assert the non-occurrence of the unpleasant event, they are usually bound to past tense and not to perspectives posterior to speech time. Nonetheless, apprehensives and apprehensional strategies are not automatically future-oriented (Lichtenberk 1995: 296, 310, 312; Mitkovska & Bužarovska 2017).[11]

Finally, the internal temporal structure of the situation denoted by the apprehensional clause need not be bounded (or telic). Compare Russian *Ja bojus', kak by on ne* ***spal***$^{\text{IPFV}}$ *uže* 'I'm afraid that he **is** already **asleep**' or the Macedonian example (24). In practice, this observation is of relevance only for NegOpt, not for NegDir strategies, since it hardly makes sense to ask the addressee to take care lest they accidentally perform "a bit" of some unbounded activity (see §4).

With these premises in mind, let us now deal more closely with the two strategies NegOpt (§3) and NegDir (§4).

[11]Plungian [Plungjan] (2004: 17) argues that an inherent future-orientation of apprehensives explains their evaluative feature (included in Component 2 of Table 1). However, there is no conceptual reason why situations that already occurred (or are occurring) should not be the object of negative evaluation, too.

3 Negative-optative strategy (NegOpt)

Negative-optative strategies are related to optative or purpose clauses with different degrees of semantic transparency. I first explain the ingredients of this construction type (§3.1), before I comment on ways of its embedding (§3.2), on temporal relations between the content of the apprehensional clause and the time of its evaluation by the judging subject (§3.3), and on issues concerning structural and semantic compositionality (§3.4).

3.1 Components of the construction

In terms of the meaning components separated in Table 2 and in Tables 3 and 4, NegOpt strategies primarily communicate the speaker's fear that an undesirable situation might obtain (= component (a)). The prevention of this situation (= component (b)) is not a proper part of the meaning of this strategy; it can be inferred as a sort of perlocutionary implicature (Dobrushina 2006: 29), provided the relevant participant (speaker or another person) has control over some part of the situation. Compare examples (1) ('(I'm afraid) they might fine him') and (7):

(7) Russian

a. *Kak by/Liš' by (nam) ne opozda-t'.*
how IRR/only IRR 1PL.DAT NEG come_late[PFV]-INF

b. *Kak by/Liš' by my/oni ne opozda-l-i.*
how IRR/only IRR 1PL.NOM/3PL.NOM NEG come_late[PFV]-PST-PL
'If only we/they don't come late.'

In North Slavic, the construction consists of a negated verbal predicate and a clause-initial particle which incorporates the irrealis marker *by*. This segment has no separate morpheme status, although it keeps the grammatical requirements of the subjunctive marker *by* from which it derives (see Footnote 13). These requirements are the *l*-form (with a subject in the nominative, see (7b)) or the infinitive (with a subject in the dative, see (7a)). In the infinitive construction the dative subject is optional; without it the construction implies self-reference to the speaker (or a group including the speaker).[12] The *l*-form, in turn, can be classified as a morphone, i.e. a property of word forms that bears no effect on syntax

[12]In narrative contexts, instead of the speaker (narrator), some protagonist may become the vantage point for self-reference. Compare examples (2), (20), (21), (24), (28), (31), (32), (38), (53).

or semantic interpretation[13] (Aronoff 1994; Stump 2016). The segment *by* follows either a WH-word (Russ. *kak* 'how') or a particle; among the latter we find focus particles (Russ. *liš'*, *tol'ko* 'only') and a particle with concessive function (Russ. *xot'*). All these combinations also occur without negation and are then used as optative markers (often in exclamations). The units discussed here (with negation) can thus can be considered negative-optative markers proper.

An alternative involves connectives that are otherwise prominently employed as complementizers or subordinators of purpose clauses (e.g., Russian *čtoby*). Such connectives thus indicate a connection to negative purpose. In Russian, however, this variant has become obsolete (see §5.1.2), while its equivalent in Polish (*żeby*) is used more widely. West Slavic languages also exhibit particles in which *by* has fused with coordinative conjunctions; compare Polish/Czech *aby* (Lichtenberk 1995: 310 for Czech) as in (8).

(8) Czech
Aby *nám* *ne-prše-l-o* *na cest-u.*
CON.IRR 1PL.DAT NEG-rain[IPFV]-PST-N on road-ACC
'[I fear] it might rain while we're on the road.'

In West Slavic, in turn, apprehensional constructions with focus or concessive particles appear to be lacking. The variant with infinitive and dative subject (7a) is altogether lacking (Czech, Slovak) or avoided (Polish), at least in independent clauses (see §3.2).

In the contemporary languages, the relation to optative and purpose clauses differs. In Russian, constructions of the type (7a)–(7b) are compositional inasmuch as the negation is included in the scope of an optative structure (cf. Dobrushina 2016: 339–342); for non-embedded clauses compare (7a)–(7b) with (9a)–(9b) and (10a)–(10b). However, clause-initial *kak by* (+ *l*-form/infinitive) can induce other

[13]The *l*-form is originally an active anteriority participle which has turned into the marker of generalized past tense in North Slavic and western South Slavic. It takes part in subjunctive marking only together with *by*, which is the paradigmatically isolated 3SG-form of the former aorist of *byti* 'be', originally used as an auxiliary to mark the pluperfect in combination with the *l*-participle. After the aorist and, consequently, the pluperfect were lost, the remnants of *by+l*-participle combinations were reinterpreted as subjunctive (i.e. an irrealis mood). In this already intransparent structure *by* still behaved as a Wackernagel clitic, it therefore continued attaching to clause-initial units (WH-words and connectives). With many of them it has coalesced, among others with Russian *kak* 'how, that' and other connectives to become complementizers or 'optative particles'. *By* and *l*-form need not appear adjacent to each other; the *l*-form requirement of *by* has persisted even after coalescence and the concomitant loss of morpheme status of the latter. For the stages of this process and some diachronic background cf. Wiemer (2015).

illocutions (combined with corresponding intonations) than optatives (or mirative exclamation). First of all, it can be used as a question, with *by* weakening epistemic commitment (see interpretation (i) for (10)).

(9) Russian

a. *Liš' by (nam) uspe-t'.*
only IRR 1PL.DAT manage_in_time[PFV]-INF

b. *Liš' by my uspe-l-i.*
only IRR 1PL.NOM manage_in_time[PFV]-PST-PL
'If only we come in time.' / 'May we come in time.'

(10) Russian

a. *Kak by (nam) uspe-t'?*
how IRR 1PL.DAT manage_in_time[PFV]-INF

b. *Kak by my uspe-l-i?*
how IRR 1PL.NOM manage_in_time[PFV]-PST-PL
(i) 'How might it be possible (for us) to come in time?',
(ii) 'How can we manage to come in time?', 'Can we really (manage to) come in time?'

In modern West Slavic, constructions like (8) show no obvious relation to manner or exclamative clauses, nor to optative clauses (contrary to Russian *kak by ne*). The relation to purpose is evident only for embedded apprehensional clauses (see §3.2). This is to be expected since, contrary to optatives, purpose clauses usually imply embedding (see §5.1.2).

Note that optatives and apprehensionals are polar opposites only insofar as Component 2 of Table 1 is concerned: wish and apprehension contrast in whether the speaker considers the situation desirable or undesirable (Dobrushina 2016: 339–341, Wiemer 2017: 298f.), but optatives are indifferent to epistemic attitudes (a wish may be realistic or not), i.e. the speaker need not have any specific commitment to the likelihood of the desired situation. On another level, namely in terms of emotional evaluation, wishes are polar to curses, whereas curses and apprehension are, in turn, polar to each other when we consider the relation to the addressee:[14] curses are uttered if the speaker wishes something bad to the addressee, whereas apprehension implies the speaker's concern for the addressee's benefit (see also §4.1).

[14]Moreover, curses are always (and wishes can be) uttered in performative speech acts, while apprehension can hardly be imagined as a performative speech act (probably because of its epistemic component).

Regardless of this, in both the apprehensional and the corresponding optative structure all involved elements exist as separate units, but only their combination yields these specific illocutionary functions. The same applies, *mutatis mutandis*, for apprehensional structures and their relation to purpose clauses in West Slavic.

In Balkan Slavic the situation differs, since clauses with initial *da ne* can be variably used either as negative-optatives (→ focus on component (a); see (11)) or as negative-directives (→ focus on component (b); see (4) from Bulgarian and (12) from Macedonian). Apart from their negative-optative function, *da ne*-clauses can also be used as negative-purpose clauses (see 13), and this may be taken as an indication that for NegOpt strategies there is no clear distinction between main and subordinate clause (see further in §3.2 and §5.1). The specific function is usually determined only by the discourse context. However, the negative-purpose function results from a transparent compositional reading of *da* as a general irrealis morpheme (with purpose being one possibility) and the negation.

(11) Macedonian (Mitkovska et al. 2017: 68)
Aj dosta, Boško, ***da ne*** *me pobara tatko …*
ptc enough pn irr neg 1sg.acc call[pfv](prs.3sg) dad
'Boško, I must go, **lest** my father call for me …'
→ (i) negative-optative; (ii) ~~preventive~~

(12) Macedonian (Mitkovska et al. 2017: 74)
Da ne *nastine-š!*
irr neg get_cold[pfv]-prs.2sg
'May you not get a cold.' (ii) '(Be careful!) Don't catch a cold!'
→ (i) negative-optative (ii) preventive [preferred]

(13) Macedonian (Mitkovska et al. 2017: 66)
Duva-j mil-o, ***da ne*** *se popari-š.*
blow[ipfv]-imp.sg dear-voc.sg.m irr neg refl scald[pfv]-prs.2sg
'Run, dear, (i) or you might get scalded; (ii) so as not to be scalded.'
(i) negative-optative; (ii) negative-purpose (> preventive)

Negative purpose and preventive readings can become indistinguishable (see (13), also (35d) below), and this applies to pertinent constructions in West Slavic as well (see, for instance, the Polish examples (21) and (22) below). The reasons for this are of a conceptual nature: (negative) purpose and prevention both presuppose control, while (negative) optatives do not (see §2). Since, in addition, South Slavic constructions based on *da ne* (despite their different opaque vs transparent structure) do not discriminate between negative purpose and negative optative function (and they are anyway very vague in their meaning range; see §3.4), they can in principle cover also preventive functions. Whether diachronically this is to be seen as an extension from negative purpose, is an open issue (see §5.1.3).

The "flexible" behavior of *da ne* is, to a certain extent, inherited from the vague functional values of *da*, which from the earliest attested times (in Old Church Slavonic) was a general irrealis-marker; optative use was among its salient uses (Večerka 1993: 79, 1996: 65; Grković-Major [Grković-Mejdžor] 2020). Balkan Slavic continues this situation (see (14)–(16)), but, in addition, *da* has become part of a cluster of verb-adjacent proclitics and it constrains the admissible range of TAM forms. Depending on the construction and the context of speech the illocutionary function of *da* is vague, and its syntactic status in clause combining is very versatile (Wiemer 2017: 300–305, 325–330, 2018: 295–306, 2021: 58–80). This carries over to combinations with negation (*ne*); among other things, *da ne* can be used as an adverbial subordinator (conjunction) with purpose clauses. *Da ne* shows semantically transparent usage types, in particular of [optative+negation > apprehensional], although in other environments (e.g. as an epistemic modifier) *da ne* loses practically all the aforementioned properties, and this correlates with the loss of morphological autonomy and semantic transparency (see §3.4). The relation between apprehensional and optative function becomes obvious when we look at equivalent utterances without negation, for instance[15] (see also (35d) vs. (35e) in §3.4):

(14) Macedonian (Tomić 2012: 371)
Da gi prečeka-te!
IRR 3PL.ACC await[PFV]-PRS.2PL
'(You should) Welcome them!'

(15) Macedonian (Tomić 2012: 371)
Golem da raste-š!
big[SG.M] IRR grow[IPFV]-PRS.2SG
'May you grow big!'

(16) Bulgarian (Bužarovska & Mitkovska 2015: 2)
Da bǎd-eš zdrav i vesel!
IRR be[PFV]-PRS.2SG healthy[SG.M] and cheerful[SG.M]
'May you be healthy and cheerful!'

[15] See Tomić (2012) for a comprehensive survey of imperatives and "bare subjunctives"; cf. also Bužarovska & Mitkovska (2015); Mitkovska et al. (2017), with further references concerning imperative-like uses.

These utterances have an optative (15)–(16) or hortative (14) value. In general, Balkan Slavic *da*+indicative can be used as an equivalent of the imperative in its different context-conditioned functions (see Footnote 15).

Basically the same comments are justified for Bulgarian *da ne bi* (etymologically irr+neg+sbjv); compare (2) and the negative purpose clause in (17):

(17) Bulgarian (Mitkovska et al. 2017: 64)

(…) *tutaksi*	*ugasi*	*fener-a*	***da***	***ne***	***bi***
immediately	extinguish[pfv].aor.3sg	flashlight-def.do	irr	neg	sbjv

njakoj	*ot*	*reka-ta*	*da*	*go*	*zabeleži.*
somebody	from	river-def	irr	3m.acc	notice[pfv].prs.3sg

'He immediately turned the flashlight off **so that** no one could notice him from the river.'

However, *da ne bi* differs from *da ne* in that it normally requires another *da* proclitic to the verb (see further §3.4). In addition, it includes another irrealis marker *bi* deriving from an older Slavic subjunctive (or conditional). The form *bi* is fossilized (originally 3sg) and does not show agreement, in contrast to the subjunctive marker proper[16] (Ivanova 2014: 147). *Da ne bi* acquires apprehensional meaning preferably in embedded clauses (Ivanova 2014: 143). However, *da ne bi* differs from *da ne* in that it normally requires another *da* proclitic to the verb (see further §3.4). In addition, it includes another irrealis marker *bi* deriving from an older Slavic subjunctive (or conditional). *Da ne bi* acquires apprehensional meaning preferably in embedded clauses (Ivanova 2014: 143). However, *da ne bi* differs from *da ne* in that it normally requires another *da* proclitic to the verb (see further §3.4). In addition, it includes another irrealis marker *bi* deriving from an older Slavic subjunctive (or conditional). The form *bi* is fossilized (originally 3sg) and does not show agreement, in contrast to the subjunctive marker proper[17] (Ivanova 2014: 147). *Da ne bi* acquires apprehensional meaning preferably in embedded clauses (Ivanova 2014: 143).

[16] Both *bi* and *by* (used in North Slavic) relate to copular *byti* 'be', but *bi* does not derive from a former aorist (see Footnote 13). Moreover, in contrast to the North Slavic *by+l* form, the subjunctive marked with the *bi+l* form has become marginal particularly in Balkan Slavic, being more and more superseded by *da*+(pfv)present. Combinations of *da+bi* have found lexicalized niches in formulaic expressions (e.g., curses).

[17] Both *bi* and *by* (used in North Slavic) relate to copular *byti* 'be', but *bi* does not derive from a former aorist (see Footnote 13). Moreover, in contrast to the North Slavic *by* + *l*-form, the subjunctive marked with the *bi*+ *l*-form has become marginal particularly in Balkan Slavic, being more and more superseded by *da*+(pfv)present. Combinations of *da+bi* have found lexicalized niches in formulaic expressions (e.g., curses).

Both Macedonian *da ne* and Bulgarian *da ne bi* allow for 'in-case' readings (see 18); these are excluded for NegOpt in North Slavic:

(18) Macedonian (Mitkovska et al. 2017: 67)
Ḱe mu se java-m na Marko ***da ne*** *ne*
FUT 3M.DAT REFL call[PFV]-PRS.1SG on PN IRR NEG NEG
zna-e za sostanok-ot.
know[IPFV]-PRS.3SG for meeting[SG.M]-DEF.SG.M
'I'll call Marko, **in case** he doesn't know about the meeting.'

In sum, Balkan Slavic clause-initial particles used in NegOpt show a wider range of uses than their North Slavic counterparts. Among other things, they are employed as rhetoric discourse markers (with mirative, epistemic and other functions). Also from a morphological point of view the NegOpt strategies are much less distinguishable from NegDir strategies (preventives) than they are in North Slavic (see §3.4).

Moreover, in Russian the structure of NegOpt varies depending on whether the clausal subject differs from the speaker or whether both coincide. If it differs, the sentence has a morphologically finite predicate (see 1, (7b)); if it coincides, the predicate is predominantly an infinitive with an (optional) dative subject (see 7a). In West Slavic the infinitival construction is avoided (in main clauses), while in Balkan Slavic the infinitive is lost altogether,[18] so that surface distinctions between the two constellations are precluded. Compare (2) and (11) (no coincidence) with (19) (coincidence):

(19) Macedonian (Mitkovska et al. 2017: 64)
Najmnogu se plaš-ev ***da ne*** *ja*
most of all REFL fear[IPFV]-IMPF.1SG IRR NEG 3F.ACC
razočara-m.
disappoint[PFV]-PRS.1SG
'Most of all I was afraid not to let her down.' (lit. '… lest I disappoint her.')

3.2 Embedding

NegOpt can occur as a clausal complement of verbs denoting fear (or related notions), of related nouns (for the latter see (25)–(26)) or of predicates of adjectival origin (e.g., Russian *bojazno* '(it is) fearful, frightening').[19] Embedding implies

[18] Remnants of a shortened infinitive in Bulgarian are rare and irrelevant for apprehensional constructions.

[19] For lists of such verbal and nominal predicates in contemporary Russian cf. Zorikhina Nilsson (2012: 56), Dobrushina (2016: 338).

that clause-initial units like Russian *kak by … ne*, Bulgarian *da ne bi*, Polish *żeby … nie* function as complementizers (see §3.3 and §5.1). Not all clause-initial connectives with apprehensional function occur as complementizers; for instance, Russian *liš' by ne*, *tol'ko by ne* and *xot' by ne* are not attested in the RNC as complementizers, while *kak by … ne* is more frequent in clauses that can be qualified as complements than in independent clauses (Dobrushina 2016: 335 and the Table on p. 330). However, *kak by … ne* still reveals more properties of autonomy than canonical complementizers (Dobrushina 2016: 345f.); this correlates with its probably very recent complementizer status (§5.1).

Embedding turns the distinction of referential identity vs difference between speaker and clausal subject into a distinction of different-subject (DS) vs same-subject (SS) between the subjects of the main and the subordinate clause. Example (20) illustrates DS:

(20) Bulgarian (Ivanova 2014: 150)
Toj se panik'osa ***da ne bi*** *gen. Borisov da*
3m.nom refl panic[pfv].aor.3sg irr neg sbjv general pn irr
potărsi văzmezdie za minalo-to.
want[pfv].prs.3sg revenge for past[n]-def.n
'He *panicked* **lest** general Borisov wants to take revenge for the past.'

For embedded NegOpt clauses, contrasts of morphological finiteness, as encountered in Russian, correspond to a distinction of balanced (for DS) vs deranked (for SS) clause-combining (following Cristofaro 2003). With SS, infinitival, i.e. deranked, complements are practically the only option in Russian (see §3.3). However, as a complementizer of apprehensive clauses *kak by ne* is overall more frequent with the *l*-form (Dobrushina 2016: 338, based on the RNC), and this pattern is strongly associated with DS (see §3.3).

As for West Slavic, Polish allows for infinitival complements in case of SS (see 21), although they are not very frequent. Moreover, the negative-purpose and preventive readings are often hardly distinguishable (see 22):

(21) Polish
Wymy-ł-a puszk-ę piask-iem i
wash[pfv]-pst-3.sg.f can[f]-acc.sg sand[m]-ins.sg and
napełni-ł-a ją wod-ą, uważa-jąc,
fill[pfv]-pst-3.sg.f 3.acc.sg.f water[f]-ins.sg take_care[ipfv]-cvb
żeby jej nie zmąci*-ć.*
con.irr 3f.gen neg disturb[pfv]-inf
'She washed the can with sand and filled it with water, *taking care* **not to**

disturb her.' ('… lest she disturb her.')
[PNC; I. Jurgielewiczowa: *Ten obcy.* 1990 [1961]]

(22) Polish

Pilnowa-ł-em	*się,*	***żeby***	***nieopatrznie***	***cz-egoś***	***nie***
beware[IPFV]-PST-1SG.M	REFL	CON.IRR	unattentively	something-GEN	NEG

chlapną-ć	*o*	*jej*	*męż-u.*
blurt_out[PFV]-INF	about	her	husband-LOC

'*I was careful* **not to blurt out anything inadvertently** about her husband.'
(i) '… in order to not blurt out …'; (ii) '… lest I blurt out …'
[PNC; J. Grzegorczyk: *Chaszcze.* 2009.]

Czech and Slovak, contrary to Polish, do not allow the irrealis connective (*aby*; see (8)) to introduce infinitival complements (M. Giger, p.c.); the use of infinitival clauses in these two languages is generally very restricted. Thus, the only opposition is between *aby*, which must combine with (so-called expletive) negation and the *l*-form, and a default complementizer (Czech/Slovak *že*), which goes without expletive negation and does not restrict the TAM-choice in the verb. These options exist in Russian and Polish as well, but their choice does not affect the SS—DS distinction. It is however indicative of the balanced—deranked distinction, provided grammatical distribution is taken into account: the option with irrealis connective and negation (*aby … ne*) restricts the choice of the verb form (for reasons given in Footnote 13), while the default complementizer does not (see §3.3). However, these restrictions only follow from the general collocational properties of clause-initial connectives with incorporated irrealis marker (*by*), which are the same in independent clauses.

Analogous remarks apply to Balkan Slavic embedded NEGOPT clauses. Although a finite-infinitive contrast is not available, deranking is based on grammatical distribution: the predicate of an embedded apprehensional clause very rarely occurs in the aorist; example (23) is constructed, since even an internet search does not provide good examples. Instead, the predicate of an embedded apprehensional clause predominantly takes a present tense form; more occasionally it occurs in the perfect.[20] See (24), which shows simultaneous reference, whereas (25) demonstrates anterior reference:

[20] A future marker (Bulg. *šte*, Mac. *ḱe*) is excluded, since it never co-occurs with *da* in the same cluster.

(23) Macedonian (constructed, by courtesy of E. Bužarovska and L. Mitkovska)
Se plaš-am ***da ne*** *zapadn-aa vo nevolja.*
REFL be_afraid[IPFV]-PRS.1SG IRR NEG fall[PFV]-AOR.3PL into trouble
'I fear **that** they have got into trouble.'

(24) Macedonian (Mitkovska et al. 2017: 67)
Tina molče-še, isplaše-n-a ***da ne***
PN be_quiet[IPFV]-IMPF.3SG frighten[PFV]-PP-SG.F IRR NEG
ima i taa takov virus.
have[IPFV].PRS.3SG also 3F.NOM such virus[SG.M]
'Tina kept quiet, fearing **that** she **might** have the same virus.'

(25) Bulgarian (Mitkovska et al. 2017: 67)
Strax me e ***da ne bi*** *da ne e*
fear 1SG.ACC be.PRS.3SG IRR NEG SBJV IRR NEG be.PRS.3SG
doš-l-a.
arrive[PFV]-LF-SG.F
'I fear **that** she **might** not have come.'

Deranking even in this extended sense does not seem to apply in the western part of South Slavic (Serbian-Croatian, Slovene), as embedded apprehensional clauses allow for future and conditional mood (see §3.3). This appears to follow from the loss of irrealis restrictions otherwise imposed by *da*.

3.3 Temporal reference

Since in North Slavic clause-initial connectives with an incorporated irrealis marker (*by*) restrict the form of the verb (*l*-participle or, in Russian and Polish, infinitive), clauses containing them cannot distinguish temporal reference, either to speaker deixis (in independent use) or to the reference interval set by a matrix predicate for an embedded apprehensional clause. Nonetheless, if *by* participates in the formation of NEGOPT strategies, these show a strong tendency toward posterior reference (to either a deictic or a syntactically anchored reference interval). Russian (independent or dependent) infinitive clauses only allow for a posterior orientation (see (9a), (26b)–(26c)), but even with the *l*-form, posterior orientation seems to practically be the exclusive option. Compare the following textbook examples:

(26) Russian

a. DS: biclausal with different subjects → balanced (symmetric):
Ja *boj-u-s'*, ***kak by*** *on-a* ***ne***
1SG.NOM fear[IPFV]-PRS.1SG-REFL how IRR 3-F.NOM NEG
*prostudi-**l**-a-s'.*
catch_cold[PFV]-PST-SG.F-REFL
'I fear that she (i) ?has caught a cold. (ii) '… will catch a cold.'

b. SS: biclausal with identical subjects → deranked (asymmetric):
Ja *boj-u-s'*, ***kak by*** ***ne***
1SG.NOM fear[IPFV]-PRS.1SG-REFL how IRR NEG
prostudi-t'-sja.
catch_cold[PFV]-INF-REFL

c. = *Ja* *boj-u-s'* *prostudi-t'-sja.*
1SG.NOM fear[IPFV]-PRS.1SG-REFL catch_cold[PFV]-INF-REFL
'I fear to catch a cold / that I will catch a cold.'

While the embedded infinitive clause in (26b) and (26c) induces a posterior reading, the finite embedded clause in (26a) might in principle allow for an anterior reading (as indicated in (i)). However, native speakers are highly reluctant to accept this interpretation (Dobrushina 2016: 282 considers it inappropriate), and I have been unable to find convincing examples in the RNC (nor does Zorikhina Nilsson 2012 adduce any). Only theoretically is there an alternative to (26b) with a finite predicate:

(27) Russian
Ja *boj-u-s'*, ***kak by*** *ja* ***ne***
1SG.NOM fear[IPFV]-PRS.1SG-REFL how IRR 1SG.NOM NEG
*prostudi-**l**-∅-sja.*
catch_cold[PFV]-PST-SG.M-REFL
'I fear that I will catch a cold / ?… that I have caught a cold.'

Since informants judge such an option as artificial, speculations about its possible interpretation are senseless. We may thus presume that, in Russian, the practically complementary distribution of (DS × finite) vs (SS × non-finite) predicates is not accompanied by variability of temporal orientation (anterior – simultaneous – posterior), i.e. the structure *kak by … ne* is strongly associated with posterior reference in both dependent and independent clauses.

The situation differs in South Slavic, but differently for different subareas. In the (north)western part, i.e. in Slovene and (standard) Serbian-Croatian, *verba* *timendi* combine with *da* as default complementizer (‘that’), which has basically lost its irrealis semantics and does not function as a verb-adjacent proclitic.[21] Concomitantly, in Slovene verbs like *bati se* ‘be afraid’ do not restrict the TAM-form of the predicate in their dependent clause:

(28) Slovene
Politik ***se boji,*** …
politician[NOM.SG] REFL fear[IPFV].PRS.3SG
a. Past (with *l*-form)
da ***so*** *volilc-i* ***izvoli-l-i*** *nasprotnik-a.*
COMP BE.PRS.3PL voter-NOM.PL vote[PFV]-LF-PL opponent-ACC
b. Present
da ***volij-o*** *nasprotnik-a.*
COMP vote[IPFV]-PRS.3PL opponent-ACC
c. Future
da ***bo-do izvoli-l-i*** *nasprotnik-a.*
COMP FUT-3PL vote[PFV]-LF-PL opponent-ACC
‘The politician is afraid …’
(a) ‘that the voters have voted for the opponent.’
(b) ‘that they vote for the opponent.’
(c) ‘that they will vote for the opponent.’

Forms of the indicative future (28c) seem to predominate. Moreover, as (28a) to (28c) show, *da* tends to go without negation (*ne*). Negation is not excluded, but the factors that condition its interpretation as “ordinary” sentence negation, as in (29), as opposed to expletive negation, have not been clarified yet. In either case, *da* can occur with the future (29) or the subjunctive (30a) marker. An infinitival complement is possible, in principle, in SS-clause combining (30b). However, in cases of SS, a dependent apprehensional clause tends to be finite (other than in Russian or Polish, but as in Czech and Slovak); compare (27) with (29).

(29) Slovene
Kakšn-a suš-a, boji-m se, da ne
which-NOM.SG.F drought[F]-NOM.SG fear[IPFV]-PRS.1SG REFL IRR NEG

[21]For this process and its inner-South Slavic cline cf. Sedláček (1970); Grković-Major [Grković-Mejdžor] (2004); Grković-Major [Grković-Mejdžor] (2020). For an account of Slovene cf. Uhlik & Žele (2017: 98–103). All examples and claims on Slovene in the remainder of this subsection are taken from there.

bo *dolgo* *deževa-l-o.*
FUT.3SG long_time rain[IPFV]-LF-N
'What a drought. I'm afraid it won't rain for long.'

(30) Slovene

a. *Boji-m* *se,* *da* *(ne)* *bi* *pozabi-l-∅*
fear[IPFV]-PRS.1SG REFL COMP NEG SBJV forget[PFV]-LF-SG.M
denarnic-o.
purse-ACC
'I fear that I might / will forget my purse.'

b. *Boji-m* *se* *pozabi-ti* *denarnic-o.*
fear[IPFV]-PRS.1SG REFL forget[PFV]-INF purse-ACC
'I fear to forget my purse.'

The situation in Serbian-Croatian is similar, however here perfective present tense with negation appears to be the preferred option in dependent clauses (see 31a). Another option is modal auxiliaries in the subjunctive (see 31c); compare the examples in (31):

(31) Serbian (Alanović 2018: 163)
Boji *se,* …
fear[IPFV]-PRS.3SG REFL

a. Perfective present (+ negation)
da *ne* *zakasni.*
IRR NEG be_late[PFV].PRS.3SG

b. Perfective future
da *će* *zakasni-ti.*
IRR FUT.3SG be_late[PFV]-INF

c. Conditional with possibility modal auxiliary
da *bi* *moga-o-∅* *zakasni-ti.*
IRR SBJV can-LF-SG.M be_late[PFV]-INF
'He is afraid …'
(a) 'to arrive late' [lit. 'lest he arrive late'].
(b) 'that he will arrive late.'
(c) 'that he might arrive late.'

In Balkan Slavic, since the relevant clause-initial particles do not entail any restrictions of temporal orientation, embedded clauses may also show simultaneous or anterior time-reference; compare (23)–(25) with (32):

(32) Bulgarian (Ivanova 2014: 150)
Vrabče-ta-ta pomisli-l-i ***da ne bi*** *da e*
sparrow-PL-DEF.PL think[PFV]-LF-PL IRR NEG SBJV IRR be.PRS.3SG
izbuxna-l-a vojna-ta, (…)
break_out[PFV]-LF-SG.F war[F]-DEF.SG.F
'The sparrows thought [> were afraid] **that** the war **might** have broken out (but nowhere could any rumbling be heard).'

3.4 Loss of compositionality

All clause-initial connectives relevant for NegOpt show a high degree of coalescence (which has diminished or eliminated morphological transparency) among their components. In South Slavic, these components, first of all, cannot be separated by other (even clitic) units; in North Slavic, *by* has coalesced with the preceding component, but the negation can be separated from it. As for semantic transparency (vs opacity), the matter is more complicated.

Macedonian *da ne* (IRR+NEG) has lost segmentability and semantic transparency. This regards its use both in apprehensional main and dependent clauses as well as its use as epistemic modifier (employed, for instance, in deliberative questions; see (35a)). The evidence is, first, that in these functions *da ne* is used non-redundantly with sentence negation (Mitkovska & Bužarovska 2017, Mitkovska et al. 2017: 67, 75 et passim). See (18) for an 'in-case'-reading and (33), which can acquire an apprehensional function:

(33) Macedonian
A kade e Marko? ***Da ne*** *ne dojd-e na vreme?*
and where be.PRS.3SG PN IRR NEG NEG come[PFV]-PRS.3SG on time
'Where is Marko? (I'm afraid) maybe he won't come in time.'

Second, in these functions *da ne* never attracts stress; see the transparent use in (35c). Third, the requirement to occur adjacent to the verb (as part of a proclitic cluster) gets lost; see (34) and (35a):

(34) Macedonian
Se plaš-am ***da ne*** *slučajno dozna-at*
REFL be_afraid[IPFV]-PRS.1SG IRR NEG accidentally learn[PFV]-PRS.3PL
za toa.
about this
'I'm afraid they might *accidentally* find out about it.'

Fourth, TAM constraints get lost as well, i.e. the aorist is possible (see 35a), which, as a "confirmative tense" (Friedman 2000), is otherwise incompatible with the irrealis function of *da*. As stated in §3.2, in complementizer function this happens very rarely.

It is often difficult to distinguish *da ne* as an epistemic modifier (see 35a) from its use as a marker of apprehension (related to optative or purpose clauses). In fact, *da ne* (in both Macedonian and Bulgarian) can be used as a general epistemic downtoner void of emotional involvement; its apprehensional function can only be inferred from this general function. Compare examples (35b) to (35e); stress is indicated by bold capital letters:[22]

(35) Macedonian

a. Epistemic modifier: no apprehension
Da ne ***včera*** *ja skrši vazn-a-ta?*
IRR NEG yesterday 3F.ACC break.AOR.2SG vase[F]-SG-DEF.SG.F
'Did you by any chance break the vase **yesterday**?'

b. Epistemic modifier: apprehension
Da ne (si) ***O****di-š?*
IRR NEG REFL.DAT go_away[IPFV]-PRS.2SG
'(I'm afraid:) Are you really leaving?'
alternatively, deliberative question: 'You are not leaving, are you?'

c. Transparent collocation: prohibitive
Da ***NE*** *odi-š!*
IRR NEG go_away[IPFV]-PRS.2SG
'(I order you:) Don't go!'

d. Opaque: negative optative/negative purpose > preventive
Pobrza-j-te! Da ne go ispušti-te
hurry[PFV]-IMP-PL IRR NEG 3M.ACC miss[PFV]-PRS.2PL
avion-ot!
airplane[SG.M]-DEF.SG.M
'Hurry up! Don't miss the plane!' (≈ 'Otherwise you'll miss your plane.')

[22]For a detailed treatment cf. Bužarovska & Mitkovska (2015: 37 et passim), 2017, Mitkovska et al. (2017: 67–70).

e. Optative
Ubavo da si pomine-š na odmor!
nicely IRR REFL.DAT pass[PFV]-PRS.2SG on vacation
'Have a nice vacation!' (more literally 'May you nicely pass for a vacation!')

In (35a) and (35b), using *da ne* the speaker utters an assumption. Likewise, (35c) does not express apprehension, whereas (35d) does. That is, a deliberative question can yield apprehension only as a side-effect, given a suitable communicative purpose and corresponding intonation.[23] By contrast, (35d) is the negative equivalent of an optative sentence (as in (35e)), and it can acquire a preventive interpretation. In none of these instances can *da ne* be stressed, contrary to (35c), where *da+ne* is a transparent collocation: the negation can be stressed, but the utterance will be understood only as a prohibition. Notably, prohibition is associated rather with imperfective aspect, while a preventive interpretation of a NEGOPT construction (as in (35d)) is rather associated with perfective aspect. This is a tendency, not a rule, but it corresponds to the more clear-cut picture in North Slavic (see §4.1–§4.2).

As for Bulgarian *da ne bi* (etymologically IRR+NEG+SBJV), this unit cannot be split internally, but as a whole it can be separated from the verb by other constituents (see (2), (17)). It also non-redundantly combines with negated predicates (see 25). These properties clearly indicate loss of morphological and semantic transparency, but, contrary to Macedonian Bulgarian *da ne*, the modified clause needs to be introduced by another *da* (see (2), (17), (20), (25)).[24] This subsequent *da* can hardly be considered part of the complex marker itself, but must be analyzed as the usual verbal proclitic (sometimes within a clitic complex, as in (2), (19), (25), (25)). In any case, for *da ne bi* the irrealis requirement still applies, similarly to North Slavic units containing *by*. This behaviour applies in main clauses as well, regardless of whether it marks apprehension (as in (36)) or functions as an epistemic modifier (e.g., in rhetoric questions):[25]

[23] Cf. Mitkovska & Bužarovska (2017) and Mitkovska et al. (2017: 68f.), who call this type of question 'assumptive': the speaker wants to gain confirmation for their assumption that the proposition is true.

[24] There are exceptions to this, for instance if *da ne bi* modifies a PP to express surprise, e.g. *Pitam kelnera: "A zeleto s kakvo go režete? –* ***da ne bi*** *s nožovka?"* 'I ask the waiter: "How do you cut the cabbage? - **da ne bi (≈ really)** with a hacksaw?"' (Ivanova 2014: 148). Such mirative uses of *da ne bi* are indicative of its lexicalization as an epistemic modifier, in analogy to Macedonian *da ne* in (35a)–(35b).

[25] For details and slight differences between Macedonian *da ne* and Bulgarian *da ne bi* cf. Ivanova (2014: 147–150) and Mitkovska et al. (2017: 70).

(36) Bulgarian (Mitkovska et al. 2017: 73)

Lele,	*da*	*ne*	*bi*	*da*	*Vi*	*nastăp-ix*	*po*	*mazol-a?*
EXCL	IRR	NEG	SBJV	IRR	2PL.DAT	step_on[PFV]-AOR.1SG	over	bunion(M)-DO

'Oh dear, have I maybe stepped on your bunion?'

Turning now again to North Slavic, the irrealis morpheme *by* has become an integral part of clause-initial particles which cannot anymore be segmented into morphemes (see Footnote 13). If these connectives mark apprehension, negation must follow, although it need not be adjacent (compare, e.g., Russian *kak by* … *ne*, generalized as [X+*by* … NEG]). In semantic terms, too, the negation is transparent for those markers which have equivalents without negation in optative function (e.g., Russian *liš' by (ne)*, *kak by (ne)*). With respect to volitional attitude, optative and apprehensional meanings behave like antipodes: the judging subject (speaker or contextually determined) does not want E to occur (see Table 1, also Dobrushina 2016: 343); see §2. This holds for independent and main clauses likewise. However, in embedded clauses the negation semantically harmonizes with the 'fear'-component of the matrix predicate; we are thus dealing with "apprehensional concord" (in analogy to modal concord); see (37a) below. On the one hand, concord phenomena support syntactic tightness between the relevant parts of the utterance (here a complex sentence); on the other hand, they imply that previously these parts were more autonomous from each other. As long as the assumed dependent part (here: the clause introduced by [X+*by* … NEG]) can occur on its own, concord simply testifies to usual semantic mechanisms of collocatability. (In §5.1 we will see that this mechanism can be made responsible for embedding.)

Moreover, if in embedded apprehensional clauses the default complementizer (Russian *čto*, Polish *że*, Czech/Slovak *že*) is used to denote an undesirable situation, negation is not required. If negation is employed, its use is compositional, i.e. it marks the judging subject's fear that the respective situation might not occur. Thus, (37c) is synonymous with (37a) only without negation (indicated by #).

(37) Polish

a.

Rob-imy	*razem*	*interes-y*	(…)	*i*	*boj-ę*
do[IPFV]-PRS.1PL	together	deal[M]-ACC.PL		and	fear[IPFV]-PRS.1SG

się,	***żeby***	*nie*	*wpad-ł-∅*	*w*	*zł-e*	*towarzystw-o.*
REFL	**COMP.IRR**	NEG	fall[PFV]-PST-3SG.M	in	bad-ACC	company-ACC

'We are partners in business (…) and I'm afraid that he might fall in with the wrong crowd.' [PNC; M. Bielecki: *Osiedle prominentów*. 1997]

b. *(…) ***że*** *nie wpad-ł-**by**-∅ w zł-e towarzystw-o.*
comp neg fall[pfv]-pst-**sbjv**-3sg.m in bad-acc company-acc

c. (…) ***że*** *(#nie) wpadni-e w zł-e towarzystw-o.*
comp neg fall[pfv]-fut.3sg in bad-acc company-acc

'(…) I'm afraid that he will (#not) fall in with the wrong crowd.'

Concomitantly, loss of morphological transparency in the univerbation of *by* with other clause-initial particles has led to a situation in which complex connectives (X+*by* … neg) employed as negative-optative markers can be distinguished from free combinations of morphologically simple (and semantically neutral) connectives with the subjunctive marker (X + (… +) *by*).[26] After 'fear'-verbs such combinations are ungrammatical (see 37b). However, at least in Polish we do find examples like (38) in which the standard complementizer (*że*) introduces an apprehensional clause with an auxiliary complex and in which the subjunctive is marked on the auxiliary *móc* 'can'. Note, however, that an expletive negation is lacking and time-reference can only be posterior:

(38) Polish
*Bartek obawia się, **że** mog-l-i-**by**-śmy tam*
pn[nom] fear[ipfv].prs.3sg refl **comp** can-pst-pl-**sbjv**-1pl there
wnieś-ć ze sobą nie najmądrzejsz-e pomysł-y z
bring[pfv]-inf with refl.ins neg wisest-acc.pl idea-acc.pl from
nasz-ego świat-a.
our-gen.sg.m world(m)-gen.sg
'Bartek is afraid **that** there we **might** contribute not the best ideas of this planet.' [PNC; D. Terakowska: *Władca Lewawu.* 1989]

This pattern (standard complementizer + subjunctive of POSS auxiliary) resembles one of the options encountered in Serbian (see 31c).

4 Negative-directive strategy (NegDir)

The negative-directive strategy is realized by preventives.[27] Due to their directive illocutionary force, preventives cannot be embedded. This is a major difference from the negative-optative strategy (Dobrushina 2006: 41): directive-preventives

[26] Cf. Wiemer (2015, 2023: §3.1), also Dobrushina [Dobrušina] (2012, 2016: 322–327) for Russian.

[27] On the history of the term 'preventive' and its (quasi-)synonyms cf. Ivić (1958: 29f.); Birjulin (1992: 2, 1994: 99f.).

are used not just to express apprehension, but to alert the addressee to avoid the respective undesirable situation, i.e. the speaker does not want the addressee's actions (or lack thereof) to inadvertently cause such a situation (see §2). The inadvertent component can be emphasized by appropriate adverb(ial)s (see (6), (39), (42b)):

(39) Polish
Nie zapomn-ij (*przypadkiem*) *okular-ów!*
NEG forget[PFV].IMP.SG accidentally glasses-GEN
'Don't forget (accidentally) your glasses!'

Jointly with this, preventives are inherently future-oriented, and this further differentiates the NegDir strategy from NegOpt strategies.

Preventive utterances can include attention-catching devices. Many of them are imperatives, almost exclusively of imperfective verbs, e.g. Russian *smotri* 'look' (Xrakovskij & Volodin 1986: 153f. Xrakovskij 1990: 215f.), Polish *uważaj* 'be careful', Czech *dávej pozor* 'pay attention', Slovene *pazi* 'be careful', Bulgarian *pazi se* 'be careful', *vnimavaj* 'be attentive'. Only Russian *ne vzdumaj* ≈ 'don't happen to think' derives from a perfective verb;[28] incidentally, it represents itself a (lexicalized) preventive imperative. These forms distinguish singular-plural in accordance with the number of addressees; they can therefore still be classified as imperatives and need not be treated as particles.

4.1 Negated imperatives and the role of aspect

Most accounts of directive-preventive strategies in Slavic are based on Russian, and it has been emphasized that the preventive function of the negated imperative is bound to perfective aspect. In fact, this function usually seems the only sensible interpretation of a negated perfective verb employed in a directive speech act. The central function of perfective aspect is to put a limit on the denoted eventuality, thus negation operates on a boundary, not on the action itself. Apart from preventives, this can be illustrated with sentences in which negation has narrow scope over some quantifiable component of the situation (Holvoet 1989: 104; Łuczków 1997: 38; Laskowski 1998: 18 for Polish; Kučera 1984: 122 for Czech), e.g.

[28]Other attention catchers attested in preventives are nouns like Polish *uwaga*, Czech *pozor* 'attention' or particles like Czech *ať*, which has various directive and purpose-related uses.

(40) Polish

Nie zastaw ***cał-ego*** *pokoj-u swo-imi*
NEG obstruct[PFV].IMP.SG entire-GEN room-GEN REFL-POSS-INS.PL
grat-ami!
lumber-INS.PL
'Don't obstruct the **whole** room with your lumber!'

(41) Polish

Nie wróć ***zbyt późno!***
NEG return[PFV].IMP.SG too late
'Don't return **too late!**' (i.e. 'Return, but not too late!')

In preventives, negation acquires narrow scope since the purpose of the clause is only to indicate that a particular state-changing event (= boundary) should not be brought about, regardless of what the addressee might do to avoid it (see 42b). Thus the action itself is beyond its scope, or presupposed. By contrast, in prohibitives negation has wide scope, since the speaker urges the addressee not to undertake any activity connected to a state-changing event (see 42a). This contrast becomes particularly evident under minimal-pair conditions; compare the imperfective (42a) with the perfective (42b) negated imperative, both referring to a single time-located situation:

(42) Russian

a. *Ne otkryva-j okn-o!*
NEG open[IPFV]-IMP.SG window-ACC
'Don't open the window!'
→ prohibitive (IPFV) ≅ 'Don't undertake any action connected to opening the window (e.g., turning the handle)!'

b. (*Smotri,*) *ne otkro-j* (*slučajno*) *okn-o!*
look NEG open[PFV]-IMP.SG accidentally window-ACC
'(Look,) don't open the window (accidentally)!'
→ preventive (PFV) ≅ 'Whatever you do (e.g., with the handle), be careful not to open the window!'

The negated imperfective imperative yields a prohibitive meaning; the motivation for this is a volitional basis ('I don't want you to deal with situation S'). The preventive meaning, in contrast, is more closely associated with prediction (i.e. an epistemic function), which, in turn, often arises from inferences based on contingent circumstances ('If you don't take care, event E is going to happen'; see Table 4 in §2). This correlates with other contexts of negation, for which a tendency toward a volition-based split of modality (deontic → IPFV. vs non-deontic → PFV.) has been observed, at least in Russian and Polish.[29] The common denominator is that negated perfective aspect triggers implicatures focusing on boundaries, while negated imperfective aspect can defocus boundaries; it is thus compatible with other parts of clausal semantics, for which implicatures arise. Thus, the distinction between prohibitive (→ IPFV.) and preventive (→ PFV.) readings of the negated imperative lines up well with more general tendencies of aspect choice, and these appear to be most salient in the northeastern part of Slavic (= Russian). Inner-Slavic differences result from the different strength with which invited inferences have conventionalized and been associated with perfective and imperfective aspect in complementary distribution.

We may thus say that a preventive interpretation arises from the strengthening of implicatures resulting from a transparent combination of negation and perfective aspect in directive illocutions for time-located events. This interpretation is entirely compositional. Nonetheless, the conventionalization of invited inferences, based on the binary PFV:IPFV opposition, appears to be weaker in South Slavic, even if we account for constructions that have been ousting the inherited imperative (see §4.3). Moreover, the strict aspect-based division between prohibitive and preventive is definitely a rather recent development even in Russian (see §5.2).

Simultaneously, we have to realize that a person uttering a preventive speech act must assume that the addressee has control – not necessarily over the event denoted by the verb in the negated imperative, but over associated actions which can allow the addressee to avert the undesirable event (or its consequences); see Table 4 in §2. Otherwise we could hardly explain why preventives are frequent with controllable actions (see (42b), (44a) and below), but excluded for certain verbs, e.g. of mental activity, for which control cannot be given a limit; compare Polish **Nie zastanów*$^{\mathrm{PFV}}$ *się nad tym!* (Holvoet 1989: 103), to mean something weird like ‘Don’t (make it happen that you) accidentally think it over’. Importantly, although preventive utterances often build on non-agentive verbs, agentive verbs are not excluded (see (42b) and Russian *Ne slomaj*$^{\mathrm{PFV}}$ *stul* ‘Don’t break the chair!’, *Ne razbej*$^{\mathrm{PFV}}$ *steklo* ‘Don’t break the glass’). Preventives “denote uncontrollable events resulting from some relevant controllable actions, and it is these controllable actions that are in fact addressed by the prescriptor (e.g. the sentence *Don’t break the chair* may be interpreted as ‘Don’t rock the chair’ [so that it will not break; BW])” (Birjulin & Xrakovskij 2001: 38, also Xrakovskij 1990; Birjulin 1992).

[29]Cf. Wiemer (2001, 2008: 403–405, 2017: 281–285) for overviews, Padučeva (2008, 2013: 211–218) for Russian, and Divjak (2009) for a usage-based investigation on Russian.

Negated perfective imperatives happen to acquire other pragmatic functions as well, e.g. advice, request, or persuasion (e.g., Polish *Nie obraź*^PFV^ *się na mnie* 'Don't get offended at me', Serb. *Ne shvati*^PFV^ *krivo!* 'Don't misunderstand me!') or, to the contrary, categorical demands and pleas. Such functions are "fuzzier" than the preventive; they partially depend on the lexically determined actionality of the verb, partly they can be calculated from specific social and conversational constellations between the interlocutors (who is in control of what and over whom?).[30] Many such instances are archaic and have become petrified phrases (e.g., Russian/Bulgarian *Ne daj Bože* 'God forbid', lit. 'Don't give, God'). Here belong some of the aforementioned attention catchers (e.g. Russian *ne vzdumajte* 'don't (dare) think'), and we find them also in varieties of South Slavic which have otherwise abandoned perfective verbs from negated imperatives; e.g. Serbian *Ne uzmi*^PFV^ *za zlo* 'Don't take it for harm', against **Ne uzmi*^PFV^ *ovu knjigu* 'Don't take (inadvertently) this book' or **Ne nazebi*^PFV^ 'Don't get a cold' (Ivić 1958: 27f.), which would be truly preventive, but have been ousted by periphrastic constructions (see §4.3).

Moreover, negated perfective imperatives occur throughout Slavic as a means of threat or intimidation, although in South Slavic (except Slovene) this function is residual (see §4.3). Compare, for instance, Czech *Jenom mi to* ***nekup***!^PFV^ '**Don't (you) dare buy** it for me' (Ivić 1958: 28f. Kopečný 1962: 60f.), Macedonian ***Nedojdi*****^PFV^** *pa ḱe vidiš što te čeka!* '**Don't come**, or you'll see what you get (lit. 'what waits')' (Koneski 1976: 416, Tomić 2012: 346f. Bužarovska & Mitkovska 2015). Obviously, a warning is the verbalisation of a threat, and threats are conceptually very close to preventives; the crucial difference is that threats are uttered in the interest of the speaker, while preventive speech acts are issued for the sake of the addressee (see §2).

After all, we can say that, if negated perfective imperatives have not been ousted as such in the respective Slavic language, their preventive function predominates across the verb lexicon; other, minor functions are related to preventives and can be explained via pragmatic mechanisms. As for those Slavic languages in which negated perfective imperatives are no longer employed (basically, South Slavic except Slovene), it is an open question whether equivalent periphrastic constructions have taken over all other functions of negated perfective imperatives as well (see §4.3).

[30] Cf. the probably still most subtle and well-considered analyses to date in Holvoet (1989: 101–106) and Birjulin (1992, 1994: §4). Cf. also Ivić (1958: 36f.); Laskowski (1998: 22–26).

4.2 Areal cline of functions ruled by aspect?

It has been argued that in comparison to Russian (and Polish), other Slavic languages exhibit a less clear-cut prohibitive—preventive distinction ruled by aspect.[31] Such qualifications require differentiation. Let us look at the western periphery of Slavic, before we turn to the central and eastern part of South Slavic (see §4.3).

First, we do find negated perfective imperatives in preventive function in Slovene:

(43) Slovene
Ne prehlad-i se!
NEG get_cold[PFV]-IMP.SG REFL
'Don't get a cold!'

The question is how consistent this use (and its distinction from prohibitives) is in Slovene. Empirical research is lacking, however an educated guess is that consistency is much lower than in Russian (M. Uhlik, p.c.).

Second, Czech employs negated perfective imperatives in preventive meanings as well, and we do find an opposition to prohibitive meaning with imperfective verbs (Trávníček 1923: 322, Dokulil 1948: 79–81, Kopečný 1962: 60f. Kučera 1984, also Meyer 2010: 366f. for Slovak), e.g. 44.

(44) Czech
a. *Ne-škrábn-i se!*
NEG-scratch[PFV]-IMP.SG REFL
'Do not scratch yourself accidentally!'
b. *Ne-škrab se!*
NEG-scratch[IPFV].IMP.SG REFL
'Do not scratch yourself (any more)'; 'Stop scratching!'

The distinction can also be observed with verbs whose aspect correlates with different ontological types of objects. This, in turn, conditions different actionality interpretations of the entire clause; compare (45a, 45b).

[31] Cf. Zorikhina Nilsson (2013: 96) in general and Ivanova (2014: 156), Mitkovska et al. (2017) as for *da*-equivalents of negated imperatives in Bulgarian and Macedonian.

(45) Czech

a. *Ne-ztrať* *ten klíč!*
NEG-lose[PFV].IMP.SG this key[ACC.SG]
'Do not lose the key!'

b. *Ne-ztráce-j* *naděj-i!*
NEG-lose[IPFV]-IMP.SG hope-ACC.SG
'Do not lose hope!'

However, the distribution of perfective and imperfective verbs in the negated imperative seems to follow from a more general tendency due to which in Czech aspect choice is more consistently governed by different actionality types than in Russian. In the latter, imperfective verbs have expanded into contexts for which languages in the western half of the Slavic-speaking world rather choose perfective aspect. For instance, "Czech utilizes perfectives in negative constructions more widely than Russian does, and admits the imperfective in negative event descriptions much more rarely" (Kučera 1984: 118). This observation fits well with the west—east cline of Slavic aspect commonly known since Dickey (2000). Nonetheless, the preventive—prohibitive pattern which we discover for Czech, Russian and Polish must be based on the same considerations which were pointed out in §4.1. Kučera (1984) based his considerations on the factor of control: negated imperative and [+control] favours imperfective, negated imperative and [−control] favours perfective aspect. But these considerations relate control only to the lexical meaning of the verb used in the imperative; they do not account for the meaning of the whole construction, which implies that the speaker wants the addressee to undertake measures (see Table 4 in §2). If we take this into account, Kučera's analysis confirms the analysis proposed in §4.1, and we may conclude that the actionality-based preferences for perfective or imperfective verbs – which are generally stronger in Czech than in Russian – provide a favorable basis for the aspect-driven contrast between prohibitive and preventive interpretations in negated directive speech acts.

4.3 Analytic negated imperatives in South Slavic

In Balkan Slavic, negated imperatives of perfective verbs have receded, but for independent reasons. These cannot be discussed here, but it is intriguing to observe that the retreat of perfective verbs from negated imperatives has been accompanied by an increase of periphrastic constructions based not only on *da*, but also on semi-petrified auxiliary forms with incorporated negation: Bulgarian *nedej/ nedejte* (SG/PL) (+ *da*) < *ne děj(te)* from Common Slavic *dějati* 'do', Serbian *nemoj/*

nemojmo/nemojte (2SG/1PL/2PL), Macedonian *nemoj/nemojte* (2SG/2PL) <*ne mogi* from *moći* 'can' (Ivić 1958; Sedláček 1997). Note that these auxiliaries still distinguish number, as do attention catchers derived from verbs (see §4).

Bulgarian *nedej* combines with imperfective verbs in a form which goes back to the truncated infinitive, but is now homonymous with the 2/3.SG aorist, e.g. *Nedej(te) pisa!* 'Don't write!' (Ivić 1958; Sedláček 1997; Lindstedt 2010: 412). Important is the restriction to imperfective verbs, so that this construction blocks a prohibitive-preventive opposition; prohibitive meanings are the default. By contrast, Serbian/Macedonian *nemoj* combines with either perfective or imperfective aspect; Macedonian *nemoj* combines only with *da*+finite verb (since even the truncated infinitive is lost), its Serbian cognate optionally allows for the infinitive. However, aspect choice does not yield a reliable opposition of prohibitive (→ IPFV.) vs. preventive (→ PFV.) function. Therefore, the four possible realizations in (46) are practically synonymous; in combination with imperfective verbs, in any case, they trigger a prohibitive reading, while with a perfective verb no straightforward preventive reading arises (for Macedonian equivalents cf. Tomić 2012: 345):[32]

(46) Serbian

a. *nemoj* + pfv. infinitive
Nemoj *sada* *otvori-ti* *prozor!*
NEG.AUX.2SG now open[PFV]-INF window[ACC.SG]

b. *nemoj* + ipfv. infinitive
Nemoj *sada* *otvara-ti* *prozor!*
NEG.AUX.2SG now open[IPFV]-INF window[ACC.SG]

c. NEG + ipfv. infinitive
Ne *otvara-j* *sada* *prozor!*
NEG open[IPFV]-IMP.SG now window[ACC.SG]

d. *nemoj* + *da* + pfv./ipfv. present
Nemoj *sada* *da* *otvori-š*/*otvara-š* *prozor!*
NEG.AUX.2SG now IRR open[PFV]-PRS.2SG/open[IPFV]-PRS.2SG window[ACC.SG]
'Don't open the window now!'

[32] Cf., however, Szucsich (2010: 400) on connections between an apprehensional ('threat') reading and the choice of *da*+V_{fin} or infinitive after *nemoj*. He also notes that *nemoj(-mo/-te)* need not always agree with the finite verb after *da*.

Perfective verbs in negated imperatives have more and more been driven out by periphrastic constructions, so that the original imperative (see 46c) no longer has a perfective equivalent. A preventive reading in periphrastic constructions becomes more likely with low-agentivity verbs (J. Grković-Major, p.c.):

(47) Serbian
Nemoj da se razboli-š!
neg.aux.2sg irr refl get_ill[pfv]-prs.2sg
'Don't get ill!'

Apart from ousting negated perfective imperatives, these periphrastic constructions correspond to techniques which have replaced the infinitive. Infinitive loss is commonly known as a Balkan feature, and within South Slavic we observe a clear (south)east – (north)west cline, with standard Serbian and Croatian occupying an intermediate position. The most common replacement technique is the irrealis marker *da* which combines with the present tense or the perfect. This construction also replaces the negated perfective imperative, e.g.

(48) Bulgarian
(*Vnimava-j,*) *da ne zabravi-š*
pay_attention[ipfv]-imp.sg irr neg forget[pfv]-prs.2sg
knig-a-ta!
book[f]-sg-def.sg.f
'(Attention) Don't forget the book!'

(49) (= (35d))
Macedonian (Bužarovska & Mitkovska 2015: 30)
(*Pobrza-jte!*) *Da ne go ispušti-te*
hurry_up[pfv]-imp.pl irr neg 3m.acc miss[pfv]-prs.2pl
avion-ot!
plane[sg.m]-def.sg.m
'(Hurry up!) Don't miss the plane.'
or: '… for you in order not to miss the plane.' (connection with purpose clauses)

Superficially, this construction may be difficult to distinguish from the negative-optative strategy (see §3.1). As in Serbian, preventive meaning is claimed to obtain only if the denoted event is beyond the addressee's control (Čakărova 2009: 41). Compare Serbian in (47) with Bulgarian in (50):

(50) Bulgarian
Da ne si razvali-š stomax-a!
IRR NEG REFL.DAT destroy[PFV]-PRS.2SG stomach-DO
'Don't (accidentally) spoil your stomach.'

The claim concerning the addressee's control requires the same corrections as for North Slavic (see §4.1). After all, we come to the preliminary conclusion that not all South Slavic "analytic imperatives" allow for perfective verbs, but if they do, aspect choice does not yield a reliable distinction of prohibitive vs preventive function to the same degree as in Slavic languages further to the north and east. Moreover, a preventive reading of a perfective negated imperative appears to be more likely if it does not call precautional actions by their name.

5 Diachronic background

The diachronic background of apprehensional strategies is largely a *terra incognita*. Therefore, the following considerations are tentative, restricted to basic issues and should rather set the stage for future research.

5.1 Negative-optative strategy resulting from insubordination?

Negative-optative and negative-purpose clauses can be reinterpreted as complement clauses. However, even in cases when such clauses immediately follow predicates with an apprehensional semantics, their complement status often remains debatable and they show signs of low syntactic integration (see §5.1.1). From a synchronic perspective, clauses with apprehensional markers in initial position may sometimes look like truncated sentences, i.e. with an elliptic main clause (containing the matrix predicate). This would amount to insubordination, i.e. the "conventionalized main clause use of what, on prima facie grounds, appear to be formally subordinate clauses" (Evans 2007: 367). Evans himself is aware that such a *prima facie*-qualification can turn out to be inadequate after an assessment of the diachronic background. That is, what superficially resembles the remainder of a complex sentence (with elided main clause) can actually turn upside down the chronology, which rather testifies to a development from juxtaposition to clausal complementation.

Consequently, the question to be asked is: did Russian *kak by … ne*, Polish *żeby … nie*, Bulgarian *da ne bi*, etc., first occur as complementizers (i.e. as flags indicating that the following clause functions as an argument of a predicative expression in the matrix clause), or did they simply occur as clause-initial units able to mark apprehension? I want to argue that, in general, diachronic primacy should be assigned to usage in independent clauses, i.e. that juxtaposition preceded tighter clause combining. While this eventually might not apply to constructions deriving from negative purpose clauses (see §5.1.2), for all other apprehensional clause types discussed here we find strong conceptual and empirical arguments against insubordination.

5.1.1 Negative-optative in North Slavic

A plausible diachronic explanation for the rise of embedded negative-optative clauses looks as follows. Independent clauses introduced with negative-optative markers frequently occurred after 'fear'-verbs (or related predicative units). The latter were suitable as complement-taking predicates (CTPs) of clauses headed by negative-optative markers, so that the latter became reinterpreted as clausal complements of predicates because they "harmonized" with them in terms of apprehensional semantics. This process can be called "syntactic tightening by apprehensional harmony" (cf. Zorikhina Nilsson 2012: 54; Dobrushina 2016: 345f. for a similar reasoning). This is shown in Figure 1 for Russian *kak by … ne*, illustrated in (51).

(51) Russian (RNC; S. Taranov 1999; cited in Zorikhina Nilsson 2012: 59)
Ja *boja-l-∅-sja,* ***kak by*** *ty* ***ne***
1SG.NOM fear[IPFV]-PST-SG.M-REFL how IRR 2SG.NOM NEG
zabole-l-a *posle svo-ix* *skitani-j.*
get_ill[PFV]-PST-SG.F after REFL.POSS-GEN.PL wandering-GEN.PL
'I *was afraid* **that** you became/you could/might/would become ill after all your wanderings.' (Zorikhina Nilsson's translation)

There are quite a few cases in the history of various, largely unrelated languages in which apprehensional complements are based on "apprehensional harmony" (cf. Dobrushina 2021 for a critical survey). As for Russian *kak by … ne*, this complex connective appeared in complement clauses only in the 19th century (Dobrushina 2021: 140–142). In independent clauses the negation is transparent inasmuch as it marks the negative equivalent of an optative clause, but in clausal complementation negation is simply in concord with the meaning of the apprehensional matrix predicate (see §3). I see no other way of explaining this relation than as a result of a development from juxtaposition to subordination.

Further support for the process assumed in Figure 1 comes from the following considerations. First of all, not all clause-initial optative and apprehensional markers occur in embedded clauses (e.g., Russian *liš' by (… ne)*, Polish

	p	***q***		
[Stage A]	X fears :	*kak by* … clause-initial particle	*ne* NEG	+ V_{fin}/$V_{infinitive}$
	→ juxtaposition (*q* is related to *p* at discourse level)			
[Stage B]	X fears CTP	*kak by* … complementizer (semantically “harmonic” with CTP)	*ne*	+ V_{fin}/$V_{infinitive}$]
	→ complementation (*q* becomes embedded in *p*)			

Figure 1: Syntactic tightening of “apprehensionally harmonious” juxtaposed clauses

oby (… nie)), but all complementizers of embedded apprehensional (or, respectively, optative) clauses occur in independent clauses. This directed implicational relationship speaks in favour of the assumption that the diachronic source of complementizers resides in clause-initial particles, not vice versa. The question, then, is which of the relevant clause-initial markers eventually gets involved in clausal complementation, as illustrated in (52) for example (51).

(52) ***p*** ***q***

X is afraid $\boxed{\begin{array}{l}\checkmark\textit{kak by}\\ {}^{*}\textit{tol'ko by}\\ {}^{*}\textit{liš' by}\\ \ldots\end{array}}$ … *ty ne zabolela posle svoix skitanij*

‘that/lest … you become ill after all your wanderings.’

(Compare as independent clause: ‘If only you don’t get ill …’)

Finally, syntactic subordination between *p* and *q* remains weak. First, *q* can be separated from *p* by a pause (and a falling intonation contour at the end of *p*). Second, *p* and *q* cannot be permuted. Moreover, I am unaware of any examples in which *q* appears inserted into *p*. Third, island extraction of a constituent from *q* by a WH-word sounds very unnatural (and possibly does not exist); compare Russian ??*Čego ty boiš’sja, kak by ne slučilos’?*, to mean ‘What are you afraid of that might happen?’ For most of these properties cf. Dobrushina (2016: 344f.), who also points out that they are characteristic of positive optative clauses (with *kak by*) as well.

5.1.2 Negative purpose in North Slavic

The insubordination issue differs for connectives that introduce clauses whose structure demonstrates a close relationship to negative purpose. The diachronic relation between negative purpose clauses and complements to 'fear'-verbs in Slavic leaves many open questions, which cannot be solved here, but this relation is evident for clauses embedded by clause-initial particles which are composed of a standard complementizer (Russian *čto*, Polish *że*) or a coordinative conjunction (Polish/Czech/Slovak *a*), the irrealis morpheme *by* (Russian *čtoby*, Pol. *żeby*, Czech/Polish *aby*) and clausal negation.

In the contemporary languages the aforementioned connectives are widely used as (main or almost exclusive) adverbial subordinators of (positive and negative) purpose clauses. Only Russian *čtoby … ne* has become obsolete in apprehensional function, while the other ones are still well-attested in contemporary usage. Example (53) illustrates such an obsolete use:

(53) Russian (A. Chekhov; cited in Zorikhina Nilsson 2012: 55)

*… **iz***	***strax-a**,*	*čtoby*	*o*	*nem*	*ne*	*duma-l-i*	*durno.*
out.of	fear-GEN	COMP.IRR	about	3M.LOC	NEG	think[IPFV]-PST-PL	badly

'(He never kept maidservants) **fearing/out of fear** that they could/might/would think ill of him.'

Contrary to optative clauses and clauses with an apprehensional semantics, purpose clauses can hardly occur as independent clauses,[33] so that one wonders whether independent apprehensional clauses with the aforementioned clause-initial connectives are not to be considered instances of insubordination. Embedded clauses with an apprehensional meaning (after suitable CTPs) are often indistinguishable from negative purpose clauses (see (13), (22), (35d)). From a typological perspective, "the evidence for a close structural relationship between purpose and complement clauses is overwhelming", and there are several functional motivations for an overlap of purpose and certain complementation relations (Schmidtke-Bode 2009: 161, 164). Nonetheless, the relation of negative purpose (> apprehensional) to purpose is less clear than it might appear at first sight.[34]

[33] Cf. Dobrushina (2016: 271f.). This corresponds to a usual typological pattern (cf. Schmidtke-Bode 2009: 150f.).

[34] First of all, negative purpose clauses disfavour contexts with (directed) movement, while positive purpose clauses occur predominantly in exactly this context. Moreover, negative purpose clauses prefer different-subject constructions, whereas positive purpose clauses prefer same-subject constructions. These differences are indicative of different experiential correlations (cf. Schmidtke-Bode 2009: 129–144 for typological facts and a critical survey); probably, these correlations correspond to different illocutionary purposes.

This leaves us with the question whether the apprehensional usage is diachronically secondary, i.e. derived from negative purpose[35] (see 54a), or whether both functions developed in parallel, but independently from some other common source (see 54b):

(54) a. … → negative purpose → apprehensional

b. *??* → negative purpose
?? → apprehensional

As for Russian *čtoby … ne*, Dobrushina (2021: 142f.) firmly states "that independent wish-constructions were not the source of fear complements with *čtoby* (as they were with *kak by* clauses); independent wish-clauses with *čtoby* were quite rare, and all of the attested examples are instantiations of insubordination (see Dobrushina 2016: 337–346)." Evidently, the history of apprehensional complements with *čtoby … ne* and *kak by … ne* differed not only with respect to insubordination (yes for *čtoby … ne*, no for *kak by … ne*), but also in terms of their age and their further distributional behavior: *čtoby … ne* in apprehensional complements appeared in Russian since the 17th century, it preceded *kak by … ne* in this function by 200 years (Dobrushina 2021: 142f.). Then *čtoby … ne* and *kak by … ne* co-existed during the 19th century, also in independent clauses (Dobrushina 2016: 341), but after the early 20th century *čtoby … ne* started becoming obsolete as an apprehensional marker both in complement and in independent clauses (Zorikhina Nilsson 2012: 58–60). It is now the practically exclusive adverbial subordinator for purpose clauses, while *kak by … ne* is restricted to independent and complement apprehensional clauses. Russian has thus approached a stage of an almost complementary division between these two connectives, contrary to their equivalents in West Slavic, which are used in either function (which can often hardly be discriminated in particular discourse tokens). However, diachronic research is required to state with more certainty the direction of development (in terms of 54) and to give a qualified judgment on the insubordination issue. Among other things, we need to know when the relevant clause-initial connectives started coalescing with *by* and how this process was related to their employment in purpose and wish clauses (and their negative equivalents).

[35] In Ancient Greek *mé:* was used in apprehensional complement clauses before negative purpose clauses with the same subordinator appeared on the scene (Dobrushina 2021: 142f. referring to Goodwin 1897). The diachronic relation between these two clause types can thus be reversed, which calls for a case-by-case investigation (prior to generalizations about the direction).

5.1.3 Balkan Slavic *da ne*

Analogous reasoning applies, *ceteris paribus*, to clause-initial units employed in apprehensional contexts in South Slavic languages. As for Balkan Slavic, we saw that the association with apprehension has remained much looser than in North Slavic (see §3). Macedonian/Bulgarian *da ne* and Bulgarian *da ne bi* reveal a high degree of context–dependence for their interpretation, with apprehension usually as only one among many readings. Most probably, the contemporary situation here continues the initial vagueness of *da*-clauses which was, as it were, inherited by *da ne*-clauses even if this complex unit has become opaque to some degree (see §3.4). In particular, whether embedded apprehensional clauses derive from negative purpose or negative optative clauses, or rather arose directly under the scope of 'fear'-predicates, or whether different lines of development converged, remains an open question since no diachronic research has been conducted and reliable historical evidence is lacking. Cf. Mitkovska et al. (2017: 76f.), who also can only surmise that Bulgarian *da ne bi* could have "originated from an optative source via the combination of the inherited subjunctive *bi* and the Balkan subjunctive *da*".

5.1.4 Summary on NegOpt strategies

All relevant clausal connectives are based on irrealis morphemes with concomitant restrictions of TAM marking (North Slavic *by*, South Slavic *da*), even if morphological segmentability has been lost partially or entirely.[36] All these units lower (or suspend) assertiveness. Lowered assertiveness has been considered a typical feature of dependent clauses; however we also find it in all sorts of self-standing utterances with directive or optative illocutions, and such utterances need not be understood as descendants of more complex structures. It does not surprise, then, to find such self-standing clauses also for apprehension. We can safely assume that almost all etymologically complex units with *da* and *by*, respectively, started to head clauses with an apprehensional semantics prior to becoming reanalyzed as complementizers after suitable complement-taking predicates. The only exception is probably Russian *čtoby … ne*, whose employment in apprehensional complements is derived from negative purpose clauses and employment in independent apprehensional clauses was rare (see §5.1.2); for its West Slavic equivalents this issue needs to be investigated.

[36]The irrealis-feature of *da* has been lost in the western part of South Slavic, but the rise of relevant clause-initial markers considerably predated this process (Wiemer 2018: 295–303, with further references).

Thus, with this one exception, no "main clause ellipsis" need be assumed (*pace* Dobrushina 2006 and Evans 2007); rather, the clause-initial employment of almost all apprehensional markers in independent clauses can be regarded as diachronically primary. This alternative analysis allows the appearance of embedded negative-optative clauses to be anchored to a unified basis which explains the rise of complex sentences from discourse pragmatics and, possibly, from the frequency of juxtaposition with suitable, "semantically harmonious" predicative expressions. In fact, this explanation for the appearance of embedded negative-optative clauses has (at least) two advantages over assuming insubordination.

First, it complies with the common-sense assumption that syntactic complexity (namely, tightening) increases over time. Of course, acquired complexity can again be reduced, e.g. by insubordination. However, if the question is about diachronic primacy (what comes first?), diachronic data often shows that alleged cases of insubordination turn out to be independent clause patterns which existed from their earliest attestations. This brings us to the next point.

Second, my explanation is in keeping with analyses of the rise of clausal complementation for other semantic classes of predicates (i.e. potential CTPs). Compare, for instance, Old Serbian *da* (Grković-Major [Grković-Mejdžor] 2004: 198) and Latvian *lai* (Holvoet 2016: 237, 253, 260), which both started from juxtaposed clauses with independent speech acts, but ended up as complementizers after verbs denoting manipulative speech acts ('say'/'order'/'request' that p). Compare also, in general, the evolution of subordinate *da*-clauses in South Slavic. For a comprehensive treatment cf. Wiemer (2017: 297–300, 313–315, 2019).

5.2 Preventives: The interplay between aspect and negation

NegDir strategies are of an entirely different provenance. They strongly depend on the grammatical system, since their interpretation is conditioned by the key opposition of perfective vs imperfective verb stems. The preventive function of negated perfective imperatives follows from the continuous entrenchment and inner-Slavic differentiation of the aspect system, which is based on stem-derivation and therefore applies to any finite or non-finite form of the verb. As a consequence, the choice between members of a binary opposition (pfv. vs. ipfv. stem) is unavoidable (Wiemer 2008, Wiemer & Seržant 2017). A preventive function of negated imperatives could probably always be inferred under favourable discourse conditions, just as in English or German today.[37] However, the innova-

[37] Compare Engl. *Don't go there!*, Germ. *Geh da nicht hin!* Such utterances can be understood either as prohibitives ('because I don't want you to') or as preventives ('because otherwise undesirable consequences will follow'), possibly apart from other readings discussed in §4.

tion (Ivić 1958: 29f. and passim) consists in an increase of reliability of the choice between perfective and imperfective verb in the negated imperative, whereby perfective verbs in this grammatical context came to be more and more restricted to preventive meaning (as opposed to imperfective verbs becoming associated with prohibitive meaning).

Rather casual observations on this issue (Wiemer 2008: 396f.) are corroborated by systematic corpus studies of early stages of South and East Slavic. Thus, MacRobert (2013) points out that, although the admonitory use of the negated perfective imperative was "well attested in Old Church Slavonic" (2013: 286), its distribution was not identical with the situation in modern Slavic languages. Notably, other uses of the negated perfective imperative were all characterized by a focus on boundaries (especially in appeals to God to not do what He could do, using His might, at the present moment). The focus on boundaries is a common denominator of using perfective verbs, also in the imperative (see §4.1), but obviously the variety of functions and discourse constellations used to be broader, before it shrank in North Slavic to leave the preventive function. Furthermore, Mišina (2020a, 2020b) carried out comprehensive corpus studies of early East Slavic texts and compared the frequency of negated imperatives in these texts with those in modern Russian: the proportion between imperfective vs perfective stems for this grammatical form has remained more or less stable (3:1), however in early East Slavic the distribution between prohibitive and preventive function was much less clear-cut than in modern Russian, partly due to the elimination of perfective verbs from prohibitive contexts (2020a: 172–176). Therefore, the crucial difference from South Slavic is that perfective verbs have not been driven out altogether from negated imperatives (see below), but have found a niche in the preventive function. This process can therefore be captured as pragmatic strengthening, not of particular markers, but of a grammatical distinction as such, and this process consisted in an increasing division of labour in the choice of perfective vs imperfective stems (§4.1). As far as we know by now, this innovation proves most consistent in the northeastern part of Slavic, but also the West Slavic languages seem to be rather consistent in this respect (see §4.2).

South Slavic languages, largely with the exception of Slovene, have gone another way: perfective stems have altogether been abandoned from negated imperatives (apart from petrified phrases). Already in many Old Church Slavonic monuments[38] we can discern a preponderance of imperfective stems for negated imperatives regardless of their function (Kurzová-Ribarova 1972: 71; Večerka 1996: 70–73). The recent situation in South Slavic looks like a consequent continuation

[38]Old Church Slavonic was based on east South Slavic dialects.

of this trend, accompanied by the spread of periphrastic constructions building on *da*-clauses and "analytical imperatives" (with auxiliaries, Serbian/Macedonian *nemoj(te)*, Bulgarian *nedej(te)*). Among these newer constructions only the Balkan Slavic independent *da*-clauses (with present tense forms) under negation seem to show a more or less reliable functional distribution between perfective (→ preventive) and imperfective (→ prohibitive) aspect, while analytical constructions with auxiliaries do not discriminate very consistently between these two functions (§4.3). The *da*-clauses, in turn, do not clearly discriminate between preventive and negative-optative or negative-purpose function (see §3).

The ousting of negated perfective imperatives by periphrastic constructions probably started from the (south)eastern end (i.e. Bulgarian); cf. Ivić (1958). Slovene, which is situated at the opposite end of the South Slavic continuum, has been least affected by this process; perfective verbs are used in negated imperatives, particularly with preventive meaning, but the distribution in opposition to imperfective verbs (→ prohibitive meaning) is arguably less clear-cut than in Russian. In this sense, it is more conservative.

6 Summary and outlook

The NegDir strategies are more homogeneous than the NegOpt strategies, even if among the latter we disregard patterns with an affinity to (or provenance from) negative purpose clauses. Functionally, the homogeneity of the NegDir strategies derives from the interaction of aspect with negation: either negated perfective imperatives (known as preventives) in North Slavic and Slovene or their *da*-headed perfective present-tense equivalents in South Slavic (often considered "analytical subjunctives"). However, the Balkan Slavic combination *da ne* (irr+neg) reveals a high degree of morphological and semantic opacity exactly in apprehensional (including preventive) usage, while the preventive imperatives in North Slavic (and Slovene) are entirely compositional. Concomitantly, the aspect-based division of negated directive speech acts has grown very reliable in North Slavic, whereas South Slavic (with the exception of its northwestern periphery) has ousted negated perfective imperatives, and more recently developed periphrastic imperatives (with auxiliaries, Serbian/Macedonian *nemoj(te)*, Bulgarian *nedej(te)*) do not reliably distinguish between prohibitive and preventive function, even if they allow for perfective verbs.

Moreover, preventives in North Slavic are always distinguishable from the NegOpt strategy, which is formally marked by clause-initial connectives. In South Slavic, particularly in Balkan Slavic, the two strategies are considerably less clearly distinguishable, both in functional and in structural terms: *da ne*-clauses cover both the NegOpt and the NegDir function (although the former seems to be more salient), apart from other usage types. In addition, *da ne* marks negative purpose clauses, but as a unit that is transparently composed of *da* as a general irrealis morpheme (able to mark also purpose) and the negation.

NegOpt markers are based on irrealis markers with a very clear areal division: South Slavic has *da*, North Slavic has *by* (with Bulgarian *da ne bi* showing morphological coalescence of both etyma). In NegOpt connectives they have however lost their morpheme status; in South Slavic, but not in North Slavic, we can furthermore observe a loss of TAM-constraints. The degree to which NegOpt connectives are etymologically and functionally related to optative or purpose clauses varies. Again, for the North Slavic languages these two sources of apprehensional clauses can be discerned more easily, together with areal tendencies: in Russian, negative purpose clauses (*čtoby … ne*) have largely become dissociated from apprehensional semantics, while in West Slavic we observe an overlap between negative purpose and apprehensional complement clauses. Moreover, modern Russian demonstrates a tendency toward a syntactic division of labor between *kak by … ne*, which is more frequent in complement than in independent clauses, and NegOpt markers based on particles, which are only used in independent clauses. Among other things, West Slavic is understudied particularly in this respect. In turn, the western part of South Slavic shows a tendency for NegOpt connectives to drop the requirement for clausal negation, together with a loss of TAM-constraints. Finally, the western part of Slavic (Slovene, Czech, Slovak) disprefers or altogether avoids infinitival complements, so that in apprehensional complement clauses same-subject and different-subject clause combining cannot be distinguished. By contrast, Russian employs the finite—infinitive distinction consistently, while Polish in this respect occupies an intermediary position.

Syntactically independent employment of apprehensional clauses can be considered a result of insubordination only if such clauses diachronically derive from negative purpose clauses. Only for Russian *čtoby … ne* (not for *kak by … ne*) is there sufficient evidence in favour of this assumption, while neither for West Slavic (Polish *żeby … nie*, Polish /Czech/Slovak *aby … ne*) nor for South Slavic *da ne*-clauses any clear evidence exists and there remains a plausible alternative hypothesis that negative purpose and apprehensional clauses developed in parallel. For all other strategies that are associated closely with optative semantics, insubordination is not an issue at all.

In conclusion I want to emphasize that here I have only tried to give a comprehensive account of two types of strategies which, in my view, represent the most salient and widespread techniques across Slavic languages. However, in order to acquire a better understanding of how such strategies emerge and may become established in speaker communities, two directions of research urgently need to be pursued. First, we need more intensive research in the diachronic background of apprehensional constructions and their relation to other clause types. Second, on the contemporary level research should concentrate on the discovery of discourse-conditioned patterns which favour the emergence of apprehensional strategies.

Abbreviations

≅	precedes approximation of meaning
1/2/3	first/second/third person
ACC	accusative
ADV	adverb
AOR	aorist
AUX	auxiliary
COMP	complementizer
CON	connective (unspecific)
CTP	complement-taking predicate
CVB	converb
DAT	dative
DEF	definite article
DO	definite direct object (Bulgarian masculine nouns)
DS	different subject
EXCL	exclamation
F	feminine
FUT	future
GEN	genitive
IMP	imperative
IMPF	imperfect
INS	instrumental
IPFV	imperfective
IRR	irrealis
LF	*l*-form (< participle in compound tenses)
LOC	locative
M	masculine
N	neuter
NEG	negation
NOM	nominative
PFV	perfective
PL	plural
PN	proper name
POSS	possessive
PP	past (= anteriority) participle
PRS	present
PST	past
PTC	particle
Q	interrogative (= indefinite) pronoun
REFL	reflexive marker
SBJV	subjunctive
SG	singular
SS	same subject
VOC	vocative

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Corpora

PNC: Polish National Corpus, http://nkjp.pl/

RNC: Russian National Corpus, http://www.ruscorpora.ru/new/

Chapter 8

Apprehensives in East Caucasian

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In this paper, we describe the category of apprehensive in three languages of the East Caucasian (Nakh-Daghestanian) language family, Mehweb (Dargwa), Archi (Lezgic) and Akhvakh (Andic). The fact that the apprehensive, a cross-linguistically rare category, is of independent origin in these three related languages, may point to areal diffusion, and yet, the languages are not in direct contact, and the apprehensive has not yet been attested in other languages of the family. Further, in Mehweb and Archi, the two apprehensives show some important functional similarities. We attempt to reconstruct their morphosyntactic evolution, and argue that, after having followed very different paths of grammaticalization, the two apprehensives have stabilized in a similar functional domain.

1 Introduction

Apprehensives are usually described as a morphological expression of fear (Lichtenberk 1995, Dobrushina 2006, Vuillermet 2018, Smith-Dennis 2021). With an apprehensive, the speaker reports that there is a probability of an undesirable event and that he or she fears it. Apprehensives may simultaneously convey the speaker's emotion (they are apprehensive of a potentially negative event) and a command (they induce the addressee to take measures to avoid or prevent the undesirable situation). Due to the directive component (expression of a command), apprehensives may be part of main clause phenomena. Cross-linguistic variation in their status as forms used exclusively in the main clause or also in subordination is particularly relevant for their typology. Dedicated morphological apprehensives are cross-linguistically infrequent.

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This paper discusses dedicated apprehensives in three languages of the East Caucasian language family: Archi (Lezgic branch of the family), Mehweb (Dargwa branch), and North Akhvakh (Avar-Andic branch). Although the three languages belong to the same family and roughly the same geographic region (central to northern highlands of Daghestan, Eastern Caucasus), they are only distantly related – belonging to different branches of this relatively old family – and are not spoken in geographically adjacent areas. All three of them are minority languages. Archi and Mehweb are one-village languages, and Northern Akhvakh is spoken in only four villages (but see below). According to the 1926 Census, which reflects traditional population sizes better than later statistics, Mehweb was spoken by 710, Archi by 856, and Northern Akhvakh by 2216 people (Materials 1927, Dobrushina et al. 2017). Today, there are many more villagers living in the lowlands, but the degree of language vitality among resettlers is unclear. So far, these languages are the only languages of the family where inflectional apprehensives have been attested, but this might be due simply to the fact that apprehensives are often overlooked by descriptive grammars.

In Archi, Mehweb and Akhvakh, the apprehensives are expressed morphologically by verbal suffixes -*lk:ut*, -*ala* and -*alago*(*le*), respectively. Based on our field data as well as, to a certain extent, on the existing descriptive sources, we will describe the uses typical of these forms, review their availability with subjects of different persons, consider whether they are compatible with negation, discuss their status as finite or non-finite forms, and put forward hypotheses about the origins of the markers. In spite of the apparent phonological similarity of the three affixes, there is no evidence that they are cognate. We will argue that they result from three different paths of grammaticalization from different – even if not always known – sources. In Archi, the independent apprehensive probably arose through insubordination (expansion from syntactically dependent to main clause uses – Evans 2007). In Mehweb, on the other hand, subordinate uses of the apprehensive seem to evolve through an extension of the reported speech construction. In Akhvakh, its probable origin is univerbation of a verb with an interjection expressing warning. At least in Archi and Mehweb, these forms, although coming from very different sources, ultimately stabilize in very similar functional domains typical of the apprehensive cross-linguistically. Presently, it remains unclear how much of areal impact this process may have had.

For Mehweb, corpus data are limited due to the corpus size (own data, Magometov 1982), so we mostly illustrate the uses of the apprehensive with elicited data. For Northern Akhvakh, no corpus was available, but elicited data are supplemented by several examples from a dictionary (Magomedova & Abdulaeva 2007). For Archi, we have the by far richest attestation of apprehensives, so that, where available, corpus examples are provided together with elicited data, but even here naturalistic examples are limited to 26 contexts. Archi corpus data integrate various sources, including texts published in Dirr (1908), Mikailov (1967), Kibrik (1977), and Samedov & Magdilova (2017b,a) as well as our own unpublished data; all these texts will be collectively referred to as *corpus*, and the examples coming from them are marked as *naturalistic* as opposed to *elicited*.

§2, §3, and §4 describe Archi, Mehweb and Northern Akhvakh data, respectively. Note that the data were collected at very different times, and thus not all types of relevant information available from one language is also available from another. §5 is a comparative discussion of the data of the three languages, and §6 is a brief conclusion.

2 Archi: From 'fear' clauses to apprehensive via insubordination

The Archi form in *-lk:ut* was described by Kibrik (1977: 292) as a converb used in negative purpose constructions. While Kibrik only quotes subordinate uses (see §2.2 below), Dobrushina (2006) shows that, when used independently, this form is interpreted as apprehensive:[1]

(1) elicited
Jami *č'abu* ***esa-lk:ut!***
wolf.ERG sheep 4.slaughter-APPR
'The wolf might kill the sheep!'

The origin of the form in *-lk:ut* is unclear. The suffix is added to what Kibrik sees as the infinitive stem of the verb (most other forms are based on the perfective or the imperfective stem). It is very tempting to suggest univerbation of the lexical verb with the verb 'see' (perfective stem CL-*ak:u*) with loss of the class prefix and initial vowel. First of all, there is a cross-linguistically robust connection between 'see' and apprehensives (Lichtenberk 1995, Dobrushina 2006). There is another univerbation path that 'see' follows in Archi, which leads to a series of *verificative* forms meaning 'find out whether …' (see Maisak 2016 on the category of verificative in Aghul and several other East Caucasian languages, also Kibrik

[1]Here and below, for Archi, numbers in the glosses refer to genders. Typically, Gender 1 is for male referents, and Gender 2 is for female referents, and the rest of nouns are distributed between Gender 3 and 4. Mehweb and Northern Akhvakh only have three genders, designated as M, F and N. In all three languages, HPL stands for human plural agreement which conflates male and female referents, and NPL stands for non-human plural agreement.

1977). This may be taken as a token of the propensity of the verb ‘see’ to grammaticalize. This stipulation however remains problematic because of the status of the additional material (initial *-l-* and the final *-t*) which remains unclear. In fact, in view of our tentative conclusions on the evolution of this form below in §2.4, there might be no need to look for the source of the apprehensive suffix among the known lexical sources of its grammaticalization in other languages, because its development towards apprehensive uses might have occurred later, through an evolution of a syntactic construction.

In Archi, the apprehensive is used both in main clauses and in some subordinate clauses, including negative purpose clauses, ‘*in-case*’ clauses (as termed in Lichtenberk 1995) and the complement clauses of verbs of fear. Except ‘in-case’ clauses, all types are attested in the corpus; but, as we will see, it is not always easy to distinguish between these types, and all subordinate uses may be argued to fulfill the same general function.

2.1 Semantics and morphology

In independent uses, the Archi form in *-lk:ut* expresses the fact that the speaker is anticipating a negative situation. Out of some 20 occurrences of the apprehensive in the corpus, only one is arguably independent, while all other examples come from subordinate clauses. We do not believe this reflects anything except the nature of our corpus, which is almost exclusively narrative, and not interactional. The naturalistic example of independent use comes embedded in a speech reporting construction.

(2) naturalistic

χir	*q’ˤʷebos*	*priq’at*	*e⟨r⟩t:i*	*jas:a,*	*χir*	*wa*	*buwa,*
then	next	calm	⟨2⟩become:PFV	now	then	oh	Mom

un=ur=u	*χ:al-ši*	***e⟨r⟩χ:u-lk:ut,***	*t’o=ra=r,*
you.SG=QUOT=ADD	be.bad-CVB	**⟨2⟩remain:PFV-APPR**	NO=Q=QUOT
un=ur=u	*χ:al-ši*	***e⟨r⟩χ:u-lk:ut,***	*t’o=ra=r.*
you.SG=QUOT=ADD	be.bad-CVB	**⟨2⟩remain:PFV-APPR**	NO=Q=QUOT

‘Then she calmed down, then Oh Dear! (lit. ‘Oh mother!’), she said, may you not feel bad, right (lit. ‘is it not’)? may you not feel bad, right?’

Example (1) shows the use of apprehensive in the third person, and (2) in the second person. In elicitation, the apprehensive easily combines with subjects of all persons:

(3) 1st person, elicited
Bara zon ***d-ukːu-lkːut****.*
beware I:NOM **2-burn-APPR**
‘Careful, I might get burnt!’ (when another person carries boiling water)

(4) 2nd person, elicited
Un oˤrču-tːu-t ɬːan c’abu, bara un
you.SG(ERG) 4.freeze:PFV-ATTR-4 water drink:PFV beware you.SG.NOM
bec’ːo-ši ***kʷe-lkːut****.*
get.sick-CVB **1.become-APPR**
‘You drank cold water, you might get ill.’

(5) 3rd person, elicited
Jaˤt’i-li ***w-eq’ːi-lkːut!***
snake-OBL.ERG **1-bite-APPR**
‘The snake might bite (me)!’

With the 2nd person, the apprehensive *iwkulkːut* ‘Watch out not to fall!’ in (6) is close to the prohibitive *werkurdigi* ‘Do not fall!’ in that the apprehensive may be taken as a directive to avoid a dangerous outcome. For a comparison of the apprehensive and the prohibitive, it is not easy to obtain speakers’ explicit intuitions in a comparable form. We asked one of our consultants which of the forms implied a “higher danger”. For her, it was the apprehensive, while the prohibitive, as she had put it, suggests that even if the addressee falls, she would simply get up again, end of story (Russian *упадет и встанет*). Another consultant commented that, when using the prohibitive, the speaker is “sure that the addressee will fall down unless a precaution is taken”, while in the case of the apprehensive, she “merely suggests it might happen”. Finally, yet another consultant, when asked to compare the two forms, simply suggested examples in (6) as natural contexts:

(6) examples suggested by the consultant

a. ***I⟨w⟩ku-lkːut*** *qʷat’i-li-tːi-k w-e⟨r⟩χi-r-digi.*
⟨1⟩fall-APPR tree-OBL-SUP-LAT 1-⟨IPFV⟩climb-IPFV-PROH
‘Don’t climb up the tree, lest you fall down.’

b. *Doːˤzu-t minnat i,* ***w-e⟨r⟩ku-r-digi!***
big-4 request 4.be **1-⟨IPFV⟩fall-IPFV-PROH**
‘Let me ask you about one thing – do not fall down!’

These intuitions and suggestions may be linked to the epistemic and illocutionary components present in apprehensives vs. prohibitives. Apprehensives inform of a probability of a negative event in the future, while prohibitives, like imperatives, are more immediately connected to the moment of speech and, in these cases, seem to be used to prevent a situation. It thus seems that the apprehensive form conveys the feeling of danger in a more direct way (more of a warning), while the prohibitive is an instruction to avoid the situation (more of a command), but we admit that this difference can at least partly be in the eye of the beholder.

One of the reviewers remarks that a natural expectation of the difference between the apprehensive and the prohibitive is that of exerting control over the situation, because, as any directive, the prohibitive implies control. Indeed, even though the verb in (6a) and (6b) is the same, one may suspect that (6b) represents the situation of falling or not falling as more controllable, and the effect is probably obtained exactly by forming a prohibitive of a verb that normally describes a situation beyond control (a similar effect would probably be obtained in some languages by forming an imperative 'Fall!' as a warning of an explosion). Presence or absence of the subject's control may thus indeed be involved; however note that, in Archi, the prohibitive is compatible with uncontrollable states, as in (7). To our eyes, speculating that this context also implies control (e.g. *have fun! do something so that you are not bored!*) leaves room to extend the concept of control to any situation at all.

(7) naturalistic
Os t'i-t:u-t s:aʕal-li-t ***bazar kʷe-r-digi*** *zon os*
one small-ATTR-4 time-OBL-SUP **bored 1.become-IPFV-PROH** I:NOM one
pulan-nu-t biq'ʷ-ma-k uqˁa-na, u⟨w⟩ʟi-lkan.
definite-ATTR-4 place-LOC-LAT 1.go:PFV-CVB ⟨1⟩come-TEMP
'Take care of yourself (lit. do not become bored), while I go somewhere else and come back.'

In Archi, apprehensives can refer to situations that cannot be controlled, such as meteorological events in third person constructions:

(8) elicited
Mač'a ***ke-lk:ut,*** *nen-t'u noʟ'-i-ši xat:i-t:a.*
be.dark 4.become-APPR we-PTCL.4.INCL house-IN-ALL LOC.PL.go.FUT-TEMP
'It might become dark when we go home.'

Most verbal forms have an inflectional negative polarity counterpart, but there is no negative apprehensive. To warn about a risk that something desirable might not happen, Archi uses a periphrastic construction of a negative converb of the main verb with an apprehensive form of the verb 'remain'[2]. Cf. (9) with a regular apprehensive and (10) with a palliative 'negative apprehensive', and also (11):

(9) elicited
T'eˁ-t:u *caqˁ-a,* *χ:eˁl* ***eχmu-lk:ut.***
flower-PL NPL.cover-IMP rain **4.rain-APPR**
'Cover the flowers, it may rain.'

(10) elicited
T'eˁ-t:-e-s *ɬ:an* *ec-a,* *χ:eˁl* *eχ-di-t'uw*
flower-PL-OBL.PL-DAT water 4.pour-IMP rain **4.rain-PFV-CVB.NEG**
eχ:ə-lk:ut.
4.remain-APPR
'Water the flowers, what if it does not rain today.' (lit. 'lest it remains without the rain having rained.')

(11) elicited
T'eˁ-t:-e-s *ɬ:an* *eca,* *to-r* ***boq'ˁo-t'aw***
flower-PL-OBL.PL-DAT water 4.pour-IMP this-2 **come.back:PFV-CVB.NEG**
e⟨r⟩χ:ə-lk:ut.
4.remain-APPR
'Water the flowers, what if she does not come back.'

In Archi, most forms usually considered finite may take the negative suffix *-t'u.* The fact that, in negative contexts, the apprehensive takes a periphrastic form with a special negative converb may be indicative of the non-finite status of the apprehensive or the form it originated from. Also note that, in this construction, the apprehensive form of the main verb seems to be synonymous to the infinitive of the same periphrastic construction (see (16) in the context of negative purpose).

As these examples show, the apprehensive is often used in combinations with clauses that motivate the usage of the apprehensive by specifying the exact situation that can lead to an undesirable outcome; see also (12):

[2]One of the reviewers asks whether the construction is monoclausal periphrastic or biclausal. Though we have no relevant syntactic evidence, we observe that in the few naturalistic examples we have, 'remain' strictly follows the negative converb of what we assume to be the lexical verb in the periphrastic construction; and that the meaning of the construction ('remains not having V-ed', e.g. lit. 'lest she remains not having returned' in (10)) is not, or is not fully, compositional. Before more syntactic data are obtained, this may be taken as evidence in favor of a monoclausal interpretation.

(12) elicited
K'umk'um L*'eˁr-ši,* ***d-ok:i-lk:ut.***
pan be.hot-CCVB.4.AUX **2-burn-APPR**
'The pan is hot, you might burn yourself.'

Alternatively, the non-apprehensive part can describe a way to avoid the undesirable outcome. The constructions in (9) to (12) are close to various types discussed in the next section, and at times, their status as paratactic juxtaposition (as described in this section) or subordination (as described in the following section) is unclear.

2.2 Subordinate uses

Crosslinguistically, apprehensives are commonly used in subordination. In such contexts, the apprehensive is essentially a way to express the reason of the situation introduced by the main clause, which may describe an action undertaken to avoid the negative situation (negative purpose clauses) or, if it cannot be avoided, to reduce its negative impact (Lichtenberk's 'in-case' clauses), or to simply describe the emotional state (some but not all 'fear' clauses). In Archi, all such uses of the apprehensive are attested. In these contexts, the apprehensive may refer to the past, so that their interpretation as directives is clearly excluded. Cf. the naturalistic example in (13):

(13) naturalistic
χ:ams že-n-t'u lo=wu uƚdu=wu barha-s
bear self.OBL-GEN-PTCL child=ADD shepherd=ADD care-INF
š:ʷi-š:ʷi boχ:ˁo b-o⟨r⟩qˁi-r-ši e-b-di-li qi-qi
by.night-by.night hunt 3-⟨IPFV⟩go-IPFV-CVB be-3-PST-EVID at.day-at.day
jamu ***ƚ:ummu-lk:ut*** *s:ak'u q'a⟨b⟩q'i-r-ši e-b-di-li.*
that.1 **1.flee-APPR** on.guard ⟨3⟩sit-IPFV-CVB be-3-PST-EVID
'The she-bear, to sustain the shepherd (her husband) and child, used to go out hunting at night, and during the day she was looking after her husband **so that he might not escape**.'

Example (13) illustrates the claim in Vuillermet (2018): unlike in its independent uses, in subordinate uses of the apprehensive the judge of negative evaluation may be the subject of the main clause and not the speaker herself. Indeed, the she-bear is a negative character who took the protagonist prisoner and whose evaluations of what is negative and what is positive the narrator presumably does not share. In Archi, the evaluation probably has to be done from the viewpoint of who is the referent of the subject of the main clause (independently of whether the narrator shares it or not); at least this seems to be the case in all our examples.

In (13) above, two main clauses introduce purpose clauses, one of which is positive ('went out hunting in order to sustain') and the other is negative ('stay on guard in order that he does not escape'). Because of its built-in component of negative evaluation, the apprehensive may only be used to express negative purpose. In examples (14) and (15), the apprehensive used as negative purpose (a) is contrasted with the use of an infinitive as its positive purpose counterpart (b):

(14) elicited

a. *Buwa lo* ***čučəbo-lk:ut*** *o⟨r⟩qˤa.*
mother child **wash-APPR** ⟨2⟩go:PFV
'The mother went away in order not to wash the child' (lit. 'lest she wash the child').

b. *Buwa lo* ***čučəbo-s*** *o⟨r⟩qˤa.*
mother child **wash-INF** ⟨2⟩go:PFV
'The mother went away to wash the child.'

(15) elicited

a. *Buwa lo* ***čučəbo-lk:ut*** *o⟨r⟩qˤa-t'u.*
mother child **wash-APPR** ⟨2⟩go:PFV-NEG
'The mother didn't go in order not to wash the child (lit. 'lest she would have to wash the child').' (i.e. She stayed so as not to have to wash the child.)

b. *Buwa lo* ***čučəbo-s*** *o⟨r⟩qˤa-t'u.*
mother child **wash-INF** ⟨2⟩go:PFV-NEG
'The mother didn't go to wash the child.'

To explain the difference between the two examples in (15), consultants indicate that, in (15a) which uses the apprehensive, the reason for the fact that the mother did not leave is clear: she did not want to deal with washing the child. In (15b), the reason is not specified; it can be her choice not to go, but it can also be that she wanted to go and wash the child but for some reason could not make it there. Examples (13) to (15) show that the apprehensive is used in the clauses with a special type of purposive semantics, those that express situations that the subject of the main clause aims at avoiding, and the main clause describes an action undertaken to avoid it.[3]

Another, apparently less frequent means to express the apprehensive meaning is a combination of the negative converb with the infinitive of *e*≺cl≻*χ:a-* ‘remain’, the periphrastic construction used in (10) and (11) above. Interestingly, (16) comes from the same text and describes the same situation as (26) below, where the apprehensive ‘fear’ clause is used. Example (16) comes several clauses after the use of (26), as a kind of an afterthought:

(16) naturalistic
… adam-li-s ***ak:u-t’aw*** ***eχ:a-s****.*
… people-obl-dat **4.see:pfv-cvb.neg 4.remain-inf**
‘… so that no one sees (the cattle).’

The situation encoded by means of the apprehensive marker is not necessarily one that instills fear in the speaker. In the following examples, *-lk:ut* refers to a situation which is merely undesirable, not frightening:

(17) elicited
Durba χ:ʷalli b-aχ:a-qi, *nen-ə⟨b⟩ij=bu.* *a⟨b⟩a-s*
hort bread 3-take:pfv-fut we-⟨3⟩ptcl.incl=⟨3⟩ptcl ⟨3⟩do-inf
kʷa ⟨b⟩ šu-lk:ut*.*
⟨3⟩ must-appr
‘Let’s buy bread, so we do not have to bake it ourselves.’

(18) elicited
Durba o:k’ur herqˁu-qi, *q’ˁas* ***ke-lk:ut***
hort slowly locpl.walk:pfv-fut be.tired **locpl.become-appr**
nen-t’u.
we-ptcl.locpl.incl
‘Let’s go slowly, in order not to get tired.’

[3]This clause type is probably better termed *anti-purpose* rather than simply negative purpose clause, because, were this combination of propositional negation and purposive clause compositional, it would yield something like ‘P, and it was not in order to Q’ rather than what is usually called negative purpose, ‘P in order not to Q’. Similarly, prohibitives, often also called *negative imperatives*, are not merely negative counterparts of the imperatives and are often referred to by a dedicated term, i.e. *prohibitive*. In the case of the prohibitive, propositional negation simply cannot compositionally combine with imperative illocution. In fact, Kibrik considered the Archi apprehensive to be a *preventive-purpose* converb.

(19) naturalistic
Ak:on-ni-ɬ:u, barq e-b-di b-ak:u-t'aw, alšun-n-in
light-OBL-COMIT sun be-3-PST 3-see:PFV-CVB.NEG paradise-ATTR-PL
hurulʕen-til b-a⟨r⟩Li-r-ši e-b-di-li
angel-PL HPL-⟨IPFV⟩come-IPFV-CVB be-HPL-PST-EVID
ja-r-mi-ra-k, ja-r ***bazar de-ke-lk:ut****.*
this-2-OBL-CONT-LAT this-2 **bored 2-become-APPR**
'In the morning, immediately after the sunrise, the angels of the paradise would come to her lest she felt bored.'

As a comparison of examples (14), (15), (17) and (18), on the one hand, and (13) and (19), on the other, shows, negative purpose clauses with the apprehensive may have either the same or a different subject as the main clause. More such examples (elicited data) are available in (Kibrik 1977: 292).

The apprehensive may also be used in Lichtenberk's 'in-case' clauses. 'In-case' clauses refer to potential negative circumstances that cannot themselves be avoided, such as natural phenomena in (20); what can be avoided or at least diminished is their negative impact. There are no clear instances of 'in-case' clauses in the corpus, but they are easily obtained by elicitation, as in (20) (cf. also (9)–(11) above).

(20) elicited
č'il gʷaq':-a, χ:ˤəl ***exmu-lk:ut****.*
hay harvest-IMP rain **4.rain-APPR**
'Collect the hay, in case it rains. (So that the hay does not get wet.)'

The apprehensive is used in dependent clauses subordinate to the verb 'fear' (as in (21) and (22)) or similar verbs (as in (23)) as a matrix verb, and, judging from the meager corpus data we have, these uses are typical. In such cases, the apprehensive may provide the reason of the state of fear:

(21) naturalistic
Ǯemme-t:u-b zulmu ***a⟨b⟩Le-lk:ut*** *L'inč'a-r-š-e-b-di χalq'.*
thus-ATTR-3 sin ⟨3⟩come-APPR be.afraid-IPFV-CVB-be-HPL-PST people
'People were afraid that such wrongdoing may happen.'

(22) elicited
Zon ***L'inč'a-r****, un ħaˤtər-čaj* ***də-χə-lk:ut****.*
I **be.afraid**-IPFV you.SG.NOM river-OBL.ERG **2-carry.away-APPR**
'I'm afraid that a river will carry you away.'

(23) elicited
Buwa-n ***ik'^{w}*** ***arħa-r-ši*** *χ:^{w}ak:-a* *lo*
mother-GEN **heart** **4.think-IPFV-CVB.4.be** forest-IN/ESS child
q'^{ʕw}ek'e-lk:ut.
be.lost-APPR
'The mother worried that the child might get lost in the wood.'

As in negative purpose constructions, with verbs of fear, the form can be used in both different (21)–(23) and same (24) subject constructions:

(24) elicited
Peč-li-ra-k *hara:ši* *de-ke-s* ***L'inča-r*** *zon*
stove-OBL-CONT-LAT in.front 2-become-INF **be.afraid-IPFV** I(NOM)
du-k:u-lk:ut.
2-burn-APPR
'I am afraid to come near the stove (because) I might get burnt.'

It is difficult to decide whether all naturalistic contexts with the verb *L'inčar* 'fear' imply a true emotion, and to what extent. Example (25) seems to unequivocally suggest a strong emotion ('I am frightened because P is likely to happen'), while example (26) may be interpreted as merely providing a reason 'they did this out of fear of P'). Yet, even in (25), one cannot be sure that the emotion is a semantic component of the apprehensive rather than simply contributed by lexical means ('trembling to death').

(25) naturalistic
Hu, *d-as:ə-r-š=er* *inž* *d-ak':a-qi-ši* *d-i=r,* *χir*
yes 2-tremble-IPFV-CVB=QUOT self 2-die:PFV-FUT-CVB 2-be=QUOT after
e-r-di-na ***da:ɬə-de-ke-lk:ut*** ***L'inč'a-r-ši***.
be-2-PST-CVB **beat-2-LV-APPR** **be.afraid-IPFV-CVB**
'Yes, I am trembling to death, she says, afraid (i.e. experiencing fear?) of her getting to me and giving me a beating.'

(26) naturalistic
Məc:or-um-čij=u *ža-n-t'u* *buc'i* *č'abu*
paster-PL-OBL.IN/ESS=ADD self.PL-GEN-4.PTCL cow.PL sheep.PL
a⟨r⟩k'u-r-t:u-t *adam-li-s* ***ak:u-lk:ut*** ***L'inč'a-r-ši***.
⟨IPFV⟩drive-IPFV-ATTR-4 person-OBL-DAT 4.see-APPR be.afraid-IPFV-CVB
'And (they) were afraid (i.e. thought it was possible?) that people who drove cattle in the field would see (them).'

Weakening of the emotional component probably provides a bridge from the apprehensive in 'fear' clauses both to the independent apprehensive and to the uses of the apprehensive in negative purpose clauses which do not involve the verb *ʟ'inč'ar* 'fear' (such as (13) or (19), to take only naturalistic examples). We suggest that the semantic impact of the verb of 'fear' gradually weakens across constructions and uses, so that in the end the use of the verb becomes optional and the formerly subordinated apprehensive becomes insubordinated.

Indeed, in many naturalistic contexts, the verb 'fear' is used non-finitely, in subordination to the main verb expressing the action taken to avoid an undesirable situation, but also subordinating an apprehensive clause that describes this situation. Syntactically, these constructions are different from those in examples (13) to (19) only in that the relation between the main and the apprehensive clause is mediated by the 'fear' clause. Cf. the following example:

(27) naturalistic

Eχ:u-lkan žа-s χir kʷe⟨r⟩k:i-r-ši=wwu
4.remain:PFV-TEMP self-DAT after LOCPL.⟨IPFV⟩wander-IPFV-CVB=ADD

e-r-d-er ža-b ʟ'inč'a-r-ši tu-w qačaʁ=u
be-2-PST=QUOT self.PL-PL.NOM be.afraid-IPFV-CVB this-1 bandit=ADD

*w-i-muχur ta-w bišin neq'ʷit:u b-i-muχur, os ohˤli be-ke-**lk:ut***
1-be-TEMP that-1 alien ? 3-be-TEMP one ? 3-become-**APPR**

*ža-s **heʟ'əna-lk:ut** **ʟ'inč'a-r-ši** ža-b*
self.PL-DAT **whatchamacallit-APPR** **be.afraid-IPFV-CVB** self.PL-PL.NOM

ɬʷ:ak e⟨b⟩t:i-t'o=r.
near ⟨HPL⟩become:PFV-NEG=QUOT

'… we were afraid, there being this bandit, and us being on this land that is not ours, we were afraid that something bad may happen to us, whatchamacallit bad happens – and did not go near there.'

In (27), the finite clause describes the action (not coming near some place) that was supposed to divert the undesirable situation (getting into trouble), but while these two situations apparently could have been described by only two clauses, as in e.g. (19), here the link between the two is provided by the converb of the verb 'fear', and it is unclear whether the undesirable situation instilled true emotion or just needed to be avoided.

The following example comes close to an 'in-case' context; its interpretation depends on whether the speaker implies that settling down in small groups helps to avoid attracting outlaws (negative purpose interpretation) or that, while outlaws cannot be fully diverted, this reduces the damage from their attacks ('in-case' interpretation):

(28) naturalistic
A ⟨b⟩ ʟə-lkːut ***ʟ'inč'a-r-ši*** *jemmet bibiʁuda*
⟨HPL⟩ **come-APPR** **be.afraid-IPFV-CVB** thus here.and.there
qˤe-b-di-li, *jemmet t'inn-ən-n-ib* *boʟ* *e-b-di-li.*
sit:PFV-HPL-PST-EVID thus little-UNIV-ATTR-PL people be-HPL-PST-EVID
'Afraid that (the outlaws) would come, they were thus settling at different places, so that there were few people here and there (i.e. they were settling down in small groups not to attract attacks from gangs).'

Another naturalistic context relevant for the discussion is (29):

(29) naturalistic
Qʷat'i-li-s *jatːik* *e⟨r⟩χ-d-inč'išː=u* *e⟨r⟩ku-na* ***heʟ'əna***
tree-OBL-DAT up ⟨2⟩climb-PFV-COND=ADD ⟨2⟩fall:PFV-TEMP **thing**
ke-lkːut ***ʟ'inč'a-r*** *bo-li* *ik'-mi-s* *eku.*
4.become-APPR **be.afraid-IPFV** say:PFV-CVB heart-OBL-DAT 4.fall:PFV
'It occurred to me: in case I climb up the tree and fall, what if something (bad) happens (to me).' (the speaker was pregnant at the moment described in the narrative, so using "something bad happens" may be viewed as a figure of avoidance).

Unlike in examples (27) and (28), in (29) the verb 'fear' is in its finite form. Yet, the whole sentence is presented as a speech reporting construction which, in Archi, can extend to internal speech ('I thought that P'), and it seems unlikely that the speaker quotes her thought as saying to herself – *I thought to myself: "You know what: I am afraid I might fall and bad things might happen".* Though not totally impossible, it is highly improbable that she was explaining to herself her own feelings and frights. Indeed, the example has been translated by our native speaker consultant simply as independent apprehensive. In this case, the verb 'fear' may be interpreted as a semantically void matrix predicate used simply to host the subordinate apprehensive clause. As a result, the whole utterance introduces the undesirable situation without introducing the way to avoid it as the main clause (though this way is indirectly alluded to by the cause that may lead to this effect shown in the conditional clause 'if I climb this tree' – in other words, don't climb the tree, and bad things won't happen to you). Semantic bleaching and ultimately omission of the verb of fear is thus a possible account of how the form in *-lkːut* underwent insubordination and developed uses outside the scope of 'be afraid'.

To sum up, we see that the use of the apprehensive in subordinate constructions covers a range of meanings, from 'fear' clauses where the apprehensive shows the reason of the emotion of fear to negative purpose clauses and 'in-case' clauses introducing, in the main clause, a way to avoid a negative situation or to diminish its negative impact. What all these subordinate constructions share is that they introduce a reason for the situation described in the main clause. Constructions involving the verb 'fear' may cover this whole functional range, and one naturalistic context suggests that a 'fear' clause may be interpreted in a way similar to independent apprehensives.

2.3 Syntactic status

Above we have noticed that the syntactic status of the apprehensive as the main or dependent form is sometimes ambiguous. It is relatively uncontroversial that it is independent when used in isolation, as in (1), (2), (3) and (5). It is also relatively uncontroversially subordinate when used with the verb of 'fear', as in (21) to (29). But while in 'fear' clauses the apprehensive clause may be analyzed as the complement clause of the verb 'fear', this analysis does not work for other examples, where there is no complement taking matrix predicate and the apprehensive clause is, at best, an adverbial clause whose status is similar to purpose or reason clauses, but could sometimes just as well be viewed as a paratactic use of the independent apprehensive (cf. (30)). The position of the apprehensive clause with respect to the main clause does not provide a reliable clue. In Archi, a position of adverbial clauses preceding the main clause may be taken as the default, but postposition is also relatively widespread. In the case of the apprehensive, both orders can be elicited:[4]

[4]One of the reviewers suggested that the use of the apprehensive with the imperative (cf. (9)–(11) above) is also uncontroversially subordinate, because it has an explanatory function (her example is English *Close the window, lest we catch a cold!*). Another reviewer, on the other hand, suggested that the use of the apprehensive with 'fear' does not necessarily mean subordination. This indicates room for divergent intuitions, often based on crosslinguistic rather than language-specific evidence. We prefer to be cautious and to avoid purely functional semantic and comparative arguments in deciding upon the status of such constructions. Contra the first objection, clauses combined in a paratactical construction can easily be in explanatory relation one to another (cf. *Close the window, we might catch a cold!*). Classifying English *lest*-constructions with an imperative main clause as subordinate is based on the fact that the *lest*-construction cannot be used independently, which is not true of the Archi apprehensive. As to the second objection, at least some naturalistic examples bury the apprehensional clause together with the verb of fear deep inside subordinate syntax. In (27), for example, it occurs between converbial clauses and the main predicate to which they are related, which shows that the apprehensive *may* be a complement of the verb of fear. Of course, all this evidence does not prove that it always *is* its complement when they come together, but this is a general issue with syntactic evidence. What we do in this section is essentially admit that we have no syntactic evidence to decide what is the syntactic status of the apprehensive in many multi-clause constructions, including when combined with the imperative.

(30) elicited
W-eq'ːi-lkːut, ʷači e⟨b⟩t'ni.
1-bite-APPR dog ⟨3⟩tie:PFV
'They tied the dog lest it bites (him).'

(31) elicited
Gʷači-li-ɬːu iχtilat b-uq'i ***w-eq'ːi-lkːut.***
dog-OBL-COMIT joke 3-leave.IMP **1-bite-APPR**
'Do not tease the dog so that it doesn't bite you.'

On the basis of the limited corpus data that we have, it seems that, as compared to the uncontroversially subordinate clauses such as those with special converbs, the apprehensive clause is more likely to be postposed to the main clause. But even this cannot be taken as a token of its main clause status, because this also happens when it is introduced by the converb form of 'fear', in which case we analyze it as dependent. On the other hand, it can also be embedded into the main clause, which we also take as an unambiguous indication that the clause is subordinate (only elicited examples are available):

(32) Kibrik (1977: 292), elicited
To-w-mu-s zari, [*χːʷalli* ***b-ušbu-lkːut*** *deχˤʷ*] *LO-qi.*
this-1-OBL.1-DAT I.ERG bread 3-take-**APPR** pita 4.give:PFV-FUT
'I will give him a pita, so that he does not buy bread.'

There are also two morphosyntactic properties that group the apprehensive with non-finite forms. First, it does not have a synthetic negative counterpart. While most finite forms have a morphological way to derive negative polarity forms, most non-finite forms do not. Second, the form remains unchanged when used in uncontroversially subordinate clauses. The most common way to form subordinate constructions in Archi is the use of converbs, and none of the finite forms can be used in the context of uncontroversial subordination. We conclude that, while the syntactic status of the apprehensive clause as main vs. subordinate may vary across contexts, and remains unclear for many examples quoted above, morphosyntactic evidence suggests that the apprehensive groups with non-finite forms.

2.4 Archi apprehensive: A summary

In Archi, the apprehensive is used in independent and subordinate clauses. In all of its uses, it conveys the meaning of anticipating a negative situation, including not only a warning, but also a negative purpose, events whose negative effects one has to avoid or diminish, and complements of 'fear'. It is not limited to – though often associated with – the expression of the emotion of fear; and the lexical verb 'fear' seems to be optionally available in constructions that do not imply an emotion, in contexts similar to the independent apprehensive. The exact grammaticalization source of the apprehensive remains unclear, but these are indications that its uses might have syntactically evolved from a dependent to a main clause through insubordination of a clause subordinate to a predicate of 'fear' (33a), acquiring some properties of a directive (33b). Interestingly, in some cases the same process of insubordination is masked by the fact that the whole structure in which the main clause containing the verb of 'fear' is deleted, is itself subordinate to the main clause describing the action undertaken to avoid or diminish the negative impact (33c):

(33) Evolution of the apprehensive in Archi

a. source construction
[[negative effect] fear] = 'fear' complement

b. evolution of the independent apprehensive
[[negative effect] ~~fear~~] → warning, directive

c. evolution of the subordinate apprehensive
[[[negative effect] ~~fear.CVB~~] way-to-avoid] → negative purpose, 'in-case' clauses

The constructions shown in (33c) should not be understood as a stage of evolution subsequent to (33b). They independently result from dropping 'fear' in a special type of contexts where 'fear' itself was in subordinate function: where it introduced a converbial clause syntactically intermediate between the embedded apprehensive clause and the main clause into which the 'fear'-converb itself was embedded. If this is true, in such contexts insubordination of the apprehensive (understood as the loss of its main predicate) did not lead to its independent use. This is what we label a *masking* effect; the same process that leads to independent uses through insubordination in (33b) leads to innovative subordinate constructions in (33b').

While the origins of the form are not well understood, we suggest that it was originally a subordinate form typical of the verb 'fear' that evolved to an apprehensive through insubordination and loss of its matrix verb.

3 Mehweb: From independent apprehensive to subordinate uses via speech reporting

As in Archi, the origin of the dedicated apprehensive suffix *-ala* in Mehweb is unclear, which in itself is noteworthy: Mehweb is otherwise morphologically close to other northern Dargwa varieties, where the form seems not to be attested. The first vowel of the apprehensive affix can be interpreted as the vowel of the *-a-* stem also present in several forms with irrealis meaning, including optative, irrealis, and hypothetical conditional converb (see Daniel 2019, Dobrushina 2019). This would place the apprehensive in the domain of irrealis. Another possibility is a nominal origin, with a non-productive deverbal nominalization suffix *-ala* attested in Standard Dargwa which is based on another northern Dargwa variety.

As in Archi, in Mehweb the apprehensive can be used in subordinate clauses. Unlike in Archi, it can be argued to be an essentially finite form that acquires subordinate uses via extension of a speech reporting construction. The corpus of Mehweb is much smaller than the corpus of Archi, and is, similarly to the corpus of Archi, exclusively narrative. However small the corpus may be, the complete absence of naturalistic examples may be telling. We will briefly come back to this point in §5. All discussion below is based exclusively on elicitation.

3.1 Semantics

When used in an independent clause, the apprehensive expresses the speaker's concern that an undesirable situation may come true. Imperative clauses that go together with the apprehensive may describe an action necessary to prevent it:

(34) *D-arʔ-a* *mura,* *zab* *d-aˤq'-**ala**.*
npl-gather:pfv-imp.tr hay rain npl-go:pfv-**appr**
'Collect the hay, it might rain.'

Unlike Archi, the apprehensive in Mehweb shows regular negative morphology (35). However, this cannot be used as an argument in favor of its syntactically finite rather than subordinate nature. In Dargwa languages, all forms except commands share one morphological strategy to express polarity.

(35) *Zab* *ħa-d-aˤq'-**ala**,* *d-aˤq-a* *šin*
rain **neg**-npl-go:pfv-**appr** npl-hit:pfv-imp.tr water
agarod-le-ħe.
vegetable.garden-obl-in.lat
'Let the water go to the garden, [because/in case] it might not rain.'

In (36), the apprehensive expresses a warning about something that may happen to the addressee:

(36) *Q'ejju, w-ig^{w}-**ala*** / *ar-d-ik-**ala**.*
slow m-burn:pfv-**appr** / down-f1-fall:pfv-**appr**
'Careful, don't you get burnt / don't you fall.'

Both first and third person human subjects are also available with the apprehensive. The second and first person apprehensives typically express apprehension about a potential situation whose subject the speaker (37a) or the addressee (37b) may unintentionally become. Third person apprehensives are more generally warnings about any kind of undesirable situation, and their subject may be the potential maleficiary (37c) or the agent behind the negative situation (38), or a natural force ((34) and (35) above). In (34), (35) and (38) the maleficiary is left unspecified and is probably understood as the addressee, at least by the pragmatic default.

(37) a. first person
Nu *ʁanq'* ***uh-ala***.
I drown **m.become:pfv-appr**
'I might drown!'

b. second person
ħu *ʁanq'* ***uh-ala***.
you.sg drown **m.become:pfv-appr**
'You might drown! Beware not to drown!'

c. third person
Eli *ʁanq'* ***uh-ala***.
child drown **m.become:pfv-appr**
'The child might drown!'

(38) third person
žanawal-li-ni *maza* ***ar-b-uk-ala***.
wolf-obl-erg sheep **away-n-lead:pfv-appr**
'Beware, the wolf can steal the sheep.'

The negative evaluation is inherent to the apprehensive. When used in reference to situations which are usually viewed as positive, the evaluation necessarily changes its default value from positive to negative. Example (39) is felicitous only if the speaker wants a daughter more than a son, a preference which is culturally unexpected. Example (40) is only felicitous if the speaker does not want to recover from her illness.

(39) *Urši* ***w-aq'-ala*** *ħu-ni* *d-aq'-a* *dursi.*
boy **M-do:PFV-APPR** you.SG-ERG F1-do:PFV-IMP.TR girl
'[I am afraid that] you give birth to a boy, [better] give birth to a girl!'

(40) *Ara* ***d-uh-ala.***
health **F1-become:PFV-APPR**
'[I am afraid that] I may recover!'

3.2 Subordinate uses

Apprehensives are used in subordination to verbs of fear, in reference to both present (41) and past (42) situations. In these contexts, they are followed by the perfective converb of 'say', *ile.*

(41) *Nu uruχ* *k'u-we* *le-w-ra* *žanawal-li-ni* *maza*
I be.afraid LV:IPFV-CVB be-M-EGO wolf-OBL-ERG sheep
*ar-b-uk-**ala*** ***i-le.***
away-N-lead:PFV-**APPR** **say:PFV-CVB**
'I am afraid that a wolf will steal a sheep.'

(42) *Baba* *uruχ* *k'u-we* *le-l-le* *χʷe* *q'ac'*
grandmother fright become.frightened:IPFV-CVB AUX-F-PST dog bite
*b-ik-**ala*** ***i-le.***
3-LV:PFV-**APPR** **say:PFV-CVB**
'Grandmother feared lest the dog bite (her).'

Taking the use of *ile* as key evidence, we suggest that subordinate uses of the apprehensive are acquired through extending speech reporting. In East Caucasian languages, speech reporting has various extended uses, including introducing purpose. In (41) and (42), it introduces the cause of fear. In other words, the source constructions in (41) and (42) are literally interpreted as *I am afraid, saying: "The wolf may steal a sheep!"*, *Grandmother was afraid, saying: "What if the dog bites (me)".* It seems that, with verbs of fear, one cannot use the apprehensive omitting *ile.* An independent apprehensive is expected to refer to the moment of speech, so that, in sentences referring to the present, one could have expected to find at least a paratactic reading ('Grandmother is afraid, what if the dog bites her!'). Example (43) shows that this expectation is not borne out (omission of *ile* is judged infelicitous both in reference to the past and to the present).

(43) **Baba uruχ k'u-we le-r/le-l-le* χ^{w}*e*
grandmother fright become.frightened:IPFV-CVB AUX-F/AUX-F-PST dog
q'ac' b-ik-ala.
bite 3-LV:PFV-APPR
'Grandmother is/was afraid – what if the dog bites (her).'

In other subordinate constructions, the use of *ile* is apparently optional. This is true not only of the combinations of the apprehensive with the imperative, as in (44), which may or even must be interpreted paratactically (we have no data on whether *ile* may be used here at all), but also of negative purpose clauses as in (45) and 'in-case' clauses as in (46), where it can be used but may be omitted:

(44) *C'a-li-če ħule w-iz-e,* ***b-uš-ala.***
fire-OBL-SUP.LAT look M-LV:PFV-IMP **N-die(of.fire):PFV-APPR**
'Watch the fire, it might go out.'

(45) *It-ini ħuli b-uc-ib c'a* ***b-uš-ala*** ***(i-le).***
this-ERG look N-LV:IPFV-PST fire **N-die.out:PFV-APPR (say:PFV-CVB)**
'He was looking after the fire so it would not die out.'

(46) *Musa-ni mura d-arʔ-ib dunijal* ***ur-ala*** ***(i-le).***
Musa-ERG hay NP1-gather:PFV-AOR world **rain-APPR (say:PFV-CVB)**
'Musa collected the hay out of fear that rain would start' (lit. 'lest the world rains').

Another way to express negative purpose, alternative to the apprehensive + *ile* strategy in (46), is the use of the purposive converb in *-alis*. Unlike with the apprehensive, where the negative evaluation is hardwired into the semantics of the form itself, the negative component 'not to' is overtly expressed on the purposive converb by the prefix *ħa-* (standard negation). The converb *ile* cannot be used with the purposive converb:

(47) a. *W-aˤld-e adaj-ni ħu dam*
M-hide:PFV-IMP father-ERG you.SG.NOM beat
ħa-w-aq'-a-lis.
NEG-M-do:PFV-IRR-PURP
'Hide, so that your father does not beat you.'

b. *W-aˤld-e adaj-ni ħu dam* ***w-aq'-ala.***
M-hide:PFV-IMP father-ERG you.SG.NOM beat **M-do:PFV-APPR**
'Hide, so that your father does not beat you.'

Above, we have argued for Archi that what is common for the use of the apprehensive in all subordinate clauses – including ‘fear’ complement clauses, negative purpose clauses and ‘in-case’ clauses – is that the apprehensive provides a (negative) reason for the action described in the main clause. This becomes very clear when comparing the scope of the use of the apprehensive to that of the negative purpose converb in Mehweb. While the purpose converb competes with the apprehensive in contexts where both describe an undesirable situation that the subject of the main clause may control, the purpose converb is not available in clauses which describe situations out of her control. For example, (48), an ‘in-case’ clause, and (49), a ‘fear’ clause, are pragmatically infelicitous.

(48)	**Musa-ni*	*mura*	*d-arʔ-ib*	*dunijal*	***ħa-wr-a-lis***.
	Musa-ERG	hay	NPL-gather:PFV-AOR	world	**NEG-rain:IPFV-IRR-PURP**

Intended (same as (46)): ‘Musa collected the hay so that it would not rain.’

(49)	**Baba*	*uruχ*	*k’u-we*	*le-l-le*	*χ^{w}e*	***q’ac’***	***ħa-b-ik-a-lis***.
	grandmother	become.frightened	LV:IPFV-CVB	AUX-F-PST	dog	**bite**	**NEG-3-LV:PFV-IRR-PURP**

Intended (same as (42)): ‘Grandmother became afraid so as to make the dog not bite her.’

In (48) and (49), we see that the purpose converb is not available in some contexts where the apprehensive with *ile* can be used ((46) and (42), respectively). While the purpose reading for the apprehensive construction is only a possible contextual implication of its more general semantics of reason, the purpose converb is dedicated to expressing a purpose of an action, which neither a stimulus of fear nor a meteorological situation can (normally) be.

In addition to its uses as warning, the apprehensive thus also covers subordinate uses described above for Archi, including ‘fear’ clauses, negative purpose clauses (where it competes with the dedicated purpose converb in its negative form) and ‘in-case’ clauses. However, apprehensive ‘fear’ clauses must, and apprehensive negative purpose clauses may, be followed by the converb of the verb ‘say’, a peculiar property that will be central to the interpretation of its syntactic status in the following section.

3.3 Syntactic status

The syntactic status of the apprehensive in Mehweb is not straightforward. As in Archi, the apprehensive clause in Mehweb can be used both in isolation and in polypredicative constructions, but in the latter case it is often followed by the converb of the verb of speech, *ile*. When combined with an imperative, the apprehensive is apparently used without further morphosyntactic complications, as when used in isolation (we do not have data on whether *ile* can even be used in such contexts in the first place). Imperative-apprehensive contexts may thus be interpreted as paratactic constructions rather than constructions involving subordination. In more clear cases of subordination, an apprehensive clause may or even must be followed by *ile.* Thus, with verbs of fear, one cannot use the apprehensive omitting *ile*, while negative purpose and 'in-case' clauses allow for both constructions.

To explain the suggested evolution of the subordinate uses of the Mehweb apprehensive, it is necessary to provide a background on speech reporting in Mehweb, by and large typical of speech reporting in East Caucasian. Contexts of speech reporting are the most common use of *ile*, literally 'having said'. Cf. the following example:

(50) naturalistic
Sune-la=l hubi-ze iχ-ini ***"ma,***
self.OBL-GEN=PTCL husband-INTER that-ERG **take**
ħa-la=l urši" i-le, *kʷe-w*
you.SG.OBL-GEN=PTCL boy say:PFV-CVB in.hands-M.ESS
u-ub-i urši hub-la q'uq'u-be-če w-ix-ib.
M.become:PFV-AOR-PTCP boy husband-GEN knee-PL-SUP M-put:PFV-AOR
'She said to her husband: "Here you go, (this is) your son" and put the boy, whom she had in her hands, onto his lap.'

In (50), the speech reporting clause is embedded between the converb of the verb of speech and its argument ('she "_" having said, …'). Speech reporting thus seems to form a subordinate clause, conveying an indirect report. The converb of 'say' may combine with a lexical speech predicate, as in (51) and (52), where it may seem to function as a complementizer.

(51) naturalistic
Urši-li-ʔini ***žawab g-ib*** *k'as:i-ze "ħu-ni b-aqˤ-a"*
boy-OBL-ERG **answer give**:PFV-AOR fish-INTER.LAT you.SG-ERG N-hit-IMP
i-le*.*
say:PFV-CVB
'The youth answered to the fish: "You hit"!'

(52) naturalistic
Nu-ni ***xar*** ***b-a-i-ra*** *"hej urši, kuda-b-a heš*
I-erg **request** n-lv:pfv-aor-ego hey boy where-n.ess-intrg that
ulica", ***i-le****.*
street **say:pfv-cvb**
'I asked: "Hey guy, where is this street?"'

There are several arguments against such an analysis. First, in Mehweb, a language with otherwise strictly non-finite subordination, speech reporting would represent the only construction which involves a finite clause in an apparently subordinate function. Second, reported speech introduced by *ile* may involve imperatives, such as 'hit!' in (51) and expressions of address, such as 'hey guy!' in (52), which are all main clause phenomena, as well as preserve original deixis, e.g. 'you' in (51) and 'your' in (50). All this can be seen as indication that the speech reporting strategy using *ile* is direct rather than indirect,[5] with the reporting quote exemplifying what Spronck & Nikitina (2019) call a *dedicated syntactic domain*, which explains both the availability of main clause phenomena and the use of finite forms. The use of *ile* in speech reporting constructions, while frequent, is not obligatory, as shown in (53) (compare with (51)) and (54) (compare with (52)).

(53) naturalistic
Bars ***žawab*** ***g-ib****:* *"w-ig-an!"*
in.exchange **answer** **give:pfv-aor** m-want-prs
'They answered: "(We) want!"'

(54) naturalistic
Nu-ni hanna=ra ***xar*** ***b-a-i-ra****:* *"hej urši, heš ʁʷada-ni*
I-erg now=add **request** **3-lv:pfv-aor-ego** hey boy this street-loc.lat
saʁʷa uˤq'-es-i?"
how m-go-inf-attr
'I ask again: "Hey, boy, how should one go to this street?"'

To sum up the relevant data on speech reporting, in Mehweb, what seems a subordinate indirect speech reporting clause in (50), is in fact embedded direct speech reporting. We suggest that the uses of *ile* to introduce the apprehensive clause may be viewed as an extension of direct speech reporting constructions and do not contradict the status of the apprehensive as a finite form.

[5]Whether indirect reporting does exist in East Caucasian languages at all is in fact a controversial issue.

A collateral argument comes from reference tracking. The use of the *sa-CL-i* 'self' pronoun in (55) is indicative of the status of the construction as extended speech reporting, because the 'self' pronoun is here used in its logophoric function, under the typical reported speech condition of coreference of an argument of the dependent clause with the subject of the matrix clause.

(55) *Baba uruχ k'-uwe le-r,* ***sa⟨r⟩i***
granny be.afraid lv:ipfv-cvb cop-f **self⟨f.sg⟩**
ar-d-ik-ala i-le*.*
pv-f1-fall.down:pfv-appr say:pfv-cvb
'The grandmother is afraid of falling down.'

This supports our suggestion that apprehensive clauses with *ile* are extensions of speech reporting. It also explains how some of the restrictions on the use of the apprehensive are removed. In isolation, the apprehensive is used from the temporal and evaluative perspective of the current speaker. Speech reporting introduces a new perspective, that of the reported speaker, which includes an independent frame of temporal reference and evaluative basis, including allowing for the actual speaker not to share the evaluation of the subject of the main clause. What seems to be an apprehensive referring to the past is in fact a regular apprehensive referring to the present but in a new system of temporal reference shifted to the viewpoint of the subject of the main clause.

As yet another piece of evidence for the essentially non-subordinate nature of the apprehensive, consider the following example. In (56), where the apprehensive clause is followed by *ile*, it can be embedded.

(56) *Musa-ni [dunijal* ***ur-ala i-le****] mura d-arʔ-ib.*
Musa-erg world **rain-appr say:pfv-cvb** hay pl-gather:pfv-aor
'Musa collected the hay out of fear that rain starts.'

Without the *ile* converb, the apprehensive clause cannot be embedded. Cf. the following examples, ungrammatical with, but grammatical without embedding:

(57) a. *Eli šula-le b-uc-a [ʁadara*
child tight-advz n-hold:pfv-imp.tr dish
b-oˁrʔ-aq-ala*].*
n-break:pfv-caus-appr
'Hold the child tight so that it does not break the dish.'

b. * *Eli* [*ʁadara* ***b-oˤrʔ-aq-ala***] *šula-le* *b-uc-a.*
child dish **n-break-caus-appr** tight-advz n-hold:pfv-imp.tr
Intended: 'Hold the child tight so that it does not break the dish.'

(58) a. *Sumka b-ux-a* *mataħ* ***ar-d-uʔ-ala.***
bag n-bring:pfv-imp.tr money **pv-npl-lose:pfv-appr**
'Take the bag so you don't lose the money.'

b. * *Sumka* [*mataħ* ***ar-d-uʔ-ala***] *b-ux-a.*
bag money **pv-npl-lose:pfv-appr** n-bring:pfv-imp.tr
Intended: 'Take the bag so you don't lose the money.'

This behavior is parallel to speech reporting. When a reporting quote follows the speech clause, it may ((51), (52)) and may not ((53), (54)) use *ile.* When it is embedded, as in (50), dropping *ile* – which also serves as a delimiter for Spronck and Nikitina's dedicated syntactic domain of reporting – is supposedly less frequent (unfortunately, we do not have the relevant data to argue it is ungrammatical).

We conclude that without *ile* the form is an independent apprehensive, and in all such contexts where it is used in a polypredicative environment (probably limited to the use with imperatives, as in (57a) and (58a)) it should be interpreted as paratactic. In the contexts that are more likely to be subordinate, as 'in-case' clauses, negative purpose clauses and 'fear' clauses, its use evolves as a functional extension by adopting the syntactic pattern of direct speech reporting.

3.4 Mehweb apprehensive: A summary

As in Archi, Mehweb apprehensives have both independent and subordinate uses. In subordinate uses they cover functions similar to those in Archi, including negative purpose, 'in-case' clauses and complements of 'fear'. Unlike in Archi, subordinate uses seem to be associated with the use of the converb *ile* 'having said', optionally or obligatorily, also used to introduce reported speech with verbs of speech. We suggest that the Mehweb apprehensive originates from a form in the main clause with a directive/emotive illocution that has acquired subordinate uses by functional expansion of speech reporting. The evolution of the uses of the apprehensive in Mehweb may then be summarized as follows (note that, unlike other types of subordination, speech reporting typically places quotes after the matrix verb):

(59) Evolution of the apprehensive in Mehweb

a. source construction
[apprehensive] = warning/directive

b. evolution of subordinate uses via extended speech reporting:
[finite verb [apprehensive] *ile*] = 'fear', negative purpose and 'in-case' clauses

c. specification of some of the subordinate uses:
[finite verb [apprehensive] ~~*ile*~~] = negative purpose and 'in-case' clauses

Although the origin of the apprehensive in Archi is unclear, there are some indications that it has followed the opposite path from a dependent to main clause through insubordination, acquiring some properties of a directive (cf. the scheme in (33) above).

4 Northern Akhvakh: From a particle to a morphological apprehensive

To our knowledge, the existence of an apprehensive in Akhvakh was first reported by Denis Creissels:[6]

> I have also observed the sporadic occurrence of *an emphatic form of the prohibitive* expressing a threat in case the prohibition is not respected. This form, for which my main informant gives the Russian equivalent 'Не дай Бог …!' [God forbid – MD & ND], is marked by a suffix *-alogo* (Creissels n.d. [b]).

Creissels collected his data in the village of Axaxdərə near Zaqatala (Azerbaijan), a resettlement from Daghestanian villages speaking Northern Akhvakh. There is no reliable information about the time when the Akhvakh started migrating to Azerbaijan. The oral tradition suggests that the migration was progressive rather than punctual, and started about one century ago (p.c. with Denis Creissels). The last wave of migration occurred immediately after World War 2. According to Creissels, Axaxdərə Akhvakh is still very close to the dialect spoken in the Akhvakh district in Daghestan.

[6]While this study quotes the forms as *-alogo*, our own data and later work by Denis Creissels n.d.(a) has *-alago*.

In 2018, one of the authors of this paper worked in Tad-Magitl, a Northern Akhvakh village in the Akhvakh district, and later with Indira Abdulaeva, a native of Tad-Magitl. Most examples in this section come from these field notes.

Creissels (n.d.[a]) suggests that the apprehensive suffix *-alago* originated from the conditional converb in *-ala* in combination with the interjection *hologo ~hʷelogo* 'beware!' . The etymology of the interjection itself is unknown. Creissels provides examples of the apprehensive and the interjection used together (for the sake of consistency, we gloss his "prohibitive" as apprehensive):

(60) *Hʷelogo imaχa-ge* *ƛ'a* ***ʁad⟨u⟩k'-alago!***
beware donkey-LOC on **⟨M⟩sit-APPR**
'Beware, don't sit on the donkey! (Beware of sitting on a donkey!)' (Creissels n.d.[b])

(61) *Hʷelogo ha-be* ***gʷ-e:lago.***
beware DEM-N **do-APPR**
'Beware, don't do this!'
(Creissels n.d.[b])

Our data (in the case of Akhvakh, elicited except where indicated otherwise) confirm these examples for Tad-Magitl, a Northern Akhvakh village in Daghestan. Our consultant, Indira Abdulaeva, accepts the use of the affix both with (62), (63) and without (64) the particle, and notices that the apprehensive may have realizations "extended" by *-le*, as in (63) and (64); and that the particle may occur both as *hole* (62) and as *holego* (63).

(62) *Hole* *χʷe-de* ***q':eleč'-alago.***
beware dog-ERG **bite-APPR**
'Beware, the dog might bite you.'

(63) *Aħmadi holego* ***aq':us:-alagole!***
Ahmad beware **drown-APPR**
'Ahmad, beware, you may drown!'

(64) *C':a* ***c':-alagole/c':-alago.***
rain **rain-APPR**
'It might rain.'

In addition to *hole(go)*, the dictionary of Northern Akhvakh (Magomedova & Abdulaeva 2007) quotes another particle associated with apprehension, *čego*, translated as 'what if'.[7] In its three occurrences in the dictionary, the particle is combined with the apprehensive, as in (65); (66) is a similar example obtained in elicitation.

(65) dictionary example
čego mik'i-ga q':ʷat'ala ***b-eq'-alagole.***
beware child-ALL cold **N-come-APPR**
'What if the child catches a cold!'

(66) *Eq:-a čego χʷe-de mik'i-ge* ***q':eleč'-alagole.***
look-IMP beware dog-ERG child-LOC **bite**-APPR
'Beware lest the dog bites the child!'

4.1 Semantics and morphology

The form in *-alago*(*le*) can be used with subjects of all persons. It is thus more appropriately described as apprehensive than as emphatic prohibitive.

(67) a. 1st person
Dene ***aq':us:-alago(le).***
I **drown-APPR**
'What if I drown?'

b. 2nd person
Mene ***aq':us:-alago(le).***
you.SG **drown-APPR**
'What if you drown!'

c. 3rd person
Hudu-we ***aq':us:-alago(le).***
that-M **drown-APPR**
'What if he drowns!'

The apprehensive clause may describe either the undesirable situation itself, as in (68), or a situation that implies further negative consequences, as in (69); probably, the two uses cannot be consistently separated, because a situation that implies negative outcomes is itself perceived as negative.

[7] Reviewers note that, in this particle, the segment *go* appears once again, but even with a similar variation between *hole* and *holego* we have no evidence for its meaningful morphological segmentation. As a possible parallel, the apprehensive dedicated particle *vore* in Avar, which is the local lingua franca, also appears in the variant *voregi*, where *-gi* may be identified as the additive particle 'also', 'too', that has a wide range of discourse uses. No such evidence for *-go* in Akhvakh is currently available.

(68) *ɬ:eni goɬi:hi ba-χ:ʷ-alagole iʟ:i.*
water be:NEG:HPL:CVB **HPL-remain-APPR** INCL
'We may be left with no more water!'

(69) *ʟ':õhe **W-ux-alagole** mene.*
fall.asleep:M:CVB **M-become-APPR** you.SG
'Beware of falling asleep!'

Like in Archi, but unlike in Mehweb, the apprehensive does not have a (morphological) negative counterpart. We have no data as to how apprehension that a positively evaluated situation may not come true is expressed in Northern Akhvakh.

4.2 Syntactic status

According to our field data, the form in -*alago*(*le*) introduces an independent clause and can not be used in subordination.

In four out of the six total occurrences in Magomedova & Abdulaeva (2007), the apprehensive clause occurs in isolation. In the two other examples, it occurs in combination with an imperative (70), (71); cf. also (72) obtained in elicitation.

(70) dictionary example
*Imiχi **aq':us:-alagole** k'oli-g-une kʷari b-eq:-a.*
donkey **drown-APPR** neck-LOC-EL rope N-take.away-IMP
'Take the rope off the donkey's neck, it may suffocate.'

(71) dictionary example
*χ:eχ:i-meč'iʟa, **w-ač'aq'-alagole**.*
hurry-hurry **M-be.late-APPR**
'Hurry, or you'll be late!'

(72) *čadi b-eχ-a ãʟ'o **c':-alagole**.*
umbrella N-carry-IMP together **rain-APPR**
'Take an umbrella, it might rain!'

Example (70) looks as if the apprehensive could be interpretable as a negative purpose clause. However, this example, as well as (71) and (72), all contain imperatives. If the apprehensive were a negative purpose clause subordinate to 'take' in (70), it could also be used with the main verb in the past tense. This is not the case; (73), which is a modified version of (70) with the presumably main verb in the past, is ruled out.

(73) **Imiχi* ***aq’:us:-alagole*** *k’oli-g-une kʷari b-eq:-ari.*
donkey **drown-APPR** neck-LOC-EL rope N-take.away-PST
Intended: ‘(He) took the rope off the donkey’s neck, lest it would get strangled.’

Our consultant notices that the form conveys apprehension about a future event, and one cannot warn someone about the past. Clauses introducing negative purpose in the past are based on the negative converb (74a); this construction is also available for the main clause with reference to the moment of speech, with the main verb in the present (74b):

(74) a. *Baba-de* ***aq’:us:-iL-uk’u*** *waša ut-iLa ĩhʷari-ga.*
mother-ERG **drown-NEG-PURP** son M.let-PFV.NEG lake-LOC.LAT
‘The mother did not let the child go to the lake so that he would not drown.’

b. *Baba-de* ***aq’:us:-iL-uk’u*** *waša ut-ero guLa*
mother-ERG **drown-NEG-PURP** son M.let-M.IPFV COP.NEG
ĩhʷari-ga.
lake-LOC.LAT
‘The mother does not let the child go to the lake so that he does not drown.’

The apprehensive is equally impossible in ‘fear’ complement clauses,[8] where a locative form of the action nominal is used instead:

(75) *Dene Lo:he gudi du-ge χʷe-de* ***q’:eleč’a-ro-g-une****.*
I be.afraid.CVB COP.M you.SG-LOC dog-ERG **bite-NMLZ.OBL-LOC-EL**
‘I’m afraid that the dog might bite you.’

As for constructed contexts with a ‘fear’ verb and an apprehensive, our consultant suggested that what results in (76) are two separate sentences, which suggests a paratactic structure, similarly to the combination of the apprehensive with an imperative in (70), (71) and (72) above:

(76) *Dene Loho gudi, du-ge χʷe-de* ***q’:eleč’-alago****.*
I afraid COP.M you.SG-LOC dog-ERG **bite-APPR**
‘I’m frightened! What if the dog bites you!’

[8] One of the reviewers asks whether the apprehensive can or cannot be introduced by a quotative particle in a type of extended speech reporting construction. This would be an important piece of evidence which we very unfortunately lack.

It remains unclear how (77) should be interpreted, with the apprehensive clause sitting between the conditional clause and the imperative:

(77) *Eχ:a* *w-ano* *w-ũč-ala* ***c'ː-alagole*** *čadi* *b-eχ-a*
outside M-go.CVB M-find-COND **rain-APPR** umbrella N-carry-IMP
ãnL'o.
together
'If you plan to go outside, it might rain! Get an umbrella.'

Presumably, the condition 'if you go out' is subordinate to the imperative 'get an umbrella', and it is unclear how the apparently paratactic apprehensive makes it to the position between them. More data is needed here.

4.3 Akhvakh apprehensive: A summary

We conclude that, in Akvakh, the apprehensive is functionally similar to the independent uses of the apprehensive in Archi and Mehweb, including availability to subjects of all persons. At the same time, it is very different from the apprehensives in both languages in terms of its syntactic behavior, because it is not used in subordination. The typical contexts of use of the apprehensive in subordinate clauses in Archi and Mehweb, including negative purpose or 'fear' complement clauses, are covered by other forms, and what appears to be subordinate uses of the apprehensive has been argued to be paratactical, similar to the use of the apprehensive with imperatives in Mehweb. This is in accordance with Creissels' (n.d.(a)) suggestion that the apprehensive affix originated from univerbation of the verb with the interjection *hologo* 'beware!', as interjections are known to belong to main-clause phenomena.

(78) Evolution of the apprehensive in Akhvakh
   a. source construction:
      [verb apprehensive.particle] = warning/directive
   b. univerbation:
      [verb-APPR] = warning/directive

However, the hypothesis of the evolution of the apprehensive through univerbation with the particle needs to be carefully assessed. It is unclear why the particle would follow the verb (in all examples, the "free" particle is used sentence-initially). It is unclear why it should attach to the conditional converb. It is also unclear where the "extended" version *-alagole* comes from, and what its etymological relation to *-alago* is. Answering all these questions requires re-evaluation of the data, which may lead to reconsidering the initial hypothesis.

5 Summary discussion

The apprehensive forms described above for three languages of Daghestan, Archi, Mehweb and Akhvakh, all belonging to different branches of the family, were shown to be diverse in terms of their internal structure and possible origins, but, to some extent, convergent in their functional syntactic scope. We summarize the convergent and divergent properties of the three forms in Table 1.

Table 1: Properties of Archi, Mehweb and Akhvakh apprehensives

	Archi	**Mehweb**	**Akhvakh**
morphological source	?	?	? (conditional + interjection?)
main clause: apprehensive proper	yes	yes	yes
subordinate: ‘fear’ complement	yes	yes	no
subordinate: negative purpose	yes	yes	no
subordinate: ‘in-case’ clauses	yes	yes	no
person reference	1/2/3	1/2/3	1/2/3
finiteness	non-finite	finite	finite
negative form	periphrastic	morphological	unavailable

On the formal side, the apprehensive has a negative form in Mehweb, is negated by means of a periphrastic construction in Archi, and has no negative counterpart in Akhvakh at all. The Akhvakh apprehensive probably goes back to the combination of a verbal form with an apprehensive interjection still present in the languages, while the origins of the apprehensives in Mehweb and Archi are unclear.

All of these forms express apprehension when used in the main clause, and apply to all subjects. But unlike Akhvakh, both Archi and Mehweb forms are used in subordination, including 'fear' complements, negative purpose clauses, and what Lichtenberk (1995) has termed 'in-case' clauses. Sentential 'fear' complements provide the stimulus for fear, which may be viewed as an instantiation of the concept of cause or reason. We have suggested that what negative purpose ('so as not to') and 'in-case' ('out of fear of') clauses have in common is that the main clause suggests a way to avoid or reduce the negative impact of the event described in the apprehensive clause, so that, in these contexts too, apprehensive constructions may be seen as reason rather than purpose adverbial clauses. All apprehensive-marked subordinate clauses may thus be interpreted as expressing the reason for another event/action (expressed by the main clause) that is carried out to avoid an undesirable consequence. But with all these apparent similarities, Archi and Mehweb are different in terms of the syntactic patterns involved.

In East Caucasian languages, there is a strong morphological divide between main clauses using finite forms and dependent clauses using various types of converbs, infinitives and action nominals. Some forms considered primarily non-finite can be used as main clause predicates, but this happens under special circumstances. Thus, the perfective converb is used finitely for unwitnessed events (Kibrik 1977), and participles are sometimes used finitely, a phenomenon that is clearly understudied (Kalinina & Sumbatova 2007, Kalinina 2011). In Mehweb, too, converbs can be used in main clauses with subtle pragmatic effects (Kustova 2019). Conversely, finite forms may be used in apparently dependent contexts, but this happens in a special environment of extended speech reporting, such as in purpose clauses accompanied by the converb of the verb of speech, *boli* in Archi and *ile* in Mehweb.

In Mehweb, the use of the apprehensive in 'fear' complement clause requires the use of *ile* 'having said', and its use in negative purpose and 'in-case' clauses allows (but does not require) it. Archi has a very similar pattern of extended speech reporting, but the apprehensive shows no association with the verb of speech - instead, in both negative purpose and 'in-case' clauses the verb 'fear' can be used. In Archi, the apprehensive, unlike the finite forms but like some (probably most) non-finite forms, does not have a morphological negative counterpart. The negative correlate of the apprehensive ('lest P does *not* happen') is built periphrastically, using a negative converb of the main verb with the apprehensive of the verb 'remain'.

We hypothesize that the Archi apprehensive is essentially a non-finite form that has acquired main clause uses via insubordination (Evans 2007), by losing its main 'fear' clause. Subordinate uses may also have followed this path, but here insubordination is masked by the fact that this insubordination itself happened within a subordinate clause. In other words, the verb 'fear' was dropped as both main and subordinate matrix clause dominating the apprehensive clause, leading to different outcomes:

(79) evolution of the apprehensive in Archi (repeated from (33) above)

a. source construction
[[negative effect] fear] = 'fear' complement

b. evolution of the independent apprehensive
[[negative effect] ~~fear~~] → warning, directive

c. evolution of the subordinate apprehensive
[[[negative effect] ~~fear.cvb~~] way-to-avoid] → negative purpose, 'in-case' clauses

In Mehweb, on the other hand, the form is probably originally a finite form with a directive ('try to avoid P') or apprehensional ('beware of P happening') meaning which expands into dependent uses such as negative purpose clauses via extension of the speech reporting construction.

(80) evolution of the apprehensive in Mehweb (repeated from (59) above)

a. source construction
[apprehensive] = warning/directive

b. evolution of subordinate uses via extended speech reporting:
[finite verb [apprehensive] *ile*] = 'fear', negative purpose and 'in-case' clauses

c. specification of some of the subordinate uses:
[finite verb [apprehensive] ~~*ile*~~] = negative purpose and 'in-case' clauses

The Akhvakh apprehensive does not have subordinate uses, either because this is a recent development, as suggested by the relative transparency of its connection to the apprehensive interjection, or because it follows a different grammaticalization path.

(81) evolution of the apprehensive in Akhvakh (repeated from (78) above)

a. source construction:
[verb apprehensive particle] = warning/directive

b. univerbation:
[verb-APPR] = warning/directive

A lot remains unclear. We do not know the exact path of grammaticalization of the apprehensive in Akhvakh, and the hypothesis suggested by Denis Creissels should be considered with a pinch of salt, probably rather a point of departure for future analyses. In Archi, on syntactic grounds we posit, as a source of the apprehensive, a construction dedicated to 'fear' complement clauses, but its morphology remains obscure. Similarly, we know nothing about the morphological origins of the apprehensive in Mehweb, except a hypothetical link to the action nominal in other northern Dargwa varieties. But, again based on syntactic evidence, it seems probable that in Mehweb the form was expanding from main to subordinate clauses, while in Archi, the development was from subordinate to main clauses. And yet, in Archi and Mehweb, the two apprehensives are equally similar in their functional scope to the typological profile of apprehensives cross-linguistically. We conclude this is a case of secondary convergence by filling the same typological niche.

As we noted above in the discussion of the available data, we have quite a few naturalistic examples from the corpus of Archi, and no textual examples for Mehweb and Akhvakh. All three corpora are narrative, and the apprehensive is not expected to be frequent: as one of the authors of the Akhvakh dictionary and the author of the Akhvakh corpus, Indira Abdulaeva, notices, the Akhvakh apprehensive is after all an *interactional* form. Nevertheless, the absence of the apprehensive in the Mehweb corpus is conspicuous, even if one adjusts for the relative sizes of Archi (over 40,000 tokens) and Mehweb (10,000 tokens) corpora. Indeed, both Archi and Mehweb allow for subordinate uses of the apprehensive such as negative purpose or 'fear' complements in elicitation and could be expected to be more proportional in terms of the use of the apprehensive in the narratives. Yet, there are no textual occurrences of the apprehensive in Mehweb, and over 20 naturalistic examples from Archi. This is in line with our hypothesis: the Archi apprehensive has originated from subordinate clause uses in the first place, and is still frequent in subordination, while in Mehweb it has evolved from main clause uses and, in some of its subordinate uses, competes with older alternatives such as purpose converb with a negative suffix.

The three languages of the family where the dedicated apprehensive has so far[9] been reported belong to three different branches of the family, Lezgic (Archi),

[9]Very late in the process of revising this article, the authors became aware of the presence of forms functionally close to the apprehensive in Tsezic languages. Zaira Khalilova (Khalilova in prep.) is working on a verbal form -*dal* in Bezhta, that shows a range of uses similar to those of subordinate uses of the apprehensive in Archi, including 'fear' complement and negative purpose. Subordinate uses of the Bezhta form, described as preventative gerund and the negative counterpart of the gerund of aim, are also discussed by Kibrik & Testelec (2004: 267). Khalilova

Dargwa (Mehweb) and Andic (Akhvakh). Both Archi and Mehweb are outliers of their respective branches in central Daghestan, in strong contact with Avar and Lak but not with each other. The apprehensive does not seem to appear in other Lezgic or Dargwa languages (we have no data for other Andic languages). Akhvakh, an Andic language, also belongs to the area where Avar serves as the main language of interethnic communication (Dobrushina & Zakirova 2019). Areal convergence remains a possibility, but a thorough study of apprehensive constructions in Avar, a major language in contact with all three minority languages discussed above, as well as in Lak, another possible source for two of the three languages, would be required. This could provide a possible missing link crucial for deciding whether areal interaction might have been a (co-)factor in the evolution of the apprehensive forms in the three languages in the study, but is not an easy task because of the dialectal variation within Avar itself, and the fact that the three languages were in contact with different Avar varieties. According to the descriptions of standard varieties, and to our own field data obtained by elicitation, Avar and Lak do not have any dedicated constructions.

6 Conclusion

In this study, we have considered dedicated apprehensives in three East Caucasian languages, all belonging to different branches of the family and thus only distantly related, Archi (Lezgic), Mehweb (Dargwa) and Northern Akhvakh (Andic). The level of detail of discussion varied according to the amount of available data, and especially to the size of the corpora from which naturalistic examples could be obtained. In all three languages, the source of grammaticalization is obscure, and the path of grammaticalization is probably different. In Akhvakh, the use of the apprehensive is limited to the main clause. For Archi, it is feasible to suggest that the independent apprehensive has developed from 'fear' complement clauses via insubordination. For Mehweb, the apprehensive seems to have developed from some kind of a directive form, i.e. with main clause uses as a starting point. Yet, following different paths, Archi and Mehweb apprehensives have developed a comparable range of uses, such that they fit the typological

does not provide examples of clearly independent, directive uses, definitional for the apprehensive category. The Bezhta form shares dependent uses with the Archi apprehensive but lacks its independent use, and may correspond to an early stage of the form observed in Archi. More evidence on the origins of this form could thus close the important gap in our argumentation that has not been addressed in the study: what could be the source of a dedicated subordinate 'fear' complement form in Archi, which we suggest as the origin of its apprehensive?

range of the apprehensive. All this naturally leads to suggesting an areal convergence scenario. However, none of these languages is in direct contact, and we are unaware of any other East Caucasian language that would have a dedicated morphological apprehensive, including the two potential donors for Archi and Mehweb, namely, Avar and Lak. Apprehensive being a discourse interactional category, one could be inclined to think that areal convergence happens not, or not necessarily, on the morphological level, by developing a dedicated apprehensive form, but on the discourse level, by developing a dedicated construction to express apprehension. Here, Avar, with its dedicated particle *vore(gi)*, could indeed be the attractor for convergence. However, we are unaware of subordinate uses of the apprehensive in Avar that would explain the expansion of the apprehensive to subordinate uses in Mehweb, which led to the remarkable similarity between the two languages. We can only conclude that a more areally informed study of the apprehensives in Daghestanian languages is necessary, and remains a task for further studies.

Abbreviations

CL	agreement slot
1, 2, 3, 4	genders in Archi
M, F, N	male, female and neuter genders in Mehweb and Akhvakh
ADD	additive particle
ADVZ	adverbializer
ALL	allative (goal)
AOR	aorist
APPR	apprehensive
ATTR	attributivizer
AUX	auxiliary
CAUS	causative
COMIT	comitative
COND	conditional
CONT	localization ‘in contact’
COP	copula
CVB	converb
DAT	dative
DEM	demonstrative
EGO	egophoric
ERG	ergative
ESS	essive (absence of motion as opposed to lative and elative)
EVID	indirect evidential
F	feminine
F1	second feminine gender
FUT	future
GEN	genitive
HORT	hortative (first plural inclusive imperative)
HPL	human plural (agreement pattern)
IMP	imperative
IN	localization ‘inside’
INCL	inclusive
INF	infinitive
INTER	localization ‘between’
IPFV	imperfective stem
IRR	irrealis
LAT	lative (goal)

LOC	locative	PST	past
LOCPL	locutive plural agreement	PTCL	particle
LV	light verb	PV	preverb
N	neuter gender	PURP	purpose converb
NEG	negative	Q	interrogative particle
NOM	nominative	QUOT	quotative particle
NPL	non-human plural (agreement pattern)	SELF	pronominal stem used in reflexive and logophoric contexts
OBL	oblique	SUP	localization 'on'
PFV	perfective stem	TEMP	temporal converb
PL	plural	TR	transitive
PROH	prohibitive (negative imperative)	V	verb
PRS	present		

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Chapter 9

Inauspicious events and their outcomes in Thulung

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The Thulung (Tibeto-Burman, Kiranti sub-group, Eastern Nepal) narrative corpus yields very few examples of precautioning clauses, even though such constructions are in fact fairly easy to obtain through elicitation. The precautioning construction is characterized by a) typologically unusual clause ordering, with the precautioning clause occurring before the preemptive clause; b) a negated optative form of the verb in the precautioning clause; c) subordination of the precautioning clause by means of a converbal form of the verb 'to say'. Furthermore, precautioning clauses always show different subject from the pre-emptive clause, and a strong causal link across the two clauses, resulting in an avertive reading. A grammaticalization pathway is proposed for the development of this construction, originating from a lexical verb 'be auspicious'. Other precautioning strategies are also found: an 'otherwise' ('if not') construction, calqued on Nepali, and a coordinated construction in which the first clause contains the precautioning event in the form of reported speech, using a finite form of the verb 'to say'. In addition to precautioning strategies, Thulung also has an emphasis particle *sə* which takes on an apprehensional function (but only with 2nd person S, A or P), and can be interpreted as a prohibitive in certain situations. The apprehensive domain in Thulung is much richer than initially suggested by the corpus, making use of fixed morphosyntactic constructions and coopting other language mechanisms for the expression of different types of undesirability.

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1 Introduction

Considering the notable absence[1] of South Asian languages from descriptions of the apprehensional domain, this contribution aims to illustrate how Thulung, spoken in Eastern Nepal, expresses the domain in question. As will be seen, there is indeed no dedicated morphology which "indicates that an event will potentially occur but is undesirable, with associated pragmatics of warning or threat" (Angelo & Schultze-Berndt 2016: 255). Thulung has nonetheless developed a means of expressing apprehension through a negative purpose construction, which this chapter will explore.

Thulung is an exclusively oral language of the Kiranti subgroup (Sino-Tibetan). Its verbal morphology is predominantly suffixal, with formatives used for derivation (associated motion, Aktionsart and valence changes), for the expression of arguments[2] and mood (optative or irrealis). Apprehension, as defined by Vuillermet (2018: 262), is a cover term that "encompasses many different nuances of an emotion triggered by an undesirable, possible event". By this definition, the apprehensional domain in Thulung is expressed by means of a negative purpose construction, used to bring about the avoidance of a situation with a negative outcome. There is no morphology or construction for the expression of fear, which is instead conveyed through lexical means.[3] Considering the poorly developed apprehensional domain in Thulung, one may wonder, as did one reviewer, whether it is for good reason that languages like Thulung are not present in typological research on apprehension, especially as, as will be seen below, there is evidence that the Thulung negative purpose construction is calqued from (Indo-Aryan) contact language Nepali (this is discussed in §2.1 below). It is my opinion that a

[1]Vuillermet's (2018) Table 3 "Terminology used crosslinguistically to refer to apprehensional functions" lists languages from South America, Australia, Papunesia, North America, Eurasia (Central Siberian Yupik) but none from South Asia.

[2]The personal indexes are portmanteaux indicating person/number/clusivity and role of up to two arguments, together with tense and, when relevant, imperative mood.

[3]This is either through a verb 'to fear', as in:

(i) *Meram-ka phʌl-mu ŋis-ta.*
dist.dem-erg cut-inf fear-3sg.pst
'He was afraid to cut [the finger].' [Miyakma-Lakpa, 127]

or through the noun 'fear', as in:

(ii) *A-ŋim-ka go ne a-kʌl khrep-to.*
1sg.poss-fear-instr 1sg top 1sg.poss-face cover-1sg>3sg.pst
'I covered my face in fear.' [Ghumne Pani, 97]

better, cross-linguistic understanding of the apprehension domain necessitates input from both languages where it is well-represented, as is the case of other languages described in this volume, and those where it is a considerably more minor phenomenon, as in Thulung, the latter providing useful insights about how a language without morphology can develop a strategy to express a category.

The structure of this contribution is as follows: in §2, I shall describe the Thulung negative purpose construction, proposing a development pathway based on the elements that make up the construction. I then present some related constructions, either an alternative means of expressing a similar function (the 'if not' construction, in §3) or relevant in that they represent alternative grammaticalizations from a same lexical source verb (§4), before concluding in §5.

2 The Thulung negative purpose construction

According to Lichtenberk (1995: 302), the precautioning function in language is made up of two subtypes: an *avertive* relationship, defined by a direct causal link between the two clauses, and an *'in-case'* relationship, in which precautions are taken in case the apprehension-causing situation occurs but without seeking to avert it. The precautioning function is expressed through a negative purpose construction which – to my knowledge – does not have 'in case' readings.

The terminology for the semantic content of the two clauses in the negative purpose construction is that adopted throughout this volume: the apprehension-causing situation is the outcome that must be averted, expressed through what I shall call the *precautioning clause*; the action taken to avoid the situation is expressed in the *pre-emptive clause*.

2.1 Presentation and main characteristics

The Thulung narrative corpus which I have assembled since 1999, and which constitutes just over ten hours of interlinearized material,[4] yields the very low count of two examples of the negative purpose construction, seen in (1) and (2) (along with two examples of the equivalent positive purpose construction). The two other sources that exist for Thulung diverge considerably in terms of the presence of the construction. Allen (1975), the first grammatical description of Thulung, is accompanied by 17 pages of transcribed and glossed "straightforward narrative texts" (p. 140) (the fluent translation follows and is not part of the page

[4] Some of these annotated recordings can be found at: https://pangloss.cnrs.fr/corpus/Thulung_Rai

count) and ten pages of a transcribed and translated (but not glossed) "narrative text in ritual language" (p. 168). Allen's corpus yields not a single instance of the negative purpose construction, nor of its positive counterpart. The second source of Thulung narrative material is a recent Thulung translation of the New Testament,[5] which yields a great number of examples of negative and positive purpose constructions matching those discussed in this contribution and illustrated in examples (1) and (2). I have found negative purpose clauses to be easy to obtain through elicitation,[6] and have made use of elicited material as well in the discussion presented here.

(1) *Jakke biruwa bʌstu-ka me-po-mi-nʉ rwaksaka*
small plant.species cattle-ERG NEG-eat-3PL>3SG-OPT SUB
ur-mu ba:si.
surround-INF must
'So that the cattle does not eat the small plants, we must surround them (with bamboo sticks).' [verb dictionary]

(2) *Lam rem-ben-mu ʌpthero me-dʉm-nʉ rwaksaka*
path look-CAUS-INF difficult NEG-become.3SG-OPT SUB
prʌ-m-ri-m si-mu lam si-mu ɖʉmla homlowo gui
create-1PL>3SG.PST-NMLZ die-INF path die-INF culture even.now 1DI
thu:lu-ka bet-to-ŋa be:-mim rwak-pa lwa bu.
Thulung-ERG do-SIM.CVB-INT do-NMLZ say-NPST.PTCP story be.3SG
'So that it is not difficult to show [others] the path [for dealing with a corpse], we created a path for dying, and even today we Thulung follow this "death culture", and this is the story of that.' [Miyakma-Lakpa, 136]

The construction in (1) features two clauses: the precautioning clause [*jakke biruwa bʌstu-ka me-po-mi-nʉ rwaksaka*], which expresses a possible negative outcome, and a pre-emptive clause [*urmu ba:si*], which expresses an action to take to avoid the negative outcome. The precautioning clause contains a negated verb in the optative mood followed by the subordinator *rwaksaka,* and is always a subordinate clause. The pre-emptive clause is always the main clause, which, in (1), contains an impersonal deontic necessity construction (see §4 below). Note that while the form of the verb in the precautioning clause must be a negative

[5] https://thulungmedia.com/en/thulung-bible

[6] By elicitation, I mean that I have gotten these constructions in response to describing a situation made up of two events, one with negative consequences and one which could help avoid the negative consequences.

optative, and it must be followed by subordinator *rwaksaka*, there are fewer constraints on the pre-emptive clause: an impersonal deontic necessity construction is common, but so too are finite verbs, as in (3).

(3) *Tsɵttsɵ-mim-ka pi:masi me-bi-mi-nʉ rwaksaka*
child-PL-ERG fruit NEG-beg-3PL>3SG-OPT SUB
thʌk-tʉ.
hide-3SG>3SG.PST
'So that the children wouldn't beg for the fruit, he hid it.' [elicited]

The negative optative is also found in independent clauses, as in (4),[7] and expresses the speaker's desire to avoid a situation. The optative, which occurs in a complete paradigm, i.e. with all three persons, is found with both positive and negative verbs. For examples of positive optative-marked verbs, see (15) and (16) below. (A proposed development path for the optative marker is described in §2.2.)

(4) *Phe:ri phe:ri hopmam dzuʈho-ra mi-grɵms-i ma ʈika*
again again like.this ritual.pollution-LOC NEG-meet-1PI CONJ bindi
ba:-mu mi-ba:si-nʉ rwak-i.
wear-INF NEG-must-OPT say-1PI
'We say "Let's not meet again and again in this kind of ritual pollution, and may we not have to wear bindis again".' [Life in Mukli, 185]

The undesirability of the situation that is conveyed by the precautioning clause of the negative purpose construction, and which the pre-emptive action is meant to avoid, transparently comes from the negative optative form. The negative purpose construction is most often used to transmit expectations of what constitutes appropriate and inappropriate behavior, rather than to convey direct interdiction with respect to a specific action. This is probably why the pre-emptive clause often uses the impersonal deontic necessity construction, and not, as far as I can ascertain from elicitation, an imperative (this is also found to be the case in Upper Tanana Dene, see Lovick 2026 [this volume]). Thulung therefore does not conform to the cross-linguistic tendency, noted by Dixon (2009: 24), for the clause expressing the pre-emptive action (in his terminology, the focal clause)

[7] Thulung has, as discussed, an optative mood, used to express the desire that an event take place. Non-past verb (unmarked for optative) in 1st person non-singular inclusive forms can have a hortative reading. This is contrasted, in (4), by translating 'let's' for the hortative, and 'may we' for the optative.

of avertive-type constructions (subsumed within his possible consequence constructions) to contain positive or negative imperatives.

The formative *rwaksaka* is also used to subordinate both expressions of direct speech or thought[8] to an utterance verb, as in (5), as well as other types of complement clauses, as in (6).

(5) *Khamba-nim reʈ-to-m bu rwaksaka*
Tibetan-F bring.up-1SG>3SG.PST-NMLZ be.3SG SUB
me-sɵ-ʉ-wa.
NEG-tell-3SG>3SG-IRR
'He had not told [anyone at home] that he had brought up [=married] a Tibetan woman.' [Foundation, 122]

(6) *Thuluŋ-ka tha:_lwas-tʉ pʌila-ŋa oram rok-ta ma*
Thulung-ERG realize-3SG>3SG.PST first-INT PROX.DEM come-3SG.PST CONJ
bu-mim rwaksaka tha:_lwas-tʉ.
be-NMLZ SUB realize-3SG>3SG.PST
'The Thulung realized, he realized that this one had arrived and stayed [there] first.' (i.e. was therefore the rightful owner of the land) [Thulung Origin, 40–41]

At the same time, *rwaksaka* is also used, and synchronically analyzable, as an anterior converb for the verb *rwa:mu*, 'say', as in (7):

(7) *ʌm-tsi ma ba-tsi rwak-saka mu hoʈ-dzɵl-tsi ma lʌk-tsi*
sleep-2DU CONJ stay-2DU say-ANT.CVB fire light-ANTIC-2DU CONJ go-2DU
ʔe.
HS
'Having said "sleep and stay here" they lit a fire and left.' [Hansel, 24]

The converbal form in (7) is the presumed source for subordinator *rwaksaka* in (1), (2), (5) and (6), similarly to what is found in many Tibeto-Burman languages, where the grammaticalization of the verb 'say' to encode "causal, conditional, evidential, and purposive meanings" (Coupe 2018: 202) is often based on a converbal form (Coupe 2018: 200–201; Chappell 2008 for Sinitic languages; Matić & Pakendorf 2013 for Siberian languages).[9] This pattern is also discussed from a ty-

[8]This conflation of saying and thinking is common in languages of Nepal; see Noonan (2006). Also, note that all reported speech in Thulung is expressed as direct speech.

[9]Matić & Pakendorf raise the possibility that Siberian *non-canonical* functions for the verb 'say' (among which is grammaticalization to a marker to subordinate purpose clauses) have "historical connections with similar structures in East and Southeast Asia, the Himalayas and South Asia and the Turkic and Turkic-influenced region spreading from Central Asia to the Balkans." (Matić & Pakendorf 2013: 358).

pological perspective by Aikhenvald (2009: 387–390), who describes a pathway for the grammaticalization of verbs of reported speech into clause-chaining devices.[10] The diachronic development of the subordinator probably accounts for the order of the clauses which is observed in the (negative) purpose construction and others involving material subordinated with *rwaksaka*.

Interestingly, Nepali has a strikingly similar structure to the Thulung negative purpose construction.[11,12]

(8) Nepali
Dza:ɖo-le nʌ-mʌr-os bhʌn-era luga lʌgai-di-ẽ.
cold-INSTR NEG-die-OPT say-CVB clothes wear-APPL-1SG.PST
'So that he would not die of cold, I dressed him.' [elicited]

The place of language contact in the use of this construction is worth considering: the constructions in Thulung and Nepali are close enough that it is reasonable to think of them as calques, raising the question of direction of borrowing. Thulung is in a situation of intensive contact with Nepali, with, to my knowledge, no remaining monolingual speakers. Evidence of borrowing from Nepali is found throughout the language, most prominently in the lexical and syntactic domains, but with consequences for the morphosyntax as well (see Lahaussois 2003 for a contact-induced shift in the position of the ergative split within the personal pronoun system). The contact situation itself suggests that the likely direction of borrowing is from Nepali into Thulung, but there are other clues as well. As mentioned above, the corpus that I have made use of herein is made up of three sub-corpora: my own data, collected from 1999 onwards; some transcribed and annotated textual data from N. J. Allen, collected in the late 1960's and published in Allen (1975); and materials from recent translation efforts of the New Testament into Thulung, available online in their Devanagari transcription. The materials collected by Allen do not contain a single occurrence of the negative purpose construction; my data contains but very few; the New Testament translation contains a great many. This could of course be due to the size of the various corpora, the New Testament translation being by far the largest of the three. But the dates are possibly also of significance, with the earlier materials showing no

[10] Aikhenvald provides examples from Dolakha Newar, Chantyal (both spoken in Nepal) and Galo.

[11] The equivalent of (8), which was provided by Rajesh Khatiwada, is found in Thulung in (18).

[12] The suffix *-os* has been called an injunctive in some grammatical descriptions of Nepali, and is used to "to express a wish or desire" (Matthews 1998: 197). The purposive construction is also described in the same section of the grammar (Matthews 1998: 199).

signs of the construction, and the more recent materials revealing its presence, suggestive of increased borrowing from Nepali, as evidenced elsewhere in the language as well.[13]

The fact that such a close mirror of the construction is found in Nepali as well raises some profound questions about the descriptions of categories that may in fact be marginal, as suggested by one reviewer. If the Thulung construction is calqued on the Nepali one, then does it deserve to be described here as part of the language's apparatus for the expression of apprehension? The fact is that all of the most important elements of the construction are widely used in Thulung: the negative optative, as well as subordination with *rwaksaka.* In that sense, even if the construction itself is calqued, the elements making it up are well attested in the language (even if they in turn are possibly the result of earlier borrowing, as may be the case for subordinator *rwaksaka*), and as such the construction reveals the language's ability to use what, at least synchronically, are "native" elements in order to express something for which a need has arisen (if we assume that the negative purpose construction is a recent phenomenon in Thulung).

As mentioned above, Thulung also has a positive purpose construction (again, with only two examples in my narrative corpus), which is symmetrical with the negative purpose construction described herein, consisting of an optative-marked clause, surbordinate by *rwaksaka* to a main clause. It is exemplified in (9):

(9) *Omsaka pho-nʉ rwaksaka bui-ɖʌ:la jak-tʉ.*
like.this vomit.3SG-OPT SUB head-above hit-3SG>3SG.PST
'So that he would vomit like this, he hit him on the head.' (in order to see what he had eaten) [Miyakma-Lakpa, 87]

The only other type of purpose clause in Thulung is one involving a motion verb,[14] a construction that only occurs in an affirmative form. There is thus only a single way to express an action carried out for negative purpose, and that is the construction described above.

[13] A third possibility is of course that the New Testament suffers from translation bias, as it was presumably based on the Nepali version of the Bible. See Wälchli (2007) for more on bias and how this affects materials, particularly parallel corpora (such as the Bible), used in linguistic studies.

[14] These are of the type:

(i) *Keʈi phʌp-ɖa lʌ:-mi.*
girl pick.up-PURP go-3PL
'They go to pick up the girl.'

2.2 Order of clauses and grammaticalization pathway

One notable characteristic of the construction is the order of clauses in the construction, which goes counter to typological tendencies: Dixon (2009: 39, 48) claims that for possible consequence clauses, many languages tend to move the focal clause (ie, the pre-emptive clause) to the first position, giving prime importance to the expression of how to avert the negative consequence.[15] In Thulung, the pre-emptive clause follows the precautioning clause, something which can be posited to come about from the development of the construction.

I propose the following stages for the development of the negative purpose construction:

1. an optative mood marker develops from the lexical verb *nʉmu* ‘be good, be auspicious’ (henceforth glossed ‘be.good’);
2. a negative optative marked clause is subordinated with *rwaksaka* and interpreted as a precautioning clause;
3. the resulting precautioning clause, coupled with a main clause, yields a negative purpose construction.

The forms that make up each stage along the pathway are still productively used.

2.2.1 Stage 1: Lexical verb *nʉmu* to optative mood marker

The lexical verb *nʉmu*, ‘be good’, is illustrated in examples (10) and (11), where it appears in finite form and in participial form, respectively. The ability to occur with derivational suffixes,[16] as in (10), and as a participle, as in (11), confirms its status as a lexical verb.

(10) *Sʉ:-ka khok-ʉ-lo wo me tsokho ḍʉ:*
who-ERG cook-3SG>3SG-TEMP even DEM ritually.pure beer
nʉ*-*le *ni.*
be.good-LIM.3SG indeed
‘Whoever cooks it, that ritually pure beer will end up good.’ [Bungarma, 135]

[15]Note, however, that Schmidtke-Bode (2009: 113 ff) shows that, while VO languages have a strong preference for a purpose clause following the main clause, the data on OV languages (of which Thulung is one) is mixed as to the ordering of clauses.

[16]The derivational suffix in (10) is used to express what I call *liminal* Aktionsart, in other words a focus on the boundaries of the action. See Lahaussois (2020a,b).

(11) *Ma:ke* ***nʉ-pa*** *khal-kam lwas-tʉ ʔe.*
grain **be.good-NPST.PTCP** type-GEN see-3SG>3SG.PST HS
'He saw a good type of grain.' (when searching for a place to establish a settlement) [Miyakma-Chandra, 115]

I propose that the verb *nʉmu*, 'be good' develops into the optative marker through the reanalysis of two independent clauses into a mood-marked clause. In (12), the two independent clauses [*gu dalaŋa sʌ:*] and [*nʉ*] are in a paratactic relationship.

(12) *Gu dala-ŋa sʌ: nʉ.*
3SG fast-INT heal.3SG be.good.3SG
'He will heal fast, that is good.' [constructed]

The two clauses in a paratactic relationship are reinterpreted as a single clause (with the prosody of the construction as it currently stands suggesting this),[17] and *nʉ* is reanalyzed as an optative mood marker, as in (13).

(13) *Gu dala-ŋa sʌ:-nʉ.*
3SG fast-INT heal.3SG-OPT
'May he heal fast.' [fieldnotes]

The fact that the 3SG non-past intransitive verb form has no overt indexation on the verb, which therefore occurs as a bare stem, results in their being no inflection marker in the space between the two finite verbs in (12). This could well have contributed to facilitating a reanalysis of *sʌ:* as a verb root and *nʉ* as a mood marker. Note that in present-day Thulung, the optative mood marker can be suffixed directly to the verb root or to an inflected verb stem, with both forms currently accepted by some speakers, as shown in (14).

(14) *Ma:ke dzəmka* ***si:-rʉ-nʉ/siʈ-nʉ*** *rwaksaka go*
grain carefully **mature-3SG>3SG.PST-OPT/mature-OPT** SUB 1SG
biruwa dzəmka khor-pu.
small.weeds carefully hoe-1SG>3SG
'So that the grain matures nicely, I hoe the small weeds with care.' [fieldnotes]

[17] Another hint is that it is written as a single word in the translation of the New Testament by native Thulung speakers.

My Thulung corpus provides a number of examples of optative mood marker -*nʉ*, many of which are found in recordings of incantations and prayers. Some additional illustrations are given in (15) and (16).

(15) *Go oram-nuŋ tseŋ-si-ŋu he: dʉm-ŋu-lo* ***dʉm-ŋu-nʉ***
1SG DEM-SOC hang-REFL-1SG how become-1SG-TEMP **become-1SG-OPT**
rwak-ta ʔe.
say-3SG.PST HS
'And he said "I will hang myself with (i.e. onto) this, and however I become, may I become like that".' (man trying to escape from cave by grabbing onto monkey tail) [Redukpa, 107]

(16) *Bu:dzi-ŋa* ***me-dʉm-nʉ*** *ottha tsiri-ŋa bo:-tsi gana banta*
shy-INT **NEG-become.3SG-OPT** here tsiri.ritual-INT do-1DU 2SG where
ba-nna taŋma du: ba-nna.
be-2SG.PST wild.banana grove be-2SG.PST
'May it not be shy, we are doing tsiriŋa here, where did you stay, did you stay in the wild banana grove?' (ritual chant in which one calls to the wild ginger to reveal itself) [Bungarma, 216]

2.2.2 Stage 2: Subordination of negative optative-marked clause

Stage 1 traced the development of the optative marker. Stage 2 is that in which an optative marked clause, and specifically a negative optative marked clause, comes to be a precautioning clause, with subordinator *rwaksaka.*

In Thulung, the subordination with *rwaksaka* of a clause with an optative marked main verb yields a precautioning clause, which must then be completed by a matrix clause. Without the subordinator *rwaksaka*, we simply have a clause in the optative, which can function independently (as in (13) or (16)); with the subordinator, the interpretation is as a dependent precautioning clause. Thus in (17), which is intentionally incomplete in order to highlight the dependent status of the clause, we have what is interpreted as a precautioning clause; this needs a preemptive clause for resolution, as in (18) below:

(17) *Dzu-kka me-si:-nʉ rwaksaka …*
cold-INSTR NEG-die.3SG-OPT SUB
'So that he not die of cold, …' [elicited]

2.2.3 Stage 3: Interpretation of precautioning clause coupled with appropriate main clause as a negative purpose construction.

In the final stage of the proposed pathway, the combination of the precautioning clause with a main clause of appropriate semantics results in a negative purpose construction, with the main clause interpreted as the pre-emptive action which can be (or has been) taken in order to avoid the undesirable outcome expressed by the precautioning clause. As mentioned above, the main clause commonly contains an impersonal deontic necessity construction, but finite verbs are also found. Example (18) takes the precautioning clause from (17) and adds a main clause, while (19) is an example from the New Testament translation.[18]

(18) *Dzu-kka me-si:-nʉ rwaksaka je phʌ-beṭ-to.*
cold-INSTR NEG-die.3SG-OPT SUB clothes wear-CAUS-1SG>3SG
'So that he does not die of cold, I dressed him.' [elicited]

(19) *Mer-mim-ka paul mesinḍa me-lʌ-mi-nʉ rwaksaka sela thʉr-so-mḍi.*
DIST.DEM-PL-ERG Paul there NEG-go-3SG.POL-OPT SUB message send-3PL>3SG.POL
'They sent Paul a message so that he would not go there.' [Acts 19:31]

A few points should be made about the negative purpose construction resulting from the combination of the precautioning and pre-emptive clauses. The order of clauses found in Thulung, with the pre-emptive clause following rather than preceding the precautioning clause, can be seen to come about naturally from the development of the construction: it is unsurprising in an SOV language to find complement clauses occurring before the main verb; clauses, such as precautioning clauses, which are marked with *rwaksaka,* follow this order as well.

Regarding the issue of 'subject' identity across clauses, Schmidtke-Bode (2009: 133) has observed in his typology of purpose clauses that negative purpose clauses have a preference for "different, ie non-coreferential subjects" (36.9% of his sample, vs. 5.8% for positive purpose clauses); this contrasts with the preference of positive purpose clauses for having the same subject' (54.2%), while negative purpose clauses only have the same subject in 10.5% of cases. Examples collected through both elicitation and the very limited sample in my narrative corpus all have different subjects across the two clauses of the negative purpose

[18]The material in (19) is converted from Devanagari into my transcription system, and I have added glosses and my own translation.

construction. Even if the arguments of the clauses are not always explicitly expressed, as in (18), they are easily recoverable through indexation on the verb.[19] The occurrences of negative purpose constructions in the New Testament corpus generally concur with this same-subject tendency, but exceptions are found, such as (20) where the subject of both clauses is 1SG, as can be seen from the person indexation on the verb:

(20) *Inima nʌ me-nɵ-beʈ-pu-nʉ rwaksaka go inima-ɖa*
2SG.POSS mind NEG-hurt-CAUS-1SG>3SG-OPT SUB 1SG 2SG.POSS-LOC
athawo tsʌkthɵ me-rok-pa mulʉ be-uto.
or nearby NEG-come-NPST.PTCP decision make-1SG>3SG.PST
'In order to not exert pain on your mind, I decided not to come to your place or thereabouts.' [2 Corinthians 2:1]

To summarize, the negative purposive construction discussed in this section is the main Thulung construction in the apprehensional domain, i.e. used to express potential undesirable events, though it is not particularly frequently found in narratives outside translations. It is formed in an analogous manner to the positive purposive construction in Thulung, and consists of a clause with a (negative) verb form marked with the optative and subordinated by the marker *rwaksaka*. I have presented evidence that the optative mood grammaticalized from a lexical verb 'be good' and the subordinator grammaticalized from an anterior converb of a quotative verb.

In the remainder of this paper, I will discuss an alternative strategy to express negative outcomes (§3), and additional functions of the optative (§4).

3 Alternative strategy to express negative outcomes

Thulung can also use an 'if not' construction to link a pre-emptive clause to a situation expressing a potential negative outcome, as shown in (21). The linker between the two independent clauses, *meʔema:la*, is a combination of a negative copula and a conditional marker.

[19]Thulung argument indexation uses two sets of indexes, one for intransitives (glossed as the index for a single argument, 'S'), and one for transitives (glossed 'A>P'). In (17), the precautioning clause has a 3SG S; the pre-emptive clause has a 1SG A (and a 3SG P), as seen from the indexation on the transitive (a causativized intransitive verb) verb *phʌbeʈto*.

(21) *Semsʌŋ dzɵmka tsir-mu ba:si meʔema:la dzhar.*
firewood carefully stack-INF must otherwise fall.3SG
'Firewood must be stacked carefully; otherwise it will fall.' [verb dictionary]

Note that undesirability is not expressed through this construction; it must be inferred through an evaluation of the consequence of the realization of the 'if not' clause.

The same state of affairs can be expressed, with no discernible semantic differences, as a conditional with a negative protasis, as in (22).

(22) *Semsʌŋ dzɵmka me-tsir-ʉ-wa-ma:la dzhar.*
firewood carefully NEG-stack-3SG>3SG-IRR-COND fall.3SG
'If he doesn't stack firewood carefully, it will fall.' [verb dictionary]

In other words, the 'if not' construction functions much like a negative conditional, with the difference that it is made up of independent clauses which are linked by *meʔema:la*. The negative consequences are expressed in the second clause of both constructions, and because they are potential (rather than realized), the clause is always non-past. Another example is provided in (23), this time with an imperative verb in the first clause.

(23) *ʌbʌ mal-ni ma reʈ-ni meʔema:la go gani-lai*
now search-2PL>3SG.IMP CONJ bring-2PL>3SG.IMP otherwise 1SG 2PL-DAT
pulis-kara lʌ:-nini.
police-LOC take-1SG>2PL
'Search and bring him, otherwise I will take you to the police.' (uttered by someone in a story whose child has been kidnapped) [Leledym, 36]

The 'if not' construction, seen in (21) and (23), is not considered the primary strategy for a few reasons: the negative purpose construction presented in §2.1 exhibits a stronger causal link between the two clauses, in addition to morphologically encoding undesirability through the negative optative. In the 'if not' construction, the second clause factually expresses a likely outcome, should the condition expresses by the first clause not be met, and it is up to the addressee to evaluate the undesirability of the consequence of disregarding expected behavior.

4 The negative impersonal deontic necessity construction

In addition to the pathway described above, whereby *nʉmu* ‘be good’ is the source for the optative mood marker, the same lexical verb has followed a separate grammaticalization pathway, resulting in a negative impersonal deontic necessity construction. The construction is illustrated in examples (24) and (25):

(24) *Hapa kʌrmʌr* ***be-si menʉ.***
lots hard.work **do-IMPER.NMLZ NEG.DEON**
‘One must not do lots of hard work [while pregnant].’ [Childbirth, 4]

(25) *Athaldika-wo malo bʌne-mma nem-gonu ʦɵtʦɵ-mim-lai*
nowadays-even just make-PASS.PTCP house-inside child-PL-DAT
hu:-si menʉ.
enter-IMPER.NMLZ NEG.DEON
‘Even nowadays, when we make a new house, it is not acceptable for children to enter.’ [Jau Khleu, 232]

The bare verb root takes the impersonal nominalizer *-si*, although occasionally we find an infinitive form of the verb in this construction in the place of the *-si* nominalized form (an infinitive can be considered another type of nominalization). Marker *-si* is assumed to be related to reflexive marker *-si*, which can be used, among other functions, to express an impersonal (see Lahaussois 2016, 2023).

The construction stands in contrast to prohibitives (in other words, negative imperatives, which are inflected for the arguments involved): unlike the latter, it is impersonal, and is used to express recommendations against certain behaviors, generally those going against cultural norms. Thulung prohibitives, on the other hand, refer to specific actions, undertaken by specific agents, rather than to impersonal statements about behavior. See the contrast between (26), the negative deontic necessity construction, and (27),[20] the prohibitive:

(26) *Moʈorsaikʌl hapa* ***onbe-si menʉ.***
motorcycle lots **drive-IMPER.NMLZ NEG.DEON**
‘One shouldn’t drive a motorcycle too fast.’

[20] Examples (26) and (27) can also be contrasted with the same sentence with a negative optative:

(i) *Moʈorsaikʌl hapa me-onbeʈ-nʉ.*
motorcyle lots NEG-drive-OPT
‘May he not drive the motorcycle too fast.’ / ‘He shouldn’t drive the motorcycle too fast.’ [elicited]

(statement expressing opposition to a certain behavior, not directed at anyone in particular) [fieldnotes]

(27) *Moṭorsaikʌl hapa* ***me-onbeṭ-ni****.*
motorcycle lots **NEG-drive-2PL>3SG.IMP**
'Don't drive the motorcycle too fast!'
(direct interdiction, aimed at a particular person in a specific instance) [fieldnotes]

The development of the negative impersonal deontic necessity construction likely involves a transition from a negated 3SG form of *nʉmu* 'be good' following a verb form nominalized with *-si*, with a compositional interpretation of 'V-ing is not auspicious', to a construction expressing negative deontic necessity:

(28) Development of the nominalized verb (INF or *-si*) + NEG-*nʉ* into negative deontic necessity:
'V-ing is not good' → 'It is not good to V' → 'One shouldn't V'

This impersonal nature of the construction is reminiscent of the directive infinitive in some SAE languages. According to Van Olmen (2010: 502), "prohibitive speech acts are more face-threatening than their positive imperative counterparts. For that reason, they also have a stronger preference for the more polite directive infinitive" which… "does not require a particular addressee, [and] has a weakly directive illocutionary profile".

One factor in support of development into a full-fledged construction is that there has been phonological change: the negative marker *me~mi-* is sometimes found in this construction to have assimilated to *mʉ-*, resulting in a marker *mʉnʉ* (and a construction V-*si mʉnʉ*). This change in the negative prefix is only found in this construction.

From the same source verb *nʉmu* 'be good', we therefore have two development pathways, resulting in an optative mood marker (which can be negative) on the one hand, and a negative deontic necessity construction on the other, as illustrated in Figure 1. It is interesting that the idea of auspiciousness should feature so prominently in constructions surrounding apprehension and deontic necessity. This ties in nicely with what has been written about Rai religious practice of *mundum*, namely that it "underlies moral values and encourages the behavior that is required to uphold the order defined by the ancestors, which includes self-control, paying attention to one another, and compliance with social order" (Schlemmer 2019).

nʉmu, 'be good'

Nominalized verb + negative finite form of 'be good'
V-*mu*/V-*si* *me-nʉ*

Impersonal negative deontic necessity
V-*mu*/V-*si* *menʉ~mʉnʉ*
'One should not V'

Parataxis of two independent clauses
V_{finite}, *nʉ*
'V, and it was good'

Optative mood V-*nʉ*
(and negative Optative *me*-V-*nʉ*)

Precautionary clause
(within Avertive construction)
me-V_{finite}/$V_{\text{stem-nʉ}}$ *rwaksaka*
'So that V doesn't happen...'

Figure 1: Two development paths from lexical verb *nʉmu*

5 Conclusion

While initial searches of the Thulung corpus revealed no morphology dedicated to expressing apprehension, in concordance with the absence of South Asian languages in descriptions of the apprehensional domain, upon closer inspection it was found that Thulung has nonetheless developed a strategy, in the form of the negative purpose construction. This construction results from the coopting of morphology with other functions, namely a negative optative verb form and a subordinator, in order to yield a precautioning clause which combines with a main clause expressing a pre-emptive action to avert the undesired outcome. This contribution has attempted to provide some empirical data on how the apprehensional domain is expressed in Thulung.

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Abbreviations

1, 2, 3	first, second, third person	NMLZ	nominalizer
ADD	additive ('also')	NPST.PTCP	non-past participle
A	agent argument	OPT	optative
ANT.CVB	anterior converb	PASS	passive
ANTIC	anticipator Aktionsart (telic)	P	patient argument
APPL	applicative	PI	plural inclusive
CAUS	causative	PL	plural
COND	conditional	POL	polite (for personal pronouns)
CONJ	conjunction		
DAT	dative	POSS	possessive
DEM	demonstrative	PROX	proximal
DIST	distal	PST	past
DU	dual	PTCP	participle
ERG	ergative	PURP	purposive
F	feminine	REFL	reflexive
GEN	genitive	SAE	Standard Average Eureopean
HS	hearsay		
IMP	imperative	S	subject of intransitive verb
IMPER.NMLZ	impersonal nominalizer		
INF	infinitive	SG	singular
INSTR	instrumental	SIM.CVB	simultaneous converb
INT	intensifier	SOC	sociative
IRR	irrealis	SUB	subordinator
LIM	liminal Aktionsart (telic)	TEMP	temporal sequencer
LOC	locative	TOP	topic
NEG	negative	V	verb
NEG.DEON	negative deontic		

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Chapter 10

The apprehensive function of a final enclitic particle in Baoding (Sinitic)

Na Song
INALCO-CRLAO

This paper aims to examine the apprehensive function of a final enclitic particle *=lɛ.ia* in the Baoding dialect, a Mandarin dialect spoken in Northern China (Sinitic). This particle is restricted to future situations and its main pragmatic function is to require the addressee's involvement in order to avoid a potential undesirable situation. It mainly but not exclusively appears in independent clauses and originally has a temporal function. In this paper, we examine the semantic and syntactic constraints on *=lɛ.ia*, its various functions, and we have a detailed look at the predicate that it co-occurs with. We argue that the apprehensive function of *=lɛ.ia* is a semantic extension of its temporal function as an imminent future marker. We also investigate the possible source of the enclitic particle: *=lɛ.ia* synchronically functions as a single unit but can be traced to two final enclitic particles: the enclitic *=lɛ* which marks change of state, and the enclitic particle *=ia* marking imminent future. These data on the Baoding dialect not only fill a gap in providing detailed data on apprehensional morphology in a broad sense from a Sinitic language, but also point to subtle semantic variations and restrictions in the grammatical expression of undesirability, and to the strong pragmatic expectation on the addressee.

1 Introduction

Languages may have different ways of encoding potential undesirable events which may be more or less grammaticalized. As a general characterization, apprehensive markers are semantically specialised markers that denote a potential event that is considered undesirable. They are also known as a modality that combines both epistemic and attitudinal components (Lichtenberk 1995: 293).

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As a category, the apprehensive has not received a lot of attention cross-linguistically, apart from a handful of studies on Austronesian languages (Lichtenberk 1995), Amazonian languages (Vuillermet 2018) and Australian languages (Dixon 2002, Angelo & Schultze-Berndt 2016: 255–296, Verstraete 2005 among others).

Cross-linguistically, apprehensive marking may take the form of an independent word/particle as in To'aba'ita (Lichtenberk 1995) or northern Australian Kriol (Angelo & Schultze-Berndt 2016), or of an affix as in Yukulta (Keen 1983: 247) or in Ese Ejja (Vuillermet 2018). In Sinitic languages, the apprehensive morpheme can be a modal adverb or a clausal linker as was described for Standard Mandarin (§2.1). In Baoding the apprehensive is primarily means of expressed by a modal enclitic particle (§2.2 and following sections).

An example of a morpheme in apprehensive function as defined above is (1). As noted by Merlan (1983: 102)), in Ngalakgan, a northern non-Pama–Nyungan language in Australia, the evitative (apprehensive) prefix *mele-* is used in an independent clause expressing some possible event which is deemed undesirable, rather than in a subordinate clause.

(1) Ngalakan (Gunwinyguan/Ngalakan: Australia; Merlan 1983: 102)
Gu-wol-nowi ***mele****-bolk.*
CLF-smoke-3SG.POSS **EVIT**-get.out
'The smoke might come out.'

Green (1989: 168) contrasts the apprehensional constructions in main vs. subordinate clauses in Marrithiyel (a language of the Daly River region of Australia's Northern Territory) by highlighting that in a main clause, the apprehensional marking expresses the speaker's view that the event is undesirable, while in subordinate clauses it expresses the subject's view.

(2) Marrithiyel (Western Daly, Australia; Green 1989: 80; 170)
*Gu-n-ning-pirr-****Ø-fang****.*
3S.R-go-1SG.O-NSG.NIRR.S-leave-3PL-**APPR**
'(I'm afraid) they might leave me.'

In (2), the apprehensive verbal suffix *-fang* appears in a main clause. Leaving may not necessarily correspond to an undesirable situation, but the apprehensive markers actually encodes that, from the speaker's point of view, the event is undesirable. A similar situation is also found in Baoding (see §3.2 and §4). Verstraete (2006: 202) finds that in almost all his examples of Australian languages the apprehensional construction can be used in a main clause, most typically in warnings.

Another topic which has recently drawn attention is the origin of apprehensional constructions. In Northern Australian Kriol, the temporal particle *bambai* 'later' coexists with a homophonous grammaticalized apprehensive particle, which displays a distinct syntactic behaviour in addition to its specialized semantics.

(3) Kriol (English-lexified creole language, Northern Australia; Angelo & Schultze-Berndt 2016: 269)

a. *Ai show yu thad lilgel* ***bambai****, Denise, yu luk la im.*
1SG show 2SG DEM girl **later** NAME 2SG look LOC/ALL 3SG
'I'll show you (=introduce you to) that girl later, Denise, you'll see her.'

b. *Ai gan lad=i yu insaid.* ***Bambai*** *yu dagat mi.*
1SG CAN.NEG let=TR 2SG inside **APPR** 2SG eat 1SG
'I won't let you in. You might eat me.'

Concerning Sinitic languages, there are only few relevant studies, which concern adverbial apprehensional markers that could be grammaticalized from 'fear' verbs in Standard Mandarin (Gao 2003). This paper provides data from Baoding, a Mandarin dialect of the Sinitic group.

Figure 1: Location of Baoding, Hebei, China(Song 2020: 27)

Baoding is located in Hebei province (China) and is about 140 km south of Beijing. According to The Language Atlas of China (Maps B 1-3, Liu 2012), the

Baoding dialect belongs to the Jì-lŭ subgroup of Mandarin (a subgroup spoken in a large part of the Hebei province and in the western part of the Shandong province). Although the precise number of Baoding speakers is unknown, we can estimate that about the elder half of the population of the city are likely to speak the Baoding dialect; due to the promotion of Standard Mandarin, younger speakers under 35 have lost their skills in the local language. Despite the geographical proximity of the Baoding dialect and the Beijing dialect, on which Standard Mandarin is based, the Baoding dialect possesses some syntactic features which are quite different from Standard Mandarin, notably in its Tense, Aspect, Modality, and Evidentiality (TAME) system.

The data presented in this article are taken from the author's personal fieldwork data, gathered during a total of five months in a period extending from 2012 to 2017, which includes elicited sentences, in addition to approximately 40 hours of daily conversation. Our language consultants are around the age of 60 (two females and one male).

In this paper, I will show that besides the adverbial apprehensional markers grammaticalized from 'fear' verbs described for Standard Mandarin (see §2), Baoding may also express apprehension through a final enclitic particle in a main clause. This particle, *=lɛ.ia*, has two main functions: a temporal function, when it encodes an imminent future event, and an apprehensive function with certain types of verbs (see §3.2), which is related to speaker vs. addressee involvement: the particle is used to require the addressee's involvement in order to avoid a potential undesirable situation as illustrated in the following example:

(4) Baoding
Pã^{213} *tɕiɛ̃*$^{213\text{-}21}$ *tiɑ˞*, *tʰiɔ*$^{213\text{-}21}$*=**lɛ.ia**.*
tie fasten a.little fall=APPR
'Tie it up a little bit, it's falling!' (Context: The speaker requires her husband to tie up the flour bag (25kg) at the back of his bicycle when she sees the bag is falling.)

This paper is structured as follows: the current section has provided a general overview of the apprehensive as a cross-linguistic category. §2 presents the apprehensive morphemes in Standard Mandarin and a similar strategy in Baoding. After introducing the TAME system and the final enclitics in Baoding, §3 focuses on the apprehensive function of *=lɛ.ia*, its distribution, and the predicates with which it co-occurs. §4 analyses the semantic contribution of *=lɛ.ia*. §5 discusses how *=lɛ.ia* interacts with person and with sentence type (polar questions). §6 proposes a hypothesis on the source of *=lɛ.ia* and tries to give an account of the semantic and pragmatic links between its temporal and apprehensive functions. §7 concludes and gives some further perspectives on the motivation for this semantic change.

2 Apprehensional constructions in Sinitic Languages

2.1 Apprehensive morphemes in Standard Mandarin

In Sinitic Languages, apprehensive morphemes have been described for Standard Mandarin by Gao (2003), with reference to Lichtenberk's 1995 study. She describes three morphemes which could have an apprehensional use: *pà*, *kàn* and *bié*, whose respective sources are a 'fear' verb, a 'see' verb, and a prohibitive marker. The following examples (5a) and (5b) illustrate the lexical and grammatical uses, respectively, of *pà* in Standard Mandarin:

(5) Standard Mandarin (Sinitic, Sino-Tibetan; Gao 2003)

a. 老鼠怕猫。
Lǎoshǔ ***pà*** *māo.*
mouse **fear** cat
'A mouse fears cats.'

b. 老没见赵老师露面，怕是叫外国人请去演讲了。
Lǎo méi jiàn Zhào lǎoshī lòumiàn, ***pà*** *shì jiào wàiguórén qǐng*
always NEG see prof.Zhao show.up **POT** be PASS foreigner invite
qù yǎnjiǎng le.
go give.a.lecture COS
'(I) have not seen Prof. Zhao put in an appearance. He might be away on an invitation by foreigners to give a lecture.'

c. 红云软软的说着，怕他不明白。
Hóngyún ruǎnruǎnde shuō, ***pà*** *tā bù míngbai.*
Hongyun gently speak **APPR** 3SG NEG understand
'Hongyun speaks gently, lest he does not understand.'

Example (5c) illustrates the apprehensional use of *pà*: the morpheme has undergone a process of decategorialization (Hopper & Traugott 1993: 104) and it is on the way of the grammaticalization into a general potential modal adverb.

The prohibitive adverb *bié* meaning 'do not do something' as in (6a) can be also employed to mark the apprehensive as shown in (6b):

(6) Standard Mandarin (Sinitic, Sino-Tibetan; Gao 2003)

a. 别着急。
Bié *zháojí.*
NEG$_{PROH}$ worry
'Don't worry.'

b. 看看去，别是一楼下水道又堵了。
Kànkan qu, ***bié*** *shì yīlóu xiàshuǐdào yòu dǔ le.*
look.RED go **APPR** be first.floor sewer again block COS
'Go and have a look. Maybe the sewer of the first floor got blocked up again.' [lit: 'Let it not be the sewer of the first floor which got blocked up again.']

The last morpheme is *kàn*, which originally means 'to see'. The source of this apprehensive marker is the complement-taking verb *kán* 'see'. Examples (7a) and (7b) illustrate, respectively, its verbal use and its apprehensive use in a threat.

(7) Standard Mandarin (Sinitic, Sino-Tibetan; Gao 2003)

a. 我看他拿了几本书走了。
Wǒ ***kàn*** *tā ná le jǐ běn shū zǒu le.*
1SG **see** 3SG take PFV several CLF book leave COS
'I saw him take several books and leave.'

b. 再麻烦，看我带你到局里去!
Zài máfan, ***kàn*** *wǒ dài nǐ dào jú li qu!*
again bother **APPR** 1SG take 2SG go.to police.station LOC GO&DO
'If (you) bother (me) again, I will take you to the police station!' [lit.: 'You'll see that I will take you to the police station.']

Similar apprehensive markers derived from 'fear' verbs, 'see' verbs and a prohibitive marker are also attested in Baoding, a non-standard variety of Mandarin spoken in Northern China.

2.2 A standard Mandarin-like strategy of apprehensive morphemes in Baoding

Baoding has two strategies for expressing apprehension. Firstly, Baoding may use markers grammaticalized from complement-taking verbs such as 'to fear', 'to see' and from the prohibitive morpheme, similarly to Standard Mandarin, as illustrated in (8a)–(8b).

(8) Baoding

a. *Tʰa*^{45} ***pʰa***^{51} *tʰa*^{45} *pa*^{51}*.*
3SG **fear** 3SG father
'She is afraid of her father.'

b. *Uɤ*^{213} *te*^{51} *tso*^{51} *ti* *pɔ*^{22} ***pʰa***^{51} *ni*^{213} *tʂʰʅ*$^{45\text{-}35}$*-tʂo ɕiæ̃*^{22}*.*
1SG deliberately do PAST tasteless **APPR** 2SG eat-STA salty
'I did (the meal) with less salt deliberately lest you find it too salty.' [lit.: 'you eat it salty.']

c. *ɕiæ̃*^{45} *tɕʰi*$^{51\text{-}45}$ *kɤ* *tiæ̃*^{51}*xua*^{51}*,* ***pʰa***^{51} *tha*^{45} *pɛ*^{22} *pʰɔ*^{213} *i*$^{45\text{-}35}$ *tʰɑ̃.*
first make CLF phone.call **APPR** 3SG in.vain run one trip
'Give (him) a phone call first, lest he goes there in vain (lest he has a wasted journey).'

As mentioned earlier, the verb *pà* or *kongpa* 'fear' in Standard Mandarin is grammaticalizing into a general potential modal adverb and is not a dedicated apprehensional marker. By contrast, grammaticalized *pʰa*^{51} in (8b–8c), expresses both probability and undesirability, and functions as a dedicated apprehensional, namely a precautioning ('lest') clause.

The examples in (9) parallel the Standard Mandarin examples in (6). In example (9b), *piɛ*^{51} is no longer a prohibitive adverb as in (9a). Instead, it is used to express the speaker's wish or hope that an undesirable situation will not turn out to be true. In the same way, it concerns the speaker's fear that the undesirable event will take place.

(9) Baoding

a. ***Piɛ***^{51} *lɔ*^{213} *nɔ*^{22}*iɑ˞*$^{213\text{-}21}$*=lɛ.*
NEG$_{\textbf{PROH}}$ always stay.late.at.night=COS
'Don't always stay late at night.'

b. *Tsæ̃*^{22}*mɛ̃ mɑ̃*^{22}*xuɤ-lɛ* *pæ̃*^{51}*tʰiæ̃*^{45} ***piɛ***^{51} *io*^{51} *pu*^{45} *tɕʰi*$^{51\text{-}45}$*=lo.*
1INCL be.busy.with long.time **APPR** again NEG go=IRR
'We were busy with it for a long time. (I hope that it) won't be cancelled again.' [lit.: 'Let it not be the case that we don't go again.'] (Context: the two sisters have prepared their travel for a long time, however, the news announces storms in the next days. One of the sisters is getting worried.)

In a broad sense, *piɛ*^{51} in example (9b) has a 'negative optative' interpretation, that is, 'hope that it is not the case…'. The key factor for this interpretation is the semantic feature of ‘control’: if the event is non-controllable, it means that the speaker worries about the potential occurrence of the undesirable event. If it is controllable, the marker has a prohibitive interpretation. The semantic interpretation of prohibitive vs. negative optative thus depends on whether the event is controllable or not.

Like Standard Mandarin, Baoding can also employ *khæ̃*^{51} meaning ‘to see’ to express an apprehensional function (in the broad sense). Example (10a) illustrates the use of *khæ̃*^{51} as a main verb, and (10b) illustrates its use in a threat:

(10) Baoding

a. *Uɤ*^{213} *tsɛ*^{51} ***kʰæ̃***^{51} *xuɤ˞*^{213} *tɕio*^{51} *ʂue*^{51}.
1SG still **see**/read moment then sleep
‘I will do some more reading before going to sleep.’

b. *Tsɛ*^{51} *pu*^{45} *lɔ*$^{213\text{-}21}$*ʂʅ* ***kʰæ̃***^{51} *uɤ*^{213} *tsɛ̃*^{213}*mɤ* *ʂo*$^{45\text{-}35}$*ʂʅ* *ni*^{213}*!*
continue NEG well-behaved **APPR** 1SG how punish 2SG
‘If you continue to be not well-behaved, (you will see) how I punish you!’

This function differs from the apprehensional-epistemic function and the precautionary function identified in Lichtenberk (1995), and may be termed a threat function (see Vuillermet 2018: 273). That is, if a condition is met, then the speaker will take a certain action that will be undesirable for the addressee. In that case, that action is usually controllable by the speaker.

Similar cases of apprehensional morphemes originating in morphemes belonging to comparable semantic domains, such as ‘see’, ‘look at’ or ‘watch’, ‘fear’ and prohibitive markers, are attested in some Australian languages and some languages in Austronesian family (as well as in classical Greek for the prohibitive marker, see Lichtenberk 1995). The link between prohibitive and apprehensive is also confirmed by Dobrushina (2006: 50–63), Pakendorf & Schalley (2007), and Smith-Dennis (2021), as well as by several contributions in this volume (Lahaussois, Lahaussois, Lahaussois).

In the following sections, we focus on another strategy of marking the apprehensive attested in Baoding: a final enclitic particle *=lɛ.ia*, which is related to speaker vs. addressee involvement.

3 The apprehensive enclitic particle *=lɛ.ia* in Baoding and its functions

In my corpus of Baoding, the most common apprehensional strategy is the enclitic particle *=lɛ.ia* in clause final position, as shown in Table 1.

Table 1: Apprehensive expressions in Baoding

	Occurrence	Proportion
=lɛ.ia	40	81.63%
Other expressions	9	18.37%
Total	49	100%

3.1 Final enclitic particles in Baoding

Baoding, as many other Sinitic languages, possesses a large inventory of sentence-final or clause-final enclitic particles. These particles are often called *yuqici* 'mood/modal particles' in Chinese. They are referred to in English as "final particles" (Chao 1968: 149), "modal particles" (Chappell 1991), "utterance particles" (Luke 1990) or "sentence-final particles" (Li & Thompson 1981: 238). Some of these particles are considered discourse markers which do not carry any semantic content, and whose function is to convey a speaker's attitude or other pragmatic information in a certain discourse context. However, the Baoding final enclitic particles also encode various TAME categories.

Before examining in detail the enclitic particle *=lɛ.ia*, we give below an overview of final enclitic particles in Baoding (Song 2020). Final enclitic particles are organized in three distinct paradigms. The first paradigm, situated at the position which is closest to the VP, includes two particles of associated motion. The second paradigm contains particles which are relevant to the TAME system. The third paradigm, in outermost position, is composed of particles that relate to illocutionary notions or to sentence types such as interrogative particles, as illustrated in example (11).

(11) *Ni*^{213} *kæ̃*^{51}*ma*$^{213\text{-}21}$***=tɕʰi=lɛtʂ̩o=ɛ****?* *tsɤ̃*$^{51\text{-}45}$*mɤ pæ̃*^{51}*tiæ̃*^{45}.
2SG do.what=GO&DO=IPFV$_{\text{PAST}}$=INT$_{\text{WH}}$ such long.time
'What have you been doing for such a long time (away from here)?'

In (11), the enclitic particle *=tɕʰi* conveys the function of associated motion (motion-cum-purpose), and is followed by the enclitic particle *=lɛtʂo* which denotes an ongoing action in the past or a past state. The final interrogative particle *=ɛ* typically appears in a wh-interrogative. These particles appear in a fixed order (slot 1, slot 2 and slot 3).

The apprehensive enclitic particle *=lɛ.ia* is found in the second slot. In example (12), *=lɛ.ia* is coalesced with the polar question particle *=ã*, which belongs to the 3rd paradigm. *=lɛ.iã* is the result of the coalescent of *=lɛ.ia* and the interrogative *=ã*, marking unexpectedness on part of the speaker.

(12) *Ni*^{213} *tso*$^{213\text{-}21}$*=**lɛ.iã**?*
2SG leave=**APPR**.INT$_{\text{P.UNEX}}$
'Are you leaving already?' (Context: the speaker sees the addressee is packing his stuff and the speaker is surprised at his leaving, suggesting e.g. 'Are you leaving without finishing your work?')

Note that all the particles shown in Table 2 below are unstressed, i.e. they do not bear any contrastive tone. Furthermore, they trigger a tone sandhi on the preceding syllable. These prosodic features are typical of grammatical morphemes in Baoding.

3.2 The apprehensive enclitic particle *=lɛ.ia*

Apprehensive is mainly expressed in Baoding through the final enclitic particle *=lɛ.ia*. Its main function in terms of frequency is to solicit the addressee's involvement in order to avoid a potential undesirable situation. The enclitic particle *=lɛ.ia* may indicate that the possible situation is unpleasant from the speaker's point of view as illustrated in (13) below, a sentence uttered to warn the addressee that the bag of flour on the back of his bike is about to fall down.

(13) a. *Tʰiɔ*$^{213\text{-}21}$*=**lɛ.ia**, tʰiɔ*$^{213\text{-}21}$*=**lɛ.ia** …*
fall=**APPR** fall=**APPR**
'It's falling!'

b. *Pã*^{213} *tɕiɛ̃*$^{213\text{-}21}$ *tiaʶ, tʰiɔ*$^{213\text{-}21}$*=**lɛ.ia**.*
tie fasten a.little fall=**APPR**
'Tie it up a little bit, it's falling!'

In (13a) the imminent event of 'falling down' involves a high degree of certainty of occurring (unless someone does something about it) in the speaker's view

Table 2: Final enclitic particles in Baoding

Paradigm 1: Associated motion		Paradigm 2: TAME		Paradigm 3: Illocution	
tɕʰi GO&DO	Itive	*ni* CONT	Progressive aspect: ongoing action or state	*ā* $\text{INT}_{\text{P.UNEX}}$	Polar question: asking for confirmation of an unexpected event, mirative
lɛ COME&DO	Ventive	*lɛ/lɛtʂo* $\text{IPFV}_{\text{PAST}}$	Ongoing action or state in the past	*ɛ* INT_{WH}	WH-question
		lɛ COS	Change of state	*pɛ* $\text{INT}_{\text{P.N}}$	Neutral polar question
		ia EGO	Imminent event (intention)	*pɛ* OBV	Obviousness
		=lɛ.ia **APPR**	Imminent event (undesirable)	*pa* $\text{INT}_{\text{P.CONF}}$	Polar question: expectation of confirmation
		tʂo PRI	Prioritive: do X first before the beginning of doing Y	*pa* ADVI	Advisative
				ni CTR.ASR	Counter assertion

and undesirability. With regard to the semantics of the apprehensive marker, the event denoted by the verb of the clause ended by *=lɛ.ia* is always the apprehension-causing situation; the precautionary situation ('tie it up a little bit') is not expressed as part of the same clause as *=lɛ.ia*. The enclitic particle *=le.ia* requests the addressee to do something. This can be expressed overtly, as in (13b), or this needs to be inferred, as in (14).

(14) *ɕia*^{51}*y*$^{213\text{-}42}$***=lɛ.ia***.
rain-APPR
'It is going to rain.' (If you don't hurry up, you might get wet.)

One of the main questions discussed in the literature is whether constructions such as (13b) should be considered complex sentences. Even though a pre-emptive clause can be added, as in example (13b), there is no reason to consider this as a complex sentence, since both clauses can appear independently. As claimed by Verstraete (2006: 222) based on a sample of twenty Australian languages, in constructions without relation markers, intra-clausal markers such as mood markers do not encode the complex sentence relation but contribute to a pragmatic interpretation of coherence between the clauses. Similarly, *=lɛ.ia* in Baoding does not encode a complex sentence relation, and it only occurs in main clauses.

In addition to the apprehensive function, *=lɛ.ia* also has a temporal function. The temporal function is observed when it co-occurs with activity verbs: it encodes an imminent future event. For instance:

(15) *Tʰa*^{45} *tsuɤ*^{51} *kuɤ*$^{45\text{-}35}$*=ni* *tʂu*^{213} *tɕiɔ*$^{213\text{-}42}$*tsɿ****=lɛ.ia***.
3SG put cooking.pot=CONT boil ravioli=$\mathrm{FUT}_{\mathrm{IMM}}$
'She is putting the cooking pot on the fire. (She) is going to cook some ravioli.'

In the example above, *=lɛ.ia* functions as an imminent future marker without any evaluative force of apprehension. The enclitic particle *=lɛ.ia* only indicates that the event of 'cooking some ravioli' will occur in an imminent future after the utterance time. The apprehensive function is a semantic extension of the temporal one; this issue will be further discussed in §6.2.

One may think that in (13) the apprehensive interpretation is the result of the inherently negative semantics of the verb 'fall', rather than of the use of *=lɛ.ia*. Nevertheless, we argue that it is the enclitic particle *=lɛ.ia* that conveys the undesirability: the aspectual type of predicate generates differences in interpretation, and an interpretation of undesirability with semantically neutral events is possible (see §4).

The aspectual type of predicate is one important parameter to distinguish the two functions of *=lɛ.ia*. Predicates that co-occur with *=lɛ.ia* can be activity verbs, change of state verbs, or some types of stative predicates such as nominal predicates. When the predicate is an activity verb, a temporal interpretation and an apprehensive interpretation are both possible, as in (15) and (16). The difference of interpretation is then triggered by the grammatical person, that is, first-person subjects trigger the apprehensive reading. This issue will be discussed in detail in §5.1.

(16) *Uɤ*^{213} *tʂu*^{213} *tɕiɔ*$^{213\text{-}42}$*tsɿ=**lɛ.ia**, ni*^{213} *io*^{51}*ʂʅ*^{51} *pu*^{45} *xue*^{22}*-lɛ* *tʂʰʅ*^{45}.
1SG boil ravioli=**APPR** 2SG if NEG come-DIR$_{CTP}$ eat
'If you don't come back for dinner, I am going to cook some ravioli.' (If you don't come back for dinner, then I am going to cook only the ravioli, there will be no other dishes.) (Context: the speaker talks to her daughter on the phone. She implicitly requests that her daughter has dinner at home.)

Through this sentence, the speaker is warning the addressee about an imminent undesirable event. Note that the event of cooking ravioli, is not intrinsically undesirable. However, it is uttered in order to trigger a feeling of guilt in the addressee in (16): because of her absence for dinner, the speaker will only have ravioli for dinner instead of a more substantial meal. The speaker requires the addressee's involvement (i.e., of not being absent) to avoid the undesirable situation, which is, cooking only ravioli for dinner. Note that the enclitic particle *=lɛ.ia* in Baoding shares some similarities with the particle *na* in Ainu (see Tamura 2000: 176-177) in terms of requesting some actions from the addressee as shown in example (17).

(17) Ainu (Isolate; Tamura 2000: 176)
Ape us na.
fire extinguish SGST
'The fire went out (so relight it).'

With change of state verbs, *=le.ia* systematically expresses undesirability. In example (18), we see that change of state verbs are compatible with the final enclitic particle *=lɛ*, whose presence denotes the completeness of the change of state.

(18) *Tʂo*^{45} *ɕi*$^{51\text{-}45}$*=lɛ.*
porridge run.over=COS
'The porridge has boiled over.'

In fact, predicates that are compatible with the final enclitic particle *=lɛ* are also compatible with *=lɛ.ia*, and systematically trigger an apprehensive interpretation. (The issues of how *=lɛ* relates to *=lɛ.ia* and is even a possible source is discussed in §6.1.)

(19) *Kuæ̃*$^{45\text{-}35}$*-lo!* *kuæ̃*$^{45\text{-}35}$*-lo!* *ɕi*$^{51\text{-}45}$***=lɛ.ia.***
turn.off-COMPL turn.off-COMPL run.over=**APPR**
'Turn it off! It is going to run over.' (Context: the speaker asks her husband to turn off the fire, the porridge of millet is going to run over.)

With nominal predicates denoting a time reference of the day or special events or festivals on the calendar, such as *sæ̃*^{45}*ʂʅ*^{22} 'thirty' (20), *ʂʅ*^{22}*ɤ˞*^{51}*tiæ̃*$^{213\text{-}21}$ '12am' (21), or *sæ̃*^{45}*ʂɤ˞*^{22} 'eve of the Chinese New Year' (22), the only possible interpretation is apprehensive.

(20) *Sæ̃*^{45}*ʂʅ*^{22}***=lɛ.ia.***
thirty=**APPR**
'Beware you are going to be 30 years old.' (If you don't get married, it will be too late!) (Context: in this region, women are expected to be married before 30 / implied by the speaker: you will soon not be able to meet someone to get married.)

(21) *ʂʅ*^{22}*ɤ˞*^{51} *tiæ̃*$^{213\text{-}21}$***=lɛ.ia.***
twelve o'clock=**APPR**
'It will be 12am.' (If you don't prepare lunch, it will be too late.)

(22) *Sæ̃*^{45}*ʂɤ˞*^{22}***=lɛ.ia,*** *xɛ*^{22} *me*^{22} *mɛ*$^{213\text{-}21}$*-ʂã piɔ*$^{51\text{-}45}$*.*
Chinese.New.Year's.Eve=**APPR** yet NEG buy-RES ticket
'It will soon be Chinese New Year's Eve, (I) have not bought the train ticket (back home) yet.'

Example (22) seems to be an apparent counterexample to the overall claim that they solicit the involvement of the addressee. Here the addressee and the speaker are the same person: the speaker is talking to herself and solicits her own involvement.

Table 3 below shows the distribution of *=lɛ.ia* collocating with different types of predicates in my corpus. Occurrences with activity verbs are infrequent: there are only six examples of this type, but as many as five of them do not have an apprehensional interpretation. This suggests that the occurrence with activity verbs, allowing for the most non-apprehensional interpretation, is much rarer.

Change of state verbs and nominal predicates are probably the bridging context in Baoding that allow for the semantic reanalysis of *=lɛ.ia* from an imminent future meaning to an apprehensive meaning.

Table 3: The distribution of *=lɛ.ia* collocating with different types of predicates in my 40 hour corpus.

=lɛ.ia collocating with …	Occurrence	Proportion
… activity verbs	6	13.33%
… change of state verbs	30	66.67%
… nominal predicates	9	20%
Total	45	100%

The meaning added by *=lɛ.ia* can also be described through a comparison with the egophoric enclitic particle *=ia*, used to mark a future event in (23a). Egophoricity is related to personal knowledge, experience or intention of the speaker. Baoding does not allow the speaker to talk directly about the internal state of others, such as their intentions, on the account of the fact that these are not directly perceivable to the speaker (see Song 2019).

(23) a. *Uɤ*^{213} *ʂɑ̃*^{51} *pæ̃*^{51}*kõ*^{45}*ʂʅ*$^{213\text{-}21}$*=tɕʰi=**lɛ.ia***.
1SG go.to office=GO&DO=**APPR**
'I am going to the office (I tell you in case you need me).' (Context: the speaker is going to his office, before leaving, he talks to his wife.)

b. *Uɤ*^{213} *ʂɑ̃*^{51} *pæ̃*^{51}*kõ*^{45}*ʂʅ*$^{213\text{-}21}$*=tɕʰ**ia***.
1SG go.to office=GO&DO.**EGO**
'I am going to the office.' (lit. 'I am telling you that I am going to the office.'

In certain cases, the addressee is meant to do something to prevent the apprehension-causing situation or mitigate for it. The example (23a), with *=lɛ.ia*, implies a kind reminder 'beware': the speaker thinks that his leaving may be undesirable, because the addressee might need his presence. In other words, he calls for a reaction of the addressee on his leaving, for instance by asking him to not leave. By contrast, the speaker using the particle *=ia* in (23b) only informs the addressee about his intention of leaving, without expected it would cause inconvenience to his addressee. Example (23b) has a neutral imminent future interpretation, while (23a) does not. But it is the first-person subject which triggers the apprehensive reading with *=le.ia*. As will be discussed in §5.1, person is thus the second crucial factor in determining whether a sentence with *=le.ia* has an imminent future reading (just as the particle *=ia*) or an apprehensive reading.

4 Apprehensive particle *=lɛ.ia* and relevant semantic features

This section analyses three main semantic features of *=lɛ.ia* which support the analysis of *=lɛ.ia* as an apprehensive marker: UNDESIRABILITY, INFERENTIAL EVIDENCE and IMMINENCE. The first and most important is undesirability. By adding *=lɛ.ia*, even a semantically neutral or positive event can be presented as undesirable from the speaker's point of view.

(24) *ʂɣ̃45-35*=***lɛ.ia****!* *ʂɣ̃45-35*=***lɛ.ia****!* *Ni213* *xɛ22* *lɣ̃51-45-tʂo*
give.birth.to=**APPR** give.birth.to=**APPR** 2SG still be.in.a.daze-STA
ʂɣ̃213-21 *mɛ=niɛ.*
what=CONT.INT$_{WH}$
'(She) is giving birth to a child! How come are you still in a daze?!'

In example (25a), the speaker accesses visually some evidence on the information of 'giving birth to a child' and urges the addressee to react. The way of access to the information may also influence the use of *=lɛ.ia*; this issue will be discussed below. The event 'give birth to a child' is typically desirable. However, by adding *=lɛ.ia*, the speaker conveys that (from their point of view) the consequence of the addressee not getting involved would be undesirable. That is, if the addressee does not get involved, then the person giving birth will soon be in danger. By contrast, if 'giving birth to a child' is expected and considered good news, then *=lɛ.ia* is infelicitous (25a). One has to use an adverb is used instead and the enclitic particle *=lɛ*, indicating a completed change of state (25b).

(25) a. *#ʂɣ̃45-35*=***lɛ.ia****!*
give.birth.to=**APPR**
'(She) is giving birth to a child!' [as good news]

b. *Kʰɣ213* *kɛ45* *ʂɣ̃45-35=lɛ!*
finally should give.birth.to=COS
'Finally (she) is giving birth to a child!'

Likewise, food getting cooked is a typically positive event. However, the context of (26) is the preparation of a dish where mung bean sprouts must be only briefly soaked in boiling water before being taken out. If left too long in hot water, they will be overdone and no longer crunchy. Thus 'being cooked' is undesirable in this case. The speaker requires the involvement of the addressee, 'to take the mung bean sprout off the fire' to avoid the beans to be (over)cooked.

(26) *ʂo*^{22}*=**lɛ.ia**!* *Kæ̃*^{213}*tɕiɛ̃*^{213} *tuæ̃*^{45}*-lo!*
cooked=**APPR** quickly take.with.both.hands-COMPL
'(The mung bean sprouts) are going to be cooked. Take them off quickly!'

The second semantic feature of *=lɛ.ia* which is relevant to its apprehensive meaning is inferential evidence, that is, the use of *=lɛ.ia* is sensitive to information source. *=lɛ.ia* marks that the information on the imminent event was acquired from inferred evidence. In (25a), the apprehensive *=lɛ.ia* conveys inferential evidence which is not entailed by the imminent future semantics. By contrast, *=lɛ* in (25b) marks that they have been informed by a nurse, rather than inferred the event.

The literature on inferred evidentiality further distinguishes two subcategories, that is, the distinction between the inference made on the basis of sensory evidence and the inference made on the basis of assumption or general knowledge (see Aikhenvald 2004: 64, 174–176). In the case of Baoding, *=lɛ.ia* can be used with both types of inferential evidence. The first type of inferential evidence concerns an inference made on the basis of sensory evidence, i.e. whether the speaker saw circumstantial evidence or has previous knowledge on an actual event or state. In the following examples, the use of *=lɛ.ia* implies that the speaker's information is inferred by visual/auditory/olfactory evidence: the vibration of the pot cover in (27) (repeated from 19), the dark clouds in (28) (repeated from 14), the pipe of the whistling kettle as in (29), or even a smell of burning as in (30). In this sense, *=lɛ.ia* may be analysed as an inferential evidential.

(27) *Kuæ̃*$^{45\text{-}35}$*-lo!* *Kuæ̃*$^{45\text{-}35}$*-lo!* *ɕi*$^{51\text{-}45}$*=**lɛ.ia**.*
turn.off-COMPL turn.off-COMPL run.over=**APPR**
'Turn it off! It is going to run over.'

(28) *ɕia*^{51}*y*$^{213\text{-}42}$*=**lɛ.ia**.*
rain-**APPR**
'It is going to rain.' (If you don't hurry up, you might get wet.)

(29) *Kʰɛ$^{45-35}$=**lɛ.ia**, kʰɛ$^{45-35}$=**lɛ.ia**, tuæ̃^{45}xu^{22}=tɕi=pa.*
boil=APPR boil=APPR take.with.both.hands.kettle=GO&DO=ADVI
'(The water) is going to boil! Go and take it off the fire. (Otherwise it is going to overflow.)' (Context: the speaker hears the pipe of the whistling kettle in the kitchen, but she cannot see it. She asks her husband to take it off the fire.)

(30) *Pɔ^{45}kuɤ$^{45-35}$=**lɛ.ia**, xuɤ$^{213-21}$ʂã tsuɤ$^{51-45}$-tʂo ʂɤ̃$^{213-21}$mɤ=niɛ?*
burn.cooking.pot=APPR fire.above put-STA what=CONT.INT$_{WH}$
'The (food in the) cooking pot is going to burn up. (I can smell it). What is on the fire?'

The second type of inferential evidence conditioning the occurrence of *=lɛ.ia* concerns evidence inferred through assumption based on general knowledge or based on the speaker's experience with similar situations. In the following examples, the evidence comes from the speaker's knowledge about an actual event or state. In (31a) (repeated from 20), the speaker is the mother of the addressee and knows her age. Similarly, in (31b) (repeated from 21), the speaker knows the actual state, that is, the present time.

(31) a. *Sæ̃^{45}ʂʅ^{22}=**lɛ.ia**.*
thirty=APPR
'Beware you are going to be 30 years old.' (If you don't get married, it will be too late!) (Context: in this region, women are expected to be married before 30 / implied by the speaker: you will soon not be able to meet someone to get married.)
b. *ʂʅ^{22}ɤ˞^{51} tiæ̃$^{213-21}$=**lɛ.ia**.*
twelve o'clock=APPR
'It will be 12am.' (If you don't prepare lunch, it will be too late.)

With an unpredictable event, an utterance with apprehensive *=lɛ.ia* in addition requires an epistemic downtoner such as the epistemic modal adverbs *kɛ^{45}* 'might', as shown in (32):

(32) *Kʰɤ^{45} piɛ^{51} ʂã^{51} xe^{51} kɤ tsɔ^{213}ʂʅɤ˞$^{51-45}$=tɕʰi, ni^{213}*
absolutely NEG$_{PROH}$ go.to DEM$_{DIST}$ CLF morning.market=GO&DO 2SG
*tɕʰiæ̃^{22}pɔ˞^{45} (**kɛ^{45}**) tio$^{45-35}$=**lɛ.ia**.*
wallet might steal=APPR
'Don't go to that morning market, or your wallet might be stolen.'

In (32), the speaker has neither sensory or any other tangible base for the inference, nor any evidence about an actual event or state. The inference is made on the basis of the speaker's experience with similar situations: 'go to that market' could be the cause of 'having the wallet stolen'.

The third semantic feature relevant to the apprehensive meaning is concerned with imminence. The particle *=lɛ.ia* cannot be used with general future time reference. Example (33) is ungrammatical due to the fact that *miɣ̃˞²²* 'tomorrow' is not compatible with the semantics of an imminent future.

(33) *Miɣ̃˞²²* *ɕia⁵¹y²¹³⁻⁴²=**lɛ.ia**.*
tomorrow rain=APPR
'It will/may rain tomorrow.'[1]

To summarize, the semantic features of the enclitic particle *=lɛ.ia* are [+UNDESIRABLE],[2] [+INFERENTIAL EVIDENCE] and [+IMMINENCE].

5 Interactions of *=lɛ.ia* with other grammatical domains

In this section, we provide a brief account of how *=lɛ.ia* interacts with person and speech acts, and look into the way it is interpreted in interrogative sentences, more precisely in polar questions.

5.1 The role of person

Besides aspectual verbal class, person is another important factor that triggers the apprehensive reading of a sentence with *=lɛ.ia*. We saw that *=lɛ.ia* may encode imminent future without conveying any apprehension with activity verbs. But if they occur with a first-person/first-person plural exclusive[3] or second-person subject, then *=lɛ.ia* always conveys apprehension, as shown in example (34):

(34) *Tsʰɛ²² tɕi²¹³* *tiæ̃²¹³* *iɛ?* *Ni²¹³* *tʂʰʅ⁴⁵fæ̃⁵¹⁻⁴⁵=**lɛ.iã**?*
only how.many o'clock INT_{WH} 2SG have.meal=APPR.$\text{INT}_{\text{P.UNEX}}$
'What time is it? You are already going to have your meal?'

[1] To express 'It will/may rain tomorrow', Baoding resorts to adverbs denoting epistemic modality such as *pa⁴⁵tʂʰɣ̃˞²²ʂʅ⁵¹* / *pɔ²¹³puʐuɛ̃²¹³* 'maybe'.

[2] *=lɛ.ia* has the feature [+UNDESIRABLE] with change of state verbs and nominal predicates, but not systematically with activity predicates.

[3] First person inclusive is not compatible with *=lɛ.ia*.

(Context: the speaker sees the addressee is taking out her lunch box and the speaker is surprised that the addressee does not wait for others and have lunch together.)

The verb *tʂʰʅ^{45}fæ̃^{51}* ‘to have a meal’ is an activity verb, but because it occurs with the second-person subject, the only possible interpretation is that of an apprehensive. The speaker suggests that having a meal now is undesirable by the addressee. The speaker implicitly asks the addressee to delay her meal, in order to avoid possible negative consequences. With a 2nd person subject, the apprehensional clitic often occurs in interrogatives. In this case its function is to question the addressee’s awareness of a potential undesirable event (undesirable for the addressee or caused by them).

In (35) (repeated from 16), *tʂu^{213}* ‘to boil’ is an activity verb which occurs with a first person subject. This eliminates the purely temporal reading and the only possible interpretation is apprehensive. The reading with 1st person subject can be considered as conveying emotional blackmail.

(35) *Uɤ^{213} tʂu^{213} tɕiɔ$^{213\text{-}42}$tsɿ=**lɛ.ia**, ni^{213} io^{51}ʂʅ^{51} pu^{45} xue^{22}-lɛ tʂʰʅ^{45}.*
1SG boil ravioli=**APPR** 2SG if NEG come-DIR$_{\text{CTP}}$ eat
‘If you don’t come back for dinner, I am going to cook some ravioli.’ (If you don’t come back for dinner, then I am going to cook only the ravioli, there will be no other dishes.) (Context: the speaker talks to her daughter on the phone. She implicitly requests that her daughter has dinner at home.).

As for third person-subject sentences, both the temporal reading (36) (repeated from 15) and the apprehensive (37) reading of *=lɛ.ia* are possible with activity verbs. For instance:

(36) *Tʰa^{45} tsuɤ^{51} kuɤ$^{45\text{-}35}$=ni tʂu^{213} tɕiɔ$^{213\text{-}42}$tsɿ=**lɛ.ia**.*
3SG put cooking.pot=CONT boil ravioli=**FUT$_{\text{IMM}}$**
‘She is putting the cooking pot on the fire. (She) is going to cook some ravioli.’

(37) *Tʰa^{45} tʂu^{213} tɕiɔ$^{213\text{-}42}$tsɿ=**lɛ.ia**, kʰuɛ^{51} kʰæ̃^{51}kʰæ̃$^{51\text{-}45}$=tɕʰi.*
3SG boil ravioli=**APPR** quick look.RED=GO&DO
‘She (Grandma) is going to cook ravioli. Go and have a look quickly!’ (Context: Grandma is very old and may hurt herself in the cooking process.)

The distinction between the two interpretations in the above examples is not clear-cut. Note that in a third-person subject sentence an imminent event cannot be expressed by the egophoric enclitic particle *=ia* which encodes an intention. The reason for this is that one cannot anticipate a third person's intention (an instance of the 'anticipation rule' discussed in Tournadre & LaPolla 2014). However, there is an exception when the third person is closely associated with the speaker, i.e. belongs to the personal sphere of the speaker. Example (38) illustrates the use of *=ia* in a third-person subject sentence:

(38) **A:** *Tʰa^{45} tʂʰu$^{45\text{-}35}$-lɛ=lɛ=pɛ?*
3SG exit-DIR$_{\text{CTP}}$=COS=INT$_{\text{P.N}}$
'Has she already gone out?'

B: *Me^{22}=ni, tʰa^{45} tʂə^{51}tɕjo^{51} tʂʰu^{45}tɕʰ**ia**.*
NEG=CTR.ASR 3SG soon exit.go.**EGO**
'Not yet, she is going out soon.'

The context of the example (38) is that A is talking on the phone with B and asking her if her daughter is going out or not. As the third person in question here is B's daughter, their relationship is close enough to be considered as belonging to B's domain, the speaker B is therefore allowed to use the egophoric marker *=ia* as if she were talking about her own intention. However, this situation is relatively rare. Usually there is no opposition in third-person subject contexts as we found for first-person subject sentences in (23a) and (23b) (see the above §3.2). Unlike with a first-person subject, it is possible for *=lɛ.ia* to have both a temporal function and an apprehensive function in the case of a third-person subject.

In sum, two parameters determine whether *=lɛ.ia* has a temporal or apprehensive function: the verbal class, and the person of the subject of the sentence.

As claimed by Aikhenvald (2010: 225–226), the apprehensive may be considered as a special form of command. Table 4 lists the occurrences of *=lɛ.ia* conveying undesirability with different grammatical subjects. It shows that in Baoding, the form most frequently occurs with an impersonal subject in uncontrollable events. This conforms quite well to the semantics of an apprehensive: such situations are frequently apprehension-causing.

No matter which person is the subject, sentences with *=lɛ.ia* in its apprehensive function always convey an implicit request on the part of the speaker. In terms of pragmatic function, the speaker always expects the addressee to react in order to confront the undesirable situation or to sometimes prevent its undesirable consequences. The speaker warns or reminds the addressee about the possibility of an undesirable event from his point of view which is going to occur.

Table 4: *=lɛ.ia* and person

	Occurrence	Proportion
First person subject	2	4.54%
Second person subject	10	22.72%
Third person subject	8	18.18%
Impersonal subject (in uncontrollable events)	24	54.54%
Total	44	100%

In (14), it is about to rain so the addressee should hurry or cover himself lest he get wet; in (26), (30), (29) the food will be overcooked, burned, or the water will run over, so the addressee should act to prevent these events to happen.

Even though when the event is undesirable for the speaker rather than for the addressee, as shown in (16), the implied meaning of the speaker is still that (s)he expects the addressee 'to come back for dinner' so that the speaker can avoid the undesirable effect of the event 'cooking only the ravioli' rather than a proper meal. It is about addressee's involvement. To some extent, this is in accordance with Aikhenvald's point of view in that utterances with *=lɛ.ia* could be considered as a special form of command. More details concerning this issue will be provided in the following section.

5.2 The interaction with interrogative sentences

The behaviour of apprehensive markers in interrogative sentence types remains largely unexplored. To my knowledge, only Vallejos (2016: Chapter 12, cited in Vuillermet 2018), reports on a seemingly apprehensional morpheme in Kukama-Kukamiria (a Tupian language of Peruvian Amazonia) which mostly occurs in interrogative sentences. Apprehensives seem indeed infrequent in such sentence types and markers in both Baoding and Kukama-Kukamiria seem atypical in this regard.

In Baoding, the apprehensive *=lɛ.ia* occurs in polar questions, and then encodes an implicit request to the addressee to prevent the undesirable event, as illustrated in (39).

(39) *Ni*^{213} *tso*$^{213\text{-}21}$=***lɛ.iã****?*
2SG leave=APPR.INT$_{\text{P.UNEX}}$
'Are you leaving already?' (Context: I see you are packing your stuff and I

am surprised at your leaving: Are you leaving without finishing your work?)

In (39), the speaker considers the addressee's leaving undesirable: he is expecting the addressee to stay longer in order to finish some work planned to be done together. The speaker is thus asking for confirmation that this unexpected, undesirable event is really going to occur. Compare with the following sentence:

(40) *Ni*^{213} *tso*$^{213\text{-}21}$*=iã?*
2SG leave=**EGO**.INT$_{\text{P.UNEX}}$
'Are you going to leave?'

In example (40), the speaker is surprised to see the addressee's preparing to leave and asks him/her about his/her intention of leaving. The enclitic particle *=ia* only encodes unexpectedness, with no apprehensional overtone or evaluative force from the part of the speaker.

The enclitic *=lɛ.ia* can also occur in interrogative sentences with 3rd person-subjects (41).

(41) *T*^{h}*a*^{45} *tso*$^{213\text{-}21}$*=**lɛ.iã**?*
3SG leave=**APPR**.INT$_{\text{P.UNEX}}$
'Is he leaving already?' (without finishing his work)

In (41), the undesirable event is the inferred consequence of the person leaving. This consequence is considered undesirable for the speaker, but not necessarily for the addressee. The speaker is asking the addressee to confirm whether the leaving (and its unpleasant consequence) is going to happen. If so, then the addressee is expected to do something to avoid the negative effect likely to occur on the speaker. If the sentence is declarative, then the undesirable effect would be endured by the addressee (42).

(42) *T*^{h}*a*^{45} *tso*$^{213\text{-}21}$*=**lɛ.ia**.*
3SG leave=**APPR**
'He is going to leave.' (If you still need him, go and find him quickly.)

To conclude, the distribution of the enclitic particle *=lɛ.ia* in interrogative sentences confirms the semantics of apprehension: regardless of whether the undesirable event is caused by the addressee or a 3rd person, the undesirable effect is endured by the speaker in interrogatives, (and by the addressee in declaratives). The expectation of a reaction likely to avoid or minimize the negative effect is always from the speaker's to the addressee's point of view.

6 Hypotheses on a possible source of *=lɛ.ia* and on the semantic link between its functions

In this section, we give a hypothesis on a possible source for the apprehensive enclitic particle *=lɛ.ia*, and propose a brief account of the semantic and pragmatic links between its temporal and apprehensive functions.

6.1 A hypothesis on the source of *=lɛ.ia*

From a morphological point of view, *=lɛ.ia* is a composite particle, which is likely constituted of two separate particles *=lɛ* and *=ia*, both found in the second paradigmatic slot of final enclitic particles (see Table 2 above in §3.1). The particle *=lɛ* ‘COS’ denotes a change of state (see example (21)), and the particle *=ia* ‘EGO’ expresses the intention of the speaker about an imminent future action (see §3.2).

Semantically, *=lɛ.ia* could well be the synchronic product of the combination of the meaning of these two separate particles: a change of state in an imminent future. We can suppose that the undesirability interpretation is a later development, resulting in a request of involvement from the speaker to the addressee by adding an evaluative force or an overtone of apprehension: ‘a change of state, which might be undesirable, is going to happen’. This fits quite well with the apprehensive meaning. In addition, semantic changes are often characterized by an increase rather than a decrease of subjectivity (Traugott 1989: 49).

Concerning the sources of the two particles *=lɛ* and *=ia*, since earlier stages of the Baoding dialect are not documented, we can only base our hypothesis on what we know of the diachronic evolution of mainstream Mandarin Chinese and on a comparison with other varieties of Mandarin dialects spoken in Northern China, in particular those spoken in Shaanxi and Hebei. Although there are some divergences on the sources of final particles in Standard Mandarin, most scholars (Cao 1987: Chapter 3, Ōta 1987: 351, Liu 1985) agree that the source of the final particle ***le***[4] (cognate with *=lɛ* in Baoding) is related to a verb ***liao*** [liɔ] meaning ‘to finish’. The enclitic particle *=lɛ* in Baoding is also pronounced as *=liɛ* by some speakers of the older generation (over 60 years old), which supports the same origin. We can extrapolate that Baoding *=lɛ* has gone through similar stages of development,[5] and is the output of a monophthongization of the form *=liɛ*.

[4] The transcription in Pinyin for Standard Mandarin is in bold letters.

[5] Northern varieties of Mandarin are nowadays quite similar to the language spoken during the Yuan Dynasty (1271–1368) (see Li et al. 2009: Chapter I). Baoding is not an exception. Thus, we can suppose that there are some similarities in the development of some grammatical morphemes between Baoding and Standard Mandarin.

As for the second component *=ia*, based on a comparative study of the four editions of a colloquial Chinese textbook *Laoqida* in Joseon period of Korea published from the 14th to18th century, Takekoshi (2002) identifies two main functions of a sentence final particle *ye [ie]: a.* to mark a judgemental statement; *b.* to mark a change of state. The second function encompasses a change of state occurring in the past or the present and a change of state in the imminent future. This makes it a plausible candidate for the origin of the egophoric particle *=ia* in Baoding that expresses intention in an imminent future. A similar phenomenon is also found in the neighbouring Mandarin dialects and in Jin dialects spoken in Shanxi and Shaanxi (located at the west of Hebei), where an enclitic particle *=ia* can mark a future event (Xing 2002, 2005, Lü 1995, Shi 2004).

6.2 The semantic and pragmatic link between the temporal and apprehensive functions from a typological perspective

As shown in the literature on apprehensive constructions, the issue of the syncretism of apprehensive morphemes with other functions in many languages, and the link between their various functions, remains largely understudied.

We mentioned in the previous §2.1 that the enclitic particle *=lɛ.ia* has two functions in Baoding: the imminent future function when it combines with activity verbs, and the apprehensive function when it co-occurs with change of state verbs or nominal predicates. What we are interested in here, and try to clarify, is the semantic and pragmatic relations linking these two functions, by looking in the crosslinguistic and typological literature. Angelo & Schultze-Berndt (2016) show for instance that apprehensive morphemes can be historically related with a temporal morpheme, with evidence from a number of unrelated languages in the world. For instance, *nachher* 'afterwards, later' has developed an apprehensive function in colloquial German. A similar phenomenon is attested in Dutch with the temporal adverb *straks* 'very soon, immediately' and in several varieties of Kriol spoken across northern Australia (Boogaart 2009, 2020, Angelo & Schultze-Berndt 2016).

They argue for their observations with the 'invited inferencing theory of semantic change model' (Traugott & Dasher 2001: 34–40; Traugott 2006: 552–553) a certain lexeme often used in a particular context can acquire the function of encoding another meaning by association with the inference invited by the speaker when it is linked with a certain conceptual structure. The inferred meaning then goes through a process of conventionalization and becomes the new meaning of this lexeme.

Such a path is plausible for *=lɛ.ia* in Baoding. With regard to the semantic change from temporal to apprehensive, the inference invited by the speaker is that the event which is going to occur is considered undesirable. The change of state verb provides an ideal context for the apprehensive interpretation on account that change of state verbs convey that a new state or event is going to come about. The particle *=lɛ.ia* acquires a new function through the reanalysis by the speaker in that context. When *=lɛ.ia* is used with other types of predicates, such as activity verbs, it also acquires in some contexts and under some conditions (1st person subject) an apprehensive meaning, albeit retaining the temporal function in other contexts. As was claimed by Traugott (2006: 553), generally the new polysemy is distributionally different from the old one and a reanalysis has therefore occurred.[6]

7 Conclusion

We have shown in this paper that Baoding has a final enclitic particle *=lɛ.ia* in apprehensive function: it conveys imminence and undesirability (from the speaker's point of view) and requests the addressee's involvement to avoid a potential undesirable situation with. We also pointed out that apprehensive *=lɛ.ia* can be used in polar questions. In this case, the speaker asks the addressee for confirmation of the unexpected and undesirable event, and implicitly requests him to prevent it.

We saw the relevance of grammatical person for the apprehensive reading: Baoding first-person subjects systematically trigger it. Person is thus a crucial factor determining whether a sentence with *=le.ia* has an imminent future reading or an apprehensive reading.

The apprehensive function of *=lɛ.ia* can be considered the result of a semantic extension of its temporal (imminent future) function. The distribution of the two functions of *=lɛ.ia* is shown to depend on the aspectual features of the predicates, and to a lesser extent, on the person of the subject. Cross-linguistic evidence (see Heine & Kuteva 2002) has been presented in favour of the hypothesis that change of state verbs and nominal predicates are the bridging context in Baoding for the semantic reanalysis of *=lɛ.ia* from an imminent future to an apprehensive function. This analysis is supported by the hypothetical origin of *=lɛ.ia*: the change

[6]Traugott gives an example of epistemic *must* in Modern English available in past tense and stative contexts, whereas the older deontic *must* in late Old English does not appear in the same contexts.

of state particle *=lɛ* and the imminent future particle *=ia* have combined to denote an imminent change of state and later acquired an apprehensive meaning. Although it cannot be considered as grammaticalization in a narrow sense, what draws our attention in this process of giving birth to the apprehensive function of *=lɛ.ia* is that the semantic reanalysis is accompanied by an increase of subjectivity, which confirms the unidirectional tendency identified by Traugott (1989).

Abbreviations

1SG	first singular personal pronoun
2SG	second singular personal pronoun
3SG	third singular personal pronoun
ADVI	advisative
ALL	allative
APPR	apprehensive
CLF	classifier
COME&DO	associated motion come and do
COMPL	completive aspect
CONT	continuous aspect
COS	change of state
CTR.ASR	counter assertion
DEM	demonstrative
DEM_{DIST}	distal demonstrative pronoun
DIR_{CTP}	directional centripetal
EGO	particle expressing egophoricity
EMPH	emphatic particle
EVIT	evitative
FACT	factitive
FUT_{IMM}	imminent future
GO&DO	associated motion go and do
IPFV	imperfective aspect
INCL	inclusive pronoun
INT_{WH}	WH type interrogative particle
$\text{INT}_{\text{P.CONF}}$	polar question particle (with an expectation of confirmation)
$\text{INT}_{\text{P.N}}$	polar question particle (neutral)
$\text{INT}_{\text{P.UNEX}}$	polar question particle (asking for confirmation usually unexpected)
IRR	irrealis
LOC	locative
NEG	negative
NEG_{PROH}	prohibitive
NIRR	non irrealis
NSG	non singular
O	object
OBV	obviousness
PASS	passive
PAST	past tense
POT	potential
PRI	prioritive
R	realis
RED	reduplication
RES	resultative
S	subject
SG	singular
SEQ	sequential
SGST	suggestion particle
STA	stative

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Chapter 11

Apprehensional markers in Korean

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This paper addresses the diachronic development and synchronic distribution of thirteen markers of apprehension in Korean. They have been grammaticalized to varying degrees from diverse constructions, forming multiple layers in contemporary Korean. The sources involve the future marker, questions, cognitive verbs, and mode- or purpose-adverbializers, among others. The notion of apprehension is either fully semanticized in the constructions or is only pragmatically inferred from them. The role of pragmatic inference is also evident in the reanalysis of a sentence-ender as an apprehensional causal connective. In many cases of pragmatic inference, subjectification from objective to subjective meanings, and directionality from epistemic possibility to apprehension are prominent.

1 Introduction

1.1 Objectives

Korean has a number of grammatical forms that encode the speaker's apprehension, i.e., apprehensionals (or "preventive markers"; Ramstedt 1997[1939], Kim 1960, Ko 1989, Son 1994). Their primary function is to warn the addressee of potentially harmful consequences of an action, or to present reasons for the speaker's action or mental state, as exemplified in the following:

(1) a. Connective *-lla* 'lest'
Pelley tuleo-lla *changmwun tat-ala.*
insect come.in-APPR window close-IMP
'Close the window lest insects come inside.'

Seongha Rhee & Tania Kuteva. 2026. Apprehensional markers in Korean. In Martina Faller, Marine Vuillermet & Eva Schultze-Berndt (eds.), *Apprehensional constructions in a cross-linguistic perspective*, 381–421. Berlin: Language Science Press. DOI: ??

b. Connective *-lkkapwa* 'for fear of'
Cencayng-i na-lkkapwa cam-i an o-n-ta.
war-NOM come.out-APPR sleep-NOM not come-PRS-DECL
'(I) cannot sleep for fear of a war.'

Despite the early appearance of the notion in Korean linguistics, as in Ramstedt (1997[1939]), the markers have not yet received any attention from a grammaticalization perspective. In order to fill this research gap, this paper addresses the diachronic development and synchronic distribution of the apprehensionals in Korean from a grammaticalization perspective, and discusses a number of theoretically important issues that arise from the analysis.

Here we are using the term *apprehensional* for the overall domain of both bi-clausal and mono-clausal structures which in some way involve apprehension. Notice that while most of the markers we discuss entail undesirability, others can be used in more neutral contexts. As proposed in Kuteva et al. (2019: 865), it is justifiable to treat both bi-clausal and mono-clausal structures as instantiations of the same overall – apprehensional – domain all the more so because there exists a fairly regular pathway between the two in the process of "insubordination" (Evans 2007, Evans & Watanabe 2016) leading from the bi-clausal to the mono-clausal structure.

Since Korean has a large inventory of apprehensionals, a microscopic approach to a few select forms may prove fruitful, but we adopt a more macroscopic perspective in order to describe the system of Korean apprehensionals as a whole, and thus include both weakly and strongly grammaticalized forms. We leave more fine-grained research on individual forms for future research. This paper is organized in the following manner. After a short introduction to the data sources as well as demographic, syntactic, and typological characteristics of the Korean language, in §1, we describe the grammatical means of signaling apprehension as belonging to three major categories according to the types of lexical and grammatical sources, in §2. In §3, we discuss a few theoretical issues, such as grammaticalization parameters, layering and specialization, roles of source concepts, and a few factors involved in semasiological changes such as pragmatic inference, reinterpretation, and subjectification. We summarize the discussion and conclude the paper in §4.

1.2 Data sources

The data for analysis are taken from various sources, including the Sejong Historical Corpus for diachronic investigation and the Drama and Cinema Corpus for investigation of contemporary usage.

The Sejong Historical Corpus is a 15 million-word, historical section of the Sejong Corpus, a 200 million-word corpus developed as part of the 21st Century Sejong Project by the Korean Ministry of Culture and Tourism and the National Institute of Korean Language (1998–2006). The texts in the historical section, dating from 1446 through 1923, are of diverse genres including Buddhist scriptures and commentaries, medical advisories, poems, literature commentary, books on ethics, novels, short stories, interpreter training textbooks, reports to kings, biblical tracts, newspaper articles, diaries, personal letters, etc. Of particular importance is the inclusion of 43 pieces of *sinsosel* 'new novels' (mostly between 1906–1913), a literary genre that bridges the classical novels and modern novels in the history of Korean literature. The genre was influenced by Western literary styles including the use of colloquial language. Since grammaticalization processes can be more easily captured in colloquial data, *sinsosel* is often the data source where emerging grammatical functions are first attested.

The data for Modern Korean is taken from the Drama and Movies Corpus. The corpus is a 24 million-word contemporary corpus, a collection of 7,454 scenarios of dramas and movies dating from 1992 through 2015, compiled by Min Li of Seoul National University. To search and process the data, the Korean concordance program UNICONC, developed by Jinho Park, was used.[1]

1.3 Preliminaries on Korean

Korean is spoken by about 77.2 million people (*Ethnologue* Simons & Fennig 2018) in and around the Korean peninsula: South Korea, North Korea, China, Japan, and other Korean communities around the world. It is a head-final, agglutinating language with a SOV word order. Sentential arguments are not only omissible but such omission is often preferred when their referents are contextually recoverable. The sentence-final word is the main-clause verb with a number of morphological markers. These sentence-final verbal suffixes, collectively called "sentence-final particles" by most Korean linguists, consist of the T(ense)-A(spect)-M(odality)-Mood morphology, including such subjective and intersubjective markers as evidentiality, epistemicity, politeness, deferentiality, and honorification. These occur in ordered slots, the last of which is occupied by a sentence-ender designating the speech-act type and politeness/deferentiality.

The structure that is of particular interest for the present research is clause-embedding patterns. Clause embedding may involve direct or indirect quotation

[1]We thank the developers of the program and corpora for generously granting their use for research.

(as shown in (2) and (3), respectively). The distinction between direct and indirect quotation is often difficult because the distinction is not strictly binary. A good clue in Korean is whether the embedded clause ends with a regular sentence-ender (direct speech), such as *-ta* in (2b), or with a complementizer (indirect speech), such as *-tako* in (3b). Since Korean has a large number of regular sentence-enders (Kim 2001: 147–151 lists 381 forms), but only four to five complementizers, the distinction is relatively easy. There are rare occasions, however, when the final form is unclear as to its sentence-ender or complementizer status in case the phonetically reduced complementizer, formally identical with the sentence-ender, is used. In such cases other clues may help disambiguate its status.

(2) Direct Speech

a. (SUBJECT) "[(SUBJECT) (OBJECT) VERB]" VERB
(NP-NOM) [(NP-NOM) (NP-ACC) **V-A-T-M-END**] **V-A-T-M-END**

b. *Ku-ka* "[*atul-i* *pap-ul* *mek-koiss-ess-napo-ta*]"
he-NOM [son-NOM food-ACC eat-PROG-PST-CONJ-DECL]
chwuchukha-koiss-ess-ta.
guess-PROG-PST-DECL
'He was guessing, (saying) "(My) son seemed to be eating food (then)."

(3) Indirect Speech

a. (SUBJECT) [(SUBJECT) (OBJECT) VERB] VERB
(NP-NOM [(NP-NOM) (NP-ACC) **V-A-T-M-COMP**] **V-A-T-M-END**

b. *Ku-ka* [*atul-i* *pap-ul* *mek-koiss-ess-napo-tako*]
he-NOM [son-NOM food-ACC eat-PROG-PST-CONJ-COMP]
chwuchukha-koiss-ess-ta.
guess-PROG-PST-DECL
'He was guessing that (his) son might have been eating food (then).'

A significant feature of Korean grammatical markers in general is that they signal not only garden-variety grammatical notions, such as grammatical case, local case, coordination, subordination, etc., but also diverse stance-related notions, such as tepidity, unexpectedness, beneficiality, extremity, unrealistic hypotheticality, inordinateness, etc., which are often difficult to label. Thus, Korean grammatical markers are paradigm examples of "elaborateness" or "semantically elaborate categories" (Kuteva 2009, Kuteva et al. 2019) or markers of languages of an "esoteric" nature (Thurston 1988, Grace 1997, Crowley 2000, Wray & Grace 2007, as cited in Rice 2017, 2019).

Another important fact is that intersubjectivity marking is obligatory in Korean. Most prominently, Korean sentences, properly ended, should be marked with the speech level (*hwakyey* in Korean linguistics), depending on the relative social hierarchy between the speaker, hearer, and sentential subject. This system is complex and variable, with four to seven different levels of honorification. The rules pervasively apply to not only verbal morphologies but also nominal morphologies.

2 Grammaticalization of apprehensionals

2.1 Apprehensionals in contemporary Korean

As indicated earlier, Korean has a large number of apprehensionals with differential degrees of semanticization of the notion of apprehension and with different levels of currency. They have the status of a clausal connective (CONN), usually specifying the reason for the fear in the clause that hosts the apprehensional, or of a sentence-ender (END), simply to indicate the speaker's emotional state of apprehension with respect to the content of the proposition. When apprehensionals occur as connectives, the clauses hosted by them will function as subordinate clauses. Some apprehensionals can be used either as a connective on the subordinate clause or as a sentence-ender on the main clause, whereas some others are used only as one or the other (cf. Ramstedt 1997[1939], Kim 1960, Ko 1989, Son 1994). The forms are listed in Table 1 with their source constructions and grammatical classes.

As shown in Table 1, the forms that end with an *-e/a* (note that these two are phonologically controlled allomorphs) can function as a clausal connective or a sentence-ender due to the fact that *-e/a* are 'heteronyms' (Lichtenberk 1991), carrying dual functions in the two different grammatical categories.[2] As a connective it marks sequentiality or causality relations between the two events denoted by the two clauses (cf. "consolidating connective" (Koo 1987)), and as a

[2]The connective function is older and its use as a sentence-ender is a 19th century innovation (Suh 1993: 139–141; see, however, Park 1996 for a different view). At the time of innovation, when a sentence ended with this form (formerly a connective), it was felt that the speaker stopped an utterance without completing it. Thus, the speech style using this sentence-ender has been named *panmal* 'half-talk'. As a sentence-ender, *-a/e* lacks honorification or politeness, but is uniformly used across all speech acts, e.g., declarative, interrogative, imperative, hortative, promissive, etc.

Table 1: Apprehensional forms, source constructions, and grammatical categories

	Form	Source construction	Gloss for source construction	Class
a.	*-lkka*	< *-l-kka*	FUT-Q	CONN
b.	*-lkkapwa*	< *-l-kka-po-a*	FUT-Q-see-CONN/END	CONN/END
c.	*-lkkamwusewe*	< *-l-kka-mwusep-e*	FUT-Q-be.fearful-CONN/END	CONN/END
d.	*-lkkamolla*	< *l-kka-molu-a*	FUT-Q-not.know-CONN/END	CONN/END
e.	*-lkkasiphe*	< *-l-kka-siph-e*	FUT-Q-think-CONN/END	CONN/END
f.	*-lla*	< *-l-la*	FUT-END	END/(CONN)
g.	*-lseyla*	< *-l-*SA*-i-la*	FUT-occasion-be-END	END/(CONN)
h.	*-cimolla*	< *-ci-molu-a*	NOMZ-not.know-CONN/END	CONN/END
i.	*-cimoluni*	< *-ci-molu-ni*	NOMZ-not.know-CONN	CONN
j.	*-cimoshakey*	< *-ci-mos-ha-key*	NOMZ-cannot-do-MODE	CONN
k.	*-cimoshatolok*	< *-ci-mos-ha-tolok*	NOMZ-cannot-do-PURP	CONN
l.	*-cianhkey*	< *-ci-ani-ha-key*	NOMZ-not-do-MODE	CONN
m.	*-cianhtolok*	< *-ci-ani-ha-tolok*	NOMZ-not-do-PURP	CONN

sentence-ender it marks the end of a sentence at the "intimate" speech level. It is noteworthy, in all cases except for (a) *-lkka,* (f) *-lla* and (g) *-lseyla*, that the grammatical function of the final element of the source construction determines the grammatical category of the grammaticalized form, a point observed in Rhee (2007) (see §3.4 for discussion of the exceptional cases). Note that the markers of MODE and PURP(ose) in (j) through (m) are adverbializing connectives used to form converbs (Ramstedt 1903, 1997[1939], Haspelmath 1995), called *pwutongsa* among Korean linguists.[3]

Another aspect evident from the list is that there is a certain, limited number of concepts that are involved in the source constructions, i.e. grammatical markers of FUT(urity), Q(uestion), MODE, and PURP(ose) and lexical verbs of perception, emotion, and cognition such as 'see', 'be fearful of', 'not know', and 'think'. The majority of the apprehensionals, seven out of thirteen, developed from constructions involving future tense marker *-l* (and its allomorph *-ul*).

[3]An anonymous reviewer notes that these mode- and purpose-marked forms (j) through (m) exhibit differences from the rest in that they all involve a converb and a negative morpheme. It is true in that light that the paradigm of Korean apprehensionals exhibits a low level of "paradigmaticity" (Lehmann 2015[1982]), in that there are many members with dissimilar formal properties within a category. The category of apprehensionals is a functionally motivated paradigm.

Among these seven apprehensionals, five also involve a question marker. Six of them involve the nominalizer *-ci*. Similarly, six of them involve perception-emotion-cognition verbs. For the following more detailed discussion of these apprehensionals, the thirteen apprehensionals are classified into three groups according to their prominent source characteristics, i.e. those developed from constructions involving future and question markers (§2.2), those involving future markers without question markers (§2.3), and those involving the nominalizer (§2.4). It is evident that the three groups do not form mutually-exclusive classes, and thus inclusion of an apprehensional in a particular class is largely a matter of convenience. Likewise, those involving perception, emotion, and cognition verbs are not separately addressed, but are discussed in the three categories, when they are relevant.

2.2 Apprehensionals from future + question

The first group of apprehensionals are those originating from the source constructions involving the future tense marker *-l-* and the question marker *-kka*, i.e. *-lkka, -lkkapwa, -lkkamwusewe, -lkkamolla,* and *-lkkasiphe.* The primary function of the grammatical morpheme *-l-* is marking the future tense, indicating probability, intent, prediction, etc. but it can also indicate counterfactual or conjecture when used with a past tense marker. It is etymologically related to the prospective-marking adnominal *-l.*[4] All these functions have the irrealis meaning in common.

2.2.1 *-lkka*

Since the apprehensional *-lkka* originated from the source construction of the future *-l-* and the interrogative sentence-ender *-kka*, the source meaning is 'Will x …?', where x is a proposition, the content of which is the event or state being feared by the speaker or the sentential subject. This is the oldest apprehensional attested in the earliest *Hankul* literature (15th c., Late Middle Korean), as shown in (4):[5]

[4]Korean has different adnominal forms depending on the temporal relation of the event with respect to its referenced event (or speech time), i.e., *-(u)l* for prospective (later) as in *mek-ul pizza* 'pizza to eat', *-(u)n* for anterior (previous), as in *mek-un pizza* 'pizza eaten', and *-nun* for simultaneous (same), as in *mek-nun pizza* 'pizza being eaten'. As indicated, the prospective adnominal *–(u)l* is conceptually and historically related to the future tense marker *–(u)l(i)-*.

[5]*Hankul*, also often written as *hangeul*, is the Korean alphabet, invented in 1443 by King Sejong the Great. Prior to its invention, Chinese characters were used for their meaning (semantogram) or sound (phonogram), thus the interpretation of some historical documents dating from the pre-*hankul* period remains controversial, despite notable advances made in recent studies.

(4) *Swuthahoychen-i neki-toy* [...] *swutal-i wen-ul mot*
(name)-NOM think-CONN (name)-NOM wish-ACC cannot
ilwu-lkka hA-ya [...]
realize-APPR do-and
'As the divinity Sutahwechon (Suddhavasehi Devehi) thought, [he came to the earth and intervened ...] lest Sudal fail to accomplish his dreams, ...' (1447 *Sekposangcel* 6:25a)

The apprehensional meaning associated with *-lkka*, translated as 'lest' in the example, largely comes from the context, as it appears that the notion of apprehension was not yet fully grammaticalized at the time. In other words, Suddhavasehi Devehi's intervention is interpreted as a result of his fear that Sudal might fail to accomplish his dream of spiritual enlightenment. Since it is not likely that Suddavasehi Devehi actually uttered the question 'Will he not ...?', the form -*lkka* on the embedded clause, while appearing to involve the question marker -*kka*, is interpreted instead as a marker of fear or serious concern (see §3.4 for more discussion on *-lkka*). Indeed, the collocation pattern in the 15th century Korean data strongly supports the hypothesis that the apprehensional meaning was pragmatically inferred from the co-occurring psych-verbs in the main clause. Among the common collocates, in addition to the light verb *hA-* 'do/say' in the example above, are *neki-* 'think/regard/consider', *pwunpyelha-* 'distinguish', *silumha-* 'worry', *ceh-/ceph-* 'fear', *uysimha-* 'doubt', *hyey-* 'consider, calculate', *sulh-* 'lament', *twuli-* 'fear', *kongha-* 'fear', *sip-/siph-* 'think', *minmangha-* 'feel ashamed', *kunsimha-* 'worry', *twulyeha-* 'fear', *sAyngkakha-* 'think', *nyemnyeha-* 'worry', *acheha-* 'dislike', *mwusep-* 'be afraid', *kepulnay-* 'be afraid', *cAkepha-* 'be afraid', *kongphoha-* 'fear', *cosimha-* 'be mindful', *samkaha-* 'be wary', etc. Evidently, most of the co-occurring psych-verbs have a negative meaning. Thus, a sentence with an embedded question with the structure of, for instance, ['I think [Will a war break out?']] becomes an apprehensional sentence ['I fear that a war might break out']. Similarly, ['as I think [Will a war break out?']] becomes ['lest a war break out.'] (see §3.4 for more discussion on structural reanalysis).

Since the undesirability-of-event interpretation of *-lkka* heavily depended on the meaning of the co-occurring verb, *-lkka* sometimes occurred as a non-apprehensional marker, in neutral or even in positive (desirable) contexts, e.g. with *pAla-* 'hope'. In non-apprehensional usage, *-lkka* carries diverse modal meanings, e.g., uncertainty, possibility, conjecture, etc. In our Present-Day Korean (PDK) data, all connective usage tokens (about 991 tokens) are instances of fully developed apprehensionals.[6] Even when it is used as the sentence-ender of

[6]The statistics given here are either actual hit-records or projections based on random sampling of 10 percent of all hits, or 500 tokens.

an embedded direct quotation and thus is immediately followed by a verb that specifies the nature of the embedded question, *-lkka* is predominantly used in contexts involving undesirable events. For instance, in such cases, 69.6% of the instances (about 2,454 tokens) involve a verb with meanings related to worries and fears, while 30.4% of them (about 1,072 tokens) appear in neutral contexts (e.g., *ha-* 'say', *sayngkakha-* 'think'), and no instances are found in positive contexts. This suggests a scenario where, as *-lkka* becomes used in increasing frequency in contexts with verbs with negative connotation, it has developed into an apprehensional connective even without an accompanying verb of undesirable meaning (see §3.3 for more discussion on semantic absorption). In PDK, the apprehensional *-lkka* is moderately frequent but is yielding the functional primacy to its reinforced innovation *-lkkapwa* discussed immediately below (see §3.2 for the synchronic states of affairs regarding the different apprehensional forms).

2.2.2 *-lkkapwa*

The source construction of *-lkkapwa* is *-l-kka-po-a,* which involves the future tense marker *-l-*, the interrogative sentence-ender *-kka*, the lexical verb *po-* 'see' and the sentence-ender/connective *-a,* thus yielding the compositional meaning of '(I) see, (saying) "Will x …?"' (when *-a* is a sentence-ender) or 'as (I) see, (saying) "Will x …?", …' (when *-a* is a connective).[7] From these meanings of the source constructions, the apprehensional interpretations emerge, e.g. '(I) fear x might happen' or 'as (I) fear x might happen' or 'lest x should happen.'

The apprehensional *-lkkapwa* is first attested in the late 19th century data, i.e. the Modern Korean period. Among the earliest attestations is the following, in which *-lkapoa*, the historical form of *-lkkapwa*, occurs:

(5) *CalmoshAtAka os-ul tuley-kena phAymwul-ul sanghA-lkapoa*
by.accident dress-ACC make.dirty-or jewelry-ACC damage-APPR
cosim-i toy-ya …
concern-NOM become-and
'As (I) became worried lest my (rented silk) dress become sordid or my (rented) jewellery become scratched [at the party], (I hurried home to take them off and return them to you).' (1911 *Welhakain* 23)

[7]As an anonymous reviewer notes, the verb *po-* 'see' has a large number of semantic designations and has developed into diverse grammatical markers (see Rhee 2019, 2020, 2021a,b among many others, for discussion of semantic extension and grammaticalization of the verb).

Even though it has a short history, the apprehensional connective *-lkkapwa* is the most frequently used apprehensional in PDK. Unlike in the earlier form *-lkka*, from which it originated as a reinforced form, the apprehensional meaning is fully semanticized in the connective *-lkkapwa*, i.e., the marker is not used in neutral or positive contexts (e.g., occurring with such verbs as *ha-* 'say' (the historical descendant of hA-), *sayngkakha-* 'think', *pala-* 'hope', *kotayha-* 'look forward to', etc.). It is found in a corpus search that all occurrences of the connective *-lkkapwa* are of the apprehensional usage (about 6,697 tokens). In PDK it can also be used as the apprehensional sentence-ender, through a process of insubordination, as exemplified in the following constructed examples:[8]

(6) a. APPR connective
Cencayng-i na-lkkapwa tteli-n-ta.
war-NOM come.out-APPR shiver-PRS-DEC
'I am shivering for fear that a war might break out.'

b. APPR sentence-ender
Cencayng-i na-lkkapwa.
war-NOM come.out-APPR
'A war might break out (I'm afraid).'

2.2.3 *-lkkamwusewe*

The source construction of *-lkkamwusewe* is *-l-kka-mwusep-e,* which involves the future tense marker *-l-,* the interrogative sentence-ender *-kka*, the lexical verb *mwusep-* 'be fearful' and the sentence-ender/connective *-e,* thus yielding the compositional meaning of '(I) am fearful, (saying) "Will x …?"' (when *-e* is a sentence-ender) or 'as (I) am fearful, (saying) "Will x …?", …' (when *-e* is a connective). From these meanings of the source construction, the apprehensional interpretations emerge, e.g. '(I) fear x might happen' or 'as (I) fear x might happen' or 'lest x should happen.'

Historically, the forerunner construction of the apprehensional *-lkkamwusewe* is first attested in the 19th century data in the form of *-lkamwusewesye*, containing

[8] In this process, however, there also occurred a functional split, i.e., depending on the context, it signals apprehension (e.g., 'I fear that …'), the speaker's tentative intention (e.g., 'I might as well …'), or inference (e.g., 'It seems that …'). As a connective it only marks apprehension. As a sentence-ender, it can, in principle, have any of these three functions, but all sentence-ender instances found in the corpus are of the tentative-intentional function (i.e., non-apprehensional), a state of affairs indicating that the divergence from its apprehensional source is nearly complete in the domain of sentence-enders in PDK.

the final particle *-esye*, a tonic variant of *-e* with a fortified sequential or causal meaning, as shown in (7):

(7) [The magistrate who appeared without notice protested why there were no toast songs in the drinking party, but …]
… kisAyng-tul-un chAykmanghA-lkamwusewesye mos
entertainer-PL-TOP scold-APPR cannot
hA-te-n chA-i-la.
do-RETRO-ADN point-be-DECL
'… (in fact) the entertainers had not (sung) lest (the magistrate) might scold (them for partying).' (1896 *Toklipsinmwun*)

Example (7) describes a scene in which the magistrate who appeared without notice pretends to be unhappy about the entertainers not singing at the party thrown by the low-ranking officials. The entertainers did not sing but instead remained quiet because the magistrate was known to be stern and austere and thus refrained from any unnecessary spending of the government fund. From the context the entertainers could have been really 'afraid' (*mwusep-* 'be fearful') or they were simply 'concerned' (the context supports the former interpretation). The apprehensional *-lkkamwusewe* is still in use in PDK, though not occurring at a high frequency. A corpus search returns only 86 instances of *-lkkamwusewe/-lkkamwusep*, all carrying the apprehensional function either as a connective (31.4%) or as a sentence-ender (68.6%). In the PDK usage, in spite of the presence of *mwusep-* 'be fearful', the construction in general does not denote genuine fear but much milder concern evoked by the situation, analogous to the English preface discourse marker *I'm afraid* (*but* …). Since the early attestations invariably encoded genuine fear (cf. 7), the PDK usage of *-lkkamwusewe* is evidently the result of semantic bleaching or generalization, which is a common concomitant, or even a prerequisite, of grammaticalization (Bybee & Pagliuca 1985, Bybee et al. 1994; see also §3.1).[9]

2.2.4 *-lkkamolla*

The source construction of *-lkkamolla* is *-l-kka-molu-a,* which involves the future tense marker *-l-*, the interrogative sentence-ender *-kka*, the lexical verb *molu-*

[9]We are grateful that Eva Schultze-Berndt and Marine Vuillermet (p.c. August 5, 2020) brought to our attention Jing-Schmidt & Kapatsinski (2012) reporting similar phenomena in English, Mandarin and Russian.

'not know' and the sentence-ender/connective *-a,* thus yielding the compositional meaning of '(I) do not know, (saying) "Will x …?"' (when *-a* is a sentence-ender) or 'as (I) do not know, (saying) "Will x …?", …' (when *-a* is a connective). From these meanings of the source constructions, the apprehensional interpretations emerge, e.g. '(I) fear x might happen' or 'as (I) fear x might happen' or 'lest x should happen.' The question arises in the mind of the speaker because of absence of knowledge or uncertainty, which also may arouse some form of fear.

The apprehensional *-lkkamolla* is first attested in the 17th century data, i.e., the Early Modern Korean period, and is still in use, though not occurring at a high frequency, in PDK. The following is an example among the earliest instances of its use, in the form of *-lkamolna*:

(8) [I want to get a couple of poems and some calligraphy from him, but …]
… cyecye-y ei nyeki-si-lkamolna cacyehA-na-ita.
he-NOM how think-HON-APPR restrain.self-IND-DECL
'… (I) restrain myself from (requesting) fearing how he would react/think.' (1687 *Kwuwunmong* SNU Archive)

As *-lkkamolla* is the least frequent apprehensional, a corpus search returns only 55 hits, of which 40 are of the apprehensional usage. In non-apprehensional usage, *-lkkamolla* simply denotes absence of knowledge (as in 'I asked him because I didn't know if …'). Even though the earliest attestation is of the connective usage, as exemplified in (8), all instances of *-lkkamolla* found in the corpus are of the sentence-ender function.

2.2.5 *-lkkasiphe*

The source construction of *-lkkasiphe* is *-l-kka-siph-e,* which involves the future tense marker *-l-*, the interrogative sentence-ender *-kka*, the lexical verb *siph-* 'think, inclined to think' and the sentence-ender/connective *-e,* thus yielding the compositional meaning of '(I) think, (saying) "Will x …?"' (when *-e* is a sentence-ender) or 'as (I) think, (saying) "Will x …?", …' (when *-e* is a connective). From these meanings of the source constructions, the apprehensional interpretations emerge, e.g. '(I) fear x might happen' or 'as (I) fear x might happen' or 'lest x should happen.' The speaker is inclined to think or pose a question about a potential future event, which arouses some form of fear or serious concern.

The lexical verb *siph-* in the source construction has undergone a series of semantic changes whereby a number of grammatical forms emerged in history (see Kim 2010, Rhee 2021b,c for the verb's diverse grammaticalization scenarios including the evidentiality function). The earliest precursor forms of *-lkkasiphe* appear as *-lkosip*-END (e.g., *-lkosipeyla*) and *-lkasipe* in Late Middle Korean, but whether those forms in the examples indicated mere inquisitive thought or fear is unclear. An unequivocal example of the apprehensional use is attested in the 17th century data (Early Modern Korean) of a wartime diary written by a court lady (note that the exact year of authorship is unknown):[10]

(9) *Skwum-i ha pencaphA-ni na-y cwuk-ulkasipu-te-la.*
dream-NOM very be.disorderly-as I-NOM die-APPR-RETRO-DECL
'The dreams were so disconcerting that I feared that I might die.' (163x *Pyengcailki* 410)

The semantic change from 'think' to 'be afraid' must be the result of subjectification prompted by inferences from the context of undesirable events in which the verb *siph-* occurs. For instance, Rhee (2021c) shows that the earliest meaning of the lexeme *siph-* 'think' has undergone a series of changes largely dependent on the co-occurring words, and developed into the markers of desiderative, epistemic-evidential modality, apprehensional, tentative intention, similative and evaluative connective, and discourse markers (see also Kim 2010 and Rhee 2021b for discussion of the grammaticalization of the verb *siph-*). In PDK *siph-* no longer retains its original meaning 'think', but is present only as a part of grammatical markers including the apprehensional *-lkkasiphe*. This form is among the least frequent apprehensionals; a corpus search returns 112 instances of *-lkkasiphe*, all of which function as an apprehensional connective. The functional split between sentence-ender and connective function, discussed for *-lkkapwa* above, is also observed with *-lkkasiphe*. In other words, when it is used as a sentence-ender it marks the speaker's tentative intention, rather than apprehension.

2.3 Apprehensionals from future

The next group of apprehensionals involving the future marker *-l-* in their source constructions are *-lla* and *-lseyla*. These two are different from those illustrated in §2.2 in that they do not involve the interrogative sentence-ender *-kka*.

[10]Interpretation of historical data can be variable and often debatable. However, non-apprehensional interpretations of (9), e.g., desiderative, tentative intentional, inferential, etc., are contextually inappropriate.

2.3.1 *-lla*

The source construction of *-lla* is *-l-la,* which involves the future tense marker *-l-* and the sentence-ender *-la*. The sentence-ender *-la* is no longer in common use in PDK, and its grammatical status and function are controversial. For instance, Ramstedt (1997[1939]: 85), in line with Underwood (1915[1890]), analyzes *-lla* as consisting of the future *-l-* and the vocative *-a*, labeling it as 'preventive', which, however, is disputed in Ko (1989: 293, 341–342), who regards the form as mono-morphemic (i.e., unanalyzable) and labels it as a marker of *kyengkyeypep* 'premonitive'. Indeed, the analysis of the future marker immediately followed by the vocative is in violation of morphosyntactic rules of Korean. The final element *-la* seems to be a declarative sentence-ender with an overtone of exclamation or surprise, since its formal and functional relative, the sentence-ender *-ela,* is one of the exclamative/mirative sentence-enders in PDK. With this analysis the construction *-l-la,* future-declarative, yields the compositional meaning 'x will ...!', which is functionally reinterpreted as '(I) fear x might happen.' There is nothing that compels such reinterpretation, but such a reinterpreted meaning becomes favored, which results in the emergence of an apprehensional. Therefore, it is through reinterpretation that the mere possibility of something taking place in the future comes to signal the speaker's fear (see §3.4 for more discussion).

A forerunner construction of *-lla* is first attested in the 17th century data, but this did not have a dedicated apprehensional meaning as it does now, but only signaled possibility and conjecture. In the 18th century, *-lla* developed into a marker of tentative intention with first-person subjects. With second- or third-person subjects, it developed into the apprehensional, which then became highly frequently used. The usage of possibility- or conjecture-marking survived until the beginning of the 20th century. In PDK, it is an unequivocal apprehensional, even with first-person subjects, but its frequency has considerably declined. The decline seems to be due to the functional specialization of its competitor *-lkkapwa.* Most attestations in the corpus are those of the homophonic marker of intention, a variant of *-llyeko*. The following is one of the earliest attestations of the apprehensional *-lla*:

(10) [In preparation of funeral]
Pang tewu-lla pwul stʌy-cimal-ko ko-y tuleka-lla
room be.hot-APPR fire make-PROH-and cat-NOM enter-APPR
kwistol mak-elʌ
flue.opening block-IMP
'Don't make fire (for floor-heating) lest the room become too hot (for the corpse), and block the flue opening lest a cat enter (through it) [which is ominous according to superstitions].' (ca. 1870 *Pakhungpoka*)

In (10) the grammatical status of the apprehensional *-lla* is unclear, i.e., whether it is a sentence-ender, resulting in the meaning of 'I'm afraid the room might become too hot', or a connective, resulting in the meaning of 'lest the room become too hot' as suggested in the translation. The uncertainty of grammatical status is an intriguing issue, which will be discussed in §3.4. The apprehensional *-lla* is infrequent in PDK. For instance, a corpus search returns 184 instances of the apprehensional *-lla*, 173 tokens of which (94.0%) are of the sentence-ender function and 11 tokens (6.0%) are of the connective function.

2.3.2 *-lseyla*

The source construction of *-lseyla* is *-l-sa-i-la,* which involves the future tense marker *-l-,* the defective noun *sa* 'occasion' functioning as a nominalizer, the copula *i-* and the declarative sentence-ender *-la*, thus yielding the compositional meaning of 'this occasion is that x will occur!'[11] From this meaning of the source construction, the apprehensional interpretations emerge, e.g. '(I) fear x might happen' or 'as (I) fear x might happen.' A simple statement of a possible future event with an exclamative overtone (see *-lla* above) gives rise to the apprehensional meaning.

The forerunner construction of *-lseyla* is frequently attested in Late Middle Korean with an overtone of exclamation and at the same time signaling a causal relation between the two connected clauses. These early examples did not carry the apprehensive function. The unambiguous apprehensional function is first attested in 1703, and became very frequent in folk songs in Early Modern Korean. The following is an example from an 18th century collection of songs:

(11) [Attracted by the spring breeze I stroll to the stream, only to find countless fallen flowers in the water.]

Cye	*kos-a*	*senwen-ul*	*nam*	*al-lseyla*	*stenaka-cimal-wala.*
that	flower-VOC	fairy.land-ACC	other	know-APPR	leave-PROH-IMP

'Oh, you, flowers, don't flow down (the stream) lest others would find out about the fairyland (upstream).' (ca 1767 *Haytongkayo* Cwussipon)

[11] The source composition of the portmanteau form *-lseyla* (and its historical variant *-lsyeyla*) has been a point of controversy (see Jang 2002: 137–138). Considering the involvement of the copula *i-* and the declarative sentence-ender *-la*, we hypothesize that the preceding formant is a nominal, and thus the semantically-bleached defective noun *sa* 'time/occasion' in Late Middle Korean is the best candidate, both formally and semantically well-motivated. The defective noun *sa* 'time/occasion' originally denoted 'the sun' from which the notion 'temporal/spatial expanse or interval' was derived (Jeong 2003: 61, see also Rhee & Koo 2021).

Similar to *-lla* in (10) in the preceding discussion, the apprehensional *-lseyla* in (11) is ambiguous with respect to its grammatical status, i.e., whether it is a sentence-ender resulting in the meaning of 'I'm afraid other people might find out about this fairyland' or a connective resulting in the meaning of 'lest others would find out about this fairyland' as provided in the translation above (see §3.4 for more discussion).

Even though *-lseyla* is often considered one of the representative apprehensional forms in the history of Korean, it is not frequently used anymore in PDK and carries the flavor of archaism. A corpus search shows 233 tokens of *-lseyla*, all in the apprehensional connective usage. This indicates that *-lseyla*, formerly used as either a connective or a sentence-ender, has lost its sentence-ender function in PDK.

2.4 Apprehensionals from nominalizers

2.4.1 *-cimolla* and *-cimoluni*

The source constructions of *-cimolla* and *-cimoluni* are *-ci-molu-a* and *-ci-molu-ni.* Both involve the negative-polarity nominalizer *-ci* and the lexical verb *molu-* 'not know', which are combined with the sentence-ender/connective *-a* and the causal connective *-ni*, respectively, thus yielding the compositional meanings of '(I) do not know if …' (with the sentence-ender *-a*), 'as (I) do not know if …' (with the connective *-a*), or 'because (I) do not know if …' (with the causal connective *-ni*).

The forerunner constructions of these apprehensionals were first attested in the early 20th century data. The following is a PDK example of *-cimolla* taken from a drama:

(12) [The speaker, tempted to check out the room of a man toward whom she has amorous feelings, now in fear of being caught, speaks to herself:]
Mos ha-y na-l cincca isanghan yeca-lu po-l-cimolla.
cannot do-END I-ACC truly weird woman-as see-FUT-APPR
'(I) just can't. I'm afraid (if I get caught) he might think I am a weird woman.' (2010 Drama *Kayinuy chwihyang* Episode #9)

In PDK, the apprehensionals in this group are used at a high frequency. The apprehensional *-cimolla* is more frequently used as a sentence-ender than as a connective, and signals that the speaker fears the undesirable situation being described by the proposition, whereas *-cimoluni* is more common as a connective, sometimes occurring in fortified variant forms, *-citomolla, -cimolunikka*, or

-*citomolunikka*. A corpus search returns 5,982 hits of *-cimolla/-cimoluni* forms (including their fortified forms), of which 2,085 (34.9%) carry the apprehensional function, i.e., marking a level of apprehension in undesirable events. In other instances, they simply denote uncertainty. About 719 tokens (34.51%) of all apprehensional uses are instances of the connective function and 1,366 tokens (65.5%), of the sentence-ender function. It is also noteworthy that despite their functional similarity, their distributional patterns are quite dissimilar, i.e., for -*cimolla* it is nearly even between the connective and sentence-ender functions (49.2% vs. 50.8%), -*citomolla* occurs predominantly in the sentence-ender function (98.0%), -*cimoluni(kka)* is more specialized in the connective function (74.2%), and -*citomoluni(kka)* is found only in the connective function (100%).

2.4.2 *-cimoshakey* and *-cimoshatolok*

The source constructions of -*cimoshakey* and -*cimoshatolok* are *-ci-mos-ha-key* and -*ci-mos-ha-tolok*, which involve the negative polarity nominalizer -*ci*, negation marker *mos* 'cannot', and the light verb *ha-* 'say/do' in common, and the mode-marking adverbializer -*key* 'in the mode of' or the purpose-marking particle -*tolok*, respectively, thus yielding the compositional meanings of 'in the mode in which x cannot' (with the mode-marker -*key*), or 'in order that x cannot' (with the purpose-marker -*tolok*).[12] It is noteworthy in this context that -*ci-mos-ha-*, before it developed into the apprehensionals -*cimoshakey* and -*cimoshatolok*, was commonly used in negative causative constructions to mean 'make someone unable to do x, make someone not do x' (but not associated with apprehension), e.g., 'make your daughter not make friends with the lowly', 'make someone unable to understand', etc. In historical texts, the causer and causee in negative causative constructions are (nearly) invariably humans. These semantic features seem to have been inherited by the apprehensionals at the time of their development and then bleached as the grammaticalization processes continue (see (13) below). The conceptual motivation of these forms developing into apprehensionals is intuitively straightforward, as they carry the meaning of negative purpose, corresponding to the English apprehensional *so that … not*.

[12] Incidentally, as an anonymous reviewer notes, in addition to negative sentences involving the nominalizer -*ci* (long-form negatives), Korean has -*ci*-less negatives (short-form negatives). Connectives from short-form negatives with the mode- or purpose-marker also carry the meaning of a situation to be avoided, e.g. *an o-key* [not come-MODE] 'so that (he) not come' or *mos o-tolok* [not come-PURP] 'so that (he) not be able to come'. However, unlike those addressed in this paper, these forms lack syntagmatic cohesion by virtue of discontinuous configuration, as [NEG V-MODE/PURP], in which the verb occurs between the two grammatical markers. These peripheral apprehensionals merit future research.

The forerunner constructions of these apprehensionals were first attested in the 17th century data, with the prohibitive-apprehensive function. In PDK, these apprehensionals have not advanced much in grammaticalization morphosyntactically. However, they departed from the pure syntactic construction with a compositional meaning in that in PDK they are no longer constrained by agentivity of the implicit imposer, or by a presupposed active action of the sentential subject. The distinction between the earlier and later apprehensional connectives in this respect is illustrated in (13), in which the older (13a) involves a human subject ('lowly servants') capable of an action ('raise their head to see') whereas the newer (13b) involves an inanimate subject ('wind') incapable of voluntary action (see §3.1 for more discussion):

(13) a. *Chyenchyul-i wulelepo-cimoshᴀkey myengpwun-ul emhi*
lowly.servant-NOM look.up-APPR order-ACC sternly
hᴀ-si-ko ...
do-HON-and
'(You) give orders in a stern manner so that the lowly servants would not (dare to) raise their head to see you, and ...' (18xx *Hancwunglok* #84)

b. *Palam-i tuleo-cimoshakey tantanhi kkun-to mwukk-ecwu-mye ...*
wind-NOM enter-APPR tightly string-also tie-BEN-as
'As (he) tightly ties the string of the jacket hood so that the wind may not come into the jacket through the hood opening, ...' (2009, Drama *Khulisumasuey nwuni olkkayo* Episode #5)

The following is an example of the apprehensional -*cimoshakey* in a Modern Korean text:

(14) *Sanyangkwun-i cumsayng-ul cap-nuntey ku cumsayng*
hunter-NOM animal-ACC capture-while that animal
po-cimoshakey mom-ul swumky-e.
see-APPR body-ACC hide-and
'When a hunter is capturing an animal, (he) hides himself so that it cannot see (him).' (1903 *Sinhakwelpo* 3)

In PDK, -*cimoshakey* is moderately frequent, but -*cimoshatolok*, only occurring in literary texts, is not frequently used as an apprehensional. A corpus search returns 533 instances of -*cimoshakey*, of which 446 tokens (83.7%) are instances of the apprehensional connective usage. Other tokens are instances of manner modifier for stative verbs, many of which are idiomatized or lexicalized (e.g., *cemcanh-cimoshakey* [be.mannerly-*cimoshakey*] 'in an unmannerly way'). On the other hand, the search returns only 96 instances of -*cimoshatolok*, all of which are of the apprehensional connective usage.

2.4.3 -*cianhkey* and -*cianhtolok*

The source constructions of -*cianhkey* and -*ciahntolok* are -*ci-ani-ha-key* and -*ci-ani-ha-tolok*, which all have the negative-polarity nominalizer -*ci*, negation marker *ani* 'not', and the light verb *ha-* 'say/do' in common, and in addition involve the mode-marking adverbializer -*key* 'in the mode of' or the purpose-marking particle -*tolok*, respectively, thus yielding the compositional meanings of 'in the mode in which x does not' (with the mode-marker -*key*), or 'in order that x does not' (with the purpose-marker -*tolok*). It is clear that these are very close relatives of the apprehensionals -*cimoshakey* and -*cimoshatolok*, discussed above, the only difference being *mos* 'cannot' vs. *anh* 'do not', the former involving the semantic feature of inability. The conceptual motivation of these forms developing into apprehensionals is also intuitively straightforward, as they carry the meaning of negative purpose, corresponding to the English apprehensional *lest*.

The forerunner constructions were first attested in the 18$^{\text{th}}$ century data, with a weaker preventive-apprehensional function, i.e., 'so as one does not' (as compared to -*cimoshakey* 'so as one cannot'). Just as is the case with their relatives, in PDK, these apprehensionals have not advanced much in grammaticalization morphosyntactically, but departed from the pure syntactic construction with a compositional meaning in that in PDK they are no longer constrained by agentivity of the implicit imposer, or presupposed active action of the sentential subject (cf. §2.4; see §3.1 for more discussion). (15) is taken from a Modern Korean source:

(15) (The Jews to Prophet Samuel)
Wuli-lulwihaya cwuk-cianihakey ne-uy cwu hananim-skuy
we-for die-APPR you-GEN lord God-to
kuytoha-ap-sose.
pray-HON-OPT
'[Since we have done wrong] please pray for us to your Lord God lest we die (for our sins).' (1903 *Sinhakwelpo*)

In PDK, -*cianhkey* is highly frequent and -*cianhtolok* is moderately frequent. A corpus search returns 4,647 hits of -*cianhkey*, of which about 2,983 tokens (64.2%)

are in apprehensional use. Non-apprehensional uses involve, in parallel with *-cimoshakey* discussed above, the form occurring with stative verbs modifying them with respect to manner (e.g., *namcatap-cianhkey* [be.manly-*cianhkey*] 'in an unmanly manner; in an ungentlemanly manner'). The instances of the apprehensional use are all in the connective function, even though some of the attested examples are inversions with the overtone of after-thought addition. The search also returns 538 hits of *-cianhtolok*, all of which are instances of the apprehensional connective use.

3 Discussion on theoretical issues

3.1 Grammaticalization parameters

Since the development of apprehensionals from compositional source constructions is an instance of grammaticalization, discussion of the developmental processes with respect to grammaticalization parameters is in order. Among a number of proposed principles and hypotheses, we adopt the mechanisms presented in Heine & Kuteva (2002: 2), i.e. desemanticization, extension, decategorialization, and erosion (for general exposition of each parameter, see Heine & Kuteva 2007: 35–44).

3.1.1 Desemanticization

Also commonly called "semantic bleaching" in figurative terms, desemanticization refers to the loss of meaning content. As briefly alluded to in the preceding exposition, lexical items have undergone desemanticization, thus creating a "divergence" (Hopper 1991) between the lexical item in the source construction and the grammatical item as a component of an apprehensional. For instance, there are six lexical items that participate in the construction of an apprehensional, namely, *po-* 'see' in *-lkkapwa, mwusep-* 'be fearful' in *-lkkamwusewe, molu-* 'not know' in *-lkkamolla, -cimolla,* and *-cimoluni, siph-* 'think, inclined to think' in *-lkkasiphe*, *SA* 'occasion' in *-lseyla,* and *ha-* 'do/say' in *-cimoshakey, -cimoshatolok, -cianhkey,* and *-cianhtolok* (note that the light verb *ha-* 'do/say' in *-cimoshakey*, etc. is already only marginally lexical). These lexical items have undergone semantic bleaching to variable degrees as shown in Table 2, which lists the forms of the source lexemes, their glosses and lexical core meanings, and their meanings in the apprehensionals.

Semantic bleaching is not restricted to lexical items only. The functions associated with grammatical forms have also become weakened. For instance, the

Table 2: Lexical sources of Korean apprehensionals

po-	‘see’	visual perception	experience, be in
mwusep-	‘be fearful’	feeling of fear	be concerned
molu-	‘not know’	absence of knowledge	uncertainty
siph-	‘think’	cogitation	possibility
ha-	‘do/say’	action, locution	be
SA	‘occasion’	temporal event	be in

sentence-ender -*kka,* whose characteristic function is to mark an interrogative mood, does not carry this speech act meaning of posing a question when it is part of the apprehensional forms.[13] It simply signals indeterminacy (thus uncertainty) associated with a question.

3.1.2 Extension

Also known as context generalization, extension refers to the use of a form in new contexts. The apprehensional source constructions, before their grammaticalization, retained the semantic and grammatical functions associated with each component of the source construction. This semantic-functional retention in the constructions may constrain their usage (cf. “persistence” (Hopper 1991)).

However, when the constructions undergo desemanticization, as illustrated above, the constraints on their usage may be loosened, whereby the grammaticalizing forms are enabled to be used in new contexts. For instance, when the verb *po-* ‘see’ is used in the source construction of the apprehensional -*lkkapwa,* the usage is evidently restricted to a context in which the sentential subject actually visually perceives an object. When the construction is fully grammaticalized as an apprehensional, the form is not restricted to such situations. Rather, the event being described is typically *irrealis*, e.g., an imagined war in the future (as in (1b)), the possibility of having one’s silk dress stained (as in (5)), etc.

Furthermore, in §2.4, as we discussed the cases of -*cimoshakey*, -*cimoshatolok*, -*cianhkey*, and -*cianhtolok*, we noted that the negative purpose meaning of these source forms is largely retained even after grammaticalization, but that the fully grammaticalized apprehensionals are no longer constrained by the requirements

[13]The complete loss of the interrogative function in the interrogative sentence-ender -*kka* is only observable with apprehensionals. Rhetorical questions lack illocutionary force but are still perceived as questions by language users.

of agentivity of the implicit imposer, or a potential active action of the sentential subject. This can be illustrated with the grammaticalization of *-cimoshatolok*. When *-cimoshatolok* is used while the source meanings of the constructional components are fully operative, i.e., of the nominalizer (*-ci*), 'cannot' (*mos*), 'do' (*ha-*), and 'for the purpose of' (*-tolok*), there are restrictions imposed by these forms. For instance, the expression *o-cimoshatolok* with the verb *o-* 'come' would mean 'in order to prevent (someone) from coming', involving the purposive action of prevention on the part of the imposer (associated with the purposive *-tolok*), and the potential, but foiled, action (i.e., 'coming') of a volitive agent on the part of the sentential subject (associated with *mos ha-* 'cannot do'). However, the fully grammaticalized apprehensional *-cimoshatolok* can be used in contexts not constrained by such requirements, as in 'lest the food be exposed to the direct sunlight' (from lit. 'in order to prevent the direct sunlight from reaching the food'), etc. (see (13) above for similar examples).

3.1.3 Decategorialization

Decategorialization refers to the loss in morphosyntactic properties characteristic of lexical or other less grammaticalized forms. Grammaticalization of apprehensionals involves decategorialization. Since decategorialization typically involves, but is not restricted to, the primary categories such as nouns and verbs, we will look at such items as listed in Table 2 above, i.e. *po-* 'see', *mwusep-* 'be fearful', *molu-* 'not know', *siph-* 'think', and *sA* 'occasion'. Among the characteristic properties of the verbal category are modifiability by adverbs, inflection by tense-aspect-modality modulation, etc. and those of the nominal category are modifiability by adjectives, pluralizability, etc. However, all the lexical forms involved in the grammaticalized apprehensionals have lost their primary-category properties, as in, for example, *-*lkkacaseyhipwa* (*caseyhi* 'carefully' modifying *po-* 'see'), *-*lkkamaywumwusewe* (*maywu* 'very' modifying *mwusep-* 'be fearful'), *-*lkkacenhyemolla* (*cenhye* '(not) at all' modifying *molu-* 'not know'), *-*lkkakiphhisiphe* (*kiphhi* 'profoundly' modifying *siph-* 'think'), *-*lsAtulila* (*-tul* plural marker for *sA* 'occasion'), etc. The decategorialization evidenced by the ungrammaticality of modified forms applies to all the items listed in Table 2.

3.1.4 Erosion

Also known as phonetic reduction, erosion refers to the loss in phonetic substance. Korean has a large number of grammatical markers that have undergone extensive or drastic erosion, partly due to simplification of articulatory gestures as well as the language's agglutinating nature, e.g., *-la-ko-ha-nun-kes-un* [END-CONN-say-ADN-thing-TOP] > *-lan* [TOP] (Rhee 2011: 764). However, this parameter is not prominent in the development of apprehensionals in Korean. As shown in Table 1 in §2.1, the apprehensionals and their source constructions do not show any meaningful degrees of phonetic erosion. The only notable reduction occurs with the apprehensionals *-cianhkey* and *-cianhtolok*, in which *ani-ha-* 'not-do' changed to *anh-*. However, this change is not specific to these apprehensionals only; rather it applies to all instances of such a string in Korean.

One aspect relevant to this relatively lesser degree of phonetic reduction is that it may have to do with the fact that the grammatical notion of apprehension in many of the apprehensionals discussed above is not (yet) fully entrenched. In other words, many of them have functional flexibility and can be used in non-apprehensional function (note, for example, the tentative intention function in *-lkkapwa, -lkkasiphe,* etc. in §2.2). Among the apprehensionals that do not exhibit such functional flexibility are *-lla* and *-lseyla.* This situation is captured by the 'parallel reduction hypothesis' (Bybee et al. 1994: 19–21), according to which phonological reduction and semantic bleaching occur in tandem.[14] Another form that does not exhibit functional flexibility, i.e., one that is used only as an apprehensional marker, is *-lkkamwusewe*, which has not undergone much phonetic reduction. Obviously, the functional specialization of *-lkkamwusewe* as an apprehensional is due to the semantics of its source, involving the verb *mwusep-* 'be fearful'. Even so, the semantics of the source lexeme is not intact but has undergone bleaching from 'genuine fear' to 'concern' in the course of grammaticalization (see §2.2 above).

In this context there is one important aspect of functional change that merits our attention, i.e., pragmatic strengthening or conventionalization of pragmatic inference. In the cases discussed above, the source structures did not involve lexemes explicitly denoting 'fear', except for *-lkkamwusewe* from *mwusep-* 'be fearful'. They originated from constructions involving perceptual or cognitive verbs denoting 'see', 'not know', 'think', etc. (see §2.1 above). The apprehensional meanings that emerged from grammaticalization are the results of semanticization of the pragmatic inferences from the source constructions that involve such perceptual or cognitive verbs as well as the grammatical markers of indeterminacy, e.g.,

[14] The relationship between formal change and conceptual change is a controversial issue. Some researchers assert that formal change precedes conceptual change (e.g., Lightfoot 1989, 1991), others suggest the reverse (e.g., Sapir 1921, Givón 1975, Heine et al. 1991, Heine 1993), and still others suggest that the two cooccur (e.g., Hopper & Traugott 2003[1993], Bybee et al. 1994). It is difficult to establish a definite order, but it is evident that they interact with each other.

futurity, prospective, mode, purpose, and question. When these notions are combined, the source constructions have stronger potential to invite the inference of potential harm, which eventually becomes conventionalized (see §3.3 below for discussion on the human propensity for associating uncertainty with negative emotion, i.e., fear). Thus, the acquisition of the apprehensional meaning proceeds from a more or less neutral to a negative meaning, i.e., apprehension. This points to the fact that grammaticalization cannot be generalized as a uniformly reductive process of "erosion" or "bleaching". Rather, there is reduction, largely with respect to phonetic substance, but there is also strengthening, largely with respect to meaning or function (cf. "the loss-and-gain model" (Heine et al. 1991: 109–111)). The role of pragmatic inference is discussed in more detail in §3.4.

3.2 Multiple layers and specialization

As is shown in the preceding exposition, there are multiple apprehensionals that emerged in the history of Korean, which have been grammaticalized to varying degrees from diverse constructions, thus forming multiple layers in contemporary Korean. A historical survey shows that the earliest forms are -*lseyla* and -*lkka,* used in Late Middle Korean. Even though -*lseyla* remains infrequent in PDK, -*lkka* is itself a moderately frequent apprehensional and furthermore has served as the basis of other future-and-question-based apprehensionals in the course of history (see §2.2), creating multiple branches of "divergence" (Hopper 1991) or "split" (Heine & Reh 1984). The developmental patterns are shown in Figure 1, in which the arbitrary groupings of "infrequent" (1–10 pmw), "moderately frequent" (11–50 pmw), and "highly frequent" (>50 pmw) are used and the thickness of the line represents the level of frequency.

When multiple forms carry an identical or similar function, there occurs functional tension among them and certain forms may gain functional primacy over others, pushing other competitors outside the functional domain or into other genre- or register-specific usage within the domain. This process has been termed "specialization" (Hopper 1991). From a statistical analysis of the frequency in the Modern Korean corpus, the primary apprehensional is -*lkkapwa*, with 276 occurrences per million words. It is followed by -*cianhkey* (123 pmw) and -*cimolla* (55 pmw). The relative functional strengths of the apprehensionals in PDK are graphically presented in Figure 2.

From Figure 1 and Figure 2, one notable aspect of the divergence and specialization patterns is that the most frequent apprehensionals in PDK are not those with the longest history of grammaticalization. Conversely, those with the shortest history are not the least frequent in PDK. For instance, the most frequently

Figure 1: Historical development of apprehensionals

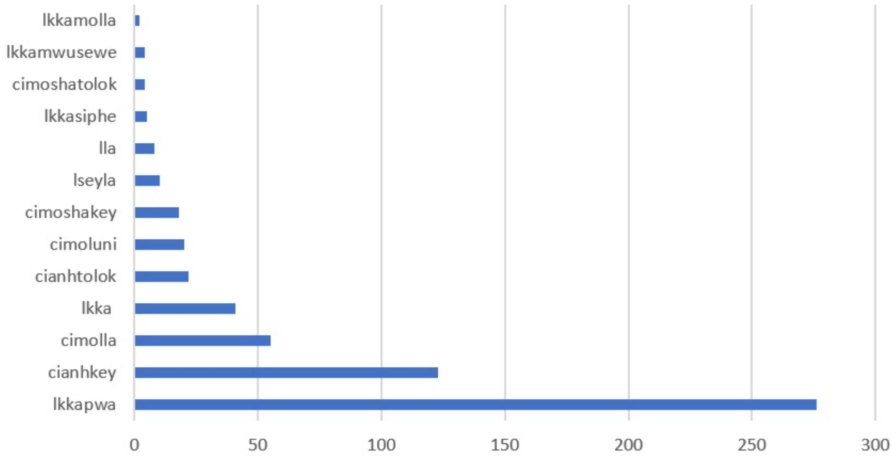

Figure 2: Frequency of apprehensionals in PDK (Drama-Movies corpus)

used apprehensionals such as -*lkkapwa*, -*cianhkey*, and -*cimolla* came to be used as apprehensionals only from the 19th, 18th, and 20th century onwards, respectively, whereas the least frequently used apprehensionals such as -*lkkamolla,* -*lkkamwusewe,* and -*cimoshatolok* developed into apprehensionals from the 17th, 19th, and 17th century, respectively.

Also important in this context is the notion of the degree of grammaticalization. Even though instances of grammaticalization are often discussed as of variable degrees, e.g. "highly" grammaticalized, "weakly" grammaticalized, etc., there is no established method to measure the degree of grammaticalization. We may use the four parameters discussed above, i.e., desemanticization, extension, decategorialization and erosion, to examine how much change has occurred along these parameters. Alternatively, Lehmann's (2015[1982]) six parameters may serve as a diagnostic, i.e., the degree of attrition (reduction of semantic content and/or phonetic substance), paradigmaticization (decrease of the paradigm size), obligatorification (use increasingly becoming obligatory), condensation (narrowing of structural scope), coalescence (increase of bondedness), and fixation (reduction of syntagmatic variability). Either way, however, since the parameters operate at different domains of grammar, such as phonology, morphology, semantics, and syntax, and since the source concepts may already be grammaticalized to some degree, different levels of change cannot be quantified for direct comparison in determining relative degrees of grammaticalization across parameters. These practical problems notwithstanding, we can hypothesize, following Hopper & Traugott (2003[1993]: 106), that "the more frequently a form occurs in texts, the more grammatical it is assumed to be" and that textual frequency may represent different degrees of grammaticalization of multiple forms in the same or similar functional paradigm (see, however, Heine 1992 for discussion of exceptions as well as Heine & Kuteva 2007: 38–39 for the lack of a clear correlation between frequency and grammaticalization).

From this perspective, the state of affairs of the Korean apprehensionals suggests that *-lkkapwa* is the most grammaticalized and *-lkkamolla* is the least grammaticalized, and further that the historical depth and the degree of grammaticalization are not coextensive.

3.3 The role of source concepts

The role of source semantics in grammaticalization is crucial because grammaticalization is basically a conceptually motivated process (Heine 1986, 1997, Heine & Claudi 1986, Heine et al. 1991, Kuteva 2001). We noted in §2.1 that there is a certain limited number of concepts that are involved in the source constructions, i.e. lexical verbs of perception and cognition such as 'see', 'be fearful', 'not know', and 'think' and the grammatical concepts of futurity, question, mode, and purpose. The relation between the source concepts and the resultant grammatical functions has been noted by grammaticalization researchers (cf. the source determination hypothesis by Bybee et al. (1994)), and we are going to consider the development of apprehensionals in this light.

The contribution of cognitive verbs such as 'fear', 'not know', and '(be inclined to) think' and the perception verb 'see' to the development of apprehension is obvious, since apprehension is an inherently cognitive/perceptual state of affairs (cf. "apprehensional-epistemic" in Lichtenberk (1995)). As we noted in §2.2, the oldest predecessor construction of apprehensional *-lkka* in Late Middle Korean was followed by verbs of cognition, e.g. 'worry', 'doubt', 'lament', etc. (see §2.2 for a more extensive list), while *-lkka* was only a combination of the future marker (*-l-*) and the interrogative sentence-ender (*-kka*). When the internal cohesion in the collocation increased, the collocational constructions became formally univerbated and conceptually integrated into the set of apprehensionals, thus leading to the emergence of multiple apprehensionals, e.g. *-lkkasiphe* (with 'think'), *-lkkamolla* (with 'not know'), *-lkkamwusewe* (with 'be fearful of'), and *-lkkapwa* (with 'see').[15]

[15] Conceptual integration cannot be easily proven with quantifiable criteria. In Korean, however, one such indication appears in the way people write. These apprehensionals are structurally

It is interesting to note that in the development of apprehensionals from *-lkka*, the 'fear'-semantics of the co-occurring verbs or propositions denoting undesirable events seems to have been absorbed by *-lkka*, a process whereby *-lkka* itself became an apprehensional around the 18th century. This is a plausible scenario, a process that Bybee et al. (1994) labeled 'absorption of contextual meaning' and included among the possible grammaticalization mechanisms (see Bybee et al. 1994: 230–236 for illustration of modal meanings). Similarly, Rhee (2017) shows that the semantics of many sentence-final particles that developed from insubordination are in fact the inferred meanings of the imagined reconstruction of an elided main clause, instances of borrowed semanticization from neighboring forms. A caveat is that the process of borrowed semanticization or absorption has not happened in full, and as a result there are non-apprehensional usages of *-lkka* in PDK (see §2.2).

Another important aspect of the contribution of source semantics is that not only the lexical semantics but also grammatical concepts associated with the source constructions are important contributors. It is evident that futurity plays an important role in the development of apprehensionals, as shown by the fact that seven out of thirteen developed from constructions containing the future tense marker *-l-*. The semantics of the particles for mode (*-key*) and purpose (*-tolok*), each involved in two apprehensionals, is also inherently futuristic, in the sense that the event marked by them is intended by the (implied) sentential subject but has not yet been realized. Interrogative marking also plays an important role.[16] Five of the apprehensional markers have the question marker *-kka* as a component. The grammatical notions of futurity and interrogative are significant because they bring forth the notions of uncertainty and indeterminateness, which, in turn, engender the notion of apprehension. It is true that the source of apprehension does not have to be uncertainty, considering that we may become particularly apprehensive when we are positive that something bad is going to happen.[17] However, it is also true that the uncertainty-anxiety relationship has long been recognized in human psychology and neuroscience. For instance, it has been noted that the human brain is an "anticipation machine" (Gilbert 2006,

compositional and thus need to be written with inter-lexical spacing according to orthographic rules, but they are often written without spaces at the word boundary in informal writing (e.g., *-lkkasiphe* instead of *-lkka siphe*; *-lkkamolla* instead of *-lkka molla*, etc.). Incidentally, disappearance of interlexical spacing is commonly observed in the grammaticalization of multiword constructions such as auxiliary verbs, complex postpositions, sentence-final particles, etc.

[16]The connective *-key*, also often referred to as a converb marker, is polyfunctional but its primary function is marking the mode of the action or state denoted by the verb (Rhee 1996). For diverse functions of *-key* in grammaticalized constructions, see Kim (2011).

[17]We thank an anonymous reviewer who brought this insightful point to our attention.

as cited in Grupe & Nitschke 2013: 488), and that people have a general propensity toward negative affective responses to uncertainty (Anderson et al. 2019, see also Carleton 2016b,a).

3.4 Pragmatic inference, reinterpretation, and subjectification

It is widely recognized that grammaticalization often results from pragmatic inference (Traugott 1989, Hopper & Traugott 2003[1993], Slobin 1994, Bybee et al. 1994, Fischer 2008, and numerous others). The process can be illustrated with the apprehensional connective *-lkkapwa*, developed from *-l-kka-po-a* [FUT-Q-see-CONN], exemplified in (1b), with modification:

(16) Connective *-lkkapwa* 'for fear of'

a. *Ku-nun cencayng-i na-lkkapwa cam-ul mos ca-n-ta.*
he-TOP war-NOM come.out-APPR sleep-ACC cannot sleep-PRS-DECL
'He cannot sleep for fear of a war.'

b. *Ku-nun cencayng-i na-l-kka pw-a cam-ul mos ca-n-ta.*
he-TOP war-NOM come.out-FUT-Q see-CONN sleep-ACC cannot sleep-PRS-DECL
Lit. 'As he sees/saw (it), (saying,) "Will a war break out?", he cannot sleep.'

Before the poly-morphemic *-lkkapwa* was fully grammaticalized, the meaning of the example (16a), based on the components constituting *-lkkapwa*, would have been (16b), 'As he sees/saw (it), (saying,) "Will a war break out?", he cannot sleep.' This interpretation, faithful to the meanings of the source components, makes reference to 'his seeing (or having seen) something'. However, since the war has not (yet) broken out, actual occurrence of visual perception of the war is not likely. Probable events may be only hearing rumors about a war or sensing the possibility in one way or another. Therefore, making reference to seeing (by using the verb *po-* 'see') may be interpreted by the hearer as either a hyperbole or a metaphor of expressing a cognitive state (i.e. knowledge) by means of bodily perception (i.e. seeing), a crosslinguistically common instance of metaphorization (see Lakoff & Johnson 1980, Sweetser 1990).

Also, it is not clear if the referent of *ku-nun* 'he' in (16a) really uttered the question "Will a war break out?" In fact, it is not likely that the question was actually asked by him, because *-lkka* as a regular future interrogative marker (not as an apprehensional) is typically a monologic sentence-ender (a type of "audience-blind" form, Rhee & Koo 2017: 105).[18] In other words, the speaker imagines that the sentential subject of an apprehensional sentence asks himself or herself if something will happen. Thus, the question exists only in the mind of the speaker. In this process, the sentence-ender of the question in the future tense is pragmatically reinterpreted as a connective of a clause that marks the cause of apprehension. This type of figurative hyperbole is a common expressive strategy in Korean lexicalization and grammaticalization, observed by Rhee (2009), who labels it a "through a borrowed mouth" phenomenon.[19] What is noteworthy is that all this expressive enrichment involves pragmatical inference, i.e., from "a question is a speech act involving the speaker's uncertainty" to "uncertainty about the future generates fear" and further to "a question about the future marks fear".

Pragmatic inference may well – but does not have to – lead to subjectification, a process whereby "meanings based in the external described situation [become] meanings based in the internal (evaluative/perceptual/cognitive) situation", or "meanings tend to become increasingly situated in the speaker's subjective belief-state/attitude toward the situation" (Traugott & König 1991: 208–209), a frequent semantic concomitant in grammaticalization (Traugott 2003). In this light, the semantic/functional change through pragmatic inference described above is a paradigm example of subjectification, i.e., the change from 'as one sees something and asks "Will x occur?"' to 'for fear of x' is a shift from a description of the world (i.e., the external described situation) to one of a mental world (i.e., the meaning situated in the speaker's subjective attitude).

Similarly, the development of apprehension from uncertainty and indeterminateness is a clear instance of subjectification. Uncertainty is a cognitive state often resulting from the indeterminateness of a situation, whereas apprehension is more subjective by virtue of being an affective state. Even when apprehensionals do not mark fear but potential undesirability, the evaluative judgment of a future event involves subjectivity.[20] Similarly, the development of the verb *molu-* 'not know' to the apprehensionals *-lkkamolla, -cimolla,* and *-cimoluni* is

[18]For this characteristic of the sentence-ender *-lkka*, the clause headed by *-lkkapwa,* embedding a question thus marked, represents a situation in which the sentential subject is asking a monologic question to himself/herself while looking at an event or state.

[19]Rhee (2009) lists descriptive adverbials that originated from quotation constructions, e.g., *cwuk-keyss-ta-ko* 'saying "I will die"' to *cwukkeysstako* 'desperately', *michyess-ta-ko* 'saying "I am insane"' to *michyesstako* 'nonsensically', etc. The quotations "I will die", "I am insane" did not originate from the person being described but from the imagination of the observer.

[20]It is noteworthy in this context that Kim (2021) notes in her analysis of *-ltheyntey* from a connective (e.g., 'while P will occur') to a final particle (e.g., 'It is regrettable that P') that the grammaticalization involves semantic extension into wishing, regretting, worrying, etc., in which conjecture and context play important roles.

an instance of subjectification, along the series of changes as 'have no knowledge' > 'be unsure' > 'be uncertain' > 'be worrisome' > 'be apprehensive'. The series of changes along the conceptual continuum strongly suggests developmental directionality from epistemic possibility to apprehension (Dobrushina 2006, Pakendorf & Schalley 2007, among others).

When pragmatic inferences occur, often involving subjectification, there also occur morphosyntactic reanalysis and functional reinterpretation, i.e., a polymorphemic construction with compositional meaning being reanalyzed as a single form with grammatical meaning. Certain types of morphosyntactic change can result in much more drastic consequences. For instance, we noted in §2.1 that the resultant grammatical category (i.e. connective or sentence-ender) of most instances of apprehensionals is identical with the grammatical category of the final element in the source construction and that there are three exceptional cases, i.e. *-lkka, -lla*, and *-lseyla* ((a), (f), and (g) in Table 1). They all end with a sentence-ender: *-lkka* with an interrogative sentence-ender *-kka*, and *-lla* and *-lseyla* with a declarative sentence-ender with some nuance of exclamative, *-la*. However, *-lkka* is now an apprehensional connective, not an apprehensional sentence-ender, and *-lla* and *-lseyla* end with a sentence-ender *-la* but now as apprehensionals they also function as connectives. This state of affairs warrants some discussion.

The source construction of *-lkka*, as noted in §2.2, is an interrogative sentence in the form of direct quotation, i.e. an embedded question. However, when the direct quote provides a cause/reason as the background of the main clause, the form at the boundary of embedding, i.e. the end of the embedded sentence, *-lkka*, is functionally reinterpreted as a causal connective (linking the state or event feared by the speaker or the sentential subject to its consequence). Let us look at the following example:

(17) a. *Cencaying-i na-l-kka cam-i an o-n-ta.*
war-NOM come.out-FUT-Q.END sleep-NOM not come-PRS-DECL
'Will a war break out? (I) cannot go to sleep.'

b. *Cencaying-i na-lkka cam-i an o-n-ta.*
war-NOM come.out-APPR.CONN sleep-NOM not come-PRS-DECL
'(I) cannot sleep for fear of a war (breaking out).'

The change from (17a) to (17b) clearly shows the inter-clausal morphosyntactic reanalysis of regarding the sentence-final *-l-kka* as a clause-final connective and functional reinterpretation of regarding the marker of a question with respect to the future as a marker of apprehension, thus from a sentence-ender to a connective. As a matter of fact, this type of change is pervasive in the grammaticalization of connectives in Korean, which involves "self-quotative constructions" (SQCs) (Koo & Rhee 2019: 256), -*lkka* being one of them. The following is an example of SQC connective -*na*, whose original function is marking an interrogative sentence, just like the apprehensional -*lkka* being discussed here:

(18) a. Connective (causal) -*na*
Amwu-to *eps-**na*** *coyongha-ta.*
anyone-even not.exist-**CONN** be.quiet-DECL
'It's quiet, perhaps because there's nobody around.'

b. Source construction -*na* '-Q'
Amwu-to *eps-**na*** *coyongha-ta.*
anyone-even not.exist-**Q** be.quiet-DECL
Lit. '"Is nobody here?" it is quiet.' (Koo & Rhee 2019: 267)

The source-construction example (18b) consists of two sentences: one monologic question and one (subject-less) declarative sentence. It is noteworthy that two juxtaposed sentences of this type can undergo coalescence (Haspelmath 2011) into a single sentence (as in (18a)). In this process, the sentence-final element (the question particle -*na* in (18b)) develops into a connective (the causality marker in (18a)). The examples in (18) show a change analogous to the one with -*lkka*, from a sentence-ender to a connective; specifically, from a question marker to a causal connective (note, however, that -*na* does not encode apprehension).

A similar pattern is also observed in the development of apprehensional connectives from apprehensional sentence-enders, such as -*lla* and -*lseyla*. The apprehensionals -*lla* and -*lseyla* end with a sentence-ender -*la* in their source constructions. Previous studies analyze them as sentence-enders (Ramstedt 1997[1939], Kim 1960, Ko 2002[1987], Jang 2002). Let us look at the example (10) illustrating the apprehensional -*lla*, repeated here for convenience as (19a):

(19) a. [In preparation of funeral]
Pang tewu-lla *pwul stʌy-cimal-ko* *ko-y* *tuleka-lla*
room be.hot-APPR fire make-PROH-and cat-NOM enter-APPR
kwistol *mak-elʌ.*
flue.opening block-IMP
'Don't make fire (for floor-heating) lest the room become too hot (for the corpse), and block the flue opening lest a cat enter (through it) [which is ominous according to superstitions].' (ca. 1870 *Pakhungpoka*)

b. *Pang tewu-**l-la*** *pwul stʌy-cimal-ko* *ko-y* *tuleka-**l-la***
room be.hot-**FUT-END** fire make-PROH-and cat-NOM enter-**FUT-END**
kwistol *mak-elʌ.*
flue.opening block-IMP
'I'm afraid the room will become too hot. Don't make fire (for floor-heating), and I'm afraid a cat will enter. Block the flue opening.'

The translation (19a) of the example is based on the interpretation of *-lla* as a connective. If we follow the analyses referenced above, the example would be translated as (19b) with the structure of: [$_{S1}$ I'm afraid the room will become too hot.] [$_{S2}$ Don't make fire (for floor-heating)], and [$_{S3}$ I'm afraid a cat will enter.] [$_{S4}$ Block the flue opening.], containing four finite sentences, with S2 and S3 constituting a coordinated sentence, combined by the coordinating connective *-ko* 'and'. Note, however, that conceptual cohesion does not exist between S2 and S3. Instead, the conceptual cohesion is found between S1 and S2, and between S3 and S4, respectively. In other words, S1 is the reason for saying S2, and S3 for saying S4. The connections between S1 and S2 and between S3 and S4 are pragmatic causality or "enablement" (cf. Sweetser 1990: 77).[21] Therefore, the example is better analyzed as a sentence constituting two sentences in coordination, each consisting of a subordinate clause and an imperative main clause: [$_{S}$[$_{S1}$[$_{SUB.CL}$ Reason-APPR.CONN] [$_{MAIN.CL}$ Command]]-COORD [$_{S2}$[$_{SUB.CL}$ Reason-APPR.CONN] [$_{MAIN.CL}$ Command]]]. In other words, the connective interpretation in (19a) is better than the sentence-ender interpretation in (19b). This situation is closely related to the crosslinguistic observation that apprehensionals are often used in the contexts of "imperative indicating the action the hearer should undertake to avoid the unpleasant outcome" and "prohibitive indicating the action the hearer should avoid" (Schultze-Berndt 2015) and that apprehensionals are linked to warning/threat and negative command (prohibitive) (Pakendorf & Schalley 2007, as cited in Schultze-Berndt 2015).

Therefore, the pragmatic motivation for attributing a causal relation between two sentences and for redefining the grammatical status of the sentences seems to have led to the reanalysis of a former sentence-ender into a connective. This type of reanalysis has occurred in the grammaticalization of former sentence-enders into connectives (as shown in (18) and (19) above, and conversely, and more frequently, in the grammaticalization of former connectives into sentence-enders through the processes of *insubordination* (Evans 2007, Evans & Watanabe

[21]In her discussion of conjunctions with an example of "What are you doing tonight, because there's a good movie on," Sweetser (1990: 77) interprets the function of the causal conjunction *because* as a means of justifying and enabling the question, thus the cause of speech act.

2016) and *incoordination* (Kuteva et al. 2015, 2017). In fact, pragmatic inferences that operate in insubordination and incoordination prompted the functional extension of some apprehensional connectives to the paradigm of sentence-final particles, marking tentative intention, doubt, indeterminateness, possibility, etc. (see Koo & Rhee 2001, 2013, Sohn 2003, Rhee & Koo 2017).

4 Summary and conclusion

Despite the fact that Korean has a large number of apprehensionals, their developmental paths have not received any serious analysis to date. As we intended to address all apprehensionals being used in PDK, we were unable to trace all the grammaticalization trajectories of the individual forms diachronically, but we have shown that Korean apprehensionals have differential historical depths and varying degrees of specialization. They form multiple layers in the functional domain of apprehensionals in PDK.

Among the four main interrelated grammaticalization mechanisms (Heine & Kuteva 2002: 2), i.e., desemanticization, extension, decategorialization, and erosion, erosion is the least prominent one. Unlike highly grammaticalized markers which tend to be short in form, many apprehensionals are still at the periphrastic stage in morpho-syntactic terms. This suggests that apprehensional markers in Korean have not acquired the status of full-fledged apprehensionals. In fact, many of them can still be used for non-apprehensional functions. This functional split is prominent in their use as an apprehensional connective and as a non-apprehensional sentence-ender. Further, some apprehensionals originated from structurally complex constructions through strengthening of pragmatic inferences based on the semantics of the components, notably epistemic modals, whereas the 'fear'-based apprehensionals incorporated a 'fear' verb and thus their apprehensional function was visible throughout their history. Multiple forms synchronically layered in the functional domain of apprehensionals have variable specializations as shown in their frequency of use.

Lexical semantics as well as the functions of grammatical forms in the source constructions contributed to the creation of the notion of apprehension. Of particular importance are (i) the perception/cognition verbs denoting 'be fearful', 'not know', 'think', and 'see' in the lexical sources, and (ii) futurity, interrogative, mode and purpose from the grammatical formants.

In the course of development of apprehensionals, pragmatic inferences, subjectification, morphosyntactic reanalysis, and functional reinterpretation are prominent driving forces. In particular, the cross-categorial flexibility between connectives and sentence-enders is largely due to the pragmatic motivations that attribute a causal relation between two sentences and redefine the grammatical status of the sentences. These developmental patterns are not isolated instances but are commonly attested in the grammaticalization scenarios involving insubordination and incoordination in the history of Korean.

Abbreviations

ACC	accusative
ADN	adnominalizer
APPR	apprehensional
BEN	benefactive
COMP	complementizer
CONJ	conjecture
CONN	connective
DECL	declarative
END	sentence-ender
FUT	future
GEN	genitive
HON	honorific
IMP	imperative
IND	indicative
MODE	mode-adverbializer
NOM	nominative
NOMZ	nominalizer
OPT	optative
PDK	present day Korean
PL	plural
PROG	progressive
PROH	prohibitive
PRS	present
PST	past
PURP	purposive
Q	question particle/marker
RETRO	retrospective
TOP	topic
VOC	vocative

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Chapter 12

Explicit apprehensions, implicit instructions: An indirect speech act in the grammar

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Among the languages that grammaticalize the apprehensive domain, some use a subordinator like English *lest* ('Don't run, *lest* you fall'); others have an apprehensive mood in their verb system (≈ 'Don't run, you *might* fall!'). The Oceanic languages of north Vanuatu, whose apprehensional strategies are quite diverse, feature both strategies. This study will focus on Mwotlap and its apprehensive mood. This modal marker appears to imply a form of interclausal dependency; yet rather than being due to syntactic subordination, this dependency effect is arguably pragmatic in nature. Indeed, by exposing an event as a risk to be avoided (e.g. 'you might fall'), the apprehensive clause serves to justify a certain request ('don't run'). Sometimes, only the apprehension is made explicit ('You might fall!'), leading the hearer to reconstruct the intended order. The apprehensive mood thus reflects the grammaticalization of an indirect speech act: one where explicit apprehensions can stand for implicit instructions. The pragmatic effect created by this indirect request is sometimes exploited for politeness strategies, or for its humorous potential.

1 Are apprehensional constructions inherently dependent?

1.1 Apprehensional constructions: Presentation

The grammatical encoding of apprehensional meanings was initially brought to light in individual language descriptions, covering various families and areas –

Alexandre François. 2026. Explicit apprehensions, implicit instructions: An indirect speech act in the grammar. In Martina Faller, Marine Vuillermet & Eva Schultze-Berndt (eds.), *Apprehensional constructions in a cross-linguistic perspective*, 423–460. Berlin: Language Science Press. DOI: ??

e.g. Austin (1981) on Diyari (Pama-Nyungan, Australia); Lichtenberk (1995) on Toqabaqita (Oceanic, Solomons); François (2003) on Mwotlap (Oceanic, Vanuatu); Pakendorf & Schalley (2007) on Sakha (Turkic, Siberia); Vuillermet (2018b) on Ese-ejja (Takanan, Bolivia); Smith-Dennis (2021) on Papapana (Oceanic, Papua New Guinea) – to cite but a few. Beyond the discrepancies in terminology and glossing ("avertive", "evitative", "apprehensive", "adversative", *lest* …), these various constructions appear to share enough properties to justify defining a new semantic domain, labelled *apprehensional.* This has led to a recent line of research in typology (see Dobrushina 2006, Vuillermet 2018a,b, Faller & Schultze-Berndt 2018) – and is the object of the present volume.

In a nutshell, apprehensional markers are grammatical morphemes that label a potential event as undesirable, and worthy of being avoided indirectly – by carrying out another action. Whereas prohibitives (such as *Don't jump!*) ask the addressee to directly refrain from an action (assuming they can control it), the semantic mechanism of an apprehensive is more complex, because it normally involves two distinct events P and Q: an event P which can be controlled, and a second event Q which cannot be controlled directly, but which can be avoided, by carrying out the action P.

Here is an example from the Oceanic language Papapana (Papua New Guinea) as described by Smith-Dennis (2021):

(1) Papapana (Oceanic; Papua New Guinea; Smith-Dennis 2021: 426)
O=nabe=i, *o=**te*** *mate=i.*
2SG.SBJ=swim=IRR 2SG.SBJ=**APPR** die=IRR
'Swim, (otherwise) you might die.' ~ 'Swim, (so that) you don't die.'

In this example, the undesirable event (Q) is {you dying}, over which the addressee has no direct control. So, instead of asking them to avoid Q directly by means of a prohibitive (**Don't die!*), the speaker requests that Q be avoided *indirectly* – namely, by carrying out another action P (*Swim!*) which should prevent Q from happening. In such a structure, the apprehensive clause (*you might die*) serves as a justification for the main clause (*Swim!*): it clarifies the nature of the undesirable event Q that this action P should help avoid.

1.2 Two main apprehensional strategies

The typological characteristics of the apprehensional domain are explained in this volume's introduction (Vuillermet et al. 2026 [this volume]); I will only mention a few points, using the same terminology as them.

Apprehensional morphemes may belong formally to two main grammatical types: (a) a subordinating conjunction, or (b) a modal marker on the predicate.

Apprehensional subordinators (e.g. English *for fear that*) encode an undesirable event as a dependent clause Q, which serves as an explanation for the main clause P. The typical structure, labelled "precautioning" (Vuillermet 2018b: 260, Vuillermet et al. 2026 [this volume]), takes the form of a twofold structure {P, SUB Q}. P is called a "pre-emptive clause", whether it encodes an order, a prohibition, or a statement. As for the undesirable situation Q that is meant to be avoided through the event P, it is called the "precautioning clause". In English, this can be rendered using the (archaic) subordinator *lest*:

(2) {*I put the knife away*}$_P$ {***lest*** *you hurt yourself*}$_Q$.

In principle (though there can be exceptions), such subordinators are restricted to dependent clauses, and never occur in main clauses.

The second major formal strategy involves a special modal marker on the verb, labelled the "apprehensive mood" (Vuillermet et al. 2026 [this volume]). As a first approximation, the apprehensive mood can be rendered in English with the modal auxiliary *might*. This can also surface as a twofold structure {P, MOD Q} – mostly parallel to (2) above:

(3) {*I put the knife away*}$_P$, (*because*) {*you* ***might*** *hurt yourself*}$_Q$.

One key difference between the subordinating strategy and the modal one is that the latter can also be used in an independent clause:

(4) *You* ***might*** *hurt yourself.*

A sentence like (4) may be syntactically well-formed, but it typically implies a reference to an intended instruction (order, prohibitive) which is either explicit, or must be inferred from the context. This question will be central to our study.

Languages differ in their degree of grammaticalization of apprehensional strategies. The reason why it took a long time to identify this domain as typologically significant is that many of the world's languages use linguistic devices that do not target that meaning specifically. For example, while English *might* can sometimes be interpreted as the equivalent of an apprehensive mood as in (3), this auxiliary has a broader semantic spectrum, which is not restricted to undesirable events (cf. *You might be able to find a good job there*). Yet more and more languages are being found to feature grammaticalized morphemes whose role is specifically to tag events as undesirable.

Most languages have only one of the two strategies described above, either the precautioning subordinator or the apprehensive mood. Ese Ejja (Bolivia) is noteworthy in having both: a subordinator *e-…kwajejje* and an apprehensive *-chana* (Vuillermet 2018b) – corresponding, respectively, to structures like (2) and (3)–(4) above. Finally, some languages have a single morpheme (like *ada* in Toqabaqita, Lichtenberk 1995) that can fill both functions, subordinating and main-clause modal.

1.3 Apprehensive mood and clause dependency in northern Vanuatu

The present study aims to describe and compare the apprehensional strategies of a group of 17 Oceanic languages spoken in the Torres and Banks Islands of north Vanuatu. In spite of their historical diversification, these related languages are often structurally parallel (François 2011): one of their commonalities is precisely to have grammaticalized strategies to encode apprehensional semantics.

However, what is also striking is the diversity of these devices: not only do the morphemes differ in their form, but they also result from different paths of grammaticalization. Some only have a 'lest' subordinator, others only an apprehensive mood, others have both. After an overview of the strategies attested in the region (§2), the second half of this study will focus on the language Mwotlap, which employs a modal marker *tiple*:[1]

(5) Mwotlap

⟨*Tēy*	*van*	*na-gayga*	*en,*⟩$_{\mathrm{P}}$	⟨*nēk*	***tiple***	*qēsdi*⟩$_{\mathrm{Q}}$*!*	
hold	DIREC	ART-rope	DEF	2SG	APPR	fall	

'Hold on to the rope, (otherwise) you *might* fall! / *lest* you fall!'

The form *tiple* fits in the slot of Tense-Aspect-Mood markers in Mwotlap, and will thus be described as its "apprehensive mood" (François 2005a: 130).

Parallel to (3) above, Mwotlap *tiple* is most often found in biclausal sentences such as (5); this can make it functionally equivalent to a subordinate structure of the type {P, *lest* Q}. As it happens, northern Vanuatu is an area where certain Tense-Aspect-Mood markers have been observed to encode, by themselves, a form of clause dependency (§4.1). Could it be the case that the apprehensive mood of Mwotlap also encodes syntactic subordination on its own?

In fact, it is not rare to hear utterances consisting only of an apprehensive clause Q, with no pre-emptive clause P:

[1]Examples are cited in their practical orthographies. Conventions for Mwotlap include: *e* = [ɛ]; *ē* = [ɪ]; *o* = [ɔ]; *ō* = [ʊ]; *y* = [j]; *g* = [ɣ]; *b* = [ᵐb]; *d* = [ⁿd]; *n̄* = [ŋ]; *q* = [k͡pʷ]; *m̄* = [ŋ͡mʷ]. Other languages of the area essentially share the same conventions.

(6) *Nēk* ***tiple*** *qēsdi!*
2SG APPR fall
[*to a boy in a tree*] 'Hey, you might fall!'

Such examples raise the question of the status of that sentence: is it fully independent? Or is it dependent in some way, either syntactic or pragmatic? This is indeed a recurring question in the typology of apprehensional constructions:

> "One of the main issues in the existing literature on apprehensives is their syntactic and/or pragmatic status as dependent or independent clauses." (Smith-Dennis 2021: 428)

The question thus arises of how best to analyze examples such as (6). If *tiple* were found to have a subordinating role, then one may want to analyze a standalone clause like (6) as an instance of *insubordination* – i.e. a subordinate clause used independently (cf. Evans 2007). Such a historical process has indeed been proposed to account for certain types of apprehensional markers, as in the Oceanic language Papapana (Smith-Dennis 2021) or in the East Caucasian language Archi (Daniel & Dobrushina 2026 [this volume]). Rather, I will propose that apprehensive clauses in Mwotlap are grammatically well-formed independent sentences. They do present a form of dependency towards an external utterance, explicit or implicit; yet that dependency is not syntactic in nature, but rather pragmatic.

In the analysis I propose, the work of the apprehensive modality is to alert the addressee to a specific risk that should be avoided. By formulating such an apprehension, the speaker yields support to a particular instruction – one that is sometimes specified as in (5), and sometimes left implicit as in (6). This interplay between *explicit apprehensions* and *implicit instructions* is the central mechanism of the apprehensive mood of Mwotlap.

1.4 Data and sources

The present study rests on primary data I collected during a number of field trips in Vanuatu since 1997, on the 17 languages of the Torres and Banks Islands (see Figure 1). For reasons of length, this article will only mention a representative sample of eight languages: Hiw, Lo-Toga, Löyöp, Lemerig, Vurës, Dorig, Lakon – and of course, Mwotlap.

My sources take the form of three main sets of data. First, I designed a conversational questionnaire (François 2019) aiming to elicit lexical, grammatical

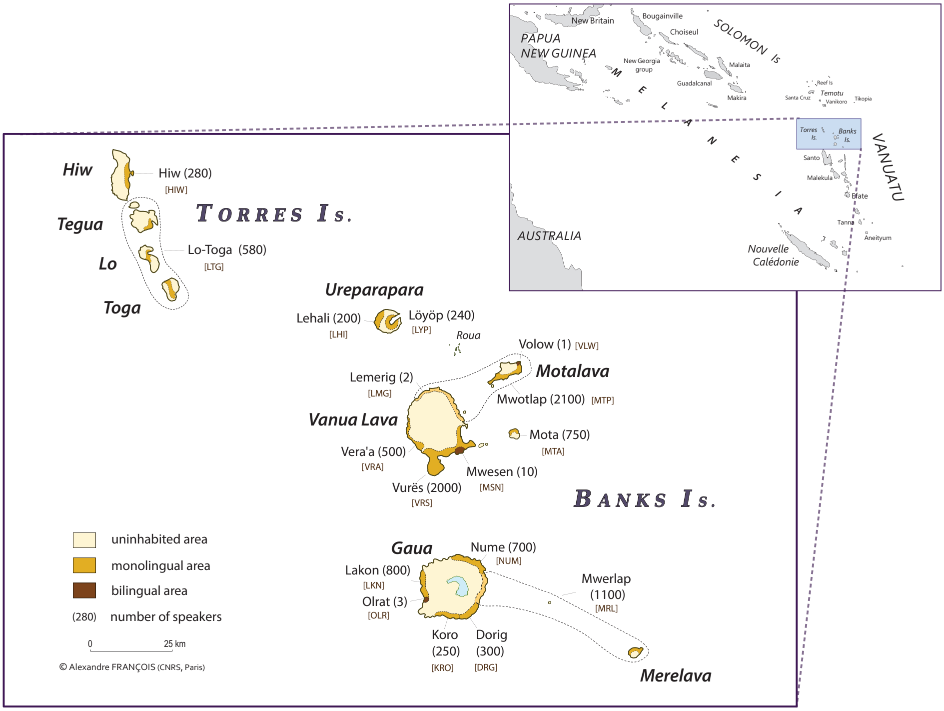

Figure 1: Location of the 17 Torres and Banks languages in Vanuatu (South Pacific)

and phraseological data in each language: this allowed me to collect essential information on apprehensional structures from all 17 languages.[2] Second, participant-observer immersion in each language community gave me the opportunity to collect snippets of spontaneous conversation in my handwritten notebooks (archived in François 2015).

Finally, I recorded 389 narratives and conversations, totalling 50 hours, in all languages. Among these, 263 texts were transcribed and annotated in the presence of native speakers; together, they form an electronic text corpus of 250,000 words – with the largest corpora being in Mwotlap (100,000 w.), Lo-Toga, Hiw, Dorig, Lakon, Lemerig. These recordings are all archived in the *Pangloss Collection* of the CoCoON archive, with open access (François 2022c). I regularly enrich them with time-aligned transcriptions and translations, which are indexed using permanent identifiers (DOI) at the sentence level.

[2]This questionnaire helped me elicit example sentences (19), (22), (23), (29) below.

The present study will cite examples either from my field notes or from my text corpora; whenever a sentence is accessible online, I will provide a permanent link to it, so it can be heard in its original context. While apprehensional markers regularly surface in the spoken narratives I recorded, I was also able to hear many tokens in the spontaneous speech of daily conversations; in such cases – arguably the most natural ones – rather than links to audio files, I will provide references to my field notes.

Several linguists have worked on northern Vanuatu languages, whether their focus was the lexicon (Codrington & Palmer 1896 for Mota; Malau 2021 for Vurës; François 2022a for Mwotlap) or the grammar (François 2001, 2003 for Mwotlap; Schnell 2011 for Vera'a; Malau 2016 for Vurës). An overview of the Torres and Banks languages can be found in François (2011, 2012). The present study finds its roots in my descriptive monograph on the Tense-Aspect-Mood system of Mwotlap (François 2003), particularly in the chapter "Évitatif" (pp. 301–312). However, I will bring here new insights and discussion, as well as a much greater amount of primary data – in Mwotlap and especially in the other languages, most of which are still undocumented.

This study will begin with an overview of apprehensive strategies in northern Vanuatu (§2), highlighting their diversity, and their links with other grammatical devices (ablative case, prohibitive modality). We will then focus on the morpheme *tiple* of Mwotlap, as it shows the most unambiguous case of an apprehensive modality, distinct from other moods, and clearly different from a precautioning subordinator. After describing the syntax and semantics of the Mwotlap apprehensive (§3), §4 will return to our discussion about the stand-alone uses of the apprehensive – as in (6) above – and to their pragmatic correlates.

2 Apprehensional semantics in north Vanuatu

2.1 Grammatical overview of north Vanuatu languages

The Vanuatu archipelago was first settled about 3,100 years ago by Austronesian navigators, speakers of the Proto Oceanic language (Bedford & Spriggs 2008, Lipson et al. 2020). This was followed by three millennia of *in situ* diversification, during which the linguistic unity of Vanuatu progressively fragmented into 138 languages (François et al. 2015). Among these, 17 formed in the northern islands of the Torres and Banks groups, through a process of internal diversification.

In spite of their divergence, the linguistic history of northern Vanuatu is also one of structural convergence, due to a sustained history of contact among communities (François 2011). Today, the Torres and Banks languages share many linguistic structures – whether this is due to their common origin, or to later re-convergence.

The next paragraphs provide a short overview of some major grammatical properties common in the area, and relevant to the discussion of apprehensional strategies. (For the sake of internal consistency, all examples in this overview will be given in Mwotlap.)

2.1.1 Word order in the verbal clause

The languages of Vanuatu have accusative alignment. They have fixed rules for word order in the clause, with a consistent order SVO (i.e. SV, AVO):

(7) Mwotlap [https://doi.org/10.24397/pangloss-0007409#S25]

Iplu-k	*mē-dēn̄*	*ēgēn.*
partner-1sg	prf-arrive	now

‘My friend has arrived.’

(8) Mwotlap [https://doi.org/10.24397/pangloss-0003282#S76]

Gēn	*tu-wuh*	*Vēnvēntey*	*talōw.*
1incl.pl	fut-kill	(V.)	tomorrow

‘We will kill Vēnvēntey tomorrow.’

Subordinators, such as the complementizer, are normally found at the left edge of the dependent clause, before its subject:

(9) Mwotlap [https://doi.org/10.24397/pangloss-0002298#S73]

No	*ne-myōs*	***so***	*nok*	*leg*	*mi*	*nēk.*
1sg	stat-want	**comp**	1sg.irr	marry	with	2sg

‘I want to marry you.’

Independent and subordinate clauses have the same internal word order.

2.1.2 Tense-Aspect-Mood-Polarity marking

The syntax of verbal clauses revolves around a constituent which the Oceanic tradition (e.g. Durie 1988, Evans 2003) calls the *verb complex* (vc).

The vc consists minimally of a verb, which is the head.[3] This head is optionally followed by one or more postverbal modifiers: e.g. a second verb in a serial pattern, or a *postverb* (a kind of adverb specialized in the postverbal position). In (10),

[3] Lexical adjectives and nouns can also head predicates inflecting in tamp, in the same way as verbs do (François 2005a: 131, 2017: 328, 2026). Thus, while most examples of apprehensives in this study will involve a verbal head, (35)–(36) will involve adjectives, and (50) a noun predicate.

the verb complex includes a verbal head *van* 'walk' and a postverb *yeghuquy* 'freely':

(10) Mwotlap [https://doi.org/10.24397/pangloss-0007411#S123]

N-et	*⟨tit-*	***VAN***	***yeghuquy***	*vēhte⟩*$_{VC}$	*van*	*lē-vētan*	*en.*
ART-person	NEG:POT$_1$-	walk	freely	NEG:POT$_2$	DIREC	LOC-land	DEF

'One cannot walk freely into that piece of land.'

Attached to the lexical elements of the verb complex are markers of Tense, Aspect, Mood, and Polarity (henceforth TAMP), which take the form of affixes or particles. Example (10) has a discontinuous TAMP marker, the Negative potential *tit- … vēhte* 'cannot'. In Torres and Banks languages, TAMP morphemes cannot combine with each other: they form a single paradigm of unanalyzable, portmanteau forms that encode the semantics of tense, aspect, mood, and polarity in a single morpheme – whether it is simple or discontinuous. The TAMP paradigm in Mwotlap has 26 members (François 2005a: 133), one of which is the apprehensive mood *tiple* (§3).

TAMP morphemes occur in two slots in the clause, labelled TAMP$_1$ and TAMP$_2$, which surround the lexical elements of the verb complex:

(11) Structure of a verbal clause in Mwotlap:
subject ⟨ TAMP$_1$ VERB (postverbs) TAMP$_2$ ⟩$_{VC}$ *object* adjuncts

One slot TAMP$_1$ follows the subject, and opens the verb complex. The second slot TAMP$_2$ closes it, preceding the object[4] or any other complement.

Some TAMP morphemes are bipartite, with components occupying both slots TAMP$_1$ and TAMP$_2$: this was illustrated in (10) with the Negative potential *tit-… vēhte*.[5] Some consist of a single element that occurs postverbally in TAMP$_2$. But the majority of TAMP morphemes occupy only the first slot TAMP$_1$ – e.g. the perfect *me-…* in (7) or the future *te-…* in (8). The apprehensive *tiple* also belongs in that TAMP$_1$ slot – as we saw in (6).

[4]Mwotlap is strict in inserting TAMP$_2$ before the object; other Torres and Banks languages sometimes place it after the object.

[5]In these lines, the ellipsis "…" corresponds to the lexical elements of the VC (i.e. the verb complex minus TAMP markers). Bipartite morphemes include discontinuous morphemes of negation, which are common in Banks languages: see Schnell (2011: 31) for Vera'a, François (forthcoming) for Dorig.

2.1.3 Subordination and TAMP

When two morphemes are homophonous, they can often be identified through their position in the sentence.

For example, (12) has a particle *so* that is located between the subject and the verb, in the TAMP$_1$ slot; therefore, it is a TAMP marker. This is an instance of the *prospective* mood, which encodes a number of irrealis values (volitional, deontic, hortative, subjunctive) in the affirmative (François 2005a: 134):

(12) Mwotlap [https://doi.org/10.24397/pangloss-0007413#S338]
Nok ***so*** *in* *nē-bē.*
1SG.IRR **PROSP** drink ART-water
'I want to drink water.'

By contrast, (9) had its complementizer *so* located before the subject; this was a subordinating particle.

Historically speaking, it is likely that there is an etymological connection between the complementizer and an irrealis marker; but synchronically, they must be analyzed as two distinct morphemes, each with properties of its own (François 2003: 249–257). The two homophones (complementizer *so*, prospective *so*) can coexist in the same clause:

(13) Mwotlap [https://doi.org/10.24397/pangloss-0007414#S36]
No *ne-myōs* ***so*** *nok* ***so*** *vētleg* *nēk* *a* *Apnōlap.*
1SG STA-want **COMP** 1SG.IRR **PROSP** send 2SG FOC Vanua.Lava
'I want to send you to Vanua Lava island.'
(lit.) 'I want <u>*that*</u> I <u>*shall*</u> send you …'

These grammatical principles, demonstrated here for Mwotlap, apply equally to other languages of north Vanuatu, whose syntactic structures are essentially parallel.

2.2 An areal typology of apprehensional strategies

The preceding notes will prove useful as we discuss the two types of apprehensional devices used in the Torres and Banks languages – respectively, the subordinator and the modal marker.

2.2.1 Unity and diversity of formal strategies

As explained in §1.4, I will provide data from a representative sample of eight languages, out of the 17 spoken in the area. Table 1 lists them in geographical order from northwest to southeast, and provides essential information on apprehensional strategies. The target markers correspond to the two rows in the middle ("*lest* subordinator", "apprehensive mood"). The shaded areas indicate when either of these apprehensive devices also has a different function in the system – as discussed in the following paragraphs.

Table 1: A sample of Apprehensional markers from the Torres-Banks area, highlighting their connections with other functions

	Hiw	Lo-Toga	Löyöp	Mwotlap	Lemerig	Vurës	Dorig	Lakon
ablative	ton	dën	nin	den	‘en	den	dēn	jen
‘lest’ SUBORDINATOR	–	–	nin	–	‘en	den	tekor	ätäwoo
APPREHENSIVE MOOD	mit, vit	mit, mik	nin	tiple	‘en	–	–	mētē … lee
prohibitive	take, tati	mit, mik tate	tet	(ni)tog	‘en +Red., ‘og ‘en	nitog, mitV-	tog … te	mētē … lee

The first observation is that all languages in the area have at least one strategy dedicated to the expression of apprehensional meanings: either the subordinating strategy, or the verbal mood, or both. One may be surprised by the diversity of the forms involved, among languages which are all closely related. This may be indicative of the diachronic instability of apprehensive morphemes, and of their propensity to disappearing, or to being renewed over time (see Vuillermet et al. 2026 [this volume]).

And yet, the existence of dedicated apprehensive strategies in all these languages confirms a more general observation, that their situation of sustained contact has given rise to grammatical systems that are generally parallel in their semantic structures. The presence of apprehensional devices across the whole area points to the existence, in this group of languages, of a "typological niche" for this particular meaning – a phenomenon already observed in other multilingual areas (Daniel & Dobrushina 2026 [this volume], in the Caucasus).

2.2.2 A note on Bislama

The areal tendency to develop specialized strategies for the expression of apprehensionals is also apparent through the influence Oceanic languages have had upon Bislama, the English-lexifier creole used as a lingua franca in the country.

In Bislama, the adjective *nogud* ‘bad’ (<Eng. *not good*) is commonly used predicatively with a variety of meanings, including ‘be bad, be immoral’:

(14) Bislama
Hem i toktok olsem lo yu? i nogud ia!
3SG 3SG talk thus OBL 2SG 3SG bad DEIC
'She spoke that way to you? That's bad!'

That word has grammaticalized as a clause-initial marker, a slot sometimes used by TAM markers:[6]

(15) Bislama (Crowley 2004: 186)
***Nogud** oli faenemaot yumitu.*
APPR 3PL discover 1INCL.DU
'They *might* discover us!' / 'What if the two of us are discovered?' (Crowley 2004)

Sentence (15) is taken from the *Bislama Reference Grammar* by Terry Crowley (2004), who describes *nogud* as an "adverb expressing a warning with the meaning 'what if' or 'I hope not'". I propose to analyze *nogud* in (15) as a marker of apprehensive modality.

Interestingly, Crowley's grammar assigns this marker to his chapter on subordinators, and notes: "*nogud* can also be used as a subordinator to introduce an adversative clause that expresses the idea of 'in case'". He then provides this example (glosses are mine):

(16) Bislama (Crowley 2004: 186)
Bae yumi karem ambrela ***nogud*** *bae i ren.*
FUT 1INC:PL carry umbrella APPR FUT 3SG rain
'We will take an umbrella *in case* it rains.'

Whether we consider *nogud* as a modal marker in (15) or as a subordinator in (16), this morpheme clearly constitutes a dedicated device for encoding apprehensive semantics. The word depicts a situation as undesirable – as its etymology suggests. Ultimately, this development of Bislama was inherited from the structures of its Oceanic substrates, all of which feature specific strategies for encoding apprehensionality.

2.2.3 Apprehensive subordinators

As Table 1 shows, the languages Vurës and Dorig lack any apprehensive mood: what encodes the apprehensional meaning is a clause-initial word that behaves

[6] In Bislama, TAM markers sometimes occur before the subject: see the future *bae* in (16).

like a subordinator. The verb of that dependent clause inflects for a general irrealis mood, which lacks any specific connection to the apprehensive:

(17) Vurës (Malau 2021)
Kōmōrōn̄ ri ēlgōr, ***den*** *kōmōrōn̄ a m̄ës.*
2DU IMP:2NSG beware **LEST** 2DU IRR fall
'Watch out, *in case* you two fall down.'

The etymology of *den* is that of an ablative preposition ('from, out of'):

(18) Vurës [https://doi.org/10.24397/pangloss-0003276#S33]
No kara mōl me ti ***den*** *taval maram.*
1SG REC.PST return hither PST ABL other.side world.of.living
'I just came back *from* the World of the Living.'

The grammaticalization of the apprehensive from an ablative adposition rests on a spatial metaphor: the risk that is to be avoided (e.g. an accident) is analogical to a place you move, or keep, away from. Vurës shares this pattern of grammaticalization with several of its neighbours: Table 1 shows that the ablative preposition and the *lest* subordinator are homophonous in three languages of the sample,[7] namely Löyöp, Lemerig, Vurës (in the other languages, the form is only an ablative, and does not receive apprehensional meanings). The connection ablative-apprehensive is also attested in other parts of the world: thus, in Upper Tanana Dene (Athabaskan, Alaska), the morpheme *ch'a'* is both a postposition 'away from' and a 'lest' subordinator (Lovick 2026 [this volume]).

The apprehensive *den* of Vurës is a good example of a precautioning subordinator that is found only in dependent clauses. In her grammar of Vurës, Malau describes it in her chapter on subordination (2016: 677ff), as a conjunction introducing "adversative or 'lest' clauses". She does mention the fact that *den* clauses are sometimes uttered with the prosody of an independent clause, but this always happens in the immediate vicinity of the pre-emptive clause:

> While the *lest* clause is clearly a dependent clause, subordinate to the main clause, often, in terms of clause intonation, the clause occurs as an independent sentence. The information given is dependent on that given in the previous, main clause, but a fall in intonation and pause can indicate that it forms a separate sentence, much as an afterthought. (Malau 2016: 679)

[7] All Torres–Banks languages have an ablative preposition whose etymon reconstructs to **dani* (François 2005b: 494). The morpheme has grammaticalized into an apprehensive subordinator ('lest') in seven languages of northern Banks (three of which are shown in Table 1): Lehali *dän*; Löyöp *nin*; Lemerig *'en*; Vera'a *den*; Vurës *den*; Mwesen *nen*; Mota *nan*.

Neither Malau's nor my corpus feature any example where Vurës *den* would be used in an independent clause, without a pre-emptive clause in its immediate neighborhood. In other terms, *den* is a pure subordinator.

A similar case is the form *tekor* in Dorig. In much the same way as Vurës *den*, Dorig *tekor* occupies the slot of subordinator, and combines with a general irrealis mood:

(19) Dorig [https://www.odsas.net/object/105864]
Nēk s-tekor o sri-n, ***tekor*** *nēk so-dlōm!*
2SG IRR-beware ART bone-3SG **LEST** 2SG IRR-swallow
'Beware the bones, *lest* you swallow them!'

Like its Vurës equivalent, a clause with *tekor* can take the apparent prosody of an independent clause; but it still comes right in the vicinity of the pre-emptive clause:

(20) Dorig [https://doi.org/10.24397/pangloss-0007437#S21]
Ar te van~van vga te vak gēn neñ!
IMP:2NSG PROH$_1$ INF~go beyond PROH$_2$ DIREC FOC DIST
Tekor *kmur s-van wōn i tbi-kmur.*
LEST 2DU IRR-go find PERS ancestor-2DU
'Don't you walk beyond that point over there! *You might come across* [the ghost of] your ancestor.'

The apprehensive subordinator *tekor* of Dorig is grammaticalized from a verb also present in (19), namely *tekor* or *tekgor* [tɛkɔr] 'watch out, beware', literally 'watch (*tek*) over (*gor*)'. All Banks languages feature a verb 'beware' that is derived from a verb 'look, watch', plus a suffix **ɣoro* that is polysemous 'around, over, against …' (François 2000, 2005b: 495); this derivation yielded such verbs as Vurës *ēlgōr* in (17), and Mwotlap *etgoy* in (40). Among the 15 Banks languages, the five spoken on Gaua have further grammaticalized that verb into an apprehensive subordinator: Nume *kērē-gor*, Dorig *tek-or*, Koro *ēl-gor*, Olrat *ēl-woy*, Lakon *ätä-woo* (Table 1), Mwerlap *(ma)ta-gor*. Outside Vanuatu, Toqabaqita (Solomon Is.) is another Oceanic language whose apprehensive marker *ada* derives from a former verb meaning 'see; look out, watch out' (Lichtenberk 2008: 780).

All these languages illustrate a pattern whereby a verb 'beware' in the imperative has grammaticalized into an apprehensive linker: 'beware you'll swallow them' → '*lest* you swallow them'. When used as a conjunction, these forms do not behave like verbs anymore (with a subject or TAMP inflection): instead, they fill the syntactic slot of a complementizer – see (9).

2.2.4 From subordinator to mood marker

The language Lemerig, like its neighbour Vurës, coexpresses the ablative (21) with the *lest* subordinator (22), through the same form *'en* [ʔɛn]:

(21) Lemerig [https://doi.org/10.24397/pangloss-0003278#S25]
Në k-van kal sag ***'en*** *naw.*
1sg irr:1sg-walk inland uphill abl sea
'I'll walk uphill, *away from* the sea.' [ablative]

(22) Lemerig [https://www.odsas.net/object/105216]
Në k-mi~mi'ir rān e ***'en*** *ē sē n-pël.*
1sg irr:1sg-hab~sleep over.it top lest pers anyone irr:3sg-steal
'I sleep on it *so* nobody steals it.' [*lest* linker]

A sentence like (22) is structurally parallel to other precautioning clauses like (17) or (19), consisting of a *lest* subordinator and a generic irrealis mood.

But Lemerig went one step further, as the same form *'en* grammaticalized into a modal marker. Thus (23) has two homophonous morphemes *'en*, one in the subordinator position (before the subject 'kava'), one in the tamp$_1$ slot:

(23) Lemerig [https://www.odsas.net/object/105201]
Gätru ge wān? — Ȯòó, ***'en*** *ga* ***'en*** *ran̄ näk!*
1incl.du fut chew.kava excLm:no lest kava appr intoxicate 2sg
'Shall we do some kava-chewing? — No way! You might get dizzy.'
(*lit.*) lest the kava plant might intoxicate you

While they are etymologically related, these two instances of *'en* are synchronically two different morphemes, with different properties – in a way reminiscent of the two forms *so* of Mwotlap (§2.1.3). The coexpression {ablative = *lest* subordinator = apprehensive mood} is found both in Lemerig and in Löyöp (Table 1).

2.2.5 From apprehensive to prohibitive

To be accurate, the modal value of Lemerig *'en* is not restricted to apprehensive meanings. When combined with reduplication on the verb, *'en* also encodes the prohibitive. In order to cover the meanings 'apprehensive' and 'prohibitive', I propose a tentative gloss prvt for 'preventive':

(24) Lemerig [https://doi.org/10.24397/pangloss-0003278#S24]
Näk ***'en*** *'eñ~'eñ!*
2SG PRVT INF~weep
'Don't cry!'

That modal marker *'en* optionally combines with other markers for prohibitive, like *'og*:

(25) Lemerig [https://doi.org/10.24397/pangloss-0003271#S76]
Näk 'og ***'en*** *vus~vus në!*
2SG PROH PRVT INF~kill 1SG
'Don't kill me!'

Among the languages cited in Table 1, Lemerig is the most extreme case of polyfunctionality for a single morpheme. The many functions of *'en* are summarised in (26). Drawing on comparative evidence from the other Torres-Banks languages, they can be represented as a grammaticalization chain.

(26) Possible grammaticalization chain of *'en*

ablative adposition	'out of, away from'
→ precautioning subordinator	'lest X happens'
→ apprehensive mood	'X might happen'
→ prohibitive constructions	'don't do X'

Other than Lemerig, the coexpression between the apprehensive mood and the prohibitive is also found in two other languages of our north Vanuatu sample: Lo-Toga and Lakon. For example, in Lo-Toga *mit* is ambiguous between an apprehensional reading (27) and a prohibitive (28):

(27) Lo-Toga [https://doi.org/10.24397/pangloss-0003292#S27]
Nike tat ho vēn o! Ne ñwië ***mit*** *kur nike.*
2SG NEG.IRR NEG.POT go out ART monster PRVT eat 2SG
[The hero's mother begs her son to stay in the cave where they live.]
'You can't possibly go out! The monster *might* eat you!' [Apprehensive]

(28) Lo-Toga [https://doi.org/10.24397/pangloss-0003289#S16]
Deñwē pe noke ve not nike nōk! – O, nike ***mit*** *not noke!*
today SUB 1SG IPFV kill 2SG now EXCLM 2SG PRVT kill 1SG
[The character, a spider, begs the hero not to carry out his threat]
'I'm killing you right now! – No! *Don't* kill me!' [Prohibitive]

In the introduction to this study (§1.1), I contrasted the semantics of apprehensives with that of prohibitives in terms of (respectively) *indirect* vs. *direct* requests to avoid a certain event, in correlation with the degree of control of the agent upon that event. The latter two examples are good illustrations of this semantic contrast:

- In (27), the hero has control over the action expressed in the pre-emptive clause, namely whether he'll exit the cave or not; but he has only *indirect* control over the second event, namely whether the monster will eat him or not. This is a typical context for an apprehensive.
- In (28), the spider begs the hero not to carry out his threat to kill him; this is clearly a request to *directly* refrain from that action, which the agent has full control of. Therefore, semantically, (28) is an unambiguous prohibitive.

In spite of their semantic differences, these two constructions are coexpressed in Lo-Toga.

Similarly, in the language Lakon, the apprehensive (29) and the prohibitive (30) share the same discontinuous modal morpheme *mētē* … *lee* (glossed prvt 'preventive'):

(29) Lakon [https://www.odsas.net/object/105973]
Na tē ñoo~ñoo tuwoo to on jaajun ***mētē*** *pal* ***lee***.
1sg ipfv$_1$ hab~sleep over.it ipfv$_2$ purp person prvt$_1$ steal prvt$_2$
'I sleep on it *so* nobody steals it.' [Apprehensive]

(30) Lakon [https://doi.org/10.24397/pangloss-0003185#S22]
Ta, nēk ***mētē*** *vuh* ***lee*** *na!*
no 2sg prvt$_1$ kill prvt$_2$ 1sg
[The character, a spider, begs the hero not to carry out his threat]
'No! *Don't* kill me!' [Prohibitive]

Table 1 cites several languages (Hiw, Löyöp, Mwotlap …) where the apprehensive mood and the prohibitive are morphologically distinct: in those languages, sentences like (27) and (28) would employ different markers (see §3.2 for Mwotlap). But in Lo-Toga and Lakon, the two meanings can be expressed by the same morpheme, thereby blurring the boundary between direct and indirect requests. The coexpression of apprehensives and prohibitives has already been observed in other language families of the world (Dobrushina 2006: 50–63; see Vuillermet et al. (2026 [this volume]) for other references). While Pakendorf & Schalley (2007) argue that this pattern of coexpression is rare, Smith-Dennis (2021: 453) finds it rather widespread, at least in the Pacific region.

Pakendorf & Schalley (2007: 525) propose a path of semantic change {*apprehension → warning → prohibition*}, in that order. This sequence is supported by the facts of Lo-Toga. Indeed, its neighbour Hiw has a cognate form *mik*/*mit*/*vit* which is purely an apprehensive mood marker (Table 1); it is likely that the original meaning of *mit* was apprehensive, and that its extension to the prohibitive was a recent innovation of Lo-Toga.

2.3 The diversity of apprehensive strategies in North Vanuatu

Let us recapitulate our findings so far. The languages of north Vanuatu all have linguistic devices dedicated to the encoding of apprehensional semantics. These devices differ across languages:

- They differ in their phonological shape (*vit, mit, nin, tiple, den, tekor, mētē…lee*, etc.).
- They differ in their etymologies: e.g. some originate in an ablative preposition, others in a verb 'watch out', etc.
- They differ in their syntactic status. Some languages only have a precautioning subordinator, others only an apprehensive mood. Others again can combine both strategies, whether expressed by the same surface form or not.

Finally, there are differences even within the set of languages that employ the modal strategy. As Table 1 shows, two languages have a mood marker (Hiw *mit*, Mwotlap *tiple*) which is dedicated solely to the coding of apprehensive modality. By contrast, in three languages, the modal marker (Lo-Toga *mit*, Lemerig *'en*, Lakon *mētē … lee*) has a broader 'preventive' meaning, which encompasses apprehensive and prohibitive semantics.

The present study set out to address a central problem, namely the behaviour of the apprehensive mood in independent clauses. Among north Vanuatu languages, some are less well suited to address this question: for example, those which only have a subordinator – like Vurës or Dorig – are less likely to use it in independent clauses (§2.2.3): we need a language where apprehensional semantics are encoded by a TAMP marker. As for those modal markers that are ambiguous between apprehensive and prohibitive, they will also be difficult to analyze: when found in an independent clause, that so-called "preventive" mood will often simply be interpreted as a prohibitive, in ways that will be difficult to distinguish from an apprehensional use.

In sum, the best configuration for tackling our problem would be a language with a "pure" apprehensive mood – one that is not a subordinator, and is formally distinct from prohibitives. Luckily, this is the case for Mwotlap, the language which has the richest corpus in our data; this will now be our main focus.

3 The apprehensive mood of Mwotlap

The previous pages discussed the diversity of apprehensional strategies in north Vanuatu. Amongst them, Mwotlap appeared to be the language best suited to answer our initial question: can the apprehensive mood be used on its own? And if so, is the clause fully independent, or does it present a form of dependency with another clause?

Before we can address this question, it will be useful to describe the characteristics of Mwotlap's apprehensive, by observing how it is used in my recorded texts and in daily conversation. We will then come back to our central discussion in §4.

3.1 Forms of the Mwotlap apprehensive

The apprehensive mood of Mwotlap is attested with a number of formal variants. Table 2 lists them together with the number of tokens of each allomorph in my text corpus of 99,800 words (which does not include my field notes).

Table 2: Free variants of the apprehensive mood marker in Mwotlap

Allomorph	*tale*	*tile*	*tele*	*taple*	*tiple*	*teple*	*tevele*	*tepele*	*vele*	Total
# tokens	7	5	5	8	20	1	2	1	5	54

All these forms are used interchangeably, even by the same speaker, without any semantic or pragmatic difference.[8] Because *tiple* is the most common variant, I will use it as the citation form for that morpheme, referring to the whole set of allomorphs.

[8]Mwotlap shows such extreme formal variation only for two morphemes, which both belong to the Tamp domain: (1) the apprehensive *tiple*, (2) the permansive ⟨l^{a}/$_{e}$v((g)e)tō⟩ 'still' → *laptō* ~*lavetō* ~*lapgetō* ~*leptō* ~*levetō* ~*lepgetō* (François 2003: 130). A third morpheme, the dilatory future, alternates freely between three forms: *qoyo* ~*tiqoyo* ~ *tiqyo* (François 2003: 38).

The etymology of *tiple* is unclear. Reconstruction is made difficult by the absence of any cognate form in any other language of the area (cf. Table 1) – except for Volow, a now extinct dialect of Mwotlap, which had *tavele* or *tivele*. A potential etymon would be the root **tavala* 'on the opposite side, beyond' (cf. Clark 2009: 194): this would suggest a pattern '*P, on-the-opposite-side Q*' – somehow reminiscent of English '(*you should do*) *P, otherwise Q* (*might happen*)'. If this etymology is correct, then we would have here another spatial metaphor, in addition to the one we saw in §2.2.3 with the ablative **dani* ('away from' → '*lest*').

The total number of tokens for the apprehensive mood, regardless of allomorphy, is 54. Considering the size of the corpus, the morpheme appears on average once every 1,850 words. This is a relatively low frequency, compared for example with the Potential *te- … vēh* which appears every 915 words. This may be because my text corpus consists mostly of narratives, which are less prone to the expression of undesirable, irrealis clauses than interactive conversation.

Besides the modal marker *tiple*, Mwotlap also shows traces of what may have been once a precautioning subordinator of the form *den* (homophonous with the ablative):

(31) Mwotlap [https://doi.org/10.24397/pangloss-0003275#S11]
Nēk tog van~van. Den nēk ***taple*** *yap na-pgal me hiy dōyō.*
2SG PROH INF~go LNK 2SG APPR pull ART-war hither DAT 1INCL.DU
[The hero wishes to go visit an enemy village.]
'Don't go there! You *might* end up attracting a conflict upon us.'

At first glance, the structure of (31) recalls that of (17) in Vurës or (23) in Lemerig. However, contrary to what happens in these two languages, Mwotlap *den* cannot encode apprehensional semantics by itself any more: it can only do so in the presence of the apprehensive mood *tiple*. In synchronic Mwotlap, *den* is now used as a (rare) coordinator between clauses, meaning 'as' or 'because':

(32) Mwotlap [https://doi.org/10.24397/pangloss-0003275#S49]
No mas kay mat kōyō. Den kōyō hole iseg na-lqōvēn mino.
1SG must shoot dead 3DU LNK 3DU talk play ART-woman my
'I have to kill them. *Because* they've been flirting with my wife.'

Based on such evidence, the function of *den* in a sentence like (31) must be understood as a causal coordinator: it introduces an argument explaining the reason for the previous sentence. In sum, apprehensive semantics in synchronic Mwotlap is encoded exclusively by the modal marker *tiple*; the language lacks any 'lest' subordinator.

3.2 A regular contrast between apprehensive and prohibitive

Importantly, Mwotlap *tiple* never encodes the prohibitive, which is expressed by a distinct marker *tog* or *nitog*. Thus compare the direct request of the prohibitive *tog* in (33) with the indirect strategy of the apprehensive construction in (34):

(33) Mwotlap [https://doi.org/10.24397/pangloss-0002300#S146]
Nēk ***tog*** *teñ~teñ!*
2SG PROH INF~cry
'Don't cry!' / 'Stop crying!'

(34) Mwotlap [https://doi.org/10.24397/pangloss-0007413#S26]
Óòó, dō mōl! Imam ***tale*** *boel dōyō.*
EXCLM:no 1INCL.DU return father APPR get.angry 3DU
'No, let's go back home! Dad might get angry at us.'

Mwotlap is strict in differentiating the prohibitive from the apprehensive. This is an important difference with languages like Lo-Toga in (27)–(28), where a single morpheme can express both a direct or an indirect request. Table 3 compares the two linguistic ways to get the addressee to avoid an undesirable event Q – namely, the direct strategy (prohibitive) vs. the indirect one (apprehensive).

Table 3: Two linguistic strategies for avoiding an event Q: Direct request (prohibitive) vs. indirect request (apprehensive)

Direct strategy	**Indirect strategy**
Undesirable event Q: controllable by addressee e.g. (33) Q = you crying	Undesirable event Q: not directly controllable by addressee e.g. (34) Q = Dad being angry at us
→ **PROHIBITIVE**	→ **APPREHENSIVE**
1) direct request to refrain from Q e.g. 'Don't cry!' / 'Stop crying!'	1) mention an event Q to be avoided e.g. 'Dad might get angry at us.' 2) point, explicitly or not, to the (controllable) action P that can avoid Q e.g. 'Let's go back home!'

Compared to other apprehensional strategies attested in the region, Mwotlap *tiple* is the clearest example of a proper apprehensive modality, as it is unambiguously distinct from other morphemes in the language – whether the ablative, the *lest* subordinator, or the prohibitive. From now on, all sentence examples will be in Mwotlap.

3.3 Avoiding an event vs. avoiding its consequences

The Mwotlap apprehensive can be used in the two configurations identified by Lichtenberk (1995: 299) in his pioneering study – respectively, the *avertive* use and the *'in-case'* construction.

All the examples we have seen so far are of the *avertive* type. That is, the apprehensive clause represents an undesirable event Q, which another action P is meant to prevent altogether. In principle, the success of P should imply that the event Q does not materialize at all: *if we go back home now, then Dad won't be angry.* In such cases, the apprehensive can also be translated as a negative purpose clause ('Let's go back home, *so* Dad *doesn't* get angry').

A much rarer pattern, known as the 'in-case' construction, is when Q describes an event that in itself cannot be avoided – e.g. a weather situation. The apprehensive here is not about preventing Q altogether, but avoiding its undesirable consequences:

(35) Mwotlap [elicited; EWH.Telefon.095]
Lep no-sot gōh, mahē ***tiple*** *momyiy!*
take ART-shirt this place APPR cold
'Take this sweater, *in case* the weather gets cold.'
#Take this sweater, so the weather doesn't get cold.

(36) Mwotlap [https://doi.org/10.24397/pangloss-0007436#S94]
Nēk vēl~vēlēgē, ne-met ***vele*** *mah!*
2SG INTSF~hurry ART-tide APPR dry
'Hurry up (fishing), ? *in case/before* it gets to low tide.'

Obviously, the actions described in the first clause P cannot, by themselves, avoid the change of ambient temperatures, or prevent the tide from going low. Rather, they indicate the behaviour that would help prevent the negative consequences of those natural events: (35) that you may catch a cold; and (36) that you may fail to catch any fish while you still could.

In both these examples, the apprehensive clause keeps its argumentative function: it exposes an undesirable situation as an argument for justifying a particular action.

3.4 The modal viewpoint behind the apprehensive

The apprehensive *tiple* entails that the event Q is "undesirable". This is a modal projection, anchored in a subjective viewpoint. But whose viewpoint is it exactly?

The typical configuration – the one we've seen in most of our examples so far – is when the modal viewpoint coincides with the speaker. Thus in (34), the younger brother asks his elder brother to go back home, because *he* (the younger one) is the one who fears their father's anger.

When the apprehensive clause comes along with an imperative or a prohibitive, the notion that Q is undesirable is usually shared by both the speaker and the addressee. Thus in (31) *You might attract a conflict upon us*, one can easily imagine that the wish to avoid a major conflict is shared by both parties. The same analysis would be true of our initial example (5) *Hold on to the rope so you don't fall*: the event to be avoided would be unfortunate both in my perception and in yours – or more exactly, in the perception that I assume you must share with me. This is all the more likely since the apprehensive is used as an argument meant to convince the addressee to act in a certain way.

Another configuration applies when the modal viewpoint is anchored not with the speaker, but with the agent of an action. This happens especially when the subject of the pre-emptive action (P) is a third person. Thus in (37), the narrator retells the actions of the hero, and adds an apprehensive clause to represent the character's motivations:

(37) Mwotlap [https://doi.org/10.24397/pangloss-0007414#S124]

Kē	*so*	*ni-van*	*vege*	*hōw,*	*ni-siok*	***taple***	*qalgoy*	*yow*
3sg	prosp	3sg-go	anyway	north	art-canoe	appr	be.stuck	out

a	*Ayveñ*	*en.*
loc	(place)	def

[the hero Qasvay is maneuvering his ship between the shallow reefs]
'He tried to force his way north, *so the canoe wouldn't get stuck* on the Ravenga rocks.'

In all cases, the apprehensive's modal viewpoint (i.e. the judge of the notion that event Q should be avoided) coincides with whoever provides the impetus towards the pre-emptive action P. In (37), the action of forcing the canoe through a certain path is carried out by the hero Qasvay, so he must be understood to be also the modal source behind the apprehensive that follows. Likewise in (34), when a young boy asks his elder brother to go back home, he (the younger brother, i.e. the speaker) is the one who provides the impetus towards the action P '[let's] go home', and so he is logically the modal anchor for the apprehensive in the following clause.

When the apprehensive is used in an independent clause – as we'll discuss in §4 – the underlying modal subject always coincides with the speaker.

3.5 The apprehensive in subordinate clauses

The apprehensive is the expected mood in a number of subordinate clauses. Thus, *tiple* is the normal TAMP marker taken by the complement clause after predicates meaning 'fear' or 'worry':

(38) Mwotlap [Emails.2014-04-22]
Nok mētēgteg ⟨na-mtewot ***tele*** *qal nēk⟩.*
1SG fear ART-injury APPR hit 2SG
'I'm afraid you might get injured.'

(39) Mwotlap [https://www.odsas.net/object/104560]
Kēy dēm~dēm meh aē ⟨so kēy ***tiple*** *qeleñ⟩.*
3PL DUR~think too.much of.it COMP 3PL APPR be.lost
'They're worrying that they might get lost.'

It is also common after matrix verbs meaning 'beware',[9] 'prevent', 'forbid' – with or without a complementizer:

(40) Mwotlap [https://doi.org/10.24397/pangloss-0002415, at 10'25"]
Nēk etgoy ⟨kēy ***tiple*** *ekas nēk⟩.*
2SG beware 3PL APPR find 2SG
'Make sure they don't find you.'

(41) Mwotlap [https://doi.org/10.24397/pangloss-0009554#S68]
Nok higoy kōmyō ⟨so kōmyō ***tele*** *van~van hep na-ñye*
1SG forbid 2DU COMP 2DU APPR INF~go beyond ART-cape
mey gēn⟩.
FOC there
'I forbid you to ever walk beyond that headland over there.'

[9]The verb in (40) *etgoy* 'beware, watch out' (< *et* 'look' + *goy* ≈'over …') is morphologically parallel to the verbs found in Gaua languages further south (§2.2.3), except it has not grammaticalized into an apprehensive complementizer itself.

In these cases, the verb in the apprehensive is clearly dependent on the matrix verb for syntactic reasons: it belongs to an object clause, sometimes with overt markers of deranking (complementizer *so*).

These examples of subordination are the only cases when the role of the apprehensive mood differs from its usual precautioning function. Thus in (39) or (41), the choice of mood is arguably due to a form of semantic concordance between the notion of undesirability inherent in the apprehensive, and the meaning of the matrix predicate ('worry', 'forbid'). This is quite different from the typical use of the apprehensive we've seen so far, which consists in representing an undesirable situation Q as an argument for a pre-emptive action P.

4 The stand-alone apprehensive: An indirect speech act

After this overview of the properties of Mwotlap's apprehensive mood, we can now turn to the initial question of our study. Can the apprehensive be used in an independent clause, just like most TAMP markers, or does it always entail a form of dependency with another clause? If so, how can we characterize that dependency?

4.1 Juxtaposed clauses

In order to examine the interclausal dependency possibly triggered by the apprehensive mood, it is methodologically wise to eliminate those cases where *tiple* occurs in a clause that is already tagged as subordinate anyway, as in (38)–(41) above, and focus instead on more ambiguous examples.

By far the most typical case is for apprehensive clauses to appear juxtaposed to the pre-emptive clause P, usually in the order {P, MOD-Q} – e.g. (36)–(37). The reverse order – with the apprehensive clause first – is rare, but is attested:

(42) Mwotlap [https://doi.org/10.24397/pangloss-0003262#S54]

Nēk tig~tig qōtō, nok van tēq. — Nēk ***tepele*** *tēq higap:*
2SG DUR~stand PROVIS 1SG.IRR go shoot 2SG APPR shoot miss

ba ino nok van tēq.
but 1SG 1SG.IRR go shoot

[two brothers go bird hunting]
'You stay there, I'll shoot it.
— But you might miss it: let me shoot it myself.'

Usually, the prosody between the two clauses shows the continuity that is typical of an asyndetic juxtaposition – as can be heard in the (linked) recording of (36); this continuity is usually rendered by a simple comma in the transcription. That said, it is also quite frequent to hear prosodic discontinuity between the clauses, in the form of a longer pause and/or a drop in F0 prosody: this is conspicuous in the audio version of sentences (31) and (34) above. Such cases are best represented typographically as separate sentences.

Occasionally, one can find two apprehensive sentences in a row, right after the pre-emptive clause:

(43) Mwotlap (https://doi.org/10.24397/pangloss-0003272#S35)

Gēn	*tog*	*van~van!*	*Kēy*	***taple***	*tēy*	*maymay*	*gēn!*	*Kēy*	***taple***
1INC:PL	PROH	INF~go	3PL	APPR	hold	strong	1INC:PL	3PL	APPR

big	*gēn!*
eat.flesh	1INC:PL

[the fish are afraid of humans]
'Let's not go out there! They might catch us! They might eat us!'

One may see these examples as positive evidence that the apprehensive mood can occur in an independent clause. However, the presence of the pre-emptive clause in the immediate context makes them somewhat ambivalent. Indeed, we saw in §2.2.3 that *den* in Vurës regularly appears in clauses which prosodically form "a separate sentence, much as an afterthought" (Malau 2016: 679) – without ceasing to be a subordinator. And in fact, Vurës does not allow *den* to start a fully independent clause: the clause Q is always immediately adjacent to the pre-emptive clause P.

In other words, examples such as (34) or (43) are not sufficient to establish that the Mwotlap apprehensive can occur on its own, in an independent clause. The only clue suggesting independence is prosody – reflected here as punctuation – but this may be too weak evidence to determine that the apprehensive can really stand alone in a fully independent clause. If our corpus only had examples like these ones, where the apprehensive clause is immediately adjacent to the pre-emptive one, we might have concluded – along the lines of Malau's analysis for Vurës – that a clause in the apprehensive mood is really subordinate; simply, it sometimes bears the prosody of an afterthought.

Such an interpretation in terms of subordination would rest on the proposal that the relation of syntactic dependency, instead of being marked by a straightforward '*lest*'-type subordinator (like Vurës *den*), could also be encoded by a TAMP marker. While such a configuration would be unusual, it would not be entirely new: indeed, the two neighboring Torres languages have been shown to encode some forms of interclausal dependency by means of certain TAMP markers – namely, the Subjunctive and the Background perfect (François 2010). Could it be the case that the Mwotlap apprehensive is also a subordinating device in and of itself? Does it always require a full pre-emptive clause to be expressed in the immediate context?

As we'll see now, the answer is negative: there is no requirement for the pre-emptive clause P to be made explicit: apprehensive clauses are in fact capable of forming independent clauses.

4.2 When the pre-emptive clause is minimal

While the twofold pattern ⟨P, MOD-Q⟩ is indeed prevalent, the pre-emptive component P is sometimes reduced to the bare minimum.

For example, the P clause is sometimes hidden in a mere interjection, such as the vocal gesture for negation *Óòó* [˥ɔ.˧ɔ.˥ɔ] 'no!, no way!' which encodes disapproval or protest:[10]

(44) Mwotlap [https://doi.org/10.24397/pangloss-0002492#S72]
Dam~dam egal tog van! — ⟨***Óòó!***⟩$_P$ ⟨*kē* ***tile*** *m̄ēt!*⟩$_Q$.
DUR~hang CONAT POLIT to.it EXCLM:no 3SG APPR break
'Go on, slide down the rope! — ⟨No way!⟩$_P$ ⟨it might break!⟩$_Q$.'

The context makes it easy to reconstruct the clause hidden behind that interjection. A wants B to slide down the rope, but B protests (using the interjection *Óòó*): in other terms, B refuses to slide down, and justifies that decision with an apprehensive clause.

A similar mechanism can take place with the adversative linker *ba* 'but':

(45) Mwotlap [https://doi.org/10.24397/pangloss-0007413#S35]
Kamyō so vasem van, ⟨***ba***⟩$_P$ ⟨*nēk* ***tale*** *boel*⟩$_Q$.
1EXCL.DU PROSP reveal DIREC but 2SG APPR get.angry
'We were going to tell you, ⟨but⟩$_P$ ⟨you might get angry⟩$_Q$.'

The pragmatic function of *ba* 'but' is to reverse the argumentative polarity of the previous sentence. The hearer can reconstruct here the action hidden behind that *ba*, namely: *We almost wanted to tell you; but* [*we finally chose to keep it secret*] *for fear you might get angry at us.*

[10] About that gesture, see the Lemerig example (23) in §2.2.4; and also François (2011: 220).

An even more subtle example is provided by (46). In this folktale, a father intends to sacrifice himself for his children, by stepping inside a large oven, in order to turn magically into food. His son, fearing the fatal consequences, protests:

(46) Mwotlap [https://doi.org/10.24397/pangloss-0007413#S130]
Nok hayveg lelo qēyēñi. — ⟨***Imam!***⟩$_P$ ⟨*nēk* ***tale*** *mat!*⟩$_Q$.
1SG enter inside oven father 2SG APPR die
'Let me get inside the oven. — Father ⟨!⟩$_P$, ⟨you might die!⟩$_Q$.'

The apprehensive clause *you might die* is provided as an argument towards the instruction *Don't do it!* Yet that instruction is not made explicit by the speaker: the only hint that helps retrieve it would be the prosodic contour of protest that comes with the vocative *imam!* '(but) Father!' (François 2003: 311).

4.3 An indirect speech act

In the three examples just discussed, the apprehensive clause came in response to a previous formulation of an intended action (*Slide down the rope*; *Tell me your secret*; *Let me get inside the oven*): this made it actually easy to reconstruct the implicit instruction, by simply reversing that scenario. But sometimes, the preemptive clause P is even more drastically reduced, down to purely contextual clues.

Because narratives mostly feature apprehensives in fictitious dialogues recreated by the storyteller, they are not ideal to observe its stand-alone instances and their pragmatic implications. Some key examples below will therefore not come from my recorded texts, but from conversations I heard on the fly during immersive fieldwork. This comes with the disadvantage that no audio link can be provided; but with the advantage that these sentences constitute, arguably, the most authentic instances of apprehensive utterances, as they build upon the pragmatic circumstances of a genuine, empirical situation.[11]

One day, a toddler was awkwardly handling a large knife around the house, and someone warned me:[12]

(47) Mwotlap [Mtp.AF-BP4-07b]
Kē ***tiple*** *tig nēk aē!*
3SG APPR injure 2SG with.it
'He might injure you with that knife!'

[11] Such empirical observations would not become more authentic if the linguist asked speakers to recreate these dialogues later, as a way to produce audio recordings. This staged procedure suggested by one of the reviewers would be, in our view, methodologically flawed.

[12] Austin (1981: 229) discusses a similar example in Diyari (Australia), in his section *'Lest' as main clauses.*

This was the first utterance after a long silence, so there was no way to simply retrieve an instruction from the discourse context. Analyzing this sentence as a subordinate clause would be far-fetched; the only way to do so would be to describe (47) as a case of *insubordination* (Evans 2007), i.e. the independent use of a formally subordinate clause.[13] While this interpretation cannot be dismissed, it would rest on the hypothesis that apprehensive clauses are inherently subordinate – yet this is precisely what I am questioning here. In fact, there is no clear indication that the apprehensive mood of Mwotlap is, nor ever was, a marker of formal syntactic subordination.

Alternatively, I propose that (47) is actually a well-formed sentence from a syntactic point of view, but that it is pragmatically elliptical. As we saw earlier (§1.1, §3.2), the semantic work of the apprehensive mood is to present a potential situation Q as a risk to be avoided; contrary to the prohibitive which forms direct requests, the apprehensive constitutes an indirect strategy (Table 3). As a result, by just formulating a risk Q using an apprehensive mood, the speaker instructs the hearer to identify a pre-emptive action (P) that would prevent that scenario, or its consequences, from happening. Most often, the speaker spells out that action P explicitly as in (34) or (43), or at least hints at it as in (44)–(47). Yet on some occasions, the pre-emptive scenario P cannot be extracted from the discourse context, and the hearer is left to infer it from situational clues, combined with their practical knowledge about the world.

In the case of the knife-wielding toddler in (47), the speaker was instructing me to mentally figure out whatever scenario P could avoid the detrimental event of being injured: e.g. *Stay away from that toddler ~ Be careful ~ Get out of the house for a moment ~ Take away the knife from his hands ~* etc. By only making the risk (Q) explicit, the speaker leaves the decision on the appropriate pre-emptive action to the addressee.

The crucial point here is that an apprehensive clause in Mwotlap is always understood as an argument for some kind of action. This makes it different from other TAMP categories such as the *hodiernal* future,[14] which could also have been used in that situation:

[13] An analysis in terms of insubordination is proposed for similar apprehensive constructions, by Smith-Dennis (2021) for Papapana; Daniel & Dobrushina (2026 [this volume]) for Archi (Caucasus); or Dąbkowski & AnderBois (2026 [this volume]) for A'ingae (Colombia).

[14] The *hodiernal* future (< Lat. *hodiernus* 'of today') is the required form of the future when referring to an irrealis event that is to take place the same day as the moment of utterance (François 2003: 258–269).

(48) Mwotlap
Kē ***ti**-tig* ***qiyig*** *nēk aē!*
3SG FUT.HOD$_1$-stab FUT.HOD$_2$ 2SG with.it
'He's going to injure you with that knife!'

A sentence in the future like (48) may be read as a threat, a prediction, or a warning, and of course, may well result in some actual reaction by the hearer. Yet it could as well be uttered "for its own sake" – e.g. as a joke to elicit laughter. In fact, no linguistic element in (48) constitutes any unambiguous appeal to action.

By contrast, the apprehensive modality in (47) implies a request. Contrary to what the label *apprehensive* might suggest, the mood marker *tiple* can never be used for its own sake, as a way to simply express one's apprehension. If I just want to convey my feelings *I'm scared that you might get injured*, then a sentence like (47) is not an appropriate strategy: for such a meaning, I would use a sentence with the verb 'fear', as in (38) above. By contrast, the apprehensive mood *tiple* can only be used as an argument for something else – namely, the need to take action. The illocutionary force it bears is arguably equal to that of an order or a prohibitive, with the only difference that the request remains indirect (Table 3).

We can even propose that apprehensive sentences constitute a form of indirect speech act (Searle 1975): they represent an apprehension as a strategy to perform an instruction. The nature of the request is sometimes made explicit through a pre-emptive clause (§4.1); yet sometimes it remains implicit, and is left to the addressee to figure out. But minimally, the apprehensive entails the request to take some form of precautionary action to avoid the undesirable event Q.

In sum, the apprehensive mood of Mwotlap reflects the conventionalization of an indirect speech act. *Tiple* encodes a certain pragmatic mechanism, yet does not imply any hypotactic relationship between two clauses; nor is there any sign that it was ever a subordinator in the past. It is quite possible that it might evolve into a subordinator in the future, e.g. if it ends up requiring the effective presence of a main clause in its immediate vicinity; but examples like (44)–(47) show that *tiple*, in fact, has not yet become a subordinator.

4.4 A politeness strategy

Languages commonly employ indirect speech acts as a politeness strategy (Searle 1975): instead of an imperative *Close the door!*, it is more polite to phrase it as an apparent question *Would you mind closing the door?*, or a statement *It's getting cold in here.* In Brown & Levinson's (1987: 70) terms, a direct order would threaten the addressee's "negative face", and a common politeness strategy consists in softening such a "face-threatening act" using speech acts that are not fully directive.

And indeed, Mwotlap exploits the indirect speech act of stand-alone apprehensives for their politeness potential. Thus if my father-in-law wants to enter the room where my child is asleep, I may fear that the noise could wake her up; yet using a simple prohibitive *Don't come in!* could be taken by my in-law as too blunt and disrespectful. In such a situation, a Mwotlap speaker may choose to merely evoke an undesirable situation Q as a way to hint at the underlying instruction P:

(49) Mwotlap [Mtp.AF-AP5-41a]
Tētē mino ***tele*** *matyak!*
baby my APPR wake.up
[stopping the father-in-law before he enters the room]
'My baby might wake up!'

While (49) is syntactically well-formed, it is pragmatically elliptical: it instructs the hearer to mentally identify the nature of a pre-emptive action that may help prevent the baby from waking up. This strategy shifts the burden of formulating an imperative from the speaker to the hearer. The apprehensive thus does an efficient work of getting a message across, while preserving the face of both participants.

4.5 The humorous potential of the apprehensive

Finally, the pragmatic mechanism at play with the Mwotlap apprehensive is perhaps most conspicuous when it is exploited for its humorous potential.

One of the favourite pastimes of teenagers in the area is to playfully tease each other about their romantic relations, real or imagined. Interestingly, humorous speech appears to be particularly prone to the use of stand-alone apprehensives, perhaps because they play on people's imaginations. I once witnessed a dialogue between two teenage boys, in the cheeky tone that is typical of friendly interactions on the island of Motalava. One boy (let's call him Stan) had just stealthily smiled at a girl who was walking in the distance. Her brother Joe caught sight of this, and said to Stan:[15]

[15]Note that in (50), the predicate head is not a verb, but a noun. Indeed, in the absence of copulas, north Vanuatu languages allow nouns to inflect for tense-aspect-mood-polarity in just the same way as verbs [see §2.1.2]. The meaning of the noun predicate is ascriptive, i.e. 'be an N'. When it is TAMP-inflected, it also receives a dynamic interpretation: that is, in (50) the noun *wulus* must be read as a dynamic event '[become] brothers-in-law' (François 2026: 1040).

(50) Mwotlap [Mtp.AF-AP8-11b]

Ēt! *Dō* ***tiple*** *wulus!*

INTJ 1INCL.DU APPR brother.in.law

'Hey! Hope we don't become in-laws!'

(lit.) 'You and I *might* (become) in-laws!'

This witty line made everyone laugh. The logic here rests on the idea that brothers-in-law owe great respect to each other, have to avoid each other or to comply with various taboos, which are central to kinship relations in this society (Codrington 1891: 43–45; Malau 2016: 12–14; François 2022b: 218). Joe and Stan were good friends, joking at each other all the time, but the prospect of one day becoming in-laws would mean the end of this casual friendship, and the beginning of a very different sort of respectful relation, filled with rules and pitfalls. Many jokes play on the contrast between casual and formal kinship relations, and a sentence like (50) was no exception: the contrast between those two different social statuses was source for laughter.

But what probably made the joke even wittier was the ellipsis triggered by the stand-alone apprehensive, as it forced the hearer to retrieve a hidden instruction behind it. Hearing (50) drove everyone to wonder what could have suddenly caused the mention of becoming in-laws. One had to rewind Joe's whole reasoning, from a new kinship relation in an imaginary future, back to … the brief smile he had just seen Stan send to his sister. Only this logical path could connect the dots between Q and P – that is, between the "apprehended" situation (Q: *you and I might end up becoming in-laws*) and the implicit instruction (P: *you'd better stop smiling at my sister!*). The sentence was all the more witty that this particular instruction P was left unsaid, and could only be retrieved through some acrobatic mental gymnastics.

During the time I spent in the island with the community, I often heard such a jocular use of the apprehensive mood in stand-alone sentences. What makes such utterances noteworthy to the linguist observer is how they exploit the pragmatic mechanism that is precisely central to the apprehensive mood. The stand-alone apprehensive not only exposes a potential "risk", it also incites the hearer to reconstruct a hidden instruction behind it, anchored in a specific discourse context. The stretched distance between the two ends of the reasoning is often key to the success of the joke.

5 Conclusion

The languages of north Vanuatu have developed different devices to encode apprehensional semantics. Some make use of a precautioning subordinator (similar to English *lest*), and make it a requirement that the pre-emptive clause should be expressed in the immediate context. But Mwotlap illustrated a different configuration: an apprehensive mood (*tiple*) which can appear in dependent and independent clauses alike.

The primary role of Mwotlap *tiple* is to flag a virtual situation as undesirable. Yet an apprehensive clause is never uttered for its own sake, as though one simply predicted an inevitable situation with a tone of regret (*Alas, we'll soon get soaked in this rain*). Rather than just predicting a problem, this modal marker also flags it as an argument towards a call for action. Ultimately, this mood arguably bears directive illocutionary force, as much as an order or a prohibitive.

Quite often, the target instruction takes the form of a separate clause P, while the apprehensive Q serves as a background justification for it: ⟨*Let's go back inside*⟩$_P$⟨[*because otherwise*] *we'll get soaked in the rain*⟩$_Q$. But our study of the Mwotlap apprehensive showed that the pre-emptive clause P is not always present, and may need to be reconstructed by the hearer based on contextual clues. An utterance consisting solely of an apprehensive clause (≈Eng. *We risk getting soaked in the rain!*) may be syntactically complete, yet it remains pragmatically elliptical, as it hints at an implicit order.

Whereas the process of insubordination normally sees a subordinate clause gain independent status, I have proposed that the Mwotlap apprehensive works in the opposite way. Its use in conversation suggests it encodes a mechanism that is inherently pragmatic, based on an indirect speech act: present an apprehension as a justification for an instruction, whether the latter is explicit or not. Only time will tell if this pragmatic dependency eventually grammaticalizes into full subordination, or if the apprehensive mood preserves its subtle brand of grammatical freedom.

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Abbreviations

1	first person	IPFV	imperfective
2	second person	IRR	irrealis
3	third person	ITER	iterative
INC	first person inclusive	LEST	*lest*-like subordinator
ABL	ablative	LNK	linker
APPR	apprehensive mood	LOC	locative
ART	article	MOD	modal
COMP	complementizer	NEG	negation
CONAT	conative	NSG	non-singular
DAT	dative	OBL	oblique
DEF	definite	PERS	personal article
DEIC	deictic	PL	plural
DIREC	directional	PRF	perfect
DIST	distal deictic	POLIT	polite imperative
DU	dual	POT	potential
DUR	durative	PROH	prohibitive
EXCL	exclusive	PROSP	prospective
EXCLM	exclamative	PROVIS	provisional
FOC	focus	PRVT	preventive
FUT	future	PST	past
FUT.HOD	hodiernal future	REC.PST	recent past
HAB	habitual	RESTR	restrictive
IMP	imperative	SG	singular
INCL	inclusive	STAT	stative aspect
INF	infinitive	SUB	subordinator
INTSF	intensifier	TOP	topicalizer

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Chapter 13

The grammatical realisation of apprehensional functions in Ngumpin-Yapa languages (Australia)

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This paper provides a survey of apprehension constructions in the Ngumpin-Yapa (Pama-Nyungan) languages of northern Australia. As in many Australian languages, there are morphosyntactically distinct strategies for encoding apprehension in Ngumpin-Yapa languages: a verbal *apprehensional* morpheme and a nominal *aversive* morpheme. Languages differ in terms of the extent to which these morphemes uniquely encode apprehension, some also encoding physical location or potentiality. We survey apprehensional morphemes in nine Ngumpin-Yapa languages: Walmajarri, Ngardi, Jaru, Wanyjirra, Gurindji, Bilinarra, Mudburra, Warlmanpa and Warlpiri. In these Ngumpin-Yapa languages, the verbal apprehensional morphemes are found in the auxiliary complex as a base for bound pronominals. The nominal aversive morphemes are oblique case suffixes that relate nominals and nominalised predicates to main predicates. The auxiliary bases form part of the system of finite tense–aspect–modality, whereas the case suffixes act as complementisers for non-finite clauses, with dependent tense features. Both classes of morphemes can appear in independent and dependent syntactic environments. Additionally, both classes of apprehensional morphemes can have non-apprehensional functions, for example in Wanyjirra the comitative case suffix introduces concomitant participants, which can also be used to express aversion to particular participants. This chapter makes a substantive contribution to our understanding of apprehension in Australian languages – a topic that hitherto has received little attention, despite the prevalence of apprehensional morphemes in Australian languages. This survey highlights the fact that even in a group of closely related languages, the grammatical encoding of undesirability can vary greatly.

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1 Introduction

In this chapter we survey and compare apprehensional constructions found in languages of the Ngumpin-Yapa subgroup of the Pama-Nyungan family (Australia). We define apprehension as referring to undesirable and avoidable eventualities or entities (since entities with apprehensional marking are employed metonymically for undesirable eventualities, we use the term outcome to encompass both eventualities and entities). Apprehensional constructions may simply comprise an expression of an avoidable and undesirable possible outcome; or may additionally express an alternate course of events which avoids the outcome. This is schematised in Figure 1.

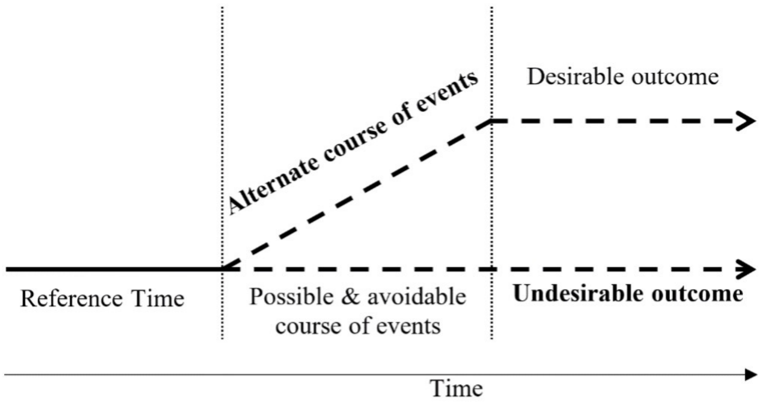

Figure 1: Schema of prototypical meaning of apprehension, where the bolded situation are those which may be overtly expressed in apprehensional constructions.

By undesirable, we refer to some event or entity that is evaluated as something that should be avoided. This evaluation is from the speaker's perspective on behalf of the person they are warning, and broadly relates to a negative outcome, whether due to an event, an event associated with an entity, or simply the coincidence of the subject and another entity. By avoidable we mean that an alternate course of action should be salient to the addressee, as in (1), where the speaker overtly specifies the available course of action through an imperative clause 'get down from there'.

(1) Mudburra (Ngumpin-Yapa; PN)

Karu	*jurd*	*wandi,*	*bi=ngku*	*linymurr*	*ka-yinarra*	*yali*
child	get_down	fall[IMP]	APPR=2SG.NS	break	be-SPEC	that

karndi=ma.
tree=TOP
'Kid, get down from there [alternate course of events], (or else) that tree might break on you [undesirable outcome]!'
[SD:DOS1-2017_017-02:2674149_2677841]

The alternate course of events allowing the undesirable outcome to be avoided may be explicit, as above in (1) or implicit, as in (2). We will refer to a clause which makes the alternate course of events explicit as PREEMPTIVE clause, following Evans (1995: 265). However, the syntactic relation between the two clauses in Ngumpin-Yapa languages is unclear (this is taken up in §2.1.3).

(2) Warlmanpa (Ngumpin-Yapa; PN)
Nga=nyanu-n=nga kuma-nma!
APPR=REFL-2SG.S=APPR cut-POT
'You might cut yourself!' [H_K01-505B, DG, 38:14]

Lichtenberk makes a distinction between situations in which the eventuality may be avoided altogether, and where the effects of the eventuality may be mitigated (Lichtenberk 1995: 297), though we have no clear data for the latter in these languages. As will be shown in the survey (§1.1), only one Australian language (Martuthunira) is known to grammatically distinguish these two functions. For the purposes of this paper, the alternate course of events may either avoid the event altogether, or mitigate the effects of the event (thus avoiding the undesirable outcome regardless).

Australian languages have a number of lexical and morphosyntactic strategies to convey apprehensional situations. Grammaticalised morphemes with apprehensional functions have been identified within verbal and nominal paradigms (i.e. morphological strategies); we also find grammaticalised multi-clausal constructions which convey apprehension (i.e. syntactic strategies).

In this chapter, we focus on investigating the degree to which apprehension has been grammaticalised in Ngumpin-Yapa languages, a subgroup of the Pama-Nyungan language family. We begin by first surveying apprehension cross-linguistically within Australia (§1.1) and present a grammatical overview of Ngumpin-Yapa languages (§1.2), leading into a systematic account of apprehensional constructions in these languages. We discuss the various functions of the aversive case (relational, subordinating, and insubordinate functions signalling apprehension, §2.1); the marking of complements of 'fear' predicates (§2.2), as well the encoding of apprehension by a morpheme known as the auxiliary (§2.3). Finally, we consider the range of functions that sentences involving apprehensionals tend to serve in discourse, divided into directives and assertives (§3).

1.1 Apprehension in Australian languages

In descriptions of Australian languages, apprehensional morphemes have been assigned a number of terminological labels, reflecting the lack of a unified approach to understanding these morphemes. Table 1 exemplifies the wide array of glosses given to apprehensional morphemes in grammars of Australian languages.

Moreover, while many Australian languages appear to have some strategy for encoding apprehensional meanings, there has been very limited literature examining how these constructions vary across the continent.[1] A singular exception is Zester's (2010) survey of apprehensional constructions in 8 Pama-Nyungan languages and 4 non-Pama-Nyungan languages, which sought to compare the category of apprehension in Australian languages and English.[2]

We categorise apprehensional morphemes into three distinct types based on morphosyntactic locus of exponence, described below:

1. nominal case inflections;
2. verbal mood inflections; and
3. non-inflecting particles.

1.1.1 Nominal case suffixes

Apprehensional suffixes in Australian languages occur within nominal paradigms, either as dedicated case suffixes or as a function of a case suffix not primarily apprehensional in nature. In Wargamay (Dyirbalic), for example, a locative suffix has a range of adjunct spatial functions, such as to mark location in space and time ('in', 'at' 'on', 'near' etc.), accompaniment ('with') and circumstance ('while'). The Wargamay locative case can also be used in apprehensional contexts, as in (3), in which the locative suffix *-nga* is used to indicate that the white man is something to be avoided.

[1] Gaby (2017) presented an unpublished typology of apprehension in Australian languages.

[2] The Pama-Nyungan languages included in Zester's study were: Yidiny, Djabugay, Pithanthathara, Martuthunira, Panyjima. The non-Pama-Nyungan languages were Ndjébbana (Arnhem), Gooniyandi (Bunuban), Wambaya (Mirndi) and Jingulu (Mirndi). No Ngumpin-Yapa languages were covered in Zester's survey.

Table 1: Sample of the various glosses given to apprehensional morphemes.

Language	Morpheme(s)	Gloss	Reference
Djabugay	*lan, -ndabi, -ndjabi*	‘aversive’	(Patz 1991: 268)
Wangkajunga	*-ngkamarra*	‘avoidance’	(Jones 2011: 95, 290–291)
Kayardild	*waalu-tha/-waal-i-ja*	‘evitative’	(Evans 1995: 173–175)
Warumungu	*-kajji*	‘lest’	(Simpson & Heath 1982: 22–23)
Alyawarra	*-ikitja*	‘negative causative’	(Yallop 1977: 69, 75–76)
Walmajarri	*-ngkamarra*	‘projected reason’	(Hudson 1978: 31)
Mangarrayi	*baḷaga*	‘evocative/anticipatory’	(Merlan 1989: 146–147)
Ngarinyman	*ngaja*	‘apprehensive’	(Angelo & Schultze-Berndt 2016)
Bilinarra	*ngaja*	‘admonitive’	(Meakins & Nordlinger 2014: 305)

(3) Wargamay (Dyirbalic; PN; Dixon 1981: 30)
Ngayba bimbirigi waybala-nga.
1SG.S run.PERF white_man-LOC
'I'm running for fear of the white man.'

Other languages possess dedicated nominal suffixes for encoding apprehension. In the Australianist tradition, these are typically labelled 'evitative' or 'aversive' cases.[3] These suffixes encode that the referent of the case-marked phrase is to be avoided, often translated with some implicit eventuality associated with the referent, as in the Yidiɲ example in (4).

(4) Yidiɲ (Yidinic; PN; Dixon 1977: 262)
*Jaja munubujun nyinany / bama-**yida**.*
child.ABS inside:STILL sit:PST person-**AVE**
'The child still sat inside [the hut] for fear of [being seen by] the people.'

In some languages, the dedicated case suffix is formally comprised of another case suffix as well as an apprehensional increment. In Pintupi, for example, to form the 'avoidance' case, the increment *-marra* is added to a locative case form which marks accompaniment, animate goals, and animate locations (Hansen & Hansen 1978: 57). Examples (5) and (6) demonstrate the same stem *mul̪ipuru* 'one obscured by the shelter' in the locative (*-ngka*) and avoidance cases (*-ngkamarra*) respectively. This type of complex case formation is cross-linguistically common in Australia, and has been termed "compound case" (Schweiger 2000: 256) or "parasitic formation" (Matthews 1991: 85–86).

(5) Pintupi (Western Desert; PN; Hansen & Hansen 1978: 99)
*Tjampitjin-ta=litju=lu watjan̪u, mul̪i-**puru-ngka**.*
name-LOC=1DU.EXCL.S=3SG.LCT said shelter-**OBSC-LOC**
'We spoke to Tjampitjinpa who was obscured by the shelter.'

(6) Pintupi (Western Desert; PN; Hansen & Hansen 1978: 99)
*Mul̪i-**puru-ngkamarra**=n̪a=lura yarka-∅ pitjangu.*
shelter-**OBSC-AVE**=1SG.S=3SG.AVE secret-NOM went
'To avoid that one who is obscured by the shelter, I kept myself hidden as I walked.'

[3]Henceforth we will generally refer to isomorphic case forms encoding apprehension as 'aversive' cases and gloss them consistently as 'AVE'.

1.1.2 Verbal inflection

Some Australian languages have a dedicated verbal inflection which signals apprehension. Martuthunira (Ngayarta, Pama-Nyungan), for example, has a 'lest' verb suffix which encodes apprehensional circumstances, exemplified in (7).

(7) Martuthunira (Ngayarda; PN; Dench 1995: 73)
Nhiyu pawulu wangka-yangu puni-layi karla-ngku ***kampa-wirri****, mir.ta*
this child tell-PASSP go-FUT fire-EFF **burn-APPR** not
waruul puni-nguru.
still go-PRS
'This child has been told to go lest he gets burnt by the fire, but he still hasn't gone.'

A less common locus of encoding apprehensional meanings across the Australian continent is that of a discontinuous inflection as in Nyangumarta (Marrngu, Pama-Nyungan) (Sharp 2004: 165, 186–187), or a clitic as in another Marrngu language, Mangarla (Agnew 2020: 335) which takes predicates as its hosts, as shown in (8). In the case of the latter, the status of the ANTICipatory in Mangarla as a clitic is affirmed by its co-occurrence with TAM inflections other than the imperative and its (rare) encliticisation to nominal predicates.[4]

(8) Mangarla (Marrngu; PN; Agnew 2020: 336)
*Pantu warlu parra~parra kampa=nyjurruny=**kulu**.*
MED fire heat~RDP burn.IMP=1PL.INCL.O=**ANTIC**
'That fire is hot and might burn us.'

There is variability across languages with respect to whether a given verbal inflection is a dedicated marker of apprehension or semantically less specific, e.g. whether it also has a general irrealis meaning denoting any possible and unrealised events which need not be negatively evaluated. For instance, Wargamay has an 'irrealis' verbal suffix which can be used in future, unreal and apprehensive contexts. The 'irrealis' inflection is exemplified in (9), where the event is not negatively evaluated, and in (10), where it is.[5]

[4]The verb in (8) is glossed as an imperative, which expresses a wide range of functions in Mangarla (Agnew 2020: 320).

[5]Further examples are: the anticipatory inflection in Ngardi (Ngumpin-Yapa, Pama-Nyungan; Ennever 2018), the subjunctive inflection in Yulparija (Western Desert, Pama-Nyungan; (Burridge 1996: 39) or the obligative and hypothetical inflections in Wangkajunga (Western Desert, Pama-Nyungan; Jones 2011: 201–202). All of these inflections share a semantic basis that the speaker is conveying that the event is likely to occur. Undesirability does not seem to be encoded. In some languages, clausal negation affects possible interpretations, as in Mangarla (Agnew 2020: 335–339).

(9) Wargamay (Dyirbalic; PN; Dixon 1981: 54)
*Ngayba nga: **walama**.*
1SG.S NEG **ascend.IRR**
'I'm not climbing (anymore, because I'm tired).'

(10) Wargamay (Dyirbalic; PN; Dixon 1981: 54)
*Ngaru gilwalga / **ba:dima**.*
NEG kick.NEG.IMP **cry.IRR**
'Don't kick him, lest he cry / he might cry.'

In most Australian languages, where a verb suffix denotes an apprehensional meaning, the clause in which it occurs is usually either juxtaposed or subordinate to a main clause encoding a preemptive event, as in (11), suggesting these verb inflections may be indicative of subordination.

(11) Yidiɲ (Yidinic; PN; Dixon 1977: 351)
*Nyundu gajidagan / giyara-nggu guba:-**nji**.*
you:ERG long_way:INCH:VBLSR:IMP stinging_tree-ERG burn-**LEST**
'You keep away, lest you get stung by the stinging tree!'

1.1.3 Non-inflecting particles

Some Australian languages have particles (often complementisers) which introduce a subordinate clause and place it in an apprehensional relation to the (preemptive) main clause. Zester (2010) discusses two non-Pama-Nyungan languages in Australia's north exhibiting apprehensional complementisers: the Mirndi languages Wambaya (Nordlinger 1998: 210) and Jingulu (Pensalfini 2003). An example from Jingulu is given in (12), where the subordinate clause is headed by the evitative complementiser *karningka* (glossed 'LEST').

(12) Jingulu (Mirndi; Pensalfini 2003: 120)
*Buba jimi-rni ila-mi jungkali marru-ngka-mbili **karningka***
firewood that(n)-FOC put-IRR far house-ALL-LOC **LEST**
buba-arndi ngunji-jiyimi marru.
fire-INST burn-come house
'Put the firewood down far from the house, lest the fire burn the house down.'

1.1.4 No apprehensional morphemes

Other languages may convey apprehension by the juxtaposition of a preemptive clause with a clause expressing a possibility, but without overt apprehensional morphology on the latter. In Warluwarra (Ngarna, PN), there are no morphemes which could be clearly associated with apprehension. Instead, apprehension is derived via pragmatic enrichment in irrealis situations, particularly when juxtaposed with a preemptive clause, as in (13).[6] In this case, if the addressee smells the meat (preemptive), it mitigates the effects of it being rotten (for example, it results in them not eating it if it is rotten).

(13) Warluwarra (Wagaya-Warluwaric; PN; Breen 1970: 234)
Jatjunma jinja karlaanha / warrya ngarlatha jiwa.
smell.IMP that[ACC] meat[ACC] rotten maybe it
'Smell that meat; it might be rotten.'

Ngumpin-Yapa languages present an opportunity to refine cross-linguistic generalisations regarding apprehensional constructions, particularly in the Australian context. Firstly, most Ngumpin-Yapa languages exhibit both a nominal case suffix and a particle-like auxiliary to signal apprehension, unlike any other language surveyed in Australia. Additionally, Ngumpin-Yapa languages, despite being a close-knit group of languages, show a very interesting diversity in the expression of complements of 'fear' predicates and apprehensional marking in general.

1.2 Ngumpin-Yapa languages

This survey focuses on nine languages of the Ngumpin-Yapa subgroup, a subgroup within the widespread Pama-Nyungan family in Australia. The languages for which data is discussed in this paper are: Walmajarri, Jaru, Ngardi, Gurindji, Wanyjirra, Bilinarra, Mudburra, Warlmanpa and Warlpiri. Our current understanding of their relative cladistic relationships within the Ngumpin-Yapa subgroup is represented in Figure 2. The areal distribution of Ngumpin-Yapa languages across northwestern Australia is shown in Figure 3.

Evidence for the unity of the Ngumpin and Yapa subgroups, and for their position within the Nyungic subgroup, is provided by Laughren & McConvell (2004), while evidence adduced from pylogenetic studies is provided by Bowern

[6]Note that this is similar to languages which have an irrealis inflection or complementiser-like particles which are used in apprehensional constructions but do not encode specifically 'apprehension'. In fact there may be no distinction between these two types. The distinction lies in how strongly the morphemes are associated with apprehension.

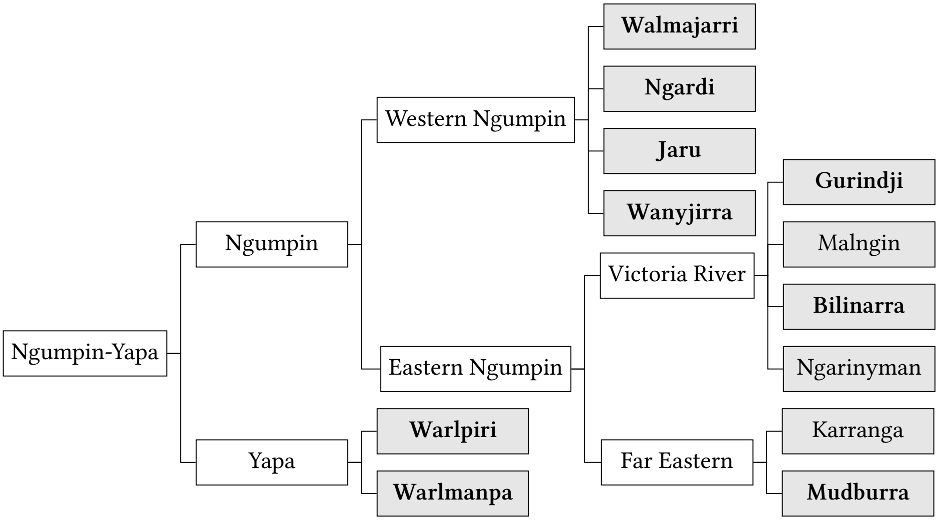

Figure 2: Phylogeny of the Ngumpin-Yapa subgroup. Languages included in this study are emphasised. Adapted from McConvell (2009: 791)

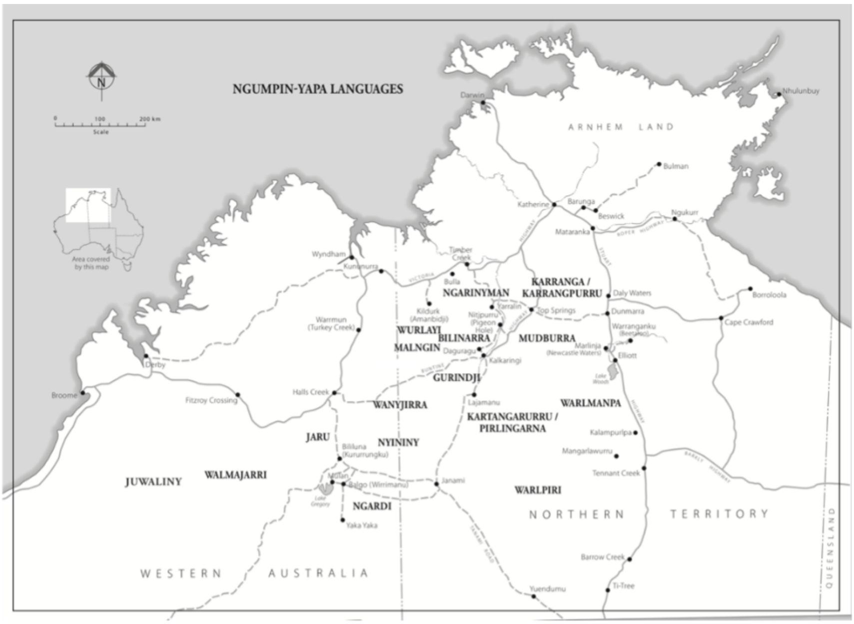

Figure 3: Map of relative locations of Ngumpin-Yapa languages across northern Australia (adapted from Meakins & Nordlinger 2017; cartographer: Brenda Thornley).

& Atkinson (2012) on the basis of lexical data, and by Macklin-Cordes & Round (2015) on the basis of phonotactic data. In this section, we briefly detail some of the relevant morphosyntactic characteristics of these languages.

A working orthography of all the languages used in this survey is presented in Table 2. The only phoneme not shared across the subgroup is a retroflex flap which is only fully phonemicised in certain dialects of Warlpiri and is otherwise an allophone of the retroflex stop (Western Warlpiri) or an allophone of the retroflex continuant (glide) phoneme (Ngardi and Jaru). As for stops, voicing is not distinctive in the languages under discussion. Across the Ngumpin-Yapa languages there are three orthographies in use which largely differ by the representation of stops.[7] We retain the orthography from source materials in examples.

Table 2: Maximal Ngumpin-Yapa consonant phoneme inventory (orthographic forms given in brackets where they differ from the expected IPA symbol)

	Peripheral		Apical		Laminal
	Bilabial	Velar	Alveolar	Retroflex	Palatal
Stop	p/b	k/g	t/d	ʈ <rt/rd>	c <j>
Nasal	m	ŋ <ng>	n	ɳ <rn>	ɲ <ny>
Lateral			l	ɭ <rl>	ʎ <ly>
Flap/Trill			r <rr>	ɽ <rd>	
Glide	w			ɻ <r>	j <y>

All Ngumpin-Yapa languages have a three-way distinction in vowel quality with most languages generally described as having a marginal length contrast with a low functional load (Table 3).

Table 3: Ngumpin-Yapa vowel inventory

	Front	Back
High	i, (ii)	u, (uu)
Low	a, (aa)	

[7] p ~t ~k: Gurindji, Warlpiri, Ngardi, Warlmanpa
b ~d ~k: Mudburra
b ~d ~g: Bilinarra, Ngarinyman, Wanyjirra

Ngumpin-Yapa languages are agglutinating and purely suffixing. Core arguments are case-marked and are also cross-referenced by bound pronominal enclitics which encode person, number and case features. These languages all demonstrate the properties of non-configurationality as described for Warlpiri (Hale 1983), including: free word order, discontinuous phrase structure and frequent ellipsis of nominals. All Ngumpin-Yapa languages share the property that the pronominal clitics generally occupy second position of the clause, so-called Wackernagel's position (Wackernagel 1892). Other features of these clitics vary considerably across the subgroup – both with respect to the range of grammatical relations that bound pronouns may cross-reference, and the types of clausal elements to which they may attach.

One type of host for the pronominal clitics are termed *auxiliary bases.* These are otherwise uninflecting particles which obligatorily host the pronominal clitics. The semantics of the auxiliary base varies considerably across languages. In some cases, the auxiliary base is essentially a meaningless host for the pronominal enclitics and has been termed "catalyst" by some authors, for example *ba* in Mudburra (Capell 1956: 68–69), *ngu* in Gurindji (McConvell 1996b), and *nga* in Ngarinyman (Jones et al. 2019: 30). In some Ngumpin-Yapa languages, there is a small inventory of semantically meaningful auxiliaries (or "modal particles"), which contribute tense, aspect, or mood information to the clause. For example, the auxiliaries in Warlpiri include *lpa* 'imperfective' and *ka* 'present' (Laughren 2013), which combine with verb inflections to encode fine-grained TAM distinctions.[8]

In nearly all Ngumpin-Yapa languages, lexemes of certain other grammatical categories also attract the pronominal enclitics (in lieu of an auxiliary base in the given construction), including complementisers, interrogatives as well as imperative-inflected verbs. In languages without a catalyst auxiliary, such as Ngardi, in the absence of any of these more typical hosts, the bound pronouns encliticise to the first phonological word (or in some cases, the first phrase) (Ennever 2021: 310–313).

These bound pronominals pattern primarily according to a nominative-accusative system, in which A and S are encoded by a subject series, and O is encoded by an object series. For many of the languages surveyed, there is syncretism between the object (o) and oblique (obl) series in all subject/number combinations except 3sg, leading to the convention of labelling these forms as either o 'object/oblique', or ns 'non-subject'. In nearly all of the Ngumpin languages, nominals in certain other (traditionally semantic) cases such as the

[8]Western Warlpiri has a future verb inflection in addition to the auxiliaries.

locative, allative and ablative can also be cross-referenced by bound pronominals (which, if distinct from the oblique, are labelled LCT 'locational' here), whereas this is not permitted in either of the Yapa languages. Language-specific systems are given in Table 4. Ngumpin-Yapa languages do not utilise the bound pronominal system for aversive case-marked adjuncts (§2.1), but do for complements of 'fear' predicates (§2.2).

Table 4: Relationship between case marking and the cross-referencing bound pronouns.

		Subject			Non-subject			
	Case	ERG	ABS	ABS	DAT	LOC	ALL	ABL[a]
Yapa	Warlpiri	S	S	O	OBL	—	—	—
	Warlmanpa	S	S	O	OBL	—	—	—
Eastern Ngumpin	Bilinarra	S	S	O	OBL	—	—	—
	Gurindji[b]	S	S	O	OBL	LCT	LCT	LCT
	Mudburra	S	S	O	OBL	OBL	OBL	—
	Wanyjirra	S	S	O	OBL	OBL	OBL	OBL
Western Ngumpin	Jaru	S	S	O	OBL	LCT	LCT	LCT
	Ngardi	S	S	O	OBL	LCT	LCT	—
	Walmajarri	S	S	O	OBL	ACS	—	—

[a] We use the case label *ablative* here, however *elative* is used for some languages.

[b] This follows McConvell's (1996b: 84) analysis of bound pronouns in Gurindji, where "affected locations … are cross-referenced by clitics, unlike other locations, and in a way distinct from either S, O, or IO clitics."

All Ngumpin-Yapa languages exhibit extensive case marking. Nominals bear no overt case marking in the S and O function, and bear overt ergative case marking in the A function (in accordance with an ergative-absolutive case-marking pattern). While free pronouns in Warlpiri, Ngardi, Jaru, Walmajarri, and Eastern Mudburra exhibit ergative-absolutive marking like other nominals, in Wanyjirra, Gurindji, Bilinarra, Western Mudburra and Warlmanpa, free pronouns are invariably unmarked in all three core functions.

All Ngumpin-Yapa languages possess two morphosyntactically distinct parts of speech with verbal semantics: inflecting verbs and coverbs. Inflecting verbs are a closed class of inflecting words (from around thirty to over a hundred) which bear tense/aspect/mood inflection; however the languages vary considerably in terms of the number and type of features encoded by verbal inflections. Variation is found, for example, in the distinctions made for past time reference (with a particularly differentiated system in Jaru and Ngardi), the presence of associated motion markers (Mudburra, Ngardi, Warlpiri, Warlmanpa), the presence of directional suffixes which can be cumulatively realised with another feature (Ngardi, Warlpiri), and patterns of reduplication (with a particularly complex system in Mudburra). An inflecting verb may constitute a simple predicate on its own, or it may act as a light verb in a complex predicate along with a coverb (also called a "preverb"; see Osgarby & Bowern 2023).

Ngumpin-Yapa languages exhibit both finite and non-finite subordination. The latter typically involves a non-finite verb form or a coverb bearing complementiser case suffixes, expressing the relative time and, in some cases, control properties of the subordinate clause (Simpson & Bresnan 1983). Finite subordination is achieved by the use of a complementiser particle within the finite subordinate clause. Such clauses may have both temporal (T-relative) or anaphoric (NP-relative) functions; however multiple interpretations for the relationship between clauses are usually possible (Hale 1976).

2 Morphosyntax of Apprehension in Ngumpin-Yapa languages

The complete inventory of grammatical markers of apprehension in Ngumpin-Yapa languages is presented in Table 4 and visually represented in Figure 4. Three loci of grammaticalised apprehensional marking have been identified. The first is what will be termed an *aversive case*, a dedicated nominal case suffix, which can mark a nominal referent as something which is feared, or a non-finite clause which denotes an undesired outcome. The second is the idiosyncratic marking of complements of 'fear' predicates. These are treated as a grammaticalised form of apprehensional marking since their marking is idiosyncratic but shows some relationship to the grammar of apprehension elsewhere in the languages. The third is an auxiliary (although in some languages these auxiliaries are more complementiser- or particle-like) which indicates that the event referred to by the finite clause is undesired. While most languages have all three constructions available, Bilinarra and Jaru lack an aversive case suffix.

The aversive case is discussed in §2.1, first exemplifying its relational function (§2.1.1), then its subordinate function (§2.1.2), before discussing the issues in accurately distinguishing the two functions (§2.1.3). Furthermore, in some languages these cases exhibit insubordinate functions (§2.1.4). 'Fear' complementation is

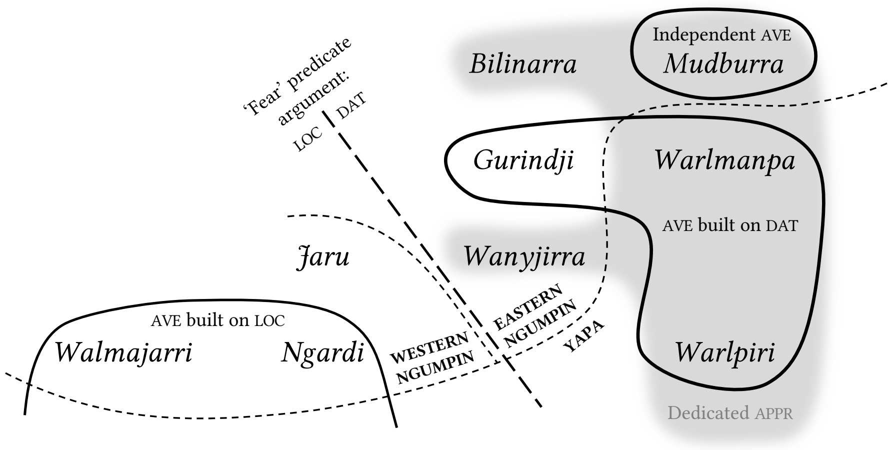

Figure 4: Geographical distribution of dedicated apprehensional markers in Ngumpin-Yapa languages.

discussed in §2.2: strategies include the locative case (§2.2.1), the dative case (§2.2.2), as well as the aversive (§2.2.3). These variable case-marking patterns are summarised in Table 5. Finally, in §2.3 we analyse the auxiliaries, discussing their syntactic properties in §2.3.1 and whether they are dedicated to encoding apprehension or not in §2.3.2.

2.1 Aversive case

Six Ngumpin-Yapa languages possess an aversive case suffix that can be classified as a dedicated apprehensional morpheme. Three languages lack a dedicated case suffix for expressing apprehension, and instead allow apprehensional meanings to be encoded by a polyfunctional case such as locative or comitative. In Figure 4 we distinguish between aversive case suffixes that are formally independent of other suffixes (Mudburra), those built on the dative case suffix (Gurindji, Warlpiri, Warlmanpa), and those built on the locative case suffix (Walmajarri, Ngardi). Warlpiri, Warlmanpa (Yapa) and Gurindji (Eastern Ngumpin) all have aversive case forms which seem to be comprised of the dative case suffix *-ku* ~*-wu* plus an increment, together marking the aversive case.

Two Western Ngumpin languages (Walmajarri and Ngardi) possess aversive suffixes formally built on a locative case allomorph *-Ca* + an increment *-marra*. This case formation (LOC + *marra*) is widespread in the Western Desert languages

Table 5: Overview of morphemes used to signal apprehension in Ngumpin-Yapa languages.

	Language	Case in aversive function	Case on 'fear' complement	Auxiliary[a]
Yapa	Warlpiri	*-kujaku* AVE	DAT, AVE	*kalaka/kajika* POT
	Warlmanpa	*-kuma* AVE	DAT	*kala(ka)* APPR *nga=...=nga* FUT=...=POT
Eastern Ngumpin	Bilinarra	*-ngka, -la, -Ca* LOC	DAT	*ngaja* ADM
	Gurindji	*-kumpalng*, *-wumpalng* AVE	DAT	*ngaja* ADM
	Mudburra	*-wirri* AVE	DAT, AVE (rare)	*bi(ya)* APPR
	Wanyjirra	*-kunyja*, *-wunyja* COM	DAT	*ngarra* ADM
Western Ngumpin	Jaru	*-ngka, -la, -Ca* LOC	LOC	*ngarra* 'possible'
	Ngardi	*-Camarra* AVE	LOC, AVE	*ngarda* HYP
	Walmajarri	*-Camarra*, *-karrarla* AVE	LOC, AVE (rare)	*nga-rta* MR1-DUB *pa-rta* MR2-DUB

[a] Note that in the *auxiliary* column we have retained the authors' glossing, however elsewhere we have glossed these auxiliaries APPR in order to allow ready comparison across languages. The *-Ca* forms in the *aversive function* column represent forms which exhibit extensive allomorphy conditioned by the preceding segment and the syllable or mora count of the stem.

to the south of Walmajarri and Ngardi, for example Kukatja and Wangkajunga. A further variant is *-karrarla* in Walmajarri. Interestingly, a locative formative *-rla* is still involved in *-karrarla* but as the right-most constituent, with a base *-karra*. An example is provided in (14).

(14) Walmajarri (Western Ngumpin; PN; Richards & Hudson 2012)
Kiwiri ma=rnalu jart-kuji-lany ***mimi-karrarla.***
wood MR1=1PL.EXCL.S burn-CAUS-CUST sick-**AVE**
'We would burn the *kiwiri* wood to keep away sickness.'

Like many other case suffixes in Ngumpin-Yapa languages, the aversive case can either mark a participant in a particular semantic relation to the clausal predicate (§2.1.1) or it may serve as a subordinating case suffix (§2.1.2). These functions have been termed *relational* and *complementiser* functions by Dench & Evans (1988) – although the two are not always easily distinguished (§2.1.3). Additionally, aversive cases may appear in insubordinate constructions (§2.1.4). These constructions lack a main clause predicate and instead the implied event associated with the participant marked by the case is insubordinated, i.e. appears in a main clause function (see Evans & Watanabe 2016). In all cases in the corpora that we are aware of, the aversive case is used to indicate an entity or situation which is to be avoided, rather than indicating a situation to be mitigated.

2.1.1 Relational function

Aversive case-marked NPs in all main clause types introduce a referent to be feared/negatively evaluated; and prototypically provide a justification for the action described by the remainder of the clause. Examples include (15) to (17).

(15) Mudburra (Eastern Ngumpin; PN)[9]
Yurrub wandi ***ngarrambalyangka-wirri.***
hide fall.IMP **police-AVE**
'Hide, for fear of the police!'[10] [MSD: AHA1-2016_027-03: 1708833_1711002]

[9] Mudburra, Ngardi, and Warlmanpa data come directly from corpora. The reference format of Mudburra and Ngardi is:
Speaker initials: Audio code: Timestamp.
The reference format of the Warlmanpa data is:
Audio code: Speaker initials: Timestamp.

[10] In Ngumpin-Yapa languages (and indeed other Australian languages with complex predicates), there is not always a straightforward link between the glosses of predicates and the given translation, as is the case in this example where the inflecting verb is glossed 'fall' yet this does not form part of the translation. It is often the case that the coverb bears the lexical information, and the inflecting verb essentially only bears grammatical (tense, mood, and aspect) information.

(16) Ngardi (Western Ngumpin; PN)
Wakurra=rna wirriri ya-ni ***mula-ngkamarra ngantany-jamarra.***
NEG=1SG.S back go-PST **this-AVE man-AVE**
'I didn't go back for fear of this man.' [PSM: TEN1-2017_008-02: 938110_946191]

(17) Walmajarri (Western Ngumpin; PN; Richards & Hudson 2012)
Jilngin-tamarra *ma=rna kaniny~kaniny yuka-lku.*
dew-AVE MR1=1SG.S inside~RDP lie-POT
'I'll sleep inside on account of the dew.'

In some languages, such as Jaru (Western Ngumpin), the locative case can encode apprehension independent of a 'fear' predicate as in (18). Further, in Warlpiri and Bilinarra, the locative-marked nominal can comprise the entire utterance, as in (19)–(20). Similarly, the comitative case (which prototypically encodes accompaniment) has an additional apprehensional function in Wanyjirra, as in (21).

(18) Jaru (Western Ngumpin; PN; Tsunoda 1981: 58)
*Wanyja-rra gardiya-**la**!*
leave-IMP white_man-**LOC**
'Leave (it) for fear of the white man (if you touch it, he might get angry).'

(19) Bilinarra (Eastern Ngumpin; PN; Meakins & Nordlinger 2014: 129)
*Gawayi! Gururij-**ja**!*
come_here car-**LOC**
'Come here! Look out—car!'

(20) Warlpiri (Yapa; PN; Mary Laughren, p.c.)
*Warlu-**ngka**!*
fire-**LOC**
'Look out for the fire!'

(21) Wanyjirra (Eastern Ngumpin; PN; Senge 2015: 200)
*Ngu=rna=yanunggula wulaj garriny-ana yalu-**wunyja** garaj-**guja***
REAL=1MIN.S=3AUG.LCT hide stay-PRS DIST2-**COM** human-**COM**
*ngaringga-**wuja**.*
woman-**COM**
'I hide from that man and woman.'

Aversive-marked NPs generally only have adjunct status in all Ngumpin-Yapa languages, and as such they are never cross-referenced in the bound pronoun.[11]

[11]See the Walmajarri example in §2.2.3 as an isolated exception.

This will be contrasted with complements of 'fear' predicates in the dative and locative cases which are treated as clausal arguments (§2.2).

2.1.2 Subordinating function

The aversive case can also mark non-finite subordinate 'lest' clauses, as in (22).

(22) Warlmanpa (Yapa; PN)
*Pingka pa-nka wa-nja-**kuma**.*
slowly go-IMP fall-INF-**AVE**
'Go slowly lest you fall.' [H_K01-505B: DG, 43:43]

However, they are not necessarily dedicated apprehensional markers, and may more broadly mark undesirable events (23), or events without a clear preemptive action (24); or in the case of Warlpiri, even impossible (yet desired) events (25).

(23) Warlpiri (Yapa; PN; Laughren 2015: 53)
*Pirtiri-ma-ni ka **ya-ninja-kujaku** janjanypa-rlu.*
pester-CAUS-NPST PRS **go-INF-AVE** demanding-ERG
'The demanding (child) is pestering her not to go.'

(24) Mudburra (Eastern Ngumpin; PN)
*Wirlarnkarra ba=rla karri-nyarra **barany-karra-wirri**, jirrbu wandi-nya-wirri.*
frightened CAT=3SG.DAT be-NARR **slip-D.LOC-AVE** drown fall-INF-AVE
'He was scared of it (the water), for fear of slipping, for fear of drowning.' [MAC: PMC1-M9-01: 2684588-2688547]

(25) Warlpiri (Yapa; PN; Laughren 2015: 6)
*Wiri ka=rna nyina-mi yalumpu=ju pirnki-kirra **yuka-nja-kujaku**.*
big PRS=1SG.S sit-NPST that=TOP cave-ALL **enter-INF-AVE**
'I'm too big to enter the cave.'

There are no clear constraints on argument co-reference between main and non-finite subordinate 'lest' clauses in any of the Ngumpin-Yapa languages examined here. The subject of the subordinate aversive clauses can be co-referent with the subject of a main clause as in (24). Conversely, there can be co-reference between the object of the main clause and the subject of the subordinate clause, as in (26).

(26) Warlpiri (Yapa; PN; Laughren 2015: 5)
[*Kurdu-kujaku pi-nja-kujaku*]*=rna yirra-rnu.*
[child-AVE bite-INF-AVE]=1SG.S place-PST
'Lest (it) bite the child I put it (the dog) away.'

2.1.3 Main clause or subordinate clause?

The previous two sections considered relational functions of aversive case (i.e. a nominal modifier to a main clause) and subordinate functions of aversive case (i.e. subordinate clauses related to the main clause by an apprehensional relation) as clearly discrete constructions. However, in Ngumpin-Yapa languages this distinction is not always straightforward.

The close formal similarity between the relational and subordinate usage of the aversive case in these languages can be exemplified with the Warlpiri data below. In (27a), there is a subordinate clause, headed by a non-finite verb marked with aversive case, and a nominal which is optionally marked with aversive case. However, either of these constituents may be elided: in (27b), the nominal does not surface, although this is still likely to be considered a subordinate clause as the non-finite verb surfaces. In (27c), the verb is elided and only the nominal surfaces, which is formally undifferentiated from any other nominal modifier, and the event associated with the nominal *karli* 'boomerang' is inferred. Thus there is no clear separation between a subordinate clause usage of the aversive case and a relational usage.

(27) Warlpiri (Yapa; PN; Laughren 2015: 2)

a. *Yapa=ka wuruly-parnka karli(-kijaku) luwa-rninja-kijaku.*
man=PRS hide-run.NPST boomerang-AVE strike-INF-AVE
'Someone's running away to not get struck by a boomerang.'

b. *Yapa=ka wuruly-parnka luwa-rninja-kijaku.*
man=PRS hide-run.NPST strike-INF-AVE
'Someone's running away to not be struck.'

c. *Yapa=ka wuruly-parnka karli-kijaku.*
man=PRS hide-run.NPST boomerang-AVE
'Someone's running away to avoid (being struck by) a boomerang.'

In (27c), despite there being no overt verbal predicate providing evidence for a non-finite subordinate clause, the aversive case here can be viewed as functioning as a dynamic predicate (Laughren 2015). The blurring of clear distinctions between relational and subordinating usages of case is something of a feature of all Ngumpin-Yapa languages and is captured neatly by Hale's (1982: 285) usage of the term "vague predicational use of case" to refer to case marking of a nominal which implies the presence of a verbal predicate. Laughren (2015) goes as far as to argue that aversive *-kujaku* always has scope over an event rather than solely a participant of a clause.

2.1.4 Insubordinate function

Finally, there are some select, additional uses of case marking in apprehensional contexts in what can be categorised as insubordinate usages of case (see Evans 2007, Evans & Watanabe 2016), that is, the use of case as a fully independent predicate. An example of this was provided in (19) involving the polyfunctional locative case in Gurindji. To this we can add independent usages of the aversive case in Mudburra. Insubordinate constructions of this type are found with both nominal (28) and non-finite verbal (29) forms. Both utterances are used with the illocutionary force of a directive and coerce the addressee to some action, which corresponds to the implied main clause in these uses of the aversive.

(28) Mudburra (Eastern Ngumpin; PN)
Kardajala-wirri!
sandhill_people-AVE
'Careful of the sandhill people!' [MLD: RGR1-T27A-01: 819656_8211120]

(29) Mudburra (Eastern Ngumpin; PN)
Janki-nyi-wirri!
burn-INF-AVE
'Careful, you might get burnt!' [MLD: RGR1-T56A-01: 836058_836794]

2.2 Complements of 'fear' predicates

In the preceding section, a type of apprehensional construction was described in which a case suffix introduces a semantic relation between an NP or an adverbial clause and the main clausal predicate. In all of the examples provided, the aversive cases mark syntactic adjuncts and are licensed by their own semantic relation which they establish between their host and the clausal predicate (or, in the case of the insubordinate function, an implied predicate).

Apprehensional meanings can also be conveyed by predicates encoding fear, for example 'to be afraid'; 'to be frightened of' and so forth. 'Fear' predicates in Ngumpin-Yapa languages all select intransitive subjects but exhibit interesting variation with respect to how the complement (the SOURCE/OBJECT OF FEAR) is grammatically encoded. This variation can be observed in the choice of case marking but also in how these arguments are (or are not) cross-referenced in the pronominal complex.

In Ngumpin-Yapa languages, 'fear'-predicates are only found as complex verbs. The lexical element expressing a state of fear is either a nominal (e.g. Walmajarri and Ngardi) or a coverb (e.g. Gurindji), in either case combining with an inflecting verb which encodes TAM information. To illustrate briefly with an example from Ngardi, the nominal *rayin* 'afraid' combines with a generic stative verb (e.g. *karri-* 'be') which is inflected for TAM. The resulting argument structure of this particular complex predicate (*rayin=karri-*) is that of an extended intransitive involving an absolutive subject and a locative case-marked "object" denoting the source of fear. Both the absolutive subject and the locative "object" are also cross-referenced in the enclitic bound pronouns which index subject, person and argument features.

Predicates of 'fear' in Ngumpin-Yapa languages may select an argument in a range of seemingly idiosyncratic case forms, including the locative (§2.2.1), dative (§2.2.2), and aversive (§2.2.3). As shown in Table 6, Ngumpin-Yapa languages use either the locative or the dative, and some languages allow the aversive case instead of the locative or dative.

Table 6: Case marking of the complement of a 'fear' predicate across Ngumpin-Yapa languages.

	Language	Locative	Dative	Aversive
Yapa	Warlpiri		✓	✓
	Warlmanpa		✓	
Eastern Ngumpin	Bilinarra		✓	N/A
	Gurindji		✓	
	Mudburra		✓	✓
	Wanyjirra		✓	N/A
Western Ngumpin	Jaru	✓		
	Ngardi	✓		✓
	Walmajarri	✓		✓

2.2.1 Locative

Those Ngumpin-Yapa languages which can mark semantic sources of fear with the locative case can also use the locative case relationally to link a complement to a 'fear' predicate. This functionality is found in the Western Ngumpin languages: Ngardi, as in (30), Jaru, as in (31), and Walmajarri, as in (32). In these languages, locative case-marked arguments of 'fear' predicates are one of a circumscribed set of locative-marked arguments which are available for cross-referencing (Ennever & Browne 2023).[12]

(30) Ngardi (Western Ngumpin; PN)
*Rayin-karri-nya=**rlanyanta** yalu-ngka, **kurrkurr-ta**.*
afraid-be-PST=**3SG.LCT** that-LOC **owl-LOC**
'He is afraid of that one, the owl.' [MMN: TEN1-2017_016-02: 212312_217107]

(31) Ngardi (Western Ngumpin; PN; Tsunoda 1981: 113)
*Yambagina nga=**nyanda** yuwa-marn-an **gunyar-a**.*
chile CAT=**3SG.LCT** afraid-talk-PRS **dog-LOC**
'A child is afraid of a dog.'

(32) Walmajarri (Western Ngumpin; PN; Hudson 1978: 23)
***Ngalijarra-rla** pa=**jarrangurla** lapa-rni rayin.*
1DU.INCL-LOC MR1=**1DU.INCL.LCT** run-PST fear
'He ran off scared for fear of us two.'

2.2.2 Dative

The dative case can be used to mark a complement of a source of fear in Warlpiri, as in (33), Warlmanpa, as in (34), Mudburra, as in (35), Bilinarra, as in (36), and Gurindji, as in (37), and Wanyjirra, as in (38) (other functions of the dative are also discussed in §2.2.4). In each of these languages, the dative case marked NPs are treated as if they were arguments and are cross-referenced by a dative/oblique bound pronoun.

(33) Warlpiri (Yapa; PN; Mary Laughren, p.c.)
***Lani-jarri** ka=npa=**rla** **warna-ku**=ju?*
afraid-INCH.NPST PRS=2S=**3SG.DAT** **snake-DAT**=TOP
'Are you afraid of the snake?'

[12]This peculiarity extends beyond what one might strictly call apprehension to sources of emotion; it is found e.g. with predicates of anger and shame.

(34) Warlmanpa (Yapa; PN)
*Ngayu=ma=**rna-rla** **lani-ja-nya** ali-ku=nya **wangani-ku**.*
1=TOP=1SG.S-**3SG.DAT** afraid-INCH-PRS that-DAT=FOC dog-DAT
'I'm afraid of that dog.' [wrl-20200221: DK, 36:12]

(35) Mudburra (Eastern Ngumpin; PN)
*Yali **birrard** ka-yini **dimana-wu** ba=**rla**, yali.*
that **frightened** be-PST **horse-DAT** CAT=**3SG.DAT** that
'That one is scared of horses, that one.' [MSD: DOS1-2017_020-01: 1266566_1268319]

(36) Bilinarra (Eastern Ngumpin; Meakins & Nordlinger 2014: 367)
*Nyawa yabagaru, **wuugarra**=rningan=ba=**rla** garra **warrija-wu**.*
this little **scared**=AGAIN=EP=**3SG.DAT** be.PRS **crocodile-DAT**
'This kid is also afraid of the crocodile.'

(37) Gurindji (Eastern Ngumpin; Meakins & McConvell 2021: 516–517)
*Wuukarra ngu=lu-**rla** karri-nyana **nyanawu-wu***
afraid CAT=3PL.S=3DAT be.PRS aforementioned-DAT
***kuliyan-ku**, nyampayirla-wu, crocodile-ku.*
aggressive-DAT whatsitsname-DAT crocodile-DAT
'They're too afraid of those dangerous ones, whatsitsname, crocodiles.'

(38) Wanyjirra (Eastern Ngumpin; Senge 2015: 319)
Ngu=nda=yanunggula ribaying garriny-ana gardiya-wu.
REAL=2AUG.S=3AUG.OBL frighten BE-PRS white_man-DAT
'You are frightened of white men.'

In Gurindji Kriol – a mixed language arising from Gurindji and Kriol codeswitching (McConvell & Meakins 2005) – complements of the same 'fear' predicate found in Gurindji (i.e. *wuukarra*) select a complement headed by the dative preposition *bo*, as in (39), analogous to Gurindji using the dative case to mark complements of 'fear' (see Meakins 2008: 430–431).

(39) Gurindji Kriol (Mixed language; Eastern Ngumpin and English-lexified Creole)
*Dat ting dat yapakayi ngakparn i bin wukarra-karra **bo***
that thing that small frog 3SG.S PST afraid-CONT **DAT.PREP**
dat jangkarni-wan i bin jeya tumaji.
that big-NMLZ 3SG.S PST there because
'The small frog was afraid of the big one, because it was there.' [FSS: FM08_a077: 218275_222408]

2.2.3 Aversive

Mudburra, Warlpiri, Jaru, Walmajarri, and Ngardi can use the aversive case to mark complements of 'fear' predicates as exemplified in (40) for Mudburra and in (41) for Warlpiri. However, none of these languages exclusively require the aversive case in these constructions: each language may alternatively either use the locative or the dative. We are unaware of any meaningful distinctions attributable to the differential patterns of case marking.

(40) Mudburra (Eastern Ngumpin; PN; Osgarby 2018: 227)
Nyamba-wirri wirlarnkarra ka-yini?
what-AVE scared be-PRS
'What is he scared of?'

(41) Warlpiri (Yapa; PN; Laughren 2015: 4)
Kula=lpa nantuwu-rla wirriya warrka-karla, ***laninji***
NEG=IPFV horse-LOC boy climb-IRR **frightened**
*wanti-nja-**kujaku**.*
fall-INF-**AVE**
'The boy can't get up on a horse, being scared, lest (he) fall off.'

Interestingly, aversive marked NPs, while occasionally co-occuring with 'fear' predicates, are never cross-referenced in the pronominal complex, unlike dative or locative marked NPs. For example, in Ngardi, it is possible to mark sources of fear in one of two ways: with the locative or the aversive. It is only the former (locative case-marked NPs) that may be cross-referenced in the bound pronoun. Thus compare (30) with (42).[13]

(42) Ngardi (Western Ngumpin; PN)
Karrarda-karri-nyanta=rnalu maju kunyarr-amarra kuliny-jamarra.
fright-be-PRS=1PL.EXCL.S indeed **dog-AVE** **aggressive-AVE**
'We are afraid of that vicious dog.' [DMN: LC42a: 1262081_1266907]

2.2.4 Summary of idiosyncratic case marking

In this section, we have seen that predicates of 'fear' in different Ngumpin-Yapa languages select complements in dative, locative or aversive cases. 'Fear' predi-

[13]Interestingly, an analogous form of variation in marking complements of 'fear' predicates is found in Warrongo which can either mark a complement of fear in the locative, or else the 'fear' predicate can select an entire subordinate clause whose predicate will take a dedicated 'apprehensive' inflection (Tsunoda 2011: 619).

cates which select for locative or aversive case fall into a restricted set of predicates which select for non-core case (i.e. case other than absolutive, ergative, or dative). It is also surprising to see such considerable variation across closely related languages.

Nevertheless, when one considers the wider polyfunctionality of these case forms, these idiosyncrasies fall into wider patterns of marking of certain semi-transitive predicates. In the case of datives, it has been noted that various emotion predicates tend to select dative objects. The range of predicates cover the semantic space of: 'afraid of', 'sulk over', 'angry with', 'happy for', 'worry about' (Wanyjirra: Senge 2015: 571). Datives also encode topics of speech activities and certain cognitive processes (e.g. 'talk about', 'forget', 'laugh at'). Therefore, while the widely attested functions of the dative appear to be generally more centered around benefactives or affected participants, the object marking of 'fear' predicates seems to fit more straightforwardly into a category of marking topics of cognition or emotional states.

Locative case marking of "objects" of semi-transitives is somewhat less reported in Ngumpin-Yapa languages than dative marking. However, in at least Jaru, Walmajarri and Ngardi, a small number of emotion predicates appear to subcategorise locative objects, including 'fear', 'shame' and 'anger' predicates (Tsunoda 1981: 58; Ennever 2021: 529). The use of locative case to mark the conceptual source of these emotional states perhaps has its origin in the marking of circumstance (i.e. 'I am afraid in the presence of *x*'). The marking of circumstance, be it a temporal or spatial location, is a more general semantic property of the locative case and is also extended somewhat transparently to the subordinating functions of the locative which (in some Ngumpin-Yapa languages) marks simultaneous tense (i.e. circumstances temporally coincident with the event depicted by the main clause).

Finally, we saw that some Ngumpin-Yapa languages exhibit aversive case marking of complements of 'fear' predicates. That is, a 'fear' predicate (which encodes apprehension) takes an aversive-marked nominal as its complement (where the aversive case also encodes apprehension independently). Lichtenberk (2016) shows that this is a typologically common construction to encode apprehension.

2.3 Auxiliaries

In addition to the encoding of apprehension via case suffixes (§2.1), all Ngumpin-Yapa languages possess an auxiliary that occurs in apprehensional constructions.[14] These auxiliaries are variously analysed as modal particles or complementisers and exhibit co-occurrence restrictions with certain TAM inflections. The auxiliaries which are used in finite apprehensional constructions are catalogued in Table 7. With the exception of *kajika* (Warlpiri, marks general irrealis), each are analysed in their respective grammar as an apprehensional marker (most commonly glossed 'admonitive'). However, as we discuss in §2.3.2, the situation may not be this straightforward.

Table 7: Auxiliaries which can signal apprehension in Ngumpin-Yapa languages.

	Language	Auxiliary(ies)
Yapa	Warlpiri	*kalaka*
		kajika
	Warlmanpa	*kala*
		kalaka
		nga=…=nga
Ngumpin	Bilinarra	*ngaja*
	Gurindji	*ngaja*
	Mudburra	*bi(ya)*
	Wanyjirra	*ngarra*
	Ngardi	*ngarra*
	Walmajarri	*nga-rta*
		pa-rta

Taking Ngardi as an example, the subordinating aversive case *-Camarra* in (43) can be contrasted with the auxiliary *ngarra* in (44). In the former, the apprehension clause is non-finite, and so each constituent in the non-finite clause takes *-ngkamarra*; and in the latter, the apprehension clause is finite, and so is headed by the auxiliary *ngarra*. In both examples a preemptive clause expresses the action to be taken to avoid the undesirable outcome. The status of the preemptive clause in examples such as (44) as matrix clause or in a paratactic relationship with an apprehensional finite clause will be discussed in §2.3.1.

(43) Ngardi (Western Ngumpin; PN)

Ya-nanta	*ma=rna*	*kayirra-purda,*	***nya-ngu-ngkamarra***
go-PRS	TOP.AUX=1SG.S	north-ALL	**see-INF-AVE**

[14]In some languages, the auxiliary-like morpheme may be analysed as a particle. Across Ngumpin-Yapa languages, it is often difficult to distinguish the two parts of speech.

ngaringka-rlamarra.
woman-AVE
'I am going north, for fear of the woman seeing (me).' [PSM: TEN1-2018_034-01: 1028530_1038329]

(44) Ngardi (Western Ngumpin; PN)
Wurna=rna ya-nanta kayirra ***ngarra****=yi ngaringka-rlu nya-nngi.*
travel=1SG.S GO-PRS north **APPR**=1SG.NS woman-ERG see-IRR
'I am going north, lest that woman see me.' [PSM: TEN1-2018_034-01: 1008445_1010489]

There is variation in the languages across two dimensions: whether the auxiliaries are permissible in matrix clauses or not (which is taken as a distinction between auxiliaries and complementisers, in that by definition complementisers cannot occur in matrix clauses, and conversely auxiliaries are not permitted in non-finite subordinate clauses), and whether the apprehension component is encoded or not.[15]

In this section, we review the evidence for both their complementiser status and whether the auxiliaries in question are used exclusively in apprehensional constructions or if the auxiliary has apprehensional functions as just a subset of wider irrealis functions. Generally speaking the preponderance of evidence is for nearly all Ngumpin-Yapa languages to: i) have a main clause auxiliary rather than a *bona fide* complementiser; and ii) have non-dedicated apprehensionals that nevertheless show clear evidence of developing towards dedicated apprehensives.

2.3.1 Syntactic status: Main clause or subordinate?

As noted by Vuillermet & Schultze-Berndt (2018) "the relationship between precautioning and preemptive clauses may be one of subordination, coordination or simple juxtaposition." So far, we have observed non-finite subordinate apprehensional constructions (involving case suffixes) which were clearly subordinate in that the non-finite clauses rely on the main clause for the interpretation of their tense values and generally do not occur as independent clauses. However, in the case of finite clauses with apprehensional auxiliaries, there is no clear evidence that they are subordinate to a matrix clause.

[15] See McConvell (2006) for a diachronic analysis of Ngumpin-Yapa complementisers, including a brief discussion of whether some auxiliaries have undergone insubordination, being historically derived from complementisers.

Clauses with the apprehensional auxiliaries do not rely on any main clause for the interpretation of their tense values or any control properties, nor do they exhibit different morphosyntax depending on whether they occur on their own or accompanied by a preemptive clause. For example, Warlmanpa uses *nga=… =nga* regardless of whether there is a preemptive clause, as in (45), or not, as in (46). In both cases, the clause is finite and marked by the same auxiliary form.

(45) Warlmanpa (Yapa; PN)
Ngurra-ka=rna pina pana, kula ***nga****=ju=****nga*** *yarnunju*
camp-ALL=1SG.S return GO.POT NEG **APPR**=1SG.NS=**APPR** food
ji-nnya.
burn-PRS
'I should return home, lest he not cook food for me.' [N_D03-007868, BN, 36:57]

(46) Warlmanpa (Yapa; PN)
Ngarrka-ngu ***nga****=ngu=****nga*** *karta pi-nya*
man-ERG **APPR**=2SG.NS=**APPR** spear act_on-POT
'Careful, the man might spear you.' [H_K01-505B, DG, 19:51]

Similarly in Mudburra, the same modal auxiliaries occur with and without an additional clause expressing the preemptive event, as in (47) and (48) respectively. In (47), as with the Warlmanpa examples above, both clauses have their own set of argument relations expressed in the pronominal complex and there are no constraints on co-reference properties between the two clauses.

(47) Mudburra (Eastern Ngumpin; McConvell 1980: 72)
Karu=ma=yina-ngu-lu warra nya-ngka,
child=TOP=3PL.NS-LINK-[3]PL.S care see-IMP
bi*=yina-ngu-lu bi-rnarra kaya-li=ma!*
APPR=3PL.NS-LINK-[3]PL.S bite-SPEC ghost-ERG=TOP
'Take care of the children, in case ghosts bite them!'

(48) Mudburra (Eastern Ngumpin; Osgarby 2018: 59)
Bu*=ngku biya-la warlaku-lu.*
APPR=2SG.NS bite-SBJV dog-ERG
'The dog might bite you.'

It is therefore necessary to propose that the syntactic relationship between preemptive clauses and apprehensive clauses is simply juxtaposition and the relationship between the two clauses is likely at a discourse level, rather than a syntactic dependency.

2.3.2 Functional status: Dedicated apprehensional?

The Ngumpin-Yapa languages vary in the extent to which the auxiliary used in apprehensional utterances can be argued to be a dedicated marker of apprehension. Mudburra *bi* and Warlmanpa *nga=…=nga* are both analysed as dedicated apprehensionals, as will be discussed below. The form *ngaja* in Bilinarra and Gurindji is also analysed as a dedicated apprehensional (Meakins & Nordlinger 2014: 419); however evidence from at least Gurindji suggests that it is possibly better treated alongside *ngarra*/*ngarda* ('hypothetical') in Wanyjirra, Jaru, Walmajarri and Ngardi, rather than as a dedicated marker of apprehension.[16] Each of these will be discussed in turn.

2.3.2.1 Mudburra

In Mudburra, the auxiliary base *bi* (allomorphs: *bi*, *bu*, and *biya*) is dedicated to encoding apprehensive situations (Osgarby 2018: 104), irrespective of whether it occurs with a preemptive clause, as in (49), or without one, as in (50). This auxiliary base can be contrasted with 'hypothetical' *nya-* in (51) which is used for possible events that are not negatively evaluated.

(49) Mudburra (Eastern Ngumpin; Osgarby 2018: 178)
Biya *wurruja ya-yinarra, lurrbu=rni yuwa-rra!*
APPR dry become-SPC return=RSTR put-IMP
'Go and put it back, in case it dries out!'

(50) Mudburra (Eastern Ngumpin; Osgarby 2018: 105)
Biya *wari-li bi-rnarra.*
APPR snake-ERG bite-SPEC
'The snake might bite her/him/it.'

When used in a conditional protasis clause, the condition is negatively evaluated, as in (51).

(51) Mudburra (Eastern Ngumpin; Osgarby 2018: 107)
[*Kawarraj* ***bi****=n=****baa*** *warnd-arda,*]$_{\text{CONDITION}}$
forget **APPR**=2SG.S=**COND** get-SBJV

[16] Meakins & Nordlinger (2014: 305) provide the following description of the function of *ngaja*: "(it) usually refers to something potentially dangerous occurring in the immediate discourse context."

[*nya=n durd ma-rrarnda wumangku=ma.*]RESULT
HYP=2SG.S hold do-SBJV dreaming=TOP
'**If** you **were in danger of** forgetting them, you would have to hold on to your totemic designs.'

2.3.2.2 Warlmanpa

Warlmanpa's apprehensional auxiliary *nga=…=nga* could be decomposed into two independent auxiliaries. The base *nga=* encodes a situation with future time reference, as in (52), often with a 'result' meaning, as in (53), when juxtaposed alongside another clause.[17]

(52) Warlmanpa (Yapa; PN)
***Nga**=rna=ngu=lu turru ma-nji-nmi jaru.*
FUT=1EXCL.S=2SG.NS=PL.S tell do-INC-FUT message
'We're going to start telling a story for you.' [N_D29-023128, MFN, 01:15]

(53) Warlmanpa (Yapa; PN)
*Pingka wangka, **nga**=rna purtu-ka-mi*
slowly speak.IMP **FUT**=1SG.S hear-be-FUT
'Speak slowly, **so** I can hear you.' [H_K01-506A, DG, 32:04]

The auxiliary clitic *=nga* encodes epistemic possibility as illustrated by (54)–(55):

(54) Warlmanpa (Yapa; PN)
*Witawita=rnalu=**nga** ngayu, but ngayu-jarra=ma=ja, little bit wiri,*
small=1EXCL.S=**POT** 1 but 1-DU=TOP=1DU.EXCL little bit big
yumpa=ma wita one.
this=FOC small one
'Maybe we were all small, but us two were a bit bigger, this one was small.' [N_D29-023128, MFN, 05:34]

(55) Warlmanpa (Yapa; PN)
*Maliki yimpa=ma milpayi, kula=jana=**nga** nya-nganya.*
dog this=TOP blind NEG=3PL.NS=**POT** see-PRS
'This dog is blind, he probably can't see them.' [N_D03-007868: BN: 36:01]

[17] Interestingly, Warlmanpa has a future tense inflection which does not require the use of the auxiliary base *nga=*, however *nga=* 'FUT' is only used in combination with the future inflection. It is not clear whether the auxiliary base *nga=* in these contexts is contributing further information, or has become a default auxiliary base.

In Warlmanpa, an apprehensional construction is formed via the use of these two morphemes in combination with a future/potential verb inflection. Synchronically, this could be captured in a compositional analysis of *nga=* 'FUT' and *=nga* 'POT', as apprehensional situations typically occur posterior to reference time, and expect an alternative course of events to avoid the situation, presupposing a potential modality (that is, the event is uncertain). However, the compositional analysis is set back by the fact that *nga=…=nga* in combination with a future or potential verb inflection invariably results in an apprehensional reading, as in (56).

(56) Warlmanpa (Yapa; PN)
Ngurra-ka=rna pina pa-na, kula ***nga****=ju=****nga*** *yarnunju ji-nnya.*
camp-ALL=1SG.S return go-POT NEG **FUT**=1SG.NS=**POT** food burn-PRS
'I should return home, lest he not cook food for me.' [H_K01-505B, DG: 19:51]

That is, there are no noted cases of *nga=…=nga* with a potential or future verb inflection which do not force an apprehensional reading.[18] However, when combined with a counterfactual verb inflection, the clause does not involve apprehension:

(57) Warlmanpa (Yapa; PN)
Kari=n pirraku-ja-rrarla=ma, ***nga****=rna-ngu=****nga*** *ngapa*
COND=2SG.S thirsty-INCH-IRR=TOP **FUT**=1SG.S-2SG.NS=**POT** water
*yi-****njakurla****.*
give-**CFACT**
'If you were to become thirsty, I would give you water.' [H_K01 506A, DG, 07:31]

Furthermore, non-compositional modal systems are found elsewhere in Australia. For example, Caudal et al. (2019) suggest the Iwaidja and Anindilyakwa TAM paradigms are best treated as non-compositional circumfixes, which is likely the best analysis for the Warlmanpa construction. Verstraete (2005) also investigates the functions of mood having multiple exponents in a survey of Australian languages (see also Legate 2003, Iatridou 2000, Romero 2014, Caudal &

[18]Nash (1979) analyses the meaning of auxiliaries in Warlmanpa as being dependent on the verb inflection. Hence, under this analysis, *nga=…=nga* combining with most verb inflections encodes an apprehensional meaning, but with the counterfactual inflection, *nga=…=nga* is interpreted as a conditional (result).

Bednall 2016).[19] Interestingly, Warlmanpa also has an auxiliary *kala(ka)* 'apprehensive', which is homophonous with *kala* 'habitual' (*kala* is also analysed as a past habitual auxiliary in Warlpiri). The contrast between *kala* 'APPR' and *kala* 'HAB' in Warlmanpa is demonstrated in (58)–(59). The choice of *nga=…=nga* or *kala(ka)* is speaker-dependent, and both combine with the same range of verb inflections.

(58) Warlmanpa (Yapa; PN; Browne 2021: 286)
Kula=jana pa-nka-pa, kala=ngu-lu pi-nyi.
NEG=3PL.NS go-IMP-AWAY APPR=2SG.NS-3PL.S bite-FUT
'Don't go near them, lest they bite you!'

(59) Warlmanpa (Yapa; PN; Browne 2021: 287)
Kala=lu parti-nja-na work-ku, ngayu kala=rnalu ka-ngu.
HAB=3PL.S leave-MOT-PST.AWAY work-DAT 1 HAB=1PL.EXCL.S be-PST
'They would leave for work, and we would stay (home).'

2.3.2.3 Bilinarra and Gurindji

Bilinarra *ngaja* is also described as exclusively denoting apprehensive situations, occurring both as a particle in matrix clauses and as a complementiser heading finite subordinate clauses (Meakins & Nordlinger 2014: 305, 419), exemplified in (60).

(60) Bilinarra (Eastern Ngumpin; Meakins & Nordlinger 2014: 305)
Gula ga-ngga bin.ga-gurra=ma nyila=ma garu=ma, ***ngaja*** *nyiny ya-ngu.*
NEG take-IMP river-ALL=TOP that=TOP child=TOP **APPR** drown go-POT
'Don't take the kid to the river in case he drowns.'

Likewise, in the majority of textual examples available for Gurindji, *ngaja* denotes apprehensive situations (McConvell 1996a: 94–95; Meakins & McConvell 2021: 548–550). There are just a few examples, such as (61), however, that suggest it may have a more general function as a marker of epistemic possibility similar to the *ngarra* auxiliary found in Wanyjirra, Jaru, Walmajarri and Ngardi.

[19] In Wardaman for instance, a single irrealis prefix can be used to mark either deontic or apprehensive moods depending on whether it is combined with a present tense suffix or not (Merlan 1994: 179; and see Verstraete 2005: 224–225.

(61) Gurindji (Eastern Ngumpin; Meakins & McConvell 2021: 206)
Nyila=ma walaparr na ma-ni. ***Ngaja****=nga=rla kayikayi pa-nana*
that-TOP jump now do-PST **maybe**=POT=3SG.DAT chase hit-PRS
maitbi pungayarri-lu-wayin.
maybe bull-ERG-including
'It made [the horse] jump up suddenly. Maybe a bull started chasing it.'

2.3.2.4 Wanyjirra, Jaru, Walmajarri and Ngardi

While Mudburra *bi*, Bilinarra *ngaja*, and Warlmanpa *nga=…=nga* are analysed as being fully dedicated to apprehensional situations (or, at least there is a part of the construction which unambiguously conveys apprehensional situations), the other auxiliary bases in Ngumpin-Yapa languages have either a more general irrealis meaning, or a small set of meanings associated with epistemic uncertainty which include apprehension.[20]

In Ngardi the auxiliary base *ngarda* can be used in apprehensional circumstances, as in (62), but has a more general irrealis meaning, exemplified in (63). In Wanyjirra (Senge 2015: 643), *ngarra* – in addition to more frequent admonitive uses – can sometimes simply convey epistemic possibility, "irrespective of whether it refers to a pleasant or unpleasant event" – as for example in (64) where the event is in fact favourable for the participants. Similarly, Tsunoda (1981: 204) states that *ngarra* 'possibly' (denoting epistemic possibility) in Jaru often implies that the event will have an unpleasant context, but in a small number of other uses it has also been found to encode capability and is glossed as 'can', as in (65).

(62) Ngardi (Western Ngumpin; PN)
Wakurra=n nga-lku puka, ***ngarda****=n nyurnu=wanti-ju.*
NEG=2SG.S EAT-POT rotten **HYP**=2SG.S sick=FALL-POT
'Don't eat the rotten (food), you might fall ill.' [MAM: TEN1-2016_001-01: 329550_346102]

(63) Ngardi (Western Ngumpin; PN)
Ngarda*=ngala ngari ka-ngi.*
APPR=1PL.EXCL.NS belongings carry-IRR
'I can carry the stuff for us.' [MDN: LC22b: 831428_835666]

[20] For comparability, we will gloss each of these auxiliaries as 'APPR'

(64) Wanyjirra (Eastern Ngumpin; Senge 2015: 643)
*Nyannga maarn yan-gu marri, gangirriny **ngarra**=ngaliwa*
COND cloud.ABS go-FUT off sun.ABS **APPR**=1AUG.INCL.NS
garru-wu.
stay-FUT
'When the clouds go away, there will be sun for us.'

(65) Jaru (Western Ngumpin; Tsunoda 1981: 165)
*Ngaju-nggu **ngarra**=rna bali=wu-nggu guyu nyangga langga-muwa*
1SG-ERG **APPR**=1SG.S find=hit-PURP game if head-ONLY
bid-nyin-ang-gu.
stick-sit-CONT-PURP
'I can find game if only a head is sticking out (of grass).'

Notably, however, these languages all have other auxiliary bases or particles with epistemic possibility, in addition. For example, Bilinarra has a particle *gudigada*, glossed as 'maybe', as in (66).

(66) Bilinarra (Eastern Ngumpin; Meakins & Nordlinger 2014: 409)
***Gudigada** jubu=lu=nga ya-ni-rni jarragab.*
maybe just=3AUG.S=DUB GO-PST-HITH talk
'Maybe they just came to talk.'

2.3.2.5 Warlpiri

In Warlpiri, the cognate *ngarra* can be used to indicate future tense or past irrealis, depending on the verb inflection it co-occurs with (Laughren et al. 2007: 788). With the non-past inflection, it has a future meaning, as in (67), and with the irrealis inflection, it indicates a situation with past time reference that did not eventuate, as in (68). In either case, the situation may or may not be negatively evaluated, but the form does not mark apprehension.

(67) Warlpiri (Yapa; Laughren et al. 2007: 78)
*Ngaka **ngarra**=rna-jana ma-**ni**-rra jukurra-rlu.*
soon **IRR**=1SG.S-3PL.NS get-**NPST**-AWAY tomorrow-ERG
'I will go and get them tomorrow.'

(68) Warlpiri (Yapa; Laughren et al. 2007: 788)
*Nama=ju langa-ngka yuka-ja **ngarra**=rna pali-**yarla**.*
ant=1SG.NS ear-LOC enter-NPST **IRR**=1SG.S die-**IRR**
'An ant got into my ear and I almost died.'

This inter-language variation may have a historical explanation. Bybee et al. (1994: 211) suggest that:

> "A rather unexpected set of examples of a shift to a speaker-oriented modality is found in the development of an admonitive mood out of a marker of possibility. In most of these cases, the sense of possibility (either root or epistemic) carries with it the notion that the possible situation is unpleasant, dangerous or deleterious in some way" (see also Pakendorf & Schalley 2007).

If the *ngaCa* forms are reflexes of a single protoform, then the protoform was likely a general irrealis marker, and these reflexes synchronically may be at various stages of a shift towards dedicated apprehensional markers (i.e. apprehension may be an "invited inference" of varying strength, in terms of Traugott & Dasher 2001: 34–40).

Furthermore, Warlpiri also exhibits variation between *kalaka* and *kajika*, both of which can be used in apprehensional circumstances. In most varieties of Warlpiri, *kalaka* is a dedicated apprehensional; and *kajika* marks a potential event, as in (69), though it can also be used for apprehension, as in (70).

(69) Warlpiri (Yapa; Laughren et al. 2007: 115)
Jaja-nyanu-rlu ***kajika****=rla* *manyu* *wangka-mi* *jiliwirri*
MM-anaph.prop-erg **pot**=3sg.dat fun speak-npst funny
mirntirdi-nyanu-ku *wita-ku:* [...]
DD-anaph.prop-erg child-dat
'A child's maternal grandmother might say jokingly to her little grandchild: [...]'

(70) Warlpiri (Yapa; Simpson 1991: 388)
Kula=lpa=npa=jana *paka-karla* — *lawa.* ***Kajika****=npa* *marlaja-wanti*
neg=ipfv=2sg.s=3pl.ns hit-irr — no **pot**=2sg.s cause-fall.npst
tarnnga, *pali-mi* ***kajika****=npa*
always die-npst **pot**-2sg.s
'You cannot hit them. You might fall for good because of it, you might die.'

Laughren et al. (2007: 231) note that some southern Warlpiri speakers use *kalaka* to indicate the apodosis clause of conditional constructions, and as such, is not a dedicated apprehensional in these languages, further highlighting the apparent instability of apprehensional markers in Ngumpin-Yapa languages.

Table 8 summarises the extent to which auxiliaries can be seen as dedicated apprehension markers in Ngumpin-Yapa languages. The rightmost form(s) for a given language can be seen as those which are most strongly associated with apprehension. Forms such as *nga=…=nga* in Warlmanpa (in combination with future or potential inflections) encode an apprehensional reading without any other (known) possible reading. In Jaru, *ngarra* can be used to encode either apprehension, or ability, whereas *warri* is analysed as a broad irrealis marker not necessarily associated with apprehension. Whereas *ngarra* in Warlpiri, Warlmanpa, Wanyjirra, and Walmajarri can be used to indicate apprehension, the same morpheme in Jaru must be analysed as a broader irrealis marker.

In addition to this semantic categorisation of auxiliaries, there is an interesting correlation between the semantic category and its corresponding syntactic status. The auxiliaries denoting epistemic uncertainty (i.e. a broad irrealis meaning) which do not invite an apprehensional reading are also not associated with juxtaposition of another clause. This is the case for *ngarra* in Warlpiri and Warlmanpa (in Warlpiri, *ngarra* specifically has future temporal reference), which are indicated by diamonds in Figure 5. The next semantic categorisation presented above in Table 8 is implied apprehension, meaning that the auxiliary can signal epistemic uncertainty, but is more often associated specifically with apprehension – these auxiliaries are frequently (but not necessarily) juxtaposed with a preemptive clause.[21] This is the case for *ngarda* in Ngardi and *ngarra* in Wanyjirra, as represented by the circles in Figure 5.

The next logical syntactic status along this cline is for the auxiliary to only occur in subordinate clauses (due to the high frequency of being juxtaposed with a preemptive clause). Interestingly, no auxiliaries in Ngumpin-Yapa languages have reached this stage, although this can be seen in other Australian languages, such as *karningka* in Jingulu which is a dedicated apprehensional complementiser (see §1.1) – represented by the squares. While this syntactic status is not instantiated in Ngumpin-Yapa languages, the semantic function is: *ngaja* in Bilinarra and *bi* in Mudburra, for example, are analysed as dedicated apprehensional markers, however they can still surface without an additional juxtaposed clause, suggesting they are not bona fide complementisers (represented with triangles). Thus, while there is a clear correlation between syntactic status and semantic function, this is not a perfect correlation. Moreover, the correlation suggests a diachronic path towards these auxiliaries and particles becoming complementisers on the one hand, and dedicated markers of apprehension on the other.

[21] Given the syntactic status of adjoined clauses in Ngumpin-Yapa languages, and Australian languages more generally (Nordlinger 2006, Hale 1976), we use the term *juxtaposed* to mean that a clause is frequently found preceding/following another clause.

Table 8: Categorisation of irrealis auxiliaries in Ngumpin-Yapa languages by their association with apprehensional readings. Different kinds of underlining indicate cognate forms.

		Apprehension			
		None	Implied		Encoded
	Invites apprehensional reading	N	Y	Y	Y
	Other possible reading(s)	Y	Y	Y	N
	Broad irrealis meaning	Y	Y	N	N
Yapa	Warlpiri	*ngarra, marda*		*kajika kalaka*[a]	
	Warlmanpa	*marta*	*ngarra*		*nga=... =nga kala(ka)*
Eastern Ngumpin	Bilinarra	*gudigada*			*ngaja*
	Gurindji			*ngaja*	
	Mudburra	*nya*	*barra*		*bi*
	Wanyjirra	*wayi, marri*	*ngarra*		
Western Ngumpin	Jaru	*warri*		*ngarra*	
	Ngardi	*parda, mayi*	*ngarda*		
	Walmajarri	*kuyungurla*	*ngarta, parta*		

[a] As noted earlier, some speakers of Warlpiri alternatively use *kalaka* in conditional constructions without apprehension.

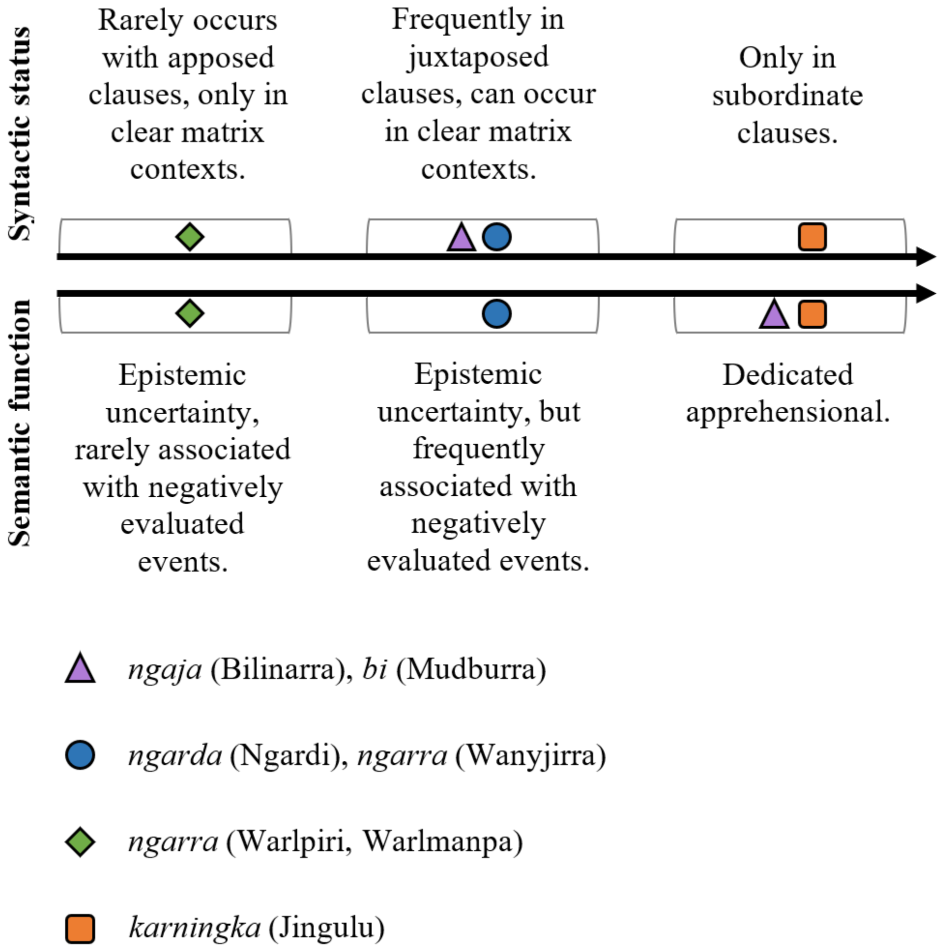

Figure 5: Correspondences between syntactic status and semantic function of auxiliaries/particles.

Finally, there are some cases where both a dedicated case morpheme and a non-dedicated auxiliary construction can co-occur as in (71). In these cases, it seems the apprehensive clause expands upon the negative event associated with the referent – here the addressees are being taken away for fear of the man, which is a nominal taking the aversive case, followed by a clause which expands upon the negative event associated with the men.

(71) Ngardi (Western Ngumpin; PN)

Ka-ngku=rna=nyurra ***ngantany-jamarra ngarda****=lu=nganpa*
carry-POT=1SG.S=2PL.NS **man-AVE** **APPR**=3PL.S=1PL.INCL.NS
luwa-rnngi.
spear-IRR
'I will take all of you for fear of the men, lest they (the men) spear us.'
[MRN: LC35b: 322091_328446]

3 Discourse functions

To conclude this paper, we now discuss the discourse functions (speech acts) instantiated by the different apprehensional constructions we have presented. Apprehensional constructions (i.e. clauses involving apprehensional morphemes, with or without an overt preemptive clause) in Ngumpin-Yapa are frequently associated with two kinds of discourse functions: directives (§3.1) and assertives (§3.2). Apprehensional constructions in directive function compel an addressee to some action and entail a negative outcome should the addressee not comply. Apprehensional constructions in assertive function on the other hand are a type of representative speech act that motivates a course of events, again by expression of a less desirable course of events. The primary morphosyntactic correspondences are that apprehensional directives make use of irrealis modalities often with second person subjects (implied or overt), whereas apprehensional assertives make use of realis modalities and describe events which affect participants with any personhood but most frequently first persons – summarised in Table 9.

Table 9: Typical modality and person values for the types of apprehensional constructions in Ngumpin-Yapa languages.

Speech act type	Typical modality	Typical affected participant
Directive	Irrealis	2nd (1st person occasionally)
Assertives	Realis	Any (1st person most common)

3.1 Directives

Apprehensional constructions are often associated with directives, either instructing (commands) an addressee to perform some action (or avoid performing some action) in order to avoid an undesirable outcome, or suggesting (entreaties) an action to this effect. There are three main morphosyntactic strategies for this. The first two involve either a direct command (imperative clause) or indirect command (potential/irrealis clause) denoting the instructions and an associated apprehensional construction (either a full clause, or a nominal with relational function marked for apprehension). The third is a conditional preemptive clause which signals an event which will lead to an undesired event. These strategies are summarised in Table 10, and discussed in more detail below. Interestingly, while the first two types of preemptive clauses can be negated with an associated change of illocutionary meaning ('do/do not do a certain action'), the conditional clause strategy is inherently intended not to be heeded in its positive polarity form (i.e. do not perform this action) even without the presence of any clausal negation.

Table 10: Summary of directives involving apprehension.

Speech act	Morphosyntax	Expected response
Direct command	Imperative construction and either (i) aversive dependent; or (ii) associated finite apprehensional clause.	Addressee is expected to perform this action to avoid something undesired.
Indirect command	Irrealis verbal inflection and either (i) aversive dependent; or (ii) associated finite apprehensional clause.	Addressee is encouraged to perform this action to avoid something undesired.
Conditional warning	Conditional complementiser with finite matrix apprehensional clause.	Addressee should avoid allowing some action occurring to avoid something undesired.

3.1.1 Direct command

An apprehensional construction may involve an overt preemptive clause which serves as a command; its morphosyntactic realisation involves a verb in the imperative inflection. These constructions convey that the addressee is expected (i.e. deontic modality) to perform the action denoted by the imperative clause, which will then avoid the undesired entity (for a case marked nominal in an aversive function, as in (72)), or avoid an undesired event, as in (73), where the clause containing the predicate is headed by the auxiliary *kala* 'HAB'.

1. Aversive dependent

(72) Jaru (Western Ngumpin; PN; Tsunoda 1981: 58)
Wanyja-rra gardiya-rla!
leave-IMP white_man-LOC
'Leave it for fear of the white man!'

2. Finite apprehensional clause

(73) Warlmanpa (Yapa; PN)
Lamarta-ka jawurra=nya, kala=nga ngapa walikwalik-wa-nmi.
hold-IMP cup=FOC APPR=POT water spill-fall-FUT
'Hold the cup, lest water spill.' [wrl-20190205, DK, 34:01]

3.1.2 Indirect command

Apprehensional constructions may also occur with main clauses that comprise a verb in an irrealis mood and take the form of indirect commands. In these cases, a preemptive clause is realised with irrealis modality (typically through an irrealis inflection, though this is language-dependent), with either an aversive-marked nominal (or non-finite subordinate clause), as in (74), or with a finite clause with an apprehensional auxiliary base, as in (75) and (76). Unlike imperatives, the addressee is not deontically *expected* to perform the action referred to by the irrealis clause, but the speaker believes it would be beneficial (typically for the addressee) to do so.

(74) Walmajarri (Western Ngumpin; PN; Richards & Hudson 2012)
Ngajirta nga-n jatpara-karra ***yan-ta purangu-rlamarra****.*
NEG MR2-2SG.S stop-MANN **go-IRR sun-AVE**
'Don't keep stopping, go, for fear of the sun (getting too hot).'

(75) Ngardi (Western Ngumpin; PN)
Wanja-ku=n ngarda=ngkurla wanti-ngi jina-ngka.
leave-POT=2SG.S APPR=2SG.LCT fall-IRR foot-LOC
'You should leave it (the axe) lest it fall on your foot.' [PSM: TEN1-2020_004-01: 26146_37777]

(76) Ngardi (Western Ngumpin; PN)
Wakurra=n walya-ngka yirra-ku yawiyi, ***ngarda wanti-ngi****.*
NEG=2SG.S ground-LOC place-POT poor_thing **APPR** **fall-IRR**
'You shouldn't put (the kid) down on the ground, he might fall over.'
[PSM: TEN1-2020_006-01: 245249_252587]

Finally, there are apprehensional constructions where the preemptive clauses are protases of conditionals. In these cases, the matrix clause is marked with the apprehensive auxiliary, and a finite subordinate clause, headed by a conditional, signals an event which will lead to the apprehensive situation eventuating. That is, if the event depicted by the protasis clause becomes true, the undesired event will occur. The inferred directive of these constructions is to *prevent* the event of the conditional clause from unfolding. Unlike the commands detailed above, the conditional warning has a purely inferred directive component. Furthermore, conditional warnings must have a finite apprehensional component (i.e. using the apprehensional auxiliary). These can either serve as warnings in specific circumstances, as in (77), or more general warnings about the world, as in (78).

(77) Warlmanpa (Yapa; PN)
Kari*=n kuyu nga-rninjakurla,* ***nga****=n=****nga***
COND=2SG.S meat eat-CFACT **APPR**=2SG.S=**APPR**
murrumurru-ja-ma.
sick-INCH-POT
'If you were to eat this meat, you might become sick (i.e. don't eat the meat).' [H_K01 506A, DG, 04:42]

(78) Warlpiri (Yapa; PN; Laughren et al. 2007: 518)
Nyunypa-ngku ka luwa-rni kiwinyi=ji kuyu nyanungu-ku. Kala
saliva-ERG PRS shoot-NPST mosquito=TOP meat that_one-DAT but
kaji*=lpa yapa luwa-karla warlura-rlu=ju, ngula=ju* ***kalaka*** *milpa*
COND=IPFV person shoot-IRR gecko-ERG=TOP that=TOP **APPR** eye
karltara-jarri-mi.
blind-INCH-NPST
'It spits on mosquitoes which it catches thus to eat. If the gecko hits a person (with its spittle), then the person can go temporarily blind in the eyes.'

Some Ngumpin languages do not have this conditional apprehensional construction: for example, in Wanyjirra and Ngardi, for a combination of a conditional subordinate clause with a *ngarda*/*ngarra* marked main clause, the main clause denotes only possibility, and is rarely associated with apprehension, as exemplified in (79)–(80).

(79) Ngardi (Western Ngumpin; PN)
Kaji=*rna* *kuyi* *ka-ngku-rni* *kupa-ku* ***ngarra***=*yi*=*n.*
COND=1SG.S meat carry-POT-HITH cook-POT **IRR**=1SG.NS=2SG.S
'If I bring back this meat, you might cook it for me.' [PSM: TEN1-2020_006-01: 245249_252587]

(80) Wanyjirra (Western Ngumpin; PN; Senge 2015: 643)
Nyanga *maarn* *yan-gu* *marri,* *gangirriny* ***ngarra***=*ngaliwa* *garru-wu.*
COND cloud go-FUT off sun **IRR**=1AUG.INC.NS stay-FUT
'When the clouds go away, there will be the sun for us.'

Mudburra allows a similar construction, in which the protasis can be apprehensional (marked by the *bi*= auxiliary base). In this case, the condition is something to be avoided, and the main clause signals an action to be undertaken to revert the undesirable event. This is exemplified in (81), where the condition is the addressee forgetting a referent (totemic designs), which is undesired, and the main clause is that the addressee would have to be sure to hold on to the totemic designs. By performing the main clause event under the appropriate conditions (undesired conditions), the addressee reverts to a state where the undesired conditions are no longer met.

(81) Mudburra (Eastern Ngumpin; PN; Osgarby 2018: 107)
[*Kawarraj* ***bi***=*n*=***baa*** *warnd-arda,*]$_{\text{CONDITION}}$
forget **APPR**=2SG.S=**COND** get-SBJV
[*nya*=*n* *durd* *ma-rrarnda* *wumangku*=*ma.*]$_{\text{RESULT}}$
APPR=2SG.S hold do-SBJV dreaming=TOP
'**If** you **were in danger of** forgetting them, you would have to hold on to your totemic designs.'

3.2 Assertives

Apprehensional constructions in Ngumpin-Yapa languages are also used as assertives, in particular providing the reasoning behind a speaker's actions. Examples are given in (82)–(85) which demonstrate the wide range of temporal references permitted in the preemptive clause: the example in (82) has present reference; (83) has past reference; and finally (84)–(86) have future reference. The apprehensional component, either a finite clause with an apprehensive auxiliary and irrealis modality, or a nominal with aversive case marking (or a non-finite subordinate clause with aversive case marking), obtains its temporal reference from the preemptive clause.

(82) Ngardi (Western Ngumpin; PN)
Ngu=rna jaku=ya-nanta, ***ngarda****=yi=lu paja-rnngi kunyarr-u.*
SEQ=1SG.S depart=go-PRS **APPR**=1SG.NS=3PL.S bite-IRR dog-ERG
'I'm getting out of here, **lest** those dogs bite me!' [PSM: TEN1-2017_010-01: 954694_959466]

(83) Ngardi (Western Ngumpin; PN)
Ngu=lu=yanu wiriny=ma-ni yard-ta=wali ngantany jayin-kurlu
SEQ=3PL.S=3PL.NS bind=get-PST yard-LOC=then man chains-PROP
jina ***ngarda****=lu yaparti-ngi.*
foot **APPR**=3PL.S run-IRR
'They tied the aboriginal men up in the yard (by their) feet with chains, lest they flee.' [PSM: TEN1-2017_007-02: 72766_82623]

(84) Ngardi (Western Ngumpin; PN)
Ya-nku=rna yalu-rlamarra, ***ngarda****=yirla wangka-nyanku.*
go-POT=1SG.S that-AVE **APPR**=1SG.LCT speak-IPFV.POT
'I will go on account of that one, lest she speak with me.' [MMN: TEN1-2017_026-01: 1168486 _1207635]

(85) Warlmanpa (Yapa; PN)
Ngapa-kuma*=rna mulpu-nga purluny-ya-n.*
water-AVE=1SG.S shelter-LOC enter-go-FUT
'I will enter shelter for fear of the rain.' [H_K01 506A, DG, 39:05]

Apprehensional constructions can be used not in reference to a specific event but rather to justify claims about generally observed behaviours or phenomena. In (86), a speaker claims that cars cannot go over sandhills, the justification being that the car will get bogged, which is marked with apprehensional morphology.

(86) Warlpiri (Yapa; PN; Laughren et al. 2007: 116)
Jilja ka pirli=piya parntarri-nja-ya-ni kula=lpa murduka=yi
sandhill PRS rock=SEMBL crouch-INF-go-NPST NEG=IPFV car=CONT
jilja-ngka ya-ntarla, lawa — ***kalaka yuka-mi****.*
sandhill-LOC go-IRR NEG **APPR** **sink-NPST**
'A sandhill stretches out like a rocky mesa—a car can't go over a sandhill, as it's likely to get bogged.'

4 Conclusion

This chapter has provided the first in-depth analysis of apprehensional constructions of Ngumpin-Yapa languages. These languages exhibit a case suffix which can mark a nominal denoting a feared referent (or a feared event metonymically associated with a referent), or it can mark a non-finite subordinate clause denoting a feared event. In some cases it is difficult to distinguish which particular function is instantiated in a given utterance, which is a common analytical problem in Australian languages (Hale 1982). While most Ngumpin-Yapa languages have this construction available, few use the aversive case suffix to mark the complement of a 'fear' predicate, and it is more common to mark these complements using dative or locative suffixes which still trigger verbal indexation (thus distinct from adjunct uses of the dative and locative cases). In most Ngumpin-Yapa languages, the form of the aversive suffix is an increment to another case (particularly dative, as in Warlmanpa, or locative, as in Ngardi), which can be seen in other Australian languages as well (e.g. Pintupi).

Most Ngumpin-Yapa languages also have an auxiliary which marks finite clauses as denoting undesired eventualities – while this has been found in non-Pama-Nyungan languages, this is rare for Pama-Nyungan languages. There is great interlanguage variation as to whether these auxiliaries are more like complementisers (i.e. in these languages, these constructions rely on a matrix preemptive clause), and to the extent to which these auxiliaries exclusively encode apprehension or whether other interpretations are available. Generally, these other interpretations are general irrealis meanings, which leaves available an apprehensive interpretation. This variation indicates that within these languages, apprehension is not a diachronically stable phenomenon, particularly striking in one form which appears to be a shared inherited form *ngaCa* which is dedicated to apprehensional meanings in some languages but has a more general irrealis function in others, as well as exhibiting variance regarding its syntactic status. Furthermore, the interpretation of apprehension is tied to the verb inflection, as well as the auxiliary. Finally, the chapter discussed the most common usage of apprehensional constructions: directives of varying strength, and assertives, which primarily provide the reasoning for a speaker's actions.

Abbreviations

1/2/3	1st/2nd/3rd person	HAB	habitual
A	Agent	HITH	hither
ABL	ablative	HYP	hypothetical
ABS	absolutive	IMP	imperative
ACC	accusative	INCL	inclusive
ACS	accessory	INCH	inchoative
ADM	admonitive	INF	infinitive
ALL	allative	INST	instrumental
ANAPH.PROP	anaphoric/reflexive pronoun	IPFV	imperfective
		LEST	'lest'; apprehensional marker
ANTIC	anticipatory	IRR	irrealis
APPR	apprehensive	LCT	locational
AUG	augmented	LINK	linker
AUX	auxiliary	LOC	locative
AVE	aversive	MANN	manner
CAT	catalyst	MIN	minimal
CAUS	causative	MM	mother's mother
CFACT	counterfactual	MOT	associated motion
COM	comitative	MR1/2/3	modal root 1/2/3
COND	conditional	NARR	narrative
CONT	continuous	NEG	negative
CUST	customary aspect	NMLZ	nominalizer
DD	daughter's daughter	NPST	non-past
DAT	dative	NOM	nominative
DIST	distal demonstrative	NS	non-subject
D.LOC	derivational locative	O	object
DU	dual	OBL	oblique
DUB	dubitative	OBSC	obscured
EFF	effector	PASSP	passive perfective
ERG	ergative	PERF	perfect
EXCL	exclusive	PL	plural
FOC	focus	PN	Pama-Nyungan
FUT	future	POT	potential

PREP	preparative	RSTR	restrictive
PROP	proprietive ('having')	S	subject
PRS	present	SBJV	subjunctive
PST	past	SEMBL	semblative
RDP	reduplication	SG	singular
PURP	purposive	SPEC	speculative
REAL	realis	TOP	topic
REFL	reflexive	VBLSR	verbaliser

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Chapter 14

A fieldwork questionnaire on apprehensionals

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This questionnaire is intended as a guide to fieldworkers for systematically surveying the constructions used in the apprehensional functions known so far, namely apprehensive, precautioning, ‘fear’-complementation and timitive. It builds on the parameters of variation attested so far crosslinguistically, at the semantic, syntactic and pragmatic levels, and comes in the form of brief explanations of the variation targeted, illustrated by prompts in English to be translated into the language of study. A systematic investigation of the fine-grained semantics of the different functions of apprehensionals may uncover subfunctions that have not been clearly identified yet, and/or offer an aid in establishing whether or not these expressions are dedicated apprehensionals.

Introduction

Apprehensionals is a cover term that refers to any grammatical construction expressing the (high) *possibility* of an *undesirable* event, or expressing the undesirability of an entity. This questionnaire considers in turn the following four functions: Apprehensive, Precautioning, ‘fear’-complementation, and Timitive. Their similarities and differences are summarized in the table below, and clarified in the introduction of each (sub)section.

Languages differ a lot in the number of apprehensional constructions they have – up to three in Ese Ejja (Pano-Tacanan; Bolivia & Peru; Vuillermet 2018a) – as well as in the number of functions (synchronically) available to their apprehensional construction(s) – up to three functions for a single apprehensional marker in Marrithiyel (Western Daly; Australia; Green 1989).

Marine Vuillermet. 2026. A fieldwork questionnaire on apprehensionals. In Martina Faller, Marine Vuillermet & Eva Schultze-Berndt (eds.), *Apprehensional constructions in a cross-linguistic perspective*, 515–537. Berlin: Language Science Press. DOI: ??

Table 1: Four functions of apprehensionals

	Scope	Obligatory presence of	Typical function	Section
Apprehensive ***watch out** you **might** fall*	clause	NA	warning, threat	§1
Precautioning *tie your laces **lest** you fall*	clause	a preemptive clause to which the undesirable event is associated	justification of a preemptive action	§2
‘fear’-complementation ***I fear lest** you fall*	clause	the undesirable event is the non-subject complement of a transitive predicate of fear	account of the feeling of fear	§2.6
Timitive *she ran away **for fear** of the dog*	NP	a main clause encoding the reaction wrt the undesirable entity	justification of a reaction	§3

(1) Ese Ejja and its three apprehensionals (Apprehensive, Precautioning, Timitive)

a. Apprehensive
*’Biya ’biya ’biya ’biya! Kekwa-ka-**chana** miya!*
bee bee bee bee pierce-3A-**APPR** 2SG.ABS
‘Bee, bee, bee, bee! **Watch out** it **might** sting you!’[1]

b. Precautioning
Owaya ekowijji shijja-ka-ani [***e**-jja-saja-ki* ***kwajejje***].
3ERG rifle clean-3A-PRS **PREC**-MID-block-MID **PREC**
‘He cleans his rifle [**lest** it get blocked].’

[1]The combination of *watch out* or *be careful* with the modal *might* is probably the best proxy in English for an apprehensive sentence. The modal *might* expresses potentiality only, without further specifying whether it is positively or negatively evaluated: *he might come* might be positive, e.g. in *that he might come makes me so happy!* The round brackets signal that equivalents of these two phrases are not necessary in languages with dedicated apprehensive markers, although they might often co-occur.

c. TIMITIVE
*Iñawewa kwaji~kwaji-ani ’biya=**yajjajo**.*
dog run~RDP-PRS bee=**TIM**
‘The dog is running **for fear of** the bees.’

(2) Marrithiyel and its multifunctional apprehensional marker (Green 1989: 80, 166, 58, MV’s emphasis)

a. APPREHENSIVE function
*Gu-n-ning-pirr-Ø-**fang**.*
3S.REAL-go-1SG.O.NSG.NIRR.S-leave-3PL-**APPRL**
‘(**I am afraid**) they might leave me.’

b. PRECAUTIONING function
(…) [*atjan ambi gu-iwinj-sjang-Ø-**fang***].
dog NEG 3NSG.S.REAL-3NSG-listen-3PL-**APPRL**
‘(He turned the generator off), [**lest** they not hear the dogs].’

c. TIMITIVE function
*Wanthi ngun-wa ngaka yigin-**fang**.*
behind 1SG.SBJ.IRR-go.FUT sister 1SBJ.POSS-**APPRL**
‘I will come afterwards, **to avoid my sister**.’

The questionnaire builds on a previous manuscript (Vuillermet 2018b) but includes substantial revisions and additions. It is intended as a guide to fieldworkers for surveying the constructions used in the target language in the apprehensional functions listed above, and as an aid in establishing whether or not these expressions are dedicated apprehensionals. It systematically tests for the parameters of variation attested so far crosslinguistically, at the semantic, syntactic and pragmatic levels.

It consists of prompts in English to be translated into the language of study. The targeted grammatical morpheme is paraphrased in English if no equivalent exists (as for the timitive examples in 1c and 2c). Prompts are preceded by an explanation of the targeted parameter(s) and, sometimes, by a reference to a specific language illustrating the parameter(s) and/or suggestions of possible contexts to facilitate the elicitation task. Ideally, the researcher should suggest the form in the target language, and ask the speaker to:

1. Give a context when the sentence could be uttered – contexts should be suggested if the speaker cannot find any;

2. (Where applicable) translate it back into a national or regional language and/or the language of work.

Ideally again, the sentences should be checked with different speakers.

To avoid cultural or situational incongruities, several prompts are offered, among which the researcher might choose. They may also create their own prompt with the same variation target in mind to adapt the questions to the consultant, language, and/or situation. As an example, restrictions on the grammatical persons available are frequent with apprehensives: for a language where the apprehensive is restricted to 2nd person, the researcher will change the grammatical person to 2nd person when other grammatical persons are suggested in the prompts (except, of course, where restriction on grammatical person itself is the target).

A systematic investigation of the fine-grained semantics of the different functions may uncover subfunctions that have not been clearly identified yet. Users should be aware of the dynamic nature of questionnaires (Lahaussois 2019): this questionnaire is likely to evolve in the next decade, and updates will be posted on the TULQUEST website (http://tulquest.huma-num.fr/en).

1 Apprehensive

This section targets the apprehensive function: the marker occurs in main clauses, which are not required to be associated with a preemptive clause (a sentence that gives direction on how to avoid the undesirable event). As mentioned above, one and the same marker might be available for different functions, e.g. both for the apprehensive and the precautioning function, the latter being necessarily associated (and often syntactically subordinated) to a preemptive clause (see §2). Note that it might be very tricky to determine whether the marker occurs in subordinated clauses and also in main clauses if an insubordination process is in progress. (See §2.1 in the introduction to this volume, Vuillermet et al. 2026 [this volume])

1.1 Intrinsic undesirability

For apprehensionals, the expectation is that they are most frequently found in descriptions of events that are universally or culturally judged to be undesirable, which in turn makes warnings a natural context for their occurrence pragmatically (1.1.1). The following three questions check for the presence of the undesirability component in the potential candidates for an apprehensive marker.

Prompts (3)–(5) illustrate occurrences with typically undesirable events, while (6)–(7) and (8)–(10) illustrate occurrences with typically positive and neutral events, respectively. If a language has no dedicated marker, the questions in (1.1.1) will help to find out what is/are the non-dedicated construction/s to express a (highly) potential, undesirable event.

1.1.1 Is there a construction frequently used in warnings, or in justification/ avoidance of an action that invokes a potential negative consequence?

(3) (*Watch out / be careful*) *you might fall / slip!*
Possible context: The addressee is walking very fast on a muddy path.

(4) *Their dog / a snake might bite me!*
Possible context: I am not going to their house / I am not going onto that path in the dark.

(5) *He might hit the baby!*
Possible context: A toddler might hit his younger sibling.

1.1.2 Is the reading with typically positive situations like 'smile', 'laugh', 'help', 'find a 100 dollar bill' necessarily negative?

(6) (*Watch out*) *you might laugh!*
Possible context: During an official discourse when one should stay serious and quiet. Impossible context:[2] pure potentiality, you might laugh, and that would actually be desirable, or at least not undesirable.

(7) (*Watch out*) *Thomas might help!*
Possible context: Thomas is so clumsy that the speaker would prefer him to not help.
Impossible context: Thomas' helping is a desirable possibility.

1.1.3 Is the reading with typically neutral situations like 'walk', 'talk', 'sleep', 'come', 'cook' necessarily negative?

This question checks again whether the apprehensional candidate necessarily encodes undesirability.

[2] An impossible context is a context in which the target sentence is expected to be (semantically) infelicitous if the marker has the meaning component tested for, namely undesirability in the present case.

(8) (*Watch out*) *you might fall asleep!*
Possible context: People being on watch, people watching a movie.
Impossible context: The addressee's sleeping is desirable, e.g. if s/he usually has trouble sleeping.

(9) (*Watch out*) *Zoe might come!*
Possible context: The speaker (and addressee) fears Zoe could hear them speaking or see them doing something which would make her angry.
Impossible context: The speaker and/or addressee will rejoice to possibly meet Zoe again.

(10) (*Watch out*) *Dan might cook!*
Possible context: Dan is a bad cook and it is undesirable that he cooks.
Impossible context: Dan is a great cook and the perspective of him possibly cooking is a reason for rejoicing.

1.2 Degree of preventability of an event

This question and the next one test whether the target language restricts the apprehensional morphology to situations where the undesirable event is fully preventable: while one can prevent someone from cooking, one cannot prevent rain or thunder. Some languages allow apprehensives to occur with raining or thundering events since one can still prevent the *consequences* of such an undesirable situation, e.g. getting wet (see also Vuillermet 2018a: 274, and Fedotov 2026 [this volume]). In Mwotlap (Austronesian; Vanuatu) however, the equivalent of 'watch out there will be an earthquake' is ungrammatical (p.c. A. François, May 2014).

Note that there might be a continuum in the preventability of the undesirable event. Nobody can fully prevent the event of falling, but being aware of the risk of falling should lead you to be more careful and potentially control for that risk. Also, the addition of a semantic participant to a verb phrase can facilitate the preventability of the event: while 'he might vomit' is an event difficult to prevent per se, 'he might vomit **on you**' is more readily preventable, e.g. by moving out of the way.

Note also that the degree of preventability may well be linked to the origin of the marker: if '*beware* you might fall' and '*make sure* you do *not* fall' seem fairly interchangeable, the 'beware' formulation focuses on the potential risk *the addressee is not aware of*, while the 'make sure not X' formulation focuses on the *actual preventability* of the undesirable event.

1.2.1 Are non-preventable events compatible with the apprehensive function, e.g. events typically expressed by impersonal (like meteorological) verbs and [−agentive] verbs?

1.2.1.1 **Impersonal verbs** like 'rain', 'snow', 'wind.blow', 'thunder', 'be.an.earthquake'; existential verbs

(11) (*Watch out*) *it might rain/thunder!*

(12) (*Watch out*) *there might be snakes on the path!*

1.2.1.2 **[−agentive] verbs**

Note that the (un)availability of such verbs might also be linked to the different semantics available with different grammatical persons, see §1.3.

(13) (*Watch out*) *you might fall asleep/fall/cough/vomit!*[3]

(14) (*Watch out*) *Alice might fall asleep/fall /cough/vomit!*

1.3 Availability of the grammatical persons

The availability of the grammatical persons in an apprehensive construction varies very much from one language to the other. In several languages, not all grammatical persons are available. The questions test whether core argument roles in a given construction are available to all or just a subset of grammatical persons.

An extreme case is Matsés (Pano-Tacanan, Peru & Brazil), where not only the markers do not exist for all grammatical persons, but there are different forms for different person configurations (Fleck 2003: 439ff). In such a case, the researcher might want to test all possible configurations.

1.3.1 What grammatical persons are available with the apprehensive marker?

(15) (*Watch out*) *I/he/she/it/we/they might fall on/hurt you!*

(16) (*Watch out*) *you might fall on/hurt me/her/him/it/us/them!*

(17) (*Watch out*) *you/he/she/it/we/they might get lost!* or (*OMG*) *I might get lost!*

[3]Some languages may differentiate between an agentive and non-agentive use of some verbs like 'cough', e.g. by case or person marking (split S/active-inactive); in that case make sure to test the non-agentive version.

1.4 Interaction of grammatical persons with communicative functions

While expressing a warning or a general concern is one of the most common functions of the apprehensive, expressing a threat is especially common with 1st person, e.g. *watch out I'll hit you (if you keep bothering me)*, as pointed out by Epps (2008: 631) for Hup (Nadahup; Colombia & Brazil). Interestingly, there is at least one language (Matsés) where the apprehensional expression corresponding to 'watch out I'll hit you' necessarily means that the speaker will *accidentally* or *unwittingly* hit the addressee: only the warning or general concern reading is available, while the threat reading is not (Fleck 2003: 439; p.c. 2014).

In other languages like in Ese Ejja (Vuillermet 2018a: 274), the same expression can be used in both warnings and threats. Note that first person subject and full agentivity are typical correlates of the necessary broader conditions that must hold for a threat reading (Searle 1975, Salgueiro 2010), but not the only ones (see the discussion in the Introduction, §3.3, Vuillermet et al. 2026 [this volume]).

1.4.1 What is the reading of [+agentive] verbs with 1st person?

Note that a warning reading in (18) is (almost) impossible, while both readings (threat and warning) are available in (19), where the hitting event can be [−agentive] if non-intentional.

(18) (*Watch out*) *I might follow you!*

(19) (*Watch out*) *I might hit you!*

1.4.2 What is the reading of [−agentive] verbs with 1st person?

If the verb is truly non-agentive (i.e. no "voluntary coughing" reading available), could it be anything else than a warning?

(20) (*Watch out*) *I might fall* (*over you*)*!*

(21) *(Watch out*) *I might cough* (*on you*)*!*

1.5 Semantics: Illocutionary act or expression of general concern

In some languages, the morpheme might express an undesirable possibility (from the perspective of the speaker), and not necessarily expect a reaction of the addressee. The absence of expectation towards the addressee might be best explained to the consultant by letting them imagine talking to themselves, or comment on a movie where a protagonist is taking risks.

(22) (*OMG*) *I might lose my temper!*
Context: The speaker is in an important discussion where they should not lose their temper.

(23) (*OMG*) *I might not understand a thing …*
Possible context: the speaker fears that they might not understand what a teacher or an authority person (e.g. a police officer) means.

(24) (*OMG*) *Jessica might fall!*
Possible context: The speaker is suddenly seeing Jessica climbing up a tree, but cannot reach her as they are too far apart. Or the speaker is watching a film where a protagonist is taking risks.

1.6 Compatibility with different temporal settings

Apprehensive markers are usually available in immediately undesirable situations, i.e. with future orientation, but in some languages different temporal settings are also available.

1.6.1 Test the grammaticality with temporal adverbs like *today* vs. *tomorrow* vs. *yesterday*.

1.6.2 Is the marker available for past events?

Apprehensive markers are sometimes available in a past setting, which places them into the domain of epistemic markers: the undesirable event has potentially happened already, but one will not immediately know if it really did or not. In To'abaita (Austronesian; Solomon Islands; Lichtenberk 1995: 296), 'be.sick+APPR' may mean 'you may have been sick' / 'you may be sick' / 'you may get sick'.

(25) *They might have eaten already.*

(26) *He might have died yesterday.*

(27) *I might have gotten lost!*

1.6.3 Can the event only be about to happen or also happen some time in the future?

Apprehensive markers might only be available with immediate future/prospective orientation but not a general future orientation: in some languages it is possible to warn of an immediate danger, but not of a potential event later in the future. In Hup (Epps 2008: 609), an additional future marker emphasizes the possibility of the event to happen in a less immediate future.

(28) (*Watch out*) *you might hurt yourself!*
Possible context: if you continue playing with the machete.

(29) (*Watch out*) *you might hurt yourself one day!*
Possible context: if you keep driving when you are drunk as you did last night.

1.7 Compatibility with negation

In some languages, the apprehensive marker is not compatible with a negation marker (possibly because of an origin in a negative marker).

1.7.1 Is the apprehensive construction compatible with overt negation?

(30) *We might not hear the baby crying!*

(31) (*OMG*) *the luggage might not all fit in the trunk.*

(32) (*Watch out*) *Dan might not come!* (although he said so)

(33) (*Beware*) *it might not rain!* (when the speaker or addressee wants the rain to water her/his plants)

(34) (*Watch out*) *the dog might not bark at the thieves!* (and so not advise when there are thieves)

(35) (*Watch out*) *the cake might not fit into the bag!* (and the cake would be ruined if pushed too hard into the bag)

1.8 Compatibility with interrogatives

Many apprehensives are proper mood markers and are impossible in an interrogative sentence.

1.8.1 Is the apprehensive function also available in a question?

(36) *Might Bob leave/laugh/fall?* (and I don't want that)

A marker that expresses (among other meanings) undesirability in Kukama-Kukamiría (Tupían; Brazil, Colombia & Peru) is very frequent in questions (Vallejos Yopán 2016: 415ff.). This marker also expresses surprise, in the sense that the undesirable event cannot be easily explained: in (37), the presence of a new tree that has appeared overnight makes the situation scary. Note that the questions in (37) and (38) are not typical questions expecting an answer from the addressee, but rather rhetorical ones expressing incredulity in the face of an event which is so unexpected that it becomes scary.

(37) *How might this tree have grown overnight?*

(38) *How can this bill possibly be in my pocket?*
Possible context: the speaker has very unexpectedly just found a bill in their pocket and they fear someone could take them as a thief.

1.9 Optative uses

An optative is used to express a wish, regret, hope or desire of the speaker (Grosz 2012, Dobrushina et al. 2013). A negated optative could then potentially express the opposite of a desire, i.e. an undesirable event.

1.9.1 Does the language have an optative construction, and if so, can it be negated? If a language displays both an apprehensive and an optative that can be negated, how do the speakers explain the difference between the two forms?

Hypothesis: there is an expectation for the addressee to act against the potential undesirable event in apprehensives, vs. absence of expectation for the optative.

(39) (*Watch out*) *John might fall!* *vs.* *may John not fall!*

(40) (*Watch out*) *you might fall!* *vs.* *may you not fall!*

1.10 Scope/distribution

Apprehensives might occur with an NP forming a complete utterance which warns of an undesirable event metonymically associated with the tagged undesirable referent. Note that §3 on the timitive function is entirely dedicated to apprehensionals which have scope over NPs.

1.10.1 Does the apprehensive necessarily attach to a verb or can it also attach to an NP?

This possibility might depend on the morphosyntactic nature of the morpheme, as in Urarina (isolate; Peru; Olawsky 2006) where it is a clitic.

(41) (*Watch out*) *you'll fall!*

(42) (*Watch out for*) *the crocodile!*

1.11 Compatibility with the evidential system

In Hup (Epps 2008: 632), only the reportative is compatible with the apprehensive marker, while no other evidential specifications are.

1.11.1 Is the apprehensive compatible with the evidential markers of the language (if any)?

(43) (*I heard*) *Rob might die.*

(44) (*I witnessed*) *Rob might die.*

2 Precautioning

This section targets the precautioning function, in which the apprehensional clause (PRECautioning) is obligatorily associated with a clause encoding a pre-emptive action, which seeks to prevent the (consequences of the) apprehensional clause. In many languages, the precautioning clause is syntactically subordinated to the preemptive clause. The description of the precautioning construction in the target language should indicate the nature of the link between the preemptive and the precautioning clauses:

- At the syntactic level: the precautioning clause is a syntactically subordinate construction if language-internal criteria for subordinate status apply. Alternatively, the link between the two clauses may rather be of a pragmatic nature;
- At the semantic level: the precautioning clause is causally linked to the preemptive clause (Lichtenberk's PREVENTIVE aka. NEGATIVE PURPOSE subtype), or not (aka. 'IN-CASE' subtype), see §2.2. for a detailed explanation of the distinction.

2.1 Intrinsic undesirability

The following two questions check for the presence of the undesirability component in the potential candidates for a precautioning marker. §2.1.1 is about intrinsically undesirable events, and §2.1.2 and §2.1.3 about semantically positive and neutral events, respectively. As mentioned above in §1.1, if a language has no dedicated marker, the first questions in §2.1.1 might help to find out what construction(s) the target language uses to express the precautioning function.

2.1.1 Is there a construction used to express both a potential undesirable event and an associated preemptive clause?

(45) *I quickly picked up the child from the ground* [*lest the cow trample on him*].

(46) *Alex should repair the damaged fence* [*lest people fall off the cliff*].

2.1.2 Is the reading with typically positive situations like 'smile', 'laugh', 'help', 'find a 100 dollar bill' necessarily negative?

(47) *I put my hand on Martina's mouth lest she laugh.*
Possible context: The protagonists are hiding and making fun of someone, and should not be discovered.
Impossible context: The laughing is desirable.

(48) *They chased Dan away lest he help.*
Possible context: Dan is so clumsy that they prefer her/him not to help.
Impossible context: The helping is desirable.

2.1.3 Is the reading with typically neutral situations like 'walk', 'talk', 'sleep', 'come', 'cook' necessarily negative?

(49) *I kept pinching Marine lest she fall asleep.*
Possible context: An important event where sleeping is not welcome.
Impossible context: The falling asleep is desirable.

(50) *Eva hurried up to leave lest Peter come with her.*
Possible context: Eva doesn't like Peter / want Peter to join.
Impossible context: Peter's coming is desirable to Eva.

2.2 Degree of preventability of an event

Lichtenberk (1995: 298) distinguishes between

- the PREVENTIVE (aka. NEGATIVE PURPOSE, e.g. *He gave them his umbrella* ***so that*** *they do* ***not*** *get wet*)
- and the 'IN-CASE' subfunctions (e.g. *He gave them his umbrella* ***in case*** *it might rain / ?****so that*** *it does not rain*).

In the first type, there is a causal link between the preemptive event and the non-occurrence (prevention) of the undesirable event encoded by the precautionary clause. In the second type, the undesirable event cannot be prevented itself, but its potentially negative consequences can. According to Lichtenberk (1995: 308), at least one language, Martuthunira (Pama-Nyungan; Australia) distinguishes morphologically between the two subfunctions.

2.2.1 Must the undesirable event be fully preventable?

2.2.1.1 **Impersonal verbs** like 'rain', 'snow', 'wind.blow', 'thunder', 'be.an.earth-quake'; existential verbs

(51) *Take your coat lest it rain (#so that it does not rain).*
[vs. *Take your coat lest you get wet (= so as not to get wet).*]

(52) *Check your shoes (before you put them on) lest there be a scorpion inside. (#so that there will be no scorpion inside.)*

2.3 Who is the judge of the undesirability?

An important parameter considered by Green (1989) to distinguish between apprehensive and precautioning constructions is the identity of the judge of the undesirable potential event: in apprehensive constructions, only the speaker can judge the event to be undesirable (and the apprehensional construction has thus a modal value), while in precautioning constructions, the subject of the preemptive clause can, or, in other words, "the speaker simply reports the apprehension of the main clause subject, without himself endorsing or otherwise commenting on the feeling" (p. 168). In precautioning clauses, the undesirability judge is in fact always the subject of the preemptive clause if it is an assertive clause. However, if the preemptive clause is a directive or a modal construction with directive force, then the speaker rather than the subject can be the judge of the potential undesirability, as in *take your coat lest you catch a cold.*

The (non-)availability of the subject to be the judge of undesirability might be considered as a proxy for the distinction between apprehensive and precautioning construction, although this parameter and the obligatory presence of a preemptive clause are theoretically orthogonal. This apparent contradiction comes from the fact that apprehensives tend to be modal (and thus allow only the speaker to be the judge of undesirability) *and* not require a preemptive clause, while, conversely, precautioning markers tend to allow the subject of assertive preemptive clauses to be the judge. The alignment of the two parameters is however not obligatory, so it should not be considered as a test to distinguish apprehensives from precautioning constructions. Once again, the origin of the marker plays a big role in the functions and properties it acquires.

One way to verify the availability of the subject as the judge is to identify the kind of preemptive clause available. If the subject can be the judge, assertive preemptive clauses with third person subject like (54) should be available. When only the speaker can be the judge, the preemptive clause are directives as in (53a)–(53c). In (53a)–(53c), the speaker is the one who anticipates the potential falling event, and the apprehensional has some modal value as in apprehensive constructions. In (54), Peter, who is the subject of the preemptive clause, is also the one anticipating that she could fall, and the example has no modal value.

(53) a. *Tie her shoelaces lest she fall!*
b. *You should tie her shoelaces lest she fall!*
c. *Don't let her go in socks lest she fall!*

(54) *Peter tied Natalia's shoelaces lest she fall.*

2.4 Compatibility with negation

Probably due to diachronic reasons, the precautioning marker might be incompatible with negation markers.

(55) *Keep the door open, otherwise the children will not go out* / ?*lest the children should not go out.*

(56) *Turn off the music, otherwise we might not hear the dogs barking* / *the children calling* / ?*lest we not hear the dogs barking* / *the children calling.*

2.5 Morphosyntax: Subordinating device with apprehensives

Some languages make use of a reported speech construction to use the apprehensive marker in the precautioning function, e.g. in Tariana (Arawakan; Brazil & Colombia; Aikhenvald 2003: 386).

2.5.1 Is the/one of the precautioning construction(s) in the language identical to an apprehensive (§1) with a quotation construction (see the equivalent literal translation in square brackets below the examples)?

(57) *You covered my mouth with your hand, lest they hear me.*
[equivalent literal translation: You covered my mouth, **thinking/saying**: “they might hear you!”]

(58) *You ran away lest he see you.*
[equ. lit. tr.: You ran away, **thinking/saying**: “he might see me!”]

(59) *They don’t eat sugar lest they get bad teeth.*
[equ. lit. tr.: They don’t eat sugar, **thinking/saying**: “we might get bad teeth!”]

2.6 ‘Fear’-complementation function

Lichtenberk (1995) has observed that apprehensionals available in the precautioning function could also be available for what he calls the ‘fear’ function: apprehensional complementizers which appear with clauses that serve as the non-subject complement of a transitive predicate of fear or concern. For instance, (archaic) English *lest* may mark subordinate clauses that complement main clause predicates expressing events of concern like ‘be frightened’, ‘be anxious’, ‘be worried’, etc., as in *I am frightened* LEST *he come.* Expectedly, its combination with main predicates expressing positive emotional states is impossible, e.g. **I am happy* LEST *he come.* Note that most languages have a general complementizer (e.g. *that* in English) available with main predicates expressing fear or concern, as well as any other emotional states (and often other predicate types expressing for instance cognition or perception). English has thus both a general complementizer as in *(I’m happy/I’m worried/I heard)* THAT *he will/might come*, in addition to its apprehensional marker *lest* available with fear or concern predicates only.

The precautioning and the ‘fear’-complementation functions are unified by the semantics of undesirability, and by their subordinated syntax. While there exist three logically possible constellations of available functions for an apprehensional construction, only the first two are attested so far:

- both the precautioning and the 'fear'-complementation functions;
- the precautioning function only;
- the 'fear'-complementation function only.

A language might allow both the precautioning and the 'fear'-complementation function despite the redundancy of the information on the undesirability of the event in the latter function. This is actually precisely the semantic harmony which makes the collocation of a 'fear' predicate and an apprehensional clause so common in languages (see e.g. López-Couso 2007 on English *lest*). By contrast, having an apprehensional marker dedicated to complementizer function with predicates of fear or concern seems non-economical, as a general subordinate marker could fulfill that function, especially if no precautioning marker is available in the language.

It is still worth checking whether unexpected diachronic reasons could have given rise to an apprehensional marker that would only be used in this 'fear' function. The questions below examine the availability of the 'fear' function to the precautioning marker.

2.6.1 Does the precautioning marker – or, very unexpectedly, a marker dedicated to 'fear'-complementation – exclusively introduce clauses that serve as the non-subject complement of a predicate of fear or concern?

(60) *I am afraid that/lest he come.*

(61) *I am happy that/*lest he come(s).*

(62) *I heard that/*lest he come(s).*

Here the researcher will have to use language-internal criteria to determine whether the apprehensional clause is a syntactic complement of the predicate of fear or concern. For instance, in *I'm afraid, he might come* (with an intonational break), the apprehensional clause is not a syntactic complement, as opposed to *I'm afraid, he might come* (with no intonational break), where it is.

2.6.2 Are events of prevention (expressed by verbs like 'forbid', 'impede') also available with the precautioning (or 'fear'-complementation) marker?

(63) *I forbid you to go there* (~*I forbid you* LEST *you go there*).[4]

[4] In Mwotlap (François 2003: 304–305), such a sentence is possible, while it is not in English.

3 Timitive

This section targets the timitive function, i.e. when an apprehensional marker is associated with an NP (and possibly also with a VP, see §3.6). Here, the undesirability is about an entity, and its anticipated dangerous or at least unwelcome behavior.

3.1 Intrinsic undesirability

Undesirable entities may be marked by a general causal marker as in *He ran away* **because of** *the bees*, or by a dedicated causal marker for undesirability, here referred to as timitive, as in *He ran away* **for fear of** *the bees*, with a grammaticalized construction corresponding to English *for fear of* (cf. example 1c). The following two questions thus check for the contexts where the timitive marker may appear: in §3.1.1, with an intrinsically undesirable entity, and in §3.1.2 with typically positively evaluated entities. Prompt (64) might help to find out what the (non)-dedicated construction(s) are that express that an undesirable/feared entity is the cause or justification for an action.

3.1.1 Is there a construction frequently used in contexts where an undesirable entity is mentioned?

(64) *I ran away* ***for fear of/because of*** *the fierce dog.*

3.1.2 Is the reading with a typically positively evaluated entity like 'mother', 'husband' or 'friend' necessarily negative?

(65) *I cooked* ***for fear of/because of****mother/husband.*
Possible context: my mother/husband will hit me if I have not cooked food for her/him.
Impossible context: It is my mother/husband's birthday and I am happy to cook for her/him.

(66) *He ran away* ***for fear of/because of*** *his friend.*
Possible context: the friend has become aggressive.
Impossible context: the friend has always been a good friend and it has not changed.

3.2 Verb types

3.2.1 Semantic role of the entity: Entity to be avoided or stimulus of fear

This question and the next one test the detailed semantics of the timitive. The marker might encode *avoidance* or *fear* of an entity. In both cases, the entity is undesirable, but each has its own very distinct role:

- in the case of *avoidance*, its role is to be the avoided entity. As a consequence, the main event is a preemptive action to avoid that undesirable entity, as in *I ran away* ***for fear of*** *the snake/****to avoid*** *the snake*;
- in the case of *fear*, by contrast, its role is to be a stimulus for a behaviour or reaction. This is why the main event is not necessarily construed as a preemptive action, and might for instance be a physiological reaction triggered by fear, as in *I fainted* ***for fear of*** *the snake/*****to avoid*** *the snake.*

That semantic distinction can thus be tested with verbs of physiological reactions indicative of fear but not conducive to avoiding the entity. In Marrithiyel, the timitive seems to require a preemptive action (Green 1989: 58), unlike in Ese Ejja, where a physiological reaction is also available (Vuillermet 2018a: 284). The prompts in (69) specifically test whether it can also encode disgust, since the emotion of disgust is close to fear.[5]

(67) a. *He ran away* ***for fear of/to avoid*** *the bees.*
  b. *He hid* ***for fear of/to avoid*** *the bear.*

(68) a. *He burst into tears* ***for fear of/*to avoid*** *the bees*
  b. *He fainted* ***for fear of/*to avoid*** *the bear.*

(69) a. *I went away* ***out of disgust for****/?****for fear of*** *the vomit.*
  b. *I vomited* ***out of disgust for****/?****for fear of*** *the stinky fish.*

3.3 Possible entities

Up to now the literature has not reported restrictions on the nature of the undesirable entity, but it is not excluded that a timitive in a given language could be unavailable with certain types of entities. The following questions target the availability of (in)animate and (non-)human entities.

[5]Fear and disgust are both human reactions triggered by a threatening situation or entity.

(70) *Rebecca left* ***for fear of*** *her mother-in-law.*

(71) *He crossed the bridge* ***for fear of*** *the plants* (e.g. *nettle*).

(72) *I went this way* ***for fear of*** *the thorny scrub /the excrement.*

(73) *He stayed in* ***for fear of*** *the rain/thunder/snow/wind/cold.*

3.4 Possible TAM situations

Up to now the literature has not reported restrictions on the TAM values of the main event, but it is not excluded that a timitive in a given language could be unavailable with some TAM markers, which is what the following prompts test for. (The previous prompts are all in the past to test for only one parameter at a time.)

(74) *I am preparing the dinner* ***for fear of*** *my husband/mother.*

(75) *He will come back quickly* ***for fear of*** *the night.*

3.5 Compatibility with interrogative words

There should be no restrictions on whether interrogative words are available with any NP markers, but it cannot be excluded; this is what the following prompt tests for.

(76) ***For fear of*** *what did you run away?*

3.6 Multifunctional apprehensionals

The last two questions are potentially redundant with §1.10 above, but one can imagine a language with more than one apprehensional, where the various functions of each apprehensional might overlap. Note that §3.6.2 might be the necessary step for a timitive to acquire the apprehensive function.

3.6.1 Can the timitive attach to NPs only or can it also attach to nominalized/subordinated verbs/VPs/clauses?

(77) *He screamed* ***for fear of*** *falling / [the baby falling].*

(78) *He ran away* ***for fear of*** *[being eaten/attacked/harmed.]*

3.6.2 Can the timitive NP form a non-verbal clause? (cf. in Manambu [mana1298] (Ndu; Papua New Guinea), Aikhenvald 2008: 153–154)

Note that such non-verbal clauses are similar to apprehensive clauses in their illocutionary reading (warning or threat).

(79) ***Watch out for*** *the tiger/the excrement!*

(80) ***Beware of*** *my strength!*

Abbreviations

3A	3rd agent
ABS	absolutive
APPR	apprehensive
APPRL	(general) apprehensional marker
ERG	ergative
FUT	future
IRR	irrealis
MID	middle
NEG	negation
NIRR	non-irrealis
NSG	non-singular
O	object
OMG	oh my God
POSS	possessive
PREC	precautioning
PRES	present
RDP	reduplication
REAL	realis
S	subject
SBJ	subject
SG	singular
TIM	timitive

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Apprehensional constructions in a cross-linguistic perspective

The functional domain of Apprehensionality encompasses grammatical markers or constructions which conventionally encode or pragmatically implicate that the situation described by the clause is an undesirable possibility. Its main manifestations discussed in this volume are (i) apprehensives, which are modal markers which can occur in main clauses, and thus could serve to translate English *might* in *Don't go near this dog, it might bite* (but also add a component of undesirability), and (ii) precautionary subordinators such as English *lest* in *Don't go near this dog, lest it bite.* As just illustrated, such elements tend to occur in warnings and are often but not necessarily accompanied by an instruction as to how to avoid the undesirable situation. Although wide-spread cross-linguistically, they have received very little attention in the linguistic literature and their discussion has been mostly restricted to brief sections in language-specific discussions of some lesser studied languages. This volume addresses this gap by presenting in-depth discussions of apprehensional markers in individual languages (and in some contributions, language groups), by experts on these languages and based on first-hand data. The editors' introduction offers a review of the existing literature, including of the varied terminology used and of the main research questions arising. The individual papers present detailed investigations of the semantic and pragmatic properties of the construction(s) in question, their syntactic status, and in many cases also their diachrony. The final contribution is a questionnaire for the in-depth exploration of variation within apprehensional constructions. The volume is aimed at typologists, semanticists interested in modality, and general linguists.